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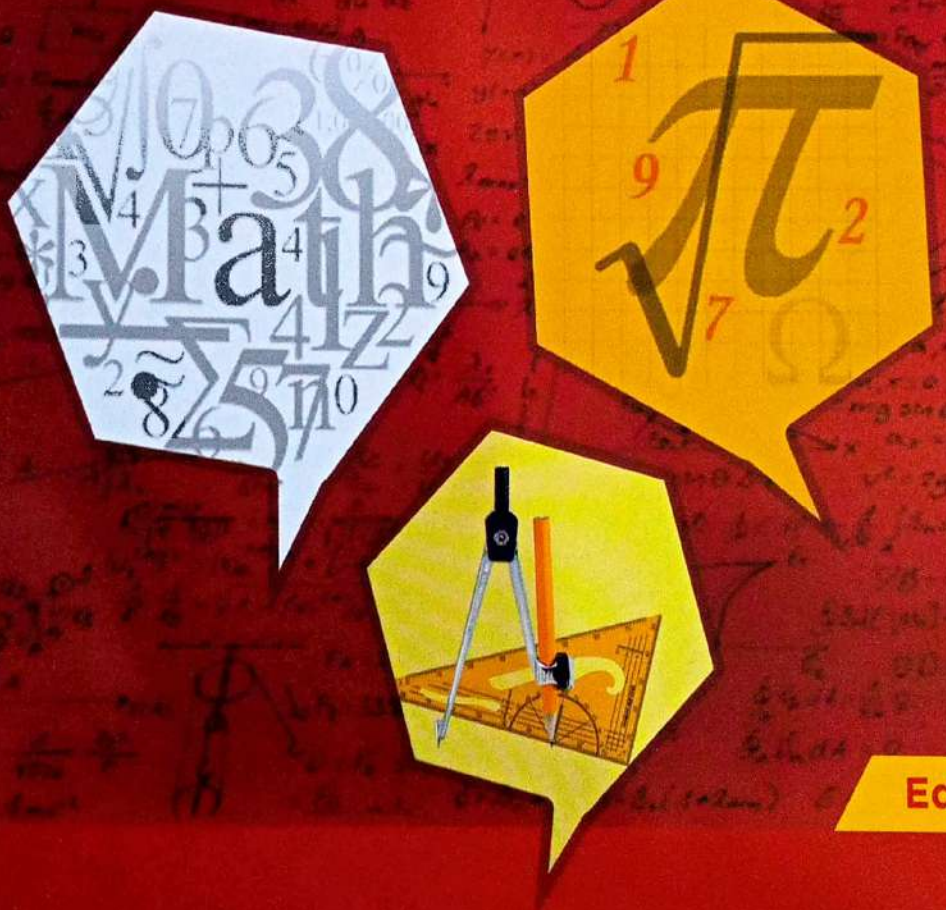


Edition 2020

MATHEMATICS

FUNCTIONS

1

Function can be easily defined with the help of the concept of mapping. Let X and Y be any two non-empty sets. "A function from X to Y is a rule or correspondence that assigns to each element of set X, one and only one element of set Y". Let the correspondence be 'f' then mathematically we write $f : X \to Y$ where $y = f(x)$, $x \in X$ and $y \in Y$. We say that 'y' is the image of 'x' under 'f'(or x is the pre image of y). Two things should always be kept in mind:

- A mapping $f : X \to Y$ is said to be a function if each element in the set X has it's image in set Y. It is possible that a few elements in the set Y are present, which are not the images of any element in set X.
- Every element in set X should have **one and only one image**. That means it is impossible to have more than one image for a specific element in set X. Functions can't be multi-valued (A mapping that is multi-valued is called a relation from X to Y).

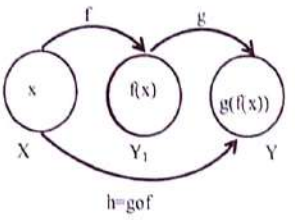
DOMAIN, CO-DOMAIN AND RANGE

Set 'X' is called domain of the function 'f'. Set 'Y' is called the co-domain of the function 'f'.

Set of images of different elements of set X is called the range of the function 'f'.

Composite Functions

Let us consider two functions, $f : X \to Y_1$ and $g : Y_1 \to Y$. We define function $h : X \to Y$ such that $h(x) = g(f(x))$. To obtain h(x), we first take the f-image of an element $x \in X$ so that $f(x) \in Y_1$, which is the domain of g(x). Then take g-image of f(x), i.e g(f(x)) which would be an element of Y. The adjacent figure clearly shows the steps to be taken.



The function 'h' defined above is called the composition of f and g and is denoted by gof.

Thus $(gof)x = g(f(x))$.Clearly Domain (gof) = $\{x : x \in$ Domain (f) , $f(x) \in$ Domain(g)$\}$

Similarly we can define,$(fog)x = f(g(x))$ and Domain (fog) = $\{x : x \in$ Domain (g) , $g(x) \in$ Domain (f)$\}$. In general $fog \neq gof$.

Inverse Function

If $f : X \to Y$ be a function defined by $y = f(x)$ such that f is both one-one and onto, then there exists a unique function $g : Y \to X$ such that for each $y \in Y$, $g(y) = x$ if and only if $y = f(x)$. The function g so defined is called the **inverse of f** and denoted by f^{-1}. Also if g is the inverse of f, then f is the inverse of g and the two functions f and g are said to be inverses of each other. The condition for existence of inverse of a function is that the function must be **one-one and onto**. Whenever an inverse function is defined, the range of the original function becomes the domain of the inverse function and domain of the original function becomes the range of the inverse function. Note that $fof^{-1}(x) = f^{-1}of(x) = x$ always and roots of the equation $f(x) = f^{-1}(x)$ would always lie on the line $y = x$. f and f^{-1} are symmetric about the line $y = x$.

Even and Odd Functions

If $f : X \to Y$ is a real valued function such that for all $x \in D \Rightarrow -x \in D$ (where D = domain of f) and if $f(-x) = f(x)$ for every $x \in D$, then f is said to be an even function and if $f(-x) = -f(x)$, then f is said to be odd function. Even functions are symmetric about the y-axis (i.e. if (x, y) lies on the curve, then (-x, y) also lies on the curve) and odd functions are symmetric about the origin (i.e. if (x, y) lies on the curve, then (-x, -y) also lies on the curve).

Periodic Function

A function $f : X \to Y$ is said to be a periodic function if there exists a least positive real number p such that $f(x + p) = f(x)$, for all $x \in X$. The least of all such positive numbers p is called the **principal period** or simply period of f. All periodic functions can be analyzed over an interval of one period within the domain as the same pattern shall be repetitive over the entire domain.

The following points are to be remembered:

If f(x) is periodic with period p, then af(x) + b where $a, b \in R$ ($a \neq 0$) is also periodic with period p.

- If f(x) is periodic with period p, then f(ax + b) where $a, b \in R$ ($a \neq 0$) is also period with period $\frac{p}{|a|}$.
- Let f(x) has period p = m/n (m, n ∈ N and co-prime) and g(x) has period q = r/s (r, s ∈ N and co-prime) and let t be the LCM of p and q i.e. $t = \frac{\text{LCM of } (m, r)}{\text{HCF of } (n, s)}$.

Then t shall be the period of f + g provided there does not exist a positive number k (< t) for which $f(k + x) = g(x)$ and $g(k + x) = f(x)$, else k will be the period. The same rule is applicable for any other algebraic combination of f(x) and g(x).

ILLUSTRATIVE PROBLEMS

1. If A = {a, b, c} and B = {b, c, d} then evaluate
(i) A∪B, A∩B, A – B and B – A
(ii) A × B and B × A

Sol. (i) $A \cup B = \{x: x \in A \text{ or } x \in B\} = \{a, b, c, d\}$

$A \cap B = \{x: x \in A \text{ and } x \in B\} = \{b, c\}$

$A - B = \{x: x \in A \text{ and } x \notin B\} = \{a\}$

$B - A = \{x: x \in B \text{ and } x \notin A\} = \{d\}$

(ii) $A \times B = \{(x, y): x \in A \text{ and } y \in B\}$

$= \{(a, b), (a, c), (a, d), (b, b), (b, c), (b, d), (c, b), (c, c), (c, d)\}$

$B \times A = \{(x, y): x \in B \text{ and } y \in A\}$

$= \{(b, a), (b, b), (b, c), (c, a), (c, b), (c, c), (d, a), (d, b), (d, c)\}$

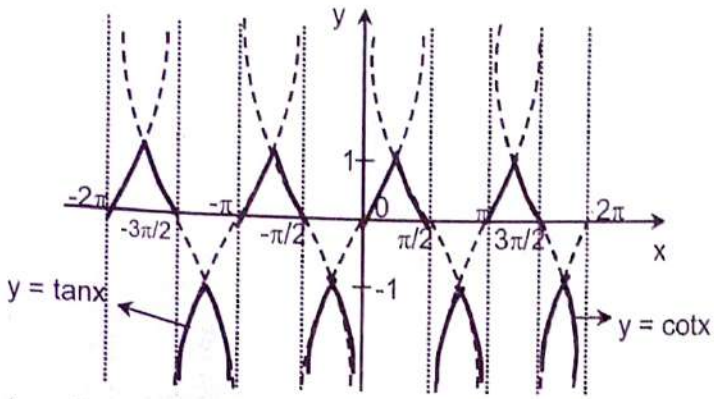
Note that $A \times B \neq B \times A$.

2. Find the range of the following functions,

(i) $f(x) = \min\{\tan x, \cot x\}$, $\forall\, x \in R$

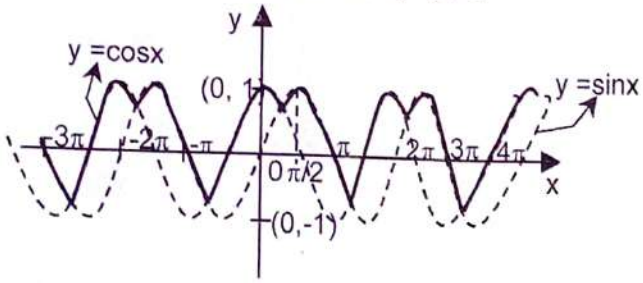
(ii) $g(x) = \max\{\sin x, \cos x\}$, $\forall\, x \in R$.

Sol. (i)



Bold line shows f(x), hence range is $(-\infty, -1] \cup [0, 1]$.

(ii)



Bold line shows f(x). Hence range of f(x) is $\left[-\frac{1}{\sqrt{2}},\ 1\right]$.

3. Find the number of solutions of $x \in [0, \pi]$ satisfying the equation $\left(\log_{\sqrt{3}} \tan x\right)\left(\sqrt{\log_{\sqrt{3}} 3\sqrt{3} + \log_{\tan x} 3}\right) = -1$ is/are

Sol. $\tan x > 0,\ \tan x \neq 1$

$$\left(\log_{\sqrt{3}} \tan x\right)\left(\sqrt{\frac{\log_{\sqrt{3}} (\sqrt{3})^3}{\log_{\sqrt{3}} \sqrt{3}} + \frac{\log_{\sqrt{3}} 3}{\log_{\sqrt{3}} \tan x}}\right) = -1$$

Let $\log_{\sqrt{3}} \tan x = t < 0$

$$t\sqrt{3 + \frac{2}{t}} = -1$$

$$t^2\left(3 + \frac{2}{t}\right) = 1$$

$$3t^2 + 2t - 1 = 0$$

$\Rightarrow t = -1$ and $t = \frac{1}{3}$ (not possible)

$$\log_{\sqrt{3}} \tan x = -1$$

$\Rightarrow \tan x = \frac{1}{\sqrt{3}}$

$\Rightarrow x = \frac{\pi}{6}$ only in $[0, \pi]$.

Hence, only one solution is possible

4. If $f(x) = (x-\alpha)^3 (x-\beta)^3$ then find the nature of roots of $f'(x) = f''(x)$.

Sol. $f(x) = (x-\alpha)^3 (x-\beta)^3$

$f'(x) = 3(x-\alpha)^2 (x-\beta)^2 [2x - (\alpha+\beta)]$

$f(x) - f'(x) = (x-\alpha)^2 (x-\beta)^2 [x^2 - (\alpha+\beta)x + \alpha\beta - 3(2x - (\alpha+\beta))]$

$\Rightarrow$ all the roots of $f(x) = f'(x)$ are real

$\Rightarrow$ all the roots of $f'(x) - f''(x)$ are real and distinct.

5. If $f(x)$ be a polynomial of degree n such that $1 + f(x) = \frac{f(x-1)+f(x+1)}{2}$ $\forall x \in R$, then the possible value of n is

Sol. Suppose $f(x)$ has degree < 2

$\Rightarrow f(x) = bx + c$

$\Rightarrow bx + c + 1 = bx + c$ not possible

Suppose $f(x)$ has degree > 2

$\Rightarrow f(x) = ax^n + bx^{n-1} + cx^{n-2} + \ldots\ldots$ $(a \neq 0)$

Comparing coefficients of x^{n-2}, we get $c = \frac{a(n)(n-1)+2c}{2}$

$\Rightarrow a = 0$

$\Rightarrow f(x)$ has degree 2.

Hence, the degree of polynomial is 2

6. If $f : I \to I$ be a function (I is set of integers) such that $f(0) = 1$, $f(f(n)) = f(f(n+2)+2) = n$. Then find the value of $f(3)$

Sol. $f(n) = 1 - n$ using $f(-n+2) = f(f(n-1)) = n - 1$

and $f(n+1) = f(f(-n+2)+2) = 1 - (n+1)$

and the results that $f(1) = 0$, $f(2) = -1$

Hence, $f(3) = -2$

7. Consider the function g(x) defined as

$g(x).\left(x^{\left(2^{2011}-1\right)}-1\right) = (x+1)(x^2+1)(x^4+1)\ldots\left(x^{2^{2010}}+1\right)-1$. Then find the value of g(2)

Sol. $g(x).\left(x^{\left(2^{2011}-1\right)}-1\right) = \frac{(x-1)(x+1)(x^2+1)(x^4+1)\ldots\left(x^{2^{2010}}+1\right)}{(x-1)} - 1$

After simplification $g(x) = \frac{x}{x-1}$

$\Rightarrow g(2) = 2$

SOLVED PROBLEMS

1. The domain of $\dfrac{\left[\sin^{-1}x^4\right]+\left[\left[x-2\tan^{-1}x\right]\right]+\sqrt{\sin(\cos x)}}{\left\{3x^2-7\right\}+a^{\sqrt{\sin x}}+\ln\cos\dfrac{1}{\sqrt{1-x^2}}}$, (where {.} and [.] denotes fractional and integral part of x), is

(A) $\left(-2,\ \sqrt{2}\right)$ (B) (0, 1)

(C) (–1, 1) (D) (0, 5/4)

Sol. **(B)** $-1 \le x^4 \le 1$ ..... (1)

$\sin(\cos x) \ge 0$ ..... (2)

$\sin x \ge 0$ ..... (3)

$\cos\dfrac{1}{\sqrt{1-x^2}} > 0$ ..... (4)

On solving (1) and (2), we get $0 < x < 1$ which satisfies (3) and (4)

2. If $f(x) = \sin x + \cos ax$ is periodic then a is

(A) 2 (B) π

(C) $\dfrac{\pi}{2}$ (D) $\sqrt{2}$

Sol. **(A)** Let λ be the period of $\sin x + \cos ax$ then

$\sin(\lambda + x) + \cos a(\lambda + x) = \sin x + \cos ax$ for all x

In this identity, putting $x = 0$ and $x = -\lambda$, we get

$\sin\lambda + \cos a\lambda = 1$ and $1 = -\sin\lambda + \cos\lambda$

solving these equation, we get $\sin\lambda = 0$ and $\cos a\lambda = 1$

hence $\lambda = n\pi$ and $a\lambda = 2m\pi$. Where m, n are non-zero integers

hence $\dfrac{a\lambda}{\lambda} = \dfrac{2m\pi}{n\pi}$ or $a = \dfrac{2m}{n}$ (since $\lambda \ne 0$)

3. If $y = f(x)$ is symmetric about the lines $3x + 4y + 1 = 0$ and $4x - 3y - 7 = 0$, then it must be symmetric about

(A) (1, 1) (B) (1, –1)

(C) (0, 0) (D) (1, 0)

Sol. **(B)** If a function is symmetric about two mutually perpendicular lines, it must be symmetric about their point of intersection.

4. Let f(x) and g(x) be bijective functions where $f: \{a, b, c, d\} \to \{1, 2, 3, 4\}$ and $g: \{3, 4, 5, 6\} \to \{w, x, y, z\}$ respectively. The number of elements in the range set of g (f (x)) is

(A) 1 (B) 2

(C) 3 (D) 4

Sol. **(B)** Range of f(x) for which g(f(x)) is defined is {3, 4}

Hence domain of g(f(x)) has two elements

$\therefore$ Range of g(f(x)) also has two elements

5. Solution set of $f^{-1}(x) = x$ is a proper subset of the solution set of $f(x) = f^{-1}(x)$; then

(A) solution set of $f^{-1}(x) = x$ is identical to solution set of $f(x) = x$

(B) $f(x) = f^{-1}(x)$ has infinite solutions

(C) f(x) and $f^{-1}(x)$ are identical

(D) all of the above

Sol. **(C).** Curves of $y = f(x)$ and $y = f^{-1}(x)$ will either intersect on $y = x$ line or will coincide.
Since solution set of $f(x) = f^{-1}(x)$ is a proper superset of solutions of $f(x) = x$.
$\therefore$ $f(x)$ and $f^{-1}(x)$ must be identical.

6. Let $f(x) = \ln\left(\frac{x^2+e}{x^2+1}\right)$ then the range of $g(x) = \sqrt{\sin f(x)} + \sqrt{\cos f(x)}$ is

(A) $\left(1, 2^{3/4}\right]$ (B) $\left[1, 2^{1/2}\right)$

(C) $\left[1, 2^2\right]$ (D) $(1, 2)$

Sol. **(A)** $\frac{x^2+e}{x^2+1} \in (1, e]$, $f(x) = \ln\left(\frac{x^2+e}{x^2+1}\right) \in (0, 1]$

$g(\theta) = \sqrt{\sin\theta} + \sqrt{\cos\theta}$ where $\theta \in (0, 1]$

$g'(\theta) = \frac{\cos\theta}{2\sqrt{\sin\theta}} - \frac{\sin\theta}{2\sqrt{\cos\theta}} = 0 \Rightarrow \theta = \frac{\pi}{4}$

$g(\theta)$ has an extremum point $\theta = \frac{\pi}{4}$

$g(0) = 1 \Rightarrow g(1) = \sqrt{\sin 1} + \sqrt{\cos 1}$

While $g\left(\frac{\pi}{4}\right) = 2^{3/4}$

$\Rightarrow g(1) > g(0) = 1$

Since $g\left(\frac{\pi}{2}\right) = g(0) = 1$

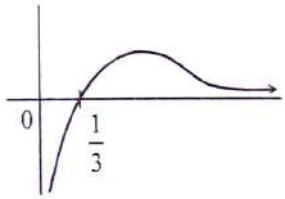
$g(x) = \left(1, 2^{3/4}\right]$

7. The set A, for which the function $f: (0, \infty) \to A$, $f(x) = \frac{3x-1}{x^2+x}$ is a surjective function is

(A) $(0, 1]$ (B) $(-\infty, 0)$

(C) $(-\infty, \infty)$ (D) none of these

Sol. **(D)** See in the figure



8. Let $f: R \to R$ and $f\left(x+\frac{5}{6}\right) + f(x) = f\left(x+\frac{1}{2}\right) + f\left(x+\frac{1}{3}\right)$, then

(A) $f(x)$ is periodic (B) $f(x)$ is even

(C) $f(x+2) - f(x+1) = f(x+1) - f(x)$ (D) none of these

Sol. **(C).** We have observed that this function has a property that if A, B, C and D are four real numbers such that $A + B = C + D$,
Then $f(A) + f(B) = f(C) + f(D)$
Now $x + 2 - (x + 1) = x + 1 - x$

9. Given that $f(x)+f\left(\frac{1}{1-x}\right)=\frac{2(1-2x)}{x(1-x)}$, $x \in R, x \neq 0, 1$ then the given f(x) is

(A) one-one
(B) many one
(C) f(0) = 1
(D) none of these

Sol. **(A)** $f(x)+f\left(\frac{1}{1-x}\right)=\frac{2}{x}-\frac{2}{1-x}$ ..... (1)

Now replace x by $\frac{1}{1-x}$ we get

$f\left(\frac{1}{1-x}\right)+f\left(1-\frac{1}{x}\right)=2(1-x)-2\left(1-\frac{1}{x}\right)=-2x+\frac{2}{x}$ ..... (2)

Replace x by $1-\frac{1}{x}$ we get

$f\left(1-\frac{1}{x}\right)+f(x)=\frac{2x}{x-1}-2x$ ..... (3)

From equation (1) – (2), we get

Now, $f(x)-f\left(1-\frac{1}{x}\right)=2x-\frac{2}{1-x}$ ..... (4)

From equation (3) + (4), we get

Again, $2f(x)=\frac{2x}{x-1}-\frac{2}{1-x}$

$f(x)=\frac{x+1}{x-1}$ which is a one-one function

10. The polynomial P(x) is such that for any polynomial q(x) we have p(q(x)) = q(p(x)). Then p(x) is

(A) even
(B) odd
(C) of even degree
(D) of odd degree

Sol. **(B, D).** Let q (x) = k, where $k \in R$

and $p(x) = a_0 + a_1x + a_2x^2 + \ldots + a_nx^n$

$\Rightarrow a_0 + a_1k + a_2k^2 + \ldots + a_nk^n = k$

$\Rightarrow a_0, a_2, a_3 - a_n = 0$ and $a_1 = 1$

$\Rightarrow p(x) = X.$

11. Let $f(x) = \ln(2x - x^2) + \sin\frac{\pi x}{2}$, then

(A) Graph of f is symmetrical about the line x = 1
(B) Graph of f is symmetrical about the line x= 2
(C) maximum value of f is 1
(D) minimum value of f does not exist .

Sol. **(A), (C), (D)**

Obviously $f(1 + \alpha) = f(1 - \alpha) \quad \forall \alpha \in (0, 1)$.

Domain of f is (0, 2)

Hence (A) is true.

Maxima of $\ln(2x - x^2)$ as well as that of $\sin\left(\frac{\pi x}{2}\right)$ occur at x = 1.

Hence (C) is true.

As $x \to 0+$ or $x \to 2-$, $f(x) \to -\infty$.

Hence (D) is true.

12. If $f(x) = \log_{[x-1]}\frac{|x|}{x}$, where [.] denotes greatest integer function, then

(A) domain of $f = (2, \infty)$
(B) range of $f = \{0, 1\}$
(C) domain of $f = [3, \infty)$
(D) range of $f = \{0\}$

Sol. **(C), (D)**

$f(x) = \log_{[x-1]} \operatorname{sgn} x$. Domain of $f = [3, \infty)$. Range of $f = \{0\}$.

13. $f: R \to R$ be a periodic function such that $\{f(x) | x \in N\}$ is an infinite set, then period of the function cannot be

(A) rational
(B) irrational
(C) integer
(D) π

Sol. **(A, C)**

Let $f(T + x) = f(x)$ and $T \in R$

Suppose $T = \frac{p}{q} \Rightarrow f(qT + x) = f(x) \Rightarrow f(p + x) = f(x)$

$\Rightarrow \{f(n) | n \in N\}$ cannot contain infinite elements.

Comprehension

Let $f(x)$ be a real valued function satisfying the functional equation $f(x) + f(1 - x) = k$ for all $x \in Q$, where k is a constant quantity. Let m be a positive integer. Put $x = \frac{r}{m+1}$ in the given equation, we get

$$f\left(\frac{r}{m+1}\right) + f\left(\frac{m+1-r}{m+1}\right) = k$$

$$\Rightarrow \sum_{r=1}^{m} f\left(\frac{r}{m+1}\right) + \sum_{r=1}^{m} f\left(\frac{m+1-r}{m+1}\right) = mk$$

$$\Rightarrow \sum_{r=1}^{m} f\left(\frac{r}{m+1}\right) + \sum_{t=1}^{m} f\left(\frac{t}{m+1}\right) = mk \qquad \text{(Putting } m + 1 - r = t\text{)}$$

$$\therefore 2\sum_{r=1}^{m} f\left(\frac{r}{m+1}\right) = mk \Rightarrow \sum_{r=1}^{m} f\left(\frac{r}{m+1}\right) = \frac{mk}{2}.$$

14. If $f(x) = \frac{4^x}{4^x + 2}$, where $x \in Q$, then $f\left(\frac{1}{2007}\right) + f\left(\frac{2}{2007}\right) + \ldots + f\left(\frac{2006}{2007}\right)$ equals to

(A) 1003
(B) 2006
(C) 2007
(D) none of these

Sol. **(A)**

15. If $f(x) = \frac{3^{x-3}}{3^{1-x} + 3^x}$ for all $x \in Q$, then the value of the sum $f\left(\frac{1}{55}\right) + f\left(\frac{2}{55}\right) + \ldots + f\left(\frac{54}{55}\right)$ is

(A) 1
(B) 27
(C) 54
(D) 55

Sol. **(A)**

16. If $f(x) = \frac{a^x}{a^x + \sqrt{a}}$ $(a > 0)$, then $\sum_{r=1}^{2n-1} 2f\left(\frac{r}{2n}\right)$ is equal to

(A) 1

(B) 2n

(C) 2n – 1

(D) $\frac{(2n-1)a}{2}$

Sol. **(C)**

Assertion Reason

17. Let f: X → Y be a function defined by $f(x) = 2\sin\left(|x| + \frac{\pi}{4}\right) - \sqrt{2}\cos|x|$, then

Statement 1: The set X is $\left[0, \frac{\pi}{2}\right] \cup \left[\pi, \frac{3\pi}{2}\right]$ for f(x) to be one-one function.

because

Statement 2: $f'(x) \le 0 \ \forall \ x \in \left[\pi, \frac{3\pi}{2}\right]$.

(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1
(B) Statement -1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement -1 is True, Statement -2 is False
(D) Statement -1 is False, Statement -2 is True

Sol. **(D)**

Since $f(0) = f(\pi)$, hence f(x) is many one function.

For $x \in \left[\pi, \frac{3\pi}{2}\right]$, $f(x) = \sqrt{2}\sin x \Rightarrow f'(x) = \sqrt{2}\cos x \le 0 \ \forall \ x \in \left[\pi, \frac{3\pi}{2}\right]$.

18. Match the following:

	Column – I		Column – II
(A)	f: $R \to \left[\frac{\pi}{4}, \pi\right)$ and $f(x) = \cot^{-1}(2x - x^2 - 2)$, then f(x) is	(p)	one-one
(B)	f: $R \to R$ and $f(x) = e^{ax}\sin bx$ where $a, b \in R^+$, then f(x) is	(q)	into
(C)	f: $R^+ \to [2, \infty]$ and $f(x) = 2 + 3x^2$, then f(x) is	(r)	many-one
(D)	f: $X \to X$ and $f(f(x)) = x \ \forall \ x \in X$, then f(x) is	(s)	onto
		(t)	bijective

Sol. **(A) → (q, r); (B) → (r, s); (C) → (p, q); (D) → (p, s, t)**

(A) f'(x) changes sign in the neighbourhood of x = 2.

$2x - x^2 - 2 = -1 - (x-1)^2 \le -1$

$\Rightarrow \cot^{-1}(2x - x^2 - 2) \in \left[\frac{3\pi}{4}, \pi\right)$.

(B) $-e^{ax} \le e^{ax}\sin bx \le e^{ax} \Rightarrow$ (iii), (iv).

(C) $f'(x) > 0$ in $x \in R^+ \Rightarrow$ (i)

$f(x) \in (2, \infty)$ in $x \in R^+ \Rightarrow$ (ii)

(D) Let $X \equiv (x_1, x_2, \ldots, x_n)$

Let $f(x_1) = x_2 \Rightarrow f(f(x_1)) = f(x_2)$

$\Rightarrow f(x_2) = x_1 \Rightarrow$ (i), (iv).

Answer Q.19, Q.20 and Q.21 by appropriately matching the information given in the three columns of the following table.

Match the following Column(s)

Column–I (Real valued function)	Column–II (Domain)	Column–III (Range)
(I) $f(x)=\sin\left(\log\left(2x+\sqrt{1+x^2}\right)\right)$	(i) $R-\{2\}$	(P) $[-1, 1]$
(II) $f(x)=\frac{(x^3-3x+1)(x-2)}{(x-2)}$	(ii) $\{-1, 1\}$	(Q) $\left[0, \sqrt{3}\right)$
(III) $f(x)=\sqrt{24\{x\}-5-16\{x\}^2}$	(iii) $\left[\frac{1}{4}, 1\right)$	(R) $\left\{\frac{\pi}{2}\right\}$
(IV) $f(x)=\sin^{-1}[x]+\sec^{-1}\left(\frac{2x}{1+x^2}\right)$ (where [.] denotes the greatest integer function)	(iv) $[-3, 3]$	(S) $[-17, 19]$

19. Which of the following combination is **CORRECT**?
(A) (I) (i) (P) (B) (I) (iv) (P)
(C) (I) (i) (S) (D) (IV) (i) (P)

Sol. **(B)**

20. Which of the following combination is **CORRECT**?
(A) (III) (iii) (Q) (B) (III) (iii) (S)
(C) (IV) (i) (Q) (D) (III) (i) (R)

Sol. **(A)**

21. Which of the following combination is **INCORRECT**?
(A) (II) (iv) (S) (B) (IV) (ii) (R)
(C) (I) (iv) (S) (D) (III) (iv) (S)

Sol. **(D)**

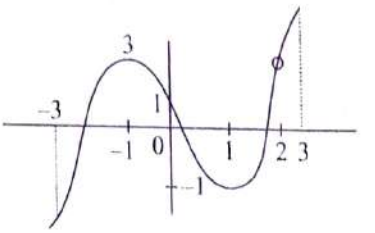
19.-21. (I) $-\infty < \ln\left(2x+\sqrt{1+x^2}\right) < \infty$

So, $-1 \le \sin\left(\ln\left(2x+\sqrt{1+x^2}\right)\right) \le 1$

(II) $y = x^3 - 3x + 1$

$y' = 3x^2 - 3$

Range $= [f(-3), f(3)] = [-17, 19]$



(III) Let $\{x\} + t$

So, $f(t) = \sqrt{24t-5-16+2}$

$\because\ 24t - 5 - 16t^2 \ge 0$

$\frac{1}{4} \le t \le \frac{5}{4}$ and $0 \le t < 1$

So, $\frac{1}{4} \le t < 1$

$\because$ f(t) is an increasing function, so range $\in \left[f\left(\frac{1}{4}\right), f(1)\right], \left[0, \sqrt{3}\right)$

$[x] = -1, 0, 1$ and $\frac{2x}{1+x^2} = \pm 1$

$\therefore$ Df $\in \{-1, 1\}$

Range $\in \{f(-1), f(1)\} = \left\{-\frac{\pi}{2}+\pi, 0+\frac{\pi}{2}\right\} = \left\{\frac{\pi}{2}\right\}$

22. Let $f(x) = \begin{cases} x^3 & ; \ -1 \le x \le -\frac{1}{2} \\ x^2+1 & ; \ -\frac{1}{2} < x < 0 \end{cases}$, if the number of solution of $f(x) = f^{-1}(x)$ is k, then the value of k is _____

Sol. **(0.00)**

Graph of f(x) and $f^{-1}(x)$ do not intersect each other

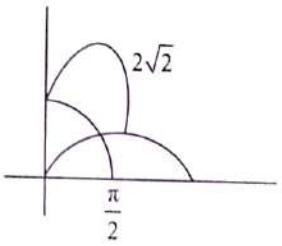
23. If ℓ_1 be the minimum value of $\frac{1}{|\sin x|}+\frac{1}{|\cos x|}$, then $[\ell_1]$ is _____ (where [.] denotes the greatest integer function)

Sol. **(2.00)**

Using AM $\ge$ HM

$$\frac{|\sin x|+|\cos x|}{2} \ge \frac{2}{\frac{1}{|\sin x|}+\frac{1}{|\cos x|}} \Rightarrow \frac{|\sin x|+|\cos x|}{4} \ge \frac{1}{\frac{1}{|\sin x|}+\frac{1}{|\cos x|}}$$

$$\frac{1}{|\sin x|}+\frac{1}{|\cos x|} \ge \frac{4}{|\sin x|+|\cos x|} \Rightarrow \text{So, } \left(\frac{1}{|\sin x|}+\frac{1}{|\cos x|}\right)_{\min} = 2\sqrt{2}$$



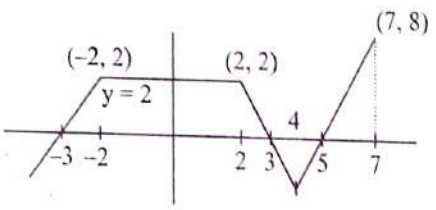
24. If f(x) is a polynomial of degree 4 with real co-efficients such that f(x) = 0 is satisfied by x = 1, 2, 3 only, then $\frac{3+f'(1)f'(2)f'(3)}{100}$ is _____

Sol. **(0.03)**

Clearly one of the roots are repeated twice

So, one of f'(1), f'(2) or f'(3) is 0

25. The graph of the function y = f(x) is given below, then if number of solution of the equation f(f(x)) = 0 is ℓ, then $\frac{\ell}{100}$ is _____



Sol. **(0.04)**

f(x) = 0 → −3, 3, 5

So, f(x) = −3, 3, 5 number of solutions

2 + 1 + 1 = 4

2

LIMIT CONTINUITY AND DIFFERENTIABILITY

LEFT AND RIGHT LIMITS

Let $y = f(x)$ be a given function, and $x = a$ is the point under consideration. Left tendency of $f(x)$ at $x = a$ is called it's left limit and right tendency is called it's right limit.

Left tendency (left limit) is denoted by $f(a-0)$ or $f(a-)$ and right tendency (right limit) is denoted by $f(a+0)$ or $f(a+)$ and are written as

$$\left.\begin{aligned} f(a-0) &= \lim_{h\to 0} f(a-h) \\ f(a+0) &= \lim_{h\to 0} f(a+h) \end{aligned}\right\} \text{ where 'h' is a small positive number.}$$

Thus for the existence of the limit of $f(x)$ at $x = a$, it is necessary and sufficient that $f(a-0) = f(a+0)$, if these are finite or $f(a-0)$ and $f(a+0)$ both should be either $+\infty$ or $-\infty$.

Reasons for non-existence of the limit

From the discussion so far, it is clear that $\lim_{x\to a} f(x)$ will not exist due to any of these three reasons; (a). $f(x)$ is not defined in the neighborhood of $x = a$. For example, consider $\lim_{x\to \pi/2} \sec^{-1}(\sin x)$

(b) $f(x)$ doesn't have a unique tendency. For example, consider $\lim_{x\to\infty} (\cos(1/x))$.

(c) Left and right tendencies of $f(x)$ are not the same. For example, consider $\lim_{x\to 1} \{x\}$, where $\{.\}$ denotes the fractional part.

Methods for Calculating the Limits of the Form $\lim_{x\to a} (f(x))^{g(x)}$

Here we discuss two different cases,

(i) When $\lim_{x\to a} f(x) = 1$. In this case, $\lim_{x\to a} (f(x))^{g(x)} = \lim_{x\to a} (1 + f(x) - 1)^{g(x)}$

$$= \lim_{x\to a} \left((1+f(x)-1)^{\frac{1}{f(x)-1}} \right)^{g(x)(f(x)-1)} = e^{\lim g(x)(f(x)-1)} \text{ because, } \lim_{x\to a} \left((1+f(x)-1)^{\frac{1}{f(x)-1}} \right) = e$$

(ii) When $\lim_{x\to a} f(x) \neq 1$ but $f(x)$ is positive in the neigbourhood of $x = a$.

In this case we write, $(f(x))^{g(x)} = e^{g(x)\ln f(x)} \Rightarrow \lim_{x\to a} (f(x))^{g(x)} = e^{\lim g(x).\ln(f(x))}$

Limits of the type $\lim_{x\to\infty} a^x$:

Here three different possibilities may arise,

(i) When $0 \le a < 1$

In this case $\lim_{x\to\infty} a^x = 0$, because a^x will keep on decreasing with an increase in x.

(ii) When $a = 1$. In this case, $\lim_{x\to\infty} a^x = 1$, because in this case the given function becomes a constant function.

(iii) When $a > 1$. In this case $\lim_{x\to\infty} a^x = \infty$, Because the value of the function will keep on increasing with an increase in the values of x.

Remarks:

- $\lim_{x\to\infty} a^x$ does not exist if $a < 0$, since the function is not defined for $a < 0$.
- Normally students get confused between $\lim_{x\to\infty} (1+1/x)^x$ and $\lim_{x\to\infty} a^x$, when $a > 1$. Because $1 + 1/x > 1$ $\forall\ x \in R^+$ and students start arguing that $\lim_{x\to\infty} (1+1/x)^x$ should be infinite. But we have already seen that $\lim_{x\to\infty} (1+1/x)^x = e$.

  Actually, these two results can not be compared as in the case of $\lim_{x\to\infty} a^x$, 'a' is a fixed number. On the other hand in case of $\lim_{x\to\infty} \left(1+\frac{1}{x}\right)^x$, $\left(1+\frac{1}{x}\right)$ is clearly a variable number, decreasing with an increase in x.

CONTINUITY OF A FUNCTION

A function f(x) is said to be continuous at $x = a$ if $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = f(a)$ i.e. L.H.L.=R.H.L. = value of the function at a i.e. $\lim_{x\to a} f(x) = f(a)$.

If f(x) is not continuous at $x = a$, we say that f(x) is discontinuous at $x = a$.

f(x) will be discontinuous at $x = a$ in any of the following cases:

(i) $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$ exist but are not equal.

(ii) $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$ exist and are equal but not equal to f(a).

(iii) f(a) is not defined.

(iv) At least one of the limits does not exist.

Continuity of a Function in an Interval

Continuity in an Open Interval:

A function f(x) is said to be continuous in an open interval (a, b) if it is continuous at each point of (a, b).

Continuity in a Closed Interval

A function f(x) is said to be continuous in a closed interval [a, b] if

(i) f(x) is continuous from right at $x = a$ i.e. $\lim_{x\to a^+} f(x) = f(a)$.

(ii) f(x) is continuous from left at $x = b$ i.e. $\lim_{x\to b^-} f(x) = f(b)$ and;

(iii) f(x) is continuous at each point of the interval (a, b).

Geometrical Meaning of Continuity

(i) The function 'f' will be continuous at $x = a$ if there is no break in the graph of the function $y = f(x)$ at the point (a, f(a)).

(ii) The function f(x) will be continuous in the closed interval [a,b] if the graph of $y = f(x)$ is an unbroken line (curved or straight) from the point (a,f(a)) to (b,f(b)).

Continuity of Composite Functions

If the function $u = f(x)$ is continuous at the point $x = a$, and the function $y = g(u)$ is continuous at the point $u = f(a)$, then the composite function $y = (g \circ f)(x) = g(f(x))$ is continuous at the point $x = a$.

Differentiability

Let $y = f(x)$ be continuous in (a, b). Then the derivative or differential of $f(x)$ at $x \in (a, b)$, denoted by dy/dx or $f'(x)$ and is defined as $\frac{dy}{dx} = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$ ...(1)

provided the limit exists and is finite.

Right Hand Derivative

Right hand derivative of $f(x)$ at $x = a$ is denoted by, $Rf'(a)$ or $f'(a^+)$ and defined as

$Rf'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$, $h > 0$.

Left Hand Derivative

Left hand derivative of $f(x)$ at $x = a$ is denoted by $Lf'(a)$ or $f'(a^-)$ and is defined as

$Lf'(a) = \lim_{h\to 0} \frac{f(a-h)-f(a)}{-h}$, $h > 0$. Clearly, $f(x)$ is differentiable at $x = \frac{1}{u^2+u-2}$ if and only if $R = Lf'(a)$.

Note: If a function $f(x)$ is differentiable at $x = a$ then it is also continuous at $x = a$. But if a function is continuous at a point, it is not necessarily differentiable at that point. Let us consider the function $f(x) = |x|$.

Hence $f(x) = \begin{cases} x & , \ x > 0 \\ -x & , \ x < 0 \\ 0 & , \ x = 0 \end{cases}$. $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} x = 0 \Rightarrow \lim_{x\to 0^-} f(x) = 0$

Also $f(0) = 0$. Hence, $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^-} f(x) = f(0) = 0$

$\Rightarrow f(x)$ is continuous at $x = 0$

Now, $Rf'(0) = \lim_{h\to 0} \frac{f(0+h)-f(0)}{h} = \lim_{h\to 0} \frac{f(h)-f(0)}{h} = \frac{h-0}{h} = 1$

$Lf'(0) = \lim_{h\to 0} \frac{f(0-h)-f(0)}{-h} = \lim_{h\to 0} \frac{f(-h)-f(0)}{-h} = \lim_{h\to 0} \frac{h-0}{-h} = -1$

$Rf'(0) \neq Lf'(0)$. Hence $f(x)$ is not differentiable at $x = 0$.

ILLUSTRATIVE PROBLEMS

1. **Evaluate** $\lim_{x\to 0}\sin^{-1}\left(\frac{\cos^{-1}x+\cos^{-1}x^2}{\pi}\right)$.

Sol. $\lim_{x\to 0^-}\left(\frac{\cos^{-1}x+\cos^{-1}x^2}{\pi}\right)=1$

$\Rightarrow \lim_{x\to 0^-}\sin^{-1}\left(\frac{\cos^{-1}x+\cos^{-1}x^2}{\pi}\right)$

does not exist as LHL does not exist.

2. **Evaluate** $\lim_{x\to 0}\frac{\ln\left(\tan\left(\frac{\pi}{4}+ax\right)\right)}{\sin bx}$.

Sol. $\lim_{x\to 0}\frac{\ln\left(\tan\left(\frac{\pi}{4}+ax\right)\right)}{\sin bx}$

$=\lim_{x\to 0}\frac{\cot\left(\frac{\pi}{4}+ax\right).a.\sec^2\left(\frac{\pi}{4}+ax\right)}{b\cos bx}=\frac{2a}{b}$.

3. **Evaluate** $\lim_{x\to 0}\frac{x}{a}\left[\frac{b}{x}\right]$, **(where [.] denotes the greatest integer function).**

Sol. $\lim_{x\to 0}\frac{x}{a}\left(\frac{b}{x}-\left\{\frac{b}{x}\right\}\right)=\lim_{x\to a}\left(\frac{b}{a}-\frac{x}{a}\left\{\frac{b}{x}\right\}\right)=\frac{b}{a}$.

4. **Find the points of discontinuity of** $y=g(u)=\frac{1}{u^2+u-2}$, **where** $u=\frac{1}{x-1}$.

Sol. The function $u=f(x)=\frac{1}{x-1}$ is discontinuous at the point $x=1$.

The function $y=g(u)=\frac{1}{u^2+u-2}=\frac{1}{(u+2)(u-1)}$ is discontinuous at $u=-2$ and $u=1$.

When $u=-2$, $\frac{1}{x-1}=u=-2$

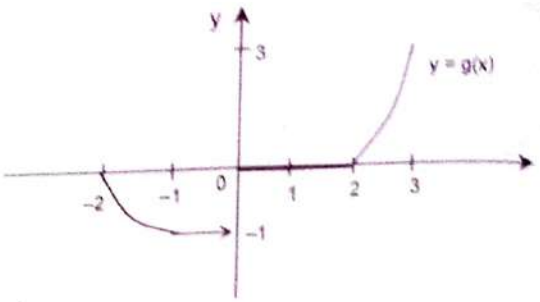
$\Rightarrow x=\frac{1}{2}$.

When $u=1$, $\frac{1}{x-1}=1$

$\Rightarrow x=2$. Hence, the composite function $y=g(f(x))$ is discontinuous at three points $x=1/2, 1, 2$.

5. **If $f(x) = x^2 - 2|x|$, then test the continuity and differentiability of $g(x)$ in the interval $[-2, 3]$, where $g(x) = \begin{cases} \min \{f(t);\ -2 \le t \le x\} & -2 \le x < 0, \\ \max \{f(t);\ 0 \le t \le x\} & 0 \le x \le 3. \end{cases}$**

Sol. $f(x) = \begin{cases} x^2 + 2x, & x < 0 \\ 0, & x = 0 \\ x^2 - 2x, & x > 0 \end{cases}$



Case I: $-2 \le x < -1$, $f(x)$ decreases

$\Rightarrow g(x) = x^2 + 2x$

Case II: $-1 \le x < 0$, $f(x)$ increases

$\Rightarrow g(x) = -1$

Case III: $0 \le x < 1$, $f(x)$ decreases $\Rightarrow g(x) = 0$

Case IV: $1 \le x < 2$, $f(x)$ increases but $f(x) < 0$

$\Rightarrow g(x) = f(0) = 0$

Case V: $2 \le x \le 3$, $f(x)$ increases and $f'(x) \ge 0\ \forall\ x \in [2, 3)$

$\Rightarrow g(x) = f(x) = x^2 - 2x,\ 2 \le x \le 3$

Therefore $g(x) = \begin{cases} x^2 + 2x, & -2 \le x < -1 \\ -1, & -1 \le x < 0 \\ 0, & 0 \le x < 2 \\ x^2 - 2x, & 2 \le x \le 3 \end{cases}$

Clearly $g(x)$ is continuous everywhere except at $x = 0$.

Also $g(x)$ is non-differentiable at $x = 0$ and 2.

SOLVED PROBLEMS

1. $\lim_{x\to 2}\dfrac{(10-x)^{1/3}-2}{x-2}$ is equal to

(A) $\dfrac{1}{12}$ (B) $-\dfrac{1}{12}$

(C) $-\dfrac{1}{10}$ (D) $\dfrac{1}{10}$

Sol. **(B)**

$$\lim_{x\to 2}\frac{(10-x)^{1/3}-2}{x-2}=\lim_{h\to 0}\frac{(8-h)^{1/3}-2}{h}$$

$$=\lim_{h\to 0}\frac{1}{3}(8-h)^{-2/3}(-1)=-\frac{1}{3}\times(8)^{-2/3}=-\frac{1}{3\times 4}=-\frac{1}{12}.$$

2. If $\lim_{x\to 0}\dfrac{a\sin x-bx+cx^2+x^3}{2x^2\ln(1+x)-2x^3+x^4}$ exists and is finite, then it is equal to

(A) $\dfrac{1}{20}$ (B) $\dfrac{2}{45}$

(C) $\dfrac{3}{40}$ (D) none of these

Sol. **(C)**

$$\text{Given limit}=\lim_{x\to 0}\frac{a\left(x-\frac{x^3}{3!}+\frac{x^5}{5!}-\ldots\right)-bx+cx^2+x^3}{2x^2\left(x-\frac{x^2}{2}+\frac{x^3}{3}-\ldots\right)-2x^3+x^4}=\lim_{x\to 0}\frac{(a-b)x+cx^2+\left(1-\frac{a}{6}\right)x^3+\frac{ax^5}{120}\ldots}{2\frac{x^5}{3}-\frac{x^6}{2}+\ldots}$$

For this limit to exist, we must have $a=b$, $c=0$, $a=6$

and given limit $=\dfrac{a}{120}\times\dfrac{3}{2}=\dfrac{6\times 3}{120\times 2}=\dfrac{3}{40}$.

3. If $f(x)=\dfrac{1}{(x-1)(x-2)}$ and $g(x)=\dfrac{1}{x^2}$, then points of discontinuity of $f(g(x))$ are

(A) $\left\{-1,\ 0,\ 1,\ \dfrac{1}{\sqrt{2}}\right\}$ (B) $\left\{\dfrac{-1}{\sqrt{2}},\ -1,\ 0,\ 1,\ \dfrac{1}{\sqrt{2}}\right\}$

(C) $\{0,\ 1\}$ (D) $\left\{0,\ 1,\ \dfrac{1}{\sqrt{2}}\right\}$

Sol. **(B).** $f(g(x))=\dfrac{1}{\left(\frac{1}{x^2}-1\right)\left(\frac{1}{x^2}-2\right)}=\dfrac{x^4}{(1-x^2)(1-2x^2)}$

$\Rightarrow f(g(x))$ is discontinuous at $x=\pm 1$, $x=\pm\dfrac{1}{\sqrt{2}}$ and $x=0$

Since $g(x)$ is discontinuous at $x=0$.

4. If α, β are the roots of the equation $ax^2 + bx + c = 0$, then $\lim_{x\to\alpha}\left(ax^2+bx+c+1\right)^{2/x-\alpha}$ is

(A) $2a(\alpha - \beta)$ (B) $2\ln|a(\alpha - \beta)|$

(C) $e^{2a(\alpha-\beta)}$ (D) $e^{a^2|\alpha-\beta|}$

Sol. **(C).** $\lim_{x\to\alpha}\left(ax^2+bx+c+1\right)^{2/x-\alpha} = e^{\lim_{x\to\alpha}(1+a(x-\alpha)(x-\beta)-1)\frac{2}{x-\alpha}} = e^{2a(\alpha-\beta)}$.

5. The integer n for which the $\lim_{x\to0}\dfrac{\cos^2 x - \cos x - e^x\cos x + e^x - \frac{x^3}{2}}{x^n}$ is a finite non-zero number is

(A) 2 (B) 3

(C) 4 (D) 5

Sol. **(C).** Given that $\lim_{x\to0}\dfrac{\cos^2 x - \cos x - e^x\cos x + e^x - \frac{x^3}{2}}{x^n} = \lim_{x\to0}\dfrac{(\cos x - 1)(\cos x - e^x) - \frac{x^3}{2}}{x^n}$

$$= \lim_{x\to0}\frac{\left(1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+\ldots\ldots-1\right)\left[\left(1-\frac{x^2}{2!}+\frac{x^4}{4!}-\ldots\ldots\right)-\left(1+x+\frac{x^2}{2!}+\frac{x^3}{3!}-\ldots\ldots\right)\right]-\frac{x^3}{2}}{x^n}$$

$$= \lim_{x\to0}\frac{\left(-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+\ldots\ldots\right)\left[\left(-x-x^2-\frac{x^3}{3!}-\frac{2x^5}{5!}-\ldots\ldots\right)\right]-\frac{x^3}{2}}{x^n}$$

$$= \lim_{x\to0}\frac{\left(\frac{x^3}{2}+\frac{x^4}{2}+\frac{x^5}{12}-\frac{x^5}{24}\ldots\ldots\right)-\frac{x^3}{2}}{x^n} = \text{non-zero if } n = 4.$$

6. The graph of the function $y = f(x)$ has a unique tangent at $(e^a, 0)$ through which the graph passes then $\lim_{x\to e^a}\dfrac{\ln(1+7f(x))-\sin(f(x))}{3f(x)}$ equals to

(A) 1 (B) 2

(C) 7 (D) none of these

Sol. **(B).** $\lim_{x\to e^a}\dfrac{\ln(1+7f(x))-\sin(f(x))}{3f(x)}\left(\dfrac{0}{0}\text{form}\right)$

Using L' Hospitals rule

$$= \lim_{x\to e^a}\frac{7f'(x)-\{\cos(f(x))\cdot f'(x)\}\{1+7f(x)\}}{3f'(x)(1+7f(x))} = \lim_{x\to e^a}\frac{7-\cos(f(x))\{1+7f(x)\}}{3(1+7f(x))} = 2.$$

7. Consider the following statements S and R:

S : If f(x) is a differentiable function with $f'(x) \neq 0$ in (a, b) and f (x) is increasing in (a, b), then $\dfrac{f(x)}{f'(x)}$ is also increasing in (a, b).

R : Both sinx and cos x are decreasing functions in the interval $\left(\frac{\pi}{2}, \pi\right)$.

Which of the following is true?

(A) both S and R are wrong
(B) S is wrong and R is correct
(C) both S and R are correct
(D) S is correct and R is wrong

Sol. **(B).**

8. If $f(x) = (1 + x)(2 + x^2)^{1/2}(3 + |x^3|)^{1/3}$ then $f'(-1)$ is

(A) $\sqrt{3}.2^{1/3}$
(B) $\sqrt{2}3^{1/3}$
(C) 0
(D) does not exist

Sol. **(A).** $f'(x) = (2 + x^2)^{1/2}(3 + |x|^3)^{1/3} + (1 + x).[g(x)]$

$\Rightarrow f'(-1) = \sqrt{3}.2^{1/3}$.

where $g(x) = \left((3+|x^3|^{1/3}\right)^1 (2+x)^{1/2} + \left((2+x)^{1/2}\right)^1 \left(3+|x^3|\right)^{1/3}$.

9. If $f(x) = \lim\limits_{n\to\infty} \sum\limits_{r=0}^{n} \dfrac{\tan\frac{x}{2^{r+1}} + \tan^3\frac{x}{2^{r+1}}}{1-\tan^2\frac{x}{2^{r+1}}}$ then $\lim\limits_{n\to 0}\dfrac{f(x)}{x}$ is

(A) 1
(B) 0
(C) −1
(D) none of these

Sol. **(A).** $\lim\limits_{n\to\infty} \sum\limits_{r=0}^{n} \dfrac{\tan\frac{x}{2^{r+1}} + \tan^3\frac{x}{2^{r+1}}}{1-\tan^2\frac{x}{2^{r+1}}} = \lim\limits_{n\to\infty} \sum\limits_{r=0}^{n} \left(\tan\dfrac{x}{2^r} - \tan\dfrac{x}{2^{r+1}}\right)$

$= \tan x \Rightarrow \lim\limits_{x\to 0}\dfrac{f(x)}{x} = 1$.

10. Let f: R → R be a differentiable function at x = 0 satisfying f(0) = 0 and f'(0) = 1, then the value of $\lim\limits_{x\to 0}\dfrac{1}{x}\sum\limits_{n=1}^{\infty}(-1)^n f\left(\dfrac{x}{n}\right) =$

(A) 0
(B) − ln2
(C) 1
(D) e

Sol. **(B).** $L \equiv \lim\limits_{x\to 0}\dfrac{1}{x} f\left(\dfrac{x}{n}\right) = \lim\limits_{x\to 0}\dfrac{f(x/n) - f(0)}{x - 0}$

Let $\dfrac{x}{n} = p$

$\Rightarrow L = \lim\limits_{p\to 0}\dfrac{f(p) - f(0)}{np} = \dfrac{1}{n}f'(0) = \dfrac{1}{n}$

$\therefore \lim\limits_{x\to 0}\dfrac{1}{x}\sum\limits_{n=1}^{\infty}(-1)^n f\left(\dfrac{x}{n}\right) = \sum\limits_{n=1}^{\infty}(-1)^n\dfrac{1}{n} = -\sum\limits_{n=1}^{\infty}(-1)^{n-1}\dfrac{1}{n} = -\ln 2$

11. If $f(x)$ is continuous in $[0, 2]$ and $f(0) = f(2)$, then the equation $f(x) = f(x + 1)$ has

(A) non real root in $[0, 2]$ (B) at least one real root in $[0, 1]$

(C) at least one real root in $[0, 2]$ (D) at least one real root in $[1, 2]$

Sol. **(B, C).** Let $G(x) = f(x) - f(x + 1)$

$G(0) = f(0) - f(1)$

$G(1) = f(1) - f(2)$

$\Rightarrow G(0) + G(1) = 0$

$\Rightarrow G(0)$ and $G(1)$ are of opposite sign

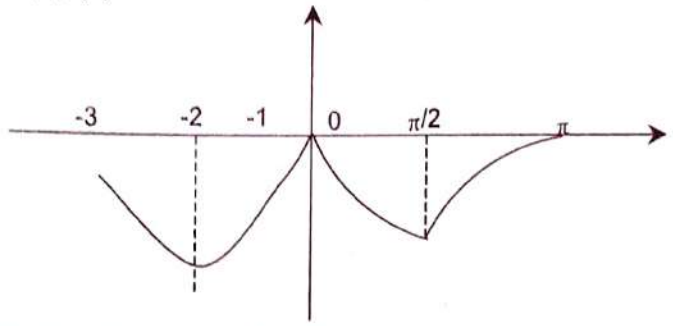
$\Rightarrow f(x) = f(x + 1)$ at least one in $[0, 1]$.

12. Let $f(x) = \begin{cases} x^2 + 4x, & -3 \le x \le 0 \\ -\sin x, & 0 < x \le \pi/2 \\ -\cos x - 1, & \pi/2 < x \le \pi \end{cases}$. Then

(A) $x = -2$ is the point of global minima (B) $x = \pi$ is the point of global maxima

(C) $f(x)$ is non-differentiable at $x = \pi/2$ (D) $f(x)$ is discontinuous at $x = 0$.

Sol. **(A), (B), (C)**



From above fig. clearly A, B, and C are correct.

13. If $\lim_{x \to a} f(x) = \lim_{x \to a} [f(x)]$ ([.] denotes the greatest integer function) and $f(x)$ is non-constant continuous function, then

(A) $\lim_{x \to a} f(x)$ is an integer (B) $\lim_{x \to a} f(x)$ is non-integer

(C) $f(x)$ has local maximum at $x = a$ (D) $f(x)$ has local minimum at $x = a$

Sol. **(A), (D)**

$\lim_{x \to a} [f(x)]$ can exist only when $f(x)$ either increases or decreases at both sides of the point $x = a$. Since $\lim_{x \to a} f(x) = \lim_{x \to a} [f(x)]$. So this can occur only when $\lim_{x \to a} f(x)$ is an integer.

Comprehension

If α, β (where $\alpha < \beta$) are the points of discontinuity of the function $g(x) = f(f(f(x)))$, where $f(x) = \dfrac{1}{1-x}$, and $P(a, a^2)$ is any point on xy-plane.

Now answers the following questions:

14. The points of discontinuity of $g(x)$ is

(A) $x = 0, -1$ (B) $x = 1$ only

(C) $x = 0$ only (D) $x = 0, 1$

Sol. **(D).** $f(x) = \frac{1}{1-x} \Rightarrow x \neq 1 \Rightarrow f(f(x)) = \frac{x-1}{x} \Rightarrow x \neq 0, 1.$

And $f(f(f(x))) = x,\ x \in R - \{0, 1\}$.

Hence points of discontinuity are $x = 0, 1$.

15. The domain of $f(g(x))$ is

(A) $x \in R$

(B) $x \in R - \{1\}$

(C) $x \in R - \{0, 1\}$

(D) $x \in R - \{0, 1, -1\}$

Sol. **(C).** $g(x) = f(f(f(x))) = x,\ x \in R - \{0, 1\}$

$\Rightarrow f(g(x)) = \frac{1}{1-x} \Rightarrow x \neq 0, 1.$

16. If point $P(a, a^2)$ lies on the same side as that of (α, β) with respect to line $x + 2y - 3 = 0$ then

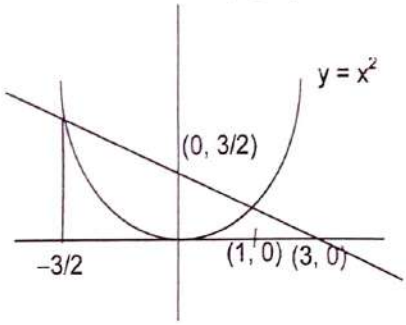
(A) $a \in \left(-\frac{3}{2}, 1\right)$

(B) $a \in R$

(C) $a \in \left(-\frac{3}{2}, 0\right)$

(D) $a \in (0, 1)$

Sol. **(A).** From graph $a \in \left(-\frac{3}{2}, 1\right)$



Assertion Reason

17. Let $(\sin y)^{\sin\left(\frac{\pi x}{2}\right)} + \frac{\sqrt{3}}{2}\sec^{-1}(2x) + 2^x \tan(x+2) = 0$

Statement –I: $\frac{dy}{dx}$ at $x = -1$ will not exist.

Because

Statement –II: $(f(x))^{g(x)}$ is discontinuous if $f(x) < 0$.

(A) Statement-1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement -1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement -1 is True, Statement -2 is False

(D) Statement -1 is False, Statement -2 is True

Sol. **(A)**

Since $y < 0$ for $x = -1$, hence $(\sin y)^{\sin\left(\frac{\pi x}{2}\right)}$ does not exist in the neighbourhood of $x = -1$.

18. Using the fact $\frac{x}{1+x} < \ln(1+x) < x \quad \forall\, x > 0.$

We can deduce $\left(1+\frac{1}{k}\right)^k < e < \left(1+\frac{1}{k}\right)^{k+1} \quad \forall\, k \in$ the integer.

Now taking the product for $k = 1, 2, \ldots, n-1$, conclude by induction that

$\frac{n^{n-1}}{(n-1)!} < e^{n-1} < \frac{n^n}{(n-1)!}$ and consequently

$en^n e^{-n} < n! < en^{n+1}e^{-n}$

So $e < \frac{n!}{n^n e^{-n}} < ne \Rightarrow e^{1/n} < \left(\frac{n!}{n^n e^{-n}}\right)^{1/n} < n^{1/n} e^{1/n}$

But $\lim_{n\to\infty} e^{1/n} = \lim_{n\to\infty} n^{1/n} = 1$

$\therefore$ for $n \to \infty$; $n! \approx n^n e^{-n}$

Now match the column:

Column – I		Column – II	
(A)	$\lim_{n\to\infty}\left(\frac{3(n!)}{n^{3n}}\right)^{1/n}$ is equal to	(p)	$e^2/4$
(B)	$\lim_{n\to\infty}\left(\frac{(n!)^3}{n^{3n}e^{-n}}\right)^{1/n}$ is equal to	(q)	e^{-3}
(C)	$\lim_{n\to\infty}\left(\frac{(n!)^2}{n^{2n}}\right)^{1/n}$ is equal to	(r)	0
(D)	$\lim_{n\to\infty}\left(\frac{n^{2n}}{(2n)!}\right)^{1/n}$ is equal to	(s)	e^{-2}
		(t)	e^2

Sol. **(A) → (r), (B) → (s), (C) → (s), (D) → (p)**

(A) $\because\ n! \approx n^n e^{-n}$

$$\lim_{n\to\infty}\left(\frac{3n!}{n^{3n}}\right)^{1/n} \approx \lim_{n\to\infty}\left(3n^{-2n}e^{-n}\right)^{1/n} = 0$$

(B) $(n!)^3 \approx n^{3n}e^{-3n}$

$$\therefore \lim_{n\to\infty}\left(\frac{(n!)^3}{n^{3n}e^{-n}}\right)^{1/n} \approx \lim_{n\to\infty}\left(e^{-2n}\right)^{1/n} = e^{-2}$$

(C) $(n!)^2 \approx n^{2n}e^{-2n}$

$$\therefore \lim_{n\to\infty}\left(\frac{(n!)^2}{n^{2n}}\right)^{1/n} \approx \lim_{n\to\infty}\left(e^{-2n}\right)^{1/n} = e^{-2}$$

(D) $(2n!) \approx (2n)^{2n}e^{-2n}$

$$\therefore \lim_{n\to\infty}\left(\frac{n^{2n}}{(2n)!}\right)^{1/n} \approx \lim_{n\to\infty}\left(\frac{1}{2^{2n}e^{-2n}}\right)^{1/n} = \frac{e^2}{4}$$

Answer Q.19, Q.20 and Q.21 by appropriately matching the information given in the three columns of the following table.

Match the following Column(s)

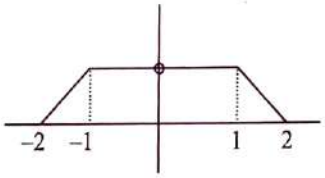
Column–I (Function)	Column–II Number of points of discontinuity	Column–III Number of points of differentiability
(I) $f(x)=\begin{cases}\lfloor [x] \rfloor & , \ 0 \le \{x\} < \frac{1}{2} \\ |x| & , \ \frac{1}{2} \le \{x\} < 1\end{cases} \ \forall \ x \in \left[-\frac{7}{2}, \frac{7}{2}\right]$ (where [.] denotes the greatest integer function)	(i) 5	(P) 4
(II) $f(x) = [3x] - \{2x\} \ \forall \ x \in \left[0, \frac{3}{2}\right)$ (where [.] denotes the greatest integer function and {.} fractional part)	(ii) 4	(Q) 14
(III) $f(x) = \text{sgn}(3 \sin x + 4 \cos x) \ \forall \ x \in [-2\pi, 2\pi]$	(iii) 7	(R) 3
(IV) $f(x)=\begin{cases}\{x\} & , \ -2 \le x < -1 \\ \lvert \text{sgn}(x) \rvert & , \ -1 \le x \le 1 \\ \{-x\} & , \ 1 < x \le 2\end{cases} \ \forall \ x \in (-2, 2)$ (where {.} denotes fractional part of function)	(iv) 1	(S) 5

19. Which of the following combination is **CORRECT**?

(A) (I) (i) (R)
(B) (III) (ii) (R)
(C) (IV) (iv) (P)
(D) (I) (iii) (Q)

Sol. **(D)**

Graph of $y = f(x)$



20. Which of the following combination is **CORRECT**?

(A) (II) (i) (S)
(B) (III) (i) (P)
(C) (II) (iii) (R)
(D) (III) (i) (R)

Sol. **(A)**

21. Which of the following combination is **INCORRECT**?

(A) (I) (iii) (Q)
(B) (IV) (iv) (R)
(C) (II) (ii) (S)
(D) (II) (i) (S)

Sol. **(C)**

22. The number of values of $x \in (2, \infty)$ where $f(x)=\sqrt{x+\sqrt{8x-16}}+\sqrt{x-\sqrt{8x-16}}$ is not differentiable is equal to _____

Sol. **(4.00)**

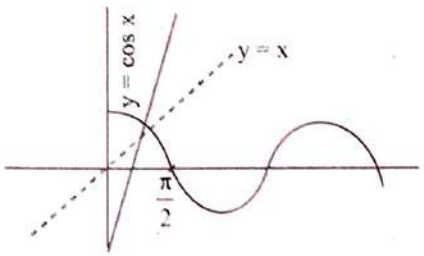
$$f(x)=\sqrt{\left(\sqrt{(x-2)}\right)^2+\left(\sqrt{2}\right)^2+2\sqrt{2}\cdot\sqrt{x-2}}+\sqrt{\left(\sqrt{x-2}\right)^2+\sqrt{2}-2\sqrt{2}\sqrt{x-2}}$$

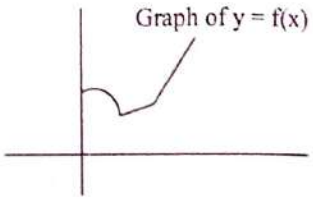
$$f(x)=\left|\sqrt{x-2}+\sqrt{2}\right|+\left|\sqrt{x-2}-\sqrt{2}\right|$$

23. Let $f(x) = \max(\cos x, x, 2x - 1)$ where $x \geq 0$, then number of points of non-differentiability of $f(x)$ is ____

Sol. **(2.00)**

Shown in the figure





24. If $f(x)=\begin{cases}\dfrac{\tan\left[x^2+x-100\right]\pi}{x^2+1}+\left[\left\{e^{\sin x^3}\right\}\right]+ax^3+b & , \ 0<x\leq 1\\ 2\cos\pi x+\tan^{-1}x & , \ 1<x\leq 2\end{cases}$ is differentiable in (0, 2), then $a-b+\frac{\pi}{4}$ is equal to ____

Sol. **(2.33)**

$$f(x)=\begin{cases}ax^3+b & , \ 0<x\leq 1\\ 2\cos(\pi x)+\tan^{-1}x & , \ 1<x\leq 2\end{cases}$$

$\because$ $f(x)$ is continuous at $x = 1$

$$\Rightarrow a+b=-2+\frac{\pi}{4}$$

$$f'(x)=\begin{cases}3ax^2 & 0<x\leq 1\\ -2\pi\sin\pi x+\dfrac{1}{1+x^2} & 1<x\leq 2\end{cases}$$

$f(x)$ is differentiable at $x = 1$

$$\Rightarrow 3a=\frac{1}{2}\Rightarrow a=\frac{1}{6}\Rightarrow b=\frac{\pi}{4}-\frac{13}{6}\Rightarrow a-b+\frac{\pi}{4}=\frac{1}{6}+\frac{13}{6}=\frac{7}{3}=2.33$$

25. If $\lim\limits_{x\to 0}\left(\dfrac{\sin(ax)-\ln\left(e^x\cos x\right)}{x\sin bx}\right)=\dfrac{1}{2}$, then the value of $\dfrac{3a+2b}{2}$ is equal to ____

25. **(2.50)**

$$\lim_{x\to 0}\left(\frac{a\cos(ax)-\dfrac{e^x(\cos x-\sin x)}{e^x\cos x}}{\sin bx+bx\cos bx}\right)=\frac{1}{2}\Rightarrow a=1\Rightarrow\lim_{x\to 0}\left(\frac{\cos^2 x-\cos x+\sin x}{\cos x(\sin bx+bx\cos bx)}\right)=\frac{1}{2}$$

$$\Rightarrow b=1$$

APPLICATION OF DERIVATIVES 3

Tangent and Normal

The derivative of a function $y = f(x)$ represents the slope of the tangent to the curve at the general point (x, y).

Equation of Tangent and Normal

Let $y = f(x)$ be the given curve. We already know that $\frac{dy}{dx}$ at any point lying on the curve would give us the slope of the tangent that can be drawn at that point. Let (x_1, y_1) be any point on the curve, that means, $y_1 = f(x_1)$. Now the slope of the tangent that can be drawn to the curve at (x_1, y_1) will be $\left(\frac{dy}{dx}\right)_{(x_1, y_1)}$. Thus the equation of the tangent at (x_1, y_1) would be,

$$(y - y_1) = \left(\frac{dy}{dx}\right)_{(x_1, y_1)}(x - x_1) \text{ or, } y - f(x_1) = \left(\frac{dy}{dx}\right)_{(x_1, y_1)}(x - x_1)$$

Similarly, the equation of the normal at (x_1, y_1) would be

$$y - y_1 = -\frac{1}{\left(\frac{dy}{dx}\right)_{(x_1, y_1)}}(x - x_1) \text{ provided that } \left(\frac{dy}{dx}\right)_{(x_1, y_1)} \neq 0.$$

Angle of Intersection of Two Curves

Let $y = f(x)$ and $y = g(x)$ be two given intersecting curves. Angle of intersection of these curves is defined as the acute angle between the tangents that can be drawn to the given curves at the point of intersection. Let (x_1, y_1) be the point of intersection $\Rightarrow y_1 = f(x_1) = g(x_1)$. Slope of the tangent drawn to the curve $y = f(x)$ at (x_1, y_1), $m_1 = \left(\frac{df}{dx}\right)_{(x_1, y_1)}$.

Similarly slope of the tangent drawn to the curve $y = g(x)$ at (x_1, y_1), $m_2 = \left(\frac{dg}{dx}\right)_{(x_1, y_1)}$.

If 'α' be the angle (acute) of intersection, then $\tan\alpha = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$.

If $\alpha = 0$, then $m_1 = m_2$. Thus the given curves would touch each other at (x_1, y_1).

If $\alpha = \frac{\pi}{2}$, then $m_1 m_2 = -1$. Thus the given curves would meet at right angle at (x_1, y_1) (or curves cut orthogonally at (x_1, y_1)).

Monotonicity

Let $y = f(x)$ be a given function with 'D' as it's domain. Let $D_1 \subseteq D$ then;

APPLICATION OF DERIVATIVES 3

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Monotonicity

Let $y = f(x)$ be a given function with 'D' as it's domain. Let $D_1 \subseteq D$ then;

Increasing Function

f(x) is said to be increasing in D_1 if for every $x_1, x_2 \in D_1, x_1 > x_2$
$\Rightarrow f(x_1) > f(x_2)$. It means that the value of f (x) will keep on increasing with an increase in the value of x. Refer to Fig.2

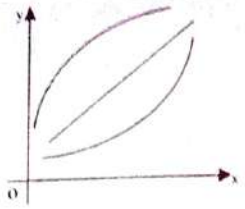


Fig. 2 - Increasing Functions

Non-Decreasing Function

f(x) is said to be non-decreasing in D_1 if for every $x_1, x_2 \in D_1, x_1 > x_2$
$\Rightarrow f(x_1) \geq f(x_2)$. It means that the value of f (x) would never decrease with an increase in the value of x. Refer to Fig.3.

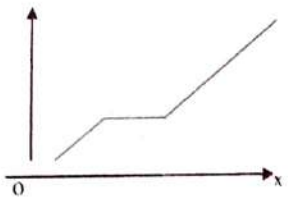


Fig. 3 -Non-decreasing Functions

Decreasing Function

f(x) is said to be decreasing in D_1 if for every $x_1, x_2 \in D_1, x_1 > x_2$
$\Rightarrow f(x_1) < f(x_2)$. It means that the value of f(x) would decrease with an increase in the value of x. Refer to Fig.4.

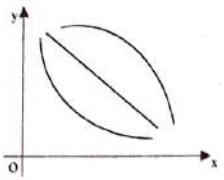


Fig. 4 -Decreasing Functions

Non-Increasing Function

f(x) is said to be non-increasing in D_1 if for every $x_1, x_2 \in D_1, x_1 > x_2 \Rightarrow f(x_1) \leq f(x_2)$. It means that the value of f(x) would never increase with an increase in the value of x. Refer to Fig. 5.

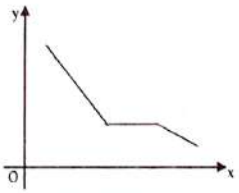


Fig. 5 -Non-increasing Functions

Basic Theorems

Let y = f(x) be a given function, continuous in [a, b] and differentiable in (a, b). Then

- f(x) is increasing in [a , b] if $f'(x) > 0 \ \forall \ x \in (a, b)$.
- f(x) is non-decreasing in [a , b] if $f'(x) \geq 0 \ \forall \ x \in (a, b)$.
- f(x) is decreasing in [a , b] if $f'(x) < 0 \ \forall \ x \in (a, b)$.
- f(x) is non-increasing in [a , b] if $f'(x) \leq 0 \ \forall \ x \in (a, b)$.

Remarks

- If $f'(x) \geq 0 \ \forall \ x \in (a, b)$ and points which make $f'(x)$ equal to zero (in between (a, b)) don't form an interval, then f (x) would be increasing in [a, b].
- If $f'(x) \leq 0 \ \forall \ x \in (a, b)$ and points which make $f'(x)$ equal to zero (in between (a, b)) don't form an interval, f (x) would be decreasing in [a, b].
- If $f(0) = 0$ and $f'(x) \geq 0 \ \forall \ x \in R$, then $f(x) \leq 0 \ \forall \ x \in (-\infty, 0)$ and $f(x) \geq 0 \ \forall \ x \in (0, \infty)$.

- If $f(0) = 0$ and $f'(x) \leq 0 \ \forall \ x \in R$, then $f(x) \geq 0 \ \forall \ x \in (-\infty, 0)$ and $f(x) \leq 0 \ \forall \ x \in (0, \infty)$.
- A function is said to be monotonic if it's either increasing or decreasing.
- The points for which $f'(x)$ is equal to zero or doesn't exist are called **critical points.** Here it should also be noted that critical points are the interior points of an interval.
- The stationary points are the points where $f'(x) = 0$ in the domain.

Concept of Local Maximum and Local Minimum

Let $y = f(x)$ be a function defined at $x = a$ and also in the vicinity of the point $x = a$. Then, $f(x)$ is said to have a local maximum at $x = a$, if the value of the function at $x = a$ is greater than the value of the function at the neighbouring points of $x = a$.

Mathematically, $f(a) > f(a - h)$ and $f(a) > f(a + h)$, where $h > 0$,

Similarly $f(x)$ is said to have a local minimum at $x = a$, if the value of the function at $x = a$ is less than the value of the function at the neighbouring points of $x = a$.

Mathematically, $f(a) < f(a - h)$ and $f(a) < f(a + h)$ where $h > 0$.

A local maximum or a local minimum is also called a local extremum.

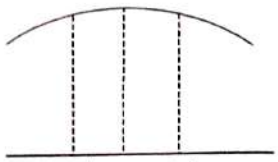


Fig. 6 - Local Maxima

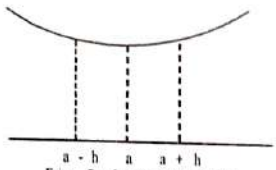


Fig. 7 - Local Minima

Test for Local Maximum/ Minimum

We have two cases to consider

Test for Local Maximum/Minimum at x = a if f (x) is Differentiable at x = a:

If $f(x)$ is differentiable at $x = a$ and if it is a critical point of the function (i.e. $f'(a) = 0$) then we have the following three tests to decide whether $f(x)$ has a local maximum or local minimum or neither at $x = a$, If $f'(a) = 0$ and $f'(x)$ changes it's sign while passing through the point $x = a$, then

(i) $f(x)$ would have a local maximum at $x = a$ if $f'(a-0) > 0$ and $f'(a+0) < 0$. It means that $f'(x)$ should change it's sign from positive to negative.

(ii) $f(x)$ would have local minimum at $x = a$ if $f'(a-0) < 0$ and $f'(a+0) > 0$. It means that $f'(x)$ should change it's sign from negative to positive.

(iii) If $f(x)$ doesn't change it's sign while passing through $x = a$, then $f(x)$ would have neither a maximum nor minimum at $x = a$.

Test for Local Maximum/Minimum at x = a if f (x) is not Differentiable at x = a:

Case I: When $f(x)$ is continuous at $x = a$ and $f'(a - h)$ and $f'(a + h)$ exist and are non-zero, then $f(x)$ has a local maximum or minimum at $x = a$ if $f'(a - h)$ and $f'(a + h)$ are of opposite signs.

If $f'(a - h) > 0$ and $f'(a + h) < 0$ then $x = a$ will be a point of local maximum.

If $f'(a - h) < 0$ and $f'(a + h) > 0$ then $x = a$ will be a point of local minimum

Case II: When f(x) is continuous and f'(a – h) and f'(a + h) exist but one of them is zero, we should infer the information about the existence of local maximum / minimum from the basic definition of local maximum / minimum.

Case III: If f(x) is not continuous at x = a and f'(a – h) and/or f'(a + h) are not finite, then compare the values of f(x) at the neighbouring points of x = a.

Remark:

- It is advisable to draw the graph of the function in the vicinity of the point x = a because the graph would give us the clear picture about the existence of local maximum / minimum at x = a.

Concept of Global Maximum/Minimum

Let y = f(x) be a given function with domain D. Let [a, b] $\subseteq$ D. Global maximum/minimum of f(x) in [a, b] is basically the greatest/least value of f(x) in [a, b].

Global maximum and minimum in [a, b] would always occur at critical points of f(x) within [a, b] or at the end points of the interval.

Global Maximum / Minimum in [a, b]

In order to find the global maximum and minimum of f(x) in [a, b], find out all the critical points of f(x) in (a, b). Let $c_1, c_2, \ldots c_n$ be the different critical points. Find the value of the function at these critical points. Let $f(c_1)$, $f(c_2)$, …, $f(c_n)$ be the values of the function at critical points. Say, $M_1 = \max \{f(a), f(c_1), f(c_2), \ldots, f(c_n), f(b)\}$ and $M_2 = \min \{f(a), f(c_1), f(c_2), \ldots, f(c_n), f(b)\}$.

Then M_1 is the greatest value of f (x) in [a, b] and M_2 is the least value of f (x) in [a, b].

Global Maximum / Minimum in (a, b)

Method for obtaining the greatest and least values of f(x) in (a, b) is almost same as the method used for obtaining the greatest and least values in [a, b], however with a caution.

Let y = f(x) be a function and $c_1, c_2, \ldots c_n$ be the different critical points of the function in (a, b).

Let $M_1 = \max\{f(c_1), f(c_2), f(c_3) \ldots, f(c_n)\}$

and $M_2 = \min\{f(c_1), f(c_2), f(c_3) \ldots, f(c_n)\}$

Now if $\lim\limits_{\substack{x \to a+0 \\ (\text{or} x \to b-0)}}$ f(x)>M_1 or < M_2, f(x) would not have global maximum (or global minimum) in (a, b)

This means that if the limiting values at the end points are greater than M_1 or less than M_2, then f (x) would not have global maximum/minimum in (a, b). On the other hand if $M_1 > \lim\limits_{\substack{x \to a+0 \\ (\text{and} x \to b-0)}}$ f(x) and $M_2 < \lim\limits_{\substack{x \to a+0 \\ (\text{and} x \to b-0)}}$ f(x), then M_1 and M_2 would respectively be the global maximum and global minimum of f(x) in (a, b).

Rolle's Theorem

It states that if y = f (x) be a given function and satisfies following conditions

- f(x) be continuous in [a , b]
- f(x) be differentiable in (a , b)
- f(a) = f(b)

then f'(c) = 0 at least once for some c $\in$ (a, b).

Applications of Rolle's Theorem

If $y = f(x)$ satisfies the Rolle's theorem in [a , b], then $f'(x) = 0$ for some $x \in (a, b)$. As any solution of $f'(x) = 0$ will give us a root of $f'(x) = 0$, hence we can say that at least one root of $f'(x) = 0$ will belong to (a, b) if $f(x)$ satisfies all conditions of Rolle's Theorem.

- Let $x = a$ and $x = b$ be the roots of $f(x) = 0$ and $y = f(x)$ satisfies the condition of Rolle's theorem in [a, b]. Here $f(a) = f(b) = 0$. Hence we can say that between two roots of $f(x) = 0$ at least one root of $f'(x) = 0$ will lie.
- Let $y = f(x)$ be a polynomial function of degree n. If $f(x) = 0$ has real roots only, then $f'(x) = 0$, $f''(x) = 0, \ldots f^{n-1}(x) = 0$ will have real roots. It is in fact the general version of above mentioned application, because if $f(x) = 0$ have all real roots, then between two consecutive roots of $f(x) = 0$, exactly one root of $f'(x) = 0$ will lie.

Lagrange's Mean Value Theorem

It says that if y = f (x) be a given function which is;

- Continuous in [a , b]
- Differentiable in (a , b)

then $f'(c) = \dfrac{f(b)-f(a)}{b-a}$ at least once for some $c \in (a, b)$.

ILLUSTRATIVE PROBLEMS

1. Which normal to the curve $y = x^2$ forms the shortest chord?

Sol. Let $P(t, t^2)$ be the parametric co-ordinates of any point on the parabola $y = x^2$

$\Rightarrow \dfrac{dy}{dx} = 2x = 2t$

Hence Normal at P is $(y - t^2) = \left(-\dfrac{1}{2t}\right)(x - t)$

If this line meets the curve again at $Q(t', t'^2)$, then

$(t'^2 - t^2) = \left(-\dfrac{1}{2t}\right)(t' - t) \Rightarrow t' + t = -\dfrac{1}{2t}$ as $(t = t') \Rightarrow t' = -t - \dfrac{1}{2t}$

If l is length of the normal chord then $z = (l)^2 = PQ^2$

$z = (t - t')^2 [1 + (t + t')^2]$

Putting the value of t from (ii) we have

$$z = \left(t + t + \frac{1}{2t}\right)^2 \left(1 + \left(-\frac{1}{2t}\right)^2\right)$$

$$z = \frac{(4t^2 + 1)^3}{16t^4}$$

Put $4t^2 = p$ where $p > 0$

$$z = \frac{(p+1)^3}{p^3} = (p + 3) + \frac{3}{p} + \frac{1}{p^2}$$

$$\Rightarrow \frac{dz}{dp} = \left(1 - \frac{3}{p^2}\right) - \frac{2}{p^3} = 0 \Rightarrow (p+1)^2(p-2) = 0 \Rightarrow p = 2 \text{ as } p \neq -1$$

and $\frac{d^2z}{dp^2} = \frac{6}{p^3} + \frac{6}{p^4} > 0$

Hence z is minimum when $p = 2 = 4t^2$.

2. Use Lagranges mean value theorem to prove that $1 + x < e^x < 1 + xe^x \; \forall \; x > 0$.

Sol. Let $f(x) = e^x$ in $[0, x]$.

Then f is continuous in $[0, x]$ and differentiable in $(0, x)$.

So by Lagranges mean value theorem there exists $c \in (0, x)$ such that

$$f'(c) = \frac{f(x) - f(0)}{x - 0}$$

$$\Rightarrow e^c = \frac{e^x - 1}{x}$$

Now $0 < c < x \Rightarrow e^0 < e^c < e^x$ or $1 < e^c < e^x$ or $1 < \frac{e^x - 1}{x} < e^x$

$\Rightarrow 1 + x < e^x < 1 + xe^x$

Hence proved.

3. Investigate the behaviour of the following functions for monotonicity in the given intervals,

(i) $f(x) = -\sin^3 x + 3\sin^2 x + 5$, $x \in [-\pi/2, \pi/2]$

(ii) $f(x) = \sec x - \text{cosec } x$, $x \in (0, \pi/2)$

Sol. (i) $f(x) = -\sin^3 x + 3\sin^2 x + 5$, $x \in [-\pi/2, \pi/2]$

$\Rightarrow f'(x) = -3\sin^2 x \cdot \cos x + 6 \sin x \cdot \cos x = 3 \sin x \cos x (2 - \sin x)$

As $\cos x > 0$ and $2 - \sin x > 0 \; \forall \; x \in (-\pi/2, \pi/2)$

and $\sin x > 0 \; \forall \; x \in (0, \pi/2)$, $\sin x < 0 \; \forall \; x \in (-\pi/2, 0)$

$f'(x) > 0$, $x \in (0, \pi/2)$ and $f'(x) < 0$, $x \in (-\pi/2, 0)$

$\Rightarrow f(x)$ is increasing in $(0, \pi/2)$ and decreasing in $(-\pi/2, 0)$.

(ii) $f(x) = \sec x - \text{cosec } x$, $x \in (0, \pi/2)$

$\Rightarrow f'(x) = \sec x \tan x + \text{cosec } x \cot x > 0 \; \forall \; x \in (0, \pi/2)$

Thus $f(x)$ is increasing in $(0, \pi/2)$.

4. Let $f(x) = \begin{cases} x^3 - x^2 + 10x - 5, & x \leq 1 \\ -2x + \log_2(b^2 - 2), & x > 1 \end{cases}$. Find all possible real values of b such that f(x) has greatest value at x = 1.

Sol. for $x \leq 1$, $f'(x) = 3x^2 - 2x + 10$

$$= 3\left\{x^2 - \frac{2x}{3} + \frac{10}{3}\right\}$$

$$= 3\left\{\left(x - \frac{1}{3}\right)^2 + \frac{29}{9}\right\} > 0$$

$\therefore f(x)$ is an increasing function for $x \leq 1$.

Now for $x > 1$, $f'(x) = -2 < 0$

So $f(x)$ is deceasing function for $x > 1$.

Now $f(x)$ will have greatest value at $x = 1$ if $\lim_{x\to 1^+} f(x) \le f(1)$

$\Rightarrow \lim_{h\to 0} f(1+h) \le 5$

$\Rightarrow \lim_{h\to 0} \{-2(1+h) + \log_2(b^2-2)\} \le 5$

$\Rightarrow -2 + \log_2(b^2-2) \le 5$

$\Rightarrow (b^2-2) \le 2^7$

$\Rightarrow 2 < b^2 \le 130 \ (\because b^2 - 2 > 0)$

$\therefore b \in [-\sqrt{130}, -\sqrt{2}) \cup (\sqrt{2}, \sqrt{130}]$

5. If $f(x)$ is continuous in $[a, b]$, $a, b > 0$ and differentiable in (a, b), prove that there exists at least one $c \in [a, b]$ such that $\frac{f(b)-f(a)}{b-a} \ge \left(\frac{ab}{c^2}\right)^{\frac{n-1}{2}} f'(c)$.

Sol. Consider the function $H(x) = f(x) - f(a) - A(x^n - a^n)$, where A is to be obtained from the condition that $H(b) = 0$. Also $H(a) = 0$. Moreover, $H(x)$ is continuous in $[a, b]$ and differentiable in (a, b).

Hence by Rolle's theorem, there exists a $c \in [a, b]$, such that $H'(c) = 0$.

$\Rightarrow 0 = f'(c) - nAc^{n-1}$ or $f'(c) = Anc^{n-1}$ i.e. $f'(c) = \frac{f(b)-f(a)}{b^n - a^n} \cdot nc^{n-1}$.

or $f'(c) = \frac{[f(b)-f(a)]}{(b-a)} \frac{[nc^{n-1}]}{\left[b^{n-1} + b^{n-2}a + b^{n-3}a^2 + \ldots\ldots + ba^{n-2} + a^{n-1}\right]}$

$\Rightarrow \frac{f(b)-f(a)}{b-a} = \frac{f'(c)}{c^{n-1}} \cdot \frac{b^{n-1} + b^{n-2}a + \ldots\ldots + ba^{n-2} + a^{n-1}}{n}$

$\ge \frac{f'(c)}{c^{n-1}} \left[b^{1+2+\ldots\ldots+(n-1)} \cdot a^{1+2+\ldots\ldots+(n-1)}\right]^{\frac{1}{n}}$

or $\frac{f(b)-f(a)}{b-a} \ge \frac{f'(c)}{c^{n-1}} \left((ab)^{\frac{(n-1)n}{2}}\right)^{1/n} = \frac{f'(c)}{c^{n-1}} (ab)^{\frac{n-1}{2}}$.

Hence $\frac{f(b)-f(a)}{b-a} \ge \left(\frac{ab}{c^2}\right)^{\frac{n-1}{2}} f'(c)$.

6. If the sides and angles of a plane triangle vary in such a way that its circumradius remains constant, prove that $\frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C} = 0$ where da, db and dc are small increments in the sides a, b, c respectively.

Sol. Since in a triangle $\angle A + \angle B + \angle C = \pi$

$\therefore dA + dB + dC = 0$

If R is circumradius then $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$

$a = 2R\sin A$

on differentiating $da = 2R \cos A \, dA \Rightarrow \frac{da}{\cos A} = 2R\,dA$

Similarly $\frac{db}{\cos B} = 2RdB$ and $\frac{dc}{\cos C} = 2RdC$

$\therefore \frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C} = 2R(dA + dB + dC) = 0.$

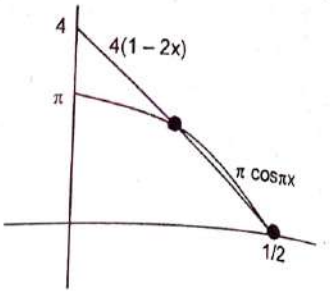
7. Prove that $\frac{\sin \pi x}{x(1-x)} \le 4$, when $0 < x < 1$.

Sol. Let $g(x) = \sin \pi x - 4x(1 - x)$

since $g(x) = g(1-x)$

$\Rightarrow$ g(x) is symmetric about $x = \frac{1}{2}$,

Hence it is sufficient to investigate the function in (0, 1/2].



$g'(x) = \pi \cos \pi x - 4(1 - 2x)$

g(x) has only one critical point in $\left(0, \frac{1}{2}\right)$

$g(0) = 0, \; g\left(\frac{1}{2}\right) = 0$

$\Rightarrow$ either $g(x) \ge 0$ or $g(x) \le 0, \forall x \in \left(0, \frac{1}{2}\right)$

but $g\left(\frac{1}{6}\right) = \frac{1}{2} - 4\left(\frac{1}{6}\right)\left(\frac{5}{6}\right) = \frac{9-10}{18} = -\frac{1}{18} < 0$

$\Rightarrow g(x) \le 0, \forall x \in \left(0, \frac{1}{2}\right) \Rightarrow \sin \pi x \le 4x(1 - x), \; \forall x \in \left(0, \frac{1}{2}\right)$

8. Find the absolute maximum value of $f(x) = \frac{1}{|x-4|+1} + \frac{1}{|x+8|+1}$.

Sol. Say $g(x) = \frac{1}{|x-4|+1} \Rightarrow \frac{1}{|x+8|+1} = g(x + 12)$

Now for $x \in (-\infty, -8)$ both g (x) and g (x + 12) are increasing hence maximum value can't occur in this interval.

Similarly for $x \in (4, \infty)$ both g (x) and g (x + 12) are decreasing hence maximum value can't occur in this interval.

So, now for all values of $x \in (-8, 4)$

$f(x) = \frac{1}{x+9} + \frac{1}{5-x} = \frac{14}{(5-x)(x+9)}$

which will have maximum either at x = 4 or x = –8 or at minima of (5 – x)(x + 9).

Also minima of (5 – x)(x + 9) does not exist.

$\Rightarrow f(4) = 1 + \frac{1}{13} = \frac{14}{13} = f(-8)$. So, absolute maximum is $\frac{14}{13}$.

9. **A point P (x, y) moves on the curve $x^{2/3}+y^{2/3}=a^{2/3}$, a > 0. For each position (x, y) of P, perpendiculars are drawn from origin upon the tangent and normal at P. the length (absolute value) of them being p_1 (x) and p_2 (x) respectively. Prove that $\frac{dp_1}{dx}.\frac{dp_2}{dx}\le 0$.**

Sol. Any point (x, y) on the curve can be represented by using parameter θ, as ($a\cos^3\theta$, $a\sin^3\theta$).

$$\therefore \frac{dy}{dx}=\frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}=\frac{3a\sin^2\theta\cos\theta}{-3a\cos^2\theta\sin\theta}=-\tan\theta$$

∴ Equation of the tangent at P is

$(y-a\sin^3\theta)=-\tan\theta(x-a\cos^3\theta)$

or, $\tan\theta\, x+y=a\sin^3\theta+a\cos^3\theta\tan\theta$ .....(1)

$$p_1=\left|\frac{a\sin^3\theta+a\cos^3\theta\tan\theta}{\sec\theta}\right|=\frac{1}{2}|a\sin 2\theta|$$

Also equation of the normal at P is

$(y-a\sin^3\theta)=\cot\theta(x-a\cos^3\theta)$

or, $x\cot\theta-y=a\cos^3\theta\cot\theta-a\sin^3\theta$ .....(2)

$$p_2=\left|\frac{a\cos^3\theta\cot\theta-a\sin^3\theta}{\operatorname{cosec}\theta}\right|=|a\cos 2\theta|$$

$\Rightarrow 4p_1^2+p_2^2=a^2=$ constant $\forall$ point P (x, y).

$\Rightarrow$ If p_1 increases, p_2 decreases and conversely.

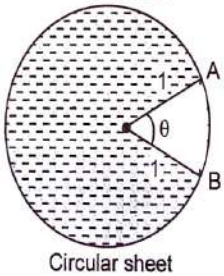
$\Rightarrow \frac{dp_1}{dx}$ and $\frac{dp_2}{dx}$ are of opposite signs.

$$\Rightarrow \frac{dp_1}{dx}.\frac{dp_2}{dx}\le 0$$

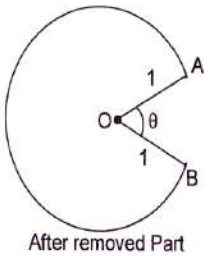
10. **A conical vessel is to be prepared out of a circular sheet of gold of unit radius. How much sectorial area is to be removed from the sheet so that the vessel has maximum volume?**

Sol. Sectorial area AOB is removed and the remaining part be folded in to a cone of height h and radius r.

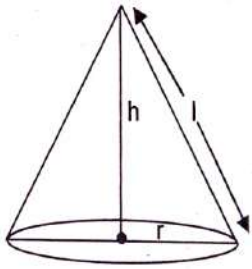
$$\because \theta=\text{Angle}=\frac{\text{Arc AB}}{\text{radius}}=\frac{\text{Arc AB}}{1}=\text{Arc AB}$$



Circular sheet



After removed Part



∴ $2\pi r=\theta$ and $r^2+h^2=1$

∴ volume of cone $v=\frac{1}{3}\pi r^2h=\frac{1}{3}\pi r^2\left(\sqrt{1-r^2}\right)$

$$\Rightarrow V^2=\frac{\pi^2r^4}{9}(1-r^2)=y\ (\text{Let})$$

$\therefore \frac{dy}{dr} = \frac{\pi^2}{9}(4r^3 - 6r^5)$ and $\frac{d^2y}{dr^2} = \frac{\pi^2}{9}(12r^2 - 30r^4)$

for maximum or minimum of y, $\frac{dy}{dr} = 0 \Rightarrow r = \sqrt{\frac{2}{3}}$ for which $\frac{d^2y}{dx^2} = -\frac{16}{27}\pi^2$

$\Rightarrow$ y is maximum then V is also maximum at $r = \sqrt{\frac{2}{3}}$

$\Rightarrow$ Required sectorial area $= \frac{1}{2} \times (1)^2 \times \theta = \pi\sqrt{2/3}$.

SOLVED PROBLEMS

1. Let $f'(\sin x) < 0$ and $f''(\sin x) > 0$, $\forall x \in (0, \frac{\pi}{2})$ and $g(x) = f(\sin x) + f(\cos x)$, then $g(x)$ is decreasing in

(A) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ (B) $\left(0, \frac{\pi}{4}\right)$

(C) $\left(0, \frac{\pi}{2}\right)$ (D) $\left(\frac{\pi}{6}, \frac{\pi}{2}\right)$

Sol. **(B).** $g'(x) = f'(\sin x).\cos x - f'(\cos x).\sin x$

$\Rightarrow g''(x) = -f'(\sin x)\sin x + \cos^2 x \,.\, f''(\sin x) + f''(\cos x).\sin^2 x - f'(\cos x).\cos x$

$> 0 \ \forall x \in (0, \frac{\pi}{2})$

$\Rightarrow g'(x)$ is increasing in $\left(0, \frac{\pi}{2}\right)$. Also $g'\left(\frac{\pi}{4}\right) = 0$

$\Rightarrow g'(x) > 0 \ \forall x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ and $g'(x) < 0 \ \forall x \in \left(0, \frac{\pi}{4}\right)$. Thus $g(x)$ is decreasing in $\left(0, \frac{\pi}{4}\right)$.

2. Let $f(x)$ be a monotonic polynomial of $2m - 1$ degree where $m \in N$ then the equation $f(x) + f(3x) + f(5x) + \ldots + f((2m-1)x) = 2m - 1$ has

(A) atleast one real root (B) $(2m - 1)$ real roots

(C) exactly one real root (D) none of these

Sol. **(C).** $f(x)$ is monotonic $\Rightarrow f'(x) < 0$ or $f'(x) > 0 \quad \forall x \in R$

$\Rightarrow f'(px) < 0$ or $f'(px) > 0 \quad \forall x \in R$

$\Rightarrow f(px)$ is also monotonic

$\Rightarrow f(x) + f(3x) + \ldots + f[(2m-1)x]$ is a monotonic polynomial of odd degree $(2m - 1)$, so it will attain all real values only once.

3. Given that $f'(x) > g'(x)$ for all real x, and $f(0) = g(0)$, then $f(x) < g(x)$ for all n belonging to

(A) $(0, \infty)$ (B) $(-\infty, 0)$

(C) $(-\infty, \infty)$ (D) none of these.

Sol. **(B).** Let $h(x) = f(x) - g(x)$

$h'(x) = f'(x) - g'(x) > 0 \ \forall x \in R$

$\Rightarrow h(x)$ is an increasing function and $h(0) = f(0) - g(0) = 0$.

Therefore, $h(x) > 0 \ \forall x \in (0, \infty)$ and $h(x) < 0 \ \forall x \in (-\infty, 0)$.

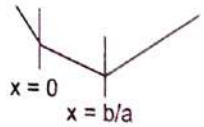
4. The function $f(x) = |ax - b| + c|x| \ \forall\ x \in (-\infty, \infty)$, where $a > 0, b > 0, c > 0$, assumes it's minimum only at one point if

(A) $a \neq b$ (B) $a \neq c$

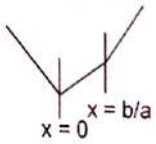
(C) $b \neq c$ (D) $a = b = c$

Sol. **(B).** $f(x) = \begin{cases} b-(a+c)x, & x < 0 \\ b+(c-a)x, & 0 \le x < \dfrac{b}{a} \\ (a+c)x+b, & x \ge \dfrac{b}{a} \end{cases}$

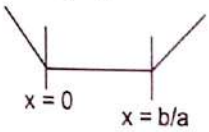
Case (i): when $c < a$



Case (ii): when $c > a$



Case (iii): when $c = a$



5. Let domain and range of $f(x)$ and $g(x)$ are $[0, \infty)$. If $f(x)$ be an increasing and $g(x)$ be a decreasing function. Also $h(x) = f(g(x))$, $h(0) = 0$ and $p(x) = h(x^3 - 2x^2 + 2x) - h(4)$ then for all x belonging to $(0, 2)$

(A) $p(x) \in (0, -h(4))$ (B) $p(x) \in (-h(4), 0)$

(C) $p(x) \in (-h(4), h(4))$ (D) $p(x) \in (h(4), -h(4))$

Sol. **(A).** Here $h(x) = f(g(x))$

and $h'(x) = f'(g(x))g'(x) < 0 \ \forall\ x \in [0, \infty)$

Since $g'(x) < 0 \ \forall\ x \in [0, \infty)$

and $f'(g(x)) > 0 \ \forall\ x \in [0, \infty)$

Also, $h(0) = 0$.

Therefore, $h(x) < 0 \ \forall\ x \in [0, \infty)$

Now, $p(x) = h(x^3 - 2x^2 + 2x) - h(4)$

or $p'(x) = h'(x^3 - 2x^2 + 2x)(3x^2 - 4x + 2) < 0 \ \forall\ x \in (0, 2)$

Since $h'(x^3 - 2x^2 + 2x) < 0 \ \forall\ x \in (0, \infty)$

and $3x^2 - 4x + 2 > 0 \ \forall\ x \in R$

$\Rightarrow p(x)$ is a decreasing function

$p(2) < p(x) < p(0)$ for all $x \in (0, 2)$

$h(4) - h(4) < p(x) < h(0) - h(4)$

$\Rightarrow 0 < p(x) < -h(4)$.

6. If the equation $x^5 - 10a^3x^2 + b^4x + c^5 = 0$ has three equal roots, then

(A) $2b^2 - 10a^3b^2 + c^5 = 0$ (B) $6a^5 + c^5 = 0$

(C) $2c^5 - 10a^3b^2 + b^4c^5 = 0$ (D) $b^4 = 15a^4$

Sol. **(B, D).** $f(x) = x^5 - 10a^3x^2 + b^4x + c^5$

$f'(x) = 5x^4 - 20a^3x + b^4$

$f''(x) = 20x^3 - 20a^3$

If $x = \alpha$ be a root that is repeated three times

$\Rightarrow f''(\alpha) = 0,\ f'(\alpha) = 0,\ f(\alpha) = 0$

$\Rightarrow \alpha = a,\ 5a^4 - 20a^4 + b^4 = 0,\ a^5 - 10a^5 + ab^4 + c^5 = 0$

$\Rightarrow \alpha = a,\ b^4 = 15a^4$ and $c^5 + 15a^5 - 9a^5 = 0$ or $6a^5 + c^5 = 0$

7. The equation $8x^3 - ax^2 + bx - 1 = 0$ has three real roots in G.P. If $\lambda_1 \le a \le \lambda_2$, then ordered pair (λ_1, λ_2) can be

(A) $(-2, 2)$ (B) $(24, 29)$

(C) $(-10, -8)$ (D) none of these.

Sol. **(B, C).** $8x^3 - ax^2 + bx - 1 = 0$

Let the roots be $\frac{\alpha}{r},\ \alpha,\ \alpha r$

$$\Rightarrow \alpha\left(\frac{1}{r} + 1 + r\right) = \frac{a}{8},\ \alpha^3 = \frac{1}{8} \Rightarrow \alpha = \frac{1}{2}$$

$$\Rightarrow r^2 + 1 + r = \frac{ar}{4} \text{ or } r^2 + r\left(1 - \frac{a}{4}\right) + 1 = 0$$

$$\Rightarrow \left(1 - \frac{a}{4}\right)^2 - 4 \ge 0 \text{ or } (a+4)(a-12) \ge 0 \text{ or } a \le -4 \text{ or } a \ge 12.$$

8. If $a, b > 0$, let $f(a, b) = \frac{a^3b}{(a+b)^4}$, then

(A) absolute maximum value of $f(a, b)$ is $\frac{81}{512}$

(B) absolute maximum value of $f(a, b)$ is $\frac{27}{256}$

(C) The maximum value when $a = b$ is smaller than the maximum value when $a \ne b$.

(D) Maximum value is same for all a and b.

Sol. **(B, C)**

Using AM $\ge$ GM for a, a, a, 3b we get $\frac{a+a+a+3b}{4} \ge (3a^3b)^{1/4}$

$$\Rightarrow \frac{3}{4}(a+b) \ge (3a^3b)^{1/4} \Rightarrow \frac{3^4}{256}(a+b)^4 \ge 3a^3b \Rightarrow \frac{a^3b}{(a+b)^4} \le \frac{27}{256}$$

Choice (C) is obviously correct.

9. If $f(x) = \int_0^x \frac{\sin t}{t}\,dt,\ x > 0$, then

(A) $f(x)$ has local maxima at $x = n\pi$ $(n = 2k, k \in I^+)$

(B) $f(x)$ has local minima at $x = n\pi$ $(n = 2k, k \in I^+)$

(C) $f(x)$ has neither maxima nor minima at $x = n\pi$ $(n \in I^+)$

(D) $f(x)$ has local maxima at $x = n\pi$ $(n = 2k - 1, k \in I^+)$

Sol. **(B, D).** $f'(x) = \dfrac{\sin x}{x}$

For $f'(x) = 0$, $\dfrac{\sin x}{x} = 0 \Rightarrow x = n\pi (n \in I, n \neq 0)$

$f''(x) = \dfrac{x\cos x - \sin x}{x^2}$; $f''(n\pi) = \dfrac{\cos n\pi}{n\pi} < 0$ if $n = 2k - 1 > 0$ if $n = 2k, k \in I^+$.

Hence $f(x)$ has local maxima at $x = n\pi$, where $n = 2k - 1$ and local minima at $x = n\pi$, $n = 2k$, where $k \in I^+$.

10. The values of parameter 'a' for which the point of minimum of the function $f(x) = 1 + a^2x - x^3$ satisfies the inequality $\dfrac{x^2 + x + 2}{x^2 + 5x + 6} < 0$ are

(A) $(2\sqrt{3}, 3\sqrt{3})$ (B) $(-3\sqrt{3}, -2\sqrt{3})$

(C) $(-2\sqrt{3}, 3\sqrt{3})$ (D) $(-3\sqrt{2}, 2\sqrt{3})$

Sol. **(A, B).** Given that $\dfrac{x^2 + x + 2}{x^2 + 5x + 6} < 0 \Rightarrow x \in (-3, -2)$

We have to find the extrema of the function $f(x) = 1 + a^2x - x^3$

For maximum or minimum, $f'(x) = 0$

$\Rightarrow a^2 - 3x^2 = 0$ or $x = \pm\dfrac{a}{\sqrt{3}}$ and $f''(x) = -6x$ is +ve when x is negative.

If a is positive then point of minima is $-\dfrac{a}{\sqrt{3}}$ i.e. $-3 < -\dfrac{a}{\sqrt{3}} < -2$ or $2\sqrt{3} < a < 3\sqrt{3}$

and if a is negative then point of minima is $\dfrac{a}{\sqrt{3}}$,

i.e. $-3 < \dfrac{a}{\sqrt{3}} < -2$ or $-3\sqrt{3} < a < -2\sqrt{3}$

Then, $a \in (-3\sqrt{3}, -2\sqrt{3}) \cup (2\sqrt{3}, 3\sqrt{3})$.

Comprehension

If $f(x) = (x - \alpha)^n g(x)$, then $f(\alpha) = f'(\alpha) = f''(\alpha) = \ldots = f^{n-1}(\alpha) = 0$ where $f(x)$ and $g(x)$ are polynomials.

For a polynomial $f(x)$ with rational coefficients, answer the following questions.

Answer the following questions:

11. If $f(x)$ touches x-axis at only one point, then the point of touching is

(A) always a rational number (B) may or may not be a rational number

(C) never a rational number (D) none of these

Sol. **(A)**

If $f(x)$ touches x-axis at only one irrational point, then $f(x) = (x - \alpha)^2 g(x)$, where α is irrational.

$\Rightarrow$ coefficients of $f(x)$ can't be rational

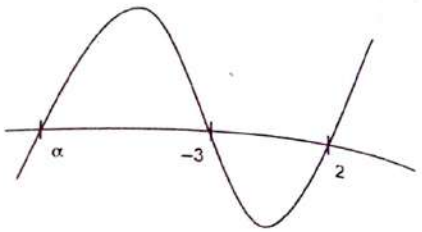
$\Rightarrow$ for $f(x)$ with rational coefficients, then point of touching is rational.

12. If f(x) is of degree 3 and touches x-axis, then
(A) all the roots of f(x) are rational
(B) only one root is rational
(C) both (A) and (B) may be possible
(D) none of these

Sol. **(A)**
The point of touching has to be rational
$\Rightarrow$ the two roots of f(x) = 0 are rational
$\Rightarrow$ third root is also rational.

13. If f(−3) = f(2) = 0 & f'(−3) < 0, then the largest integral value of c is (where $f(x) = x^3 + ax^2 + bx + c$)
(A) −18
(B) −19
(C) −12
(D) −6

Sol. **(B)**
The graph of f(x) is shown.
The third root α is lesser than − 3.
$\Rightarrow c < -18$.



Assertion Reason

14. Statement 1: The function $f(x) = (x^2 + x - 2)(x^2 + 2x - 3)$ has local extremum at x = 1.
because
Statement 2: f(x) is continuous and differentiable and f'(1) = 0.
(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1
(B) Statement -1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement -1 is True, Statement -2 is False
(D) Statement -1 is False, Statement -2 is True

Sol. **(B)**
Statement 1 is correct because $f(1)^- > f(1) < f(1)^+$.
Statement 2 is correct as f(x) has a repeated root at x = 1.
Statement 2 is not correct explanation of Statement 1 as f'(c) = 0 does not imply that f has an extrema at x = c.

15. Match the following:

	Column – I		Column – II
(A)	The number of integral solutions of the inequality $\frac{(x-1)^2(x-4)^3(x-7)}{(x-2)(x-6)} \le 0$ is/are	(p)	1
(B)	The integral values that the function $f(x) = \sqrt{x-2} + 2\sqrt{6-x}$ can take is/are	(q)	2
(C)	Let f: D $\to$ R and $f(x) = \underset{\text{n times}}{\ln\ln\ln\ldots\ln}\left(x^2 - \frac{x}{2} + \frac{49}{16} + \cos 4\pi x\right)$ then the value/s of n for which f is onto is/are	(r)	3
(D)	If f(x) is continuous and differentiable $\forall$ x $\in$ R function and f'(c) = 0 for exactly 2 real values of c then the number of real values of d for which f(d) = 0 can be	(s)	4
		(t)	7

Sol. (A) → (s, t); (B) → (q), (r), (s); (C) → (r), (s); (D) → (p), (q), (r)

(A) $f(x) \le 0$ for $x = 1, 3, 4, 7$.

(B) $f'(x) < 0\ \forall\ x \in \left(2, \frac{14}{5}\right)$ and $f'(x) > 0\ \forall\ x \in \left(\frac{14}{5}, 6\right)$

$\Rightarrow f(6) \le f(x) \le f\left(\frac{14}{5}\right) \Rightarrow f(x) \in \left[2, 2\sqrt{5}\right]$.

(C) $x^2 - \frac{x}{2} + \frac{49}{16}$ has minimum value 3 at $x = \frac{1}{4}$

$\Rightarrow \cos 4\pi x$ has minimum value -1 at $x = \frac{1}{4}$

$\Rightarrow x^2 - \frac{x}{2} + \frac{49}{16} + \cos 4\pi x \ge 2$.

Applying ln three times on both sides we get range R.

(D) $f'(x) = 0$ has n real roots $\Rightarrow f(x) = 0$ has at most $n + 1$ real roots.

Answer Q.16, Q.17 and Q.18 by appropriately matching the information given in the three columns of the following table.

Some intervals are asked in Column-I which are given in Column-II and for the function given in Column-I number of points of local maxima or minima within the related interval of Column-II are given in Column-III

Column–I	Column–II	Column–III
(I) If $f(x) = x \ln x + x - 3$ then the equation $f(x) = 0$ has atleast one root in	(i) (0, 1)	(P) number of local maximum of $f(x)$ is o
(II) If $27a + 9b + 3c + d = 0$ and $f(x) = 4ax^3 + 3bx^2 + 2cx + d = 0$, then the equation $f(x) = 0$ has atleast one root in	(ii) (1, 3)	(Q) number of local extrema for $g(x) = \int_0^x f(x)$ is atleast one
(III) If $c = \sqrt{3}$ and $f(x) = x + \frac{1}{x}$, then the interval of x for which LMVT is applicable for $f(x)$ is	(iii) (0, 3)	(R) number of local extrema of $f(x)$ is 1
(IV) If $c = \frac{1}{2}$ and $f(x) = 2x - x^2$, then the interval of x in which LMVT is applicable is	(iv) (–1, 1)	(S) number of local extrema of $f(x)$ is atleast 3

16. Which of the following combination is **CORRECT**?

(A) (I) (ii) (P) (B) (III) (ii) (R)

(C) (IV) (iv) (P) (D) (I) (i) (Q)

Sol. **(A)**

17. Which of the following combination is **CORRECT**?

(A) (II) (iii) (Q) (B) (III) (i) (P)

(C) (II) (iii) (R) (D) (III) (i) (R)

Sol. **(A)**

18. Which of the following combination is **INCORRECT**?

(A) (I) (i) (S) (B) (II) (iii) (Q)

(C) (III) (ii) (P) (D) (IV) (i) (R)

Sol. **(A)**

16.-18. (I) $f(x) = x \ln x + x - 3$

$f(1) = -2$, $f(3) = 3\ln 3 + 3 - 3 \Rightarrow f(3) > 0$

Hence, (I) $\rightarrow$ (ii)

Also $f'(x) = \frac{x}{x} + \ln x + 1 = 2 + \ln x = 0$

$x = e^{-2} \notin (1, 3)$

(II) Let $g(x) = \int_0^x f'(x) = ax^4 + bx^3 + cx^2 + dx$

$g(0) = 0$, $g(3) = 3(27a + 9b + 3c + d) = 0$

$\Rightarrow$ (II) $\rightarrow$ (iii)

Also $g'(x) = 0$ for atleast one $x \in (0, 3)$

Hence, (II) $\rightarrow$ (iii) $\rightarrow$ (Q)

(III) $f(1) = 2$, $f(3) = \frac{10}{3}$

$f'(x) = 1 - \frac{1}{x^2} \Rightarrow f'(c) = 1 - \frac{1}{c^2}$

$\Rightarrow \frac{\frac{10}{3} - 2}{3 - 1} = 1 - \frac{1}{c^2} \Rightarrow \frac{1}{c^2} = 1 - \frac{2}{3} \Rightarrow c = \sqrt{3}$

Also $f'(x) = 1 - \frac{1}{x^2} \Rightarrow f'(x) > 0$ for $x \in (1, 3)$

Hence, no extrema in (1, 3)

So (C) is also correct

(IV) $f(0) = 0$

$f(1) = 1$

$f'(x) = 2 - 2x \Rightarrow f'(c) = 2 - 2c$

$2 - 2c = \frac{1 - 0}{1 - 0} \Rightarrow 2c = 1 \Rightarrow c = \frac{1}{2} \Rightarrow$ (IV) $\rightarrow$ (i)

Also, $f'(x) = 0$ at $x = \frac{1}{2}$

Hence, (IV) $\rightarrow$ (i) $\rightarrow$ (R)

19. Let the function $f(x) = ax^6 + bx^5 + cx^4 + dx^3 + ex^2 + fx + g$ $(a \neq 0)$ touches the line $y = g(x) = x + p$ at $x = 1$, 2, and 3 & $h(x) = f(x) - g(x)$. If $\int_1^3 h(x)dx = \frac{32}{105}$. Find the value of a _____

Sol. **(2.00)**

$f(1) = 1 + p$, $f(2) = 2 + p$, $f(3) = 3 + p$

Also $f'(1) = 1$, $f'(2) = 1$, $f'(3) = 1$

Hence, $f(x) = a(x - 1)^2(x - 2)^2(x - 3)^2 + x + p$

$h(x) = f(x) - g(x) = a(x - 1)^2(x - 2)^2(x - 3)^2$

$$\int_1^3 h(x)dx = \int_1^3 a(x-1)^2(x-2)^2(x-3)^2 dx$$

Put $x - 2 = t$

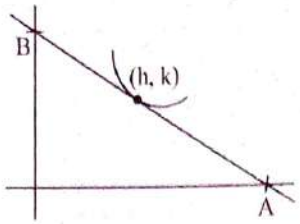
$$\Rightarrow dx = dt \Rightarrow \int_1^3 h(x) = a\int_{-1}^{1} t^2\left(t^2-1\right)^2 dt = 2a\int_0^1 t^2\left(t^2-1\right)^2 dt = 2a\left(\frac{8}{105}\right) = \frac{32}{105} \Rightarrow a = 2$$

20. Let $f(x)$ be a differentiable function on R defined by $f(x) = 5 - (x + 1)^2$ and A be the point of intersection where the tangent line drawn to the graph of $y = f(x)$ at the point $P(x, f(x))$ intersect with the x-axis and B be the intersection point where the tangent line intersect with y-axis. If $S(x)$ denotes the area of ΔOAB where O is the origin then the minimum value of $S(x)$ in the interval $(0, 1)$ is equal to $\frac{p}{q}$ (where p and q are possible numerically smallest integer). Find the value of $\frac{10q - p}{55}$ ____

Sol. **(2.00)**

$f'(x) = -2(x + 1) \Rightarrow m_T = -2(h + 1)$

Equation of tangent $(y - k) = -2(h + 1)(x - h)$



For A, $y = 0$, $x = \frac{k}{2(h+1)} + h$

For B, $x = 0, \Rightarrow y = 2h(h + 1) + k$

$$S(h) = \left(\frac{k+2h(h+1)}{2(h+1)}\left(k+2h(h+1)\right)\right)$$

$$S(h) = \frac{\left(5-(h+1)^2+2h(h+1)\right)^2}{2(h+1)} \Rightarrow S(h) = \frac{\left(h^2+4\right)^2}{2(h+1)}$$

$$S'(h) = \frac{1}{2}\left[\frac{(h+1)2\left(h^2+4\right)(2h)-\left(h^2+4\right)^2}{2(h+1)^2}\right] = \frac{1}{4}\frac{\left(h^2+4\right)}{(h+1)^2}\left[4h^2+4h-h^2-4\right]$$

$S'(h) = 0$

$\Rightarrow 3h^2 + 4h - 4 = 0 \Rightarrow 3h^2 + 6h - 2h - 4 = 0 \Rightarrow (3h - 2)(h + 2) = 0$

$h = \frac{2}{3}$ since $h \in (0, 1)$. Minimum $S(h) = \frac{\left(\frac{4}{9}+4\right)^2}{2\left(\frac{4}{9}+1\right)} = \frac{1600}{2(13)(9)}$

$\Rightarrow S(h) = \frac{800}{91} \Rightarrow p + q = 891$

21. If the range of parameter to in the interval $[0, 2\pi]$ satisfying $\frac{-2x^2+5x-10}{(\sin t)x^2+2(1+\sin t)x+9\sin t+4} > 0$ for all real values of x is $(a + b)$, then $(a+b) = \frac{4k\pi}{3}$. Find k ____

Sol. **(2.25)**

Here $-2x^{20} + 5x - 10 < 0 \ \forall \ x \in R$

$\Rightarrow (\sin t)x^2 + 2(1 + \sin t)x + 9 \sin t + 4 < 0 \ \forall \ x \in R$

Let $\sin t = a$ so $ax^2 + 2(1 + a)x + (9a + 4) < 0$

$\Rightarrow a < 0$ and $(4a - 1)(2a + 1) > 0$

$\Rightarrow a > \frac{1}{4}$ or $a < -\frac{1}{2}$

$\Rightarrow a < -\frac{1}{2} \Rightarrow \sin t < -\frac{1}{2} \Rightarrow t \in \left(\frac{7\pi}{6}, \frac{11\pi}{6}\right)$

$\Rightarrow a + b = 3\pi \ k = \frac{9}{4} = 2.25$

22. Let f be a real function defined on R (the set of real numbers) such that $f'(x) = 100(x-1)(x-2)^2(x-3)^3 \ldots\ldots (x-100)^{100}$ for all $x \in R$. If g is a function defined on R such that $\int_a^x e^{f(t)}dt = \int_0^x g(x-t)dt + 2x + 3$, then find the sum of all the values of x for which g(x) has local extremum

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Sol. **(2500)**

Put $x - t = u$

$\Rightarrow -dt = du$

$\Rightarrow \int_a^x e^{f(t)}dt = -\int_x^0 g(u)du + 2x + 3$

$\Rightarrow e^{f(x)} = g(x) + 2$

$\Rightarrow e^{f(x)}f'(x) = g'(x)$ hence the point of local extrema are x = 1, 3, 5, ..., 99.

$\text{Sum} = \frac{100}{2}(1+99) = 2500$

INDEFINITE INTEGRALS 4

Derived Substitution

Sometimes it is useful to write the integral as a sum of two related integrals which can be evaluated by making suitable substitutions. Examples of such integrals are:

A. Algebraic Twins

$$\int \frac{2x^2}{x^4+1}dx = \int \frac{x^2+1}{x^4+1}dx + \int \frac{x^2-1}{x^4+1}dx \ , \ \int \frac{2}{x^4+1}dx = \int \frac{x^2+1}{x^4+1}dx - \int \frac{x^2-1}{x^4+1}dx \ ,$$

$$\int \frac{2x^2}{\left(x^4+1+kx^2\right)}dx, \int \frac{2}{\left(x^4+1+kx^2\right)}dx \ , \text{ etc.}$$

B. Trigonometric twins

$$\int \sqrt{\tan x}\ dx, \int \sqrt{\cot x}\ dx \ , \int \frac{1}{\left(\sin^4 x+\cos^4 x\right)}dx \ , \int \frac{1}{\left(\sin^6 x+\cos^6 x\right)}dx \ , \int \frac{\pm\sin x \pm \cos x}{a+b\sin x\cos x}dx \ \text{ etc.}$$

Method of evaluating these integrals are illustrated by mean of the following examples:

Integration of Irrational Algebraic functions

1. Rational functions of $(ax + b)^{1/n}$ and x can be easily evaluated by the substitution $t^n = ax+ b$. Thus
$$\int f(x,(ax+b)^{1/n})dx = \int f\left(\frac{t^n-b}{a},t\right)\frac{nt^{n-1}}{a}dt \ .$$

2. $\int \frac{dx}{(x-k)^r\sqrt{ax^2+bx+c}}$. Here we substitute, $x - k = 1/t$.

This substitution will reduce the given integral in to $-\int \frac{t^{r-1}dt}{\sqrt{At^2+Bt+c}}$.

3. To integrate $\frac{1}{(Ax^2+B)\sqrt{Cx^2+D}}$, we first put $x = 1/t$, so that

$$\int \frac{dx}{(Ax^2+B)\sqrt{Cx^2+D}} = \int \frac{-1/t^2 dt}{(A/t^2+B)\sqrt{C/t^2+D}} = -\int \frac{tdt}{(A+Bt^2)\sqrt{C+Dt^2}} \ .$$

Now the substitution $C + Dt^2 = u^2$ reduces it to the form $\int \frac{du}{u^2 \pm a^2}$.

4. $\int \frac{\left(ax^2+bx+c\right)}{(dx+e)\sqrt{fx^2+gx+h}}$ Here, we write,

$ax^2 + bx + c = A_1(dx + e)(2fx + g) + B_1(dx + e) + C_1$

Where A_1, B_1 and C_1 are constants which can be obtained by comparing the coefficient of powers of x on both sides. And given integral will reduce to the form

$$A_1\int\frac{(2fx+g)}{\sqrt{fx^2+gx+h}}dx+B_1\int\frac{dx}{\sqrt{fx^2+gx+h}}+C_1\int\frac{dx}{(dx+e)\sqrt{fx^2+gx+h}}$$

Integration of the Functions of the Type $x^m(a+bx^n)^p$:

Case-I: If $p \in N$ (Natural number), we expand the integral with the help of binomial theorem and integrate.

Case-II: If $p \in I^-$, write $x = p^k$, where k is the L.C.M. of denominators of m and n.

Case-III: If $\frac{m+1}{n}$ is an integer, we put $(a+bx^n) = r^k$, where k is the denominator of the fraction p.

Case-IV: If $\frac{m+1}{n}+p$ is an integer, we put $a+bx^n = r^kx^n$, where k is the denominator of the fraction p.

ILLUSTRATIVE PROBLEMS

1. Evaluate $\int\frac{1}{x^4+1+5x^2}dx$.

Sol. $I=\frac{1}{2}\int\frac{2}{x^4+1+5x^2}dx=\frac{1}{2}\int\frac{1+x^2}{x^4+1+5x^2}dx-\frac{1}{2}\int\frac{x^2-1}{x^4+1+5x^2}dx$

$=\frac{1}{2}\int\frac{1+1/x^2}{x^2+1/x^2+5}dx-\frac{1}{2}\int\frac{1-1/x^2}{x^2+1/x^2+5}dx=(I_1-I_2)/2$

For I_1, we write $x-\frac{1}{x}=t\Rightarrow(1+\frac{1}{x^2})dx=dt$

$\Rightarrow I_1=\int\frac{dt}{t^2+7}=\frac{1}{\sqrt{7}}\tan^{-1}\frac{t}{\sqrt{7}}=\frac{1}{\sqrt{7}}\tan^{-1}\left(\frac{x-1/x}{\sqrt{7}}\right)$

For I_2, we write $x+\frac{1}{x}=t$.

$\Rightarrow(1-\frac{1}{x^2})dx=dt$

$I_2=\int\frac{dt}{t^2+3}=\frac{1}{\sqrt{3}}\tan^{-1}\frac{t}{\sqrt{3}}=\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{x+1/x}{\sqrt{3}}\right)+c$

Combining the two results, we get $I=\frac{1}{2}(I_1-I_2)=\frac{1}{2\sqrt{7}}\tan^{-1}\left(\frac{x-1/x}{\sqrt{7}}\right)-\frac{1}{2\sqrt{3}}\tan^{-1}\left(\frac{x+1/x}{\sqrt{3}}\right)+c$.

2. Evaluate $\int\frac{x^4-3x^2-3x-2}{x^3-x^2-2x}dx$.

Sol. Since the power of the numerator is higher than that of the denominator, i.e., the fraction is improper, we have to single out the integral part. Dividing the numerator by the denominator we obtain:

$$\frac{x^4-3x^2-3x-2}{x^3-x^2-2x}=x+1-\frac{x+2}{x(x^2-x-2)}$$

Hence $I=\int\frac{x^4-3x^2-3x-2}{x^3-x^2-2x}dx=\int(x+1)\,dx-\int\frac{(x+2)\,dx}{x(x-2)(x+1)}$

Expand the remaining proper fraction into simple ones i.e.

$$\frac{x+2}{x(x-2)(x+1)} = \frac{A}{x} + \frac{B}{x-2} + \frac{C}{x+1}$$

Hence, $x + 2 = A(x - 2)(x + 1) + Bx(x + 1) + C(x)(x - 2)$.

Substituting, in turn, the values 0, 2 and -1 of x

(the roots of the denominator) into both sides of the equality, we obtain $A = -1$, $B = 2/3$, $C = 1/3$

$$\Rightarrow \quad I = \int (x+1)\,dx + \int \frac{dx}{x} - \frac{2}{3}\int \frac{dx}{x-2} - \frac{1}{3}\int \frac{dx}{x+1}$$

$$= \frac{x^2}{2} + x + \ln|x| - \frac{2}{3}\ln|x-2| - \frac{1}{3}\ln|x+1| + c.$$

3. Evaluate $\int x^{-2/3}\left(1+x^{2/3}\right)^{-1} dx$.

Sol. If we substitute $x = p^3$, then $dx = 3p^2\,dp$.

$$I = \int \frac{3p^2\,dp}{p^2\left(1+p^2\right)} = 3\tan^{-1} p = 3\tan^{-1} x^{1/3} + c$$

4. Evaluate $\int x^{-2/3}\left(1+x^{1/3}\right)^{1/2} dx$.

Sol. If we substitute $1+x^{1/3} = r^2$ then, $\frac{1}{3x^{2/3}}dx = 2r\,dr$

$$I = \int 6r^2 dr = \left[\frac{6}{3}r^3\right] = 2\left(1+x^{1/3}\right)^{3/2} + c$$

5. Evaluate $\int x^{-11}\left(1+x^4\right)^{-1/2} dx$.

Sol. Here $\left(\frac{m+1}{n} + P\right) = \left[\frac{-11+1}{4} - \frac{1}{2}\right] = -3$

If we substitute $(1+x^4) = r^2 x^4$, then $\left(1+\frac{1}{x^4}\right) = r^2$

and $\frac{-4}{x^5}dx = 2r\,dr$

$$\text{Now } I = \int \frac{dx}{x^{13}\left(1+\frac{1}{x^4}\right)^{1/2}} = \frac{-1}{4}\int \frac{2r\,dr}{x^8 r} = \frac{-1}{2}\int \left(r^2-1\right)^2 dr$$

$$= \frac{-1}{2}\int \left(r^4 - 2r^2 + 1\right) dr = -\frac{r^5}{10} + \frac{r^3}{3} - \frac{r}{2} + c, \text{ where } r = \sqrt{1+\frac{1}{x^4}}.$$

SOLVED PROBLEMS

1. $\int 3^x (g'(x) + g(x) \cdot \ln 3)dx$ is equal to

(A) $3^x g(x) + c$ (B) $3^x \ln 3\, g(x) + c$

(C) $\frac{3^x}{\ln 3} g(x) + c$ (D) $3^x + c$

Sol. **(A).** $\int 3^x (g'(x) + g(x) \cdot \ln 3)dx$

$= \int 3^x g'(x)dx + \ln 3 \int 3^x g(x)dx = 3^x g(x) - \int 3^x \ln 3 g(x)dx + \int \ln 3 \cdot 3^x \cdot g(x)dx$

$I = 3^x g(x) + c.$

2. The value of $\int \sin x \log\left(\cot\frac{x}{2}\right)dx$ is

(A) $-\cos x \log\cot\frac{x}{2} - \log\sin\frac{x}{2} + \log\sec\frac{x}{2} + c$ (B) $-\sin x \log\cot\frac{x}{2} + \log\tan\frac{x}{2} + c$

(C) $-\cos x \log\tan\frac{x}{2} - \log\sin\frac{x}{2} - \log\sec\frac{x}{2} + c$ (D) $-\sin x \log\left(\cot\frac{x}{2}\right) + \cos x \log\left(\tan\frac{x}{2}\right) + c$

Sol. **(A).** $\int \sin x \log\left(\cot\frac{x}{2}\right)dx$

$$I = -\cos x \log\left(\cot\frac{x}{2}\right) - \int \frac{1}{\cot\frac{x}{2}} \cdot \left(\frac{-\operatorname{cosec}^2 x/2}{2}\right) \cdot (-\cos x)dx$$

$$= -\cos x \log\left(\cot\frac{x}{2}\right) - \frac{1}{2}\int \frac{\cos x}{\cos x/2 \sin x/2}dx$$

$$= -\cos x \log\left(\cot\frac{x}{2}\right) - \frac{1}{2}\int \frac{\left(\cos^2 x/2 - \sin^2 x/2\right)}{\cos x/2 \sin x/2}dx$$

$$= -\cos x \log\left(\cot\frac{x}{2}\right) - \frac{1}{2}\left[\int \cot\frac{x}{2}dx - \int \tan\frac{x}{2}dx\right]$$

$$= -\cos x \log\left(\cot\frac{x}{2}\right) - \frac{1}{2}\left[2\log\left(\sin\frac{x}{2}\right) - 2\log\sec\frac{x}{2}\right] + c$$

$$= -\cos x \log\left(\cot\frac{x}{2}\right) - \log\left(\sin\frac{x}{2}\right) + \log\left(\sec\frac{x}{2}\right) + c.$$

3. The value of $\int \frac{\sin x}{\sin\left(x - \frac{\pi}{6}\right)\sin\left(x + \frac{\pi}{6}\right)}dx$ is

(A) $\frac{1}{\sqrt{3}}\left[\log\tan\left(\frac{x}{2}\right) + \log\tan\left(\frac{x}{2} - \frac{\pi}{12}\right)\right] + c$ (B) $\frac{1}{\sqrt{3}}\left[\log\tan\left(\frac{x}{2} + \frac{x}{12}\right) + \log\tan\left(\frac{x}{2} - \frac{\pi}{12}\right)\right] + c$

(C) $\frac{1}{\sqrt{3}}\left[\log\cot\left(\frac{x}{2} + \frac{\pi}{12}\right)\right] + c$ (D) None of these

(where c is the integration constant)

Sol. **(B).** $\int \frac{\sin x}{\sin\left(x-\frac{\pi}{6}\right)\sin\left(x+\frac{\pi}{6}\right)}dx = \frac{1}{\sqrt{3}}\int \frac{2\sin x\cos\pi/6}{\sin\left(x-\frac{\pi}{6}\right)\sin\left(x+\frac{\pi}{6}\right)}dx$

$= \frac{1}{\sqrt{3}}\int \frac{\sin\left(x+\frac{\pi}{6}\right)+\sin\left(x-\frac{\pi}{6}\right)}{\sin\left(x-\frac{\pi}{6}\right)\sin\left(x+\frac{\pi}{6}\right)}dx = \frac{1}{\sqrt{3}}\int\left[\operatorname{cosec}\left(x-\frac{\pi}{6}\right)+\operatorname{cosec}\left(x+\frac{\pi}{6}\right)\right]dx$

$= \frac{1}{\sqrt{3}}\left[\log\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)+\log\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)\right]+c$ (where c is the integration constant)

4. $\int \frac{\operatorname{cosec}^2 x-2005}{\cos^{2005} x}dx$ is equal to

(A) $\frac{\cot x}{(\cos x)^{2005}}+c$

(B) $\frac{\tan x}{(\cos x)^{2005}}+c$

(C) $\frac{-(\tan x)}{(\cos x)^{2005}}+c$

(D) none of these

Sol. **(D).** $\int \frac{\operatorname{cosec}^2 x-2005}{\cos^{2005} x}dx = \int(\cos x)^{-2005}\cdot\operatorname{cosec}^2 x\,dx - 2005\int\frac{dx}{\cos^{2005} x}$.

$= (\cos x)^{-2005}.(-\cot x) - \int(-2005)(\cos x)^{-2006}(-\sin x)(-\cot x)dx - 2005\int\frac{dx}{\cos^{2005} x}$.

$= -\frac{\cot x}{(\cos x)^{2005}}+c$.

5. The value of integral $\int e^x\left(\frac{1}{\sqrt{1+x^2}}+\frac{1-2x^2}{\sqrt{(1+x^2)^5}}\right)dx$ is equal to

(A) $e^x\left(\frac{1}{\sqrt{1+x^2}}+\frac{x}{\sqrt{(1+x^2)^3}}\right)+c$

(B) $e^x\left(\frac{1}{\sqrt{1+x^2}}-\frac{x}{\sqrt{(1+x^2)^3}}\right)+c$

(C) $e^x\left(\frac{1}{\sqrt{1+x^2}}+\frac{x}{\sqrt{(1+x^2)^5}}\right)+c$

(D) none of these

Sol. **(A).** $\int e^x\left(\frac{1}{\sqrt{1+x^2}}-\frac{x}{\sqrt{(1+x^2)^3}}+\frac{x}{\sqrt{(1+x^2)^3}}+\frac{1-2x^2}{\sqrt{(1+x^2)^5}}\right)$

$= e^x\frac{1}{\sqrt{1+x^2}}+e^x\frac{x}{\sqrt{(1+x^2)^3}} = e^x\left(\frac{1}{\sqrt{1+x^2}}+\frac{x}{\sqrt{(1+x^2)^3}}\right)+c$.

Using $\int e^x\left(f(x)+f'(x)\right)dx = e^x f(x)+c$.

6. The value of the integral $\int x^2 \frac{d(x-1/x)}{x^4+1}$ is

(A) $\left(x-\frac{1}{x}\right)^2+c$ (B) $\sin^{-1}\left(x-\frac{1}{x}\right)^2+c$

(C) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x-1/x}{\sqrt{2}}\right)+c$ (D) $\frac{1}{\sqrt{2}}\sin^{-1}\left(\frac{x-1/x}{\sqrt{2}}\right)+c$

Sol. **(C).** $\int \frac{x^2}{x^4+1}\cdot d\left(x-\frac{1}{x}\right)=\int \frac{1}{x^2+1/x^2}\cdot d\left(x-\frac{1}{x}\right)$

$=\int \frac{1}{(x-1/x)^2+2}\cdot d\left(x-\frac{1}{x}\right)=\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x-1/x}{\sqrt{2}}\right)+c$.

7. The value of $\int \frac{\sin 2x+2\tan x}{(\cos^6 x+6\cos^2 x+4)}dx$ is equal to

(A) $2\sqrt{\frac{1+\cos^2 x}{\cos^7 x}}+c$ (B) $\tan^{-1}\frac{1}{\sqrt{2}}\left(\frac{1+\cos^2 x}{\cos^7 x}\right)+c$

(C) $\frac{1}{12}\ln\left(\frac{1+\cos^2 x}{\cos^7 x}\right)+c$ (D) none of these

Sol. **(C).** $2\int \frac{(\cos x+\sec x)\sin x}{(\cos^6 x+6\cos^2 x+4)}dx=-2\int \frac{(t^2+1)}{t^7+6t^3+4t}dt$. Putting $\cos x = t$

$=-2\int \frac{\frac{1}{t^5}+\frac{1}{t^7}}{1+\frac{6}{t^4}+\frac{4}{t^6}}dt=\frac{1}{12}\ln\left(\frac{t^2+1}{t^7}\right)+c=\frac{1}{12}\ln\left(\frac{1+\cos^2 x}{\cos^7 x}\right)+c$.

8. If $\int \tan^5 x\,dx = A\tan^4 x + B\tan^2 x + g(x) + c$, then

(A) $A=\frac{1}{4}, B=-\frac{1}{2}$ (B) $g(x)=-\ln|\sec x|$

(C) $g(x)=\ln|\cos x|$ (D) $A=-\frac{1}{4}, B=\frac{1}{3}$

Sol. **(A, B).** $\int \tan^3 x(\sec^2 x-1)dx=\int \tan^3 x\sec^2 x\,dx-\int \tan^3 x\,dx$

$I_1=\int \tan^3 x\sec^2 x\,dx=\frac{\tan^4 x}{4}+c_1$

$I_2=\int \tan x(\sec^2 x-1)dx=\frac{\tan^2 x}{2}-\ln|\sec x|+c_2$.

9. The value of the integral $\int e^{\sin^2 x}\left(\cos x+\cos^3 x\right)\sin x\,dx$ is

(A) $\frac{1}{2}e^{\sin^2 x}\left(3-\sin^2 x\right)+c$ (B) $e^{\sin^2 x}\left(1+\frac{1}{2}\cos^2 x\right)+c$

(C) $e^{\sin^2 x}\left(3\cos^2 x+2\sin^2 x\right)+c$ (D) $e^{\sin^2 x}\left(2\cos^2 x+3\sin^2 x\right)+c$

Sol. **(A, B)**

Put $t = \sin^2 x$

The integral reduces to

$$I = \frac{1}{2}\int e^t (2 - t)dt = \frac{3}{2}e^t - \frac{te^t}{2} + c = \frac{1}{2}e^{\sin^2 x}\left(3 - \sin^2 x\right) + c \quad \text{(option A)}$$

$$= e^{\sin^2 x}\left(1 + \frac{1}{2}\cos^2 x\right) + c \quad \text{(option B)}$$

10. If $I = \int (\sqrt{\tan x} + \sqrt{\cot x})\,dx = f(x) + c$ then $f(x)$ is equal to

(A) $\sqrt{2}\sin^{-1}(\sin x - \cos x)$

(B) $\frac{\pi}{\sqrt{2}} - \sqrt{2}\cos^{-1}(\sin x - \cos x)$

(C) $\sqrt{2}\tan^{-1}\left(\frac{\tan x - 1}{\sqrt{2}\sqrt{\tan x}}\right)$

(D) none of these

Sol. **(A, B, C)**

$$I = \int (\sqrt{\tan x} + \sqrt{\cot x})\,dx = \int \sqrt{2}\frac{\sin x + \cos x}{\sqrt{2\sin x \cos x}}\,dx$$

If $\sin x - \cos x = p$ then $(\cos x + \sin x)dx = dp$

$$I = \sqrt{2}\int \frac{dp}{\sqrt{1 - p^2}} = \sqrt{2}\sin^{-1} p + c = \sqrt{2}\sin^{-1}(\sin x - \cos x) + c = \frac{\pi}{\sqrt{2}} - \sqrt{2}\cos^{-1}(\sin x - \cos x)$$

$$= \sqrt{2}\tan^{-1}\frac{\sin x - \cos x}{\sqrt{1 - (\sin x - \cos x)^2}} = \sqrt{2}\tan^{-1}\frac{\sin x - \cos x}{\sqrt{2\sin x \cos x}} = \sqrt{2}\tan^{-1}\left(\frac{\tan x - 1}{\sqrt{2\tan x}}\right)$$

Hence A, B, C are correct answers.

Comprehension

Let us consider the integrals of the following forms $f\left(x, \sqrt{mx^2 + nx + p}\right)^{1/2}$.

Case 1: If $m > 0$, then put $\sqrt{mx^2 + nx + p} = u \pm x\sqrt{m}$

Case 2: If $p > 0$, then put $\sqrt{mx^2 + nx + p} = ux \pm \sqrt{p}$

Case 3: If quadratic equation $mx^2 + nx + p = 0$ has real roots α and β there put

$\sqrt{mx^2 + nx + p} = (x - \alpha)u$ or $(x - \beta)u$

Answer the following questions:

11. If $I = \int \frac{dx}{x - \sqrt{9x^2 + 4x + 6}}$ to evaluate I, one of the most proper substitution could be

(A) $\sqrt{9x^2 + 4x + 6} = u \pm 3x$

(B) $\sqrt{9x^2 + 4x + 6} = 3u \pm x$

(C) $x = \frac{1}{t}$

(D) $9x^2 + 4x + 6 = \frac{1}{t}$

Sol. **(A).** As $m = 9 > 0$, hence we can substitute $\sqrt{9x^2 + 4x + 6} = u \pm 3x$.

12. $\int \frac{\left(x+\sqrt{1+x^2}\right)^{15}}{\sqrt{1+x^2}}dx$ is equal to

(A) $\frac{\left(x+\sqrt{1+x^2}\right)^{16}}{10}+c$

(B) $\frac{1}{15\left(\sqrt{1+x^2}+x\right)}+c$

(C) $\frac{15}{\left(\sqrt{1+x^2}-x\right)}+c$

(D) $\frac{\left(x+\sqrt{1+x^2}\right)^{15}}{15}+c$

Sol. **(D).** Here as per notations given, we can substitute $\sqrt{1+x^2}=(u-x)$ as $m=1>0$ and $p=1>0$.

$\Rightarrow I=\int\frac{u^{15}}{u}du=\int u^{14}du=\frac{1}{15}u^{15}+c=\frac{1}{15}\left(x+\sqrt{1+x^2}\right)^{15}+c.$

13. To evaluate $\int\frac{dx}{(x-1)\sqrt{-x^2+3x-2}}$ one of the most suitable substitution could be

(A) $\sqrt{-x^2+3x-2}=u$

(B) $\sqrt{-x^2+3x-2}=(ux\sqrt{2})$

(C) $\sqrt{-x^2+3x-2}=u(1-u)$

(D) $\sqrt{-x^2+3x-2}=u(x-2)$

Sol. **(D).** Here $m=-1<0$

$p=-2<0$. Also $-x^2+3x-2=-(x-1)(x-2)$

We can use case 3

$\Rightarrow$ Putting $\sqrt{-x^2+3x-2}=u(x-2)$ or $(x-1)\,u$ or $u(1-x)$.

Assertion Reason

14. **Statement –1:** Integral of an even function is not always an odd function.

Statement –2: Integral of an odd function is an even function.

(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement -1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement -1 is True, Statement -2 is False

(D) Statement -1 is False, Statement -2 is True

Sol. **(C).** Let $g(x)=f(x)+f(-x)$

Assuming $\int f(x)dx=F(x)+c$

$\Rightarrow\int g(x)dx=\int(f(x)+f(-x))dx=\int f(x)dx+\int f(-x)dx$

$=F(x)+c+(-F(-x)+c')=F(x)-F(-x)+c+c'$ which may be an odd function if $c+c'=0$.

Similarly integral of an odd function is not always an even function.

Hence statement 1 is true and statement 2 is false.

Hence (C) is the correct answer.

15. Which of the following functions appear in integration of function in column–I, match the following

Column – I		Column – II	
(A)	$\int \frac{x^2-x+1}{x^3-4x^2+4x}dx$	(p)	$x-\log\left[1-\sqrt{1-e^{2x}}\right]+c$
(B)	$\int \frac{x^2-1}{x(x-2)}dx$	(q)	$\log\lvert x-2\rvert$
(C)	$\int \frac{e^{-x}}{1+e^{x}}dx$	(r)	$\frac{1}{(x-2)}$
(D)	$\int \frac{1}{\sqrt{1-e^{2x}}}dx$	(s)	$\log(e^x+1)-x-e^{-x}+c$
		(t)	$\log\left(\frac{e^x+1}{e^x}\right)-e^{-x}+c$

Sol. **(A) → (q, r); (B) → (q, r); (C) → (s, t); (D) → (p)**

(A) $\int \frac{x^2-x+1}{x^3-4x^2+4x}dx$

By partial fractions $\frac{x^2-x+1}{x^3-4x^2+4x}=\frac{x^2-x+1}{x(x-2)^2}=\frac{A}{x}+\frac{B}{x-2}+\frac{C}{(x-2)^2}$

On solving A = B = C. So on primitive $\ell n|x-2|$, $\ell n|x|$, $\frac{1}{(x-2)}$ will appear

(B) $\int \frac{x^2-1}{x(x-2)^3}dx=\int\left[\frac{A}{x}+\frac{B}{x-2}+\frac{C}{(x-2)}+\frac{D}{(x-2)^3}\right]dx$

On solving for A, B, C, D. We have on primitive, $\ell n|x|$, $\ell n|x-2|$, $\frac{1}{(x-2)}$, $\frac{1}{(x-2)^2}$

(C) $\int \frac{e^{-x}}{1+e^{x}}dx=\int \frac{e^{-2x}}{e^{-x}+1}dx$ {Let $(e^{-x}+1)=t$, $e^{-x}dx=-dt$}

$=\int \frac{e^{-x}\cdot e^{-x}}{(e^{-x}+1)}dx=\int \frac{(t-1)(-dt)}{t}=\int\left(\frac{1}{t}-\frac{t}{t}\right)dt=\log_e|t|-t+c$

$=\log_e|e^{-x}+1|-(e^{-x}+1)+c=\log_e\left|\frac{e^x+1}{e^x}\right|-e^{-x}+c$

$=\log_e|e^x+1|-\log_e e^x-e^{-x}+c=\log_e|e^x+1|-x-e^{-x}+c$

(D) $\int \frac{1}{\sqrt{1-e^{2x}}}dx=\int \frac{e^{-x}}{\sqrt{e^{-2x}-1}}dx$ (Let $e^{-x}=t$, $e^{-x}dx=-dt$)

$I=\int \frac{-dt}{\sqrt{t^2-1}}=-\log_e\left|t+\sqrt{t^2-1}\right|+c=-\log_e\left|e^{-x}+\sqrt{e^{-2x}-1}\right|+c$

$=-\log_e\left|\frac{1}{e^{-x}}+\sqrt{\frac{1-e^{-2x}}{e^{x}}}\right|+c=-\log_e\left|\frac{1+\sqrt{1-e^{2x}}}{e^x}\right|+c$

$=\log_e\left|\frac{e^x}{1+\sqrt{1-e^{-2x}}}\right|+c=x-\log_e\left|1+\sqrt{1-e^{-2x}}\right|+c$

Answer Q.16, Q.17 and Q.18 by appropriately matching the information given in the three columns of the following table.

Match the following Column(s)

Column–I	Column–II	Column–III
(I) $f(x)=\dfrac{1}{(x^2+1)\sqrt{x^2+2}}$	(i) $\dfrac{x^5}{A(1-x^4)^{5/2}}+c$	(P) A = 1
(II) $f(x)=\dfrac{1}{(x+2)\sqrt{x^2+6x+7}}$	(ii) $\sin^{-1}\left(\dfrac{x+1}{(x+A)\sqrt{2}}\right)+c$	(Q) A = –2
(III) $f(x)=\dfrac{x^4+x^8}{(1-x^4)^{7/2}}$	(iii) $\left(\sqrt{x}+A\right)\sqrt{1-x}+\cos^{-1}\sqrt{x}+c$	(R) A = 5
(IV) $f(x)=\sqrt{\dfrac{1-\sqrt{x}}{1+\sqrt{x}}}$	(iv) $-\tan^{-1}\sqrt{A+\dfrac{2}{x^2}}+c$	(S) A = 2

16. Which of the following combination is **CORRECT**?
(A) (II) (ii) (S) (B) (II) (i) (S)
(C) (II) (iii) (P) (D) (II) (iv) (Q)

Sol. **(A)**

17. Which of the following combination is **CORRECT**?
(A) (I) (iii) (P) (B) (I) (iv) (P)
(C) (I) (ii) (S) (D) (I) (i) (R)

Sol. **(B)**

18. Which of the following combination is **CORRECT**?
(A) (IV) (iii) (Q) (B) (IV) (ii) (Q)
(C) (IV) (iv) (S) (D) (IV) (i) (S)

Sol. **(A)**

16.-18. (I) $I=\int\dfrac{dx}{(x^2+1)\sqrt{x^2+2}}$

$=\int\dfrac{\frac{1}{x^3}dx}{\left(1+\frac{1}{x^2}\right)\sqrt{1+\frac{2}{x^2}}}$. Let $1+\dfrac{2}{x^2}=t^2$

(II) Let $x+2=\dfrac{1}{t}$

(III) $I=\int\dfrac{x^4+x^8}{(1-x^4)^{7/2}}=\int\dfrac{\left(x+\frac{1}{x^3}\right)dx}{\left(\frac{1}{x^2}-x^2\right)^{7/2}}$. Let $\dfrac{1}{x^2}-x^2=t^2$

(IV) Let $\sqrt{x}=\cos 2\theta$

19. If $\int \frac{x+\left(\cos^{-1}(3x)\right)^2}{\sqrt{1-9x^2}}dx = \frac{1}{A}\left(\sqrt{1-9x^2}+\left(\cos^{-1}(3x)\right)^B\right)+c$, then $|A+B|$ is _____

Sol. **(6.00)**

$$I=\int\frac{xdx}{\sqrt{1-9x^2}}+\int\frac{\left(\cos^{-1}(3x)\right)^2}{\sqrt{1-9x^2}}dx \Rightarrow 1-9x^2=y^2 \text{ and } \cos^{-1}(3x)=t$$

$$I=-\frac{1}{9}\left(\sqrt{1-9x^2}+\left(\cos^{-1}(3x)\right)^3\right) \Rightarrow \therefore A+B=-9+3=-6$$

20. If $\int\left(x^6+x^4+x^2\right)\sqrt{2x^4+3x^2+6}\,dx = \frac{\left(\alpha x^6+\beta x^4+\gamma x^2\right)^{3/2}}{18}+c$ where c is constant, then $|\beta+\gamma-\alpha|$ is _____

Sol. **(7.00)**

Let $2x^6+3x^4+6x^2=y^2$; $I=\frac{1}{18}\left(2x^6+3x^4+6x^2\right)^{3/2}+c$; $\alpha=2, \beta=3, \gamma=6$; $|\beta+\gamma-\alpha|=7$

21. If $\int\frac{dx}{\cos^3 x-\sin^3 x}=A\tan^{-1}(f(x))+B\ln\left|\frac{\sqrt{2}+f(x)}{\sqrt{2}-f(x)}\right|+c$, where $f(x)=\sin x+\cos x$, then the value of $\frac{\left(12A+9\sqrt{2}B\right)-3}{5}$ is _____

Sol. **(1.60)**

$$I=\int\frac{dx}{(\cos x-\sin x)(1+\sin x\cdot\cos x)}=2\int\frac{(\cos x-\sin x)}{(\cos x-\sin x)^2\left(2+(\sin x+\cos x)^2-1\right)}dx$$

$$=2\int\frac{(\cos x-\sin x)}{\left(\sin^2 x+\cos^2 x-2\sin x\cos x\right)\left(1+(\sin x+\cos x)^2\right)}dx$$

$$=2\int\frac{(\cos x-\sin x)}{\left(2-(\sin x+\cos x)^2\right)\left(1+(\sin x+\cos x)^2\right)}dx=2\int\frac{dt}{\left(2-y^2\right)\left(1+y^2\right)}\left(y=\sin x+\cos x\right)$$

$$=\frac{2}{3}\left[\tan^{-1}(y)+\frac{1}{2\sqrt{2}}\ln\left(\frac{\sqrt{2}+y}{\sqrt{2}-y}\right)\right]+c\ ;\ A=\frac{2}{3}, B=\frac{1}{3\sqrt{2}} \Rightarrow 12A+9\sqrt{2}B-3=8$$

22. If $\int\frac{\tan\theta}{\tan^2\theta+\tan\theta+1}d\theta=\theta-\frac{2}{\sqrt{A}}\tan^{-1}\left(\frac{2\tan\theta+1}{\sqrt{A}}\right)+c$, then $\frac{A}{2}$ is _____

Sol. **(1.50)**

$\tan\theta=y$

$$\therefore I=\int\frac{y}{\left(1+y+y^2\right)\left(1+y^2\right)}dy=\int\left(\frac{1}{1+y^2}-\frac{1}{1+y+y^2}\right)dy$$

$$=\tan^{-1}(y)-\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2y+1}{\sqrt{3}}\right)+c=\theta-\frac{2\tan^{-1}}{\sqrt{3}}\left(\frac{2\tan\theta+1}{\sqrt{3}}\right)+c$$

$\therefore A=3$

DEFINITE INTEGRALS

5

PROPERTIES OF DEFINITE INTEGRAL

1. Change of variable of integration is immaterial so long as limits of integration remain the same i.e.
$$\int_a^b f(x)dx = \int_a^b f(t)dt$$

2. $$\int_a^b f(x)dx = -\int_b^a f(x)dx$$

3. $$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

where the point c may lie between a and b or it may be exterior to (a, b).

4. $$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$$

5. $$\int_0^a f(x)dx = \int_0^{a/2} [f(x)+f(a-x)]dx$$

6. $$\int_{-a}^a f(x)dx = \int_0^a [f(x)+f(-x)]dx$$

7. $$\int_a^b f(x)dx = (b-a)\int_0^1 f((b-a)x+a)dx$$

8. If f (x) is a periodic function with period T, then

(a) $$\int_a^{a+nT} f(x)dx = n\int_0^T f(x)dx, \text{ where } n \in I$$

(b) $$\int_{mT}^{nT} f(x)dx = (n-m)\int_0^T f(x)dx, \text{ where } n, m \in I$$

(c) $$\int_{a+nT}^{b+nT} f(x)dx = \int_a^b f(x)dx, \text{ where } n \in I$$

Differentiation under the Integral Sign

a. Leibnitz's Rule:

If g is continuous on [a, b] and $f_1(x)$ and $f_2(x)$ are differentiable functions whose values lie in [a, b], then

$$\frac{d}{dx}\int_{f_1(x)}^{f_2(x)} g(t)dt = g(f_2(x))f_2'(x) - g(f_1(x))f_1'(x)$$

b. If $F(t) = \int_a^b g(x,t)dx$, then $\frac{dF}{dt} = \int_a^b \frac{\partial g(x,t)}{\partial t}dx$, where $\frac{\partial g}{\partial t}$ represents the differentiation of g with respect to t keeping x constant.

Definite Integral as the Limit of a Sum

An alternative way of describing $\int_a^b f(x)dx$ is that the definite integral $\int_a^b f(x)dx$ is a limiting case of the summation of an infinite series, provided f(x) is continuous on [a, b]

i.e., $\int_a^b f(x)dx = \lim_{n\to\infty} h\sum_{r=0}^{n-1} f(a+rh)$ where $h = \frac{b-a}{n}$. The converse is also true i.e., if we have an infinite series of the above form, it can be expressed as a definite integral.

Method to express the infinite series as definite integral :

- Express the given series in the form $\sum \frac{1}{n} f\left(\frac{r}{n}\right)$.
- Then the limit is its sum when $n \to \infty$, i.e. $\lim_{n\to\infty} \sum \frac{1}{n} f\left(\frac{r}{n}\right)$.
- Replace $\frac{r}{n}$ by x and $\frac{1}{n}$ by dx and $\lim_{n\to\infty} \sum$ by the sign of $\int$.
- The lower and the upper limit of integration are the limiting values of $\frac{r}{n}$ for the first and the last terms of r respectively.

Integration of Piecewise Continuous Functions

Any function f (x) which is discontinuous at finite number of points in an interval [a, b] can be made continuous in sub-intervals by breaking the intervals into these subintervals. If f (x) is discontinuous at points $x_1, x_2, x_3, \ldots, x_n$ in (a, b), then we can define subintervals $(a, x_1), (x_1, x_2) \ldots, (x_{n-1}, x_n), (x_n, b)$ such that f (x) is continuous in each of these subintervals. Such functions are called piecewise continuous functions. For integration of a piecewise continuous function, we integrate f(x) in these sub-intervals and finally add all the values.

ILLUSTRATIVE PROBLEMS

1. **Evaluate** $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}}$.

Sol. $I = \int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}} = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}dx = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin x}}{\sqrt{\cos x}+\sqrt{\sin x}}dx$

Hence, $2I = \int_{\pi/6}^{\pi/3} dx = \pi/3 - \pi/6 = \pi/6 \Rightarrow I = \pi/12.$

2. **Evaluate** $\int_{-4}^{-5} e^{(x+5)^2}dx + 3\int_{1/3}^{2/3} e^{9(x-2/3)^2}dx$.

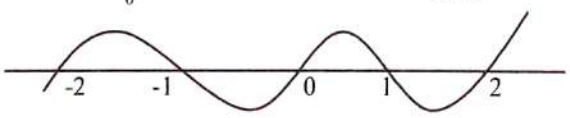
Sol. Let $I_1 = \int_{-4}^{-5} e^{(x+5)^2}dx = (-5+4)\int_0^1 e^{[(-5+4)x-4+5]^2}dx = -\int_0^1 e^{(x-1)^2}dx$.

Put $x = -t-4$ and replace t with x because it is a dummy variable.

Let $I_2 = \int_{1/3}^{2/3} e^{9(x-2/3)^2}dx = \left(\frac{2}{3}-\frac{1}{3}\right)\int_0^1 e^{9\left[\left(\frac{2}{3}-\frac{1}{3}\right)x+\frac{1}{3}-\frac{2}{3}\right]^2}dx = \frac{1}{3}\int_0^1 e^{(x-1)^2}dx \Rightarrow I_1 + 3I_2 = 0.$

3. **Find the points of maxima/minima of** $\int_0^{x^2} \frac{t^2-5t+4}{2+e^t}dt$.

Sol. Let $f(x) = \int_0^{x^2} \frac{t^2-5t+4}{2+e^t}dt \Rightarrow f'(x) = \frac{x^4-5x^2+4}{2+e^{x^2}}2x - 0 = \frac{(x-1)(x+1)(x-2)(x+2)2x}{2+e^{x^2}}$



From the wavy curve, it is clear that f '(x) changes its sign at $x = \pm 2, \pm 1, 0$ and hence the points of maxima are $-1, 1$ and of the minima are $-2, 0, 2$.

4. **Evaluate** $\int_0^1 \frac{x^b-1}{\ln x}dx$ **(b ≥ 0).**

Sol. Let $I(b) = \int_0^1 \frac{x^b-1}{\ln x}dx \Rightarrow I'(b) = \int_0^1 \frac{x^b \ln x}{\ln x}dx = \int_0^1 x^b dx$

or $I'(b) = \left[\frac{x^{b+1}}{b+1}\right]_0^1 = \frac{1}{b+1} \Rightarrow I(b) = \int \frac{db}{b+1} = \ln(b+1)+c$

If $b = 0$, then $I(b) = 0 \Rightarrow c = 0$. Hence, $I(b) = \ln(b+1)$.

5. Find the value of $\int_0^x [t+1]^3 dt$, where [.] denotes the greatest integer function and $x \in R^+$.

Sol. Let $I = \int_0^x [t+1]^3 dt$ and $n \le x < n+1 \Rightarrow [x] = n$ where $n = 0, 1, 2, \ldots$.

Now, [.] is continuous in each of (0, 1), (1, 2) ... (n, x) intervals. We can write.

$$I = \int_0^1 [t+1]^3 dt + \int_1^2 [t+1]^3 dt + \ldots + \int_{n-1}^n [t+1]^3 dt + \int_n^x [t+1]^3 dt$$

$$= \int_0^1 1^3 dt + \int_1^2 2^3 dt + \ldots + \int_{n-1}^n n^3 dt + \int_n^x (n+1)^3 dt$$

$$= 1^3 + 2^3 + \ldots + n^3 + (n+1)^3 (x-n) = \left(\frac{n(n+1)}{2}\right)^2 + (n+1)^3 (x-n)$$

$$= \left(\frac{[x]([x]+1)}{2}\right)^2 + ([x]+1)^3 \{x\}, \text{ where } \{.\} \text{ denotes fractional part.}$$

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SOLVED PROBLEMS

1. The value of $\int_0^{\pi/4} (\tan^n x + \tan^{n-2} x)\, d\left(x - \frac{[x]}{1!} + \frac{[x]^2}{2!} - \frac{[x]^3}{3!} + \ldots \right)$, where [x] is greatest integral function, is

(A) $\frac{1}{n}$ (B) $\frac{1}{n+2}$

(C) $\frac{1}{n-1}$ (D) $\frac{1}{n-2}$

Sol. **(C)**

$\because [x] = 0$ in $[0, \pi/4]$

$$I = \int_0^{\pi/4} (\tan^2 x + 1)\tan^{n-2} x\, dx = \int_0^{\pi/4} \tan^{n-2} x \sec^2 x\, dx .$$

$$= \frac{\tan^{n-1} x}{n-1}\Big]_0^{\pi/4} = \frac{1}{n-1}.$$

2. If [x] denotes the greatest integer less than or equal to x, then $\int_0^{\infty} \left[\frac{2}{e^x}\right] dx$ is equal to

(A) $\log_e 2$ (B) e^2

(C) 2/4 (D) 0

Sol. **(A)**

$$\left[\frac{2}{e^x}\right] = \begin{cases} 1 & \text{if } x \le \log_e 2 \\ 0 & \text{if } x > \log_e 2 \end{cases}$$

$$\therefore I = \int_0^{\log_e 2} \left[\frac{2}{e^x}\right] dx + \int_{\log_e 2}^{\infty} \left[\frac{2}{e^x}\right] dx = \int_0^{\log_e 2} 1.dx + 0 = \log_e 2$$

3. If $\int_{\pi/2}^{x} \sqrt{3 - 2\sin^2 t}\, dt + \int_0^{y} \cos t\, dt = 0$, then $\left(\frac{dy}{dx}\right)$ at $x = \pi$ and $y = \pi$ is

(A) $\sqrt{3}$ (B) $-\sqrt{2}$

(C) $-\sqrt{3}$ (D) none of these

Sol. **(A).** $\int_{\pi/2}^{x} \sqrt{3 - 2\sin^2 t}\, dt + \int_0^{y} \cos t\, dt = 0$, then $\left(\frac{dy}{dx}\right)$ at $x = \pi$ and $y = \pi$

$$\left(\sqrt{3 - 2\sin^2 x}\right)\cdot 1 + \cos y \frac{dy}{dx} = 0 \Rightarrow \left(\frac{dy}{dx}\right)_{\pi,\pi} = \frac{-\sqrt{3 - 2\sin^2 x}}{\cos y} = \sqrt{3}.$$

4. If $k = \int_0^1 \frac{e^t}{1+t} dt$, then $\int_0^1 e^t \ln(1+t)\, dt$ is equal to

(A) k (B) 2k

(C) e. ln 2 – k (D) none of these

Sol. **(C).** $\int_0^1 e^t \ln(1+t)\,dt = \left(\ln(1+t)e^t\right)_0^1 - \int_0^1 \frac{e^t}{1+t}\,dt$

$= e \ln 2 - k.$

5. The value of the definite integral $\int_{t+2\pi}^{t+\frac{5\pi}{2}} \left(\sin^{-1}(\cos x) + \cos^{-1}(\cos x)\right) dx$ is equal to

(A) $\frac{\pi^2}{2}$ (B) $\frac{\pi^2}{8}$

(C) $\frac{\pi^2}{4}$ (D) none of these

Sol. **(C).** Since $\sin^{-1}(\cos x) + \cos^{-1}(\cos x)$ is a periodic function with period 2π.

$\Rightarrow \int_{t+2\pi}^{t+5\pi/2} \left(\sin^{-1}(\cos x) + \cos^{-1}(\cos x)\right) dx = \int_0^{\pi/2} \left(\sin^{-1}(\cos x) + \cos^{-1}(\cos x)\right) dx.$

6. If $k \in N$ and $I_k = \int_{-2k\pi}^{2k\pi} |\sin x|[\sin x]\,dx$, (where [.] denotes the greatest integer function) then $\sum_{k=1}^{100} I_k$ equals to

(A) – 10100 (B) – 40400

(C) 20200 (D) none of these

Sol. **(D).** $\int_{-2k\pi}^{2k\pi} |\sin x|[\sin x]\,dx = \int_0^{2k\pi} |\sin x|([\sin x] + [-\sin x])\,dx$

$= -2k\int_0^{\pi} (\sin x)\,dx = -2k\left(-\cos x\right)_0^{\pi} = 2k(-2) = -4k$

$\Rightarrow \sum_{k=1}^{100} I_k = -200(101) = -20200.$

7. If f(x) is an integrable function in $\left[\frac{\pi}{6}, \frac{\pi}{3}\right]$ and $I_1 = \int_{\pi/6}^{\pi/3} \sec^2\theta\, f(2\sin 2\theta)\,d\theta$ and

$I_2 = \int_{\pi/6}^{\pi/3} \operatorname{cosec}^2\theta\, f(2\sin 2\theta)\,d\theta$, then

(A) $I_1 = 2I_2$ (B) $I_1 = 3I_2$

(C) $2I_1 = I_2$ (D) none of these

Sol. **(D).** $I_1 = \int_{\pi/6}^{\pi/3} \sec^2\theta\, f(2\sin 2\theta)\,d\theta$

Apply property $\int_a^b f(a+b-x)\,dx$

$$I_1 = \int_{\pi/6}^{\pi/3} \sec^2\left(\frac{\pi}{2}-\theta\right) f\left(2\sin 2\left(\frac{\pi}{2}-\theta\right)\right) d\theta$$

$$I_1 = \int_{\pi/6}^{\pi/3} \operatorname{cosec}^2\theta\, f(2\sin 2\theta)\, d\theta = I_2.$$

8. Given that $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{\log(n^2+r^2)-2\log n}{n} = \log 2+\frac{\pi}{2}-2$, then $\lim_{n\to\infty}\frac{1}{n^{2m}}\left[\left(n^2+1^2\right)^m\left(n^2+2^2\right)^m\ldots(2n^2)^m\right]^{1/n}$ is equal to

(A) $2^m e^{m(\pi/2-2)}$ (B) $2^m e^{m(2-\pi/2)}$

(C) $e^{m(\pi/2-2)}$ (D) $e^{2m(\pi/2-2)}$

Sol. **(A).** Given $\frac{\sum\log\left(\frac{n^2+r^2}{n^2}\right)}{n} = \log 2+\frac{\pi}{2}-2$

Let $A = \lim_{n\to\infty}\frac{1}{n^{2m}}\left[(n^2+1^2)^m(n^2+2^2)^m\cdots((2n^2)^m\right]^{1/n}$

$$\Rightarrow \log A = \frac{1}{n}\lim_{n\to\infty}\left[\log\left(1+\frac{1}{n^2}\right)^m+\log\left(1+\frac{2^2}{n^2}\right)^m+\cdots\log\left(1+\frac{n^2}{n^2}\right)^m\right]$$

$$= \frac{1}{n}\sum_{r=1}^{n}\lim_{n\to\infty}\log\left(1+\frac{r^2}{n^2}\right)^m = \frac{m}{n}\lim_{n\to\infty}\sum_{r=1}^{n}\log\left(1+\frac{r^2}{n^2}\right)$$

$$\log A = m\left(\log 2+\frac{\pi}{2}-2\right) \Rightarrow A = 2^m e^{m\left(\frac{\pi}{2}-2\right)}.$$

9. The value of $\int_{-6}^{6}\max(|2-|x||,\ 4-|x|,\ 3)dx$ is

(A) 40 (B) 50

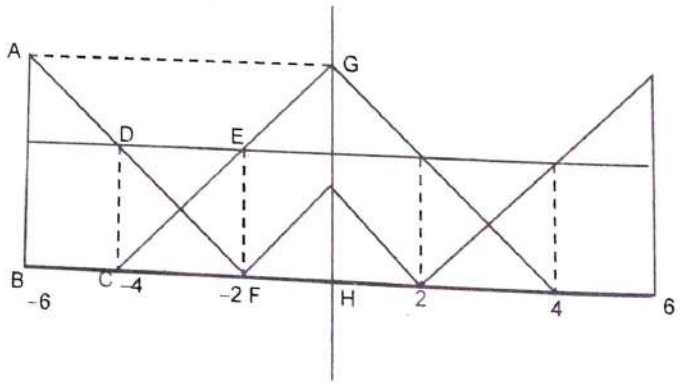
(C) 60 (D) 30

Sol. **(A).** $\int_{-6}^{6}\max(|2-|x||,\ 4-|x|,\ 3)dx$

$= 2(\Delta ABCD + \Delta DEFC + \Delta EFGH)$

$$= 2\left[\frac{1}{2}(3+4)\times 2+2\times 3+\frac{1}{2}(3+4)\times 2\right]$$

$= 2[14+6] = 40.$



10. If $I_n = \int_0^{\pi} e^x(\sin x)^n dx$, then $\frac{I_3}{I_1}$ is equal to

(A) 3/5

(C) 1 (B) 1/5

(D) 2/5

Sol. **(A).** $I_n = \int_0^{\pi} e^x (\sin x)^n \, dx = e^x (\sin x)^n \Big|_0^{\pi} - n\int_0^{\pi} (\sin x)^{n-1} \cos x \, e^x \, dx$

$\Rightarrow I_n = 0 - n\left[(\sin x)^{n-1} \cos x \, e^x\right]_0^{\pi} - \int_0^{\pi} \left\{(\sin x)^{n-1}(-\sin x) + \cos x(n-1)(\sin x)^{n-2}\right\} e^x \, dx$

$= -n\left[I_n - (n-1)I_{n-2} + (n-1)I_n\right] \Rightarrow I_n = \dfrac{n(n-1)I_{n-2}}{1+n^2}$

$\therefore \dfrac{I_3}{I_1} = \dfrac{3\times 2}{10} = \dfrac{3}{5}.$

11. $\int_{-1/2}^{1/2} \sqrt{\left\{\left(\dfrac{x+1}{x-1}\right)^2 + \left(\dfrac{x-1}{x+1}\right)^2 - 2\right\}} \, dx$ is

(A) $\ln\left(\dfrac{256}{81}\right)$ (B) $4\ln\left(\dfrac{3}{4}\right)$

(C) $4\ln\left(\dfrac{4}{3}\right)$ (D) $-\ln\left(\dfrac{81}{256}\right)$

Sol. **(A, B, C, D)**

$I = \int_{-1/2}^{1/2} \sqrt{\left(\dfrac{x+1}{x-1} - \dfrac{x-1}{x+1}\right)^2} \, dx = \int_{-1/2}^{1/2} \left|\dfrac{4x}{x^2-1}\right| dx = 2\int_0^{1/2} \dfrac{|4x|}{|x^2-1|} dx = -2\int_0^{1/2} \dfrac{4x}{x^2-1} dx$

$= -4\left\{\ln|x^2-1|\right\}_0^{1/2} = -4\ln\left(\dfrac{3}{4}\right) = 4\ln\left(\dfrac{4}{3}\right) = \ln\left(\dfrac{256}{81}\right) = -\ln\left(\dfrac{81}{256}\right).$

12. If $I = \int_0^1 \dfrac{dx}{1+x^{\pi/2}}$, then

(A) $I > \ln 2$ (B) $I < \ln 2$

(C) $I < \pi/4$ (D) $I > \pi/4$

Sol. **(A, C)**

$\dfrac{1}{1+x^2} > \dfrac{1}{1+x^{\pi/2}} > \dfrac{1}{1+x} \quad \forall\, x \in (0, 1)$

$\Rightarrow \int_0^1 \dfrac{dx}{1+x^2} > \int_0^1 \dfrac{dx}{1+x^{\pi/2}} > \int_0^1 \dfrac{dx}{1+x} \Rightarrow \pi/4 > I > \ln 2.$

13. If $\int_0^x f(t)\, dt = x^2 + 2x - \int_0^x t f(t)\, dt$, $x \in (0, \infty)$. Then $f(x)$ is

(A) periodic

(B) Periodic but fundamental period does not exists

(C) Periodic but fundamental period exists

(D) nothing can be said

Sol. **(A, B)**

Differentiating the given equation w.r.t.x we get

$f(x) = 2x + 2 - x\, f(x) \Rightarrow f(x) = 2(x+1)/(x+1) = 2 \Rightarrow f(x)$ is a constant function.

$\Rightarrow$ Periodic but fundamental period cannot be determined

Comprehension

Let f(x) is a differentiable function such that $f'(x) = f(x) + \int_0^2 f(x)dx$ if $f(0) = \frac{4-e^2}{3}$, then

14. f(x) is equal to

(A) $e^x - \left(\frac{e^2-1}{3}\right)$

(B) $e^x + \left(\frac{e^2-1}{3}\right)$

(C) $e^x - \left(\frac{e^2+1}{3}\right)$

(D) none of these

Sol. **(A)**

$f'(x) = f(x) + \int_0^2 f(x)\,dx$

Let $\int_0^2 f(x)dx = k \Rightarrow f'(x) - f(x) = k$

$\Rightarrow e^{-x} f(x) = k \int e^{-x}dx \Rightarrow f(x) = ce^x - k$

Since $\int_0^2 f(x)\,dx = k \Rightarrow k = \frac{c(e^2-1)}{3}$ & $f(o) = \frac{4-e^2}{3} \Rightarrow c = 1,\ k = \frac{e^2-1}{3}$

$f(x) = e^x - \left(\frac{e^2-1}{3}\right)$

15. The number of solution of $x + f(x) = 0$ is

(A) 0

(B) 1

(C) 2

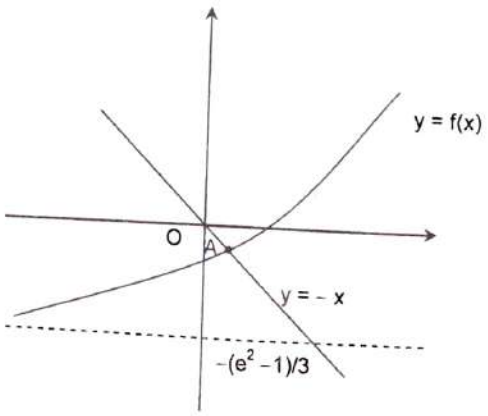
(D) none of these

Sol. **(B)**

Number of $f(x) + x = 0$

$\Rightarrow f(x) = -x$

Point A is the only solution.



16. $\int_0^1 f(x)dx$ equal to

(A) $\frac{3e-e^2-2}{3}$

(B) $\frac{3e+e^2-4}{3}$

(C) $\frac{3e-e^2-4}{3}$

(D) none of these

Sol. **(A)**

$$\int_0^1 f(x)\,dx = \int_0^1 \left(e^x - \left(\frac{e^2-1}{3}\right)\right)dx = e^x - \left(\frac{e^2-1}{3}\right)x\Big|_0^1 = \frac{3e-e^2+1}{3} - 1 = \frac{3e-e^2-2}{3}$$

Assertion Reason

17. Statement -1: $\int_{-\pi/2}^{\pi/2} \sin\left(\log\left(x+\sqrt{1+x^2}\right)\right)dx = 0$

because

Statement -2: $\int_{-a}^{a} f(x)dx = 0$ if $f(x)$ is an odd function.

(A) Statement -1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement -1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement -1 is True, Statement -2 is False

(D) Statement -1 is False, Statement -2 is True

Sol. **(A)**

Let $f(x) = \log\left(x+\sqrt{1+x^2}\right) \Rightarrow f(-x)+f(x) = 0$

Hence $f(x)$ is odd & $g(x) = \sin x \Rightarrow g(x)$ is odd

Let $h(x) = g(f(x))$

$\Rightarrow h(-x) = g(f(-x)) = -g(f(x))$ is odd.

Hence $\int_{-a}^{a} h(x)\,dx = 0$.

18. Match the following column

	Column – I		Column – II	
(A)	The value of $\frac{3}{4}\left\{\lim_{x\to 0^+}\frac{\int_0^{x^2}\sin\sqrt{t}}{x^3}dt\right\}$ is	(p)	$\ln 2$
(B)	The value of $\lim_{x\to\infty}\frac{\left\{\int_0^x e^{t^2}dt\right\}^2}{\int_0^x e^{3t^2}dt}$ is	(q)	$\lim_{x\to\infty} 2^{-x}$
(C)	If $[x] \le x$ is the largest integer function, then $\int_0^{\infty}\left[\frac{2}{e^x}\right]dx$ equals	(r)	$\frac{1}{2}$
(D)	If $\int_0^x f(t)\,dt = x + \int_x^1 tf(t)\,dt$, then the value of $f(1)$ is	(s)	$\ln\sqrt{e}$
		(t)	0

Sol. **(A) → (r, s); (B) → (q, t); (C) → (p); (D) → (r, s)**

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Answer Q.19, Q.20 and Q.21 by appropriately matching the information given in the three columns of the following table.

Match the following column(s)

Column–I	Column–II	Column–III
(I) $\left(\frac{e}{2}\right)\left(\int_{1/e}^{e}\lvert\ln x\rvert dx\right)+1$	(i) e	(P) $\left(\lim_{n\to\infty}\sum_{k=0}^{n}\frac{{}^{n}C_k}{(n^k)(k+3)}\right)+2$
(II) $\int_{-1/2}^{1/2}(\cos x)\cdot\ln\left(\frac{1+x}{1-x}\right)dx$	(ii) 0	(Q) $\int_{0}^{\infty}\frac{x\ln x}{(1+x^2)^2}dx$
(III) $16\cdot\int_{0}^{\pi/2}\frac{x\cdot\sin x\cdot\cos x}{\cos^4 x+\sin^4 x}dx$	(iii) π^2	(R) $\frac{3\sqrt{3}}{4}\cdot\int_{-\pi/4}^{\pi/4}\frac{2\pi+\sin 2\pi x}{2-\cos 2x}dx$
(IV) $\frac{8}{\pi}\cdot\int_{0}^{\pi/4}\ln(1+\tan x)dx$	(iv) ln 2	(S) $\left(\frac{-1}{\pi}\right)\int_{0}^{\pi}\ln(1+\cos x)dx$

19. Which of the following combination is **CORRECT**?
(A) (I) (i) (R) (B) (II) (ii) (R)
(C) (I) (i) (P) (D) (II) (i) (Q)

Sol. **(C)**

20. Which of the following combination is **CORRECT**?
(A) (II) (iii) (S) (B) (II) (ii) (Q)
(C) (II) (iii) (R) (D) (I) (ii) (R)

Sol. **(B)**

21. Which of the following combination is **INCORRECT**?
(A) (IV) (iv) (S) (B) (III) (iii) (R)
(C) (II) (ii) (Q) (D) (I) (ii) (P)

Sol. **(D)**

19.-21. (I) $\lim_{n\to\infty}\sum_{k=0}^{n}\frac{{}^{n}C_k}{n^k(k+3)}=\int_{0}^{1}x^2\cdot e^x dx=e-2$

(II) $I=\int_{0}^{\infty}\frac{x\ln x}{(1+x^2)^2}dx$

Put $x=\frac{1}{t}$

Then $I=-I$, hence $I=0$

(III) $\int_{0}^{\pi}\ln(1+\cos x)dx=(-)(\pi)(\ln 2)$

(IV) $\int_{0}^{\pi/4}\ln(1+\tan x)dx=\frac{(\pi)(\ln 2)}{8}$

22. If $\int_0^3 \sqrt{\frac{x^3}{3-x}}dx = \frac{9k\pi}{8}$, then k is equal to _____

Sol. **(3.00)**

Put $x = 3\sin^2\theta$

$dx = 6\sin\theta\cos\theta\, d\theta$

$$I = \int_0^{\pi/2} \sqrt{\frac{27\sin^6\theta}{3\cos^2\theta}} 6\sin\theta\cos\theta d\theta$$

$$= \int_0^{\pi/2} 3\frac{\sin^3\theta}{\cos\theta} 6\sin\theta\cos\theta d\theta$$

$$\Rightarrow I = 18\int_0^{\pi/2} \sin^4\theta d\theta$$

$$\Rightarrow I = \frac{27}{8}\pi$$

23. The value of integral $\int_{-2}^{3}([x]+[-x])dx$ is k, then $|k|$ is equal to _____ (where [.] denotes greatest integer function)

Sol. **(5.00)**

$$I = \int_{-2}^{-1}(-2+1)dx + \int_{-1}^{0}(-1+0)dx + \int_{0}^{1}(0-1)dx + \int_{1}^{2}(1-2)dx + \int_{2}^{3}(2-3)dx$$

$I = (-1) + (-1) + (-1) + (-1) + (-1)$

$I = -5$

24. Suppose $y = f(x)$ and $y = g(x)$ are two functions whose graphs intersect at the three points (0, 4), (2, 2), (4, 0) with $f(x) > g(x)$ for $0 < x < 2$ and $f(x) < g(x)$ for $2 < x < 4$. If $\int_0^4 (f(x)-g(x))\,dx = 10$ and $\int_2^4 (g(x)-f(x))dx = 5$, then area between two curves for $0 < x < 2$ is k, then $\frac{k}{2}$ is equal to _____

Sol. **(7.50)**

$$\int_0^4 (f(x)-g(x))\,dx = 10$$

$$(A_1 + A_3 + A_4) - (A_2 + A_3 + A_4) = 10$$

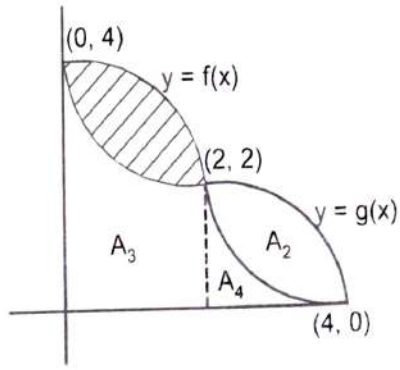
$$A_1 - A_2 = 10$$

$$\int_2^4 (g(x)-f(x))\,dx = 5$$

$$A_2 + A_4 - A_5 = 5$$

$$A_2 = 5$$

$$\Rightarrow A_1 = 15$$



25. The value of $\int_{-1}^{3}\{|x-2|+[x]\}dx$ is p, then $\frac{p}{2}$ is equal to _____ (where [x] stands for greatest integer less than or equal to x), is

Sol. **(3.50)**

$$\int_{-1}^{3} \{|x-2|+[x]\}dx$$

$$= \int_{-1}^{0} \{|x-2|+[x]\}dx + \int_{0}^{1} \{|x-2|+[x]\}dx + \int_{1}^{2} \{|x-2|+[x]\}dx + \int_{2}^{3} \{|x-2|+[x]\}dx$$

$$= \int_{-1}^{0} (2-x-1)dx + \int_{0}^{1} (2-x+0)dx + \int_{1}^{2} (2-x+1)dx + \int_{2}^{3} (x-2+2)dx$$

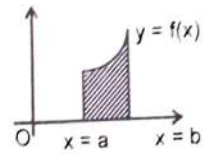
$$= \left[x - \frac{x^2}{2}\right]_{-1}^{0} + \left[2x - \frac{x^2}{2}\right]_{0}^{1} + \left[3x - \frac{x^2}{2}\right]_{1}^{2} + \left[\frac{x^2}{2}\right]_{2}^{3}$$

$$= -(-1 - \frac{1}{2}) + \left(2 - \frac{1}{2}\right) + (6-2) - (3 - (1/2)) + (9/2) - 2 = 7$$

AREA

- The area bounded by the continuous curve $y = f(x)$ above the x-axis and the ordinates at $x = a$ and $x = b$ is given by

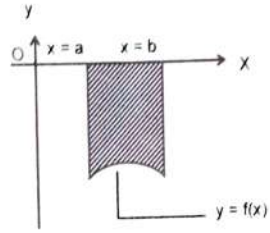
$$A = \int_a^b f(x)dx = \int_a^b y dx .$$



It is represented by the shaded portion in the graph.

- If the area is below the x - axis, then A is negative. The convention is to consider the magnitude only

$$A = \left| \int_a^b f(x)dx \right| .$$

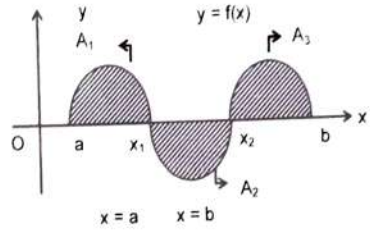


- If in general, the curve $y = f(x)$ crosses the x-axis a number of times, then the area bounded by the curve $y = f(x)$, the x - axis and the ordinates $x = a$ and $x = b$ is given by

$A = |A_1| + |A_2| + |A_3|$

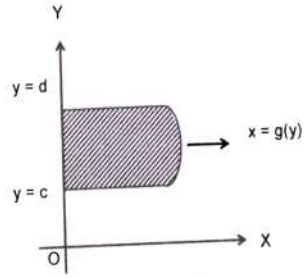
$$= \left| \int_a^{x_1} f(x)dx \right| + \left| \int_{x_1}^{x_2} f(x)dx \right| + \left| \int_{x_2}^{b} f(x)dx \right|$$

$$= \int_a^{x_1} f(x)dx - \int_{x_1}^{x_2} f(x)dx + \int_{x_2}^{b} f(x)dx$$



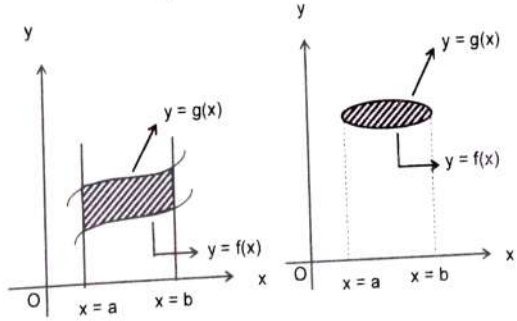
- The area bounded by the continuous curve $x = g(y)$, the y-axis and the line $y = c$ and $y = d$ is given by

$$\int_c^d x dy = \int_c^d g(y)dy .$$



- The area bounded by the curves $x = a$, $x = b (a < b)$ and the curves $y = f(x)$ and $y = g(x)$, provided $f(x) \le g(x)$ $(a \le x \le b)$, is given by $A = \int_a^b [g(x) - f(x)]dx$.

In certain case the left boundary ($x = a$) OR the right boundary may just degenerate into a point of intersection of the curves $y = f(x)$ and $y = g(x)$. Then a and b are found to be the



- The area bounded by the continuous curve $y = f(x)$ above the x-axis and the ordinates at $x = a$ and $x = b$ is given by

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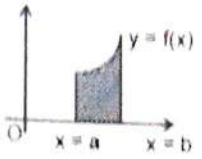

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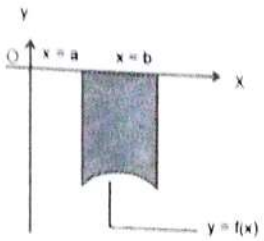
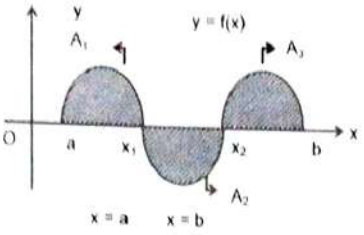


- If in general, the curve $y = f(x)$ crosses the x-axis a number of times, then the area bounded by the curve $y = f(x)$, the x - axis and the ordinates $x = a$ and $x = b$ is given by

$$A = |A_1| + |A_2| + |A_3|$$

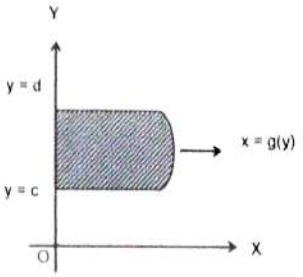
$$= \left|\int_a^{x_1} f(x)dx\right| + \left|\int_{x_1}^{x_2} f(x)dx\right| + \left|\int_{x_2}^{b} f(x)dx\right|$$

$$= \int_a^{x_1} f(x)dx - \int_{x_1}^{x_2} f(x)dx + \int_{x_2}^{b} f(x)dx$$



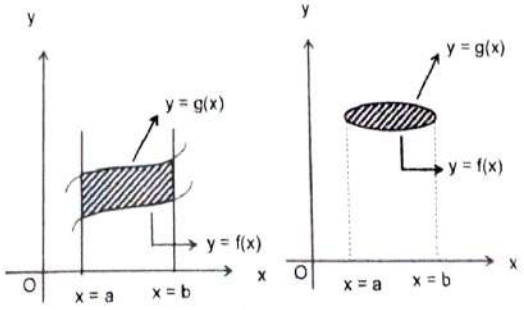
- The area bounded by the continuous curve $x = g(y)$, the y-axis and the line $y = c$ and $y = d$ is given by

$$\int_c^d xdy = \int_c^d g(y)dy .$$

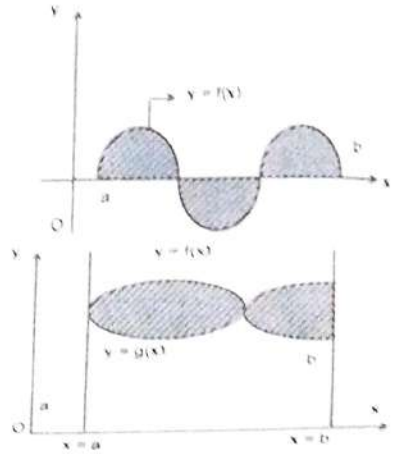


- The area bounded by the curves $x = a$, $x = b(a < b)$ and the curves $y = f(x)$ and $y = g(x)$, provided $f(x) \le g(x)$ $(a \le x \le b)$, is given by $A = \int_a^b [g(x) - f(x)]dx$.

In certain case the left boundary ($x = a$) OR the right boundary may just degenerate into a point of intersection of the curves $y = f(x)$ and $y = g(x)$. Then a and b are found to be the



abscissa of the points of intersection of the curves. Thus in general it can be said that the area bounded by a curve $y = f(x)$ and the lines $x = a$, $x = b$ and the x-axis is $\int_a^b |f(x)|\,dx$ and that bounded by the curves $y = f(x)$ and $y = g(x)$ and the lines $x = a$ and $x = b$ is $A = \int_a^b |g(x) - f(x)|\,dx$.



- It is often easy to break the interval [a, b] into several sub-intervals in which $\int f(x)\,dx$ is evaluated.

Curve Sketching for Area

- Find all x, such that $y = f(x) = 0$ $(a \le x \le b)$.
- Mark those points on the x axis.
- Find the sign of $f(x)$ in each of the subintervals b/w two successive zeroes of $f(x)$.
- Sometimes $f'(x)$ can be evaluated and the sign observed to determine whether $f(x)$ is increasing or decreasing in a particular subinterval.
- In case, two curves are to be drawn, it helps to find the point of intersection of these curves. A rough estimate of the shape of curves simplifies the problem of evaluating the area to a great extent.
- To find asymptotes for a curve, find out x in terms of y and check there exist any real y for which $x \to \pm\infty$ then y equal to that real number will be an asymptote for the curve. And also find y in terms of x and check for any real x, $y \to \pm\infty$, if there exits any such real number x, then x equal to that real number is an asymptote for the curve.

  e.g. $y = \dfrac{8a^3}{x^2 + 4a^2}$, here there is no such real x for which $y \to \infty$.

  But $y = \dfrac{8a^3}{x^2 + 4a^2}$

  $\Rightarrow y \to 0$ as $x \to \pm\infty$.

  $\therefore$ $y = 0$ is an asymptote

ILLUSTRATIVE PROBLEMS

1. Find the area of the region bounded by the curves $2^{|x|}|y| + 2^{|x|-1} \le 1$, $|x| \le \frac{1}{2}$ and $|y| \le \frac{1}{2}$.

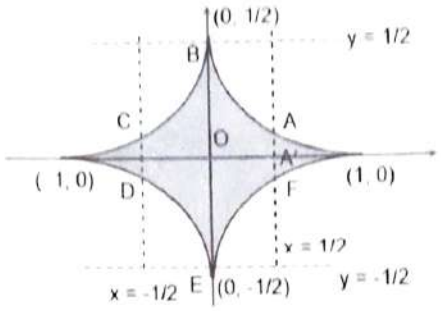
Sol. $2^{|x|}|y| + 2^{|x|-1} \le 1$ is equal to $|y| \le 2^{-|x|} - \frac{1}{2}$

The curve is symmetric about both the x and y-axes

Required region is shaded ABCDEA

Required area = 4 area of the region OA'ABO

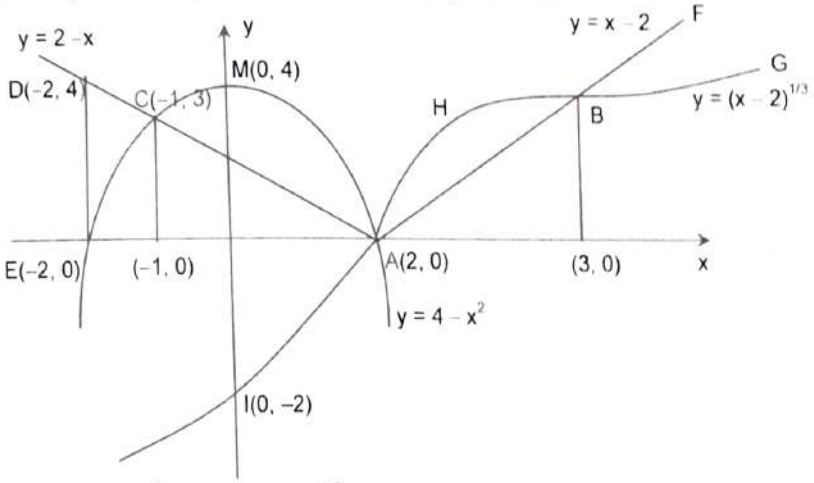
$$= 4\int_0^{1/2}\left(2^{-x} - \frac{1}{2}\right)dx = \frac{4}{\log 2}\left(1 - \frac{1}{\sqrt{2}}\right) - 1.$$



2. Find the area bounded by the curve $f(x) = \max.\{4 - x^2, |x-2|, (x-2)^{1/3}\}$ for $x \in [-2, 4]$ and x-axis.

Sol. $y = 4 - x^2$ is the parabola EMA

$y = |x - 2|$ is the pair of straight lines DCA and ABF, $y = (x-2)^{1/3}$ is the curve IAHBG



Thus $f(x) = \max.\{4 - x^2, |x-2|, (x-2)^{1/3}\}$ is

$$f(x) = \begin{cases} 2 - x, & -2 \le x < -1 \\ 4 - x^2, & -1 \le x < 2 \\ (x-2)^{1/3}, & 2 \le x < 3 \\ x - 2, & 3 \le x \le 4 \end{cases}$$

Now, area bounded by the curve and x-axis

$$= \int_{-2}^{-1}(2-x)dx + \int_{-1}^{2}\left(4 - x^2\right)dx + \int_{2}^{3}(x-2)^{1/3}dx + \int_{3}^{4}(x-2)dx$$

$$= \frac{7}{2} + 9 + \frac{3}{4} + \frac{3}{2} = \frac{59}{4} \text{ sq. units.}$$

3. Find the area bounded the curves $|y| = e^{-|x|} - \frac{1}{2}$ and $\frac{|x|+|y|}{2} + \left|\frac{|x|-|y|}{2}\right| \le 2$.

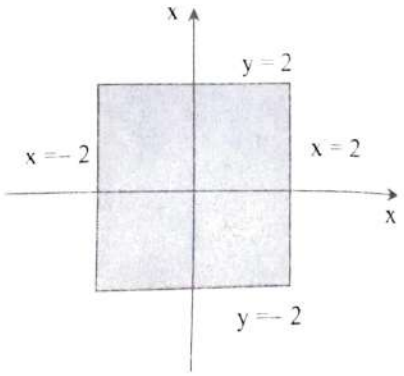
Sol. $\frac{|x|+|y|}{2}+\left|\frac{|x|-|y|}{2}\right| \le 2$

$\Rightarrow \ ||x|-|y|| \le 4-(|x|+|y|)$ (here $|x|+|y| \le 4$)

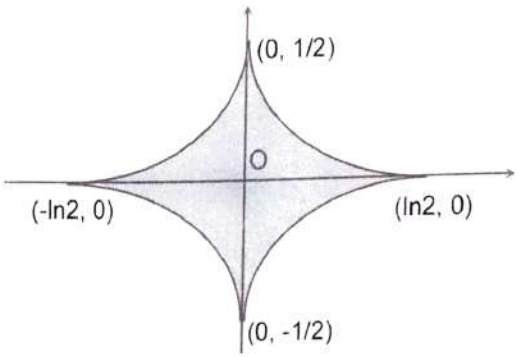
$\Rightarrow |x|+|y|-4 \le |x|-|y| \le 4-(|x|+|y|)$

$\Rightarrow |x| \le 2$ and $|y| \le 2$

so that the curve satisfying $|x| \le 2$, $|y| \le 2$ and $|x|+|y| \le 4$ is



Also curve $|y| = e^{-|x|} - \frac{1}{2}$ is symmetric about both x and y-axis. So that the curve is

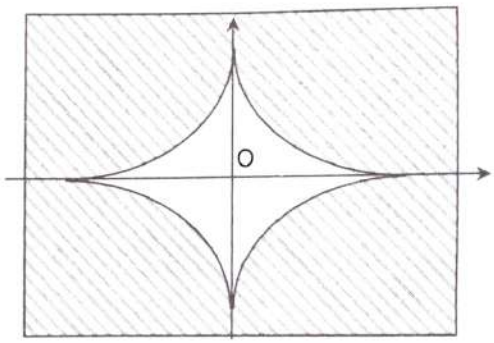


so that the required region is shaded region

= (area of square ABCD) − (area of curve A′B′C′D′)

$= 16 - 4\int_0^{\ln 2}\left(e^{-x}-\frac{1}{2}\right)dx$

$= 14 + \ln 4$ sq. units



4. Let f (x) be a continuous function given by $f(x)=\begin{cases} 2x, & |x| \le 1 \\ x^2+ax+b, & |x|>1 \end{cases}$

Find the area of the region in the third quadrant bounded by the curve $x = -2y^2$ and $y = f(x)$ lying on the left of the line $8x + 1 = 0$.

Sol. $f(x)=\begin{cases} 2x, & -1 \le x \le 1 \\ x^2+ax+b, & x<-1,\ x>1 \end{cases}$

Since f(x) is continuous. So it must be continuous at x = 1, − 1

$\Rightarrow \lim\limits_{x\to 1^-} f(x) = \lim\limits_{x\to 1^+} f(x) = f(1)$

$\Rightarrow \lim\limits_{x\to 1^-} 2x = \lim\limits_{x\to 1^+} x^2+ax+b = 2$

$\Rightarrow 2 = 1 + a + b \Rightarrow a + b = 1$ ...(1)

Also $\lim\limits_{x\to -1^-} f(x) = \lim\limits_{x\to -1^+} f(x) = f(-1)$

$\Rightarrow \lim\limits_{x\to -1^-} x^2+ax+b = \lim\limits_{x\to -1^+} 2x = -2$

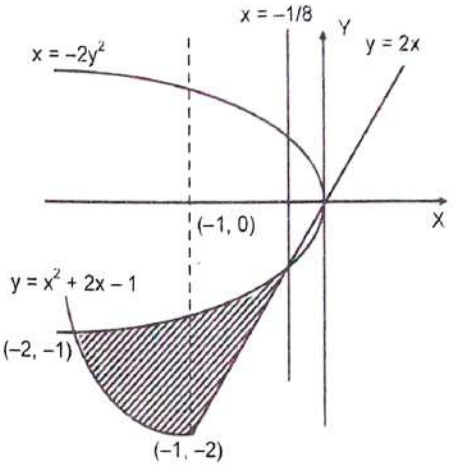
$\Rightarrow 1 - a + b = -2$

$\Rightarrow -a + b = -3 \quad \ldots(2)$

Solving (1) and (2), we have $b = -1,\ a = 2$

$\therefore\ f(x) = \begin{cases} 2x, & -1 \le x \le 1 \\ x^2 + 2x - 1, & x < 1,\ x > 1 \end{cases}$

Now drawing the given curves



$$\text{Required area} = \int_{-2}^{-1}\left[\sqrt{\frac{-x}{2}} - \left(x^2 + 2x - 1\right)\right] dx + \int_{-1}^{-1/8}\left[-\sqrt{\frac{-x}{2}} - 2x\right] dx$$

$= \frac{257}{192}$ sq. units.

5. Variable straight lines are drawn passing through the origin with gradient tanθ intersecting the circle $x^2 + y^2 - 2y = 0$ and the straight line $y = 2$ at A and B respectively. Lines drawn parallel to x-axis through and through B parallel to y-axis intersect at P. If the locus of P is $y = f(x)$ then the area enclosed between $[f(x)]$ and $|x| = 2$ where $[\cdot]$ denotes the greatest integer function is

Sol. Let P(h, k) OC = h, BC = h tan ω

$h = 2\cos\theta\ \ 2 = h\tan\theta$

In $\Delta OAQ \Rightarrow 2\cos\theta\ \ 2 = h\tan\theta$

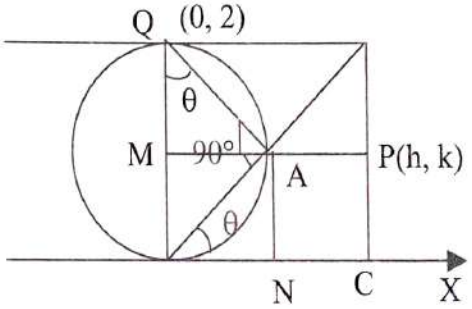
In $\Delta OAQ \Rightarrow OA = 2\sin\theta$

and in $\Delta AON\ \ AN = 2\sin^2\theta$

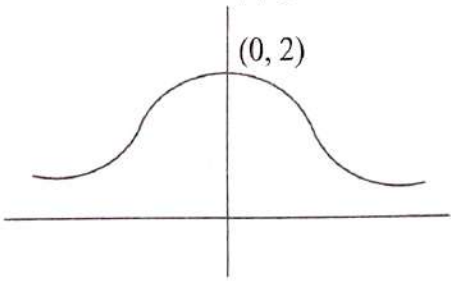
$k = 2\sin^2\theta$, P is $(2\cos\theta, 2\sin^2\theta)$

$$k = 2\sin^2\theta = \frac{2}{1+\cos^2\theta} = \frac{8}{4+h^2}$$

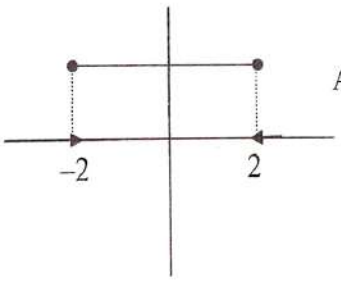
locus of P is $\left(y = \frac{8}{4+x^2}\right)$



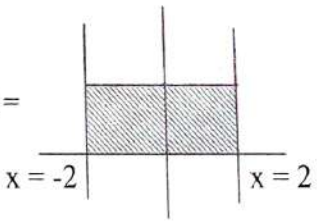
Graph of $y = \frac{8}{4+x^2}$



Graph of $y = \left[\frac{8}{4+x^2}\right]$



Area =



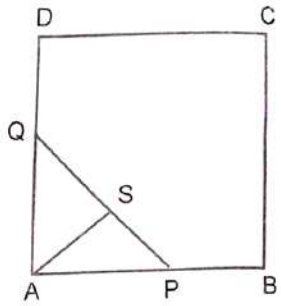
SOLVED PROBLEMS

1. Consider a square ABCD of side length 1. Let P be the set of all segments of length 1 with end points on adjacent sides of square ABCD. The mid points of segments in P enclose a region with area A, the value of A is

(A) $\frac{\pi}{4}$ (B) $1-\frac{\pi}{4}$

(C) $4-\frac{\pi}{4}$ (D) none of these

Sol. **(B).** Let S be the mid point of PQ.



Since $\angle PAQ = \frac{\pi}{2}$, we get $AS = SP = SQ = \frac{1}{2}$

$\Rightarrow$ S lies on the quarter circle of radius $\frac{1}{2}$ with centre at A.

Similarly S can also lie on quarter circle of radius $\frac{1}{2}$ with centre at B, C or D

$\Rightarrow$ area $A = 1-\frac{\pi}{4}$.

2. Let A be the area between co-ordinate axes, $y^2 = x - 1$, $x^2 = y - 1$ and the line which makes the shortest distance between two parabolas and A′ be the area between $x = 0$, $x^2 = y - 1$, $x = y$ and the shortest distance between $y^2 = x - 1$ and $x^2 = y - 1$, then

(A) A=A′ (B) $A = (A')^{1/2}$

(C) $A = 2A'$ (D) can't say anything

Sol. **(C).**

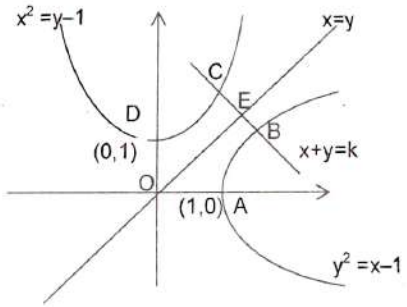
Line of shortest distance will be perpendicular to line $x = y$.

Also both parabolas are symmetric about $x = y$.

A = area OABCDO

A′ = area OECDO

$\Rightarrow A' = A/2$.



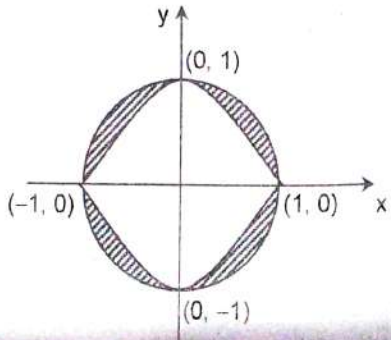
3. Area enclosed between the curves $|y| = 1 - x^2$ and $x^2 + y^2 = 1$ is

(A) $\frac{3\pi-8}{3}$ (B) $\frac{\pi-8}{3}$

(C) $\frac{2\pi-8}{3}$ (D) none of these

Sol. **(A).** Required area $= \pi - 4\left[\int_0^1 \left(1-x^2\right)dx\right]$

$= \pi - \frac{8}{3}$.



4. Area bounded by curves $y = [\cos A + \cos B + \cos C]$, $y = \left[7\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right]$ (where $[.]$ denotes the greatest integer function and A, B, C are angles of a triangle) and curve $|x-4| + |y| = 2$ is

(A) 2 (B) 3

(C) 4 (D) 6

Sol. **(B).** We know $1 < \cos A + \cos B + \cos C \le \frac{3}{2}$

and $0 < \sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} \le \frac{1}{8}$

$\Rightarrow$ curves are $y = 1$, $y = 0$

and $|x-4| + |y| = 2$, i.e. $|x| + |y| = 2$

Required area $= \frac{1}{2} \times 4 \times 2 - \frac{1}{2} \times 2 \times 1$

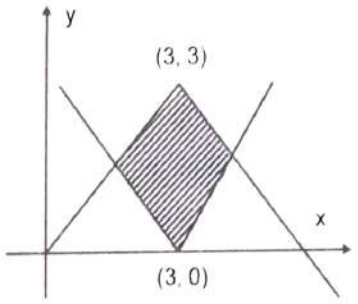
$= 4 - 1 = 3$.

5. The area bounded by $y \le 3 - |3 - x|$ and $y \ge |x - 3|$ is

(A) $\frac{5}{2}$ (B) $\frac{9}{2}$

(C) $\frac{7}{2}$ (D) none of these

Sol. **(B).** Required area $= \frac{3}{\sqrt{2}} \times \frac{3}{\sqrt{2}} = \frac{9}{2}$.



6. The area inside the parabola $5x^2 - y = 0$ but outside the parabola $2x^2 - y + 9 = 0$

(A) $12\sqrt{3}$ sq. unit (B) $6\sqrt{3}$ sq. unit

(C) $8\sqrt{3}$ sq. unit (D) $4\sqrt{3}$ sq. unit

Sol. **(A)**

$2x^2 - y + 9 = 0$

$5x^2 - (2x^2 + 9) = 0$

$3x^2 = 9$

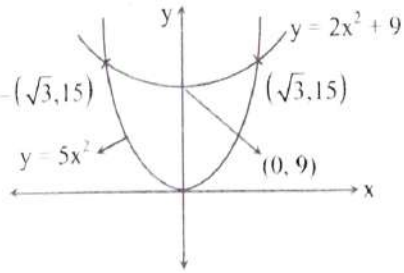
$x = -\sqrt{3}, \sqrt{3}$

Required area $= 2\int_0^{\sqrt{3}} ((2x^2 + 9) - 5x^2)dx$

$\Rightarrow 2\int_0^{\sqrt{3}} (9 - 3x^2)dx$

$\Rightarrow 2[9x - x^3]_0^{\sqrt{3}}$

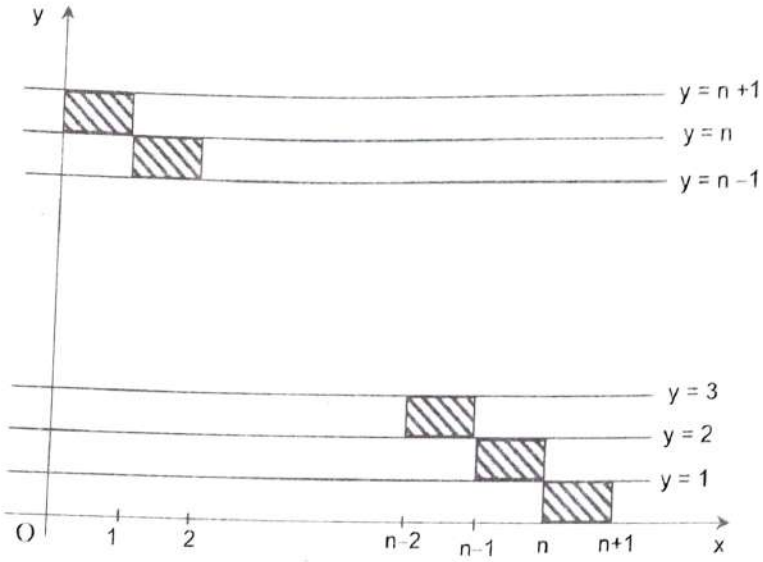
$\Rightarrow 12\sqrt{3}$ sq. unit



7. Area bounded by $[x] + [y] = n$, $n \in N$ and $y = i$ where $i \in N \ \forall \ i \le n + 1$ in the first quadrant (where $[.]$ is the greatest integer function) is

(A) n sq. units
(B) (n + 1) sq. units
(C) (n – 1) sq. units
(D) none of these

Sol. **(B).**



It is obvious that area is (n + 1) sq. units.

8. If a function $y = f(x)$ satisfying the conditions $f(x) + f(y) = f(x) f(y) + f(xy)$ where $f(1) = 0$ and $f'(1) = -2$ and the area bounded by $y = f(x)$ and $y = |\cos^{-1}(\cos x) - \sin^{-1}(\sin x)|$ from $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is then

(A) function $y = f(x)$ is $1 + x^2$
(B) $A = \frac{4\sqrt{2} - 3}{3}$ sq. units
(C) function $y = f(x)$ is $1 - x^2$
(D) $A = \frac{6 + 2\sqrt{2}}{3}$ sq. units

Sol. **(B, C).** $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

$$= \lim_{h \to 0} \frac{f\left(x\left(1 + \frac{h}{x}\right)\right) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{f(x) + f\left(1 + \frac{h}{x}\right) - f(x) f\left(1 + \frac{h}{x}\right) - f(x)}{h}$$

$$= \frac{1 - f(x)}{x} \lim_{h \to 0} \frac{f\left(1 + \frac{h}{x}\right) - f(1)}{h/x}$$

$$\Rightarrow f'(x) = \frac{1 - f(x)}{x}(-2) \qquad [\text{since } f'(1) = -2]$$

$$\Rightarrow f(x) = 1 - x^2$$

Now, $y = \cos^{-1}(\cos x) - \sin^{-1}(\sin x)$

For $x \in \left[-\frac{\pi}{2}, 0\right)$, $\cos^{-1}(\cos x) = -x$

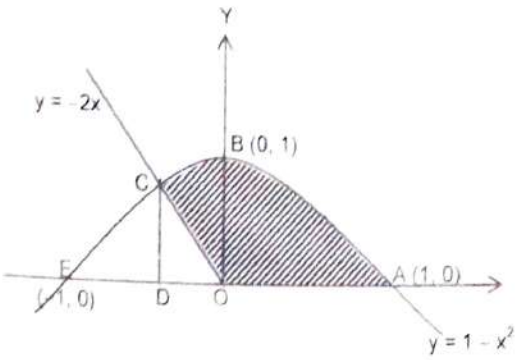
and $\sin^{-1}(\sin x) = x$

So that $y = 2x = -2x$

And for $x \in \left[0, \frac{\pi}{2}\right]$,

$\cos^{-1}(\cos x) = x$, $\sin^{-1}(\sin x) = x$ so that $y = 0$.

Required region is OABC = area of region DOABCD – area of triangle OCD



$$= \int_{-1/\sqrt{2}}^{0} (1 - x^2)\,dx - \frac{1}{2} OD \cdot CD = \frac{6 - 2\sqrt{2}}{3} - (3 - 2\sqrt{2}) = \frac{4\sqrt{2} - 3}{3} \text{ sq. units.}$$

9. If $A\left(\frac{3}{\sqrt{2}}, \sqrt{2}\right)$, $B\left(\frac{-3}{\sqrt{2}}, \sqrt{2}\right)$, $C\left(\frac{-3}{\sqrt{2}}, -\sqrt{2}\right)$ and $D(3\cos\theta, 2\sin\theta)$ are four points, then value of θ for which area of quadrilateral ABCD is maximum is, where $\theta \in \left(\frac{3\pi}{2}, 2\pi\right)$

(A) maximum area is 10 sq. units

(B) $\frac{7\pi}{4}$

(C) $2\pi - \sin^{-1}\left(\frac{3}{\sqrt{85}}\right)$

(D) maximum area is 12 sq. units.

Sol. **(B, D).** Area of quadrilateral ABCD is maximum when area of ACD is maximum
$\Rightarrow$ distance of D from AC is maximum i.e. $\cos\theta - \sin\theta$ is maximum
$= \sqrt{2}\cos\left(\theta + \frac{\pi}{4}\right)$ is maximum $\Rightarrow \theta = \frac{7\pi}{4}$ and area $= \frac{6}{\sqrt{2}} \cdot 2\sqrt{2} = 12$ sq. units
(since ABCD is a rectangle).

10. For which of the following values of m, is the area of the region bounded by the curve $y = x - x^2$ and the line $y = mx$ equals $\frac{9}{2}$?

(A) –4
(B) –2
(C) 2
(D) 4

Sol. **(B, D)**
Solving $y = x - x^2$ and $y = mx$, we get the points of intersection (0,0) and $[1-m, m(1-m)]$
Case I $m < 1$

Shaded area $= \frac{9}{2} \Rightarrow \int_0^{1-m} (x - x^2)\,dx - \int_0^{1-m} mx\,dx = \frac{9}{2}$

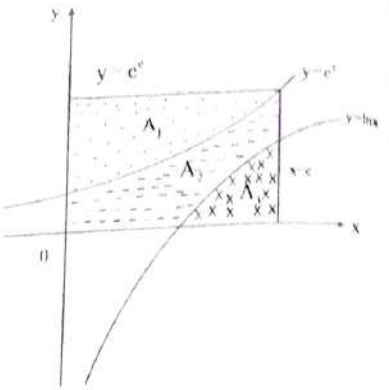
Solving this, we get $m = -2$
Case II: $m > 1$

Shaded Area $= \frac{9}{2}$

$$\int_{1-m}^{0} (-mx) - \int_{1-m}^{0} (x^2 - x)\,dx = \frac{9}{2}$$

Solving this, we get $m = 4$

11. If A_1, A_2, A_3 be the area of the regions as shown in figure given right side, then which of the following options is/are true ?



(A) $A_2 = \left(e^e - 2\right)$ SQ. units

(B) $A_2 : A_3 = \left(e^e - 2\right)$ SQ. units

(C) $A_1 - A_2 = \left[e^e(e-2)+3\right]$ SQ. units

(D) $A_2 > A_1 > A_3$

Sol. **(A, B, C)**

$$A_1 = \int_1^{e^e} \ln y\,dy = \left[y\ln y - y\right]_1^{e^e} = \left(e^e.e - e^e + 1\right) \text{ sq. units}$$

$$A_3 = \int_1^{e} \ln x\,dx = \left[x\ln x - x\right]_1^{e} = (e-e) - (0-1) = 1. \text{ sq. unit}$$

So, $A_2 = e^e.e - \left[e^e.e - e^e + 1 + 1\right] = \left(e^e - 2\right)$ sq .units

$A_1 - A_2 = e^e.e - e^e + 1 - e^e + 2 = e^e.e - 2.e^e + 3 = e^e(e-2) + 3$

Clearly $A_1 > A_2 > A_3$

Comprehension

Computing areas with parametrically represented boundaries:
If the boundary of a figure is represented by parametric equations i.e. x = x(t), y = y(t), then the area of the figure is evaluated by one of the three formulas

$$S = -\int_\alpha^\beta y(t)\cdot x'(t)dt, \qquad S = \int_\alpha^\beta x(t)\cdot y'(t)dt$$

$S = \frac{1}{2}\int_\alpha^\beta \left(xy' - yx'\right)dt$, where α and β are the values of the parameter 't' corresponding respectively to the beginning and the end of the traversal of the curve corresponding to increasing 't'.

Answer the following questions

12. The area enclosed by the astroid $\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{a}\right)^{2/3} = 1$ is

(A) $\frac{3}{4}a^2\pi$

(B) $\frac{3}{18}\pi a^2$

(C) $\frac{3}{8}\pi a^2$

(D) $\frac{3}{4}a\pi$

Sol. **(C).**

Clearly, $x = a\sin^3 t,\ y = a\cos^3 t \qquad (0 \le t \le 2\pi)$

$$S = \int_0^{2\pi} a\sin^3 t.a.3\cos^2 t(-\sin t)\,dt = -3a^2 \times 4 \times \int_0^{\pi/2} \sin^4 t\cos^2 t\,dt = -\frac{3}{8}\pi a^2 = \frac{3}{8}\pi a^2 \text{ (absolute value)}$$

13. The area of the region bounded by an arc of the cycloid $x = a(t - \sin t)$, $y = a(1 - \cos t)$ and the x–axis is

(A) $6\pi a^2$ (B) $3\pi a^2$

(C) $4\pi a^2$ (D) None of these.

Sol. **(B).**

$$S = -\int_0^{2\pi} a(1-\cos t)\,a(1-\cos t)\,dt$$

$$= -a^2\int_0^{2\pi}(1-2\cos t+\cos^2 t)\,dt$$

$$= -a^2\int_0^{2\pi}\left(1-2\cos t+\left(\frac{1+\cos 2t}{2}\right)\right)dt$$

$$= -\frac{a^2}{2}\int_0^{2\pi}(3-4\cos t+\cos 2t)\,dt$$

$$= -\frac{a^2}{2}\left[3t-4\sin t+\frac{\sin 2t}{2}\right]_0^{2\pi}$$

$= -3\pi a^2 = 3\pi a^2$ (absolute value)

14. Area of the loop described as $x = \frac{t}{3}(6-t)$, $y = \frac{t^2}{8}(6-t)$ is

(A) $\frac{27}{5}$ (B) $\frac{24}{5}$

(C) $\frac{27}{6}$ (D) $\frac{21}{5}$

Sol. **(A).**

$x = \frac{t}{3}(6-t) = \frac{1}{3}(6t-t^2)$, $y = \frac{t^2}{8}(6-t) = \frac{1}{8}(6t^2-t^3)$.

Assertion Reason

15. STATEMENT -1 : The area of region bounded by parabola $y^2 = 4x$ and $x^2 = 4y$ is $\frac{32}{3}$ sq units.

because

STATEMENT -2 : The area of region bounded by parabola $y^2 = 4ax$ and $x^2 = 4by$ is $\frac{16}{3}ab$.

(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statemen-1

(B) Statement -1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement -1 is True, Statement -2 is False

(D) Statement -1 is False, Statement -2 is True

Sol. **(D).**

As of Region bounded by parabola $y^2 = 4x$ & $x^2 = 4y$ is $= \int_0^4\left(2\sqrt{x}-\frac{x^2}{4}\right)dx$

$= \frac{4}{3}x^{\frac{3}{2}} - \frac{x^3}{12}\Big|_0^4 = \frac{32}{3} - \frac{16}{3} = \frac{16}{3}$ square units. Hence statement–1 is false.

16. Matrix matching

	Column – I		Column – II
(A)	The area bounded by the curve $y = x + \sin x$ and its inverse function between the ordinates $x = 0$ to $x = 2\pi$ is S. Then the value S is	(p)	8
(B)	The area bounded by $y = xe^{\lvert x\rvert}$ and lines $\lvert x\rvert = 1$, $y = 0$ is	(q)	1
(C)	The smaller area included between the curves $\sqrt{x} + \sqrt{\lvert y\rvert} = 1$ and $\lvert x\rvert + \lvert y\rvert = 1$ is	(r)	2
(D)	Area enclosed by $y = [x]$ and $y = \{x\}$ is	(s)	$\frac{2}{3}$
		(t)	zero

Sol. **(A) → (p), (B) → (r), (C) → (s), (D) → (q)**

Answer 17, 18 and 19 by appropriately matching the information given in the three columns of the following table.

- Column–1 contains information about equations of some curve
- Column–2 contains information about area
- Column–3 contains information about some limiting values

Column – 1		Column – 2		Column – 3	
(I)	$\frac{x^2}{9} + \frac{y^2}{16} = 1$	(i)	2	(P)	$\lim_{x\to 0} \frac{\tan 12\pi x}{2x}$
(II)	$\frac{(x+y)^2}{16} + \frac{(x-y)^2}{9} = 1$	(ii)	4/3	(Q)	$\lim_{x\to 0} \frac{\sin 2x}{x}$
(III)	$y = \lvert\log_e \lvert x\rvert\rvert$ and $y = 0$	(iii)	6π	(R)	$\lim_{x\to 0} \frac{2\left(e^{6\pi x} - 1\right)}{x}$
(IV)	$x = 2y - y^2$ and the y-axis	(iv)	12π	(S)	$\lim_{x\to 0} \frac{\sin 4x}{3x}$

17. Which of the following is correct combination?

(A) (II) (iii) (R) (B) (I) (iv) (P)
(C) (III) (i) (Q) (D) (IV) (ii) (Q)

Sol. **(C)**

18. Which of the following is correct combination?

(A) (II) (iii) (Q) (B) (IV) (ii) (P)
(C) (I) (iv) (R) (D) (III) (ii) (Q)

Sol. **(C)**

19. Which of the following is not correct combination?

(A) (I) (iv) (R) (B) (II) (iii) (P)
(C) (III) (ii) (S) (D) (IV) (ii) (S)

Sol. **(C)**

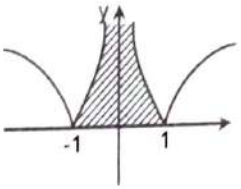
17.-19. (I) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, area $= \pi ab$; $\frac{x^2}{9} + \frac{y^2}{16} = 1$, area $= 12\pi$

(II) $\frac{(x+y)^2}{16}+\frac{(x-y)^2}{9}=1$; $\frac{\left(\frac{x+y}{\sqrt{2}}\right)^2}{\frac{16}{2}}+\frac{\left(\frac{x-y}{\sqrt{2}}\right)^2}{\frac{9}{2}}=1$; $a=\frac{4}{\sqrt{2}}$, $b=\frac{3}{\sqrt{2}}$ area $=\pi\times\frac{4}{\sqrt{2}}\times\frac{3}{\sqrt{2}}$,

Area $=6\pi$

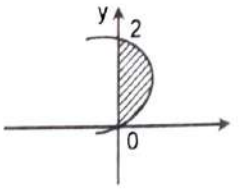
(III) $y=\left|\log_e|x|\right|$ & $y=0$

Area $=\left|2\int_0^1 \log_e x\cdot dx\right|$, Area $=2$



(IV) $x=2y-y^2$ & y-axis

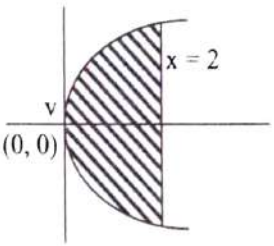
Area $=\int_0^2\left(2y-y^2\right)dx=y^2-\frac{y^3}{3}\Big|_0^2=\frac{4}{3}$



20. Let C be a curve passing through on (2, 2) such that the slope of the tangent at any point to the curve is reciprocal of the ordinate of the point. If the area bounded by curve C and line x = 2 is A, then the value of $\frac{3A}{2}$ is _____

Sol. **(8.00)**

Let P(x, y) be any point on the curve C



Now $\frac{dy}{dx}=\frac{1}{y}$

$y\,dy=dx\Rightarrow\frac{y^2}{2}=x+k$

Since the curve passes through m(2, 2) so k = 0

$y^2=2x$

Hence required area $=2\int_0^2\sqrt{2x}dx=2\sqrt{2}\times\frac{2}{3}\left(x^{\frac{3}{2}}\right)_0^2=\frac{4}{3}\sqrt{2}\times2\sqrt{2}=\frac{16}{3}$ sq. units

21. Find the value of $|a|$ for which the area of the triangle included between the axes and any tangent to the curve $x^a y=\lambda^a$ is constant _____

Sol. **(1.00)**

Given curve, $x^a y=\lambda^a$ ..... (i)

$(\lambda, 1)$ is a point on the given curve.

Now, differentiating equation (i) w.r.t. x, we get $ax^{a-1}y+x^a\frac{dy}{dx}=0$

$\Rightarrow\frac{dy}{dx}=\frac{-ax^{a-1}y}{x^a}=-\frac{ay}{x}$ at $(\lambda, 1)$ $\frac{dy}{dx}=-\frac{a}{\lambda}$. Equation of tangent at $(\lambda, 1)$; $y-1=-\frac{a}{\lambda}(x-\lambda)$,

Now, $x=0\Rightarrow y=1+a$; $y=0\Rightarrow x=\frac{\lambda}{a}+\lambda=\frac{\lambda(1+a)}{a}$

Area, $A=\frac{1}{2}\times(1+a)\frac{(1+a)\lambda}{a}$. Now, $\frac{dA}{da}=\frac{1}{2}\lambda\left[\frac{a\cdot2(1+a)-(1+a)^2}{a^2}\right]=0$

$\Rightarrow(2a-1-a)(1+a)=0\Rightarrow(a-1)(a+1)=0\Rightarrow a=1, a=-1$

22. Area bounded by the relation $[2x] + [y] = 5$ x, $y > 0$ is k, then $\frac{k}{2}$ is equal to _____ (where [.] denotes greatest integer function)

Sol. **(1.50)**

$[2x] = 0$

$2x \in [0, 1]$

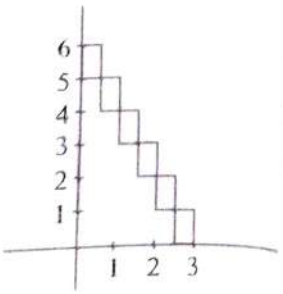
$\Rightarrow x \in \left[0, \frac{1}{2}\right]$

$[y] = 5$

$\Rightarrow y \in [5, 6]$

Similarly we can consider $[2x]$

$= 1, 2, 3, 4, 5$ from the graph area is 3 sq. unit



23. The area bounded by $y = f(x)$, y–axis and line $2y = \pi(x + 1)$, where $f(x) = \sin^{-1}x + \cos^{-1}x + \tan^{-1}\frac{1}{x} + \tan^{-1}x$, is _____

Sol. **(1.57)**

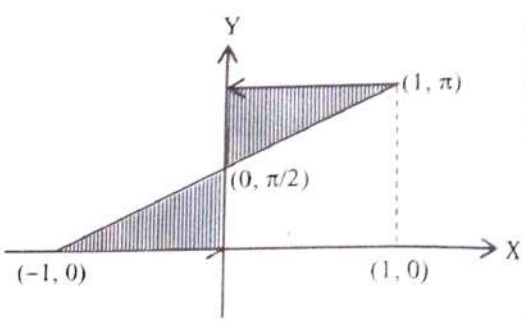
Here $f(x) = \sin^{-1}x + \cos^{-1}x + \tan^{-1}x + \tan^{-1}\frac{1}{x}$

Domain $x \in [-1, 0) \cup (0, 1]$

For, $x \in [-1, 0)$, $f(x) = \frac{\pi}{2} - \pi + \frac{\pi}{2} = 0$

And for $x \in (0, 1]$, $f(x) = \frac{\pi}{2} + \frac{\pi}{2} = \pi$

So, required area $= 2 \times \frac{1}{2} \times \frac{\pi}{2} \times 1 = \frac{\pi}{2}$.



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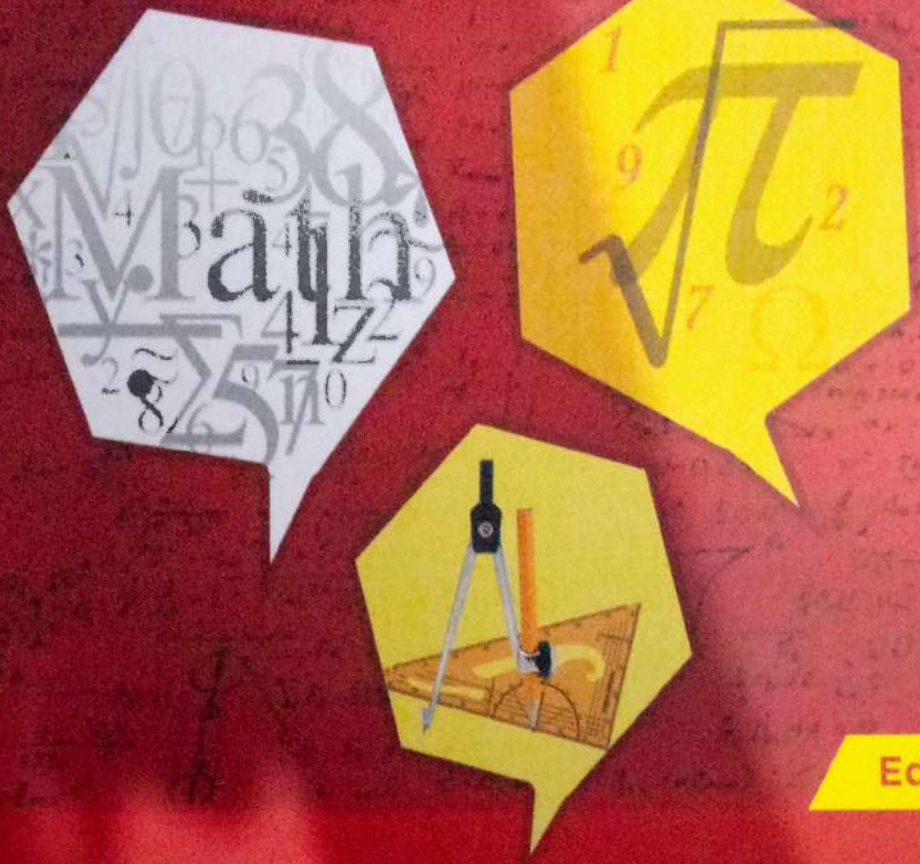


Edition 2020

MATHEMATICS

7

DIFFERENTIAL EQUATION

Order and Degree of a Differential Equation

The order of the highest differential coefficient appearing in the differential equation is called the order of the differential equation, while the exponent of the highest differential coefficient, when the differential equation is a polynomial in all the differential coefficients, is known as the degree of the differential equation.

Formation of Differential Equations

Consider a family of curves $f(x, y, \alpha_1, \alpha_2, \ldots, \alpha_n)=0$, where $\alpha_1, \alpha_2, \ldots, \alpha_n$ are n independent parameters.

Equation (1) is known as n parameter family of curves e.g. $y = mx$ is 1-parameter family of straight lines, $x^2 + y^2 + ax + bx = 0$ is two parameter family of circles.

If we differentiate equation (1) n times w. r. t. x, we will get n more relations between $x, y, \alpha_1, \alpha_2, \ldots, \alpha_n$ and derivatives of y w.r.t. x. By eliminating $\alpha_1, \alpha_2, \ldots, \alpha_n$ from these n relations and equation (1), we get a differential equation.

Clearly order of this differential equation will be n i.e. equal to the number of independent parameters in the family of curves.

e.g. consider the family of parabolas with vertex at the origin and axis as the x-axis

$y^2 = 4ax$ .... (1)

Differentiating w.r.t. x, we get $2y\frac{dy}{dx} = 4a = y^2/x$ from (1)

or, $2x\frac{dy}{dx} - y = 0$, which is the differential equation of (1) and is clearly of order 1.

Remark:

- The order of the differential equation will be equal to number of independent parameters and not equal to the number of all the parameters in the family of curves.
  e. g consider the family of the curves $y = (C_1 + C_2)e^x + C_3 e^{x+C_4}$.
  Here the number of arbitrary parameters is 4 but the order of the corresponding differential equation will not be 4 as it can be rewritten as
  $y = (C_1 + C_2 + C_3 e^{C_4})e^x$, which is of the form $y = Ae^x$. Hence the corresponding differential equation will be of order 1.

Solution of Differential Equation

First Order Differential Equations with Separable Variables

Let the differential equation be of the form $\frac{dy}{dx} = f(x, y)$ ..... (1)

where $f(x, y)$ denotes a function of x and y.

(1) is separable if $f(x, y)$ can be expressed in the form $\frac{M(x)}{N(y)}$,

M(x), N(y) are real valued functions of x and y respectively.

Then, we have $\frac{dy}{dx} = \frac{M(x)}{N(y)}$ ..... (2)

$\Rightarrow N(y)\,dy = M(x)\,dx$

Integrating both sides of (2), we get the solution *viz.* $\int N(y)dy = \int M(x)dx + c$

Differential Equations Reducible to the Separable Variable Type

Sometimes differential equation of the first order cannot be solved directly by variable separation but by some substitution we can reduce it to a differential equation with separable variables. A differential equation of the form $\frac{dy}{dx} = f(ax + by + c)$ is solved by writing $ax + by + c = t$.

Homogeneous Differential Equations

Suppose we have a differential equation of the form

$\frac{dy}{dx} = f(x,y)$, ... (1)

where f(x,y) is a function of x and y and is of the form $F\left(\frac{y}{x}\right)$ or $F\left(\frac{x}{y}\right)$.

These equations are solved by putting $y = vx$, where $v \equiv v(x)$, a function of x.

Equations Reducible to the Homogenous Form

Equations of the form $\frac{dy}{dx} = \frac{ax + by + c}{Ax + By + C}$ $(aB \neq Ab)$ can be reduced to a homogenous form by changing the variables x, y to X, Y by writing $x = X + h$ and $y = Y + k$; where h, k are constants to be chosen so as to make the given equation homogenous. We have

$\frac{dy}{dx} = \frac{d(Y+k)}{d(X+h)} = \frac{dY}{dX}$. Hence the given equation becomes $\frac{dY}{dX} = \frac{aX + bY + (ah + bk + c)}{AX + BY + (Ah + Bk + C)}$

Let h and k be so chosen as to satisfy the relation $ah + bk + c = 0$ and $Ah + Bk + C = 0$.

These give $h = \frac{bC - Bc}{aB - Ab}$ $\quad k = \frac{Ac - aC}{aB - Ab}$

which are meaningful except when $aB = Ab$.

$\frac{dY}{dX} = \frac{aX + bY}{AX + BY}$ can now be solved by means of the substitution $Y = VX$

In case $aB = Ab$, we write $ax + by = t$. This reduces the differential equation to the separable variable type.

First Order Linear Differential Equations

The most general form of this category of differential equations is $\frac{dy}{dx} + Py = Q$,, where P and Q are functions of alone.

We use here an Integrating factor, namely $e^{\int pdx}$ and the solution is obtained by

$y\,e^{\int pdx} = \int Q\,e^{\int pdx}dx + c$, where c is an arbitrary constant.

Note:

- In some cases a linear differential equation may be of the form $\frac{dx}{dy} + P_1 x = Q_1$, where P_1 and Q_1 are function of y alone. In such a case the integrating factor is $e^{\int P_1 dy}$, e.g. $\frac{dx}{dy} + x\cos y = \sin 2y$. Here the integrating factor is $e^{\int \cos y dy} = e^{\sin y}$ and the solution is given by $x.e^{\sin y} = \int \sin 2y.e^{\sin y} dy = 2e^{\sin y}(\sin y - 1) + c$.

Extended form of Linear Differential Equations

Sometimes a differential equation is not linear but it can be converted into a linear differential equation by a suitable substitution.

Bernoulli's Equation

$$\frac{dy}{dx} + Py = Qy^n \qquad \ldots (1)$$

where P and Q are function of x alone. Divide each term of (1) by y^n. We get

$$\frac{1}{y^n}\frac{dy}{dx} + \frac{P}{y^{n-1}} = Q \qquad \ldots (2)$$

Let $\frac{1}{y^{n-1}} = v$, so that $\frac{1}{y^n}\frac{dy}{dx} = \frac{1}{1-n}\frac{dv}{dx}$

Substituting in (2), we get $\frac{dv}{dx} + (1-n)v.P = Q(1-n) \qquad \ldots (3)$

(3) is a linear differential equation

General Form of Linear Differential Equation

A general form of the linear differential equation is $f'(y)\frac{dy}{dx} + P(x)\,f(y) = Q(x)$.

Here, we write $f(y) = t$. The differential equation becomes $\frac{dt}{dx} + P(x)t = Q(x)$.

General Form of Variable Separation

If we can write the differential equation in the form

$f(f_1(x, y))\,d(f_1(x, y)) + \phi(f_2(x, y))\,d(f_2(x, y)) + \ldots = 0$,

then each term can be easily integrated separately.

For this the following results must be memorized.

(i) $d(x + y) = dx + dy$

(ii) $d(xy) = y\,dx + x\,dy$

(iii) $d\left(\frac{x}{y}\right) = \frac{y\,dx - x\,dy}{y^2}$

(iv) $d\left(\frac{y}{x}\right) = \frac{x\,dy - y\,dx}{x^2}$

(v) $d(\log xy) = \frac{y\,dx + x\,dy}{xy}$

(vi) $d\left(\log\frac{y}{x}\right) = \frac{(xdy - ydx)}{xy}$

(vii) $d\left(\frac{1}{2}\log\frac{x+y}{x-y}\right) = \frac{x\,dy - y\,dx}{x^2 - y^2}$

(viii) $d\left(\tan^{-1}\frac{y}{x}\right) = \frac{x\,dy - y\,dx}{x^2 + y^2}$

(ix) $\frac{d\left[f(x,y)\right]^{1-n}}{1-n} = \frac{f'(x,y)}{\left(f(x,y)\right)^{n}}$ (x) $d\left(\sqrt{x^2+y^2}\right) = \frac{xdx+ydy}{\sqrt{x^2+y^2}}$

ORTHOGONAL TRAJECTORY

Any curve, which cuts every member of a given family of curves at right angle, is called an orthogonal trajectory the family. For example, each straight line passing through the origin, i.e. $y = kx$ is an orthogonal trajectory of family of the circles $x^2 + y^2 = a^2$

Procedure for Finding the Orthogonal Trajectory

(i). Let $f(x,y,c) = 0$ be the equation of the given family of curves, where c is an arbitrary parameter.

(ii). Differentiate $f = 0$, w.r.t.x and eliminate c i.e form a differential equation.

(iii). Substitute $-\frac{dx}{dy}$ for $\frac{dy}{dx}$ in the above differential equation. This will give the differential equation of orthogonal trajectories.

(iv). By solving this differential equation we get the required orthogonal trajectories.

APPLICATION OF DIFFERENTIAL EQUATIONS

Rate of Change Problems

Geometrical Applications

We also use differential equations for finding the family of curves for which some conditions involving derivatives are given. For this we proceed in the following way:

Equation of the tangent at a point (x, y) to the curve $y = f(x)$ is given by

$Y - y = \frac{dy}{dx}(X - x)$, At the X axis, $Y = 0$, and $X = x - y/\frac{dy}{dx}$.

At the Y axis, $X = 0$, and $Y = y - x\frac{dy}{dx}$.

Similar information can be obtained for normals by writing its equation as $(y - y)\frac{dy}{dx} + (X - x) = 0$.

ILLUSTRATIVE PROBLEMS

1. Find the order and degree (if defined) of the following differential equations:

(i) $y = 1 + \frac{dy}{dx} + \frac{1}{2!}\left(\frac{dy}{dx}\right)^2 + \frac{1}{3!}\left(\frac{dy}{dx}\right)^3 + \ldots$

(ii) $\left(\frac{d^3y}{dx^3}\right)^{2/3} = \frac{dy}{dx} + 2$ (iii) $\frac{d^2y}{dx^2} = x \ln\left(\frac{dy}{dx}\right)$

Sol. (i). The given differential equation can be re-written as $y = e^{dy/dx}$

$\Rightarrow \frac{dy}{dx} = \ln y$. Hence its order is 1 and degree 1.

(ii). The given differential equation can be re-written as

$$\left(\frac{d^3y}{dx^3}\right)^2 = \left(\frac{dy}{dx} + 2\right)^3.$$

Hence its order is 3 and degree 2.

(iii). Its order is obviously 2. Since the given differential equation cannot be written as a polynomial in all the differential coefficients, the degree of the equation is not defined.

2. Find the differential equation whose solution represents the family: $c(y + c)^2 = x^3$.

Sol. Given equation of family of curves is given by

$c(y + c)^2 = x^3$ ...(1)

Differentiating once, we get

$$c\left[2(y+c)\right]\frac{dy}{dx} = 3x^2$$

But, from (1) we have $\frac{2x^3}{(y+c)^2}(y+c)\frac{dy}{dx} = 3x^2$

$$\Rightarrow \frac{2x^3}{y+c}\frac{dy}{dx} = 3x^2$$

i.e. $$\frac{2x}{y+c}\frac{dy}{dx} = 3$$

$$\Rightarrow \frac{2x}{3}\left[\frac{dy}{dx}\right] = y + c$$

Hence $c = \frac{2x}{3}\left[\frac{dy}{dx}\right] - y$

Substituting c back into equation (1), we get

$$\left[\frac{2x}{3}\left[\frac{dy}{x}\right] - y\right]\left[\frac{2x}{3}\frac{dy}{dx}\right]^2 = x^3.$$

which is the required differential equation.

3. **The perpendicular from the origin to the tangent at any point on a curve is equal to the abscissa the point of contact. Also curve passes through the point (1, 1). Then find the length of intercept the curve on the x-axis**

Sol. Equation of tangent is $y-\beta=f'(\alpha)\{x-\alpha\}$

Also, $\alpha=\dfrac{\beta-\alpha f'(\alpha)}{\sqrt{f'(\alpha)^2+1}}$

Thus, $\dfrac{dy}{dx}=\dfrac{1}{2}\left(\dfrac{y^2-x^2}{xy}\right)$

Thus, equation of curve is $x^2+y^2-2x=0$

Then the required length of intercept is 2

4. **Find the solution of the equation, $x=1+\left(xy\dfrac{dy}{dx}\right)+\dfrac{x^2y^2}{2!}\left(\dfrac{dy}{dx}\right)^2+\dfrac{(xy)^3}{3!}\left(\dfrac{dy}{dx}\right)^3+\ldots\ldots$**

Sol. Here, $x=e^{xy\frac{dy}{dx}}$

$\Rightarrow \int ydy=\int\dfrac{\ln x}{x}dx$

$\Rightarrow y=\pm\sqrt{(\ln x)^2+c}$

Which is the required solution.

5. **If solution of equation $\dfrac{dy}{dx}=y+\int_0^2 ydx$, is y(x). Then find the value of $[|y(2)|]$ (where [.] great integer function) (given y(0) = 1)**

Sol. Given that $\dfrac{dy}{dx}=y+\int_0^2 ydx$

Say $\int_0^2 ydx=A$

$\Rightarrow \dfrac{dy}{dx}=y+A\Rightarrow\dfrac{dy}{y+A}=dx$

$\Rightarrow \ln(y+A)=x+B$

Given $y(0)=1\Rightarrow B=\ln(1+A)$

$\Rightarrow \ln\dfrac{(y+A)}{(1+A)}=x\Rightarrow y+A=(1+A)e^x$

Now find 'A'

$A=\int_0^2[(1+A)e^x-A]dx\Rightarrow A=\dfrac{e^2-1}{4-e^2}$

Hence solution $y(x)=\left(1+\dfrac{e^2-1}{4-e^2}\right)e^x-\left(\dfrac{e^2-1}{4-e^2}\right)=\dfrac{3e^x-e^2+1}{4-e^2}$

$y(2)=\dfrac{2e^2+1}{4-e^2}\Rightarrow[|y(2)|]=4.$

6. Solve $\sin y\frac{dy}{dx} = \cos y(1 - x\cos y)$.

Sol. The given differential equation is $\sin y\frac{dy}{dx} = \cos y(1 - x\cos y)$ ... (1)

or $\sin y\frac{dy}{dx} - \cos y = -x\cos^2 y$. Divide by $\cos^2 y$. We get

$\tan y \sec y\frac{dy}{dx} - \sec y = -x$

Let $\sec y = v \Rightarrow \sec y \tan y\frac{dy}{dx} = \frac{dv}{dx}$

so that $\frac{dv}{dx} - v = -x$ ... (2)

(2) is linear differential equation with $P = -1, Q = -x$

I.F $= e^{\int P dx} = e^{\int -1 dx} = e^{-x}$

The solution is given by $v.e^{-x} = \int -xe^{-x}\,dx = x\,e^{-x} + e^{-x} + c$

$= e^{-x}(x+1) + c$ or v

$= (1 + x) + ce^{x}$ or $\sec y = (1 + x) + ce^{x}$.

7. Solve $\frac{x + y\frac{dy}{dx}}{y - x\frac{dy}{dx}} = \frac{x\sin^2(x^2 + y^2)}{y^3}$.

Sol. The given differential equation can be rewritten as

$$\frac{2xdx + 2ydy}{\sin^2(x^2 + y^2)} = \frac{2x}{y}\left(\frac{ydx - xdy}{y^2}\right)$$

or, $$\int \operatorname{cosec}^2(x^2 + y^2)\,d(x^2 + y^2) = 2\int\left(\frac{x}{y}\right)d\left(\frac{x}{y}\right)$$

or, $-\cot(x^2+y^2) = \frac{x^2}{y^2} + c$ is the required solution.

8. Find the orthogonal trajectories of the hyperbola $xy = c$.

Sol. The equation of the given family of curves is

$xy = c$ ... (1)

Differentiate (1) w.r.t.x . We get $x\frac{dy}{dx} + y = 0$ ... (2)

Substitute $-\frac{dx}{dy}$ for $\frac{dy}{dx}$ in (2). We get $-x\frac{dx}{dy} + y = 0$ ... (3)

This is the differential equation for the orthogonal trajectory of given family of hyperbolas.

(3) can be rewritten as $x\,dx = y\,dy$, which on integration gives $x^2 - y^2 = c$

This is the family of required orthogonal trajectories.

9. Let $y = f(x)$ be a curve passing through (e, e^5) and satisfy the differential equation $(2ny + xy \ln x)dx - x \ln x \, dy = 0$. If $g(x) = \lim_{n\to\infty} f(x)$, then find the value of $\int_{1/e}^{e} g(x)dx$

Sol. $(2ny + xy \ln x)dx = x \ln x \, dy$

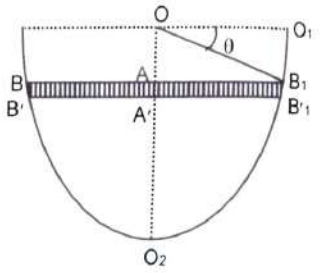
$$\frac{dy}{y} = \left(\frac{2n}{x \ln x} + 1\right)dx$$

$\Rightarrow y = e^x(\ln x)^{2n}$

Hence $g(x) = 0$.

10. A hemispherical tank of radius 2 metres is initially full of water and has an outlet of 12 cm^2 cross-sectional area at the bottom. The outlet is opened at some instant. The flow through the outlet is according to the law $v(t) = 0.6\sqrt{2gh(t)}$, where v (t) and h (t) are respectively the velocity of the flow through the outlet and the height of water level above the outlet at time t, and g is the acceleration due to gravity. Find the time it takes to empty the tank. (Hint: Form a differential equation by relating the decrease of water level to the outflow.)

Sol. Let 'O' be the centre of hemispherical tank. Let at any instant 't' water level be BAB_1 and at t + dt, water level $B'A'B'_1$.



Let $\angle O_1OB_1 = \theta$

$\Rightarrow AB_1 = r\cos\theta$ and $OA = r\sin\theta$

Decrease in the water volume in time dt,

$= \pi (AB_1)^2 . d(OA) = \pi r^2\cos^2\theta . r\cos\theta \, d\theta$

$= \pi r^3\cos^3\theta d\theta$

Also $h(t) = O_2A$

$= r - r\sin\theta = r(1-\sin\theta)$

Now outflow rate $Q = A.V(t) = A \cdot 0.6\sqrt{2gr(1-\sin\theta)}$,

where 'A' is the area of the outlet. Thus volume flowing out in time dt,

$\Rightarrow Qdt = A \cdot (0.6)\sqrt{2gr}\sqrt{(1-\sin\theta)}.dt$

We have, $\pi r^3\cos^3\theta \; d\theta = A \cdot (0.6)\sqrt{2gr}\sqrt{(1-\sin\theta)}.dt$

$$\Rightarrow \frac{\pi r^3}{A \cdot (0.6)\sqrt{2gr}} \frac{\cos^3\theta}{\sqrt{(1-\sin\theta)}} d\theta = dt$$

Let the time taken to empty the tank be T. Then

$$T = \int_0^{\pi/2} \frac{\pi r^3}{A(0.6)\sqrt{2gr}} . \frac{\cos^3\theta}{\sqrt{1-\sin\theta}} . d\theta$$

$$= \frac{-\pi r^3}{A(0.6)\sqrt{2gr}} \int_0^{\pi/2} \frac{\cos^2\theta(-\cos\theta)}{\sqrt{1-\sin\theta}} . d\theta$$

Let $t_1 = \sqrt{1-\sin\theta} \Rightarrow dt_1 = \frac{-\cos\theta}{2\sqrt{1-\sin\theta}} d\theta$

$$\Rightarrow T = \frac{-2\pi r^3}{A(0.6)\sqrt{2gr}} \int_1^0 \left(1-\left(1-t_1^2\right)^2\right) dt_1 \Rightarrow T = \frac{7\pi \times 10^5}{135\sqrt{g}} .$$

SOLVED PROBLEMS

1. If $f:(1,\infty)\to R$ is a differentiable function satisfying the differential equation $\frac{dy}{dx}=\frac{x^2-y^2}{x^2(y^2+1)}$ $\forall\ x>1$, then

(A) $\lim_{x\to\infty} f(x)=1$

(B) $\lim_{x\to\infty}\frac{f(x)}{x}\neq 1$

(C) range of f(x) is (a, b] for some a, b $\in$ R

(D) |f(x)| is bounded function

Sol. **(B)**

$$y'=\frac{1}{1+y^2}-\frac{y^2}{(1+y^2)}\cdot\frac{1}{x^2}$$

$$\Rightarrow -\frac{1}{x^2}\le y'-\frac{1}{1+y^2}\le 0$$

$$\Rightarrow \lim_{x\to\infty}\left(y'-\frac{1}{1+y^2}\right)=0$$

Let $\lim_{x\to\infty} y=L$

$$\Rightarrow \lim_{x\to\infty}\frac{1}{1+y^2}=\frac{1}{1+L^2}$$

$$\Rightarrow \lim_{x\to\infty} y'=\frac{1}{1+L^2}$$

$\Rightarrow \lim_{x\to\infty} y\neq L$ (Contradiction)

Similar contradiction arises if we take $\lim_{x\to\infty}\frac{y}{x}=L$

2. Solution of differential equation $x^3\frac{dy}{dx}=y^3+y^2\sqrt{y^2-x^2}$ is $cxy=y+k\sqrt{y^2-x^2}$ (c is arbitrary) if

(A) k = 1

(B) k = 2

(C) $k=\frac{1}{2}$

(D) none of these

Sol. **(A)**

$$\frac{dy}{dx}=\frac{y^3+y^2\sqrt{y^2-x^2}}{x^3}$$

put $\frac{y}{x}=v$

$$v+x\frac{dv}{dx}=v^3+v^2\sqrt{v^2-1}$$

$$\frac{dv}{v(v^2-1)+v^2\sqrt{v^2-1}}=\frac{dx}{x}$$

$$=\int\frac{dv}{v\sqrt{v^2-1}\left(\sqrt{v^2-1}+v\right)}$$

$$\log\left(v+\sqrt{v^2-1}\right)-\log v=\log x+c$$

3. If $y^{1/m} + y^{-1/m} = 2x$ then $(x^2-1)\frac{d^3y}{dx^3} + 3x\frac{d^2y}{dx^2} + f(m)\frac{dy}{dx} = 0$, f(m) is equal to

(A) $1 + m^2$ (B) $2 - m^2$

(C) $m + m^2$ (D) $1 - m^2$

Sol. **(D)**

$y^{2/3} - 2xy^{1/m} + 1 = 0$

$y^{1/m} = \frac{2x \pm \sqrt{4x^2-4}}{2}$

$\log y = m\log\left(x \pm \sqrt{x^2-1}\right)$ further differential w.r.t x

$\frac{y_1}{y} = \pm\frac{m}{\sqrt{x^2-1}} \Rightarrow (x^2-1)y_1^2 = m^2y^2$

$\Rightarrow (x^2-1)y_2 + xy_1 = m^2y$ (diff.....)

4. Radioactive substances A and B decay following the first order differential equation where rate of decay proportional to amount of substance. Initially at t = 0, $\frac{da}{dt} : \frac{db}{dt} = 2$ after one hour this ratio is $\frac{5}{2}$ th (where a, b are amount of substances A, B) select the most appropriate option

(A) after two hours ratio will be $\frac{25}{8}$

(B) let $f(t) = \frac{a}{b}$ at time t then $f'(x) \propto f(x)$

(C) half life of A and B is different (time taken to decay to half of the original amount)

(D) all of these

Sol. **(D)**

$\frac{da}{dt} \propto a$, $a = a_0e^{-k_1t}$

$\frac{db}{dt} \propto b$, $b = b_0e^{-k_2t}$

$\frac{a}{b} = \frac{a_0}{b_0}e^{-(k_1-k_2)t}$

5. The function f(x) satisfying $\{f(x)\}^2 + 4f(x)f'(x) + \{f'(x)\}^2 = 0$ is given by

(A) $ke^{(2+\sqrt{3})x}$ (B) $ke^{(4\pm\sqrt{5})x}$

(C) $ke^{(-2\pm\sqrt{3})x}$ (D) $k\log\left(2+\sqrt{3}\right)x$

Sol. **(C).** $\{f(x)\}^2 + 4f(x)f'(x) + \{f'(x)\}^2 = 0$

$\therefore f'(x) = \frac{-4f(x) \pm \sqrt{16\{f(x)\}^2 - 4\{f(x)\}^2}}{2} = -2f(x) \pm \sqrt{3}f(x)$

$\therefore \frac{f'(x)}{f(x)} = -2 \pm \sqrt{3}$ $\therefore \log f(x) = \left(-2 \pm \sqrt{3}\right)x + c$

$\therefore f(x) = ke^{(-2\pm\sqrt{3})x}$, where $k = e^c$.

6. The differential equation $\{xy^3(1+\cos x)-y\}dx + xdy = 0$ represents the curve

$\frac{x^2}{ay^2} = \frac{x^3}{b} + x^2\sin x + cx\cos x - d\sin x + k$. Then $a+b+c+d$ is equal to

(A) 8
(B) 9
(C) 10
(D) 11

Sol. **(B).** $\frac{dy}{dx} + y^3(1+\cos x) - \frac{y}{x} = 0$

$\frac{1}{y^3}\frac{dy}{dx} - \frac{1}{y^2 x} = -(1+\cos x)$

Let $\frac{-1}{y^2} = u$

$\therefore \frac{1}{2}\frac{du}{dx} + \frac{u}{x} = -(1+\cos x)$

$IF = e^{\int \frac{2}{x}dx} = x^2$

$\therefore ux^2 = -2\int x^2(1+\cos x)dx$

$\Rightarrow \frac{x^2}{2y^2} = \frac{x^3}{3} + x^2\sin x + 2x\cos x - 2\sin x + k$

7. If the given curve satisfies the differential equation $e^y dx + (xe^y + 2y)dy = 0$ and also passes through (0, 0) then the possible equation of curve can be

(A) $xe^y + y = 0$
(B) $x + y^2e^y = 0$
(C) $x^2e^y + ye^y = 1$
(D) none of these

Sol. **(D).** $M = e^y$, $N = x \cdot e^y + 2y$

Now, $\frac{dM}{dy}$ (keeping x constant) $= e^y = \frac{dN}{dx}$ (keeping y constant)

Hence, the given equation is exact

Now, $f = \int e^y dx + g(y) = x \cdot e^y + g(y)$

Differentiating w.r.t y we get $x \cdot e^y + \frac{dg}{dy} = N = x \cdot e^y + 2y$

i.e. $\frac{dg}{dy} = 2y$

Integrating $g(y) = y^2 + c_1 \Rightarrow f = x \cdot e^y + y^2 + c_1$

So, desired solution is $x \cdot e^y + y^2 = 0$

8. Consider curve, $r^2 = a\sin 2\theta$ then its orthogonal trajectory is

(A) $r^2 = c$
(B) $r^2 = c\sin\theta$
(C) $r^2 = c\cos 2\theta$
(D) none of these

Sol. **(C).** Differentiating w.r.t. 'r', $2r\,dr = 2a\cos 2\theta\, d\theta$, $a = \frac{r^2}{\sin 2\theta}$

$\Rightarrow rdr = \frac{r^2}{\sin 2\theta}\cos 2\theta\, d\theta \Rightarrow r \cdot \frac{d\theta}{dr} = \tan 2\theta$

So, differential equation of orthogonal trajectory, $r\dfrac{d\theta}{dr} = -\cot 2\theta$

Solving $-\tan 2\theta\, d\theta = \dfrac{dr}{r}$

$\Rightarrow \ln(\cos 2\theta) = 2\ln r + k \Rightarrow r^2 = c\cos 2\theta$

9. If $\sqrt{1-x^6} + \sqrt{1-y^6} = a\left(x^3 - y^3\right)$ and $\dfrac{dy}{dx} = f(x, y)\sqrt{\dfrac{1-y^6}{1-x^6}}$ then,

(A) $f(x, y) = \dfrac{y}{x}$ (B) $f(x, y) = \dfrac{x^2}{y^2}$

(C) $f(x, y) = \dfrac{2y^2}{x^2}$ (D) $f(x, y) = \dfrac{y^2}{x^2}$

Sol. **(B).** $-\dfrac{x^5}{\sqrt{1-x^6}} - \dfrac{y^5}{\sqrt{1-y^6}}\dfrac{dy}{dx} = a\left(x^2 - y^2\dfrac{dy}{dx}\right)$

$\Rightarrow \left(ay^2 - \dfrac{y^5}{\sqrt{1-y^6}}\right)\dfrac{dy}{dx} = ax^2 + \dfrac{x^5}{\sqrt{1-x^6}}$

$\Rightarrow \dfrac{dy}{dx} = \dfrac{\sqrt{1-y^6}}{\sqrt{1-x^6}} \times \dfrac{x^2\left(a\sqrt{1-x^6} + x^3\right)}{y^2\left(a\sqrt{1-y^6} - y^3\right)}$ ..... (1)

Also, $\dfrac{(1-x^6)-(1-y^6)}{\sqrt{1-x^6} - \sqrt{1-y^6}} = a\left(x^3 - y^3\right)$

or $x^3 + y^3 = a\left(\sqrt{1-y^6} - \sqrt{1-x^6}\right)$

$\Rightarrow a\sqrt{1-y^6} - y^3 = a\sqrt{1-x^6} + x^3$

Hence, from equation (1), $\dfrac{dy}{dx} = \dfrac{\sqrt{1-y^6}}{\sqrt{1-x^6}} \times \dfrac{x^2}{y^2}$

10. Solution of equation $y\,dx(1 + x^2 \cdot \ln xy) + x\,dy(1 + y^2\ln(xy)) = 0$ is

(A) $\ln(\ln xy) + x^2 + y^2 + c = 0$ (B) $\ln xy = ce^{\frac{(x^2+y^2)}{2}}$

(C) $2\ln(\ln xy) + x^2 + y^2 + c = 0$ (D) $\ln xy = ce^{-\frac{(x^2+y^2)}{2}}$

Sol. **(C, D).** $x\,dy + y\,dx + \ln xy(x^2y\,dx + xy^2\,dy) = 0$

$\dfrac{1}{xy}(xdy + ydx) + \ln(xy)(xdy + ydy) = 0$

$\Rightarrow d(\ln xy) + \dfrac{1}{2}\ln(xy)\,d\left(x^2 + y^2\right) = 0$

$\Rightarrow 2\ln|(\ln xy)| + x^2 + y^2 + c = 0$

$\therefore \ln(\ln xy) = -\dfrac{\left(x^2 + y^2 + c\right)}{2}$

$\ln xy = ce^{-\left(\frac{x^2+y^2}{2}\right)}$

11. The solution of the differential equation $f(x)\frac{dy}{dx} + f'(x)y = 1$ is

(A) $x = yf(x) + c$ (B) $x.f^{-1}(x) + c = 0$

(C) $y = \frac{x + c}{f(x)}$ (D) None of these

Sol. (A, C)

The given differential equation can be written as

$f(x)\,dy + f'(x)\,y\,dx = dx$ i.e. $d(f(x).\,y) = d(x)$

Integrating we get $y.f(x) = x + c$

or, $f(x) = \frac{x + c}{y}$

Comprehension

If any differential equation in the form $f(f_1(x, y))\,d(f_1(x, y)) + \phi(f_2(x, y))\,d(f_2(x, y)) + \ldots = 0$, then each term can be integrated separately.

For e.g., $\int \sin xy\,d(xy) + \int\left(\frac{x}{y}\right)d\left(\frac{x}{y}\right) = -\cos xy + \frac{1}{2}\left(\frac{x}{y}\right)^2 + c$

Answer the following questions:

12. The solution of the differential equation $xdy - ydx = \sqrt{x^2 - y^2}\,dx$ is

(A) $cx = e^{\sin^{-1}\frac{y}{x}}$ (B) $xe^{\sin^{-1}\frac{y}{x}} = c$

(C) $x + e^{\sin^{-1}\frac{y}{x}} = c$ (D) none of these

Sol. (A)

$$xdy - ydx = \sqrt{x^2 - y^2}\,dx \Rightarrow d\left(\frac{y}{x}\right) = \frac{\sqrt{1-(y/x)^2}}{x}dx$$

$$\Rightarrow \int \frac{d(y/x)}{\sqrt{1-(y/x)^2}} = \int \frac{dx}{x}$$

$$\Rightarrow \sin^{-1}\frac{y}{x} = \ln x + \ln c \text{ or } cx = e^{\sin^{-1} y/x}.$$

13. The solution of the differential equation $(xy^4 + y)\,dx - xdy = 0$ is

(A) $\frac{x^3}{4} + \frac{1}{2}\left(\frac{x}{y}\right)^2 = c$ (B) $\frac{x^4}{4} + \frac{1}{3}\left(\frac{x}{y}\right)^3 = c$

(C) $\frac{x^4}{4} - \frac{1}{2}\left(\frac{x}{y}\right)^2 = c$ (D) $\frac{x^3}{4} - \frac{1}{2}\left(\frac{x}{y}\right)^2 = c$

Sol. (B)

$(xy^4 + y)\,dx - x\,dy = 0$

$\Rightarrow xy^4dx + ydx - xdy = 0$

$$\Rightarrow \int x^3dx - \int\left(\frac{x}{y}\right)^2 d\left(\frac{x}{y}\right) = 0 \Rightarrow \frac{x^4}{4} + \frac{1}{3}\left(\frac{x}{y}\right)^3 = c.$$

14. Solution of differential equation $(2x\cos y + y^2\cos x)\,dx + (2y\sin x - x^2\sin y)dy = 0$ is
(A) $x^2\cos y + y^2\sin x = c$ (B) $x\cos y - y\sin x = c$
(C) $x^2\cos^2 y + y^2\sin^2 x = c$ (D) none of these

Sol. **(A)**

$2x\cos y\,dx - x^2\sin y\,dy + y^2\cos x\,dx + 2y\sin x\,dy = 0$

$\Rightarrow \int d(x^2\cos y) + \int d(y^2\sin x) = 0$

$\Rightarrow x^2\cos y + y^2\sin x = c.$

Assertion Reason

15. Statement -1: $y = a\sin x + b\cos x$ is the general solution of $y'' + y = 0$.

because

Statement -2: $y = a\sin x + b\cos x$ is a trigonometric function.

(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1
(B) Statement -1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement -1 is True, Statement -2 is False
(D) Statement -1 is False, Statement -2 is True

Sol. **(B)**

$y = a\sin x + b\cos x$

$\frac{dy}{dx} = a\cos x - b\sin x$

$\Rightarrow \frac{d^2y}{dx^2} = -a\cos x - b\sin x = -y$

$\Rightarrow \frac{d^2y}{dx^2} + y = 0$

But statement 2 is not the correct explanation of the statement 1.

16. Match the following

	Column – I		Column – II
(A)	Rate of increase in area of equilateral triangle of side 10 cm, when each side is increasing at the rate of 0.1 m/s	(p)	0
(B)	Solution of the equation $x\int_0^x y(t)dt = (x+1)\int_0^x ty(t)dt$	(q)	$\frac{Ae^{-1/x}}{x^3}$
(C)	Side of cube increasing by 1% then percentage increase in volume, is	(r)	$\frac{\sqrt{3}}{2}$
(D)	If, $y = f(x)$ satisfies the equation $f(x) = e^{-x} + e^{x})\cos x - 2x - \int_0^x (x-t)f'(t)dt$ then find the value of $f'(0) + f(0)$	(s)	3
		(t)	1

Sol. **(A) → (r); (B) → (q); (C) → (s); (D) → (b)**

(A) $A = \frac{\sqrt{3}}{4}\ell^2 \;\therefore\; \frac{dA}{dt} = \frac{\sqrt{3}}{4}\times 2\ell\times\frac{d\ell}{dt} = \frac{\sqrt{3}}{2}\times 10\times 0.1 = \frac{\sqrt{3}}{2}$

(B) $xy(x)+\int_0^x y(t)dt=(x+1)xy(x)+\int_0^x ty(t)dt = x^2y(x)+xy(x)+\int_0^x ty(t)dt$

$xy'(x)+y(x)+y(x)=2xy(x)+x^2y'(x)+xy'(x)+y(x)+xy(x)$

$\Rightarrow y(x)=x^2y'(x)+3xy(x) \Rightarrow x^2y'(x)=(1-3x)y(x)$

$\Rightarrow x^2\frac{d}{dx}y(x)=(1-3x)y(x)$

$\Rightarrow \int\frac{dy(x)}{y(x)}=\int\frac{(1-3x)}{x^2}dx=\int\frac{dx}{x^2}-3\int\frac{dx}{x}$

$\Rightarrow \ell ny(x)=-\frac{1}{x}-3\ell nx+c$

$\Rightarrow \ell n\{x^3y(x)\}=c-\frac{1}{x} \Rightarrow x^3y(x)=e^{c-\frac{1}{x}}=Ae^{-\frac{1}{x}} \therefore y(x)=\frac{Ae^{-\frac{1}{x}}}{x^3}$

(C) $v=x^3 \Rightarrow \delta v=3x^2\delta x$

$\therefore \frac{\delta v}{v}\times100=3x^2\frac{\delta x}{v}\times100=3x^2\cdot\frac{\delta x}{x^3}\times100=3\cdot\frac{\delta x}{x}\times100=3\times\frac{1}{100}\times100=3$

(D) $f(x)=(e^x+e^{-x})\cos x-2x-\left[x\int_0^x f'(t)dt-\int_0^x tf'(t)dt\right]$

$=(e^x+e^{-x})\cos x-2x-\left[x\{f(x)-f(0)\}-\left\{\left[tf(t)\right]_0^x-\int_0^x f(t)dt\right\}\right]$

$f(x)=(e^x+e^{-x})\cos x-2x-xf(x)+2x+\left[xf(x)-\int_0^x f(t)dt\right]$

$f(x)=(e^x+e^{-x})\cos x-\int_0^x f(t)dt$ ..... (1)

Differentiate equation (1), we get $f'(x)=e^x-e^{-x})\cos x-(e^x+e^{-x})\sin x-f(x)$

$\Rightarrow f'(x)+f(x)=(e^x-e^{-x})\cos x-(e^x+e^{-x})\sin x$

$\therefore f'(0)+f(0)=0$

Answer Q.17, Q.18 and Q.19 by appropriately matching the information given in the three columns of the following table.

Column-I: denotes the equation of the curve, Column-II: denotes the order of the differential equation of the equation of curve in Column-I, Column-III: denotes the degree of the differential equation of the equation of curve in Column-I

Column–I	Column–II	Column–III
(I) $y=cx+c^2-c^{3/2}+2$	(i) 2	(P) not defined
(II) $y=ae^{3x}+be^x$	(ii) 1	(Q) 1
(III) $(x-h)^2+(y-h)^2=r^2$	(iii) 3	(R) 4
(IV) $y^2=2k\left(x+\sqrt{k}\right)$	(iv) 4	(S) 3

17. Which of the following combination is **CORRECT**?

(A) (I) (i) (P) (B) (I) (ii) (R)

(C) (II) (ii) (P) (D) (III) (ii) (S)

Sol. **B**

18. Which of the following combination is **CORRECT**?
(A) (II) (ii) (P) (B) (II) (iii) (S)
(C) (III) (i) (Q) (D) (IV) (i) (R)

Sol. C

19. Which of the following combination is **INCORRECT**?
(A) (I) (ii) (R) (B) (II) (i) (Q)
(C) (III) (i) (S) (D) (IV) (ii) (S)

Sol. C

17.-19. (I) $\frac{dy}{dx} = c$ putting in equation of curve, we get

$y = y'x + (y')^2 + (y')^{3/2} + 2 \Rightarrow \left(y - y'x - (y')^2\right)^2 = \left((y')^{3/2} + 2\right)^2$ $\therefore$ Order = 1, degree = 4

(II) On solving, we get differential equation as $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 3y = 0$ $\therefore$ Order = 2, degree = 1

(III) Differential equation is $1 + \frac{dy}{dx} + (y - x)\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + \left(\frac{dy}{dx}\right)^3 = 0$ $\therefore$ Order = 2, degree = 1

(IV) Differential equation is $\left(y - 2x\frac{dy}{dx}\right)^2 = 4y\left(\frac{dy}{dx}\right)^3$ $\therefore$ Order = 1, degree = 3

20. If (h_1, h_2) are two solutions of the differential equation $\frac{dh}{dx} + p(x) \cdot h = Q(x)$.
Then prove that $h = h_1 + \lambda(h_1 - h_2)$ is the general solution of the equation where λ is any constant. For w relation between the constant α, β will the linear combination $\alpha h_1 + \beta h_2$ also be a solution _____

Sol. **(1.00)**
As h_1, h_2 are solutions of the differential equation;

$\frac{dh}{dx} + p(x) \cdot h = Q(x)$ ..... (1)

$\frac{dh_1}{dx} + p(x) \cdot h_1 = Q(x)$ ..... (2)

$\frac{dh_2}{dx} + p(x) \cdot h_2 = Q(x)$ ..... (3)

$\therefore$ From equation (1) and (2), we get $\left(\frac{dh}{dx} - \frac{dh_1}{dx}\right) + p(x)(h - h_1) = 0$

$\frac{d}{dx}(h - h_1) + p(x)(h - h_1) = 0$ ..... (4)

$\therefore$ From equation (2) and (3), we get

$\frac{d}{dx}(h_1 - h_2) + p(x)(h_1 - h_2) = 0$ ..... (5)

$\therefore$ From equation (4) and (5), we get $\frac{\frac{d}{dx}(h - h_1)}{\frac{d}{dx}(h_1 - h_2)} = \frac{h - h_1}{h_1 - h_2} \Rightarrow \frac{\frac{d}{dx}(h - h_1)}{h - h_1} = \frac{\frac{d}{dx}(h_1 - h_2)}{h_1 - h_2}$

Integrating, we get $\ell n(h - h_1) = \ell n(h_1 - h_2) + \ell n\lambda$. $\therefore h = h_1 + \lambda(h_1 - h_2)$

Now, $h = \alpha h_1 + \beta h_2$ will be solutions, if $\frac{d}{dx}(\alpha h_1 + \beta h_2) + p(x)(\alpha h_1 + \beta h_2) = Q(x)$

$\Rightarrow \alpha\left(\frac{dh_1}{dx} + p(x)h_1\right) + \beta\left(\frac{dh_2}{dx} + p(x)h_2\right) = Q(x) \Rightarrow \alpha \cdot Q(x) + \beta \cdot Q(x) = Q(x) \therefore \alpha + \beta = 1$

21. If n be the real value for which the substitution $y = v^n$ will transform the differential equation, $2x^4 y\frac{dy}{dx} + y^4 = 4x^6$ into a homogeneous equation. Then find the value of [n] ____

Sol. **(1.00)**

$[n] = 1$

$y = v^n \Rightarrow \frac{dy}{dx} = nv^{n-1}\frac{dv}{dx}$

$\therefore 2x^4 \cdot v^n \cdot n.v^{n-1}\frac{dv}{dx} + (v)^{4n} = 4x^6$

$\Rightarrow \frac{dv}{dx} = \frac{4x^6 - v^{4n}}{2nx^4 \cdot v^{2n-1}} \Rightarrow 4n = 6 \Rightarrow n = \frac{3}{2}$

$2n - 1 = 2 \Rightarrow n = \frac{3}{2} \therefore [n] = 1$

22. The differential equation of the family of lines situated at a constant distance k from origin has order a and degree b, then a + 2b is k, then $\frac{k}{2}$ is equal to ____

Sol. **(2.50)**

$y = mx + k\sqrt{1+m^2}$ ..... (1)

$\frac{dy}{dx} = m$. Putting in equation (1) and solving, we get $\left(y - x\frac{dy}{dx}\right)^2 = k^2\left(1 + \left(\frac{dy}{dx}\right)^2\right)$

$\therefore$ Order 1 and degree 2

23. Let y(x) be the solution of differential equation $x(x-2)\frac{dy}{dx} - 2(x-1)y = x^3(x-2)$ and y(3) = 9, then y(4) is equal to k ln $2e^2$, then $\frac{k}{5}$ is equal to ____

Sol. **(3.20)**

$\frac{dy}{dx} - \frac{2(x-1)}{x(x-2)}y = x^2$

Integrating factor = $\frac{1}{x(x-2)}$

$\therefore \frac{y}{x(x-2)} = \int \frac{x^2}{x(x-2)}dx \Rightarrow \frac{y}{x(x-2)} = x + 2\ln(x-2) + c$

$y(3) = 9 \therefore c = 0 \Rightarrow y(4) = 32 + 8 \ln 4$

$\Rightarrow 16 \ln 2e^2 \Rightarrow k = 16$

STRAIGHT LINES

Distance between Two Points

Let A and B be two given points, whose coordinates are given by $A(x_1, y_1)$ and $B(x_2, y_2)$ respectively. Then

$AB = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

Section Formula

Coordinates of the point P dividing the join of the two points $A(x_1, y_1)$ and $B(x_2, y_2)$ internally in the given ratio $\lambda_1 : \lambda_2$ are $P\left(\frac{\lambda_2 x_1 + \lambda_1 x_2}{\lambda_2 + \lambda_1}, \frac{\lambda_2 y_1 + \lambda_1 y_2}{\lambda_2 + \lambda_1}\right)$.

Coordinates of the point P dividing the join of the two points $A(x_1, y_1)$ and $B(x_2, y_2)$ externally in the ratio $\lambda_1 : \lambda_2$ are $P\left(\frac{\lambda_2 x_1 - \lambda_1 x_2}{\lambda_2 - \lambda_1}, \frac{\lambda_2 y_1 - \lambda_1 y_2}{\lambda_2 - \lambda_1}\right)$.

Centroid

The point of concurrency of the medians of a triangle is called the centroid of the triangle. The centroid of a triangle divides each median in the ratio 2:1. The coordinates of the centroid G, of the triangle with vertices $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ are $G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$.

Orthocentre

The point of concurrency of the altitudes of a triangle is called the orthocentre of the triangle. The co-ordinates of the orthocentre of the triangle $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ are

$$\left(\frac{x_1 \tan A + x_2 \tan B + x_3 \tan C}{\tan A + \tan B + \tan C}, \frac{y_1 \tan A + y_2 \tan B + y_3 \tan C}{\tan A + \tan B + \tan C}\right).$$

Incentre

The point of concurrency of the bisectors of the angles of a triangle is called the incentre of the triangle. The coordinates of the in-centre are given by

$$I = \left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c}\right).$$

Circumcentre

The point of concurrency of the perpendicular bisectors of the sides of a triangle is called circumcentre of the triangle. The coordinates of the circum-centre are

$$\left(\frac{x_1 \sin 2A + x_2 \sin 2B + x_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}, \frac{y_1 \sin 2A + y_2 \sin 2B + y_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}\right).$$

Area of a Triangle

Let (x_1, y_1), (x_2, y_2) and (x_3, y_3) respectively be the coordinates of the vertices A, B, C of a triangle ABC. Then the area of triangle ABC is

$$\frac{1}{2}\left[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right] \quad \ldots\ldots(1)$$

$$\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \quad \ldots\ldots(2)$$

LOCUS

When a point moves in a plane under certain geometrical conditions, the point traces out a path. The path of this moving point is called its locus.

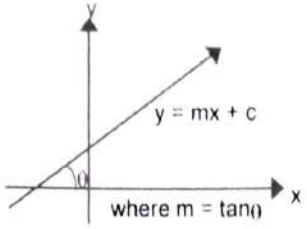
STRAIGHT LINE

Any equation of first degree of the form $Ax + By + C = 0$, where A, B, C are constants, always represents a straight line. (at least one out of A and B is non zero).

STANDARD EQUATIONS OF STRAIGHT LINES

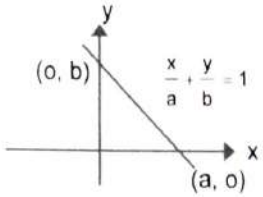
Slope Intercept Form

$y = mx + c$,
where m = slope of the line
c = y intercept



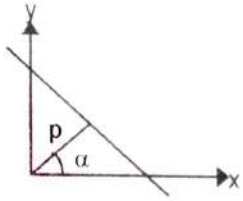
Intercept Form

$x/a + y/b = 1$
x intercept = a
y intercept = b



Normal Form

$x\cos\alpha + y\sin\alpha = p$, where α is the angle which the perpendicular to the line makes with the axis of x and p is the length of the perpendicular from the origin to the line. p is always non-negative.



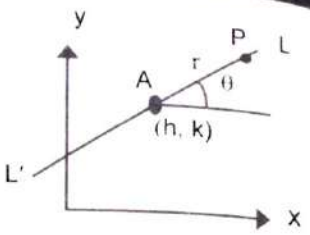
Slope Point Form

(a) One point (x_1, y_1) on the straight line.
(b) The direction of the straight line i.e., the slope m of the line
Equation: $y - y_1 = m(x - x_1)$,

Two Point Form

Equation: $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$, where (x_1, y_1) and (x_2, y_2) are the two given points. Here $m = \frac{y_2 - y_1}{x_2 - x_1}$.

Parametric Form



To find the equation of a straight line which passes through a given point A(h, k) and makes a given angle θ, with the positive direction of the x-axis. P(x, y) is any point on the line LAL'.

Let AP = r, $\Rightarrow x - h = r\cos\theta$, $y - k = r\sin\theta$.

Hence $\frac{x-h}{\cos\theta} = \frac{y-k}{\sin\theta} = r$ is the equation of the straight line LAL'.

Any point on the line will be of the form $(h + r\cos\theta, k + r\sin\theta)$ and for any value of r this will represent a point on the straight line. Here $|r|$ gives the distance of the point P from the fixed point (h, k).

If the point P is taken relatively upwards from A then r is positive otherwise negative. If the line is parallel to the x-axis then for the point right to A, r is positive and for left to A, r is negative.

FAMILY OF LINES

Let $L_1 : a_1x + b_1y + c_1 = 0$ and $L_2 : a_2x + b_2y + c_2 = 0$ be two given lines then $L_1 + \lambda L_2 = 0$ where $\lambda \in R$ gives equation of any line passing through the intersection of lines L_1 and L_2. That means $L_1 + \lambda L_2 = 0$ gives us a family of lines passing through a fixed point namely the point of intersection of L_1 and L_2.

Bisectors of the Angles between Two Given Lines

Angle bisector is the locus of a point which moves in such a way that its distance from two intersecting lines remains same.

The equations of the two bisectors of the angles between the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are

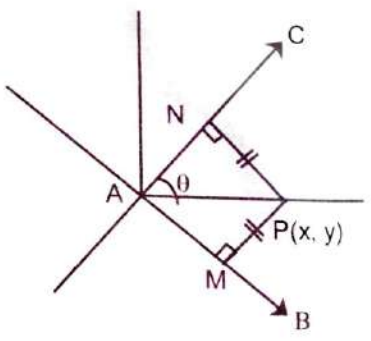
$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm\frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

If the two given lines are not perpendicular i.e. $a_1a_2 + b_1b_2 \neq 0$, then one of these equations is the equation of bisector of the acute angle and the other that of the obtuse angle.

Whether both lines are perpendicular or not the angle bisectors of these lines will be always mutually perpendicular.

The Bisectors of the Acute and the Obtuse Angles

Take one of the lines and let its slope be m_1 and take one of the bisectors and let it's slope be m_2. If θ be the angle between them, then find $\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1m_2}\right|$



If $\tan\theta > 1$, then the bisector taken is the bisector of the obtuse angle and the other one will be the bisector of the acute angle.

If $\tan\theta < 1$, then the bisector taken is the bisector of the acute angle and the other one will be the bisector of the obtuse angle.

The Equation of the Bisector of the Angle Containing the Origin

Write the equations of the two lines so that the constants c_1 and c_2 become positive. Then the equation $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$ is the equation of the bisector containing the origin.

Remarks:

(i). If $a_1a_2 + b_1b_2 < 0$, then the origin will lie in the acute angle and if $a_1a_2 + b_1b_2 > 0$, then origin will lie in the obtuse angle.

(ii). The remark (i) is helpful in finding the equation of bisector of the obtuse angle or acute angle directly.

The Equation of the Bisector of the Angle which contains a given Point

The equation of the bisector of the angle between the two lines containing the point (α,β) is

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

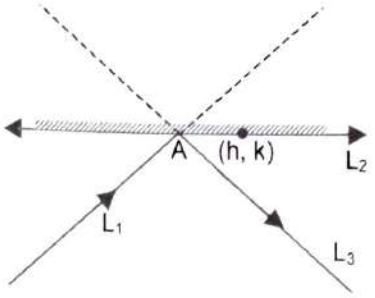
or $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = -\frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$ according as

$a_1\alpha + b_1\beta + c_1$ and $a_2\alpha + b_2\beta + c_2$ are of the same signs or of opposite signs.

The Equation of Reflected Ray

Let $L_1 \equiv a_1x + b_1y + c_1 = 0$ be the incident ray on the line mirror $L_2 \equiv a_2x + b_2y + c_2 = 0$.

Let L_3 be the reflected ray from the line L_2. Clearly L_2 will be one of the bisectors of the angles between L_1 and L_3. Since L_3 passes through A, $L_3 \equiv L_1 + \lambda L_2 = 0$.



Let (h, k) be a point on L_2. Then,

$$\frac{|a_1h + b_1k + c_1|}{\sqrt{a_1^2 + b_1^2}} = \frac{|a_1h + b_1k + c_1 + \lambda(a_2h + b_2k + c_2)|}{\sqrt{(a_1 + \lambda a_2)^2 + (b_1 + \lambda b_2)^2}}.$$

Since (h, k) lies on L_2, $a_2h + b_2k + c_2 = 0$

$\Rightarrow a_1^2 + a_2^2\lambda^2 + 2a_1a_2\lambda + b_1^2 + b_2^2\lambda^2 + 2b_1b_2\lambda = a_1^2 + b_1^2$

$\Rightarrow \lambda = 0$ or $\lambda = \frac{-2(a_1a_2 + b_1b_2)}{a_2^2 + b_2^2}$

But $\lambda = 0$ gives $L_3 = L_1$.

Hence $L_3 \equiv L_1 - \frac{2(a_1a_2 + b_1b_2)}{a_2^2 + b_2^2} L_2 = 0$.

Sometimes the reflected ray L_3 is also called the mirror image of L_1 in L_2.

Equations of Straight Lines Passing through a given Point and Equally Inclined to a given Line

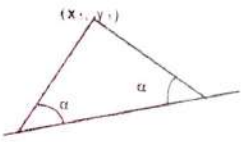
Let the straight lines passing through the point (x_1, y_1) make equal angles with the given straight line $y = m_1x + c$.

If m is the slope of the required line and α is the angle which this line makes with the given line, then

$$\tan\alpha = \pm\frac{m_1 - m}{1 + m_1m}.$$

The above expression for $\tan\alpha$, gives two values of m, say m_A and m_B. The required equations of the lines through the point (x_1, y_1) and making equal angles α with the given line are

$y - y_1 = m_A(x - x_1)$, $y - y_1 = m_B(x - x_1)$.



Pair of Straight Lines

The general equation of second degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines if

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0 \text{ and } h^2 \geq ab.$$

$\Rightarrow abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ and $h^2 \geq ab$.

The homogeneous second degree equation $ax^2 + 2hxy + by^2 = 0$ represents a pair of straight lines through the origin if $h^2 \geq ab$.

If the lines through the origin whose joint equation is $ax^2 + 2hxy + by^2 = 0$, are $y = m_1x$ and $y = m_2x$, then $y^2 - (m_1 + m_2)xy + m_1m_2x^2 = 0$ and $y^2 + \frac{2h}{b}xy + \frac{a}{b}x^2 = 0$ are identical, so that $m_1 + m_2 = -\frac{2h}{b}$, $m_1m_2 = \frac{a}{b}$.

If θ be the angle between two lines, through the origin, then

$$\tan\theta = \pm\frac{\sqrt{(m_1 + m_2)^2 - 4m_1m_2}}{1 + m_1m_2} = \pm\frac{2\sqrt{h^2 - ab}}{a + b}.$$

The lines are perpendicular if $a + b = 0$ and coincident if $h^2 = ab$.

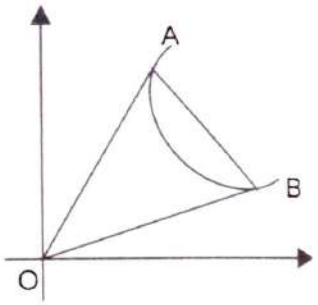
- Point of intersection of the two lines represented by $ax^2+2hxy+by^2+2gx+2fy+c=0$ is obtained by solving equations $ax + hy + g = 0$ and $hx + by + f = 0$.
- It should be noted that the lines $ax + hy + g = 0$ and $hx + by + f = 0$ are not the lines represented by $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$. These are the lines concurrent with the lines represented by the equation.

Joint Equation of Pair of Lines Joining the Origin with the Points of Intersection of a Curve and a Line

If the line $lx + my + n = 0$, $(n \neq 0)$ (i.e. the line does not pass through the origin) cuts the curve $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ at two points A and B, then the joint equation of the straight lines passing through A and B and the origin is obtained by homogenizing the equation of the curve by the equation of the line. i.e.

$$ax^2 + 2hxy + by^2 + (2gx + 2fy)\left(\frac{\ell x + my}{-n}\right) + c\left(\frac{\ell x + my}{-n}\right)^2 = 0$$

is the equation of the lines OA and OB.



ILLUSTRATIVE PROBLEMS

1. A diagonal of rhombus ABCD is a member of both the family of lines $(x + y - 1) + \lambda_1(2x + 3y - 2) = 0$ and $(x - y + 2) + \lambda_2(2x - 3y + 5) = 0$ where $\lambda_1, \lambda_2 \in R$ and one of the vertex of the rhombus is (3, 2). If area of the rhombus is $12\sqrt{5}$ square units then find the length of semi longer diagonal of the rhombus

Sol. Since diagonal is a member of both the families so it will pass through (1, 0) and (–1, 1)
Equation of diagonal AC is $x + 2y - 1 = 0$
Since one of the vertex (3, 2) which does not be on AC, so equation of BD is $2x - y = 4$

Point of intersection of AC and BD is $P\left(\frac{9}{5}, -\frac{2}{5}\right)$

If vertex B is (3, 2) then vertex D is $\left(\frac{3}{5}, -\frac{14}{5}\right)$ also $BD = \frac{12\sqrt{5}}{5}$ (say d_1)

Area of rhombus $= \frac{1}{2} \times d_1 \times d_2$

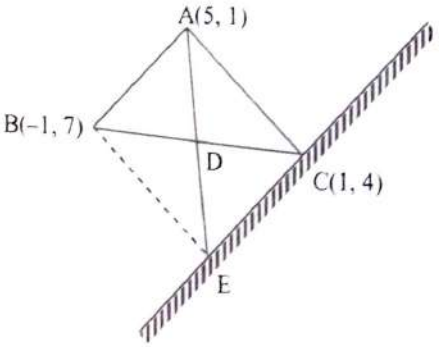
$\frac{1}{2} \times \left(\frac{12\sqrt{5}}{5}\right) \times d_2 = 12\sqrt{5}$

$d_2 = 10$

2. A triangle ABC having vertices A(5, 1), B(–1, –7) and C(1, 4) respectively. L be the line mirror passing through C and parallel to AB and a light ray emanating from point A goes along the direction of internal bisector of the angle A, which meets the mirror at BC at E, D respectively. If sum of the areas of ΔACE and ΔABC is (10 + a) sq unit then find the value of a

Sol. $\frac{AD}{DE} = \frac{BC}{DC} = \frac{AB}{AC}$

$\therefore$ point $D\left(\frac{1}{3}, \frac{1}{3}\right)$ and $E(-2, 0)$



Area of ΔACE + Area of ΔACE = 2 × Area of ΔACE

$= 2 \times \frac{19}{2} = 19$ sq. units

Then the required value of a is 9

3. The vertices of a triangle are A(m, n), B(12, 19) and C(23, 20), where m and n are integers. If its area is 70 and the slope of the median through A is –5, then find the the last digit of (m + n)

Sol. Area of $\Delta = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

$= \frac{1}{2}\begin{vmatrix} m & n & 1 \\ 12 & 19 & 1 \\ 23 & 20 & 1 \end{vmatrix} = 70 \Rightarrow -m + 11n = 197 \pm 140$ ..... (1)

Median through A has slope –5

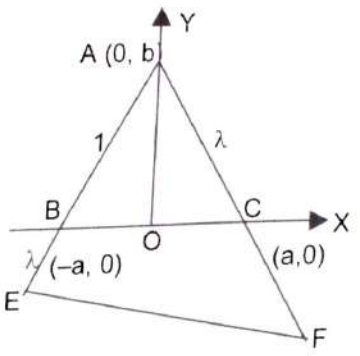
$\Rightarrow 5m + n = 107$ ..... (2)

(1), (2) $\Rightarrow m = 20, n = 7, m + n = 27$

Then the required solution is 7

4. Let ABC be a given isosceles triangle with AB = AC. Sides AB and AC are extended upto E and F respectively, such that BE. CF = AB^2. Prove that the line EF always passes through a fixed point.

Sol. Let ABC be the triangle having Vertices (–a, 0) (0, b) and (a, 0).



$BE.CF = AB^2 \Rightarrow \frac{BE}{AB} = \frac{AB}{CF}$

$\Rightarrow \frac{BE}{AB} = \frac{AC}{CF} = \frac{\lambda}{1}$

Hence the co-ordinates of E and F are $(-a(\lambda+1), -\lambda b)$ and $\left[a\left(1+\frac{1}{\lambda}\right), \frac{-b}{\lambda}\right]$

Equation of line EF is $y + \lambda b = \frac{-\lambda b + \frac{b}{\lambda}}{-a(\lambda+1) - \frac{a(\lambda+1)}{\lambda}}\left[x + a(\lambda+1)\right]$

or $y + \lambda b = \frac{\frac{b}{\lambda}(1-\lambda^2)}{-\frac{a(\lambda+1)}{\lambda}[1+\lambda]}\left[x + a(\lambda+1)\right]$

or $y + \lambda b = \frac{b(\lambda-1)}{a(\lambda+1)}\left[x + a(\lambda+1)\right]$

$a(\lambda+1)\, y + ab\lambda(\lambda+1) = b(\lambda-1)x + ab(\lambda^2-1)$

or $(bx + ay + ab) - \lambda(bx - ay - ab) = 0$

which is the equation of a family of lines passing through the point of intersection of the lines $bx + ay + ab = 0$ and $bx - ay - ab = 0$

The point of intersection being $(0, -b)$.

Hence the line EF passes through a fixed point.

5. Prove that the straight lines joining the origin to the points of intersection of the straight line hx + ky = 2hk and the curve $(x-k)^2 + (y-h)^2 = c^2$ are at right angles if $h^2 + k^2 = c^2$.

Sol. Making the equation of the curve homogeneous with the help of the equation of line, we get

$$x^2 + y^2 - 2(kx + hy)\left(\frac{hx + ky}{2hk}\right) + (h^2 + k^2 - c^2)\left(\frac{hx + ky}{2hk}\right)^2 = 0$$

or, $4h^2k^2x^2+4h^2k^2y^2-4hk^2x(hx+ky)-4h^2ky(hx+ky)+(h^2+k^2-c^2)(h^2x^2+k^2y^2+2hkxy)=0$.

This is the equation of the pair of lines joining the origin to the points of intersection of the given line and the curve. They will be at right angles if coefficient of x^2 + coefficient of $y^2 = 0$

$\Rightarrow (h^2 + k^2)(h^2 + k^2 - c^2) = 0$. Since $h^2 + k^2 \neq 0$, $h^2 + k^2 = c^2$.

SOLVED PROBLEMS

1. If the equation $ax^2 - 6xy + y^2 + 2gx + 2fy + c = 0$ represents a pair of lines whose slopes are m and m^2, then sum of all possible values of a is

(A) 17
(B) –19
(C) 19
(D) –17

Sol. **(B)**

The given lines will be parallel to lines $ax^2 - 6xy + y^2 = 0$

So, $\left(\frac{y}{x}\right)^2 - 6\left(\frac{y}{x}\right) + a = 0$

$m + m^2 = 6$

Which gives $m = -3$ or 2 ..... (1)

and $mm^2 = a$

From (2) $a = -27$ or 8 ..... (2)

Hence sum of all possible value of $a = -19$

2. A triangle ABC with vertices $A \equiv (2, 5)$, $B \equiv (5, 2)$ and $C \equiv (-1, -1)$ inscribed on the curve $xy - x - y - 3 = 0$ and $H \equiv (\alpha, \alpha)$ is orthocentre. Let D, E and F be the feet of perpendiculars from A, B and C on BC, CA and AB respectively and inradius of ΔDEF is r then

(A) orthocentre of triangle ABC is (–1, –1)
(B) equation of DE is $x + y - \frac{13}{5} = 0$
(C) $r = \frac{4}{5}\sqrt{2}$ units
(D) $r = \frac{2}{5}\sqrt{2}$ units

Sol. **(D)**

Clearly incentre of ΔDEF is orthocentre of ΔABC i.e. (α, α) which lies on hyperbola $xy - x - y - 3 = 0 \Rightarrow \alpha = 3$

Again equation of DE is $x + y - \frac{26}{5} = 0$

$\Rightarrow$ r = length of perpendicular from I(3, 3) on DE $\Rightarrow r = \frac{2}{5}\sqrt{2}$ units

3. ABC is right–angled triangle such that D is midpoint of hypotenuse AB and E is midpoint of AC. Segment BE and CD intersect at F. If BC = 1 and $AC = \sqrt{2}$ and angle $CFB = \theta$ then $\cos\theta$ is

(A) $\frac{1}{2}$
(B) 0
(C) $\frac{\sqrt{3}}{2}$
(D) $\frac{1}{\sqrt{2}}$

Sol. **(B)**

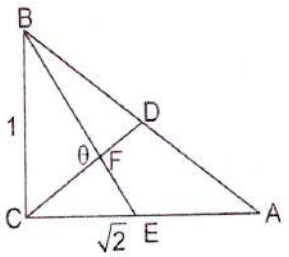
Select 'C' as origin

$\Rightarrow CD = \frac{\frac{1}{2}}{\frac{1}{\sqrt{2}}} = \frac{1}{\sqrt{2}}$

Slope $BE = -\sqrt{2}$

$\Rightarrow CD \perp BE \Rightarrow \angle BFC = 90^\circ$

$\Rightarrow \cos\theta = 0$



4. If the area of the rhombus enclosed by the lines $lx \pm my \pm n = 0$ be 2 square units, then

(A) l, m, n are in G.P
(B) l, n, m are in G.P
(C) $lm = n$
(D) $ln = m$

Sol. **(B)**

By solving the sides of the rhombus, its vertices are

$\left(0, -\frac{n}{m}\right)$,

$\left(-\frac{n}{l}, 0\right)$, $\left(0, \frac{n}{m}\right)$ and $\left(\frac{n}{l}, 0\right)$

$\therefore$ area $= \frac{1}{2} \times \frac{2n}{m} \times \frac{2n}{l} = 2$

$\Rightarrow n^2 = lm$ or l, n, m are in G.P

5. If the chord $x + 2y - 2 = 0$ of the curve $6x^2 + y^2 - 8x + 2y = 0$ subtends angle $\frac{\pi}{k}$ radians at the point (1, 1) then the value of k is

(A) 6
(B) 4
(C) 3
(D) 2

Sol. **(D)**

Shifting origin to the point (1, 1), the new equations are $x + 2y = -1$ and $6x^2 + y^2 + 4x + 4y + 1 = 0$.

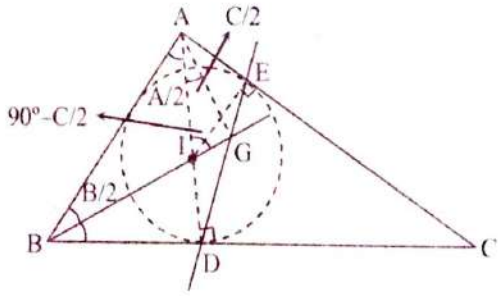
Now homogenizing the two, we see that

Coefficient of x^2 + coefficient of $y^2 = 0$

6. Let points A(7, 5), B(2, 1) and C(10, 3) the vertices of a ΔABC. I is incentre of ΔABC. E and D are points of contact of incircle and sides AC and BC of ΔABC respectively. If G is the intersection points of angle bisector of $\angle B$ and line joining ED. Then angle between line AG and BG is

(A) 30°
(B) 45°
(C) 60°
(D) 90°

Sol. **(D)**



I, EC, D are concyclic

So, $\angle EID = 180 - c$ in ΔIED

$\angle IDE = \angle IED = \frac{C}{2}$

So, $\angle AED = 90 + \frac{C}{2}$

in $\Delta AIB = 180 - \left(\frac{A+B}{2}\right)$

$\angle AIG = \frac{A+B}{2} = 90 - \frac{C}{2}$

So, A, I, G, E are concylic

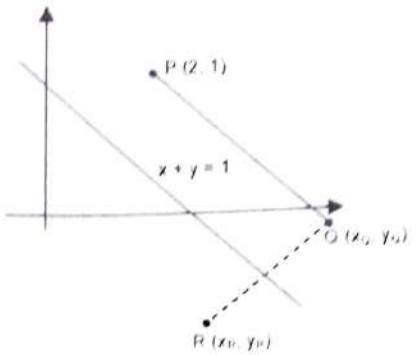
So, $\angle IAG = \angle IEG = \frac{C}{2}$

$\Rightarrow \angle AGI = 90°$

7. The point P (2, 1) is shifted through a distance $3\sqrt{2}$ units measured parallel to the line $x + y = 1$ in the direction of decreasing ordinates, to reach at Q. The image of Q with respect to given line is.

(A) (3, –4)
(B) (–3, 2)
(C) (0, –1)
(D) none of these

Sol. **(A).** $PQ = 3\sqrt{2}$



$\Rightarrow x_Q = 2 + 3\sqrt{2} \cdot \frac{1}{\sqrt{2}} = 5$

and $y_Q = 1 - 3\sqrt{2} \cdot \frac{1}{\sqrt{2}} = -2$

$\Rightarrow Q \equiv (5, -2)$

Distance of Q from the line $x + y = 1$ is

$\frac{|5 - 2 - 1|}{\sqrt{2}} = \sqrt{2} \Rightarrow QR = 2\sqrt{2}$

$\Rightarrow x_R = 5 - 2\sqrt{2} \cdot \frac{1}{\sqrt{2}} = 3$ and $y_R = -2 - 2\sqrt{2} \cdot \frac{1}{\sqrt{2}} = -4$.

Thus $R \equiv (3, -4)$.

8. The range of α for which the points $(\alpha, 2 + \alpha)$ and $\left(\frac{3\alpha}{2}, \alpha^2\right)$ lie on opposite sides of the line $2x + 3y = 6$ is

(A) (–2, 1)
(B) $(-\infty, -2) \cup (0, 1)$
(C) $(-2, 0) \cup (1, \infty)$
(D) $(-1, 0) \cup (2, \infty)$

Sol. **(B).** $f(x, y) = 2x + 3y - 6 \Rightarrow f(\alpha, 2 + \alpha) = 5\alpha$

$f\left(\frac{3}{2}\alpha, \alpha^2\right) = 3\alpha + 3\alpha^2 - 6$

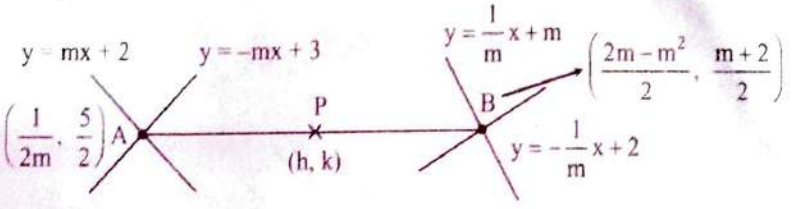
for the given condition; $5\alpha(3\alpha + 3\alpha^2 - 6) < 0$ or $\alpha(\alpha + 2)(\alpha - 1) < 0 \Rightarrow \alpha \in (-\infty, -2) \cup (0, 1)$

9. If the locus of the mid point of the intersection points of the lines $y = mx + 2$, $y = -mx + 3$ and $y = \frac{1}{m}x + m$, $y = -\frac{1}{m}x + 2$ is $4x = \frac{1}{(ay - b)} + (ay - b)(c - ay)$, then

(A) $a = 4$
(B) $b = 7$
(C) $c = 9$
(D) $a = 8$

Sol. **(A, B, C)**

So $h = \frac{\frac{1}{2m} + \frac{2m - m^2}{2}}{2}$, $k = \frac{\frac{5}{2} + \frac{m + 2}{2}}{2}$ eliminating m



$4x = \frac{1}{(4y - 7)} + (4y - 7)(9 - 4y)$

So, $a = 4, b = 7, c = 9$

10. If $6a^2 - 3b^2 - c^2 + 7ab - ac + 4bc = 0$, then the family of lines $ax + by + c = 0$ is concurrent at

(A) $(-3, 1)$ (B) $(2, 3)$

(C) $(3, -1)$ (D) $(-2, -3)$

Sol. **(C, D)**

$6a^2 + (7b - c)a - (3b^2 - 4bc + c^2) = 0$

$\Rightarrow 3a - b + c = 0$ or $2a + 3b - c = 0$

11. A line segment is divided so that the lesser part is to the greater part is equal to the greater part is to the whole. If R is the ratio of the lesser part to the greater part, then

(A) $R^2 + R - 1 = 0$ (B) $R^2 + R^{-1} = 2$

(C) $R^{(R^2+R^{-1})} + R^{-1} = 2$ (D) $R^2 + R + 1 = 0$

Sol. **(A, B, C)**

Let $y < x \Rightarrow \dfrac{y}{x} = \dfrac{x}{x+y}$

$\Rightarrow R = \dfrac{1}{1+R}$

12. Two sides of a rhombus OABC (lying entirely in first quadrant or fourth quadrant) of area equal to 2 sq. units, are $y = \dfrac{x}{\sqrt{3}}$, $y = \sqrt{3}\,x$. Then possible coordinates of B is/are ('O' being the origin)

(A) $\left(1+\sqrt{3},\ 1+\sqrt{3}\right)$ (B) $\left(-1-\sqrt{3},\ -1-\sqrt{3}\right)$

(C) $\left(\sqrt{3}-1,\ \sqrt{3}-1\right)$ (D) none of these

Sol. **(A), (B)**

Let the length of the sides of the rhombus be l

$\Rightarrow$ Area $= l^2 \sin 30° = 2 \Rightarrow l = 2$ unit

$OB^2 = OA^2 + AB^2 - 2OA.AB\cos 150°$

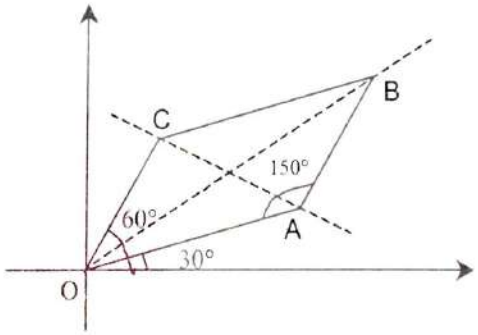
$= 4 + 4 - 2(4).\left(-\dfrac{\sqrt{3}}{2}\right) = 4(2+\sqrt{3})$

$OB = 2\sqrt{2+\sqrt{3}}$

$B \equiv \left(\sqrt{4+2\sqrt{3}},\ \sqrt{4+2\sqrt{3}}\right) \equiv \left(1+\sqrt{3},\ 1+\sqrt{3}\right)$

Hence coordinates of B can be $\left(1+\sqrt{3},\ 1+\sqrt{3}\right)$

or $\left(-1-\sqrt{3},\ -1-\sqrt{3}\right)$



Comprehension

Suppose we define the distance between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ as
$d(P, Q) = \max.\{|x_2 - x_1|, |y_2 - y_1|\}$, then

Answer the following questions

13. For a point P along the line $y = \sqrt{3}\,x$, d(0, P) is equal to

(A) x (B) y

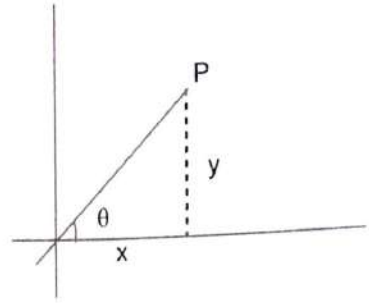
(C) x if $x < 0$ (D) y if $y \geq 0$

Sol. **(B).** Clearly

$|y| < |x|$ if $\theta < \frac{\pi}{4}$,

$|y| > |x|$ if $\theta > \frac{\pi}{4}$ and

$|y| = |x|$ if $\theta = \frac{\pi}{4}$.



14. The area of the region bounded by the locus of a point P satisfying d(P, A) = 4, where A is (1, 2) is

(A) 64 sq. units

(B) 54 sq. units

(C) 16π sq. units

(D) none of these

Sol. **(A)**

We have max.$\{|x - 1|,$
$|y - 2|\} = 4.$
If $|x - 1| \geq |y - 2|$
then $|x - 1| = 4,$
i.e. if $(x + y - 3)(x - y + 1) \geq 0$
then $x = -3$ or 5,
If $|y - 2| \geq |x - 1|$
then $|y - 2| = 4$
i.e. $(x + y - 3)(x - y + 1) \leq 0,$
then $y = -2$ or 6.
So, the locus of P bounds a square, the equation of whose sides are
$x = -3,$
$x = 5,$
$y = -2,$
$y = 6.$
Thus the area is $(8)^2 = 64.$

15. Suppose that points A and B have coordinates (1, 0) and (−1, 0) respectively, then for a variable point P on this plane the equation d(P, A) + d(P, B) = 2 represents

(A) a line segment joining A and B

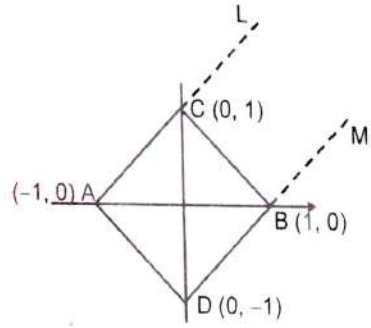
(B) an ellipse with foci at A and B

(C) region lying inside a square of area 2

(D) region inside a semicircle with AB as diameter

Sol. **(D)**

Consider a square with vertices at
(−1, 0), (1, 0), (0, 1) and (0, −1) (see figure).
If we select a point P_1 in the region BCLM,
then d(A, P) = 1 + x.
D(B, P) = y
$\therefore$ d(A, P) + d(B, P) = 1 + x + y > 2.
However for points P_2 and P_3 lying respectively on the line BC and below the line BC,
d(P, A) + d(P, B) = 2
The same argument holds for other quadrants also.
Hence, the d(P, A) + d(P, B) = 2 represents the region lying inside the square ABCD.



Assertion Reason

16. Statement-1: If the point $(2a - 5, a^2)$ is on the same side of the line $x + y - 3 = 0$ as that of the origin, then $a \in (2, 4)$.
because
Statement-2: The point (x_1, y_1) and (x_2, y_2) lies on the same or opposite sides of the line $ax + by + c = 0$ as $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ have same or opposite signs.
(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1
(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True

Sol. **(D)**
$a^2 + 2a - 8 < 0$
$(a - 2)(a + 4) < 0 \Rightarrow a \in (-4, 2)$

17. A square having sides parallel to the coordinate axes is inscribed in the region $\{(x, y): x, y > 0, y \le -x^3 + 3x\}$. If area of the square is written as $A^{1/3} + B^{1/3}$ square units where $A, B \in Z$ and $A > B$, then

Column – I		Column – II	
(A)	$\sqrt{A - B}$	(p)	6
(B)	slope of line having x-intercept as A and y-intercept as B	(q)	$\frac{1}{2}$
(C)	$\frac{A + B}{A - B}$	(r)	$\frac{1}{3}$
(D)	Circum-radius of ΔOPQ where O is origin, $P(A^{1/3}, 0)$ and $Q(0, B^{1/3})$ is	(s)	16
		(t)	18

Sol. **(A) → (t); (B) → (q); (C) → (r); (D) → (p)**
Let two x-coordinates of the square's vertices be a and b with $a > b$, then
$a^3 - 3a = b^3 - 3b = a - b$ so, $a = b^3 - 2b$ and $b = 4a - a^3 \Rightarrow a^3 = 4a - b$ and $b^3 = a + 2b$
$\Rightarrow (a - b)(a^2 + b^2 + ab) = 3(a - b) \Rightarrow a^2 + ab + b^2 = 3 \Rightarrow a = b^3 - 2b \Rightarrow b^2 = 1 + 2^{1/3}$
$\therefore$ Area of square will be $(216)^{1/3} + (-108)^{1/3}$. So $A = 216$, $B = -108$

Answer Q.18, Q.19 and Q.20 by appropriately matching the information given in the three columns of the following table.

A line which makes an acute angle θ with the positive direction of the x-axis is drawn through the point $P(x_1, y_1)$ to meet the line $x = 6$ at R and $y = 8$ at S. Column-I gives information about $P(x_1, y_1)$. Column-II gives information about distance PR. Column-III gives information about distance PS.

Column–I	Column–II	Column–III
(I) P(3, 4)	(i) PR = 2 sec θ	(P) PS = 4 cosec θ
(II) P(1, 2)	(ii) PR = 3 sec θ	(Q) PS = 5 cosec θ
(III) P(4, 3)	(iii) PR = 4 sec θ	(R) PS = 6 cosec θ
(IV) P(2, 1)	(iv) PR = 5 sec θ	(S) PS = 7 cosec θ

18. Which of the following is the only **CORRECT** combination?
(A) (I) (iii) (S)
(B) (I) (i) (Q)
(C) (III) (iii) (S)
(D) (III) (i) (Q)

Sol. **(D)**

19. Which of the following is the only **CORRECT** combination?
(A) (II) (iii) (P)
(B) (II) (iv) (R)
(C) (IV) (i) (Q)
(D) (IV) (iv) (R)

Sol. **(B)**

20. Which of the following is the only **CORRECT** combination?
(A) (I) (ii) (R)
(B) (I) (iii) (P)
(C) (IV) (iii) (S)
(D) (IV) (ii) (P)

Sol. **(C)**

18 -20. $x = x_1 + r\cos\theta,\ y = y_1 + r\sin\theta$

21. If the straight line $3x + 4y = 24$ intersects the axes at A and B and the straight line $4x + 3y = 24$ intersects the axis at C and D then the number of parabolas possible to draw through A, B, C and D is _____

Sol. **(2.00)**

Equation of the curve passing through the points A, B, C, D is $(3x + 4y - 24)(4x + 3y - 24) - \lambda xy = 0$
If it represents a parabola then the quadratic terms must form a perfect square. $\Rightarrow \lambda = 1$ or 49.

22. Distance between pair of parallel straight lines $x^2 + 2\sqrt{2}xy + 2y^2 + 4x + 4\sqrt{2}y + 1 = 0$ is _____

Sol. **(2.00)**

Lines are of the form $x + \sqrt{2}y + \lambda_1 = 0$ and $x + \sqrt{2}y + \lambda_2 = 0 \Rightarrow \lambda_1 + \lambda_2 = 4$ and $\lambda_1\lambda_2 = 1$

$$\Rightarrow \text{Distance} = \left|\frac{\lambda_1 - \lambda_2}{\sqrt{3}}\right| = \frac{\sqrt{(\lambda_1 + \lambda_2)^2 - 4\lambda_1\lambda_2}}{\sqrt{3}} = 2$$

23. Three families of lines are as follows: (I) $2x + 3y + 1 + k_1(2x + 4y + 4) = 0$
(II) $(3 + 6\tan\theta)x + (4 + 4\tan\theta)y + 4 + \tan\theta = 0$, (III) $(x + y + \lambda) + k_3(2x + y + 1) = 0$
Find the value of λ so that 3 families have a common member

Sol. **(1.21)**

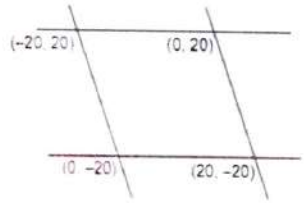
For first family the fixed point is (4, –3). For second family fixed point is $\left(1, -\frac{7}{4}\right)$ and third family has fixed point as $(\lambda - 1, 1 - 2\lambda)$. Now for three families to have a common member, these three points should be collinear so

$$\frac{1}{2}\begin{vmatrix} 4 & -3 & 1 \\ 1 & -\frac{7}{4} & 1 \\ \lambda - 1 & 1 - 2\lambda & 1 \end{vmatrix} = 0 \Rightarrow 4\left(-\frac{7}{4} - (1 - 2\lambda) + 3(1 - \lambda + 1)\right) + \left((1 - 2\lambda) - \left(-\frac{7}{4}\right)(\lambda - 1)\right) = 0 \Rightarrow \lambda = \frac{23}{19}$$

24. The graph $y^2 + 2xy + 40|x| = 400$ divides the plane into regions. The area of bounded region is _____

Sol. **(800)**

For $x \geq 0$, the equation is
$(y + 20)(y + 2x - 20) = 0$
For $x \leq 0$, the equation is
$(y - 20)(y + 2x + 20) = 0$
Area $= 20 \times 40 = 800$.



9

CIRCLE

EQUATIONS OF THE CIRCLE IN VARIOUS FORMS

- The simplest equation of the circle is $x^2 + y^2 = r^2$ whose centre is (0, 0) and radius r.
- The equation $(x - a)^2 + (y - b)^2 = r^2$ represents a circle with centre (a, b) and radius r.
- The equation $x^2 + y^2 + 2gx + 2fy + c = 0$ is the general equation of a circle with centre (-g, -f) and radius $\sqrt{g^2 + f^2 - c}$.
- Equation of the circle with points $P(x_1, y_1)$ and $Q(x_2, y_2)$ as extremities of a diameter is $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$.
- The equation of the circle through three non-collinear points $P(x_1, y_1)$, $Q(x_2, y_2)$ and $R(x_3, y_3)$ is

$$\begin{vmatrix} x^2+y^2 & x & y & 1 \\ x_1^2+y_1^2 & x_1 & y_1 & 1 \\ x_2^2+y_2^2 & x_2 & y_2 & 1 \\ x_3^2+y_3^2 & x_3 & y_3 & 1 \end{vmatrix} = 0.$$

- **Equation of a Circle under different Conditions:**

Condition	Equation
(i) Touches both the axes with centre (a, a) and radius a	$(x-a)^2 + (y-a)^2 = a^2$
(ii) Touches x-axis only with centre (α, a) and radius a	$(x-\alpha)^2 + (y-a)^2 = a^2$
(iii) Touches y-axis only with centre (a, β) and radius a	$(x-a)^2 + (y-\beta)^2 = a^2$
(iv) Passes through the origin with centre $\left(\frac{\alpha}{2}, \frac{\beta}{2}\right)$ and radius $\sqrt{\frac{\alpha^2+\beta^2}{4}}$.	$x^2 + y^2 - \alpha x - \beta x = 0$

Parametric Equation of a Circle

The equations $x = a\cos\theta$, $y = a\sin\theta$ are called parametric equations of the circle $x^2 + y^2 = a^2$ and θ is called a parameter. The point (a cos θ, a sin θ) is also referred to as point θ. The parametric coordinates of any point on the circle $(x - h)^2 + (y - k)^2 = a^2$ are given by
$(h + a\cos\theta, k + a\sin\theta)$ with $0 \le \theta < 2\pi$.

The Position of a Point with respect to a Circle

The point $P(x_1, y_1)$ lies outside, on, or inside a circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$, according as
$S_1 \equiv x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c > = \text{or} < 0$.

Equations of Tangents and Normals

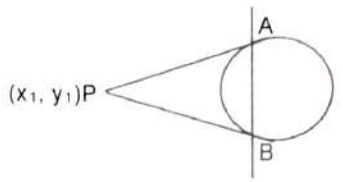
If $S = 0$ be a curve then $S_1 = 0$ indicate the equation which is obtained by substituting $x = x_1$ and $y = y_1$ in the equation of the given curve and $T = 0$ is the equation which is obtained by substituting $x^2 = xx_1$, $y^2 = yy_1$, $2xy = xy_1 + yx_1$, $2x = x + x_1$, $2y = y + y_1$ in the equation $S = 0$. If $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ then $S_1 \equiv x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$, and $T \equiv xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c$.

- Equation of the tangent to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at $A(x_1, y_1)$ is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$.
- The condition that the straight line $y = mx + c$ is a tangent to the circle $x^2 + y^2 = a^2$ is $c^2 = a^2(1 + m^2)$ and the point of contact is $(-a^2m/c, a^2/c)$ i.e. $y = mx \pm a\sqrt{1 + m^2}$ is always a tangent to the circle $x^2 + y^2 = a^2$ whatever be the value of m.
- The joint equation of a pair of tangents drawn from the point $A(x_1, y_1)$ to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $T^2 = SS_1$.
- The equation of the normal to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at any point (x_1, y_1) lying on the circle is $\frac{x - x_1}{x_1 + g} = \frac{y - y_1}{y_1 + f}$.

  In particular, equations of the tangent and the normal to the circle $x^2 + y^2 = a^2$ at (x_1, y_1) are $xx_1 + yy_1 = a^2$; and $\frac{x}{x_1} = \frac{y}{y_1}$ respectively.
- The equation of the chord of the circle $S \equiv 0$, with mid point (x_1, y_1) is $T = S_1$.
- The length of the tangent drawn from a point (x_1, y_1) outside the circle $S = 0$, to the circle is $\sqrt{S_1}$.

CHORD OF CONTACT

From a point $P(x_1, y_1)$ two tangents PA and PB can be drawn to the circle. The chord AB joining the points of contact A and B of the tangents from P is called the chord of contact of $P(x_1, y_1)$ with respect to the circle. Its equation is given by $T = 0$.



Radical Axis

The radical axis of two circles is the locus of a point from which the tangent segments to the two circles are of equal length.

Equation to the Radical Axis

In general $S - S' = 0$ represents the equation of the Radical Axis to the two circles
i.e.$\mathbf{2x(g - g') + 2y(f - f') + c - c' = 0}$ where $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ and
$S' \equiv x^2 + y^2 + 2g'x + 2f'y + c' = 0$

- If $S = 0$ and $S' = 0$ intersect in real and distinct points then $S - S' = 0$ is the equation of the common chord of the two circles.
- If $S' = 0$ and $S = 0$ touch each other, then $S - S' = 0$ is the equation of the common tangent to the two circles at the point of contact.

FAMILY OF CIRCLES

- If $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ and $S' \equiv x^2 + y^2 + 2g'x + 2f'y + c' = 0$ are two intersecting circles, then $S + \lambda S' = 0$, $\lambda \neq -1$, is the equation of the family of circles passing through the points of intersection of $S = 0$ and $S' = 0$.
- If $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ is a circle which is intersected by the straight line $\mu \equiv ax + by + c = 0$ at two real and distinct points, then $S + \lambda\mu = 0$ is the equation of the family of circles passing through the points of intersection of $S = 0$ and $\mu = 0$. If $\mu = 0$ touches $S = 0$ at P, then $S + \lambda\mu = 0$ is the equation of the family of circles, each touching $\mu = 0$ at P.

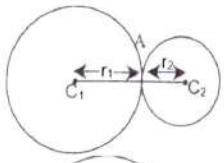
- The equation of a family of circles passing through two given points (x_1, y_1) and (x_2, y_2) can be written in the form.

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + \lambda \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$, where λ is a parameter.

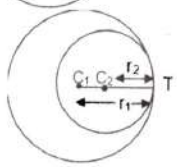
- The equation of the family of circles which touch the line $y - y_1 = m(x - x_1)$ at (x_1, y_1) for any value of m is $(x - x_1)^2 + (y - y_1)^2 + \lambda[(y - y_1) - m(x - x_1)] = 0$. If m is infinite, the equation is $(x - x_1)^2 + (y - y_1)^2 + \lambda(x - x_1) = 0$.
- Two circles are said to intersect orthogonally if the angle of intersection of the circles i.e., the angle between their tangents at the point of intersection is 90°.
  The condition for the two circles $S = 0$ and $S_1 = 0$ to cut each other orthogonally is $2gg_1 + 2ff_1 = c + c_1$.

External and Internal Contacts of Circles

If two circles with centres $C_1(x_1, y_1)$ and $C_2(x_2, y_2)$ and radii r_1 and r_2 respectively, touch each other externally, then $C_1C_2 = r_1 + r_2$. Coordinates of the point of contact are

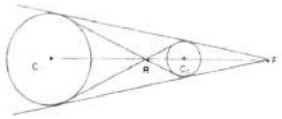


$A \equiv \left(\frac{r_1x_2 + r_2x_1}{r_1 + r_2}, \frac{r_1y_2 + r_2y_1}{r_1 + r_2} \right)$. The circles touch each other internally if $C_1C_2 = r_1 - r_2$.



Coordinates of the point of contact are $T \equiv \left(\frac{r_1x_2 - r_2x_1}{r_1 - r_2}, \frac{r_1y_2 - r_2y_1}{r_1 - r_2} \right)$.

Common Tangents to Two Circles



(a) The direct common tangents to two circles meet on the line of centres and divide it externally in the ratio of the radii.

(b) The transverse common tangents also meet on the line of centres and divide it internally in the ratio of the radii.

Notes:

- When two circles do not intersect or touch, 4 common tangents can be drawn.
- When two circles touch each other externally, 3 common tangents can be drawn to the circles.
- When two circles intersect each other, two common tangents can drawn to the circles.
- When two circles touch each other internally 1 common tangent can be drawn to the circles.

ILLUSTRATIVE PROBLEMS

1. Let circle C_1: $x^2 + (y-4)^2 = 12$ intersects circle C_2: $(x-3)^2 + y^2 = 13$ at A and B. A quadrilateral ACBD is formed by tangents at A and B to both circles. Then find the diameter of circum circle of quadrilateral ACBD

Sol. Both circles cut each other orthogonally. C and D will be centres of two circles also CD will be diameter of circumcircle of quadrilateral ACBD.

Diameter $= \sqrt{3^2 + 4^2} = 5$.

2. A circle touches a rectangle ABCD of side lengths 2a and 2b at M and N on sides AB and AD respectively, it also passes through the point C. If perpendicular distance of the line MN from points C is 6 cm. Then find the value of ab

Sol. Let radius of circle = r,

AB = 2a, AD = 2b, BM = x, ND = y

From ΔNDC

$CN^2 = ND^2 + CD^2$

$CN^2 = y^2 + (r+x)^2$

$CN^2 = y^2 + r^2 + x^2 + 2rx = r^2 + 2rx + x^2 + y^2$

$= r^2 + 2rx + r^2$

Since, ΔOTC, $x^2 + y^2 = r^2$

$CN^2 = 2r(r+x) = 2r \cdot 2a = 4ar$ ..... (1)

ΔMBC, $CM^2 = BM^2 + BC^2$

$= x^2 + (y+r)^2 = x^2 + y^2 + r^2 + 2ry$

$CM^2 = r^2 + r^2 + 2ry = 2r(r+y) = 2r \cdot 2b = 4br$

$CM^2 = 4br$ ..... (2)

$$\frac{1}{2} \cdot MN \cdot CP = \frac{1}{2} CM \cdot CN \cdot \sin 45^\circ = \frac{CM \cdot CN}{2\sqrt{2}}$$

$$\frac{1}{2} \cdot r\sqrt{2} \cdot 6 = \frac{CM \cdot CN}{2\sqrt{2}}$$

$CM \cdot CN = 12r$

$CM^2 \cdot CN^2 = 144r^2$

$4br \cdot 4ar = 144r^2$

$$ab = \frac{144}{16} = 9$$

$ab = 9$

3. Let r_1 and r_2 represent the radii of the smallest and largest circles passing through $\left(a, \frac{1}{a}\right)$ and touching the circle $4x^2 + 4y^2 - 4x - 8y - 31 = 0$, where a is the least possible value of ratio of the sides of the regular n sided ($n \geq 3$) polygons inscribed and circumscribed to a unit circle, then find the value of $r_1 + r_2$

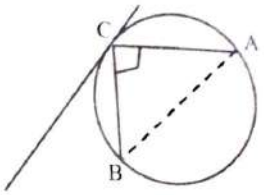
Sol. We have ratio of n sided regular polygon inside and outside unit circle $= \dfrac{2\sin\frac{\pi}{n}}{2\tan\frac{\pi}{n}} = \cos\dfrac{\pi}{n}$; which is least

when $n = 3$ (as $n \geq 3$) $\Rightarrow a = \dfrac{1}{2}$

Now $\left(\frac{1}{2}, 2\right)$ lies inside the circle with centre $\left(\frac{1}{2}, 1\right)$ and radius 3. Sum of radii of smallest and largest circle is R

Also, $2(r_1 + r_2) = 2R \Rightarrow r_1 + r_2 = 3$

4. A and B are two points in xy–plane, which are $2\sqrt{2}$ unit distance apart and subtend an angle of 90° at the point C (1, 2) on the line $x - y + 1 = 0$ which is larger than any angle subtended by the line segment AB at any other point on the line. Find the equation(s) of the circle through the points A, B and C.



Sol. AB subtends the greatest angle at C, so, the line $x - y + 1 = 0$ touches the circle at C and hence AB is the diameter.

Let the equation of the desired circle be

$(x - 1)^2 + (y - 2)^2 + \lambda (x - y + 1) = 0.$

Its radius $= \sqrt{\left(\frac{\lambda - 2}{2}\right)^2 + \left(\frac{\lambda + 4}{2}\right)^2 - (5 + \lambda)} = \left(\sqrt{2}\right)^2 \Rightarrow \lambda = \pm 2.$

$\therefore$ The equations of circle are $x^2 + y^2 - 6y + 7 = 0$ and $x^2 + y^2 - 4x - 2y + 3 = 0.$

5. If C_1, C_2 and C_3 belong to a family of circles through the points (x_1, y_1) and (x_2, y_2), prove that the ratio of the lengths of the tangent from any point on C_1 to the circles C_2 and C_3 is constant.

Sol. Equations of the circles through (x_1, y_1) and (x_2, y_2) are

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + \lambda_r \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0 \ (r = 1, 2, 3)$$

Let (h, k) be a point on C_1

$\Rightarrow \phi(h, k) + \lambda_1 \Psi(h, k) = 0$

where $\phi(h, k) = (h - x_1)(h - x_2) + (k - y_1)(1 - y_2)$ and $\psi(h, k) = \begin{vmatrix} h & k & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix}$.

Let T_2 be the length of the tangent from (h, k) to C_2 and T_3 be the length of the tangent from (h, k) to C_3.

$\Rightarrow T_2 = \sqrt{\phi(h,k) + \lambda_2 \Psi(h,k)}$, $T_3 = \sqrt{\phi(h,k) + \lambda_3 \Psi(h,k)}$

$$\Rightarrow \frac{T_2}{T_3} = \frac{\sqrt{\phi(h,k) + \lambda_2 \Psi(h,k)}}{\sqrt{\phi(h,k) + \lambda_3 \Psi(h,k)}} = \frac{\sqrt{(\lambda_2 - \lambda_1)\Psi(h,k)}}{\sqrt{(\lambda_3 - \lambda_1)\Psi(h,k)}} = \sqrt{\frac{\lambda_2 - \lambda_1}{\lambda_3 - \lambda_1}}$$ which is independent of the choice of (h, k) and hence a constant.

SOLVED PROBLEMS

1. The maximum and minimum value of $(p^2 + q^2 - 2p + 6q + 9)^{1/2}$ are I_1 and I_2 respectively if $p^2 + q^2 + 13 \geq 8p + 2q \ \forall p,q \in R$ then $\frac{I_1}{I_2}$ is equal to

(A) $\frac{2}{\sqrt{3}}$

(B) $\sqrt{\frac{3}{5}}$

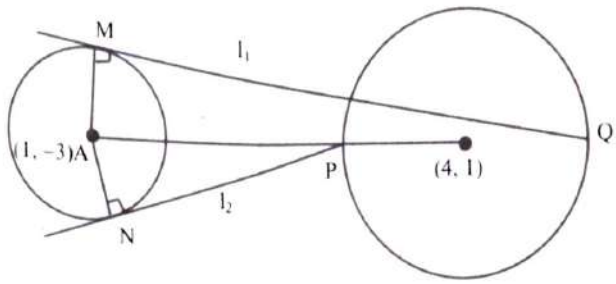
(C) $\sqrt{6}$

(D) $\frac{1}{\sqrt{2}}$

Sol. **(C)**

Let MQ be I_1 and NP be I_2

$I_1 = 4\sqrt{3}, I_2 = 2\sqrt{2}$.



2. Three parallel chords of a circle have lengths 2, 3, 4 and subtend angles α, β, $\alpha + \beta$ at the centre respectively (given $\alpha + \beta < \pi$), then $\cos\alpha$ is equal to

(A) $\frac{15}{31}$

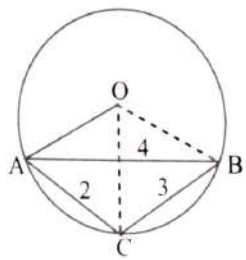
(B) $\frac{17}{35}$

(C) $\frac{17}{32}$

(D) none of these

Sol. **(C)**

Using the property that equal chords subtends equal angles at centre of circle, then problem can be converted to the diagram in adjoining figure.



$AB = 4, AC = 2, BC = 3$

$\angle ABC = \alpha/2$

$\cos(\alpha/2) = \frac{9+16-4}{2\times3\times4} = \frac{7}{8} \Rightarrow \cos\alpha = 2\cos^2(\alpha/2) - 1$

$= 2\times\frac{49}{64} - 1 = \frac{98-64}{64} \Rightarrow \cos\alpha = \frac{34}{64} = \frac{17}{32}$

3. If the circle $(x - a)^2 + y^2 = 25$ intersects the circle $x^2 + (y - b)^2 = 16$ in such a way that common chord is of maximum length, then value of $a^2 + b^2$ is

(A) 8

(B) 10

(C) 9

(D) 41

Sol. **(C).** Radical axis: $-2ax + a^2 + 2by - b^2 = 9$

which passes through (0, b)

i.e. $0 + a^2 + 2b^2 - b^2 = 9 \Rightarrow a^2 + b^2 = 9$.

4. The locus of the centre of a circle touching the circle $x^2 + y^2 - 4y - 2x = (2\sqrt{3} - 1)$ internally and tangents on which from (1, 2) is making a 60° angle with each other, will be

(A) $(x + 2)^2 + (y - 1)^2 = 7$

(B) $(x - 2)^2 + (y + 3)^2 = 5$

(C) $(x - 1)^2 + (y - 2)^2 = 3$

(D) none of these

Sol. **(C)**

Let r and R be radius of required and given circle respectively and let centre is (h, k).

By given condition $\sqrt{(h-1)^2+(k-2)^2} = R - r$

$\frac{r}{AB} = \tan 30^\circ$

$r = \frac{R-r}{\sqrt{3}} \Rightarrow AB = R - r$

$\sqrt{(h-1)^2+(k-2)^2} = R - \frac{R}{1+\sqrt{3}} = R\left(\frac{\sqrt{3}}{1+\sqrt{3}}\right)$

$R = 1+\sqrt{3}$

$\Rightarrow \sqrt{(h-1)^2+(k-2)^2} = \sqrt{3}$

so locus is $(x-1)^2 + (y-2)^2 = 3$.

5. Consider two circles $x^2 + y^2 = 1$ and $(x-1)^2 + (y-1)^2 = 3$ then the point circle which is orthogonal to the given two circles can be

(A) $\left(x-\frac{1}{2}\right)^2+\left(y-\frac{1}{2}\right)^2=0$

(B) $\left(x-\frac{1}{\sqrt{2}}\right)^2+\left(y-\frac{1}{\sqrt{2}}\right)^2=0$

(C) $(x-1)^2 + (y-1)^2 = 0$

(D) none of these

Sol. **(B)**

$x^2 + y^2 - 2x - 2y - 1 = 0$

$x^2 + y^2 - 1 = 0$

Common chord is $-2x - 2y = 0$

$\Rightarrow 2x + 2y = 0$

$y = -x$

Points of intersection of circles are $\left(\pm\frac{1}{\sqrt{2}}, \mp\frac{1}{\sqrt{2}}\right)$

6. A circle with radius 'a' and centre on y-axis slides along y-axis and a variable line through (a, 0) cuts the circle at points P and Q, then the point of intersection of tangents to the circle at P and Q will lie in the region defined by

(A) $y^2 \geq 4a(x-a)$

(B) $y^2 \leq 4ax$

(C) $x^2 + y^2 \leq 4a^2$

(D) $x^2 - y^2 \geq a^2$

Sol. **(A)**

Let the circle be $x^2 + (y-\alpha)^2 = a^2$. Let the point of intersection of tangents at P and Q be (h, k). Then equation of PQ, is $hx + (k-\alpha)(y-\alpha) - a^2 = 0$.

As it passes through $(a,0)$, so, $ha - \alpha(k-\alpha) - a^2 = 0$

$\Rightarrow \alpha^2 - k\alpha + a(h-a) = 0$. $D \geq 0 \Rightarrow k^2 - 4a(h-a) \geq 0$

i.e. $y^2 \geq 4a(x-a)$

7. Let BC be the chord of contact of the tangents from a point A to the circle $x^2 + y^2 = 1$. P is any point on the arc BC. Let PX, PY, PZ be the lengths of perpendiculars from P on the AB, BC and CA respectively then PX, PY, PZ are in

(A) A.P.

(B) G.P.

(C) H.P.

(D) none of these

Sol. (B) Let A be $(0, \alpha) \Rightarrow$ equation of BC is $\alpha y = 1$

$\Rightarrow$ B and C are $\left(\frac{-\sqrt{\alpha^2-1}}{\alpha}, \frac{1}{\alpha}\right)$ and $\left(\frac{\sqrt{\alpha^2-1}}{\alpha}, \frac{1}{\alpha}\right)$

$\Rightarrow$ Equation of AC and BC are $\frac{-\sqrt{\alpha^2-1}}{\alpha}x + \frac{y}{\alpha} = 1$ and $\frac{\sqrt{\alpha^2-1}}{\alpha}x + \frac{y}{\alpha} = 1$

and $PZ = \left|\cos\theta\frac{\sqrt{\alpha^2-1}}{\alpha} + \frac{\sin\theta}{\alpha} - 1\right|$ where $P \equiv (\sin\theta, \cos\theta)$

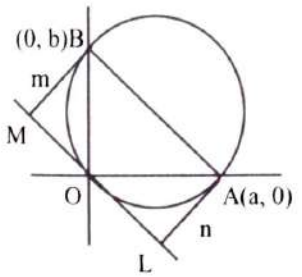
$\Rightarrow PY^2 = PX \cdot PZ$

8. A line meets the co-ordinate axes in A and B. A circle is circumscribed about ΔOAB. If the distances from A and B of the tangent to the circle at the origin be m and n, then diameter of the circle is

(A) $m(m + n)$ (B) $m + n$

(C) $n(m + n)$ (D) $m^2 + n^2$

Sol. (B) Circumcentre of $\Delta OAB \equiv \left(\frac{a}{2}, \frac{b}{2}\right)$



circumradius $= \sqrt{\frac{a^2}{4} + \frac{b^2}{4}}$

Equation of circle

$x^2 + y^2 - ax - by = 0$

equation of tangent at origin $ax + by = 0$

$AL = \left|\frac{a^2}{\sqrt{a^2+b^2}}\right|$, $BM = \left|\frac{b^2}{\sqrt{a^2+b^2}}\right|$

$AL + BM = m + n = \sqrt{a^2+b^2}$ = diameter of circle.

9. Two circles with radii 'r_1' and 'r_2', $r_1 > r_2 \geq 2$, touch each other externally. If 'θ' be the angle between the direct common tangents, then

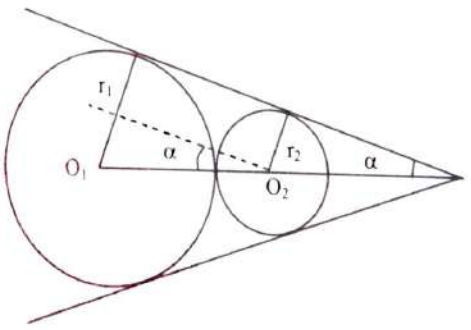
(A) $\theta = \sin^{-1}\left(\frac{r_1 + r_2}{r_1 - r_2}\right)$ (B) $\theta = 2\sin^{-1}\left(\frac{r_1 - r_2}{r_1 + r_2}\right)$

(C) $\theta = \sin^{-1}\left(\frac{r_1 - r_2}{r_1 + r_2}\right)$ (D) none of these

Sol. **(B)**

$\sin\alpha = \frac{r_1 - r_2}{r_1 + r_2}$

$\Rightarrow \theta = 2\sin^{-1}\left(\frac{r_1 - r_2}{r_1 + r_2}\right)$



10. Suppose $ax + by + c = 0$, where a, b, c are in A.P. be normal to a family of circles. The equation of the circle of the family which intersects the circle $x^2 + y^2 - 4x - 4y - 1 = 0$ orthogonally is

(A) $x^2 + y^2 - 2x + 4y - 3 = 0$ (B) $x^2 + y^2 + 2x - 4y - 3 = 0$

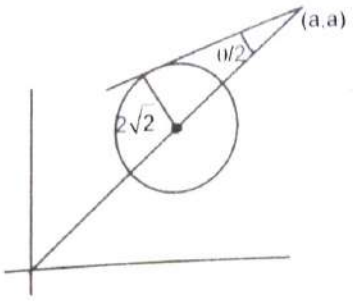
(C) $x^2 + y^2 - 2x + 4y - 5 = 0$ (D) $x^2 + y^2 - 2x - 4y + 3 = 0$

Sol. **(A).** a, b, c are in A.P., so $ax + by + c = 0$ represents a family of lines passing through the point $(1, -2)$. So, the family of circles (concentric) will be given by $x^2 + y^2 - 2x + 4y + c = 0$.
And $2(-1 \times -2 + 2 \times -2) = -1 + c \Rightarrow c = -3$.

11. The exhaustive range of values of a such that the angle between the pair of tangents drawn from (a, a) to the circle $x^2 + y^2 - 2x - 2y - 6 = 0$ lies in the range $\left(\frac{\pi}{3}, \pi\right)$, is

(A) $(1, \infty)$
(B) $(-5, -3) \cup (3, 5)$
(C) $(-\infty, -2\sqrt{2}) \cup (2\sqrt{2}, \infty)$
(D) $(-3, -1) \cup (3, 5)$

Sol. **(D).** The centre of the circle is (1, 1) and radius $2\sqrt{2}$.
Point (a, a) must lie outside the circle, so
$2a^2 - 4a - 6 > 0 \Rightarrow a < -1$ or $a > 3$.



Now, $\tan\frac{\theta}{2} = \frac{2\sqrt{2}}{\sqrt{2a^2 - 4a - 6}}$.

As $\frac{\pi}{3} < \theta < \pi \Rightarrow \frac{\pi}{6} < \frac{\theta}{2} < \frac{\pi}{2} \therefore \frac{2\sqrt{2}}{\sqrt{2a^2 - 4a - 6}} > \frac{1}{\sqrt{3}}$

$\Rightarrow \sqrt{a^2 - 2a - 3} < 2\sqrt{3} \therefore a^2 - 2a - 15 < 0 \Rightarrow -3 < a < 5 \therefore a \in (-3, -1) \cup (3, 5)$.

12. ABCD is a square side 1. P and Q lie on the side AB and R lies on the side CD. The possible values for the circumradius of triangle PQR is?

(A) 0.5
(B) 0.6
(C) 0.7
(D) 0.8

Sol. **(A, B, C)**
Let O be the circumcentre. Then $OP + OR \geq PR \geq AD = 1$, so the radius is at least 1/2. P, Q, R always lie inside or on the circle through A, B, C, D which has radius $1/\sqrt{2}$, so the radius is at most $1/\sqrt{2}$.

13. For any value of $\lambda \in R \sim \{-1\}$, the equation $S_1 + \lambda S_2 = 0$ will always represent
(The coefficients of one of the similar second degree terms in S_1 and S_2 are taken to be same)
(A) a pair of straight lines if each of $S_1 = 0$ and $S_2 = 0$ represents a pair of straight lines
(B) a circle if each of $S_1 = 0$ and $S_2 = 0$ represents a circle
(C) a parabola if each of $S_1 = 0$ and $S_2 = 0$ represents a parabola
(D) a rectangular hyperbola if each of $S_1 = 0$ and $S_2 = 0$ represents a rectangular hyperbola

Sol. **(B, D)**
Coefficient of x^2 + Coefficient of y^2 in $S_1 + \lambda S_2 = 0$ (in case of (B) or (D)) is zero $\forall \lambda$.
Remaining cases also fail at their conditions.

Comprehension

Let $S = 0$ be the equation of the locus of point P in the plane of a square ABCD such that max $(PA, PC) = \frac{1}{\sqrt{2}}(PB + PD)$.

14. The ratio of area bounded by the $S = 0$ and the given square will be

(A) $\frac{\pi}{4}$
(B) $\frac{1}{2}$
(C) $\frac{3}{2}$
(D) $\frac{\pi}{2}$

Sol. **(D)**

15. The ratio of radii of circumcircle of square ABCD and director circle of $S = 0$ is

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{4}$

(C) $\frac{1}{2}$ (D) none of these

Sol. **(D)**

16. The ratio of perimeter of $S = 0$ and the given square is

(A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{\sqrt{3}}$

(C) $\frac{1}{7}$ (D) $\frac{\pi}{2\sqrt{2}}$

Sol. **(D)**

14.-16. Let P(h, k) be a point with the given property and assume $h > 0$.

Now, as per figure $PC > PA$.

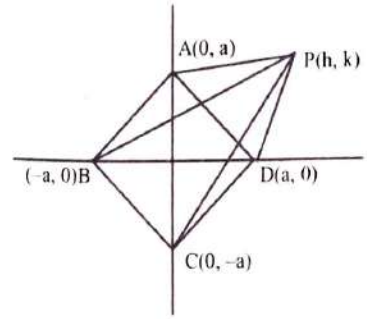
So, $\sqrt{2}\,PC = PB + PD$

$\Rightarrow 2h^2 + 2(k+a)^2 = (h-a)^2 + (h+a)^2 + 2k^2 + 2\sqrt{(h+a)^2 + k^2}$

$\Rightarrow$ locus comes out to be $x^2 + y^2 = a^2$

So, circum-circle of square ABCD will be locus of point P.

Hence results follow.



Assertion Reason

17. Let C_1 be the circle with centre $O_1(0, 0)$ and radius 1 and C_2 be the circle with centre $O_2(t, t^2 + 1)$ $(t \in R)$ and radius 2.

Statement 1: Circles C_1 and C_2 always have atleast one common tangent for any value of t.

because

Statement 2: For the two circles $O_1O_2 \geq |r_1 - r_2|$ where r_1 and r_2 are their radii for any value of t.

(A) Statement -1 is True, Statement -2 is True; Statement-2 is a correct explanation for Statement-1

(B) Statement -1 is True, Statement -2 is True; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement -1 is True, Statement -2 is False

(D) Statement -1 is False, Statement -2 is True

Sol. **(A).** Here $(O_1O_2)^2 = t^2 + (t^2+1)^2 = t^4 + 3t^2 + 1 \geq 1$

$\Rightarrow O_1O_2 \geq 1$ and $|r_1 - r_2| = 1$

$\Rightarrow O_1O_2 \geq |r_1 - r_2|$ hence the two circles have atleast one common tangent.

18. Statement-1: Two circles of radii $\sqrt{3}$ & 1 cut orthogonally then area of common region is $\frac{5\pi - 6\sqrt{3}}{6}$ square unit.

Statement-2: Area $(A \cup B)$ = Area(A) + Area(B) – Area$(A \cap B)$, where A&B represents two regions.

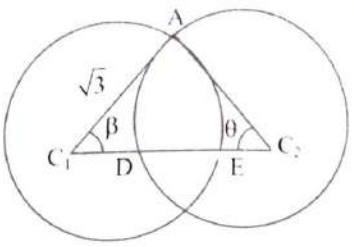
(A) Statement-1 is True, Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True, Statement-2 is not a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**



$$\text{Area}(AC_1C_2) = \frac{1}{2} \times \sqrt{3} \times 1 = \frac{\sqrt{3}}{2};\ \tan\theta = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3} \text{ so } \beta = \frac{\pi}{6}$$

$$\text{Area of arc segment } ADC_2 = \frac{\theta r^2}{2} = \frac{\frac{\pi}{3}}{2} = \frac{\pi}{6} = A_1$$

$$\text{Similarly are of arc segment } C_1EA = \frac{\pi}{6} \times \frac{3}{2} = \frac{\pi}{4} = A_2$$

$$\text{So area of common region} = A_1 + A_2 - \frac{\sqrt{3}}{2} = \frac{\pi}{6} + \frac{\pi}{4} - \frac{\sqrt{3}}{2} = \frac{5\pi - 6\sqrt{3}}{12}$$

$$\text{So total area of common region} = = 2\left(\frac{5\pi - 6\sqrt{3}}{12}\right) = \left(\frac{5\pi - 6\sqrt{3}}{6}\right).$$

19. Match the following Column–I with Column–II

Column – I		Column – II	
(A)	A circle passes through vertex D of the square ABCD and is tangent to the sides AB and BC. If AB = 1, the radius of the circle can be expressed as $p - q\sqrt{2}$, then (p + q) has the value equal to	(p)	2
(B)	A circle inscribed in triangle ABC is tangent to AB at P and its radius is 21. Given AP = 23 and PB = 27 the perimeter of the triangle is P, then $\frac{2P}{345}$ is	(q)	6
(C)	Locus of the point of intersection of the pair of perpendicular tangents to the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 7$ is the director circle of the circle with radius r, then 3r is	(r)	5
(D)	Suppose the origin (0, 5) and (a, b) are on a circle whose diameter is along the y-axis. Let L be the line that passes through the origin and (a, b). If $a^2 + b^2 = 16$ and $a > 0$, then 3 times the slope of line L is	(s)	3
		(t)	4

Sol. **(A) → (s); (B) → (p); (C) → (q); (D) → (t)**

Answer Q.20, Q.21 and Q.22 by appropriately matching the information given in the three columns of the following table.

Two fixed point A and B are 4 units apart and are on the same side of a moving line L. If perpendicular distances of A and B say P_1 and P_2 from the line L are such that $P_1 + 3P_2 = K$, K being a constant, then line L always touches a fixed circle C

Column–I	Column–II	Column–III
(I) The centre of the circle C lies on line segment joining AB	(i) if K = 4 radius of circle is r = 2	(P) if A(0, 0) and B(4, 0), then the centre of circle is (3, 0)
(II) The centre of circle C lies on one of A or B	(ii) if K = 4 radius of circle is r = 1	(Q) if A(0, 0) and B(4, 0) and K = 4 is of circle is $x^2 + y^2 - 6x + 9 = 0$
(III) If A and B are (–2, 0) and (2, 0) respectively, then the centre of the circle C is (1, 0)	(iii) if K = 8 radius of circle is r = 4	(R) if A(–2, 0) and B(2, 0) and K = 8, then origin lies inside the circle
(IV) If A and B are (–2, 0) and (2, 0) respectively, K = 4 equation circle is $x^2 + y^2 - 2x + 1 = 0$	(iv) if K = 8 radius of circle is r = 1	(S) if A(–2, 0) and B(2, 0) and K = 4, then origin lies inside the circle

20. Which of the following combination is **CORRECT**?
(A) (I) (ii) (R)
(B) (III) (ii) (R)
(C) (I) (ii) (P)
(D) (I) (i) (Q)

Sol. **(C)**

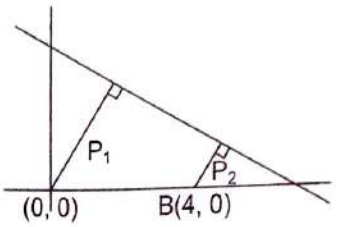
21. Which of the following combination is **CORRECT**?
(A) (II) (iii) (S)
(B) (III) (i) (P)
(C) (II) (iii) (R)
(D) (III) (iii) (Q)

Sol. **(D)**

22. Which of the following combination is **INCORRECT**?
(A) (I) (i) (S)
(B) (IV) (i) (S)
(C) (II) (ii) (S)
(D) (II) (ii) (P)

Sol. **(B)**

20.-22. Let A(0, 0) and B(4, 0) and line $ax + by = 1$



$$P_1 = \left|\frac{1}{\sqrt{a^2+b^2}}\right|$$

$$\Rightarrow P_2 = \left|\frac{4a-1}{\sqrt{a^2+b^2}}\right|$$

$P_1 + 3P_2 = k$

$$\left|\frac{1}{\sqrt{a^2+b^2}}\right| + 3\left|\frac{4a-1}{\sqrt{a^2+b^2}}\right| = k$$

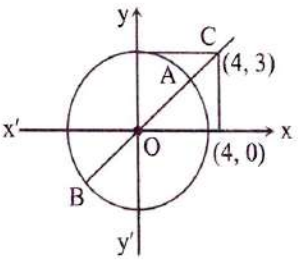
$$\Rightarrow \left|\frac{3a-1}{\sqrt{a^2+b^2}}\right| = \frac{k}{4}$$

Since (0, 0) and (4, 0) must give the same sign with respectively to line i.e. negative
So, $4a - 1 < 0$
Centre of fixed circle is (3, 0) which lies on same segment AB
If $k = 4$, $r = 1$
If A(−2, 0) and B(2, 0) $P_1 + 3P_2 = k$

$\Rightarrow \frac{|a-1|}{\sqrt{a^2+b^2}} = \frac{k}{4}$. Centre of circle is (1, 0) at $k = 4$, $r = 1$

23. Two circles whose equations are $x^2 + y^2 = 9$ and $x^2 + y^2 - 8x - 6y + m^2 = 0$, $m \in I$ having exactly two common tangents. Find the number of values of n.

Sol. **(9.00)**
For two common tangents circles must be intersecting



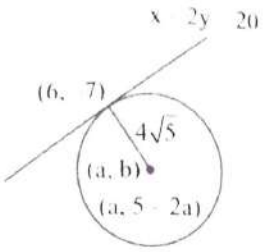
$AC < \sqrt{16+9-m^2} < BC$

$2 < \sqrt{25-m^2} < 8$

$\Rightarrow m = 0, \pm1, \pm2, \pm3, \pm4$

24. Consider the circles with radii $4\sqrt{5}$ and which are tangent to the line $x - 2y = 20$ at the point $(6, -7)$ the difference of the x-coordinates of the centers of the circle is _____

Sol. **(8.00)**



$$\frac{b+7}{a-6}\cdot\frac{1}{2} = -1$$

$\Rightarrow b = 5 - 2a$

Also, $(6-a)^2 + (12-2a) = 80$

$a^2 - 12a + 20 = 0$

$\Rightarrow a_1 + a_2 = 12,\ a_1a_2 = 20$

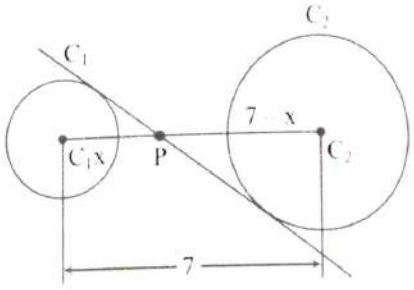
$$|a_1 - a_2| = \sqrt{(a_1+a_2)^2 - 4a_1a_2} = \sqrt{144-80} = \sqrt{64} = 8$$

25. Circle C_1 has radius 2 and circle C_2 has radius 3 and the distance between the centres of C_1 and C_2 is 7 if two lines, one tangent to both circle and the other passing through the centre of both circle, intersect at point P which lies between the centres of C_1 and C_2, then the distance between P and the centre of C_1 is _____

Sol. **(2.80)**

$3x = 14 - 2x$

$$x = \frac{14}{5}$$



26. The axes are translated so that the new equation of the circle $x^2 + y^2 - 5x + 2y - 5 = 0$ has no first degree terms, then the new equation of circle whose radius is _____

Sol. **(3.50)**

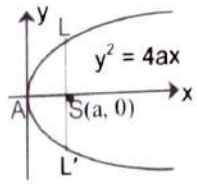
The new equation is $x^2 + y^2 = \frac{49}{4}$

PARABOLA 10

Standard equation of a parabola is $y^2 = 4ax$

LATUS RECTUM

The chord of a parabola through the focus and perpendicular to the axis is called the latus rectum.



In the figure LSL' is the latus rectum.

Also $LSL' = 2\left(\sqrt{4a.a}\right) = 4a$ = double ordinate through the focus S.

Notes:

- Any chord of the parabola $y^2 = 4ax$ perpendicular to the axis of the parabola is called double ordinate.
- Two parabolas are said to be equal when their latus recta are equal.

Parametric Coordinates

Any point on the parabola $y^2 = 4ax$ is $(at^2, 2at)$ and we refer to it as the point 't'. Here, t is a parameter, i.e., it varies from point to point.

Focal Chord

Any chord to $y^2 = 4ax$ which passes through the focus is called a focal chord of the parabola $y^2 = 4ax$.
The extremities of a focal chord of the parabola $y^2 = 4ax$ may be taken as the points t and -1/t.

TANGENT TO A PARABOLA

Tangent at the Point (x_1, y_1)

Let the equation of the parabola be $y^2 = 4ax$. Hence, value of $\frac{dy}{dx}$ at $P(x_1, y_1)$ is $\frac{2a}{y_1}$ and the equation of the tangent at P is $y - y_1 = \frac{2a}{y_1}(x - x_1)$ i.e. $yy_1 = 2a(x - x_1) + y_1^2 \Rightarrow yy_1 = 2a(x + x_1)$.

Tangent in Terms of m

Suppose that the equation of a tangent to the parabola $y^2 = 4ax$ ....(i) is $y = mx + c$ .... (ii)
The abscissae of the points of intersection of (i) and (ii) are given by the equation $(mx + c)^2 = 4ax$. But the condition that the straight line (ii) should touch the parabola is that it should meet the parabola in coincident points $\Rightarrow (mc - 2a)^2 = m^2c^2$ .... (iii).
$\Rightarrow$ **$c = a/m$.** Hence, $y = mx + a/m$ is a tangent to the parabola $y^2 = 4ax$, $\forall\, m \in R \sim \{0\}$.

Equation $(mx + c)^2 = 4ax$ now becomes $(mx - a/m)^2 = 0 \Rightarrow x = \frac{a}{m^2}$ and $y^2 = 4ax$

$\Rightarrow y = \frac{2a}{m}$. Thus the point of contact of the tangent $y = mx + a/m$ is $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$.

Tangent at the Point t

Equation of the tangent at 't' is $ty = x + at^2$.
The point of intersection of the tangents at 't_1' and 't_2' is $(at_1t_2, a(t_1 + t_2))$.

Equation of the Tangents from an External Point

Let $y^2 = 4ax$ be the equation of a parabola and (x_1, y_1) an external point P. Then, equation of the tangents is given by $SS_1 = T^2$, where $S = y^2 - 4ax$, $S_1 = y_1^2 - 4ax_1$, $T = yy_1 - 2a(x + x_1)$

CHORD OF CONTACT

Equation to the chord of contact of the tangents drawn from a point (x_1, y_1), to the parabola $y^2 = 4ax$ is $T = 0$, i.e. $yy_1 - 2a(x+x_1) = 0$.

Chord whose Mid Point is given

The equation of the chord of the parabola $y^2 = 4ax$ with mid point (x_1, y_1) is $T = S_1$
i.e. $yy_1 - 2a(x+x_1) = y_1^2 - 4ax_1$ Or $yy_1 - 2ax = y_1^2 - 2ax_1$.

NORMAL TO THE PARABOLA

Normal at the Point (x_1, y_1)

The equation of the tangent at the point (x_1, y_1) is $yy_1 = 2a(x + x_1)$. Since the slope of tangent $2a/y_1$, slope of normal is $-y_1/2a$. Also it passes through (x_1, y_1).

Hence, its equation is $y - y_1 = -\frac{y_1}{2a}(x - x_1)$ ... (i)

Normal in Terms of m

In equation (i), put $-\frac{y_1}{2a} = m$ so that $y_1 = -2am$ and $x_1 = \frac{y_1^2}{4a} = am^2$, then the equation becomes

$y = mx - 2am - am^3$ ... (ii)

where m is a parameter. Equation (ii) is the normal at the point $(am^2, -2am)$ of the parabola.

Normal at the Point t

The normal, being perpendicular to the tangent at $(at^2, 2at)$ is given by $y = -tx + 2at + at^3$.

- If normal at the point t_1 meets the parabola again at the point t_2, then $t_2 = -t_1 - \frac{2}{t_1}$.
- Point of intersection of the normals to the parabola $y^2 = 4ax$ at $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ is $(2a + a(t_1^2 + t_2^2 + t_1t_2), -at_1t_2(t_1 + t_2))$.

ILLUSTRATIVE PROBLEMS

1. Find the radius of circle touching parabola $y^2 = x$ at (1, 1) and having directrix of $y^2 = x$ as its normal.

Sol. Equation of normal at P

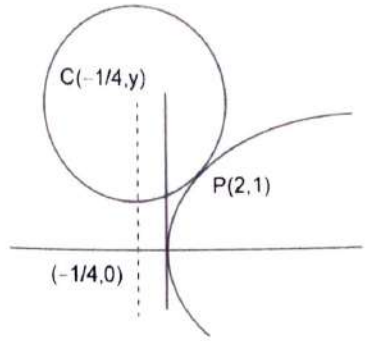
$y - 1 = -2(x - 1)$

$y + 2x = 3$

at $x = -\frac{1}{4}$

$y = 3 + \frac{1}{2} = \frac{7}{2}$.

$\Rightarrow$ radius $= \sqrt{\left(1+\frac{1}{4}\right)^2 + \left(1-\frac{7}{2}\right)^2} = \sqrt{\frac{25}{16} + \frac{25}{4}} = \frac{5\sqrt{5}}{4}$.



2. A moving variable parabola, having it's axis parallel to x-axis always touches the given equal parabola $y^2 = 4x$, the locus of the vertex of the moving parabola.

Sol. Let the moving parabola be $(y-k)^2 = -4(x-h)$ (Here we are taking –ve sign as the parabolas has to open in opposite directions). Moving parabola touches the given parabola $y^2 = 4x$,

hence $(y-k)^2 = -y^2 + 4h$ should have equal roots

$\Rightarrow 2y^2 - 2yk + (k^2 - 4h) = 0$ should have equal roots.

$\Rightarrow 4k^2 - 8(k^2 - 4h) = 0$

$\Rightarrow k^2 = 8h$

thus the required locus is $y^2 = 8x$.

3. Chords of the parabola $y^2 = 4x$ touch the hyperbola $x^2 - y^2 = 1$. Then find the locus of the point of intersection of tangents of the parabola drawn at the extremities of such chords.

Sol. Any tangent to the given hyperbola is

$x \sec\theta - y \tan\theta = 1$. ... (1)

Let (h, k) be the point of intersection of tangents to the parabola.

Then chord of contact of (h, k) is

$yk = 2(x + h)$ ... (2)

(1) and (2) are identical

$\therefore \frac{\sec\theta}{2} = \frac{\tan\theta}{k} = \frac{-1}{2h}$

$\Rightarrow \sec\theta = -\frac{1}{h}$

and $\tan\theta = -\frac{k}{2h}$

$\therefore$ Required locus is $4x^2 + y^2 = 4$.

4. Find the condition for all the three normals drawn from a given point to the parabola $y^2 = 4ax$ to be real and distinct.

Sol. Equation of the normal at the point '-m' i.e., $(am^2, -2am)$ is $y = mx - 2am - am^3$. If this normal passes through a point (h, k),

then $k = mh - 2am - am^3$. Since this is a cubic equation in m, it has three real roots (all distinct, or two equal) or one real and two imaginary roots. Hence, in general, three normals can be drawn from a given point (h, k).

Note that $f(x) \equiv ax^3 + bx^2 + cx + d = 0$, will have three distinct real roots if and only if $f'(x) = 0$ has two distinct real roots, say α and β with $f(\alpha)\, f(\beta) < 0$.

We have $f(m) \equiv am^3 + (2a-h)m + k = 0$.

$f'(m) = 3am^2 + (2a-h)$. $f'(m) = 0$, will have two distinct real roots if

$h - 2a > 0$ (as a is assumed positive) ... (1)

Here $\alpha = \sqrt{\dfrac{h-2a}{3a}}$

and $\beta = -\sqrt{\dfrac{h-2a}{3a}}$

$\Rightarrow f(\alpha)\, f(\beta) = f(\alpha).f(-\alpha)$

$= [a\alpha^3 + (2a-h)\alpha + k]\,[k - (a\alpha^3 + (2a-h)\alpha)]$

$= k^2 - \alpha^2 (a\alpha^2 + (2a-h))^2$

$= k^2 - \dfrac{h-2a}{3a}\left[\dfrac{h-2a}{3} - (h-2a)\right]^2$

$= k^2 - \dfrac{h-2a}{3a}\left[\dfrac{2}{3}(2a-h)\right]^2$

$= k^2 - \dfrac{4}{27a}(h-2a)^3 < 0$ (for three distinct real roots)

$\Rightarrow 27ak^2 < 4(h-2a)^3$ ... (2)

If $k = 0$, i.e. point (h, k) lies on the x-axis (the axis of the parabola), then condition (2) is automatically satisfied if (1) is satisfied i.e., $h > 2a$.

5. Tangent and normal are drawn at the endpoints of focal chords of parabola $y^2 = 4ax$ to form a rectangle. Then locus of circum-centre of rectangle is

Sol. Centre is midpoint of ends of focal chord

$\Rightarrow 2h = a(t_1^2 + t_2^2)$ ..... (1)

and $2k = 2a(t_1 + t_2)$ ..... (2)

with $t_1 t_2 = -1$

Eliminating t_1 and t_2 required locus is $2ax = y^2 + 2a^2$

SOLVED PROBLEMS

1. ABC is a right angle triangle (right angled at B) inscribed in the parabola $y^2 = 4x$. The minimum length of the intercept cut off by the tangents at A and C to the parabola from y-axis is
(A) 8
(B) 4
(C) 2
(D) none of these

Sol. **(B).** Let $A(t_1)$, $B(t_2)$ and $C(t_3)$ be the points on the parabola

since $AB \perp BC \Rightarrow \dfrac{4}{(t_1+t_2)(t_2+t_3)} = -1$

$t_2^2 + t_2(t_1+t_3) + t_1t_3 + 4 = 0$

$\Rightarrow (t_1+t_3)^2 \geq 4(t_1t_3+4) \Rightarrow (t_1-t_3)^2 \geq 16$

also length of intercept cut off by tangents from y-axis is $|t_1 - t| \geq 4$.

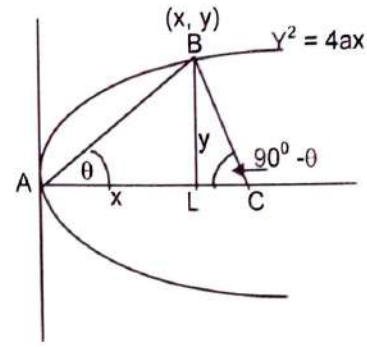
2. AB is a chord of the parabola $y^2 = 4ax$ with vertex A. BC is drawn perpendicular to AB meeting the axis at C. The projection of BC on the axis of the parabola is
(A) a
(B) 2a
(C) 4a
(D) 8a

Sol. **(C).** $\tan\theta = \dfrac{y}{x}$

Projection of BC on the x-axis

$= LC = \dfrac{y}{\tan(90^0 - \theta)} = y\tan\theta$

$= \dfrac{y^2}{x} = 4a.$



3. A circle drawn on any focal chord of the parabola $y^2 = 4ax$ as diameter cuts the parabola at two points 't' and 't'' (other than the extremity of focal chord), then
(A) $tt' = -1$
(B) $tt' = 2$
(C) $tt' = 3$
(D) none of these

Sol. **(C).** Equation of circle (t_1, t_2 are extremity of focal chord).
$(x - at_1^2)(x - at_2^2) + (y - 2at_1)(y - 2at_2) = 0$
$x^2 + y^2 - xa(t_1^2 + t_2^2) + a^2t_1^2t_2^2 - 2ay(t_1 + t_2) + 4a^2t_1t_2 = 0$
$x^2 + y^2 - ax(t_1^2 + t_2^2) - 2ay(t_1 + t_2) - 3a^2 = 0$
Let $(at^2, 2at)$ be any point on it
$a^2t^4 + 4a^2t^2 - a^2t^2(t_1^2 + t_2) - 2a.\ 2at(t_1 + t_2) - 3a^2 = 0$
this has root t_1, t_2, t_3, t_4
$t_1t_2t_3t_4 = -3$
$\Rightarrow t_3t_4 = 3 \quad (\text{as } t_1t_2 = -1)$
$\Rightarrow tt' = 3.$

4. Locus of the point $\left(\sqrt{3h}, \sqrt{3k+2}\right)$ if it lies on the line $x - y - 1 = 0$ is a
(A) Straight line
(B) Circle
(C) Parabola
(D) None of these

Sol. **(C).** $\left(\sqrt{3h}\right)^2 = \left(\sqrt{3k+2}+1\right)^2$

$3h = 3k + 2 + 1 + 2\sqrt{3k+2}$

$3^2(h - k - 1)^2 = 2^2\left(\sqrt{3k+2}\right)^2$

$\Rightarrow 9(h^2 + k^2 + 1 - 2hk - 2h - 2k) = 4(3k + 2)$

$\Rightarrow 9(h^2 + k^2 + 1 - 2hk - 2h - 2k) - 12k - 8 = 0$

$\Rightarrow 9(h^2 + k^2) - 18hk - 18h - 30k + 1 = 0$

$h^2 = ab \qquad \Delta \neq 0.$

$\therefore$ Locus is a parabola.

5. Let there be two parabolas $y^2 = 4ax$ and $y^2 = 4bx$ (where $a \neq b$ and $a, b > 0$). Then the locus of the middle points of the intercepts between the parabolas made on the lines parallel to the common axis is

(A) $y^2 = \dfrac{8ab}{b-a}x$ (B) $x^2 = \dfrac{8ab}{b-a}y$

(C) $y^2 = \dfrac{4ab}{a-b}x$ (D) none of these

Sol. **(A).** Let line parallel to common axis be $y = h$. Then coordinates of A and B are $\left(\dfrac{h^2}{4a}, h\right)$ and $\left(\dfrac{h^2}{4b}, h\right)$ respectively.

If $P \equiv (\alpha, \beta)$,

then $\alpha = \dfrac{1}{2}\left(\dfrac{h^2}{4a} + \dfrac{h^2}{4b}\right)$, $\beta = h$

$\Rightarrow 2\alpha = \dfrac{h^2}{4}\left(\dfrac{1}{a} + \dfrac{1}{b}\right)$

$\therefore$ Locus of P is $2x = \dfrac{y^2}{4}\left(\dfrac{1}{a} + \dfrac{1}{b}\right)$.

6. If three distinct normals can be drawn to the parabola $y^2 - 2y = 4x - 9$ from the point $(2a, 0)$ then range of values of a is

(A) no real values possible (B) $(2, \infty)$

(C) $(-\infty, 2)$ (D) none of these

Sol. **(B).** Given parabola is $(y-1)^2 = 4(x-2)$

Equation of any normal to the given parabola is

$y - 1 = m(x - 2) - m - m^3$

$\Rightarrow y = mx - 4m - m^3 + 1.$

Since this passes through $(2a, 0)$, $m^3 + 2m(2 - a) - 1 = 0$ will have three distinct and real values of m.

Also $3m^2 + 2(2 - a) = 0$ will have all real and distinct root

$\Rightarrow a > 2.$

7. A tangent and a normal is drawn at the point $P \equiv (16, 16)$ of the parabola $y^2 = 16x$ which cut the axis of the parabola at the points A and B respectively. If the centre of the circle through P, A and B is C then the angle between PC and the axis of x is

(A) $\tan^{-1}\dfrac{1}{2}$ (B) $\tan^{-1} 2$

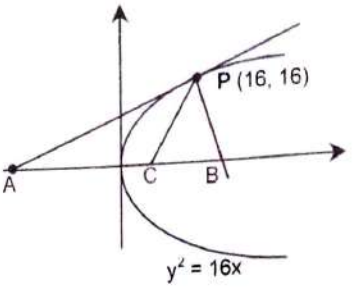
(C) $\tan^{-1}\dfrac{3}{4}$ (D) $\tan^{-1}\dfrac{4}{3}$

Sol. **(D).** By property centre of circle coincides with focus of parabola

$\Rightarrow C \equiv (4, 0)$

$\tan\alpha$ = slope of $PC = \frac{16}{12}$

$\Rightarrow \alpha = \tan^{-1}\left(\frac{4}{3}\right)$



8. A water jet form a fountain reaches its maximum height of 4m at a distance 0.5 m from the vertical passing through the point O of water outlet. The height of the jet above the horizontal OX at a distance of 0.75 m from the point O is

(A) 5 m
(B) 6 m
(C) 3 m
(D) 7 m

Sol. **(C).** The path of the water jet a parabola.

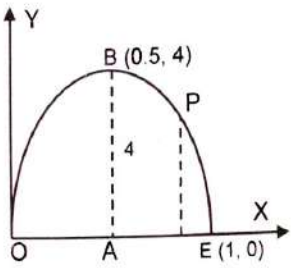
Let its equation be $y = ax^2 + bx + c$

It should pass through (0, 0), (0.5, 4), (1, 0)

$\Rightarrow c = 0, a = -16, b = 16$

$\Rightarrow y = -16x^2 + 16x$

If $x = 0.75$ we get $y = 3$.



9. Equation of a normal to the curve $y = x^2 - 6x + 6$ which is perpendicular to the straight line joining the origin to the vertex of the parabola is

(A) $4x - 4y - 11 = 0$
(B) $4x - 4y + 1 = 0$
(C) $4x - 4y - 21 = 0$
(D) $4x - 4y + 21 = 0$

Sol. **(C).** Here $y = (x-3)^2 - 3 \Rightarrow y + 3 = (x-3)^2$

Hence the vertex is (3, –3)

Slope of the line joining origin (0, 0) and vertex (3, –3) is –1.

Hence the slope of the normal is 1.

$\frac{dy}{dx} = 2x - 6$ and $-\frac{dx}{dy} = 1$

$\Rightarrow -1 = \frac{1}{2x-6}$

$\Rightarrow 2x - 6 = -1 \Rightarrow x = \frac{5}{2}$

and $y = \left(\frac{5}{2}\right)^2 - 6\times\left(\frac{5}{2}\right) + 6 = \frac{25}{4} - 15 + 6 = \frac{25-60+24}{4} = -\frac{11}{4}$

Hence the normal $y + \frac{11}{4} = 1\left(x - \frac{5}{2}\right) \Rightarrow 4x - 4y - 21 = 0.$

10. If two different tangents of $y^2 = 4x$ are the normals to $x^2 = 4by$ then

(A) $|b| > \frac{1}{2\sqrt{2}}$
(B) $|b| < \frac{1}{2\sqrt{2}}$
(C) $|b| > \frac{1}{\sqrt{2}}$
(D) $|b| < \frac{1}{\sqrt{2}}$

Sol. **(B).** Tangent to $y^2 = 4x$ in terms of 'm' is $y = mx + \frac{1}{m}$ and normal to $x^2 = 4by$ in terms of 'm' is

$$y = mx + 2b + \frac{b}{m^2}$$

If these are same lines then $\frac{1}{m} = 2b + \frac{b}{m^2}$

$\Rightarrow 2bm^2 - m + b = 0.$

For two different tangents $1 - 8b^2 > 0$

$\Rightarrow |b| < \frac{1}{\sqrt{8}} \Rightarrow |b| < \frac{1}{2\sqrt{2}}.$

11. Two parabolas with the same axis, focus of each being exterior to the other and the latus rectum being 4a and 4b. The locus of the middle points of the intercepts between the parabolas made on the lines parallel to the common axis is a

(A) straight line if $a = b$

(B) parallel if $a \neq b$

(C) parabola if all $a \neq b$

(D) ellipse if $b > a$

Sol. **(A, B).** The equation of parabola can be taken as $y^2 = 4a(x - k)$ & $y^2 = -4b(x + k)$

A line parallel to the common axis is $y = h$

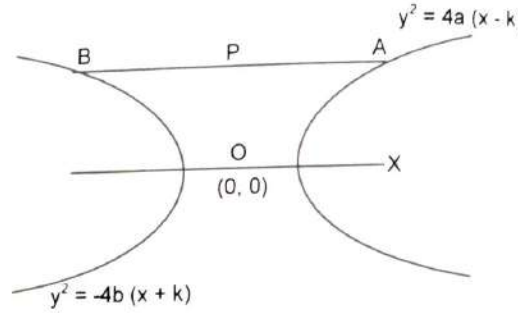
Then $A = \left(\frac{h^2}{4a} + k,\ h\right)$ and

$B = \left(-k - \frac{h^2}{4b},\ h\right)$

$\therefore$ If $P = (\alpha, \beta)$ then

$\alpha = \frac{1}{2}\left(\frac{h^2}{4a} + k - k - \frac{h^2}{4b}\right), \beta = h$

$\therefore 2\alpha = \frac{h^2}{4}\left(\frac{1}{a} - \frac{1}{b}\right)$, locus of P is $2x = \frac{y^2}{4} \cdot \frac{(b - a)}{ab}$.



12. The set of points on the axis of the parabola $2((x-1)^2 + (y-1)^2) = (x + y)^2$, from which 3 distinct normals can be drawn to the parabola, is the set of points (h, k) lying on the axis of the parabola such that

(A) $h > 3$

(B) $h > 3/2$

(C) $k > 3/2$

(D) $k > 3$

Sol. **(B), (C)**

The directrix of the parabola is $x + y = 0$ and focus is (1, 1).

The length of perpendicular from (1, 1) to the directrix is $\sqrt{2}$ $\Rightarrow$ focal length is $\frac{1}{\sqrt{2}}$

$\Rightarrow$ the point on the axis which is at a distance equal to focal length from the focus towards the right is (3/2, 3/2). All the points to the right of this point, lying on the axis will satisfy the required condition.

13. Tangents are drawn from (–2, 0) to $y^2 = 8x$, radius of circle(s) that would touch these tangents and the corresponding chord of contact, can be equal to,

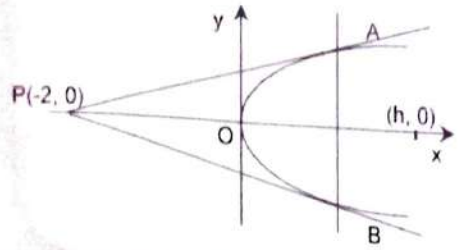
(A) $4\left(\sqrt{2} + 1\right)$

(B) $4\left(\sqrt{2} - 1\right)$

(C) $8\sqrt{2}$

(D) None of these.

Sol. **(A), (B)**



Point 'P' clearly lies on the directrix of $y^2 = 8x$.
Thus slope of PA and PB are 1 and –1 respectively.
Equation of PA : $y = x + 2$,
Equation of PB : $y = -x - 2$, equation of AB : $x = 2$.
Let the centre of the circle be (h, 0) and radius be 'r'

$$\Rightarrow \frac{|h+2|}{\sqrt{2}} = \frac{|h-2|}{1} = r$$

$$\Rightarrow h^2 + 4 + 4h = 2(h^2 + 4 - 4h)$$

$$\Rightarrow h^2 - 12h + 4 = 0$$

$$h = \frac{12 \pm 8\sqrt{2}}{2} = 6 \pm 4\sqrt{2}$$

$$\Rightarrow |h-2| = 4(\sqrt{2} - 1), 4(\sqrt{2} + 1).$$

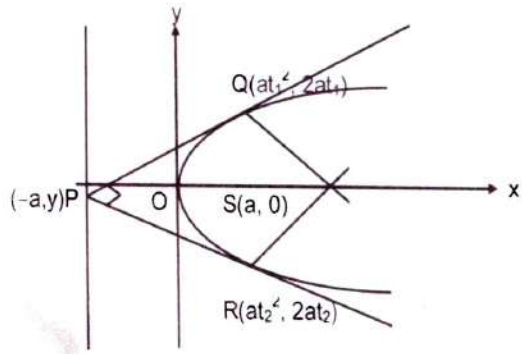
Comprehension

From a point P (h, k), in general, three normals can be drawn to the parabola $y^2 = 4ax$. If t_1, t_2, t_3 are the parameters associated with the feet of these normals, then t_1, t_2, t_3 are the roots of the equation $at^3 + (2a - h)t - k = 0$. Moreover, from the line $x = -a$, two perpendicular tangents can be drawn to the parabola.

14. If the tangents at the feet $Q\left(at_1^2, 2at_1\right)$ and $R\left(at_2^2, 2at_2\right)$ to the parabola meet on the line $x = -a$, then t_1, t_2 are the roots of the equation

(A) $t^2 - t_3 t + 1 = 0$ (B) $t^2 + t_3 t + 1 = 0$
(C) $t^2 - t_3 t - 1 = 0$ (D) $t^2 + t_3 t - 1 = 0$

Sol. **(D).** $-a = at_1t_2$
$\Rightarrow t_1t_2 = -1$
$y = a(t_1 + t_2)$

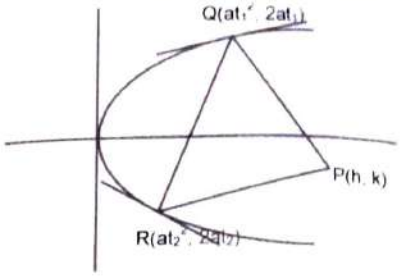


15. If the feet $Q\left(at_1^2, 2at_1\right)$ and $R\left(at_2^2, 2at_2\right)$ are the ends of a focal chord of the parabola, then the locus of P (h, k) is

(A) $y^2 = a(x - 2a)$ (B) $y^2 = a(x - a)$
(C) $y^2 = a(x - 3a)$ (D) $y^2 = 3a(x - a)$

Sol. **(C).** Slope of $PQ = \dfrac{k - 2at_1}{h - at_1^2}$

Slope of $RP = \dfrac{k - 2at_2}{h - at_2^2}$.



16. If P (h, k) is a vertex of the square comprising normals to the parabola from P and tangents from the directrix, then (h, k) is the same as

(A) (a, 0) (B) (2a, 0)

(C) (3a, 0) (D) (4a, 0)

Sol **(C).**

Assertion Reason

17. Statement-1: The locus of mid points of the chords of the parabola $y^2 = 4ax$ which subtends a right angle at the vertex is $y^2 = 2a(x - 4a)$.

because

Statement-2: Chord PQ joining points t_1 and t_2 subtends a right angle at the vertex of a parabola if $t_1t_2 = -2$.

(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement -1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement -1 is True, Statement -2 is False

(D) Statement -1 is False, Statement -2 is True

Sol. **(C).** Statement 1 is true but statement 2 is false because $t_1t_2 = -4$.

18. The chord AC of the parabola $y^2 = 4x$ subtends 90° angle at points B and D on the parabola. If A, B, C and D are represented by t_1, t_2, t_3 and t_4, then match the following

	Column – I		Column – II
(A)	$\left\|\dfrac{t_2 + t_4}{t_1 + t_3}\right\|$	(p)	0
(B)	$\|t_2t_4 - t_1t_3\|$	(q)	1
(C)	minimum value of $\|t_1 - t_3\|$	(r)	2
(D)	The y-coordinate of the mid point of the points of intersection of the tangents at A, C and B, D is	(s)	4
		(t)	1/2

Sol. **(A) → (q); (B) → (s); (C) → (s); (D) → (p)**

We have $(t_2 + t_3)(t_2 + t_1) = -4$

$\Rightarrow t_2^2 + t_2(t_1 + t_3) + t_1t_3 + 4 = 0$ ... (1)

$\Rightarrow (t_1 + t_3)^2 - 4(t_1t_3 + 4) \geq 0$

$\Rightarrow (t_1 - t_3)^2 \geq 16$

$\Rightarrow t_2 + t_4 = -(t_1 + t_3)$ and $t_2t_4 = t_1t_3 + 4$

(Because t_2 and t_4 are the roots of the equation (1)) also let y_1 and y_2 are y-coordinates of intersection points of the tangents at (A, C) and (B, D) then $y_1 = t_1 + t_3$ and $y_2 = t_2 + t_4$

$\Rightarrow y_1 + y_2 = 0$.

Answer Q.19, Q.20 and Q.21 by appropriately matching the information given in the three columns of the following table.

Column-I and Column-III: describe region in xy plane while Column-II: give the value of the area

Column–I	Column–II	Column–III
(I) The area enclosed by parabola $y^2 = 4x$ and $x^2 = 8y$	(i) 27	(P) area enclose by $y = 2x - x^3$ and $y + 3 = 0$
(II) Area cut off from $4y = 3x^2$ by $2y = 3x + 12$	(ii) 50	(Q) area enclosed by $(y-2)^2 = 8(x-3)$ and $(x-3)^2 = 2(y-2)$
(III) Area enclosed by $y^2 = 4(x-4)$, ordinate $x = 4$, and line $y = 4$	(iii) $\frac{32}{3}$	(R) area enclosed by $y^2 = 9x$ and $x^2 = 9y$
(IV) Area enclosed by latus rectum and $y^2 = 4[x]x$, $\sqrt{5} \le x < 3$ (where [.] denotes the greatest integer function)	(iv) $\frac{16}{3}$	(S) area enclosed by $y^2 = 16x$, $y^2 = -32y$ and $y = 3$

19. Which of the following combination is **CORRECT**?
(A) (I) (ii) (P)
(B) (I) (iii) (R)
(C) (I) (iv) (S)
(D) (I) (iii) (P)

Sol. **(D)**

20. Which of the following combination is **CORRECT**?
(A) (II) (i) (R)
(B) (II) (i) (P)
(C) (II) (iii) (P)
(D) (I) (i) (P)

Sol. **(A)**

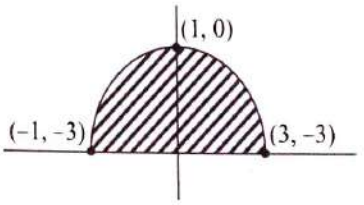
21. Which of the following combination is **CORRECT**?
(A) (III) (i) (R)
(B) (III) (iv) (Q)
(C) (IV) (iii) (R)
(D) (II) (i) (S)

Sol. **(B)**

19.-21. (I) Area enclosed by $y^2 = 4ax$ and $x^2 = 4by$ is $\frac{16ab}{3}$

Area enclosed by $y^2 = 4x$ and $x^2 = 8y$ is $\frac{32}{3}$

Now, area enclosed by $y = 2x - x^2$ and $y + 3 = 0$



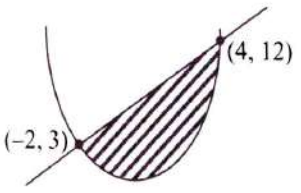
$$2\int_1^3 \left(\left(2x - x^2\right) - (-3)\right)dx = \frac{32}{3}$$

(II) $4y = 3x^2$

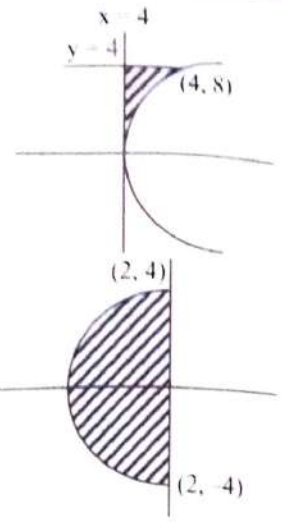
$\Rightarrow 2y = 3x + 12$

$6x + 24 = 3x^2$

Requried area $\int_{-2}^{4}\left(\frac{3x+12}{2} - \frac{3}{4}x^2\right)dx = 27$



(III) Required area $= \frac{16}{3}$



(IV) $y^2 = 4[x]x$

$\sqrt{5} \le x < 3$

$y^2 = 8x$

Required area $= \frac{32}{3}$

22. Two parabola C and D intersect at two different points, where C is $y = x^2 - 3$ and D is $y = kx^2$, the intersection at which the x value is positive is designated point A, and $x = a$ at this intersection the tangent line ℓ at A to the curve D intersects curve C at point B, other than A. If x value of point B is 1, then a is equal to _____

Sol. **(3.00)**

$C: y = x^2 - 3$, $D: y = kx^2$

Solving C and D, then $kx^2 = x^2 - 3$ ..... (1)

$x^2 = \frac{3}{1-k}$; $y = \frac{3k}{1-k}$;

$A \equiv \left(\sqrt{\frac{3}{1-k}}, \frac{3k}{1-k}\right)$

$a = \sqrt{\frac{3}{1-k}}$;

$A \equiv (a, ka^2) \equiv (a, a^2 - 3)$ tangent at curved at A is ..... (2)

from equation (1) and (2), we get

$\therefore B \equiv (1, -2)$

From equation (2), we get

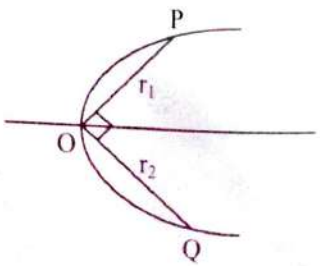
$a^3 - a = 2a^2 - 6$; $(a - 1)(a - 3)(a + 2) = 0$

$\therefore a = 3$ (as $a \neq 1, -2$)

23. Let P and Q be the points on the parabola $y^2 = 4x$ with vertex O, such that OP is perpendicular to OQ and have lengths r_1 and r_2 respectively, then the value of $\frac{r_1^{4/3} \cdot r_2^{4/3}}{r_1^{2/3} + r_2^{2/3}}$ is equal to _____

Sol. **(4.00)**

We know that $\frac{r_1^{4/3} \cdot r_2^{4/3}}{r_1^{2/3} + r_2^{2/3}} = (4a)^2$



24. Normal chord to a parabola $y^2 = 4ax$ at the point whose ordinate is equal to abscissa subtends angle θ at focus then the value of $[3 \sin\theta + 4 \cos\theta]$ is k, then $\frac{k}{2}$ is equal to _____ (where [.] denotes the greatest integer function)

Sol. **(1.50)**

Normal chord through the point whose abscissa and ordinates an equal substaints right angle at focus

25. The number of real common normal between $y^2 = \frac{4}{\sqrt{3}}x$ and $\frac{x^2}{9} + \frac{y^2}{3} = 1$ is k, then $\frac{k}{2}$ is equal to _____

Sol. **(1.50)**

Equation of normal of parabola is

$$y = mx - \frac{2}{\sqrt{3}}m - \frac{1}{\sqrt{3}}m^3 \quad \ldots\ldots (1)$$

Equation of normal of ellipse is

$$y = mx - \frac{6m}{\sqrt{9+3m^2}} \quad \ldots\ldots (2)$$

Equation (1) and (2) are identical

$$\Rightarrow \frac{2}{\sqrt{3}}m + \frac{1}{\sqrt{3}}m^3 = \frac{6m}{\sqrt{9+3m^2}}$$

$$\Rightarrow m = 0, \pm 1$$

ELLIPSE

Standard Equation of a ellipse is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, where $b^2 = a^2(1 - e^2)$, and e is the eccentricity of the ellipse.

LATUS RECTUM

The chord through a focus at right angle to the major axis is called the latus rectum.

Notes:

- The major axis AA′ is of length 2a and the minor axis BB′ is of length 2b.
- The foci are (−ae, 0) and (ae, 0)
- The equations of directrices are $x = a/e$ and $x = -a/e$.
- The length of the semi latus rectum $= b^2 / a$
- Circle is a particular case of an ellipse with $e = 0$.

Focal Distance of a Point

Sum of the focal distances of any point on the ellipse is equal to its major axis.

Position of a Point Relative to an Ellipse

The point $P(x_1, y_1)$ is outside or inside or on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, according as the quantity $S_1 \equiv \left(\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}-1\right)$ is positive or negative or zero.

Parametric Equation of an Ellipse

Clearly, $x = a\cos\theta$, $y = b\sin\theta$ satisfy the equation $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ for all real values of θ.

Hence $(a\cos\theta, b\sin\theta)$ is always a point on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ for any values of θ.

The point $(a\cos\theta, b\sin\theta)$ is also called the point θ. The angle θ is called the eccentric angle $(0 \le \theta < 2\pi)$ of the point $P(a\cos\theta, b\sin\theta)$ on the ellipse.

TANGENTS AND NORMALS

Equation of the Tangent at a Point of an Ellipse

(a) At the point (x_1, y_1) is $\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$ i.e. $T = 0$

(b) Equation of tangent at the point θ i.e. $(a\cos\theta, b\sin\theta)$ is obtained by putting $x_1 = a\cos\theta$, $y_1 = b\sin\theta$

$\Rightarrow \frac{x\cos\theta}{a}+\frac{y\sin\theta}{b}=1$

Equation of the Tangent in Terms of its Slope; Condition of Tangency

The equations of the tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with slope m are

$y = mx \pm \sqrt{a^2m^2 + b^2}$ for all finite values of m.

Notes:

- Equation of the pair of tangents from an external point (x_1, y_1):

  Let the equation of the ellipse be $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. Then the pair of tangents from (x_1, y_1) to the ellipse is given by $SS_1 = T^2$.
- Equation of the chord with mid point (x_1, y_1) : $T = S_1$.

Equation of the Normal at a Point of an Ellipse

(a) The equation of normal to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at (x_1, y_1) is $\frac{x-x_1}{x_1/a^2}=\frac{y-y_1}{y_1/b^2}$.

(b) Equation of normal at $(a\cos\theta, b\sin\theta)$ is $ax\sec\theta - by\operatorname{cosec}\theta = a^2 - b^2$.

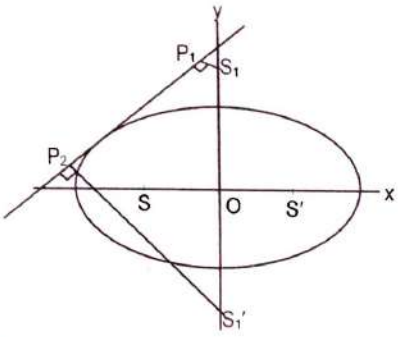
Equation of the Chord Joining the Points (a cos α, b sin α), (a cos β, b sin β)

$$\frac{x}{a}\cos\left(\frac{\alpha+\beta}{2}\right)+\frac{y}{b}\sin\left(\frac{\alpha+\beta}{2}\right)=\cos\left(\frac{\alpha-\beta}{2}\right).$$

ILLUSTRATIVE PROBLEMS

1. If two points are taken on the minor axis of an ellipse at the same distance from the centre as the foci, prove that the sum of the squares of the perpendiculars from these points on any tangent to the ellipse is constant.

Sol. Let the equation of the ellipse be $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.



Then, the distance of a focus from the centre

$= ae = a\sqrt{1-b^2/a^2} = \sqrt{a^2-b^2}$

so that the two points on minor axis are

$S_1\left(0, \sqrt{a^2-b^2}\right)$ and $S_1'\left(0, -\sqrt{a^2-b^2}\right)$.

Now any tangent to the ellipse is

$y = mx + \sqrt{a^2m^2 + b^2}$, where m is a parameter.

The sum of the squares of the perpendiculars on this tangent from the two points S_1 and S_1' is

$$\left(\frac{\sqrt{a^2-b^2}-\sqrt{a^2m^2+b^2}}{\sqrt{1+m^2}}\right)^2+\left(\frac{-\sqrt{a^2-b^2}-\sqrt{a^2m^2+b^2}}{\sqrt{1+m^2}}\right)^2$$

$= 2(a^2 - b^2 + a^2m^2 + b^2) / (1 + m^2) = 2a^2$(constant)

2. Prove that in general four normals can be drawn to an ellipse from any point, and the sum of eccentric angles of the co-normal points is equal to an odd multiple of two right angles.

Sol. Let the ellipse be $x^2/a^2 + y^2/b^2 = 1$ .....(1)

The normal at any point $(a\cos\theta, b\sin\theta)$ on (1) is $ax\sec\theta - by\operatorname{cosec}\theta = a^2 - b^2$

If the normal passes through a fixed point (x_1, y_1) then $ax_1\sec\theta - by_1\operatorname{cosec}\theta = a^2 - b^2$

or $ax_1\left(\frac{1+t^2}{1-t^2}\right) - by_1\left(\frac{1+t^2}{2t}\right) = a^2 - b^2$, where $t = \tan\frac{1}{2}\theta$

or $2ax_1(1+t^2)t - by_1(1+t^2)(1-t^2) = 2(a^2-b^2)t(1-t^2)$

or $by_1t^4 + 2(ax_1 + a^2 - b^2)t^3 + 2(ax_1 - a^2 + b^2)t - by_1 = 0$ .....(2)

This being an equation of 4th degree in t, will give in general four values of t i.e., of $\tan\frac{1}{2}\theta$ i.e., of θ

Corresponding to each value of θ we shall get one point (as $0 \le \theta < 2\pi \Rightarrow 0 \le \theta/2 < \pi$

$\Rightarrow \tan\theta/2$ will be distinct) on the ellipse, normal at which will pass through the fixed point (x_1, y_1).

Hence from any fixed point (x_1, y_1) four normals can be drawn to the ellipse $x^2/a^2 + y^2/b^2 = 1$. The four points on the ellipse, the normals at which pass through the fixed point (x_1, y_1), are called conormal points.

Let the eccentric angles of these conormal points be $\theta_1, \theta_2, \theta_3$ and θ_4 and the corresponding values of t be t_1, t_2, t_3 and t_4 respectively. Then t_1, t_2, t_3 and t_4 are the roots of the equation (2).

Hence $s_1 = \Sigma t_1 = -2(ax_1 + a^2 - b^2)/by_1$, $s_2 = \Sigma t_1t_2 = 0$,

$s_3 = \Sigma t_1t_2t_3 = -2(ax_1 - a^2 + b^2)/by_1$ and $s_4 = t_1t_2t_3t_4 = -by_1/by_1 = -1$

$\Rightarrow \tan\frac{1}{2}(\theta_1 + \theta_2 + \theta_3 + \theta_4) = \frac{s_1 - s_3}{1 - s_2 + s_4} = \frac{s_1 - s_3}{1 - 0 - 1}$.

$= \infty$ or $\frac{1}{2}(\theta_1 + \theta_2 + \theta_3 + \theta_4) = n\pi + \frac{1}{2}\pi$ or $\theta_1 + \theta_2 + \theta_3 + \theta_4 = 2n\pi + \pi$

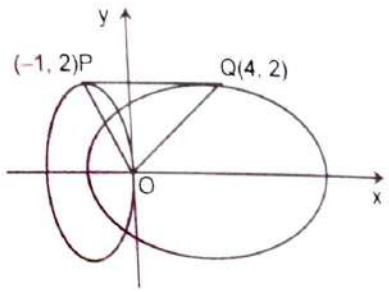
$= (2n+1)\pi$ = odd multiple of two right angles.

3. Angle subtended by common tangents of two ellipses $4(x-4)^2 + 25y^2 = 100$ and $4(x+1)^2 + y^2 = 4$ at origin is

Sol. $\frac{(x-4)^2}{25} + \frac{y^2}{4} = 1$ and $(x+1)^2 + \frac{y^2}{4} = 1$.

Slope of OP × slope of OQ $= \frac{2}{-1} \times \frac{2}{4} = -1$.

$\therefore$ OP and OQ are perpendicular to each other.



4. If the ellipse $\frac{x^2}{a^2-7} + \frac{y^2}{13-5a} = 1$ is inscribed in a square of side length $\sqrt{2}$ a then a is equal to

Sol. Since sides of square are tangent and perpendicular to each other so vertices lie on director circle

$\Rightarrow x^2 + y^2 = (a^2 - 7) + (13 - 5a)$

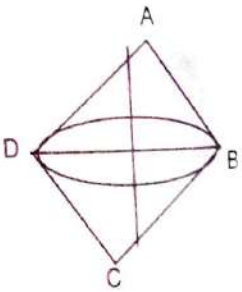
since $\sqrt{2}$ a is a side of a square.

$\Rightarrow (a^2 - 7) + (13 - 5a) = a^2$

$\Rightarrow a = \frac{6}{5}$

but for an ellipse to exist $a^2 - 7 > 0$ and $13 - 5a > 0$

$\Rightarrow a \in (-\infty, -\sqrt{7}) \cup (13/5, \sqrt{7})$



$a \neq \frac{6}{5}$.

Hence no such 'a' exists.

5. If PQR be an equilateral triangle inscribed in the auxiliary circle of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ $(a > b)$ and P'Q'R' be corresponding triangle inscribed within the ellipse then centroid of the triangle P'Q'R' lies at

Sol. Let $P(\theta)$, $Q\left(\theta+\frac{2\pi}{3}\right)$, $R\left(\theta+\frac{4\pi}{3}\right)$

then $P' \equiv (a\cos\theta, b\sin\theta)$,

$Q' \equiv \left(a\cos\left(\theta+\frac{2\pi}{3}\right), b\sin\left(\theta+\frac{2\pi}{3}\right)\right)$

$R' \left(a\cos\left(\theta+\frac{4\pi}{3}\right), b\sin\left(\theta+\frac{4\pi}{3}\right)\right)$

Let centroid of $\Delta P'Q'R' \equiv (x', y')$

$$x' = a\left[\frac{\cos\theta+\cos\left(\theta+\frac{2\pi}{3}\right)+\cos\left(\theta+\frac{4\pi}{3}\right)}{3}\right]$$

$$= \frac{a}{3}\left[\cos\theta+2\cos(\theta+\pi)\cos\frac{\pi}{3}\right] = 0$$

$$y' \equiv \frac{a}{3}\left[\sin\theta+\sin\left(\theta+\frac{2\pi}{3}\right)+\sin\left(\theta+\frac{4\pi}{3}\right)\right]$$

$$= \frac{a}{3}\left[\sin\theta+2\sin(\theta+\pi)\cos\frac{\pi}{3}\right] = 0.$$

SOLVED PROBLEMS

1. The number of points outside the ellipse on its major axis from which a normal (other than the x-axis) can be drawn to the ellipse is
(A) 0 (B) 2
(C) 4 (D) none of these

Sol. **(A).** Equation of any normal is $ax\sec\theta - by\operatorname{cosec}\theta = a^2 - b^2$.
If the normal intersects x-axis at $(\alpha, 0)$

$\Rightarrow \alpha = \dfrac{a^2 - b^2}{a\sec\theta} = \dfrac{ae}{\sec\theta} \Rightarrow |\alpha| < |a|$

$\Rightarrow (\alpha, 0)$ lies inside the ellipse.

2. Consider the ellipse $\dfrac{x^2}{3} + \dfrac{y^2}{1} = 1$. Let P, Q, R S be four points on this ellipse such that the normals drawn from these points are concurrent at (2, 2) then the centre of the conic (apart from the given ellipse) on which these 4 points lie is
(A) P, Q, R, S lie only on the given ellipse and on no other conic (C) (3, 1)
(B) (3, – 1)
(D) (0, 0)

Sol. **(B).** The equation of the normal at (x_1, y_1) on the ellipse $\dfrac{a^2x}{x_1} - \dfrac{b^2y}{y_1} = a^2 - b^2$.

It passes through (h, k)

$\dfrac{a^2h}{x_1} - \dfrac{b^2k}{y_1} = a^2 - b^2$ i.e. the four co-normal points on the ellipse satisfy $\dfrac{a^2h}{x} - \dfrac{b^2k}{y} = a^2 - b^2$

$xy(a^2 - b^2) - (a^2h)y + b^2kx = 0$

$$\left(x - \frac{a^2h}{a^2 - b^2}\right)\left(y + \frac{b^2k}{a^2 - b^2}\right) + \frac{a^2b^2hk}{(a^2 - b^2)^2} = 0$$

which is a rectangular hyperbola whose centre is $\left(\dfrac{a^2h}{a^2 - b^2}, \dfrac{-b^2k}{a^2 - b^2}\right)$.

Here (h, k) ≡ (2, 2) from equation of ellipse $\dfrac{x^2}{3} + \dfrac{y^2}{1} = 1$

$a^2 = 3, b^2 = 1$

$\therefore$ centre ≡ (3, – 1).

3. From the point A(4, 3) tangents are drawn to the ellipse $\dfrac{x^2}{16} + \dfrac{y^2}{9} = 1$ to touch the ellipse at B and C. EF is a tangent to the ellipse parallel to the line BC and towards the point A. The distance of A from EF is equal to

(A) $\dfrac{4\sqrt{18} - 12}{5}$ (B) $\dfrac{24 - 4\sqrt{18}}{5}$

(C) $\dfrac{12 - 2\sqrt{18}}{5}$ (D) none of these

Sol. **(B).** The equation of the chord of contact is $\dfrac{x}{4} + \dfrac{y}{3} = 1 \Rightarrow$ slope of line EF is $-\dfrac{3}{4}$.

$\Rightarrow$ EF is $y = -\dfrac{3}{4}x + \sqrt{18} \Rightarrow$ distance of (4, 3) from EF is $\dfrac{|24 - 4\sqrt{18}|}{5}$.

4. The length of the major axis of the ellipse $(5x-10)^2+(5y+15)^2=\frac{(3x-4y+7)^2}{4}$ is

(A) 10
(B) 20/3
(C) 20/7
(D) 4

Sol. **(B).** $(x-2)^2+(y+3)^2=\left(\frac{1}{2}\ \frac{3x-4y+7}{5}\right)^2$ is an ellipse, whose focus is (2, -3), directrix $3x-4y+7=0$ and eccentricity is $\frac{1}{2}$.

Length of $\perp$ from focus to directrix is $\frac{3\times2-4\times(-3)+7}{5}=5$

$\frac{a}{e}-ae=5\Rightarrow 2a-\frac{a}{2}=5\Rightarrow a=\frac{10}{3}$.

So length of major axis is $\frac{20}{3}$.

5. If the curves $\frac{x^2}{4}+y^2=1$ and $\frac{x^2}{a^2}+y^2=1$ for suitable value of 'a' cut on four concyclic points, the equation of circle passing through these four points is

(A) $x^2+y^2=2$
(B) $x^2+y^2=1$
(C) $x^2+y^2=4$
(D) none of these

Sol. **(B).** Let equation of the circle be $\left(\frac{x^2}{4}+y^2-1\right)+\lambda\left(\frac{x^2}{a^2}+y^2-1\right)=0$

$\Rightarrow x^2\left(\frac{1}{4}+\frac{\lambda}{a^2}\right)+y^2(1+\lambda)=1+\lambda\Rightarrow x^2\left(\frac{a^2+4\lambda}{4a^2}\right)+y^2(1+\lambda)=1+\lambda$

$\Rightarrow x^2\left(\frac{a^2+4\lambda}{4a^2(1+\lambda)}\right)+y^2=1$. Clearly the circle is $x^2+y^2=1$.

6. If tangents PQ and PR are drawn from a point on the circle $x^2+y^2=25$ to the ellipse $\frac{x^2}{16}+\frac{y^2}{b^2}=1$, $(b<4)$, so that the fourth vertex S of parallelogram PQSR lies on the circumcircle of triangle PQR, then eccentricity of the ellipse is

(A) $\frac{\sqrt{5}}{4}$
(B) $\frac{\sqrt{7}}{3}$
(C) $\frac{\sqrt{7}}{4}$
(D) $\frac{\sqrt{5}}{3}$

Sol. **(C).** A cyclic parallelogram will be a rectangle or square.
So, $\angle QPR=90^\circ$

$\Rightarrow$ P lies on director circle of ellipse $\frac{x^2}{16}+\frac{y^2}{b^2}=1$

i.e. $x^2+y^2=25$ is director circle of $\frac{x^2}{16}+\frac{y^2}{b^2}=1\Rightarrow 16+b^2=25$

$b^2=9\Rightarrow a^2(1-e^2)=9\Rightarrow 1-e^2=\frac{9}{16}\Rightarrow e^2=\frac{7}{16}\Rightarrow e=\frac{\sqrt{7}}{4}$.

7. The normal at a variable point P on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $a > b$ of eccentricity 'e' meets the axes of the ellipse at Q and R then the locus of the mid point of QR is conic with a eccentricity e' such that

(A) e' is independent of e
(B) $e' = 1$
(C) $e' = e$
(D) $e' = \frac{1}{e}$

Sol. **(C).** Equation of normal is $ax \sec\theta - by \operatorname{cosec}\theta = a^2 - b^2$

$Q \equiv \left(\frac{a^2-b^2}{a}\cos\theta,\ 0\right)$, $R \equiv \left(0,\ \frac{b^2-a^2}{b}\sin\theta\right)$

$\therefore h = \frac{a^2-b^2}{2a}\cos\theta$, $k = \frac{b^2-a^2}{2b}\sin\theta$.

Eliminating θ we get the equation of locus as

$\frac{x^2}{\frac{(a^2-b^2)^2}{4a^2}}+\frac{y^2}{\frac{(a^2-b^2)^2}{4b^2}}=1$ which is on ellipse with $\frac{(a^2-b^2)^2}{4b^2} > \frac{(a^2-b^2)^2}{4a^2}$.

So, $\frac{(a^2-b^2)^2}{4a^2}=\frac{(a^2-b^2)^2}{4b^2}(1-e'^2)$

$\Rightarrow e^4=\frac{e^4}{(1-e^2)}(1-e'^2) \Rightarrow e = e'$.

8. If a tangent of slope 2 of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is normal to the circle $x^2 + y^2 + 4x + 1 = 0$, then the maximum value of ab is

(A) 4
(B) 2
(C) 1
(D) none of these

Sol. **(A).** Equation of tangent is $y = 2x \pm \sqrt{4a^2+b^2}$

$\Rightarrow$ this is normal to the circle $x^2 + y^2 + 4x + 1 = 0$

$\Rightarrow$ this tangent passes through $(-2, 0)$.

$\Rightarrow 0 = -4 \pm \sqrt{4a^2+b^2} \Rightarrow 4a^2 + b^2 = 16$

$\Rightarrow$ using A.M $\geq$ G.M, we get $\frac{4a^2+b^2}{2} \geq \sqrt{4a^2b^2} \Rightarrow ab \leq 4$.

9. A series of ellipses are described with a given focus and corresponding directrix. The locus of the extremities of their minor axis is a

(A) circle
(B) ellipse
(C) parabola
(D) hyperbola

Sol. **(C).** Let focus S be $(0, 0)$ and ZM be a directrix, then ZS = constant, for all ellipses

i.e. $\frac{a}{e} - ae$ = constant

or, $\frac{b^2}{ae} = \lambda$ (say) ... (1)

But, if (x, y) be the end of minor axis, then $x = ae$, $y = b$ (or $-b$)

$\therefore$ From (1), locus of the end of minor axis is $y^2 = \lambda x$, which is a parabola.

10. A circle is inscribed in the ellipse $\frac{x^2}{16}+\frac{y^2}{25}=1$, where centre is one of the focii. If circle is tangent to the ellipse, then the radius of the circle is

(A) 1
(B) 2
(C) 3
(D) None of these

Sol. **(B).** Let F_1 and F_2 be the focus of the ellipse. Let F_1 be the centre of the circle C and it touches the ellipse at P.

Now consider a ray emanating from F_1 and bouncing off at P on the tangent L_1(say). Now as L_1 is tangent to the circle it must return to its center (F_1). Also L_1 is tangent to ellipse

$\Rightarrow$ it must pass through F_2.

The two case is possible iff PF_1F_2 are collinear.

$\therefore$ radius $b - be = b - \sqrt{b^2 - a^2} = 5 - 3 = 2$ units.

11. AB and CD are two equal and parallel chords of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. Tangents to the ellipse at A and B intersect at P and at C and D at Q. The line PQ

(A) passes through the origin
(B) is bisected at the origin
(C) cannot pass through the origin
(D) is not bisected at the origin

Sol. **(A, B).** Let P and Q be the points (α, β) and (α_1, β_1)

$\Rightarrow$ equations of AB and CD are $\frac{x}{a}\alpha+\frac{y}{b}\beta=1$ and $\frac{x}{a}\alpha_1+\frac{y}{b}\beta_1=1 \Rightarrow \frac{\alpha}{\alpha_1}=\frac{\beta}{\beta_1}=k$

also $\frac{\alpha^2}{a^2}+\frac{\beta^2}{b^2}=\frac{\alpha_1^2}{a^2}+\frac{\beta_1^2}{b^2} \Rightarrow \frac{\alpha}{\alpha_1}=\frac{\beta}{\beta_1}=-1 \Rightarrow$ PQ passes through origin and is bisected at the origin.

12. If the straight line $3x + 4y = 24$ intersects the axes at A and B and the straight line $4x + 3y = 24$ at C and D, then points A, B, C, D lies on

(A) circle
(B) parabola
(C) ellipse
(D) hyperbola

Sol. **(A), (B), (C), (D)**

Equation of the curve passing through all four points A, B, C, D can be written as

$(3x + 4y - 24)(4x + 3y - 24) + \lambda xy = 0$.

Now for different value of λ, we will get different curves.

Comprehension

If $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$ represents an ellipse, then $h^2 < ab$ and $abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$. If for every point (x_1, y_1) satisfying above equation $(2h - x_1, 2k - y_1)$ also satisfy it, then (h, k) is centre of it. The length of semi major axis and minor axis is nothing but the maximum and minimum value of the distance of a point lying on the curve from its centre.

Answer the following questions

13. For the ellipse $2x^2 - 2xy + 4y^2 - (3 + \sqrt{2}) = 0$, the inclination of major axis of it with x-axis is

(A) $\frac{\pi}{12}$
(B) $\frac{\pi}{8}$
(C) $\frac{3\pi}{8}$
(D) $\frac{5\pi}{8}$

Sol. **(B)**

Taking point $(r\cos\theta, r\sin\theta)$, we get

$$r^2 = \frac{3+\sqrt{2}}{3-\sqrt{2}\sin\left(2\theta+\frac{\pi}{4}\right)}$$

$$r_{max} = \sqrt{\frac{3+\sqrt{2}}{3-\sqrt{2}}}$$

$$r_{min} = 1$$

max exist when $2\theta + \frac{\pi}{4} = \frac{\pi}{2}$

$$\Rightarrow \theta = \frac{\pi}{8}$$

14. The equation of tangent to $2x^2 - 2xy + 4y^2 - (3+\sqrt{2}) = 0$ such that sum of perpendiculars dropped from foci is 2 units, is

(A) $y\cos\frac{3\pi}{4} - x\sin\frac{3\pi}{4} = 1$

(B) $y\sin\frac{3\pi}{8} - x\cos\frac{3\pi}{8} = 1$

(C) $x\cos\frac{\pi}{8} - y\sin\frac{\pi}{8} = 1$

(D) $y\cos\frac{5\pi}{8} + x\sin\frac{5\pi}{8} = 1$

Sol. **(B)**

Length of semi minor axis = 1 unit

$\Rightarrow$ tangent is at the end of minor axis

so its equation can be

$$x\sin\frac{\pi}{8} - y\cos\frac{\pi}{8} = \pm 1.$$

15. The product of perpendicular from the foci to any tangent to above given ellipse is

(A) 4

(B) 2

(C) 1

(D) $\frac{1}{2}$

Sol. **(C)**

Product of length of perpendiculars from foci = b^2 (b is length of semi minor axis) = 1.

Assertion Reason

16. Statement 1: Diagonals of any parallelogram inscribed in an ellipse always intersect at the centre of the ellipse.

because

Statement 2: Centre of the ellipse is the only point at which two chords can bisect each other and every chord passing through the centre of the ellipse gets bisected at the centre.

(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement -1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement -1 is True, Statement -2 is False

(D) Statement -1 is False, Statement -2 is True

Sol. **(A).** Statement 2 is correct as ellipse is a central conic and it also explains statement 1.

17. Match the following

Column – I		Column – I	
(A)	The number of rational points on the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ is	(p)	∞
(B)	Number of integral points on the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ is	(q)	4
(C)	Number of rational points on the curve $\frac{x^2}{3}+y^2=1$ is	(r)	0
(D)	Number of integral points on the curve $\frac{x^2}{3}+y^2=1$ is	(s)	2
		(t)	1/2

Sol. **(A) → (p); (B) → (q); (C) → (p); (D) → (s)**

(A) $\frac{x^2}{9}+\frac{y^2}{4}=1$

Let any point on ellipse be $(3\cos\theta, 2\sin\theta)$
since $\sin\theta$ and $\cos\theta$ can be rational for infinite many value of $\theta \in [0, 2\pi]$
$\Rightarrow$ there are infinite rational points.

(B) There are four integral points $(\pm 3, 0), (\pm 2, 0)$.

(C) There are infinite number of rational points on $\frac{x^2}{3}+\frac{y^2}{1}=1$ (same as (a)).

(D) Number of integral points on $\frac{x^2}{3}+\frac{y^2}{1}=1$ is two.

Answer Q.18, Q.19 and Q.20 by appropriately matching the information given in the three columns of the following table.

In Column-I: equation of conic is given, Column-II: equation of tangent is given and Column: the value of k is given where $k=\frac{\text{distance of tangent from centre}}{\text{eccentricity}}$

Column–I	Column–II	Column–III
(I) $\frac{x^2}{9}+\frac{y^2}{25}=1$	(i) $x+y=\sqrt{7}$	(P) $\sqrt{\frac{21}{2}}$
(II) $\frac{(x-1)^2}{4}+\frac{(y-3)^2}{1}=1$	(ii) $x+y=3\sqrt{2}$	(Q) $\frac{75}{4\sqrt{17}}$
(III) $\frac{(x+y-1)^2}{18}+\frac{(x-y+3)^2}{32}=1$	(iii) $\frac{x}{3}+\frac{y}{5}=\sqrt{2}$	(R) $\sqrt{\frac{10}{3}}$
(IV) $\frac{x^2}{3}+\frac{y^2}{4}=1$	(iv) $x+y=4+\sqrt{5}$	(S) $\frac{9}{\sqrt{7}}$

18. Which of the following combination is **CORRECT**?

(A) (I) (i) (Q) (B) (II) (iv) (R)
(C) (II) (ii) (P) (D) (I) (i) (P)

Sol. **(B)**

19. Which of the following combination is **CORRECT**?
(A) (III) (iv) (R) (B) (IV) (iv) (Q)
(C) (IV) (i) (Q) (D) (III) (ii) (S)

Sol. **(D)**

20. Which of the following combination is **CORRECT**?
(A) (IV) (i) (R) (B) (II) (iii) (Q)
(C) (II) (ii) (S) (D) (IV) (i) (P)

Sol. **(D)**

For ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

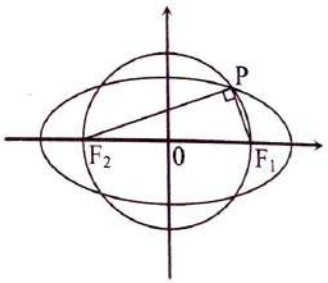
Length of tangent from origin is $\frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}}$

Equation of ellipse whose major axis is $ax + by + c = 0$ and minor axis $bx - ay + k = 0$, and length of major axis r_1 and length of minor axis is r_2, will be

$$\frac{\left(\frac{ax+by+c}{\sqrt{a^2+b^2}}\right)^2}{r_2^2}+\frac{\left(\frac{bx-ay+k}{\sqrt{a^2+b^2}}\right)^2}{r_1^2}=1$$

21. A circle has same centre as an ellipse and pass through the foci F_1 and F_2 of the ellipse. The two curves cut in 4 points. Let P be one of the points. If the area of ΔPF_1F_2 is 50 and major axis of the ellipse is 15, then the eccentricity of the ellipse is k, then 3k is equal to _____

Sol. **(1.00)**



Ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

and circle $x^2 + y^2 = a^2e^2$

Given area of $\Delta PF_1F_2 = 50$

$\frac{1}{2}PF_1 \times PF_2 = 50$

$\Rightarrow PF_1\ PF_2 = 100$

Also, $PF_1 + PF_2 = 2a = 15$

$PF_1^2 + PF_2^2 + 2PF_1PF_2 = 225$

$(F_1F_2)^2 + 2 \times 100 = 225$

$(2ae)^2 = 25$

$e^2 = \frac{25}{15\times 15}$

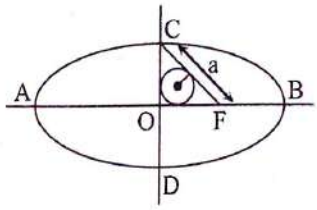
$\therefore\ e = \frac{1}{3}$

22. Point 'O' is the centre of the ellipse with major axis AB and minor axis CD. Point F is one of the focus of the ellipse. If OF = 6 and the diameter of the inscribed circle of ΔOCF is 2, then the product (AB)(CD) is p, then $\frac{p}{13}$ equal to _____

Sol. **(5.00)**

$a^2e^2 = 36$

$\Rightarrow a^2 - b^2 = 36$ ..... (1)



Using $r = (s - a)\tan\frac{A}{2}$ in ΔOCF

$\because$ r = 1 as diameter of circle is 2 and CF = a

$\therefore 1 = (s - a)\tan 45°$ when a = CF

$2 = 2s - 2a = 2s - AB = (OF + FC + OC) - AB$

$\Rightarrow 2 = 6 + \frac{AB}{2} + \frac{CD}{2} - AB$

$\frac{AB - CD}{2} = 4 \Rightarrow 2(a - b) = 8$

$\Rightarrow a - b = 4$ ..... (2)

From (1) and (2) $a + b = 9 \Rightarrow 2a = 13 \Rightarrow 2b = 5$

(AB)(CD) = 65

23. If $\sqrt{3}x + 2y = 2$ touches the ellipse $\frac{x^2}{4} + \frac{y^2}{2} = 1$, then eccentric angle of point of contact is $\frac{k\pi}{3}$, then k is equal to _____

Sol. **(0.50)**

Let point $(2\cos\alpha, \sin\alpha)$

$\sqrt{3}\cos\alpha + \sin\alpha = 1$

$\alpha = \frac{\pi}{6}$

24. If P and Q are the feet of the perpendiculars from the foci S_1 and S_2 of an ellipse $\frac{x^2}{5} + \frac{y^2}{3} = 1$ on the tangent at any point P on the ellipse, then $(S_1P)(S_2Q)$ is k, then $\frac{k}{2}$ is equal to _____

Sol. **(1.50)**

Using properties that product of length of perpendicular on any tangent from the foci of ellipse is equal to 'b^2', where b is semi-minor axis

12

HYPERBOLA

STANDARD EQUATION AND BASIC DEFINITIONS

The standard equation of a hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

- The eccentricity e of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is given by the relation $e^2 = \left(1 + \frac{b^2}{a^2}\right)$.
- Since the curve is symmetrical about the y - axis, it is clear that there exists another focus S′ at (-ae, 0) and a corresponding directirx Z′M′ with the equation x = −a/e, such that the same hyperbola is described if a point moves so that its distance from S′ is e times its distance from Z′M′.
- The points A and A′ where the straight line joining the two foci cuts the hyperbola are called the **vertices** of the hyperbola.
- The straight line joining the vertices is called the **transverse** axis of the hyperbola, its length AA′ is 2a.
- The middle point C of AA′ possesses the property that it bisects every chord of the hyperbola passing through it. It can be proved by taking $P(x_1, y_1)$ as any point on the hyperbola. If (x_1, y_1) lies on the hyperbola then so does $P'(-x_1, -y_1)$ because hyperbola is symmetric about x and y axes. Therefore PP′ is a chord whose middle point is (0, 0), i.e. the origin O. On account of this property the middle point of the straight line joining the vertices of the hyperbola is called the **centre** of the hyperbola.
- The straight line through the centre of a hyperbola which is perpendicular to the transverse axis does not meet the hyperbola in real points. If B and B′ be the points on this line such that BC = CB′ = b, the line BB′ is called the **conjugate axis**.
- The latus rectum is the chord through a focus at right angle to the **transverse axis**.
- The length of the semi - latus rectum obtained by putting x = ae in the equation of the hyperbola is $y = b\sqrt{\frac{a^2e^2}{a^2} - 1} = b\sqrt{e^2 - 1} = b.b/a = b^2/a$.

Relation between Focal Distances

The difference of the focal distances of a point on the hyperbola is constant and is equal to 2a.

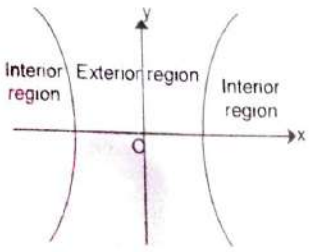
Remark:

- Using this property hyperbola can be defined in another way-Locus of a moving point such that the difference of its distances from two fixed points is constant, would be a hyperbola. For example, interference fringes formed in Young's Double Slit experiment are hyperbolic in nature.

Relative Position of a Point with respect to the Hyperbola

The quantity $\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$ is positive,

zero or negative, according as the point (x_1, y_1) lies within, upon or outside of the curve.



Parametric Coordinates

$(a \sec\theta, b \tan\theta)$ is a point on the hyperbola for all values of θ. The point $(a \sec\theta, b \tan\theta)$ is briefly called the point 'θ'

Important Properties of Hyperbola

Since the fundamental equation of the hyperbola only differs from that of the ellipse in having $-b^2$ instead of b^2, it will be found that many propositions for the hyperbola are derived from those for the ellipse by changing the sign of b^2. Some results for the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are:

(i) The tangent at any point (x_1, y_1) on the curve is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$.

(ii) The tangent at the point 'θ' is $\frac{x \sec\theta}{a} - \frac{y \tan\theta}{b} = 1$.

(iii) The straight line $y = mx + c$ is a tangent to the curve, if $c^2 = a^2 m^2 - b^2$. In other words, $y = mx \pm \sqrt{a^2 m^2 - b^2}$ touches the curve, for all those values of m when $m > b/a$ or $m < -b/a$.

(iv) Equation of the normal at any point (x_1, y_1) to the curve is $\frac{x - x_1}{x_1/a^2} = \frac{y - y_1}{y_1/(-b^2)}$.

(v) The chord through the points θ_1 and θ_2 is $\begin{vmatrix} x & y & 1 \\ a\sec\theta_1 & b\tan\theta_1 & 1 \\ a\sec\theta_2 & b\tan\theta_2 & 1 \end{vmatrix} = 0$.

(vi) The equation of the normal at θ is $ax\cos\theta + by\cot\theta = a^2 + b^2$.

(vii) Through a given point, four normals can be drawn to a hyperbola (real or imaginary).

(viii) Tangent drawn at any point bisects the angle between the lines, joining the point to the foci, whereas normal bisects the supplementary angle between the lines.

(ix) Equation of the director circle is $x^2 + y^2 = a^2 - b^2$. That means if $a^2 > b^2$, there would exist several points such that tangents drawn from them would be mutually perpendicular. If $a^2 < b^2$, no such point exist. For $a^2 = b^2$, centre is the only point from which two perpendicular tangents (asymptotes) to the hyperbola can be drawn.

Conjugate Hyperbola

The hyperbola, whose transverse and conjugate axes are respectively the conjugate and transverse axes of a given hyperbola, is called the conjugate hyperbola of the given hyperbola, and the two hyperbolas are conjugate to one another. Thus, the hyperbolas $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$ are conjugate hyperbolas.

ASYMPTOTE

We have the equations of the asymptotes as $\frac{x}{a} \pm \frac{y}{b} = 0$.

The lines $\frac{x}{a} \pm \frac{y}{b} = 0$ are also asymptotes to the conjugate hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$.

Remarks:

- The equation of the hyperbola and that of its pair of asymptotes differ by a constant. For example, if $S = 0$ is the equation of the hyperbola, then the combined equation of the asymptotes is given by $S + K = 0$. The constant K is obtained from the condition that the equation $S + K = 0$ represents a pair of lines. Finally the equation of the corresponding conjugate hyperbola is $S + 2K = 0$.
- Any line drawn parallel to the asymptote of the hyperbola would meet the curve only at one point.

Rectangular Hyperbola

A hyperbola whose asymptotes are at right angles to each other is called a **rectangular hyperbola**.

The angle between the asymptotes of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, is $2 \tan^{-1}(b/a)$. This is a right angle if $\tan^{-1} b/a = \pi/4$, i.e., if $b/a = 1 \Rightarrow b = a$.

(i) The equation of the rectangular hyperbola referred to its transverse and conjugate axes as coordinate axes is therefore $x^2 - y^2 = a^2$.

(ii) Equation of the rectangular hyperbola having asymptotes as coordinate axes: To obtain this equation we rotate the axes of reference through an angle of 45° in the clockwise direction. Thus we have to write

$\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}}$ for x and $-\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}}$ for y.

The equation of the hyperbola becomes

$(1/2)(x + y)^2 - (1/2)(x - y)^2 = a^2$ i.e. $xy = 1/2\, a^2$ or $xy = c^2$ where $c^2 = a^2/2$.

The Tangent and Normal

For the rectangular hyperbola, $xy = c^2$, the coordinates of any point on the curve may be taken as $(cp, c/p)$, where p is a parameter. We shall call the point with these coordinates, the point 'p'

(i) Equation of tangent at (x_1, y_1) is $xy_1 + yx_1 = 2c^2$.

(ii) Tangent at $(cp, c/p)$ to $xy = c^2$ is $\frac{cx}{p} + cpy = 2c^2$.

(iii) Equation of normal at $(cp, c/p)$ is $xp^3 - py - cp^4 + c = 0$.

Remarks:

- The slope of the tangent at the point $(cp, c/p)$ is $-1/p^2$, which is always negative. Hence tangents drawn at any point to $xy = c^2$ would always make an obtuse angle with the x-axis.
- Slope of normal at any point $(cp, c/p)$ is p^2, which is always positive. Hence normal drawn to $xy = c^2$ at any point would always make an acute angle with the x-axis.

Intersection of a Circle and a Rectangular Hyperbola

A rectangular hyperbola and a circle meet in four points. The mean of these four points is the middle point of the centres of the hyperbola and that of the circle.

ILLUSTRATIVE PROBLEMS

1. A series of hyperbolas is drawn having a common transverse axis of length 2a. Prove that the locus of a point P on each hyperbola, such that its distance from the transverse axis is equal to its distance from an asymptote, is the curve $(x^2 - y^2)^2 = 4x^2(x^2 - a^2)$.

Sol. The equation of a hyperbola of the series is $\frac{x^2}{a^2} - \frac{y^2}{\lambda^2} = 1$, where, λ is a parameter.

The asymptotes of this hyperbola are $x/a = \pm y/\lambda$.

Suppose (x', y') is a point P on the hyperbola which is equidistant from the transverse axis and an asymptote. Then

$$\frac{x'^2}{a^2} - \frac{y'^2}{\lambda^2} = 1$$

and $y' = \dfrac{x'/a - y'/\lambda}{\sqrt{\frac{1}{a^2} + \frac{1}{\lambda^2}}}$

i.e., $\frac{y'^2}{\lambda^2} = \frac{x'^2}{a^2} - 1$

and $y'^2\left(\frac{1}{a^2} + \frac{1}{\lambda^2}\right) = \frac{x'^2}{a^2} + \frac{y'^2}{\lambda^2} - \frac{2x'y'}{a\lambda}$

The second relation gives on simplification,

$(y'^2 - x'^2)^2 = \frac{4x'^2y'^2a^2}{\lambda^2} = 4x'^2\left(x'^2 - a^2\right)$ by the first relation.

Hence, the locus of P is $(y^2 - x^2)^2 = 4x^2(x^2 - a^2)$.

2. A variable straight line of slope 4 intersects the hyperbola $xy = 1$ at two points. Find the locus of the point which divides the line segment between these points in the ratio 1 : 2.

Sol. Let the line be $y = 4x + c$. It meets the curve $xy = 1$ at

$x(4x + c) = 1$

$\Rightarrow 4x^2 + cx - 1 = 0 \Rightarrow x_1 + x_2 = -c/4$

Also $y(y - c) = 4$

$\Rightarrow y^2 - cy - 4 = 0 \Rightarrow y_1 + y_2 = c$

Let the point which divides the line segment in the ratio 1 : 2 be (h, k)

$\Rightarrow \frac{x_1 + 2x_2}{3} = h$

$\Rightarrow x_2 = 3h + c/4 \Rightarrow x_1 = -c/2 - 3h$

Also $\frac{y_1 + 2y_2}{3} = k$

$\Rightarrow y_2 = 3k - c \Rightarrow y_1 = -3k + 2c$

Now (h, k) lies on the line $y = 4x + c$

$\Rightarrow k = 4h + c$

$\Rightarrow c = k - 4h$

$\Rightarrow x_1 = -k/2 + 2h - 3h = -h - k/2$ and $y_1 = -3k + 2k - 8h = -k - 8h$

$\Rightarrow (h + k/2)(k + 8h) = 1$

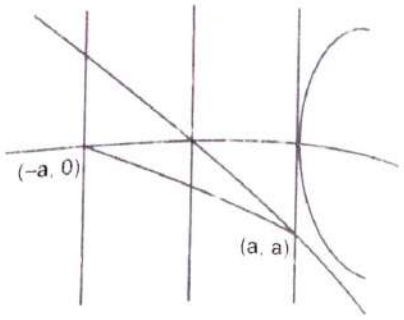
$\Rightarrow hk + 8h^2 + \frac{k^2}{2} + 4hk = 1 \Rightarrow 16h^2 + k^2 + 10hk = 2.$

Hence locus of (h, k) is $16x^2 + y^2 + 10xy = 2$.

3. If the portion of the asymptotes between centre and the tangent at the vertex of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ in the third quadrant is cut by the line $y + \lambda(x + a) = 0$, λ being parameter, then

Sol. Slope of cutting line will lie between $\left(-\frac{1}{2}, 0\right)$

$\Rightarrow \lambda \in \left(0, \frac{1}{2}\right)$.



4. Let any double ordinate PNP′ of the hyperbola $\frac{x^2}{25} - \frac{y^2}{16} = 1$ be produced both sides to meet the asymptotes in Q and Q′, then PQ. P′Q is equal to

Sol. $NP = \frac{4}{5} = \sqrt{x_1^2 - 25}$

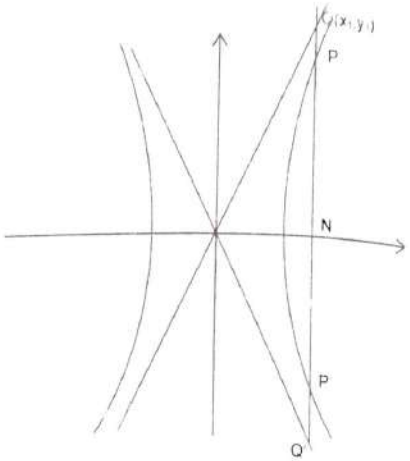
Q is on $y = \frac{4}{5}x$

$NQ = \frac{4}{5}x_1$

$PQ = NQ - NP = \frac{4}{5}\left(x_1 - \sqrt{x_1^2 - 25}\right)$

$P'Q = \frac{4}{5}\left(x_1 + \sqrt{x_1^2 - 25}\right)$

$PQ.\ P'Q = 16.$



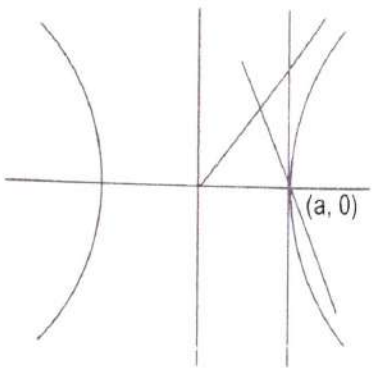
5. Portion of asymptote of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (between centre and the tangent at vertex) in the first quadrant is cut by the line $y + \lambda(x - a) = 0$ (λ is a parameter) then

Sol. The line $y + \lambda(x - b) = 0$ will intersect the portion of the asymptote in the 1st quadrant only if its slope is negative.

$\Rightarrow -\lambda < 0$

$\Rightarrow \lambda > 0$

$\therefore \lambda \in (0, \infty)$



SOLVED PROBLEMS

1. If PQ is a double ordinate of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ such that OPQ is an equilateral triangle, O being the centre of the hyperbola. Then the eccentricity e of the hyperbola satisfies

(A) $1 < e < \frac{2}{\sqrt{3}}$ (B) $e = \frac{2}{\sqrt{3}}$

(C) $e = \frac{\sqrt{3}}{2}$ (D) $e > \frac{2}{\sqrt{3}}$

Sol. **(D).** Let the coordinates of P be (α, β), then $PQ = 2\beta$

$\Rightarrow OP^2 = PQ^2 \Rightarrow \alpha = \pm \sqrt{3}\beta$ since $P(\alpha, \beta)$ also lie on hyperbola

$\Rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \Rightarrow \frac{3}{a^2} - \frac{1}{b^2} = \frac{1}{\beta^2} > 0 \Rightarrow \frac{b^2}{a^2} > \frac{1}{3} \Rightarrow e^2 > \frac{4}{3} \Rightarrow e > \frac{2}{\sqrt{3}}$.

2. A rectangular hyperbola passes through the points A(1, 1), B(1, 5) and C(3, 1). The equation of normal to the hyperbola at A(1, 1) is

(A) $2x + y = 3$ (B) $2x - y = 1$

(C) $x + y = 2$ (D) none of these

Sol. **(A).** Normal at A is parallel to BC $\Rightarrow$ equation of normal at A is $2x + y = 3$.

3. If a variable line $x \cos\alpha + y \sin\alpha = p$, which is a chord of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 (b > a)$, subtends a right angle at the centre of the hyperbola then it always touches a fixed circle whose radius is

(A) $\frac{ab}{\sqrt{b-2a}}$ (B) $\frac{a}{\sqrt{a-b}}$

(C) $\frac{ab}{\sqrt{b^2-a^2}}$ (D) $\frac{ab}{b\sqrt{b+a}}$.

Sol. **(C).** $\because$ $x \cos\alpha + y \sin\alpha = p$ subtends a right angle at centre i.e. (0, 0)

$\therefore$ making homogeneous equation of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

with the help of $x \cos\alpha + y \sin\alpha = p$ and putting coefficient of x^2 + coefficient of $y^2 = 0$

$\Rightarrow \frac{1}{a^2} - \frac{1}{b^2} = \frac{1}{p^2} \Rightarrow p = \frac{ab}{\sqrt{b^2-a^2}}$

'p' is also the length of perpendicular drawn from (0, 0) to the line $x \cos\alpha + y \sin\alpha = p$,

then radius of circle $= p = \frac{ab}{\sqrt{b^2-a^2}}$.

4. If values of m for which the line $y = mx + 2\sqrt{5}$ touches the hyperbola $16x^2 - 9y^2 = 144$ are the roots of the equation $x^2 - (a + b)x - 4 = 0$, then value of $(a + b)$ is equal to

(A) 2 (B) 4

(C) zero (D) none of these

Sol. **(C).** Equation of hyperbola is $\frac{x^2}{9} - \frac{y^2}{16} = 1$. Equation of tangent is $y = mx + \sqrt{9m^2 - 16}$

$\Rightarrow \sqrt{9m^2 - 16} = 2\sqrt{5} \Rightarrow m = \pm 2 \Rightarrow a + b =$ sum of roots $= 0$.

5. The equation of the line of latus rectum of the rectangular hyperbola $xy = c^2$ is

(A) $x - y = \sqrt{2}c$ (B) $x + y = 2\sqrt{2}c$

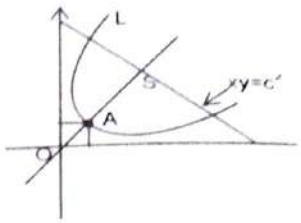
(C) $x + y = \sqrt{2}c$ (D) $x + y = 0$

Sol. **(B).** $OA = c \qquad A = (c, c)$

$AS = eOA \quad (\because e = \sqrt{2})$

$AS = \sqrt{2}c$

$\therefore$ Co-ordinates of $S = (\sqrt{2}c, \sqrt{2}c)$



Equation of line through LL'

Slope of line $= -1$ and passing through $(\sqrt{2}c, \sqrt{2}c)$ is

$(y - \sqrt{2}c) = -1(x - \sqrt{2}c) \Rightarrow y - \sqrt{2}c = -x + \sqrt{2}c \Rightarrow x + y = 2\sqrt{2}c$.

6. The equation of normal to the rectangular hyperbola $xy = 4$ at the point P on the hyperbola which is parallel to the line $2x - y = 5$ is

(A) $2x - y + 2\sqrt{3} = 0$ (B) $2x - y - 2\sqrt{3} = 0$

(C) $2x - y - 3\sqrt{2} = 0$ (D) $\sqrt{2}x + y - 3 = 0$

Sol. **(C).** Slope of the normal to the rectangular hyperbola $xy = 4$ is $m = t^2 = 2$ and $2x - y = 5$ has slope $= 2$.

$\therefore t = \sqrt{2}$.

Equation of the normal is $\left(y - \frac{2}{t}\right) = 2(x - 2t)$

$\Rightarrow ty - 2 = 2t(x - 2t) \Rightarrow ty - 2 = 2tx - 4t^2 \Rightarrow 2tx - ty + 2 - 4t^2 = 0$

$\pm\sqrt{2}(2x - y) + 2 - 8 = 0\ ;\ \pm\sqrt{2}(2x - y) - 6 = 0 \Rightarrow 2x - y \pm 3\sqrt{2} = 0$.

7. A straight line intersects the same branch of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ in P_1 and P_2 and meets its asymptotes in Q_1 and Q_2. Then $P_1Q_2 - P_2Q_1$ is equal to

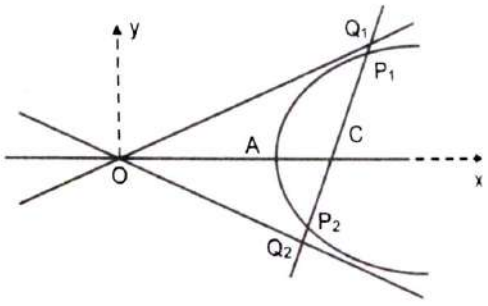
(A) $(a - b)$ (B) $\sqrt{a^2 + b^2 - ab}$

(C) 0 (D) $\sqrt{a^2 - b^2}$

Sol. **(C).**

Let C (x_1, y_1) be the mid-point of P_1P_2, so that the equation of the line P_1P_2 is

$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2}$.



It meets the asymptotes $y = \pm \frac{b}{a}x$

at $Q_1\left(x_1 + \frac{ay_1}{b}, \frac{b}{a}x_1 + y_1\right)$

and $Q_2\left(x_1 - \frac{ay_1}{b}, -\frac{b}{a}x_1 + y_1\right)$

$\Rightarrow$ mid-point of Q_1Q_2 is (x_1, y_1)

$\Rightarrow CQ_1 = CQ_2 \Rightarrow P_1Q_1 = P_1C + CQ_2 = P_2C + CQ_1 = P_2Q_1$.

8. From a point on the line $y = x + c$, c (parameter), tangents are drawn to the hyperbola $\frac{x^2}{2} - \frac{y^2}{1} = 1$ such that chords of contact pass through a fixed point (x_1, y_1). Then $\frac{x_1}{y_1}$ is equal to

(A) 2
(B) 3
(C) 4
(D) none of these

Sol. **(A).** Let the point be (α, β)

$\Rightarrow \beta = \alpha + c$

Chord of contact of hyperbola $T = 0$

$\therefore \frac{x\alpha}{2} - \frac{y\beta}{1} = 1$

$\Rightarrow \frac{x\alpha}{2} - y(\alpha + c) = 1$

$\Rightarrow \left(\frac{x}{2} - y\right)\alpha - (yc + 1) = 0$

Since this passes through point (x_1, y_1)

$\therefore x_1 = 2y_1$ and $y_1c + 1 = 0$

$\therefore y_1 = \frac{x_1}{2}$

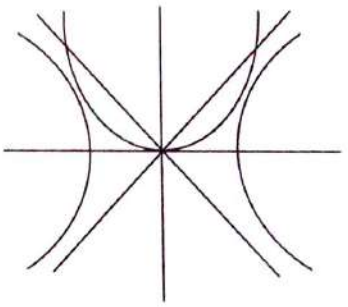
Hence $\frac{x_1}{y_1} = 2$.

9. If two tangents can be drawn to the different branches of hyperbola $\frac{x^2}{1} - \frac{y^2}{4} = 1$ from the point (α, α^2), then

(A) $\alpha \in (-2, 0)$
(B) $\alpha \in (0, 2)$
(C) $\alpha \in (-\infty, -2)$
(D) $\alpha \in (2, \infty)$

Sol. **(C, D).** (α, α^2) lie on the parabola $y = x^2$

(α, α^2) must lie between the asymptotes of hyperbola $\frac{x^2}{1} - \frac{y^2}{4} = 1$ in 1st and 2nd quadrant.



$\therefore$ asymptotes are $y = \pm 2x$

$\Rightarrow 2\alpha < \alpha^2$

$\Rightarrow \alpha < 0$ or $\alpha > 2$

and $-2\alpha < \alpha^2$

$\alpha < -2$ or $\alpha > 0$

$\therefore \alpha \in (-\infty, -2)$ or $\alpha \in (2, \infty)$.

10. The lines parallel to normal to the curve $xy = 1$ is/are

(A) $3x + 4y + 5 = 0$
(B) $3x - 4y + 5 = 0$
(C) $4x + 3y + 5 = 0$
(D) $3y - 4x + 5 = 0$

Sol. **(B, D).** We know, line $ax + by + c = 0$ is normal to curve $xy = 1$ when $ab < 0$.

Comprehension

If a circle with centre C(α, β) intersects a rectangular hyperbola with centre L (h, k) at four points P(x_1, y_1), Q(x_2, y_2), R(x_3, y_3) and S(x_4, y_4), then the mean of the four points P, Q, R, S is the mean of the points C and L. In other words, the mid-point of CL coincides with the mean point of P, Q, R, S. Analytically

$$\frac{x_1+x_2+x_3+x_4}{4}=\frac{\alpha+h}{2};\ \frac{y_1+y_2+y_3+y_4}{4}=\frac{\beta+k}{2}.$$

Answer the following questions

11. Five points are selected on a circle of radius a. The centres of the rectangular hyperbolas, each passing through four of these points lie on a circle of radius

(A) a (B) 2a

(C) $\frac{a}{\sqrt{2}}$ (D) $\frac{a}{2}$

Sol. **(D)**

Let the circle be $x^2+y^2=a^2$ and let the centres of a rectangular hyperbola be (h, k). Let the given points on the circle be (a cos θ_i, a sin θ_i), i = 1, 2, 3, ..., so that

$$\frac{a\cos\theta_1+a\cos\theta_2+a\cos\theta_3+a\cos\theta_4}{4}=\frac{0+h}{2}=\frac{h}{2}$$

$$\Rightarrow\sum_{i=1}^{5}a\cos\theta_i-a\cos\theta_5=2h.$$

Similarly, $\sum_{i=1}^{5}a\sin\theta_i-a\sin\theta_5=2k$

Let $\sum_{i=1}^{5}a\cos\theta_i=L$ and $\sum_{i=1}^{5}a\sin\theta_i=M$.

Since the points are given, L and M are known.

Hence we have $a\cos\theta_5=L-2h$, $a\sin\theta_5=M-2k$.

$\Rightarrow a^2=(2h-L)^2+(2k-M)^2$ and hence (h, k) lies on the curve $\left[x-\frac{L}{2}\right]^2+\left[y-\frac{M}{2}\right]^2=\left[\frac{a}{2}\right]^2$ which is a circle with radius $\frac{a}{2}$.

12. A, B, C and D are the points of intersection of a circle and a rectangular hyperbola which have different centres. If AB passes through the centre of the hyperbola, then CD passes through

(A) centre of the hyperbola

(B) centre of the circle

(C) mid-point of the centres of the circle and hyperbola

(D) none of these

Sol. **(B)**

If the centres of the circle and the hyperbola are (α, β) and (h, k) and the points are A (x_1, y_1), B (x_2, y_2), C (x_3, y_3) and D (x_4, y_4), then

$$h+\alpha=\frac{x_1+x_2+x_3+x_4}{2}$$

$$\text{and } k+\beta=\frac{y_1+y_2+y_3+y_4}{2} \quad \ldots\ldots(1)$$

(h, k) lies on AB, then chords of the hyperbola, passing through its centre, are bisected at the centre, so

$\Rightarrow h = \frac{x_1 + x_2}{2}$

$k = \frac{y_1 + y_2}{2}$

and hence $\frac{x_1 + x_2}{2} + \alpha = \frac{x_1 + x_2 + x_3 + x_4}{2}$

$\frac{y_1 + y_2}{2} + \beta = \frac{y_1 + y_2 + y_3 + y_4}{2}$

$\Rightarrow \alpha = \frac{x_3 + x_4}{2}, \beta = \frac{y_3 + y_4}{2}$

$\Rightarrow (\alpha, \beta)$ is the mid-point of CD and lies on CD.

13. A circle with fixed centre (3h, 3k) and of variable radius cuts the rectangular hyperbola $x^2 - y^2 = 9a^2$ at the points A, B, C , D. The locus of the centroid of the triangle ABC is given by

(A) $(x - 2h)^2 - (y - 2k)^2 = a^2$

(B) $(x - h)^2 - (y - k)^2 = a^2$

(C) $\frac{x^2}{h^2} - \frac{y^2}{k^2} = a^2$

(D) $\frac{x^2}{h^2} + \frac{y^2}{k^2} = a^2$

Sol. **(A)**

Let A, B, C and D have coordinates (x_i, y_i), i = 1, 2, 3, 4 then

$\frac{x_1 + x_2 + x_3 + x_4}{4} = \frac{3h}{2}$ ...(i)

and $\frac{y_1 + y_2 + y_3 + y_4}{4} = \frac{3k}{2}$ ...(ii)

Let the centroid of ΔABC be (α, β),

then $\alpha = \frac{x_1 + x_2 + x_3}{3}$

and $\beta = \frac{y_1 + y_2 + y_3}{3}$.

So, from (i) and (ii),

$3\alpha + x_4 = 6h$ and $3\beta + y_4 = 6k$

But (x_4, y_4) lies on $x^2 - y^2 = 9a^2$

So, $(6h - 3\alpha)^2 - (6k - 3\beta)^2 = 9a^2$

or $(\alpha - 2h)^2 - (\beta - 2k)^2 = a^2$

$\therefore$ Locus of (α, β) is $(x - 2h)^2 - (y - 2k)^2 = a^2$.

Assertion Reason

14. Statement-1: Hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} - 1 = 0$ is known as a central curve.

because

Statement-2: Every chord passing through its centre gets bisected.

(A) Statement -1 is True, Statement -2 is True; Statement-2 is a correct explanation for Statement-1

(B) Statement -1 is True, Statement -2 is True; Statement-2 is **NOT** a correct explanation for Statement-1

(C) Statement -1 is True, Statement -2 is False

(D) Statement -1 is False, Statement -2 is True

Sol. **(A)**

15. Let the foci of the hyperbola $\frac{x^2}{A^2}-\frac{y^2}{B^2}=1$ be the vertices of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and the foci of the ellipse be the vertices of the hyperbola. Let the eccentricities of the ellipse and hyperbola be e_E and e_H respectively, then match the following

	Column – I		Column – II
(A)	$\frac{b}{B}$ is equal to	(p)	1
(B)	$e_H + e_E$ is always greater than	(q)	2
(C)	If angle between the asymptotes of hyperbola is $\frac{2\pi}{3}$, then $4e_E$ is equal to	(r)	3
(D)	If $e_E^2=\frac{1}{2}$ and (x, y) is point of intersection of ellipse and the hyperbola then $\frac{9x^2}{2y^2}$ is	(s)	4
		(t)	∞

Sol. **(A) → (p); (B) → (p, q); (C) → (q); (D) → (s)**

We have $A = ae_E$

and $a = Ae_H$

$\Rightarrow e_E \cdot e_H = 1$

$\Rightarrow e_E + e_H > 2$

$B^2 = A^2(e_H^2 - 1)$

$= a^2(1 - e_E^2) = b^2$

$\Rightarrow \frac{b}{B} = 1$ also the angle between the asymptotes is

$2\tan^{-1}\frac{B}{A}=\frac{2\pi}{3}$

also, $\frac{B}{A}=\sqrt{3}$

$\Rightarrow \frac{b}{ae_E}=\sqrt{3}$

$\Rightarrow e_E^2=\frac{1}{4}$

Solving $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

and $\frac{x^2}{a^2e_E^2}-\frac{y^2}{b^2}=1$

we get $x^2=\frac{2a^2e_E^2}{1+e_E^2}$

and $y^2=\frac{b^2(1-e_E^2)}{1+e_E^2}$

$\Rightarrow \frac{9x^2}{2y^2}=\frac{2a^2e_E^2}{b^2(1-e_E^2)}=4$

Answer Q.16, Q.17 and Q.18 by appropriately matching the information given in the three columns of the following table.

Column-I: contains equation of hyperbola, Column-II: contains eccentricity of hyperbola and Column-III: contains length of latus rectum of hyperbola.

Column–I	Column–II	Column–III
(I) $7y^2 - 9x^2 + 54x - 28y - 116 = 0$	(i) $\frac{\sqrt{19}}{\sqrt{3}}$	(P) $\frac{14}{3}$
(II) $12x^2 - 4y^2 - 24x + 32y - 127 = 0$	(ii) $\frac{4}{3}$	(Q) $\frac{32}{\sqrt{3}}$
(III) $x^2 - y^2 - 4x + 4y + 16 = 0$	(iii) $\sqrt{2}$	(R) 15
(IV) $16x^2 - 3y^2 - 32x - 12y - 44 = 0$	(iv) 2	(S) 8

16. Which of the following combination is **CORRECT**?
(A) (I) (iv) (P)
(B) (II) (i) (R)
(C) (III) (iii) (S)
(D) (IV) (i) (S)

Sol. **(C)**

17. Which of the following combination is **CORRECT**?
(A) (I) (ii) (Q)
(B) (II) (iv) (R)
(C) (III) (iii) (P)
(D) (IV) (i) (S)

Sol. **(B)**

18. Which of the following combination is **CORRECT**?
(A) (I) (iii) (P)
(B) (II) (iv) (P)
(C) (III) (i) (S)
(D) (IV) (i) (Q)

Sol. **(D)**

16.-18. (I) $\frac{(y-2)^2}{(3)^2} - \frac{(x-3)^2}{(\sqrt{7})^2} = 1$

(II) $\frac{(x-1)^2}{\left(\frac{5}{2}\right)^2} - \frac{(y-4)^2}{\left(\frac{5\sqrt{3}}{2}\right)^2} = 1$

(III) $\frac{(y-2)^2}{(4)^2} - \frac{(x-2)^2}{(4)^2} = 1$

(IV) $\frac{(x-1)^2}{(\sqrt{3})^2} - \frac{(y+2)^2}{(4)^2} = 1$

19. If the chord $x\cos\alpha + y\sin\alpha = p$ of the hyperbola $\frac{x^2}{16} - \frac{y^2}{18} = 1$ subtends a right angle at the centre and the diameter of the circle, concentric with the hyperbola, to which the given chord is a tangent is d, then $\frac{d}{4}$ is equal to _____

Sol. **(6.00)**

$$\frac{x^2}{16}-\frac{y^2}{18}-\left(\frac{x\cos\alpha+y\sin\alpha}{P}\right)=0$$

$\Rightarrow$ Coefficient of x^2 + Coefficient of $y^2 = 0$

$\Rightarrow P = \pm 12$

The chord $x\cos\alpha + y\sin\alpha \pm 12 = 0$ is tangent to the circle $x^2+y^2=\left(\frac{d}{2}\right)^2$

$$\Rightarrow \frac{d}{4}=6$$

20. If 2 distinct tangents can be drawn from $(\alpha, 2)$ on different branches of $\frac{x^2}{9}-\frac{y^2}{16}=1$ and $\alpha \in (a, b)$, then $(b - a)$ is _____

Sol. **(3.00)**

Asymptotes are $y=\pm\frac{4x}{3}$

$$\pm\frac{4x}{3}=2 \Rightarrow x=\pm\frac{3}{2}$$

21. If a chord of a rectangular hyperbola parallel to its conjugate axis subtends angles θ_1 and θ_2 at its vertices, then $(\theta_1 + \theta_2)$ is _____

Sol. **(3.14)**

Let H: $x^2 - y^2 = a^2$

Let L: $x = k$

$$\tan\left(\frac{\theta_1}{2}\right)=\frac{\sqrt{k^2-a^2}}{k-a},\ \tan\left(\frac{\theta_2}{2}\right)=\frac{\sqrt{k^2-a^2}}{k+a}$$

$$\therefore \tan\left(\frac{\theta_1+\theta_2}{2}\right)\to\infty \Rightarrow \theta_1+\theta_2=\pi$$

22. The curves $(x - 1)(y - 2) = 5$ and $(x - 1)^2 + (y + 2)^2 = r^2$ intersect at 4 points A, B, C and D. If centroid of ΔABC lies on $y = 3x - 4$, and locus of D is $x = \lambda y$, then λ is _____

Sol. **(0.33)**

$$\frac{\sum_{i=1}^{4}x_i}{4}=1\ ;\ \sum_{i=1}^{4}y_i=0$$

Let $D(x_4, y_4)$

$$\therefore G\equiv\left(\frac{4-x_4}{3},\frac{-y_4}{3}\right)$$

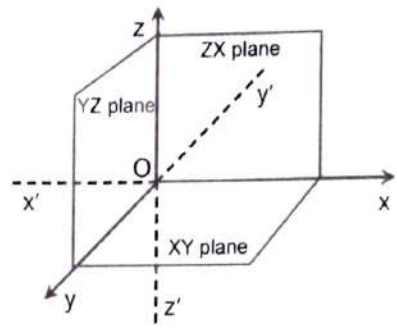
$$\therefore \text{Locus is } -\frac{y}{3}=4-x-4$$

$$\Rightarrow y = 3x$$

THREE DIMENSIONAL GEOMETRY 13

In the three-dimensional space any point on

- x-axis $= \{(x, y, z) \mid y = z = 0\}$
- y-axis $= \{(x, y, z) \mid x = z = 0\}$
- z-axis $= \{(x, y, z) \mid x = y = 0\}$
- xy-plane $= \{(x, y, z) \mid z = 0\}$
- yz-plane $= \{(x, y, z) \mid x = 0\}$
- zx-plane $= \{(x, y, z) \mid y = 0\}$



Notes:

- Distance between the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$
- The point dividing the line joining $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ in m : n ratio is $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}, \frac{mz_2 + nz_1}{m+n}\right)$ where $m + n \neq 0$.
- The coordinates of centroid of a triangle having vertices A (x_1, y_1, z_1), B (x_2, y_2, z_2) and C (x_3, y_3, z_3) is G $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3}\right)$.
- $l^2 + m^2 + n^2 = 1$, where l, m, n are the direction cosines.
- The unit vector along the line can be written as $l\hat{i} + m\hat{j} + n\hat{k}$.
- If direction cosine of line AB is (l, m, n), then direction cosine of line BA will be (−l, −m, −n).

DIRECTION RATIOS

If a, b, c are direction ratios, $l = \pm \frac{a}{\sqrt{\sum a^2}}$.

Similarly, $m = \pm \frac{b}{\sqrt{\sum a^2}}$ and $n = \pm \frac{c}{\sqrt{\sum a^2}}$ where the same sign either positive or negative is to be chosen throughout.

Notes:

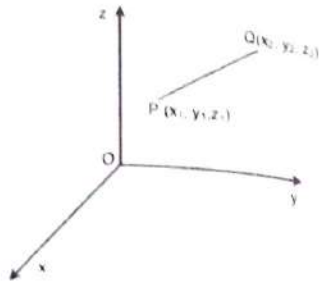
- Direction cosines of a line are unique but direction ratios of a line in no way unique but can be infinite.
- Any vector along the given direction a, b, c is $a\hat{i} + b\hat{j} + c\hat{k}$.

Direction Cosines of a Line joining two given Points

The direction ratios of the line PQ joining $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ are $x_2 - x_1 = a$(say), $y_2 - y_1 = b$ (say) and $z_2 - z_1 = c$ (say).

Then the direction cosines are

$$l = \frac{(x_2 - x_1)}{\sqrt{\Sigma(x_2 - x_1)^2}}, m = \frac{(y_2 - y_1)}{\sqrt{\Sigma(x_2 - x_1)^2}}, n = \frac{(z_2 - z_1)}{\sqrt{\Sigma(x_2 - x_1)^2}}$$

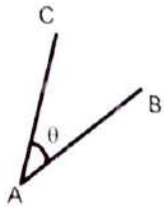


Angle between two Lines

Let θ be the angle between two straight lines AB and AC whose direction cosines are l_1, m_1, n_1 and l_2, m_2, n_2 respectively, is given by $\cos\theta = l_1l_2 + m_1m_2 + n_1n_2$.

If direction ratios of two lines are a_1, b_1, c_1 and a_2, b_2, c_2 are given, then the angle between two lines is given by

$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \cdot \sqrt{a_2^2 + b_2^2 + c_2^2}}$$



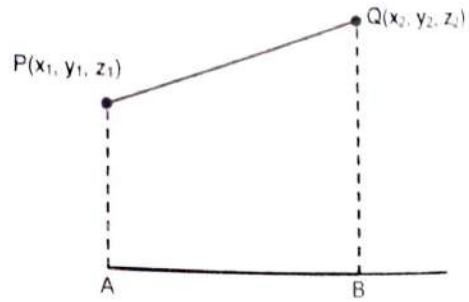
Particular Results

- We have, $\sin^2\theta = 1 - \cos^2\theta = (l_1^2 + m_1^2 + n_1^2)(l_2^2 + m_2^2 + n_2^2) - (l_1l_2 + m_1m_2 + n_1n_2)^2$
  $= (l_1m_2 - l_2m_1)^2 + (m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2 \Rightarrow \sin\theta = \pm\sqrt{\sum(l_1m_2 - l_2m_1)^2}$.
- **Condition of perpendicularity:**
  If the given lines are perpendicular, then $\theta = 90°$ i.e. $\cos\theta = 0$
  $\Rightarrow l_1l_2 + m_1m_2 + n_1n_2 = 0$ or $a_1a_2 + b_1b_2 + c_1c_2 = 0$.
- **Condition of parallelism:**
  If the given lines are parallel, then $\theta = 0°$ i.e. $\sin\theta = 0$
  $\Rightarrow (l_1m_2 - l_2m_1)^2 + (m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2 = 0$ which is true, only when
  $l_1m_2 - l_2m_1 = 0$, $m_1n_2 - m_2n_1 = 0$ and $n_1l_2 - n_2l_1 = 0$
  $\Rightarrow \frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2}$. Similarly, $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

Projection of a Line segment on a given Line

Projection of the line joining two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ on another line whose direction cosines are l, m, n is $AB = l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)$

Here PQ and AB need not be coplanar.

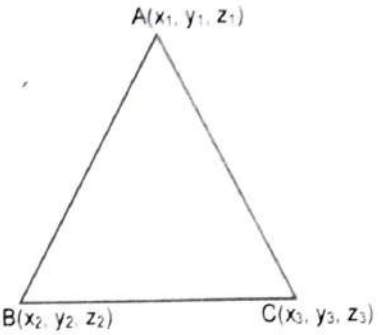


AREA OF A TRIANGLE

$$\Delta_x = \frac{1}{2}\begin{vmatrix} y_1 & z_1 & 1 \\ y_2 & z_2 & 1 \\ y_3 & z_3 & 1 \end{vmatrix}, \Delta_y = \frac{1}{2}\begin{vmatrix} x_1 & z_1 & 1 \\ x_2 & z_2 & 1 \\ x_3 & z_3 & 1 \end{vmatrix}, \Delta_z = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

So, area of ΔABC is given by the relation

$\Delta^2 = \Delta_x^2 + \Delta_y^2 + \Delta_z^2$



Equation of Plane in Different Forms

- General equation of a plane is $ax + by + cz + d = 0$. (where a, b, c gives the direction ratios of the normal to the plane).
- Equation of the plane in normal form is $lx + my + nz = p$ where p is the length of the normal from the origin to the plane and (l, m, n) be the direction cosines of the normal.
- The equation to the plane passing through $P(x_1, y_1, z_1)$ and is perpendicular to the given line having direction ratios (a, b, c) is $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$.
- The equation of the plane passing through three non-collinear points (x_1, y_1, z_1), (x_2, y_2, z_2) and (x_3, y_3, z_3) is $\begin{vmatrix} (x-x_1) & (y-y_1) & (z-z_1) \\ (x_2-x_1) & (y_2-y_1) & (z_2-z_1) \\ (x_3-x_1) & (y_3-y_1) & (z_3-z_1) \end{vmatrix} = 0$.
- The equation of the plane whose intercepts are a, b, c on the x, y, z axes respectively, is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ (a, b, $c \neq 0$).
- Equation of YZ plane is $x = 0$, equation of the plane parallel to YZ-plane is $x = d$.
  Equation of ZX plane is $y = 0$, equation of the plane parallel to ZX-plane is $y = d$.
  Equation of XY plane is $z = 0$, equation of the plane parallel to XY-plane is $z = d$.
- Four points namely A (x_1, y_1, z_1), B (x_2, y_2, z_2), C (x_3, y_3, z_3) and D (x_4, y_4, z_4) will be coplanar if one point lies on the plane passing through the other three points.

Angle between the Planes

Angle between the planes is defined as the angle between normals to the planes drawn from any point. Angle between the planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ is

$$\cos^{-1}\left(\frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{\left(a_1^2 + b_1^2 + c_1^2\right)\left(a_2^2 + b_2^2 + c_2^2\right)}}\right).$$

Notes:

- If $a_1a_2 + b_1b_2 + c_1c_2 = 0$, then the planes are perpendicular to each other.
- If $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ then the planes are parallel to each other.

Perpendicular Distance

The length of the perpendicular from the point $P(x_1, y_1, z_1)$ to the plane $ax + by + cz + d = 0$ is $\left|\frac{ax_1 + by_1 + cz_1 + d}{\sqrt{a^2 + b^2 + c^2}}\right|$

Note:

- The distance between two parallel planes is the algebraic difference of perpendicular distances on the planes from the origin.

Family of Planes

Equation of the plane passing through the line of intersection of two planes $u = 0$ and $v = 0$ is $u + \lambda v = 0$

Bisector Planes of Angle between two Planes

The equation of the planes bisecting the angles between two given planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ is $\frac{a_1x + b_1y + c_1z + d_1}{\sqrt{\left(a_1^2 + b_1^2 + c_1^2\right)}} = \pm\frac{a_2x + b_2y + c_2z + d_2}{\sqrt{\left(a_2^2 + b_2^2 + c_2^2\right)}}$.

Notes:

- If the angle between bisector plane and one of the plane is less than 45° then it is an acute angle bisector otherwise it is an obtuse angle bisector.
- If $a_1a_2 + b_1b_2 + c_1c_2$ is negative, then the origin lies in the acute angle between the given planes provided d_1 and d_2 are of same sign and if $a_1a_2 + b_1b_2 + c_1c_2$ is positive, then the origin lies in the obtuse angle between the given planes.

EQUATION OF STRAIGHT LINE IN DIFFERENT FORMS

Symmetrical Form

- Equation of straight line passing through point $P(x_1, y_1, z_1)$ and whose direction cosines are l, m, n is $\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} = r$
- Equation of straight line passing through two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is $\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$.

Note:

- The general coordinates of a point on a line is given by $(x_1 + lr, y_1 + mr, z_1 + nr)$ where r is distance between point (x_1, y_1, z_1) and the point whose coordinates is to be written.

Angle between a Line and Plane

If the equation of a plane is $ax + by + cz + d = 0$, then direction ratios of normal to this plane are a, b, c. The equation of straight line is $\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n}$, then the angle between normal to the plane and straight line is given by $\cos\theta = \frac{al + bm + cn}{\sqrt{a^2 + b^2 + c^2}}$.

Thus angle between the plane and the straight line is $\frac{\pi}{2} - 0$.

- Plane and straight line will be parallel if $al + bm + cn = 0$.
- Plane and straight line will be perpendicular if $\frac{a}{l} = \frac{b}{m} = \frac{c}{n}$.
- The line may lie in the plane if $al + bm + cn = 0$ and $ax_1 + by_1 + cz_1 = 0$.

Equation of the Plane through a given Line

(i) If the equation of line is given in general form as $a_1x + b_1y + c_1z + d_1 = 0 = a_2x + b_2y + c_2z + d_2$ the equation of plane passing through this line is
$(a_1x + b_1y + c_1z + d_1) + \lambda(a_2x + b_2y + c_2z + d_2) = 0$.

(ii) If equation of the line is given in symmetrical form as $\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n}$, then the equation of plane is $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$, where a, b, c are given by $al + bm + cn = 0$.

Notes:

- The shortest distance between two lines can be given by
$$\frac{1}{\sqrt{\Sigma(l_1m_2 - l_2m_1)^2}} \begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix}.$$
- If any straight line is given in general form then it can be transformed into symmetrical form and we can further proceed.

ILLUSTRATIVE PROBLEMS

1. Prove that the three points A (3, – 2, 4), B (1, 1, 1) and C (– 1, 4, – 2) are collinear.

Sol. The general coordinates of a point R which divides the line joining A (3, – 2, 4) and B (1, 1, 1) in the ratio $\mu : 1$ are

$$\left(\frac{\mu+3}{\mu+1}, \frac{\mu-2}{\mu+1}, \frac{\mu+4}{\mu+1}\right) \quad \ldots(1)$$

If C (–1, 4, –2) lies on the line AB, then for some values of μ the coordinates of the point R will be the same as those of C.

Let the x-coordinate of point R = x-coordinate of point C.

Then, $\frac{\mu+3}{\mu+1} = -1$

$\Rightarrow \mu = -2$

Putting $\mu = -2$ in (1) the coordinates of R are (–1, 4, –2) which are also the coordinates of C.

Hence the points A, B, C are collinear.

2. If l_1, m_1, n_1 and l_2, m_2, n_2 are the direction cosine of two concurrent lines, show that the direction cosine of two lines bisecting the angles between them are proportional to $l_1 \pm l_2, m_1 \pm m_2, n_1 \pm n_2$.

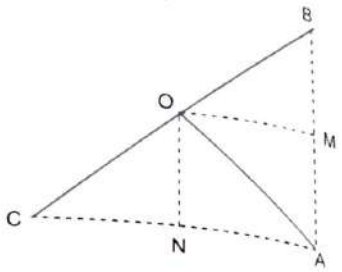
Sol. Let O be the origin. Draw OA and OB parallel to the given concurrent lines. Cut off OA = OB = r. Then the coordinate of A and B are (l_1r, m_1r, n_1r) and (l_2r, m_2r, n_2r) respectively.

Again take a point C on BO produced such that OB = OC = r.

Thus the coordinates of C are

$(-l_2r, -m_2r, -n_2r)$.

Suppose M and N are the middle points of AB and CA. Then the coordinates of M and N are



$$\left(\frac{l_1r + l_2r}{2}, \frac{m_1r + m_2r}{2}, \frac{n_1r + n_2r}{2}\right)$$

and $\left(\frac{l_1r - l_2r}{2}, \frac{m_1r - m_2r}{2}, \frac{n_1r - n_2r}{2}\right)$ respectively.

Clearly OM and ON are the internal and external bisectors of the angle AOB. Hence the direction ratios of OM and ON are

$\frac{1}{2}(l_1 + l_2)r, \frac{1}{2}(m_1 + m_2)r, \frac{1}{2}(n_1 + n_2)r$

and $\frac{1}{2}(l_1 - l_2)r, \frac{1}{2}(m_1 - m_2)r,$

$\frac{1}{2}(n_1 - n_2)r$ respectively

i.e. the d.c's of OM and ON are proportional to $l_1 + l_2, m_1 + m_2, n_1 + n_2$ and $l_1 - l_2, m_1 - m_2, n_1 - n_2$.

3. Find the equation of the plane which contains the two parallel lines $\frac{x+1}{3} = \frac{y-2}{2} = \frac{z}{1}$ and $\frac{x-3}{3} = \frac{y+4}{2} = \frac{z-1}{1}$.

Sol. The equations of the two parallel lines are

$$\frac{x+1}{3} = \frac{y-2}{2} = \frac{z-0}{1} \quad \ldots (1)$$

and $$\frac{x-3}{3} = \frac{y+4}{2} = \frac{z-1}{1} \quad \ldots (2)$$

the equation of any plane through the line (1) is

$a(x + 1) + b(y - 2) + cz = 0$ .... (3)

where $3a + 2b + c = 0$ .... (4)

the line (2) will also lie on the plane (3) if the point (3, –4, 1) lies on the plane (3), and for this we have

$a(3 + 1) + b(-4 - 2) + c = 0$ or $4a - 6b + c = 0$ .... (5)

Solving (4) and (5), we get $\frac{a}{8} = \frac{b}{1} = \frac{c}{-26}$.

Putting the values of a, b, c in (3), the required equation of the plane is $8x + y - 26z + 6 = 0$.

4. **A line with direction cosines proportional to (2, 7, −5) is drawn to intersect the lines $\frac{x-5}{3}=\frac{y-7}{-1}=\frac{z+2}{1}$ and $\frac{x+3}{-3}=\frac{y-3}{2}=\frac{z-6}{4}$. Find the coordinate of the points of intersection and the length intercepted on it. Also find the equation of intersecting straight line.**

Sol. The given equations

$$\frac{x-5}{3}=\frac{y-7}{-1}=\frac{z+2}{1} \quad ...(1)$$

and $$\frac{x+3}{-3}=\frac{y-3}{2}=\frac{z-6}{4} \quad ...(2)$$

any point P on (1) is $(3r_1 + 5, -r_1 + 7, r_1 - 2)$ and any point Q on (2) is $(-3r_2 - 3, 2r_2 + 3, 4r_2 + 6)$

the direction ratios of PQ are

$(3r_1 + 3r_2 + 8, -r_1 - 2r_2 + 4, r_1 - 4r_2 - 8)$ ...(3)

Suppose the line with d.r's 2,7,−5 will be proportional to the d.r's given by (3)

$$\therefore \quad \frac{3r_1+3r_2+8}{2}=\frac{-r_1-2r_2+4}{7}=\frac{r_1-4r_2-8}{-5} \quad ...(4)$$

Solving (4), we get $r_1 = r_2 = -1$

So the point of intersection are P(2, 8, −3) and Q(0, 1, 2)

and intercepted length

$$= PQ = \sqrt{(2-0)^2+(8-1)^2+(-3-2)^2} = \sqrt{78}$$

and equation of PQ is $\frac{x-2}{2}=\frac{y-8}{7}=\frac{z+3}{-5}$.

5. **Find the shortest distance between the lines $\frac{x-3}{3}=\frac{y-8}{-1}=\frac{z-3}{1}, \frac{x+3}{-3}=\frac{y+7}{2}=\frac{z-6}{4}$. Also find the equation of line of the shortest distance.**

Sol. Given lines are $\frac{x-3}{3}=\frac{y-8}{-1}=\frac{z-3}{1}=r_1$ (say) ... (1)

$$\frac{x+3}{-3}=\frac{y+7}{2}=\frac{z-6}{4}=r_2 \text{ (say)} \quad ... (2)$$

Any point on line (1) is P $(3r_1 + 3, 8 - r_1, r_1 + 3)$ and on line (2) is Q $(-3 - 3r_2, 2r_2 - 7, 4r_2 + 6)$. If PQ is line of shortest distance, then

Direction ratios of PQ = $(3r_1 + 3) - (-3 - 3r_2), (8 - r_1) - (2r_2 - 7), (r_1 + 3) - (4r_2 + 6)$

i.e. $3r_1 + 3r_2 + 6, -r_1 - 3r_2 + 15, r_1 - 4r_2 - 3$.

As PQ is perpendicular to lines (1) and (2)

$\therefore 3(3r_1 + 3r_2 + 6) - 1(-r_1 - 2r_2 + 15) + 1(r_1 - 4r_2 - 3) = 0$

$\Rightarrow 11r_1 + 7r_2 = 0$ ... (3)

and $-3(3r_1 + 3r_2 + 6) + 2(-r_1 - 2r_2 + 15) + 4(r_1 - 4r_2 - 3) = 0$

i.e. $7r_1 + 11r_2 = 0$ ... (4)

On solving equations (3) and (4), we get $r_1 = r_2 = 0$.

So, point P (3, 8,3) and Q (−3, −7, 6)

$\therefore$ Length of shortest distance PQ = $\sqrt{\{(-3-3)^2+(-7-8)^2+(6-3)^2\}} = 3\sqrt{30}$

Direction ratios of the shortest distance line is 2, 5, −1

$\therefore$ Equation of the shortest distance line $\frac{x-3}{2}=\frac{y-8}{5}=\frac{z-3}{-1}$.

SOLVED PROBLEMS

1. The shortest distance between the lines $2x + y + z - 1 = 0 = 3x + y + 2z - 2$ and $x = y = z$

(A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$

(C) $\frac{3}{\sqrt{2}}$ (D) $\frac{\sqrt{3}}{2}$

Sol. **(A).** Any plane passing through first line $2x + y + z - 1 + \lambda(3x + y + 2z - 2) = 0$, if it is parallel to second line

$(2 + 3\lambda)1 + (1 + \lambda)1 + (1 + 2\lambda)1 = 0 \Rightarrow \lambda = -\frac{2}{3}$.

Plane is $y - z + 1 = 0$

Distance from $(0, 0, 0) = \frac{1}{\sqrt{2}}$.

2. The distance of point of intersection of lines $\frac{x-4}{1} = \frac{y+3}{-4} = \frac{z+1}{7}$ and $\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8}$ from $(1, -4, 7)$

(A) 6 (B) $\sqrt{24}$

(C) $\sqrt{26}$ (D) none of these

Sol. **(C).** $4 + \lambda_1 = 1 + 2\lambda_2 \Rightarrow \lambda_1 - 2\lambda_2 = -3$ ... (1)

$-3 - 4\lambda_1 = -1 - 3\lambda_2 \Rightarrow 4\lambda_1 - 3\lambda_2 = -2$ ... (2)

From (1) and (2),

$-5 \times 2 = -10$

$\lambda_2 = 2, \lambda_1 = 1$

$\Rightarrow$ point of intersection $(4 + 1, -3 - 4, -1 + 7)$

i.e. $(5, -7, 6)$.

Distance of $(5, -7, 6)$ from $(1, -4, 7)$

$= \sqrt{16 + 9 + 1} = \sqrt{26}$.

3. If lines $\frac{x-1}{2} = \frac{y-2}{x_1} = \frac{z-3}{x_2}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ lies in the same plane then for equation $x_1t^2 + (x_2 + 2)t + a = 0$

(A) $2x_1 - x_2 = 1$ (B) sum of roots of above equation $= -2$

(C) $2x_1 + x_2 = -4$ (D) sum of root is 0

Sol. **(B).** Lines are coplanar if $\begin{vmatrix} 1 & 1 & 1 \\ 2 & x_1 & x_2 \\ 3 & 4 & 5 \end{vmatrix} = 0$

$\Rightarrow 2x_1 - x_2 = 2$

Also sum of roots $= \frac{-(x_2 + 2)}{x_1} = \frac{-2x_1}{x_1} = -2$.

4. The distance of the point $(2, -1, 3)$ from the plane $x - 2y + 3z - 3 = 0$, measured parallel to the line $\frac{x}{1} = \frac{y}{2} = \frac{z-1}{3}$ is equal to

(A) 1 unit (B) 3 unit

(C) 2 unit (D) none of these

Sol. **(D).** Equation of line through (2, – 1, 3) and parallel to given line is $\frac{x-2}{1}=\frac{y+1}{2}=\frac{z-3}{3}$

Any point on it can be taken as (r + 2, 2r – 1, 3r + 3)
Putting this in equation of plane, we get r + 2 – 2(2r – 1) + 3(3r + 3) – 3 = 0
r + 2 – 4r + 2 + 9r + 9 – 3 = 0
6r + 10 = 0

$6r=-10 \Rightarrow r=-\frac{5}{3}$

$\left(-\frac{5}{3}+2, 2\left(-\frac{5}{3}\right)-1, 3\left(-\frac{5}{3}\right)+3\right) \Rightarrow \left(\frac{1}{3}, -\frac{13}{3}, -2\right)$

Required distance $=\sqrt{\left(\frac{1}{3}-2\right)^2+\left(-\frac{13}{3}+1\right)^2+(-2-3)^2}$

$=\sqrt{\frac{350}{9}}=\sqrt{\frac{350}{3}}=\frac{5\sqrt{14}}{3}$ unit.

5. A mirror and a source of light are situated at the origin O and at a point on OX respectively. A ray of light from the source along the x–axis strikes the mirror and is reflected. If the direction ratios of the normal to the plane of mirror are 1 , – 1, 1, the direction cosines of the reflected ray are

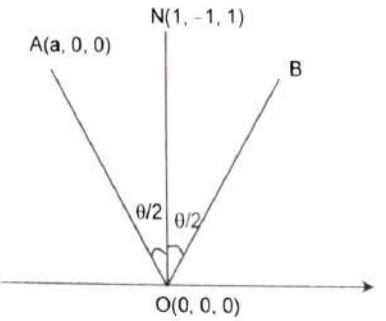
(A) $\frac{1}{3}, \frac{2}{3}, \frac{2}{3}$ (B) $-\frac{1}{3}, -\frac{2}{3}, \frac{2}{3}$

(C) $-\frac{1}{3}, -\frac{2}{3}, -\frac{2}{3}$ (D) none of these

Sol. **(B).** Direction ratios of OA = a, 0, 0
$\Rightarrow$ direction cosine's of OA = 1, 0, 0

Direction cosines of ON = $\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

$\therefore \cos\frac{\theta}{2}=\frac{1}{\sqrt{3}}$



Let l, m, n be the direction cosines of the reflected ray OB.

Then $\frac{l+1}{2\cos\frac{\theta}{2}}=\frac{1}{\sqrt{3}}$, $\frac{m+0}{2\cos\frac{\theta}{2}}=-\frac{1}{\sqrt{3}}$ and $\frac{n+0}{2\cos\frac{\theta}{2}}=\frac{1}{\sqrt{3}}$

$\Rightarrow l=-\frac{1}{3}, m=-\frac{2}{3}, n=\frac{2}{3}$.

6. The equation of the line passing through the point (1, – 2, 3) and parallel to the line x – y + 2z – 5 = 3x + y + z – 6 = 0 is

(A) $\frac{x-1}{-3}=\frac{y+2}{5}=\frac{z-3}{4}$ (B) $\frac{x+2}{-6}=\frac{y-3}{10}=\frac{z-7}{8}$

(C) $\frac{x-1}{3}=\frac{y+2}{1}=\frac{z-3}{6}$ (D) $\frac{x-1}{3}=\frac{y+2}{-1}=\frac{z-3}{2}$

Sol. **(A), (B)**
If l, m, n be the direction cosines of the line, then as it lies in both the given planes it is perpendicular to their normals.

i.e. l – m + 2n = 0 and 3l + m + n = 0 or, $\frac{l}{-3}=\frac{m}{5}=\frac{n}{4}$

$\therefore$ equation of the line is $\frac{x-1}{-3}=\frac{y+2}{5}=\frac{z-3}{4}$ and $\frac{x+2}{-6}=\frac{y-3}{10}=\frac{z-7}{8}$

7. The line $\frac{x-2}{3}=\frac{y+1}{2}=\frac{z-1}{-1}$ intersects the curve $xy = c^2, z = 0$ if c is equal to

(A) ± 1 (B) $\pm \frac{1}{3}$

(C) $\pm\sqrt{5}$ (D) none of these

Sol. **(C).** For the points where the line intersects the curve, we have $z = 0$. Therefore

$\frac{x-2}{3}=\frac{y+1}{2}=\frac{0-1}{-1} \Rightarrow \frac{x-2}{3}=1$ and $\frac{y+1}{2}=1$ $x = 5$ and $y = 1$

put these values in $xy = c^2$, we get $c = \pm\sqrt{5}$.

8. The equation of a plane through the line of intersection of $2x + 3y + z - 1 = 0$ and $x + 5y - 2z + 7 = 0$ and parallel to the line $y = 0 = z$ is

(A) $7x - 5y + 15 = 0$ (B) $7y - 5z + 15 = 0$

(C) $4x + 7y - 5z + 15 = 0$ (D) $13y - 3z + 13 = 0$

Sol. **(B).** The equation of the plane passing through the point of intersection of the given planes is

$(2x + 3y + z - 1) + k(x + 5y - 2z + 7) = 0$

$\Rightarrow x(2 + k) + y(3 + 5k) + z(1 - 2k) + (7k - 1) = 0$

This plane is parallel to x–axis

Therefore $1(2 + k) + 0(3 + 5k) + 0(1 - 2k) = 0$ $k = -2$

Hence the required equation is $-7y + 5z - 15 = 0$.

9. The lines $x = ay + b$, $z = cy + d$ and $x = a_1y + b_1$; $z = c_1y + d_1$ are perpendicular if

(A) $aa_1 + cc_1 = 1$ (B) $aa_1 + cc_1 = -1$

(C) $bb_1 + dd_1 = 1$ (D) $bb_1 + dd_1 = -1$

Sol. **(B).** The lines are $x = ay + b$; $z = cy + d$

$x = a_1y + b_1$; $z = c_1y + d_1$ $\Rightarrow$ the given equations can be written as

$\frac{x-b}{a}=\frac{y}{1}=\frac{z-d}{c}$ ... (1)

and $\frac{x-b_1}{a_1}=\frac{y}{1}=\frac{z-d_1}{c_1}$ ... (2)

The line (1) and (2) will be perpendicular if $aa_1 + 1 + cc_1 = 0 \Rightarrow aa_1 + cc_1 = -1$.

10. A line with direction cosines proportional to 2, 1, 2 meets each of the lines given by the equation $x = y + 2 = z$; $x + 2 = 2y = 2z$. The coordinates of the points of intersection are

(A) (6, 4, 6), (2, 4, 2) (B) (6, 6, 6), (2, 6, 2)

(C) (6, 4, 6), (2, 2, 0) (D) none of these

Sol. **(D).** Let P(r, r – 2, r) and Q(2k – 2, k, k) are the general coordinates of points on the two given lines.

$\therefore$ Dr's of PQ are $(r - 2k + 2, r - k - 2, r - k)$

$\therefore \frac{r-2k+2}{2}=\frac{r-k-2}{1}=\frac{r-k}{2} \Rightarrow r = 6, k = 2$

$\therefore$ Points of intersection are (6, 4, 6) and (2, 2, 2).

11. The line $\frac{x}{k}=\frac{y}{2}=\frac{z}{-12}$ makes an isosceles triangle with the planes $2x + y + 3z - 1 = 0$ and $x + 2y - 3z - 1 = 0$, then the value of k is

(A) 1 (B) –2

(C) 3 (D) 4

Sol. **(B).** The given line will be parallel to one of the bisector planes of the given planes hence equations of bisector planes $= \frac{2x+y+3z-1}{\sqrt{14}} = \pm\frac{(x+2y-3z-1)}{\sqrt{14}}$

$\Rightarrow 3x + 3y - 2 = 0$ and $x - y + 6z = 0$

$\Rightarrow k = -2.$

12. Direction cosine of normal to the plane containing lines $x = y = z$ and $x - 1 = y - 1 = \frac{z-1}{d}$ (where $d \in R - \{1\}$), are

(A) $\left\{\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right\}$

(B) $\left\{\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right\}$

(C) $\left\{0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right\}$

(D) none of these

Sol. **(A).** Let l, m, n be the direction ratios of the normal

$l + m + n = 0$

$l + m + nd = 0$

$\Rightarrow n(1 - d) = 0$

$\Rightarrow n = 0$

$\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right), \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right).$

13. If the line $x = y = z$ intersect the line

$\sin A\, x + \sin B\, y + \sin C\, z = 2d^2$, $\sin 2A\, x + \sin 2B\, y + \sin 2C\, z = d^2$, then $\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$ is equal to (where $A + B + C = \pi$)

(A) $\frac{1}{16}$

(B) $\frac{1}{8}$

(C) $\frac{1}{32}$

(D) $\frac{1}{12}$

Sol. **(A).** Let $(\lambda, \lambda, \lambda)$ be the point of intersection of two lines.

$\Rightarrow \lambda(\sin A + \sin B + \sin C) = 2d^2$ and $\lambda(\sin 2A + \sin 2B + \sin 2C) = d^2$

$\Rightarrow \frac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C} = \frac{1}{2}$

$\Rightarrow \sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} = \frac{1}{16}.$

Comprehension

A circle is the locus of a point in a plane such that its distance from a fixed point in the plane is constant. Anologously, a sphere is the locus of a point in space such that its distance from a fixed point in space is constant. The fixed point is called the centre and the constant distance is called the radius of the circle/sphere. In anology with the equation of the circle $|z - c| = a$, the equation of a sphere of radius a is $|\vec{r} - \vec{c}| = a$, where $\vec{c}$ is the position vector of the centre and $\vec{r}$ is the position vector of any point on the surface of the sphere. In Cartesian system, the equation of the sphere, with centre at $(-g, -f, -h)$ is

$x^2 + y^2 + y^2 + 2gx + 2fy + 2hz + c = 0$ and its radius is $\sqrt{f^2 + g^2 + h^2 - c}$.

Answer the following questions

14. Radius of the sphere, with $(2, -3, 4)$ and $(-5, 6, -7)$ as extremities of a diameter, is

(A) $\sqrt{\frac{251}{2}}$ (B) $\sqrt{\frac{251}{3}}$

(C) $\sqrt{\frac{251}{4}}$ (D) $\sqrt{\frac{251}{5}}$

Sol. **(C)**

The centre of the sphere is at the mid-point of the extremities of a diameter

$\Rightarrow$ the centre is $\left(-\frac{3}{2}, \frac{3}{2}, -\frac{3}{2}\right)$

and hence the radius $= \sqrt{\left(\frac{7}{2}\right)^2 + \left(\frac{9}{2}\right)^2 + \left(\frac{11}{2}\right)^2}$.

15. The centre of the sphere $(x - 4)(x + 4) + (y - 3)(y + 3) + z^2 = 0$ is

(A) $(4, 3, 0)$ (B) $(-4, -3, 0)$

(C) $(0, 0, 0)$ (D) none of these

Sol. **(C)**

Equation of the circle can be written as

$x^2 - 16 + y^2 - 9 + z^2 = 0$

or $x^2 + y^2 + z^2 = 25$

$\Rightarrow$ the centre is at $(0, 0, 0)$.

16. Equation of the sphere having centre at $(3, 6, -4)$ and touching the plane $\vec{r} \cdot \left(2\hat{i} - 2\hat{j} - \hat{k}\right) = 10$, is $(x - 3)^2 + (y - 6)^2 + (z + 4)^2 = k^2$, where k is equal to

(A) 3 (B) 4

(C) 6 (D) $\sqrt{17}$

Sol. **(B)**

Distance of the point $(3, 6, -4)$ from the given plane is equal to the radius of the sphere

$\Rightarrow$ the radius of the sphere $= \left|\frac{\left(3\hat{i} + 6\hat{j} - 4\hat{k}\right) \cdot \left(2\hat{i} - 2\hat{j} - \hat{k}\right) - 10}{\sqrt{4+4+1}}\right|$

$= \left|\frac{6 - 12 + 4 - 10}{3}\right| = 4$.

Assertion Reason

17. Statement 1: If point (α, β, γ) lies above the plane $(a^2 + 1)x + (b + 1)y + (c^2 + c + 1)z + d = 0$, then $(a^2 + 1)\alpha + (b + 1)\beta + (c^2 + c + 1)\gamma + d > 0$.

because

Statement 2: If the point (α, β, γ) lies above the plane $ax + by + cz + d = 0$ then $\frac{a\alpha + b\beta + c\gamma + d}{c} > 0$.

(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement -1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement -1 is True, Statement -2 is False

(D) Statement -1 is False, Statement -2 is True

Sol. Let the feet of perpendicular from (α, β, γ) to the xy plane intersects $ax + by + cz + d = 0$ at $(\alpha, \beta, \gamma_1)$ then $a\alpha + b\beta + c\gamma_1 + d = 0$

$\Rightarrow \gamma - \gamma_1 = \dfrac{a\alpha + b\beta + c\gamma + d}{c}$

$\Rightarrow$ The point (α, β, γ) lies above the plane if $\gamma - \gamma_1 > 0$

i.e. $\dfrac{a\alpha + b\beta + c\gamma + d}{c} > 0$

since $c^2 + c + 1 > 0$

$\Rightarrow$ statement 1 is true and 2 also explains 1.

18. Match the following:

	Column – I		Column – II
(A)	If the line $\dfrac{x-1}{1} = \dfrac{y+1}{-2} = \dfrac{z+1}{\lambda}$ lies in the plane $3x - 2y + 5z = 0$, then λ is equal to	(p)	$\sin^{-1}\sqrt{\dfrac{6}{25}}$
(B)	If $(3, \lambda, \mu)$ is a point on the line $2x + y + z - 3 = 0 = x - 2y + z - 1$, then $\lambda + \mu$ is equal to	(q)	$-\dfrac{7}{5}$
(C)	The angle between the line $x = y = z$ and the plane $4x - 3y + 5z = 2$ is	(r)	-3
(D)	The angle between the planes $x + y + z = 0$ and $3x - 4y + 5z = 0$ is	(s)	$\cos^{-1}\sqrt{\dfrac{8}{75}}$

Sol. **(A) → (q); (B) → (r); (C) → (p); (D) → (s)**

(A) $3.1 - 2(-2) + 5(\lambda) = 0 \Rightarrow \lambda = -\dfrac{7}{5}$.

(B) Point $(3, \lambda, \mu)$ lies on $2x + y + z - 3 = x - 2y + z - 1 \Rightarrow 3.2 + \lambda + \mu - 3 = 0$ and $3 - 2\lambda + \mu - 1 = 0 \Rightarrow \lambda + \mu + 3 = 0$ and $2\lambda - \mu - 2 = 0$

so $\lambda + \mu = -3$.

(C) $\sin\theta = \dfrac{1\cdot4 + 1(-3) + 1\cdot5}{\sqrt{1^2+1^2+1^2}\sqrt{16+9+25}} = \dfrac{6}{\sqrt{3}\sqrt{50}} = \sin^{-1}\sqrt{\dfrac{6}{25}}$.

D) $\cos\theta = \dfrac{1\cdot3 + 1(-4) + 1\cdot5}{\sqrt{3}\sqrt{16+9+25}} = \dfrac{4}{\sqrt{3}\sqrt{50}}$

$\theta = \cos^{-1}\sqrt{\dfrac{8}{75}}$.

Answer Q.19, Q.20 and Q.21 by appropriately matching the information given in the three columns of the following table.

Column-I: given two lines, Column-II: angle between lines and Column-III: shortest distance between lines

Column–I	Column–II	Column–III
(I) Lines $\vec{r} = (4\hat{i} - \hat{j}) + \lambda(\hat{i} + 2\hat{j} - 3\hat{k})$ and $\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(2\hat{i} + 4\hat{j} - 5\hat{k})$	(i) 0°	(P) $\dfrac{1}{\sqrt{6}}$
(II) Lines $\dfrac{x-1}{2} = \dfrac{y-2}{3} = \dfrac{z-3}{4}$ and $\dfrac{x-2}{3} = \dfrac{y-4}{4} = \dfrac{z-5}{5}$	(ii) $\cos^{-1}\left(\dfrac{23}{\sqrt{532}}\right)$	(Q) $\sqrt{\dfrac{101}{3}}$

(III) Lines $\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \lambda(2\hat{i} + \hat{j} + 2\hat{k})$ and $\vec{r} = 2\hat{i} - \hat{j} - \hat{k} + \mu(2\hat{i} + \hat{j} + 2\hat{k})$	(iii) $\cos^{-1}\left(\frac{25}{3\sqrt{70}}\right)$	(R) $\frac{1}{\sqrt{3}}$
(IV) Lines $\frac{x}{2} = \frac{y}{-3} = \frac{z}{1}$ and $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$	(iv) $\cos^{-1}\left(\frac{38}{5\sqrt{76}}\right)$	(S) $\frac{6}{\sqrt{5}}$

19. Which of the following combination is **CORRECT**?
(A) (I) (i) (R) (B) (III) (ii) (R)
(C) (IV) (iv) (P) (D) (I) (iii) (S)

Sol. **(D)**

20. Which of the following combination is **CORRECT**?
(A) (II) (iii) (S) (B) (II) (iv) (P)
(C) (IV) (iii) (R) (D) (III) (i) (R)

Sol. **(B)**

21. Which of the following combination is **INCORRECT**?
(A) (I) (iii) (S) (B) (II) (iv) (P)
(C) (III) (i) (Q) (D) (IV) (i) (P)

Sol. **(D)**

19.-21. (I) $\cos\theta = \frac{1\times2+2\times4-3\times-5}{\sqrt{1+4+9}\sqrt{4+16+25}} = \frac{2+8+15}{\sqrt{14}\sqrt{45}} = \frac{25}{3\sqrt{70}}$

$\theta = \cos^{-1}\left(\frac{25}{3\sqrt{70}}\right) \Rightarrow$ S.D. $= \frac{6}{\sqrt{5}}$

(II) $\cos\theta = \frac{2\times3+3\times4+4\times5}{\sqrt{4+9+25}\sqrt{9+16+25}} = \frac{6+12+20}{\sqrt{38}\sqrt{50}} = \frac{38}{5\sqrt{38}\sqrt{2}} = \frac{38}{5\sqrt{76}}$

$\theta = \cos^{-1}\left(\frac{38}{5\sqrt{76}}\right)$, S.D. $= \frac{1}{\sqrt{6}}$

(III) $\cos\theta = 0^\circ$, S.D. $= \sqrt{\frac{101}{3}}$

(IV) $\cos\theta = \frac{2\times3-3\times-5+1\times2}{\sqrt{4+9+1}\sqrt{9+25+4}} = \frac{6+15+2}{\sqrt{14}\sqrt{38}} = \frac{23}{\sqrt{532}}$

$\Rightarrow \cos^{-1}\left(\frac{23}{\sqrt{532}}\right) \Rightarrow$ S.D. $= \frac{1}{\sqrt{3}}$

22. Let P(a, b, c) be any point on the plane $3x + 2y + z = 7$, then the least value of $2(a^2 + b^2 + c^2)$ is ____

Sol. **(7.00)**
Minimum value of $a^2 + b^2 + c^2$ means line OP will be perpendicular to the plane $3x + 2y + z - 7 = 0$ where O is origin

So, $OP = \left|\frac{3\times0+2\times0+0-7}{\sqrt{9+4+1}}\right| = \frac{7}{\sqrt{14}} = \frac{\sqrt{7}}{\sqrt{2}}$

$OP = \sqrt{a^2+b^2+c^2} = \sqrt{\frac{7}{2}}$

$\Rightarrow a^2 + b^2 + c^2 = \frac{7}{2} \Rightarrow 2(a^2 + b^2 + c^2) = 7$

23. A variable plane which remains at a constant distance p from the origin cuts the coordinate axes in A, B, C. The locus of the centroid of the tetrahedron OABC is $x^2y^2 + y^2z^2 + x^2z^2 = \frac{k}{p^2}x^2y^2z^2$. Then $\sqrt[5]{2k}$ is _____ (where O is origin)

Sol. **(2.00)**

Let the equation of plane is $\ell x + my + nz = p$

$A\left(\frac{p}{\ell}, 0, 0\right)$, $B\left(0, \frac{p}{m}, 0\right)$, $C\left(0, 0, \frac{p}{n}\right)$ centroid of OABC $\left(\frac{p}{4\ell}, \frac{p}{4m}, \frac{p}{4n}\right) = (\alpha, \beta, \gamma)$ as, we know that

$\ell^2 + m^2 + n^2 = 1 \Rightarrow \ell = \frac{p}{4\alpha}$, $m = \frac{p}{4\beta}$, $n = \frac{p}{4\gamma}$

$\frac{p^2}{16\alpha^2} + \frac{p^2}{16\beta^2} + \frac{p^2}{16\gamma^2} = 1 \Rightarrow \frac{p^2}{16}\left(\alpha^2 \times \beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2\right) = \alpha^2\beta^2\gamma^2$

So, locus is $x^2y^2 + y^2z^2 + z^2x^2 = \frac{16}{p^2}x^2y^2z^2 \Rightarrow k = 16 \Rightarrow (2k)^{1/5} = (32)^{1/5} = 2$

24. If the line $x = y = z$ intersect the line $(\sin A)x + (\sin B)y + (\sin C)z = 2d^2$; $(\sin 2A)x + (\sin 2B)y + (\sin 2C)z = d^2$; $(A + B + C = \pi)$, then $\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} = \frac{1}{4k}$ where k is _____

Sol. **(4.00)**

Let $(\lambda, \lambda, \lambda)$ be the point of intersection of two lines

$\lambda(\sin A + \sin B + \sin C) = 2d^2$

$\lambda(\sin 2A + \sin 2B + \sin 2C) = d^2$

$\therefore k = 4$

25. Let A(+1, –1, 1) and B(–1, 1, –1) be vertices of ΔABC such that $\angle A = \angle B$. Then, locus of C is $\alpha x + \beta y + \gamma z + \delta = 0$ where, $\alpha + \beta + \gamma + \delta$ is _____

Sol. **(1.00)**

Let C(a, b, c)

$AC = BC$

$\therefore (a-1)^2 + (b+1)^2 + (c-1)^2 = (a+1)^2 + (b-1)^2 + (c+1)^2$

$\therefore x - y + z = 0$

So, $\alpha + \beta + \gamma + \delta = 1$

QUADRATIC EQUATIONS 14

An equation of the form $ax^2 + bx + c = 0$, where $a \neq 0$ and a, b, c are real numbers, is called a quadratic equation.

The roots of the quadratic equation are given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

The quantity D ($D = b^2 - 4ac$) is known as the discriminant of the equation.

BASIC RESULTS

- Let α and β be two roots of the given quadratic equation. Then $\alpha+\beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$.
- A quadratic equation, whose roots are α and β can be written as $(x - \alpha)(x - \beta) = 0$ i.e., $ax^2 + bx + c \equiv a(x - \alpha)(x - \beta)$.
- If the quadratic equation is satisfied by more than two distinct numbers (real or complex), then it becomes an identity, i.e. $a = b = c = 0$.
- The quadratic equation has real and equal roots if and only if $D = 0$ i.e. $b^2 - 4ac = 0$.
- The quadratic equation has real and distinct roots if and only if $D > 0$ i.e. $b^2 - 4ac > 0$
- The quadratic equation has complex roots with non-zero imaginary parts if and only if $D < 0$ i.e. $b^2 - 4ac < 0$.
- If $p + iq$ (p and q being real) is a root of the quadratic equation where $i = \sqrt{-1}$, then $p - iq$ is also a root of the quadratic equation. Provided a, b, c are real.
- If $p + \sqrt{q}$ is an irrational root of the quadratic equation, then $p - \sqrt{q}$ is also a root of the quadratic equation provided that all the coefficients are rational.
- The quadratic equation has rational roots if D is a perfect square and a, b, c are rational.
- If $a = 1$ and b, c are integers and the roots of the quadratic equation are rational, then the roots must be integers.

Condition for Common Root(s)

Let $ax^2 + bx + c = 0$ and $dx^2 + ex + f = 0$ have a common root α(say). Then $a\alpha^2 + b\alpha + c = 0$ and $d\alpha^2 + e\alpha + f = 0$.

Solving for α^2 and α, we get

$$\frac{\alpha^2}{bf - ce} = \frac{\alpha}{dc - af} = \frac{1}{ae - bd} \Rightarrow \alpha^2 = \frac{bf - ce}{ae - bd} \text{ and } \alpha = \frac{dc - af}{ae - bd} \Rightarrow (dc - af)^2 = (bf - ce)(ae - bd)$$

which is the required condition for the two equations to have a common root.

Note:

- Condition for both the roots to be common is $\frac{a}{d} = \frac{b}{e} = \frac{c}{f}$.

The Method of Intervals (Wavy Curve Method)

The Method of intervals (or wavy curve) is used for solving inequalities of the form

$$f(x) = \frac{(x - a_1)^{n_1}(x - a_2)^{n_2} \ldots (x - a_k)^{n_k}}{(x - b_1)^{m_1}(x - b_2)^{m_2} \ldots (x - b_p)^{m_p}} > 0 \ (< 0, \leq 0, \text{ or } \geq 0)$$ where $n_1, n_2, \ldots, n_k, m_1, m_2, \ldots, m_p$ are natural numbers and the numbers $a_1, a_2, \ldots, a_k; b_1, b_2, \ldots b_p$ are any real numbers such that $a_i \neq b_j$, where $i = 1, 2, 3, \ldots k$ and $j = 1, 2, 3, \ldots, p$.

It consists of the following statements:

- All zeros[1] of the function f(x) contained on the left hand side of the inequality should be marked on the number line with inked (black) circles.
- All points of discontinuities[2] of the function f(x) contained on the left hand side of the inequality should be marked on the number line with un-inked (white) circles.
- Check the value of f(x) for any real number greater than the right most marked number on the number line.
- From right to left, beginning above the number line (in case of value of f(x) is positive in step (iii) otherwise from below the number line), a wavy curve should be drawn which passes through all the marked points so that when passes through a simple point[3], the curve intersects the number line, and, when passing through a double point[4], the curve remains located on one side of the number line.
- The appropriate intervals are chosen in accordance with the sign of inequality (the function f(x) is positive whenever the curve is situated above the number line, it is negative if the curve is found below the number line). Their union just represents the solution of inequality.

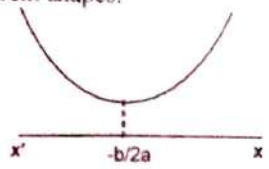
Remarks:

- Points of discontinuity will never be included in the answer.
- If you are asked to find the intervals where f(x) is non-negative or non-positive then make the intervals closed corresponding to the roots of the numerator and let it remain open corresponding to the roots of denominator.

QUADRATIC EXPRESSION

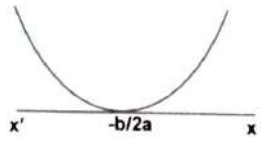
The expression $ax^2 + bx + c$ is said to be a real quadratic expression in x where a, b, c are real and $a \neq 0$. Let $f(x) = ax^2 + bx + c$ where a, b, c, $\in R$ $(a \neq 0)$.

f(x) can be rewritten as $f(x) = a\left\{\left(x+\frac{b}{2a}\right)^2 + \frac{4ac-b^2}{4a^2}\right\} = a\left[\left(x+\frac{b}{2a}\right)^2 - \frac{D}{4a^2}\right]$, where $D = b^2 - 4ac$ is the discriminant of the quadratic expression. Then $y = f(x)$ represents a parabola whose axis is parallel to the y-axis, with vertex at $A\left(-\frac{b}{2a}, \frac{-D}{4a}\right)$. Now depending upon the values of 'a' and D the parabola will have following different shapes:



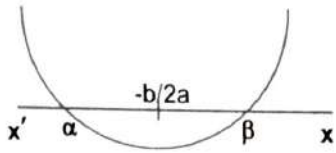
$a > 0, D < 0$

$f(x) \geq -D/4a \ \forall \ x \in R$



$a > 0, D = 0$

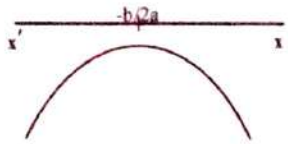
$f(x) \geq 0 \ \forall \ x \in R$



$a > 0, D > 0$

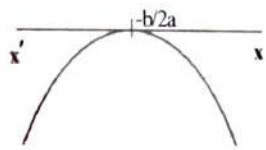
$-\frac{D}{4a} \leq f(x) < 0 \ \forall \ x \in (\alpha, \beta)$

$f(x) > 0 \ \forall \ x \in (-\infty, \alpha) \cup (\beta, \infty)$



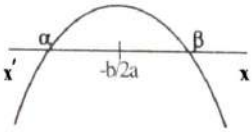
$a < 0, D < 0$

$f(x) \leq -D/4a \ \forall \ x \in R$



$a < 0, D = 0$

$f(x) \leq 0 \ \forall \ x \in R$



$a < 0, D > 0$

$-\frac{D}{4a} \geq f(x) > 0 \ \forall \ x \in (\alpha, \beta)$

$f(x) < 0 \ \forall \ x \in (-\infty, \alpha) \cup (\beta, \infty)$

[1] *The point for which f(x) vanishes (becomes zero)is called zeros of the function e.g. $x = a_i$*
[2] *The points $x = b_j$ are the points of the discontinuity of the function f(x)*
[3] *If the exponents of a factor is odd then the point is called simple point*
[4] *If the exponent of a factor is even then the point is called double point*

Interval in which the Roots Lie

Let $f(x) = ax^2 + bx + c$, where a, b, $c \in R$ and $a \neq 0$. Suppose $\lambda, \lambda_1, \lambda_2 \in R$ and $\lambda_1 < \lambda_2$. Then the following results hold good;

- Both roots of $f(x) = 0$ will be greater than λ, if $D \geq 0$, $af(\lambda) > 0$ and $\lambda < -b/2a$.
- Both roots of $f(x) = 0$ will be less than λ, if $D \geq 0$, $af(\lambda) > 0$ and $-b/2a < \lambda$.
- If λ lies between the roots of $f(x) = 0$ then $af(\lambda) < 0$.
- Both roots of $f(x) = 0$ will belong to the interval (λ_1, λ_2) if $D \geq 0$, $af(\lambda_1) > 0$, $af(\lambda_2) > 0$ and $\lambda_1 < -\frac{b}{2a} < \lambda_2$.
- Exactly one root of $f(x) = 0$ will lie in (λ_1, λ_2) if $f(\lambda_1).f(\lambda_2) < 0$.
- Interval (λ_1, λ_2) will be contained between the roots of $f(x) = 0$ if $af(\lambda_1) < 0$, $af(\lambda_2) < 0$.
- 'λ' will be the repeated root of $f(x) = 0$ if $f(\lambda) = 0$ and $f'(\lambda) = 0$.

Quadratic Inequations

Let $f(x) = ax^2 + bx + c$ be a quadratic expression. Then inequations of the type $f(x) \leq 0$ or $f(x) \geq 0$ are known as quadratic inequations. The study of these can be easily done by taking the corresponding quadratic expression and by applying the basic results of quadratic expression.

ILLUSTRATIVE PROBLEMS

1. If a, b and c are odd integers, then prove that the roots of $ax^2 + bx + c = 0$ can not be rational.

Sol. Discriminant $D = b^2 - 4ac$.

Suppose the roots are rational. Then D will be a perfect square.

Let $b^2 - 4ac = d^2$.

Since a, b and c are odd integers, d will be odd.

Now, $b^2 - d^2 = 4ac$.

Since b and d are odd integers, $b^2 - d^2$ will be a multiple of 8 **(note).** But 4ac is only a multiple of 4 (not of 8), which is a contradiction. Hence the roots of $ax^2 + bx + c = 0$ can not be rational.

2. Let f (x) is continuous function and attains only rational values If f (0) = 3, then find the roots of equation $f(1)\,x^2 + 2f(2)x + f(3) = 0$.

Sol. f (x) is continuous function and attain only rational values $\Rightarrow$ f (x) = constant = 3.

Hence given equation is $3(x^2 + 2x + 1) = 0$

$\Rightarrow x = -1$ is the only solution.

3. Find out the range in which the value of the function $\frac{x^2 + 34x - 71}{x^2 + 2x - 7}$ lies for all real values of x.

Sol. Let $\frac{x^2 + 34x - 71}{x^2 + 2x - 7} = y \Rightarrow x^2(y - 1) - 2x(17 - y) + 71 - 7y = 0$

For real values of x, $(17 - y)^2 - (y - 1)(71 - 7y) \geq 0$

$\Rightarrow 8y^2 - 112y + 360 \geq 0 \Rightarrow y^2 - 14y + 45 \geq 0 \Rightarrow (y - 9)(y - 5) \geq 0$. Either $y \leq 5$ or $y \geq 9$.

Thus for real values of x, range is $(-\infty, 5] \cup [9, \infty)$.

4. Find the value of 'a' for which $ax^2 + (a-3)x + 1 < 0$ for at least one positive real x.

Sol. Let $f(x) = ax^2 + (a-3)x + 1$.

Case (i) If $a < 0$, then $f(x)$ will be positive only for those values of x which lie between the roots. As the interval between the roots can not cover all the positive real numbers implies $f(x) < 0$ for at least one positive real x

Case(ii) If $a > 0$, then $f(x)$ will be negative only for those values of x which lie between the roots. From the graphs we can see that $f(x)$ will be less than zero for at least one positive real x, when $f(x) = 0$ has distinct roots and at least one of these roots is positive real root.

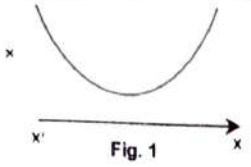


Fig. 1

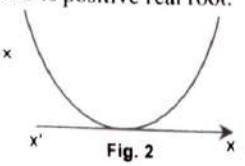


Fig. 2

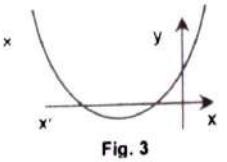


Fig. 3

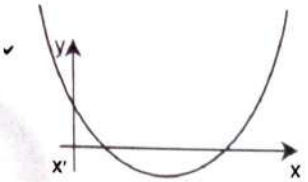


Fig. 4

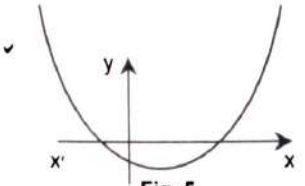


Fig. 5

For this $D > 0$, i.e. $(a-3)^2 - 4a > 0$

$\Rightarrow a < 1$ or $a > 9$ ... (1)

Both the roots are non-positive $\Rightarrow$ sum ≤ 0 and product ≥ 0

$\Rightarrow a \ge 3$ and (1) is satisfied ...(2)

$\Rightarrow$ at least one roots is positive if $a < 3$, and (1) is satisfied.

Combining (1) and (2), we get $a < 1$ so that, $0 < a < 1$.

Case (iii) Obviously $a = 0$ will satisfies this condition.

Hence the required set of values of a is $(-\infty, 1)$

5. If $\alpha, \beta; \beta, \gamma; \gamma, \alpha$ are the roots of $a_i x^2 + b_i x + c_i = 0$, for i = 1, 2, 3, (given that $\alpha, \beta, \gamma > 0$) if $\sum\alpha + \sum\alpha\beta + \alpha\beta\gamma = \left(\prod_{i=1}^{3}\left(\frac{a_i - b_i + c_i}{a_i}\right)\right)^{1/2} + k$, then k is

Sol. $\alpha + \beta = \frac{-b_1}{a_1}$ and $\alpha\beta = \frac{c_1}{a_1}$

$\Rightarrow (1+\alpha)(1+\beta) = 1 + (\alpha+\beta) + \alpha\beta$

$= 1 - \frac{b_1}{a_1} + \frac{c_1}{a_1} = \frac{a_1 - b_1 + c_1}{a_1}$

$\Rightarrow (1+\alpha)^2 (1+\beta)^2 (1+\gamma)^2 = \prod_{i=1}^{3}\left(\frac{a_i - b_1 + c_i}{a_i}\right) \Rightarrow 1 + \sum\alpha + \sum\alpha\beta + \alpha\beta\gamma = \left(\prod_{i=1}^{3}\frac{a_1 - b_i + c_1}{a_i}\right)^{1/2}$

$\Rightarrow \sum\alpha + \sum\alpha\beta + \alpha\beta\gamma = \left(\prod_{i=1}^{3}\frac{a_i - b_i + c_i}{a_i}\right)^{1/2} - 1$

$\Rightarrow k = -1.$

SOLVED PROBLEMS

1. If unity is double repeated root of $px^3 + q(x^2 + x) + r = 0$, then

(A) $p \cdot r < 0$ (B) $p \cdot q < 0$

(C) $p \cdot q \cdot r > 0$ (D) none of these

Sol. **(B).** Since $x = 1$ is double root of $px^3 + q(x^2 + x) + r = 0$

let $f(x) = px^3 + q(x^2 + x) + r$

then $f(1) = f'(1) = 0$ ...(1)

$\Rightarrow p + 2q + r = 0$ ...(2)

$p = -q$

conclusively from (1) and (2) we get

$p = -q = r$.

2. The quadratic $x^2 + ax + b + 1 = 0$ has roots which are positive integers, then $(a^2 + b^2)$ can be equal to

(A) 50 (B) 37

(C) 61 (D) 19

Sol. **(A).** $x^2 + ax + b + 1 = 0$ has possible roots α and β

$\Rightarrow \alpha + \beta = -a$ and $\alpha\beta = b + 1$

$\Rightarrow (\alpha + \beta)^2 + (\alpha\beta - 1)^2 = a^2 + b^2$

$= (\alpha^2 + 1)(\beta^2 + 1) = a^2 + b^2$

$\Rightarrow a^2 + b^2$ can be equal to 50.

3. The equation $2^{2x} + (a - 1)2^{x+1} + a = 0$ has roots of opposite signs then exhaustive set of values of 'a' is

(A) $a < 0$ (B) $a \in (-1, 0)$

(C) $a \in (-\infty, 1/3)$ (D) $a \in (0, 1/3)$

Sol. **(C).** Let $2^x = t$

$\Rightarrow t^2 + 2(a - 1)t + a = 0$.

Now $t = 1$ should lie between the roots of this equation

$\Rightarrow 1 + 2(a - 1) + a < 0 \Rightarrow a < \frac{1}{3}$.

4. If the roots of the equation $x^2 + ax + b = 0$ are c and d then one of the roots of the equation $x^2 + (2c + a)x + c^2 + ac + b = 0$ is

(A) c (B) d –c

(C) 2c (D) 2d

Sol. **(B).** Let $f(x) = x^2 + ax + b$, then $x^2 + (2c + a)x + c^2 + ac + b = f(x + c)$.

Thus roots of $f(x + c) = 0$ will be 0, d – c.

5. If the equation $|x^2 + bx + c| = k$ has four real roots, then

(A) $b^2 - 4c > 0$ and $0 < k < \frac{4c - b^2}{4}$ (B) $b^2 - 4c < 0$ and $0 < k < \frac{4c - b^2}{4}$

(C) $b^2 - 4c > 0$ and $k > \frac{4c - b^2}{4}$ (D) none of these

Sol. **(A).** For the equation to have four real roots the line $y = k$ must intersect $|x^2 + bx + c|$ at four points

$\Rightarrow D > 0$ and $k \in \left(0, \frac{-D}{4a}\right)$.

6. If $|\sin^2 x + 17 - x^2| = |16 - x^2| + 2\sin^2 x + \cos^2 x$, then x lies in
(A) $[-8, 8]$
(B) $[-4, 4]$
(C) $[-\sqrt{17}, \sqrt{17}]$
(D) none of these

Sol. **(B).** $|\sin^2 x + 17 - x^2| = |16 - x^2| + \sin^2 x + 1$
$|x + y + z| = |x| + |y| + |z|$ (when all x, y, z are ≥ 0, or ≤ 0)
$\Rightarrow 16 - x^2 \geq 0$
$\Rightarrow -4 \leq x \leq 4$.

7. The equation $\cos^8 x + b\cos^4 x + 1 = 0$ will have a solution if b belongs to
(A) $(-\infty, 2]$
(B) $[2, \infty)$
(C) $(-\infty, -2]$
(D) none of these

Sol. **(C).** Given equation can be written as

$$b = -\left(\cos^4 x + \frac{1}{\cos^4 x}\right)$$

$\leq -2\ \forall\ x \in R$
$\Rightarrow b \in (-\infty, -2]$.

8. If x, m satisfy the inequations $\log_{1/2} x^2 \geq \log_{1/2}(x + 2)$ and $49x^2 - 4m^4 \leq 0$, then
(A) $m \in (-\infty - a]$
(B) $m \in (-\infty, a)$
(C) $m \in \left(-\infty, -\sqrt{7}\right] \cup \left[\sqrt{7}, \infty\right)$
(D) none of these

Sol. **(D).** $\log_{1/2} x^2 \geq \log_{1/2}(x + 2)$
$\Rightarrow x^2 - x - 2 \leq 0 \Rightarrow x \in [-1, 2] - \{0\}$
and $49x^2 \leq 4m^4$

$$\Rightarrow -\frac{2}{7}m^2 \leq x \leq \frac{2}{7}m^2$$

$$\Rightarrow m \in \left[-\frac{\sqrt{7}}{2}, \frac{\sqrt{7}}{2}\right] - \{0\}.$$

9. If $3^{x+2} - 9^{-1/x} > 0$ then the interval of x can be
(A) $x \in (0, \infty)$
(B) $x \in (0, 250)$
(C) $x \in R$
(D) $x \in (-250, 250)$

Sol. **(A, B).** $3^{x+2} > 9^{-1/x} \Rightarrow x + 2 > -\frac{2}{x} \Rightarrow \frac{x^2 + 2x + 2}{x} > 0$

$$\frac{(x+1)^2 + 1}{x} > 0 \Rightarrow x > 0.$$

10. If $ax^2 - bx + c = 0$ has two distinct roots lying in the interval $(0, 1)$, a, b, $c \in N$, then
(A) $\log_5 abc = 1$
(B) $\log_5 abc = 2$
(C) $\log_5 abc = 3$
(D) $\log_5 abc = 4$

Sol. **(B, C, D).** $\alpha + \beta = \frac{b}{a}$, $\alpha\beta = \frac{c}{a}$ $0 < \alpha\beta < 1 \Rightarrow 0 < (1 - \alpha), (1 - \beta) < 1$

$$\Rightarrow 0 < \alpha(1 - \alpha) \leq \frac{1}{4} \text{ and } 0 < \beta(1 - \beta) \leq \frac{1}{4}$$

$0 < \alpha\beta(1-(\alpha+\beta)+\alpha\beta) \le \frac{1}{16}; 0 < \frac{c}{a}\left(1-\frac{b}{a}+\frac{c}{a}\right) \le \frac{1}{16} \Rightarrow 0 < c(a-b+c) < \frac{a^2}{16}$

$\Rightarrow a = b, c = 1 \Rightarrow a^2 > 16 \Rightarrow a > 4 \Rightarrow a = 5 \Rightarrow b = 5 \Rightarrow$ min. (abc) = 25.

11. If $ax^2 + bx + c = 0$ and $cx^2 + bx + a = 0$ $(a, b, c \in R)$ have a common non-real root, then

(A) $-2|a| < b < 2|a|$ (B) $-2|c| < |b| < 2|c|$

(C) $a = \pm c$ (D) $a = c$

Sol. **(A), (B), (D)**

$D_1 = b^2 - 4ac < 0$, $D_2 = b^2 - 4ac < 0$, as the root is non-real

$\Rightarrow$ Both roots will be common.

$\Rightarrow \frac{a}{c} = \frac{b}{b} = \frac{c}{a} = 1 \Rightarrow a = c$

Now, $b^2 - 4ac < 0 \Rightarrow b^2 - 4a^2$ (or $4c^2$) < 0

$\Rightarrow |b| < 2|a|$ (or $2|c|$).

12. Consider the equation $x^2 + x - a = 0$, $a \in N$. If equation has integral roots then

(A) $a = 2$ (B) $a = 6$

(C) $a = 12$ (D) $a = 20$

Sol. **(A), (B), (C) and (D)**

Discriminant, $D = \sqrt{1+4a}$

$\Rightarrow 1+4a$ should be a perfect square.

As $1+4a$ is always odd

$\Rightarrow 1+4a = (2\lambda+1)^2, \lambda \in I^+$

$\Rightarrow a = \lambda(\lambda+1)$.

13. The equation $a_8x^8 + a_7x^7 + a_6x^6 + \ldots + a_0 = 0$ has all its roots positive and real (where $a_8 = 1$, $a_7 = -4$, $a_0 = 1/2^8$), then

(A) $a_1 = \frac{1}{2^8}$ (B) $a_1 = -\frac{1}{2^4}$

(C) $a_2 = \frac{7}{2^4}$ (D) $a_3 = \frac{1}{2^2}$

Sol. **(B, C).** Let the roots be $\alpha_1, \alpha_2, \ldots, \alpha_8$

$\Rightarrow \alpha_1 + \alpha_2 + \ldots + \alpha_8 = 4$

$\alpha_1 . \alpha_2 \ldots \alpha_8 = \frac{1}{2^8}$

$\Rightarrow (\alpha_1 . \alpha_2 \ldots \alpha_8)^{1/8} = \frac{1}{2} = \frac{\alpha_1+\alpha_2+\ldots+\alpha_8}{8}$

$\Rightarrow$ A.M. = G.M. $\Rightarrow$ all the roots are equal to $\frac{1}{2}$

$\Rightarrow a_1 = -{}^8C_7\left(\frac{1}{2}\right)^7 = -\frac{1}{2^4}$

$a_2 = {}^8C_6\left(\frac{1}{2}\right)^6 = \frac{7}{2^4}$

$a_3 = -{}^8C_5\left(\frac{1}{2}\right)^5$.

Comprehension

If α, β, γ be the roots of a cubic equation $ax^3 + bx^2 + cx + d = 0$ then we have $\alpha + \beta + \gamma = -\frac{b}{a}$, $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$ and $\alpha\ \beta\gamma = -\frac{d}{a}$. Suppose a and b are two positive real numbers such that the roots of the cubic equation $x^3 - ax + b = 0$ are all real. Let α is a root of this cubic equation with minimal absolute value

Answer the following questions

14. The roots of the given cubic equation must be
(A) all positive
(B) all negative
(C) two positive and one negative
(D) two negative and one positive

Sol. **(C)**

15. α must be
(A) positive
(B) negative
(C) positive only if $b < a$
(D) negative only if $b < a$

Sol. **(A)**

16. The range of values of α is given by
(A) $\frac{2b}{3a} < \alpha \le \frac{b}{a}$
(B) $\frac{b}{a} < \alpha \le \frac{3b}{2a}$
(C) $-\frac{b}{a} < \alpha \le \frac{b}{a}$
(D) $-\frac{b}{a} < \alpha \le \frac{2b}{3a}$

Sol. **(B)**

Let the roots be $\alpha, \beta\ \gamma$ then
$\alpha + \beta + \gamma = 0$, $\alpha\beta + \beta\gamma + \gamma\alpha = -a$ and $\alpha\ \beta\gamma = -b$.
Product of the roots is –ve and sum is zero
$\Rightarrow$ equation has two positive and one negative root.
Let γ be the negative root then $\alpha + \beta + \gamma = 0$
$\Rightarrow$ magnitude of γ must be larger of α and β. In particular $\gamma < 0 < \alpha \le \beta \le |\gamma|$

Now $b - a\alpha = -\alpha\beta\gamma + (\alpha\beta + \beta\gamma + \gamma\alpha)\alpha = \alpha^2(\beta + \gamma) = -\alpha^3 < 0 \ \therefore \alpha > \frac{b}{a}$

$2b - 3a\alpha = -2\alpha\beta\gamma + 3(\alpha\beta + \beta\gamma + \gamma\alpha)\alpha$
$= \alpha\beta\gamma + 3\alpha^2(\beta + \gamma)$
$= \alpha\beta\gamma - 3\alpha^3 = -b - 3\alpha^3 < 0$
$\therefore\ \alpha > \frac{2b}{3a}$.

$3b - 2a\alpha = -3\alpha\beta\gamma + 2(\alpha\beta + \beta\gamma + \gamma\alpha)\alpha$
$= -\alpha\beta\gamma + 2\alpha^2(\beta + \gamma)$
$= -\alpha\beta\gamma - 2\alpha^3$
$= \alpha[2\alpha(\beta + \gamma) - \beta\gamma]$
$= \alpha[-2(\beta + \gamma)^2 - \beta\gamma]$
$= -\alpha[2\beta^2 + 5\beta\gamma + 2\gamma^2]$
$= -\alpha[(2\beta + \gamma)(\beta + 2\gamma)]$
$= -\alpha(\beta - \alpha)(\gamma - \alpha) \ge 0$.
$\therefore\ \alpha \le \frac{3b}{2a}$.

Assertion Reason

17. STATEMENT – I: The equation $2\sin^2\theta\, x^2 - 2\sin\theta\, x + 1 = 0, \theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ has one root lying in the interval (0, 1) and the other root lying in the interval (1, 2).
STATEMENT – II: If f(x) is a polynomial such that f(a) and f(b) are of opposite sign, where a < b then f(x) = 0 has exactly one root in the interval (a, b).
(A) Statement-1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1
(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement-1 is True, Statement -2 is False
(D) Statement-1 is False, Statement -2 is True

Sol. **(C)**
Clearly statement II is false because $f(a)\, f(b) < 0$
$\Rightarrow f(x) = 0$ has at least one root in (a, b).
Also, if $g(x) = 2\sin^2\theta\, x^2 - 3\sin\theta\, x + 1$
Then $g(0) = 1 > 0$

$g(1) = (2\sin\theta - 1)(\sin\theta - 1) < 0$ of $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

$g(2) = (4\sin\theta - 1)(2\sin\theta - 1) > 0$ of $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

$\therefore$ $g(x) = 0$ has a root in (0, 1) and other in (1, 2).

18. For the equation $z^6 + 6z + 20 = 0$, match the following:

	Column – I		Column – II
(A)	The number of the roots in the first quadrant can be	(p)	1
(B)	The number of the roots in the second quadrant can be	(q)	2
(C)	The number of the roots in the Third quadrant can be	(r)	3
(D)	The number of the roots in the Fourth quadrant can be	(s)	4
		(t)	∞

Sol. **(A) → (p, q); (B) → (p, q); (C) → (p, q); (D) → (p, q)**
There are no real roots of the equation
$z^6 + 6z + 20 = 0$.
If $x + iy$ is a root then $x - iy$ is also a root
Let the roots be
$x_1 \pm iy_1, x_2 \pm iy_2, x_3 \pm iy_3$
Sum of the roots
$= 2(x_1 + x_2 + x_3) = 0$
$\Rightarrow x_1 + x_2 + x_3 = 0$
$\Rightarrow$ one of x_1, x_2, x_3 is negative and other two are positive or vice-versa
$\Rightarrow$ The number of roots in each of the quadrant is either 1 or 2.

Answer Q.19, Q.20 and Q.21 by appropriately matching the information given in the three columns of the following table.

Column-I: contains some quadratic equation with variable 'k', Column-II: contains value of 'k' for which '1' lies between roots of $p(x) = 0$ and Column-III: contains value of 'k' for which exactly one root of $p(x) = 0$ lies on (0, 1)

Column–I	Column–II	Column–III
(I) $p(x) = x^2 - 2kx - 1$	(i) $k > -1$	(P) $k > \frac{1}{2}$
(II) $p(x) = x^2 + (k-1)x - 2$	(ii) $k > 0$	(Q) $k < -1$
(III) $p(x) = x^2 + (2k-1)x - 1$	(iii) $k < 2$	(R) $k < 0$
(IV) $p(x) = x^2 - kx - 2$	(iv) $k < \frac{1}{2}$	(S) $k > 2$

19. Which of the following combination is **CORRECT**?
(A) (I) (ii) (Q)
(B) (II) (i) (S)
(C) (III) (ii) (R)
(D) (I) (ii) (R)

Sol. **(D)**

20. Which of the following combination is **CORRECT**?
(A) (III) (ii) (Q)
(B) (II) (iii) (S)
(C) (II) (iii) (R)
(D) (I) (ii) (Q)

Sol. **(B)**

21. Which of the following combination is **INCORRECT**?
(A) (III) (iv) (P)
(B) (IV) (i) (P)
(C) (IV) (i) (R)
(D) (III) (iv) (Q)

Sol. **(A)**

19.-21. For 2nd column $p(1) < 0$ will get the answer and for 3rd column $p(0) \cdot p(1) < 0$
$\Rightarrow p(1) > 0$

22. Let a, b, c be three real numbers such that $a + b + c = 0$ and $a^2 + b^2 + c^2 = 2$, then the $a^4 + b^4 + c^4$ is equal to _____

Sol. **(2.00)**

Let $c = 0, a = 1, b = -1 \Rightarrow a^4 + b^4 + c^4 = 2$

23. The number of positive integral values of 'k' for which $kx^2 + (k-3)x + 1 < 0$ for atleast one positive 'x' is equal to _____

Sol. **(0.00)**

$k > 0$ also, $D > 0 \cap -\frac{b}{2a} > 0$

$\Rightarrow k^2 - 10k + 9 > 0 \cap \frac{k-3}{k} < 0$

$\Rightarrow k \in (-\infty, 1) \cup (9, \infty) \cap (0, 3)$

$\Rightarrow k \in (0, 1)$

$\therefore x = 0$

24. Consider, two numbers a, b, sum of which is 3 and the sum of their cubes is 7. Then, the sum of all possible distinct values of $\frac{a}{2}$ is equal to ____

Sol. **(1.50)**

$a + b = 3$,

$a^3 + b^3 = 7$

$\Rightarrow a^3 + (3 - a)^3 = 7$

$\Rightarrow 9a^2 - 27a + 20 = 0$

$\Rightarrow$ Sum of distinct values of $\frac{a}{2}$ is 1.50

25. Number of integral values of 'a' such that the quadratic equation $x^2 + ax + a + 1 = 0$ has integral roots is ____

Sol. **(2.00)**

$D \geq 0$

$\Rightarrow a \leq 2 - 2\sqrt{2}$

and $a \geq 2 + 2\sqrt{2}$

$\alpha + \beta = -a$

and $\alpha \cdot \beta = a + 1$

$\Rightarrow \alpha + \beta + \alpha\beta + 1 = 2$

$\Rightarrow (\alpha + 1)(\beta + 1) = 2 = 1 \times 2 = 2 \times 1 = (-1) \times (-2) = (-2) \times (-1)$

$\therefore$ a can be 1, –5

PROGRESSION & SERIES

15

ARITHMETIC PROGRESSION (A.P.)

nth Term and Sum of n Terms

If a is the first term and d the common difference, the A.P. can be written as a, a + d, a + 2d, ... The nth term a_n is given by $a_n = a + (n - 1)d$. The sum S_n of the first n terms of such an A.P. is given by $S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(a + l)$ where l is the last term (i.e. the nth term of the A.P.).

- If three terms are in A.P., then the middle term is called the arithmetic mean (A.M.) between the other two i.e. if a, b, c are in A.P., then $b = \frac{a+c}{2}$ is the A.M. of a and c.
- If $a_1, a_2, ...a_n$ are n numbers, then the arithmetic mean (A) of these numbers is
  $A = \frac{1}{n}(a_1 + a_2 + a_3 + ... + a_n)$.
  The n numbers $A_1, A_2, ..., A_n$ are said to be A.M.'s between the numbers a and b if a, $A_1, A_2, ...A_n$ b are in A.P. $\Rightarrow A_n = a + \frac{n(b-a)}{n+1} = \frac{a+nb}{n+1}$.

GEOMETRIC PROGRESSION (G.P.)

nth Term and Sum of n Terms

If a is the first term and r the common ratio, then the G.P. can be written as a, ar, ar^2, ... The nth term, a_n, is given by $a_n = ar^{n-1}$. The sum S_n of the first n terms of the G.P. is

$$S_n = \frac{a(r^n - 1)}{r - 1}, \quad r \neq 1$$
$$= na, \quad r = 1$$

If $-1 < r < 1$, then the sum of the infinite G.P. is $a + ar + ar^2 + ... = \frac{a}{1-r}$

- If three terms are in G.P., then the middle term is called the geometric mean (G.M.) between the two. So if a, b, c are in G.P., then $b = \sqrt{ac}$ is the geometric mean of a and c.
- If $a_1, a_2, ...a_n$ are non-zero positive numbers, then their G.M (G) is given by $G = (a_1a_2a_3.....a_n)^{1/n}$.
- If $G_1, G_2, ...G_n$ are n geometric means, between a and b, then a, $G_1, G_2, ..., G_n$, b will be a G.P.
  $\Rightarrow G_n = a\left(\sqrt[n+1]{\frac{b}{a}}\right)^n$.

HARMONIC PROGRESSION (H.P.)

The sequence $a_1, a_2, a_3, ...a_n, ...(a_i \neq 0)$ is said to be an H.P. if the sequence $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, ... \frac{1}{a_n}, ...$ is an A.P.

The nth term, a_n, of the H.P. is $a_n = \frac{1}{a + (n-1)d}$, where $a = \frac{1}{a_1}$, and $d = \frac{1}{a_2} - \frac{1}{a_1}$.

Harmonic Means

- If a and b are two non-zero numbers, then the harmonic mean of a and b is a number H such that the numbers a, H, b are in H.P. We have $\frac{1}{H}=\frac{1}{2}\left(\frac{1}{a}+\frac{1}{b}\right)\Rightarrow H=\frac{2ab}{a+b}$.
- There is no formula for the sum of n terms of an H.P.
- If $a_1, a_2, \ldots\ldots a_n$ are 'n' non-zero numbers, then the harmonic mean H of these numbers is given by $\frac{1}{H}=\frac{1}{n}\left(\frac{1}{a_1}+\frac{1}{a_2}+\ldots\ldots+\frac{1}{a_n}\right)$.
- The n numbers $H_1, H_2, \ldots H_n$ are said to be n-harmonic means between a and b, if $a, H_1, H_2, \ldots, H_n, b$ are in H.P. i.e. if $\frac{1}{a},\frac{1}{H_1},\frac{1}{H_2},\ldots\ldots\frac{1}{H_n},\frac{1}{b}$ are in A.P. $\frac{1}{H_n}=\frac{1}{a}+\frac{n(a-b)}{(n+1)ab}=\frac{b+na}{ab(n+1)}$.

ARITHMETICO-GEOMETRIC PROGRESSION

The sum S_n of first n terms of an A.G. P. is obtained in the following way :

$S_n = ab + (a + d)br + (a + 2d)br^2 + \ldots\ldots + (a + (n - 2)d)br^{n-2} + (a + (n - 1)d)br^{n-1}$

Multiply both sides by r, so that

$r\,S_n = abr + (a + d)br^2 + \ldots + (a + (n - 3)d)br^{n-2} + (a + (n - 2)d)br^{n-1} + (a + (n - 1)d)br^n$

Subtracting, we get, $(1 - r)S_n = ab + dbr + dbr^2 + \ldots\ldots + dbr^{n-2} + dbr^{n-1} - (a + (n - 1)d)br^n$

$= ab + \frac{dbr(1-r^{n-1})}{(1-r)} - (a+(n-1)d)br^n \Rightarrow S_n = \frac{ab}{1-r} + \frac{dbr(1-r^{n-1})}{(1-r)^2} - \frac{(a+(n-1)d)br^n}{1-r}$

If $-1 < r < 1$, the sum of the infinite number of terms of the progression is $\lim_{n\to\infty} S_n = S = \frac{ab}{1-r} + \frac{dbr}{(1-r)^2}$.

MISCELLANEOUS PROGRESSIONS

Some Important Results

- $1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}$ (sum of the first n natural numbers)
- $1^2 + 2^2 + 3^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$ (sum of squares of the first n natural numbers)
- $1^3+2^3+3^3+\ldots+n^3=\frac{n^2(n+1)^2}{4}=(1+2+3+\ldots+n)^2$ (sum of cubes of first n natural numbers)
- $1 + x + x^2 + x^3 + \ldots\ldots = (1 - x)^{-1}$, if $-1 < x < 1$
- $1 + 2x + 3x^2 + \ldots\ldots = (1 - x)^{-2}$, if $-1 < x < 1$

Method of Differences

Suppose $a_1, a_2, a_3, \ldots\ldots$ is a sequence such that the sequence $a_2 - a_1, a_3 - a_2, \ldots\ldots$ is either an A.P. or a G.P. Then th term 'a_n' of this sequence is obtained as follows :

$S = a_1 + a_2 + a_3 + \ldots\ldots + a_{n-1} + a_n$

$S = \quad a_1 + a_2 + \ldots\ldots + a_{n-2} + a_{n-1} + a_n \Rightarrow a_n = a_1 + [(a_2 - a_1) + (a_3 - a_2) + \ldots\ldots + (a_n - a_{n-1})]$

Since the terms within the brackets are either in an A.P. or in a G.P., we can find the value of a_n, the nth term. We can now find the sum of the n terms of the sequence as $S = \sum_{k=1}^{n} a_k$.

INEQUALITIES

A.M. ≥ G. M. ≥ H. M.

Let $a_1, a_2, ..., a_n$ be n positive real numbers, then we define their arithmatic mean (A), geometric mean (G) and harmonic mean (H) as $A = \frac{a_1 + a_2 + ... + a_n}{n}$,

$G = (a_1 a_2 ... a_n)^{1/n}$ and $H = \frac{n}{\left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + ... \frac{1}{a_n}\right)}$.

It can be shown that $A \geq G \geq H$. Moreover equality holds at either place if and only if $a_1 = a_2 = ... = a_n$.

Weighted Means

Let $a_1, a_2, ..., a_n$ be n positive real numbers and $m_1, m_2, ..., m_n$ be n positive rational numbers. Then we define weighted Arithmetic mean (A*), weighted Geometric mean (G*) and weighted harmonic mean (H*) as

$A^* = \frac{m_1 a_1 + ... + m_n a_n}{m_1 + m_2 + ...m_n}$, $G^* = \left(a_1^{m_1} . a_2^{m_2} . \quad ..... \quad . a_n^{m_n}\right)^{\frac{1}{(m_1 + m_2 + ... + m_n)}}$ and

$H^* = \frac{m_1 + m_2 + ... + m_n}{\left(\frac{m_1}{a_1} + \frac{m_2}{a_2} + ... + \frac{m_n}{a_n}\right)}$.

It can be shown that $A^* \geq G^* \geq H^*$. Moreover equality holds at either place if and only if $a_1 = a_2 = ... = a_n$.

Arithmetic Mean of mth Power

Let $a_1, a_2, ..., a_n$ be n positive real numbers (not all equal) and let m be a real number , then

$\frac{a_1^m + a_2^m + ... + a_n^m}{n} > \left(\frac{a_1 + a_2 + ... + a_n}{n}\right)^m$ if $m \in R - [0, 1]$.

However if $m \in (0, 1)$, then $\frac{a_1^m + a_2^m + ... + a_n^m}{n} < \left(\frac{a_1 + a_2 + ... + a_n}{n}\right)^m$.

Obviously if $m \in \{0, 1\}$, then $\frac{a_1^m + a_2^m + ... + a_n^m}{n} = \left(\frac{a_1 + a_2 + ... + a_n}{n}\right)^m$.

ILLUSTRATIVE PROBLEMS

1. Find the positive integral values of n such that $1.2^1 + 2.2^2 + 3.2^3 + 4.2^4 + 5.2^5 + \ldots + n.2^n = 2^{(n+10)} + 2$.

Sol. Series can be summed as

$2^1 + 2^2 + 2^3 + \ldots + 2^n = 2^{n+1} - 2$

$2^2 + 2^3 + \ldots + 2^n = 2^{n+1} - 2^2$

$2^3 + \ldots + 2^n = 2^{n+1} - 2^3$

$\ldots \quad \ldots \quad \ldots \quad \ldots$

$\ldots \quad \ldots \quad \ldots \quad \ldots$

$+ 2^n = 2^{n+1} - 2^n$

$= n(2^{n+1}) - (2^{n+1} - 2)$

$= 2^{n+1}(n-1) + 2.$

Given that

$2^{n+1}(n-1) + 2 = 2^{n+10} + 2$

$\Rightarrow (n-1)2^{n+1} = 2^{n+10}$

$\Rightarrow n - 1 = 2^9 \Rightarrow n = 2^9 + 1 = 513$

2. The fourth power of the common difference of an arithmetic progression with integer entries is added to the product of any four consecutive terms of it. Prove that the resulting sum is the square of an integer.

Sol. Let $a - 3d$, $a - d$, $a + d$ and $a + 3d$ be any four consecutive terms of an A.P. with integer entries and with common difference $2d$.

Consider $(2d)^4 + (a - 3d)(a - d)(a + d)(a + 3d)$

$= 16d^4 + (a^2 - 9d^2)(a^2 - d^2)$

$= 16d^4 + a^4 - 10a^2d^2 + 9d^4 = a^4 - 10a^2d^2 + 25d^4$

$= (a^2 - 5d^2)^2 = [(a - d)(a + d) - (2d)^2]^2$

which is the square of an integer since $a - d$, $a + d$ and $2d$ are integers.

3. The sum of the squares of three distinct real numbers which are in G.P. is S^2. If their sum is αS, show that $\alpha^2 \in \left(\frac{1}{3}, 1\right) \cup (1, 3)$.

Sol. We take the three numbers in G.P. to be $\frac{a}{r}$, a, ar, so that

$$\frac{a}{r} + a + ar = \alpha S \qquad (1)$$

and $$\frac{a^2}{r^2} + a^2 + a^2r^2 = S^2 \qquad (2)$$

From (1) and (2) we have $\left[1 + \left(r + \frac{1}{r}\right)\right]^2$

$$= \alpha^2\left[1 + r^2 + \frac{1}{r^2}\right] = \alpha^2\left[\left(r + \frac{1}{r}\right)^2 - 1\right]$$

$$\Rightarrow \quad 1 + r + \frac{1}{r} = \alpha^2\left[r + \frac{1}{r} - 1\right] \Rightarrow \quad \left(\alpha^2 - 1\right) r^2 - \left(\alpha^2 + 1\right) r + \left(\alpha^2 - 1\right) = 0$$

$\Rightarrow \quad r^2 - \frac{\alpha^2+1}{\alpha^2-1} r + 1 = 0.$

Since r is real, $\left(\frac{\alpha^2+1}{\alpha^2-1}\right)^2 - 4 > 0$ [The numbers in G.P. are distinct].

$\Rightarrow \left(\frac{\alpha^2+1}{\alpha^2-1} - 2\right)\left(\frac{\alpha^2+1}{\alpha^2-1} + 2\right) > 0$ (either terms in both the brackets on L.H.S. are positive or both are negative)

$\Rightarrow$ Either $\frac{\alpha^2+1}{\alpha^2-1} > 2$ or $\frac{\alpha^2+1}{\alpha^2-1} < -2$

$\Rightarrow 1 < \alpha^2 < 3$ or $\frac{1}{3} < \alpha^2 < 1.$

Hence $\alpha^2 \in \left(\frac{1}{3}, 1\right) \cup (1, 3).$

4. **Let a_1, a_2, ... be positive real numbers in geometric progression. For each n, let A_n, G_n, H_n be, respectively, the arithmetic mean, geometric mean and harmonic mean of a_1, a_2, ... a_n. Find an expression for the geometric mean of $G_1, G_2, \ldots, G_n$ in terms of $A_1, A_2, \ldots, A_n, H_1, H_2, \ldots, H_n$.**

Sol. $G_k = (a_1 a_2 \ldots a_k)^{1/k} = a_1 (r^{1+2+\ldots+(k-1)})^{1/k} = a_1 r^{\frac{k-1}{2}}$ .....(1)

$A_k = \frac{a_1 + a_2 + \ldots + a_k}{k} = \frac{a_1\left(1 + r + \ldots + r^{k-1}\right)}{k} = \frac{a_1\left(r^k - 1\right)}{(r-1)\cdot k}$ .....(2)

$H_k = \frac{k}{\frac{1}{a_1} + \frac{1}{a_2} + \ldots + \frac{1}{a_k}} = \frac{k}{\frac{1}{a_1}\left(1 + \frac{1}{r} + \ldots + \frac{1}{r^{k-1}}\right)} = \frac{a_1 k (r-1) r^{k-1}}{r^k - 1}$ .....(3)

From (1), (2) and (3), we get $G_k = (A_k H_k)^{1/2}$

$\Rightarrow \prod_{k=1}^{n} G_k = \prod_{k=1}^{n} (A_k H_k)^{1/2} \Rightarrow \left(\prod_{k=1}^{n} G_k\right)^{1/n} = (A_1 A_2 \ldots A_n \cdot H_1 H_2 \ldots H_n)^{1/2n}$

5. **Show that $\frac{1^4}{1.3} + \frac{2^4}{3.5} + \frac{3^4}{5.7} + \ldots + \frac{n^4}{(2n-1)(2n+1)} = \frac{n(4n^2+6n+5)}{48} + \frac{n}{16(2n+1)}$.**

Sol. The n^{th} term T_n of the given series is

$T_n = \frac{n^4}{(2n-1)(2n+1)} = \frac{4n^2+1}{16} + \frac{1}{32}\left[\frac{1}{2n-1} - \frac{1}{2n+1}\right]$

The sum S_n of the given series is

$S_n = \sum_{1}^{n} T_n = \frac{1}{16}\sum_{n=1}^{n} (4n^2+1) + \frac{1}{32}\sum_{n=1}^{n}\left(\frac{1}{2n-1} - \frac{1}{2n+1}\right)$

$S_n = \frac{1}{16}\left[\frac{4n(n+1)(2n+1)+6n}{6}\right] + \frac{1}{32}\left[1 - \frac{1}{3} + \frac{1}{3} - \frac{1}{5} + \frac{1}{5} - \frac{1}{7} + \ldots \frac{1}{2n-1} - \frac{1}{2n+1}\right]$

$= \frac{n}{48}\left[4n^2+6n+5\right] + \frac{1}{32}\left[1 - \frac{1}{2n+1}\right] = n\left[\frac{4n^2+6n+5}{48}\right] + \frac{n}{16(2n+1)}.$

6. Find the sum of Ist n terms of the series 5, 7, 11, 17, 25,...

Sol. Let $S = 5 + 7 + 11 + 17 + 25 + \ldots\ldots + t_r$

$S = \quad 5 + 7 + 11 + 17 + \ldots\ldots + t_{r-1} + t_r$

Subtracting, we get $0 = 5 + 2 + 4 + 6 + 8 + \ldots\ldots +$ rth term $- t_r$

$$\Rightarrow t_r = 5 + 2\left\{\frac{r(r-1)}{2}\right\} \Rightarrow t_r = r^2 - r + 5$$

$$S_n = \sum_{r=1}^{n} r^2 - \sum_{r=1}^{n} r + 5\sum_{r=1}^{n} 1 = \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} + 5n$$

$$= \frac{n}{6}\{(n+1)(2n+1) - 3(n+1) + 30\} = \frac{n}{3}(n^2 + 14).$$

7. If a, b, c are the sides of a triangle and $s = \frac{a+b+c}{2}$. Prove that $8(s-a)(s-b)(s-c) \le abc$.

Sol. Let $x = s - a$, $y = s - b$, $z = s - c$, then $a = y + z$, $b = x + z$, $c = x + y$

then the inequality reduces to

$8xyz \le (x + y)(y + z)(x + z)$ $x,y,z \ge 0$

which follows easily from A.M. $\ge$ G.M. inequality

$x + y \ge 2\sqrt{xy}$, $\quad y + z \ge 2\sqrt{yz}$, $\quad x + z \ge 2\sqrt{xz}$

$\therefore (x + y)(x + z)(x + z) \ge 8xyz$. Hence proved.

SOLVED PROBLEMS

1. a, b, c are positive numbers in G.P. and the equation $(a + d)x^2 + 2(b + e)x + (c + f) = 0$ have one real root then $\frac{a}{d}, \frac{b}{e}, \frac{c}{f}$ are in, $(a, b, c, d, e, f \in R)$

(A) AP
(B) GP
(C) HP
(D) none of these

Sol. **(C).** Let α be the real root then $(a\alpha^2 + 2b\alpha + c) + i(d\alpha^2 + 2e\alpha + f) = 0 \Rightarrow a\alpha^2 + 2b\alpha + c = 0 \Rightarrow \alpha = \frac{-\sqrt{c}}{\sqrt{a}}$

$\Rightarrow d\alpha^2 + 2e\alpha + f = 0 \Rightarrow d.\frac{c}{a} - 2e\frac{\sqrt{c}}{\sqrt{a}} + f = 0 \Rightarrow \frac{d}{a} - 2.\frac{e}{b} + \frac{f}{c} = 0$

2. If $H_1, H_2, H_3, \ldots\ldots, H_{2n-1}$ are in H.P., then $\sum_{i=1}^{2n}(-1)^i\left(\frac{H_i + H_{i+1}}{H_i - H_{i+1}}\right)$ is equal to

(A) $2n - 1$
(B) $2n + 1$
(C) $2n$
(D) $2n + 2$

Sol. **(C).** Let $\frac{1}{H_{i+1}} - \frac{1}{H_i} = k$

$$\sum_{i=1}^{2n}(-1)^i\left(\frac{H_i + H_{i+1}}{H_i - H_{i+1}}\right) = \sum_{i=1}^{2n}\frac{(-1)^i}{k}\left(\frac{1}{H_{i+1}} + \frac{1}{H_i}\right) = 2n$$

3. If a, b, c are in A.P., b, c, d in G.P. and c, d, e in H.P. then a, c, e are in
(A) A.P.
(B) G.P.
(C) H.P.
(D) none of these

Sol. **(B).** We have $2b = a + c$

$c^2 = bd$

$d = \frac{2ec}{e+c}$

$\Rightarrow \frac{c^2}{b} = \frac{2ce}{e+c} \Rightarrow c(e + c) = e(a + c)$

$\Rightarrow c^2 = ae.$

4. If x, y, z are in G.P. $(x, y, z > 1)$ then $\frac{1}{2x + \ln x}, \frac{1}{4x + \ln y}, \frac{1}{6x + \ln z}$ are in

(A) A.P.
(B) G.P.
(C) H.P.
(D) none of these

Sol. **(C).** As x, y, z are in G.P., $\Rightarrow e^{2x}, e^{4x}, e^{6x}$ are in G.P.

$\Rightarrow \ln(e^{2x}), \ln(e^{4x}), \ln(e^{6x})$ are in A.P.

$\Rightarrow \frac{1}{2x + \ln x}, \frac{1}{4x + \ln y}, \frac{1}{6x + \ln z}$ are in H.P.

5. Let $a_1, a_2, \ldots$ be in A.P and $q_1, q_2, \ldots$ be in G.P if $a_1 = q_1 = 2$ and $a_{10} = q_{10} = 3$ then
(A) a_9q_9 is not an integer
(B) $a_{19}q_9$ is an integer
(C) $a_9q_{19} = a_{19}q_{10}$
(D) none of these

Sol. **(C).** If d is the common difference of the A.P and r is the common ratio of the G.P. then
$2 + 9d = 3$ and $2r^9 = 3$
$\Rightarrow d = \frac{1}{9}$ and $r = \sqrt[9]{\frac{3}{2}}$
$a_9q_{10} = (2 + 6.1/9)\, 2.r^{18} = 12$
$a_{19}q_{10} = (2 + 18.1/9)\, 2.r^9 = 12$
$a_{19}q_9 = 4.2.r^8$ is not an integer.

6. If $S_n = \sum_{r=1}^{n} \frac{2r+1}{r^4 + 2r^3 + r^2}$ then S_{20} is equal to
(A) $\frac{220}{221}$
(B) $\frac{420}{441}$
(C) $\frac{439}{221}$
(D) $\frac{440}{441}$

Sol. **(D).** $S_n = \sum_{r=0}^{n} \frac{2r+1}{r^2(r+1)^2}$
$\Rightarrow S_{20} = \sum_{r=1}^{20} \left[\frac{1}{r^2} - \frac{1}{(r+1)^2} \right] = \frac{440}{441}$.

7. Number of ordered triplets (p, q, r), where $1 \le p, q, r \le 10$, such that $2^p + 3^q + 5^r$ is a multiple of 4 is
$(p, q, r \in N)$
(A) 1000
(B) 500
(C) 250
(D) 125

Sol. **(B).** $2^p + 3^q + 5^r = 2^p + (4-1)^q + (4+1)^r = 2^p + 4\lambda_1 + (-1)^q + 4\lambda_2 + 1^r$ (λ_1, λ_2 are integers)
If $p = 1$, q should be even and r can be any number. On the other hand if $p \neq 1$, q should be odd and r can be any number.
$\Rightarrow$ total number of ordered triplets $= 5 \times 10 + 9 \times 5 \times 10 = 500$.

8. If a, b, c are positive numbers in A.P. then roots of the equation $ax^2 - 4bx + c = 0$ are
(A) always real
(B) always rational
(C) unreal
(D) data insufficient

Sol. **(A).** $D = 16b^2 - 4ac = 4(4b^2 - ac) = 16\left(b^2 - \frac{ac}{4}\right) > 0$
as $b = \frac{a+c}{2} > \sqrt{ac}$
$\Rightarrow b^2 > ac > \frac{ac}{4}$
$\Rightarrow$ roots are always real.

9. If a, b, c, d are four unequal positive numbers which are in A.P then

(A) $\frac{1}{a}+\frac{1}{d}=\frac{1}{b}+\frac{1}{c}$

(B) $\frac{1}{a}+\frac{1}{d}<\frac{1}{b}+\frac{1}{c}$

(C) $\frac{1}{a}+\frac{1}{d}>\frac{1}{b}+\frac{1}{c}$

(D) $\frac{1}{b}+\frac{1}{c}>\frac{4}{a+d}$

Sol. **(C, D).** Let $b = a + p$, $c = a + 2p$, $d = a + 3p$

$$\frac{\frac{1}{a}+\frac{1}{d}}{\frac{1}{b}+\frac{1}{c}}=\frac{\frac{1}{a}+\frac{1}{a+3p}}{\frac{1}{a+p}+\frac{1}{a+2p}}=\frac{(a+p)(a+2p)}{a(a+3p)}=\frac{a^2+3ap+2p^2}{a^2+3ap}>1$$

$$\therefore \frac{1}{a}+\frac{1}{d}>\frac{1}{b}+\frac{1}{c}$$

$$\left(\frac{1}{b}+\frac{1}{c}\right)(a+d)=\left(\frac{1}{a+p}+\frac{1}{a+2p}\right)(a+a+3p)$$

$$=\frac{(2a+3p)^2}{a^2+3ap+2p^2}=4+\frac{p^2}{a^2+3ap+2p^2}>4.$$

10. The sum to 'n' terms of the series $\tan^{-1}\left(\frac{1}{2}\right)+\tan^{-1}\left(\frac{2}{9}\right)+\tan^{-1}\left(\frac{1}{8}\right)+\tan^{-1}\left(\frac{2}{25}\right)+\tan^{-1}\left(\frac{1}{18}\right)+\ldots$ n terms

(A) $\tan^{-1}3$

(B) $\cot^{-1}\left(\frac{1}{3}\right)$

(C) $\tan^{-1}\left(\frac{1}{3}\right)$

(D) none of these

Sol. **(A, B).** $T_r=\tan^{-1}\left(\frac{\sqrt{2}}{r+1}\right)^2$

$$S_n=\sum_{r=1}^{n}\tan^{-1}\left(\frac{2}{r^2+2r+1}\right)=\tan^{-1}\frac{r+2-r}{1+r(r+2)}=\tan^{-1}(r+2)-\tan^{-1}r$$

$$S_n=\tan^{-1}(n+2)+\tan^{-1}(n+1)-\tan^{-1}2-\tan^{-1}1=\tan^{-1}\left(\frac{3n^2+7n}{n^2+9n+10}\right)$$

$$\therefore S_\infty=\tan^{-1}(3)=\cot^{-1}\left(\frac{1}{3}\right).$$

11. If three positive unequal numbers a, b, c are in H. P., then

(A) $a + c > 2b$

(B) $a^2 + c^2 > 2b^2$

(C) $a^2 + c^2 > 2ac$

(D) none of these.

Sol. **(A), (B), (C)** $\frac{a+b+c}{3}>b$ (H.M)

$\Rightarrow a + c > 2b$

Also $\frac{a^2+c^2}{2}>\sqrt{a^2c^2}$ $\Rightarrow a^2 + c^2 > 2ac > 2b^2$ (G.M. > H.M)

$\Rightarrow a^2 + c^2 > 2b^2$.

12. If a, b, c be three unequal positive quantities in H.P. then
(A) $a^{100} + c^{100} > 2b^{100}$ (B) $a^3 + c^3 > 2b^3$
(C) $a^5 + c^5 > 2b^5$ (D) $a^2 + c^2 > 2b^2$

Sol. **(A), (B), (C), (D)**
b is the H.M. of a and c and their G.M. = $\sqrt{ac}$
G.M > H.M $\Rightarrow$ $\sqrt{ac} > b$
And A.M. of a^n and c^n > G.M. of a^n and c^n

$$\frac{a^n + c^n}{2} > \sqrt{a^n c^n}$$

$$a^n + c^n > 2\left(\sqrt{ac}\right)^n > 2b^n$$

Put n = 100, 3, 5, 2

Comprehension

We know that, if $a_1, a_2, \ldots, a_n$ are in H.P., then $\frac{1}{a_1}, \frac{1}{a_2}, \ldots\ldots, \frac{1}{a_n}$ are in A.P. and vice versa. If $a_1, a_2, \ldots, a_n$ are in A.P. with common difference d, then for any b (>0), the numbers $b^{a_1}, b^{a_2}, b^{a_3}, \ldots\ldots, b^{a_n}$ are in G.P. with common ratio b^d. If $a_1, a_2, \ldots, a_n$ are positive and in G.P. with common ratio r, then for any base b (b > 0), $\log_b a_1, \log_b a_2, \ldots\ldots, \log_b a_n$ are in A.P. with common difference $\log_b r$.

Answer the following questions.

13. If x, y, z are respectively the pth, qth and the rth terms of an A.P., as well as of a G.P., then $x^{y-z} \cdot y^{z-x} \cdot z^{x-y}$ is equal to
(A) p (B) q
(C) r (D) 1

Sol. **(D).** Let the base be taken as e. Since x, y, z are terms of a G.P.(say with common ratio t), lnx, lny, lnz are in A.P. with common difference lnt. Also, x, y, z are in A.P. (say with common difference d). Hence x − y = (p − q)d etc.
and lnx − lny = (p − q)lnt etc.
Let $E = x^{y-z} \cdot y^{z-x} \cdot z^{x-y}$ so that
$\ln E = (y - z)\ln x + (z - x)\ln y + (x - y)\ln z = (q - r)d \ln x + (r - p)d \ln y + (p - q)d \ln z$
$= d[p(\ln z - \ln y) + q(\ln x - \ln z) + r(\ln y - \ln x)]$
$= d \ln t[p(r - q) + q(p - r) + r(q - p)] = 0$
$\Rightarrow E = 1.$

14. If a, b, c, d are in G.P. and $a^x = b^y = c^z = d^v$, then x, y, z, v are in
(A) A.P. (B) G.P.
(C) H.P (D) None of these.

Sol. **(C).** Since a, b, c, d are in G.P. (say with common ratio r), lna, lnb, lnc, lnd are in A.P. with common difference lnr. Also $a^x = b^y = c^z = d^v$
$\Rightarrow x\ln a = y\ln b = z\ln c = v\ln d$
$\Rightarrow x\ln a = y(\ln a + \ln r) = z(\ln a + 2\ln r) = v(\ln a + 3\ln r)$
$\Rightarrow \frac{x - z}{z} = 2\frac{x - y}{y}$ and $\frac{y - v}{v} = 2\frac{y - z}{z} \Rightarrow \frac{1}{x} + \frac{1}{z} = \frac{2}{y}$ and $\frac{1}{y} + \frac{1}{v} = \frac{2}{z} \Rightarrow$ x, y, z, v are in H.P.

$$\frac{2a}{p} = 2A + (p-1)d \quad \ldots (1)$$

$$\frac{2b}{q} = 2A + (q-1)d \quad \ldots (2)$$

$$\frac{2c}{r} = 2A + (r-1)d \quad \ldots (3)$$

Multiply (1), (2) and (3) by $(q - r)$, $(r - p)$ and $(p - q)$ and add

$$\therefore \sum \frac{a}{p}(q-r) = 0.$$

(B) Let common ratio be taken as k and a be the first term

$R = T_r = ak^{r-1}$

$R^{s-t} = a^{s-t} k^{(r-1)(s-t)}$

Similarly, $S^{t-r} = a^{t-r} k^{(s-1)(t-r)}$

$T^{r-s} = a^{r-s} k^{(t-1)(r-s)}$.

Multiplying the above three and knowing that

$A^m A^n A^p = A^{m+n-p}$

$R^{s-t} S^{t-r} T^{r-s} = a^0 k^0 = 1.$

(C) $x = a + (m - 1)d = AR^{m-1}$

$y = a + (n - 1)d = AR^{n-1}$

$z = a + (p - 1)d = AR^{p-1}$

$y - z = (n - p)d$, $z - x$

$= (p - m)d$, $x - y = (m - n)d$

$x^{y-z} \cdot y^{z-x} \cdot z^{x-y}$

$= \left(AR^{m-1}\right)^{(n-p)d} \cdot \left(AR^{n-1}\right)^{(p-m)d} \cdot \left(AR^{p-1}\right)^{(m-n)d}$

$= A^0 R^0 = 1.$

(D) $a = AR^{p-1}$, $b = AR^{q-1}$, $c = AR^{r-1}$ for G.P.

$$a = \frac{1}{A_1 + (p-1)D}$$

$$b = \frac{1}{A_1 + (q-1)D}$$

$$c = \frac{1}{A_1 + (r-1)D} \text{ for H.P.}$$

$$\frac{1}{b} - \frac{1}{c} = (q-r)D \quad \ldots (1)$$

$\log a = \log A + (p - 1) \log R$

$= (\log A - \log R) + (\log R)P \quad \ldots (2)$

$$\sum \left(\frac{1}{b} - \frac{1}{c}\right) \log a = D(\log A - \log R) \sum (q-r) + D \log R \sum p(q-r) = 0$$

$$\therefore \sum \frac{c-b}{bc} \log a = 0$$

multiplying by $-abc$

$$\sum a(b-c) \log a = 0.$$

Answer Q.18, Q.19 and Q.20 by appropriately matching the information given in the three columns of the following table.

Column-I contains sum of a sequence, Column-II contains sum of n terms (S_n) and Column-III contains sum of infinite terms

Column–I	Column–II	Column–III
(I) $S_n = \frac{1}{4\cdot 7} + \frac{1}{7\cdot 10} + \frac{1}{10\cdot 13} + \ldots$ n terms	(i) $S_n = \frac{n^2-1}{2n^2}$	(P) $S_\infty = \frac{7}{6}$
(II) $S_n = \sum_{r=1}^{n} \frac{8r}{r^4+4r^2+16}$	(ii) $S_n = \frac{2n}{n+1}$	(Q) $S_\infty = \frac{1}{2}$
(III) $S_n = 1 + \frac{1+2}{1^3+2^3} + \frac{1+2+3}{1^3+2^3+3^3} + \ldots$ n terms	(iii) $S_n = \frac{2}{3}\left(\frac{n^2+2n+1}{n^2+2n+4}\right)$	(R) $S_\infty = \frac{1}{12}$
(IV) $S_n = \sum_{r=2}^{n} \frac{r-\frac{1}{2}}{\left(r^2-r\right)^2}$	(iv) $S_n = \frac{n}{4(3n+4)}$	(S) $S_\infty = 2$

18. Which of the following combination is **CORRECT**?
(A) (IV) (i) (S) (B) (III) (ii) (P)
(C) (I) (iv) (R) (D) (I) (i) (Q)

Sol. **(C)**

19. Which of the following combination is **CORRECT**?
(A) (III) (i) (S) (B) (I) (iv) (Q)
(C) (IV) (i) (Q) (D) (II) (iii) (P)

Sol. **(C)**

20. Which of the following combination is **CORRECT**?
(A) (I) (i) (R) (B) (III) (ii) (S)
(C) (IV) (ii) (Q) (D) (II) (iii) (P)

Sol. **(B)**

18.-20. (I) $T_r = \frac{1}{(3r+1)(3r+4)} = \frac{1}{3}\left(\frac{1}{3r+1} - \frac{1}{3r+4}\right)$

$$S_n = \sum_{r=1}^{n} T_r = \frac{1}{3}\left(\frac{1}{4} - \frac{1}{3n+4}\right) = \frac{n}{4(3n+4)}$$

$$S_\infty = \frac{1}{12}$$

(II) $T_r = \frac{8}{16}\left(\frac{1}{1+\frac{r^2}{4}-\frac{r}{2}} - \frac{1}{1+\frac{r^2}{4}+\frac{r}{2}}\right)$

$$S_n = \sum_{r=1}^{n} T_r = 2\left(\frac{1}{3} + \frac{1}{4} - \frac{1}{n^2+3} - \frac{1}{n^2+2n+4}\right)$$

$$S_\infty = \frac{7}{6}$$

(III) $T_r = \frac{2}{r(r+1)} \therefore S_n = 2\sum_{r=1}^{n}\left(\frac{1}{r} - \frac{1}{r+1}\right) = \frac{2n}{n+1}$

(IV) $T_r = \frac{1}{2}\left(\frac{1}{(r-1)^2} - \frac{1}{r^2}\right)$

$\therefore S_n = \frac{1}{2}\left(1 - \frac{1}{n^2}\right) = \frac{n^2-1}{2n^2}$

$S_\infty = \frac{1}{2}$

21. Let a_n be the n^{th} term of an A.P. If $\sum_{r=1}^{10^{99}} a_{2r} = 10^{100}$ and $\sum_{r=1}^{10^{99}} a_{2r-1} = 10^{99}$, then the common difference of the A.P. is _____

Sol. **(9.00)**

Let d be the common difference of the A.P. then $a_{2r} = a_{2r-1} + d$

$\Rightarrow \sum_{r=1}^{10^{99}} a_{2r} = \sum_{r=1}^{10^{99}} (a_{2r-1} + d)$

$\Rightarrow \sum_{r=1}^{10^{99}} a_{2r} = \sum_{r=1}^{10^{99}} a_{2r-1} + 10^{99} d$.

$\Rightarrow 10^{100} = 10^{99} + 10^{99} d$

$\Rightarrow d = \frac{10^{100} - 10^{99}}{10^{99}} = \frac{10^{99}(10-1)}{10^{99}} = 9$.

22. Let S(n) denotes the sum of first 'n' terms of an A.P. Then the value of $S = \lim_{n\to\infty} \sum_{r=1}^{n} \frac{f(r)}{n}$, where $f(n) = \frac{S(3n)}{S(2n) - S(n)}$ _____

Sol. **(3.00)**

$S(3n) = \frac{3n}{2}\left[2a + (3n-1)d\right]$

$S(2n) = \frac{2n}{2}\left[2a + (2n-1)d\right]$

$S(n) = \frac{n}{2}\left[2a + (n-1)d\right]$

$\therefore S(2n) - S(n)$

$= \frac{n}{2}\left[2(2a + 2n - 1)d - 2a - (n-1)d\right]$

$= \frac{n}{2}\left[2a + (3n-1)d\right] = \frac{S(3n)}{3}$

$\therefore \frac{S(3n)}{S(2n) - S(n)} = 3 = f(n)$.

23. Let a, b and c form a G.P. of common ratio r, with $0 < r < 1$. If a, 2b and 3c form an A.P. then r equals _____

Sol. **(0.33)**

$b = ar, c = ar^2$

a, 2b, 3c are in A.P

$\Rightarrow 4b = a + 3c$

$\Rightarrow 4ar = a + 3ar^2$

$\Rightarrow 3r^2 - 4r + 1 = 0$

$\Rightarrow r = 1, \frac{1}{3}$ (where $r \neq 1$)

So, $r = \frac{1}{3}$

24. The minimum value of the quantity $\frac{(a^2+3a+1)(b^2+3b+1)(c^2+3c+1)}{abc}$ where $a, b, c \in R^+$ is k, then $\frac{k}{20}$ is equal to _____

Sol. **(6.25)**

$\frac{(a^2+3a+1)(b^2+3b+1)(c^2+3c+1)}{abc} \geq 5^3$

BINOMIAL THEOREM

BASIC RESULTS

- $(a+x)^n = {}^nC_0a^n + {}^nC_1a^{n-1}x + {}^nC_2a^{n-2}x^2 + \ldots + {}^nC_ra^{n-r}x^r + \ldots + {}^nC_nx^n = \sum_{r=0}^{n} {}^nC_r x^r a^{n-r}$
- $(a-x)^n = {}^nC_0a^n - {}^nC_1a^{n-1}x + {}^nC_2a^{n-2}x^2 - \ldots + (-1)^r {}^nC_ra^{n-r}x^r + \ldots + (-1)^n {}^nC_nx^n = a^{n-r}$

  $= \sum_{r=0}^{n}(-1)^r {}^nC_r x^r$
- $(a+x)^n + (a-x)^n = \sum_{r=0}^{n} {}^nC_r x^r\left[1+(-1)^r\right] = 2({}^nC_0a^n + {}^nC_2a^{n-2}x^2 \ldots\ldots)$
- $(a+x)^n - (a-x)^n = \sum_{r=0}^{n} {}^nC_r x^r\left[1-(-1)^r\right] = 2({}^nC_1a^{n-1}x + {}^nC_3a^{n-3}x^3 \ldots\ldots)$

Properties of Binomial Coefficient

For the sake of convenience the coefficients ${}^nC_0$, ${}^nC_1$, ...,${}^nC_r$, ...,${}^nC_n$ are usually denoted by $C_0, C_1, \ldots, C_r, \ldots, C_n$ respectively

- $C_0 + C_1 + C_2 + \ldots + C_n = 2^n$
- $C_0 - C_1 + C_2 - \ldots + C_n = 0$
- $C_0 + C_2 + C_4 + \ldots = C_1 + C_3 + C_5 + \ldots = 2^{n-1}$
- ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$
- $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$
- $r\,{}^nC_r = n\,{}^{n-1}C_{r-1}$
- $\frac{{}^nC_r}{r+1} = \frac{{}^{n+1}C_{r+1}}{n+1}$.

Basic Properties of Binomial Expansion of $(a+x)^n$

- There are (n+1) terms in expansion.
- Binomial coefficients of terms equidistant from beginning and end are equal.
- The general term of the expansion is ${}^nC_r a^{n-r}x^r$, this is infact the (r+1)th term from the beginning.
- If n is even, there is only one middle term namely $\left(\frac{n}{2}+1\right)$th and is equal to ${}^nC_{\frac{n}{2}}a^{\frac{n}{2}}.x^{\frac{n}{2}}$.
- If n is odd, there are two middle terms namely $\left(\frac{n+1}{2}\right)$th and $\left(\frac{n+3}{2}\right)$th and are equal to ${}^nC_{\frac{n-1}{2}}a^{\frac{n-1}{2}}.x^{\frac{n+1}{2}}$ and ${}^nC_{\frac{n+1}{2}}a^{\frac{n-1}{2}}.x^{\frac{n+1}{2}}$ respectively.
- Binomial coefficient of middle term is the greatest binomial coefficient occurring in the expansion.

Multinomial Expansion

In the expansion of $(x_1+x_2+\ldots+x_n)^m$ where $m, n \in N$ and $x_1, x_2, \ldots, x_n$ are independent variables, we have

- Total number of terms in the expansion $= {}^{m+n-1}C_{n-1}$.
- Coefficient of $x_1^{r_1} x_2^{r_2} x_3^{r_3} \cdots x_n^{r_n}$ (where $r_1+r_2+\ldots+r_n = m, r_i \in N \cup [0]$ is $\frac{m!}{r_1!r_2!\cdots r_n!}$.
- Sum of all the coefficient is obtained by putting all the variables x_i equal to 1 and is n^m.

ILLUSTRATIVE PROBLEMS

1. If in the expansion of $(2x+5)^{10}$, the greatest term is equal to the middle term, then find the value of x.

Sol. In the expansion of $(2x+5)^{10}$, the middle term $= T_6$

$$\Rightarrow \frac{n+1}{1+\left|\frac{2x}{5}\right|} > 5$$

$$\Rightarrow \frac{11}{5} - 1 > \left|\frac{2x}{5}\right| \Rightarrow |x| < 3 \Rightarrow x \in (-3, 3).$$

2. Show that there will be no term containing x^{2r} in the expansion of $(x+x^{-2})^{n-3}$ if $n-2r$ is positive but not a multiple of 3.

Sol. If possible suppose that the k^{th} term contains x^{2r} in the expansion of $(x+x^{-2})^{n-3}$, k^{th} term is given by

$$t_k = {}^{n-3}C_{k-1}(x)^{n-3-k+1}\left(x^{-2}\right)^{k-1} = {}^{n-3}C_{k-1}(x)^{n-k-2}(x)^{-2k+2}$$

$$= {}^{n-3}C_{k-1}(x)^{n-k-2-2k+2} = {}^{n-3}C_{k-1}(x)^{n-3k}$$

Since the k^{th} term contains x^{2r}

$\therefore 2r = n-3k$ ....(1)

or, $3k = n-2r$

$\therefore k = \frac{n-2r}{3}$ .....(2)

Now if $n-2r$ is positive and not divisible by 3 then it is obvious from eqn. (2) that k will not be a positive integer.

Hence if $(n-2r)$ is not divisible by 3 there will be no term containing x^{2r} in the given expansion.

3. If $(1+x)^n = \sum_{r=0}^{n} {}^nC_r x^r$, then prove that

$${}^mC_r\,{}^nC_0 + {}^mC_{r-1}\,{}^nC_1 + {}^mC_{r-2}\,{}^nC_2 + \cdots + {}^mC_1\,{}^nC_{r-1} + {}^mC_0\,{}^nC_r = {}^{m+n}C_r$$

where m, n, r are positive integers and $r < m$ and $r < n$.

Sol. $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \cdots + {}^nC_rx^r + \cdots + {}^nC_nx^n$ ... (1)

and also

$(1+x)^m = {}^mC_0 + {}^mC_1x + {}^mC_2x^2 + \cdots + {}^mC_rx^r + \cdots + {}^mC_mx^m$ ... (2)

Multiplying (1) and (2), we get

$({}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \cdots + {}^nC_rx^r + \cdots + {}^nC_nx^n)$

$\times ({}^mC_0 + {}^mC_1x + {}^mC_2x^2 + \cdots + {}^mC_rx^r + \cdots + {}^mC_mx^m) = (1+x)^{m+n}$

$= {}^{m+n}C_0 + {}^{m+n}C_1x + {}^{m+n}C_2x^2 + \cdots + {}^{m+n}C_rx^r + \cdots + {}^{m+n}C_{m+n}x^{m+n}$

Equating the coefficient of x^r, we get

${}^mC_r\,{}^nC_0 + {}^mC_{r-1}\,{}^nC_1 + {}^mC_{r-2}\,{}^nC_2 + \cdots + {}^mC_1\,{}^nC_{r-1} + {}^mC_0\,{}^nC_r = {}^{m+n}C_r$.

4. If $(1+x)^n = \sum_{r=0}^{n} {}^nC_r x^r$, then prove that $C_0^2 + \frac{C_1^2}{2} + \frac{C_2^2}{3} + \cdots + \frac{C_n^2}{n+1} = \frac{(2n+1)!}{[(n+1)!]^2}$.

Sol. Given $(1+x)^n = C_0 + C_1x + C_2x^2 + \cdots + C_nx^n$ ... (1)

Integrating w.r.t. x between the limits 0 and x we get

$$\left[\frac{(1+x)^{n+1}}{n+1}\right]_0^x = \left[C_0x + C_1\frac{x^2}{2} + C_2\frac{x^3}{3} + \cdots + C_n\frac{x^{n+1}}{n+1}\right]_0^x$$

$$\frac{(1+x)^{n+1}}{n+1} - \frac{1}{n+1} = C_0x + C_1\frac{x^2}{2} + C_2\frac{x^3}{3} + \cdots + C_n\frac{x^{n+1}}{n+1} \quad \ldots (2)$$

Also $(1+x)^n = C_0x^n + C_1x^{n-1} + C_2x^{n-2} + \cdots + C_n$ ... (3)

Multiplying (2) and (3) and equating coefficients of x^{n+1} of both sides we get,

$$C_0^2 + \frac{C_1^2}{2} + \frac{C_2^2}{3} + \cdots + \frac{C_n^2}{n+1} = \frac{{}^{2n+1}C_{n+1} - 0}{n+1} = \frac{(2n+1)!}{[(n+1)!]^2}$$

5. If f (x) is periodic with period 't' such that f(2x + 3) + f(2x + 7) = 2, then find the coefficient of m^{-3t} in expansion of $\left(m + \frac{b}{m^3}\right)^{4t}$.

Sol. $t = 4$

$T_{r+1} = {}^{16}C_r.m^{16-r}\left(\frac{b}{m^3}\right)^r$

$= {}^{16}C_rm^{16-r}.\,m^{-3r}.\,b^r$

$= {}^{16}C_rm^{16-4r}.\,b^r$

$16 - 4r = -12$

$4r = 28 \Rightarrow r = 7.$

Coefficient is ${}^{16}C_7b^7$.

SOLVED PROBLEMS

1. $\sum_{i=1}^{\infty}\sum_{j=1}^{i}\sum_{k=1}^{j}\frac{1}{a^{i+j+k}}$ is equal to where $|a| > 1$

(A) $(a-1)^{-3}$

(B) $\frac{3}{a-1}$

(C) $\frac{1}{a^3-1}$

(D) none of these

Sol. **(A).** $\frac{1}{a-1}\cdot\frac{1}{a-1}\cdot\frac{1}{a-1}=\frac{1}{(a-1)^3}$.

2. If the sum of the coefficients in the expansion of $(l^2x^2 - 2lx + 1)^{51}$ vanishes then l is equal to

(A) 2

(B) – 1

(C) 1

(D) – 2

Sol. **(C).** $[(1-lx)^2]^{51} \Rightarrow (1-lx)^{102} = A_0 + A_1x + A_2x^2 + \ldots$

put $x = 1$

$(1-l)^{102} = A_0 + A_1 + A_2 + \ldots$

Therefore $l = 1$.

3. The coefficient of $a^8b^4c^9d^9$ in $(abc + abd + acd + bcd)^{10}$ is

(A) 10!

(B) $\frac{10!}{8!4!9!9!}$

(C) 2520

(D) None of these.

Sol. **(C).** $a^{10}b^{10}c^{10}d^{10}\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}\right)^{10}$

$\therefore$ Required coefficient = coefficient of

$a^{-2}b^{-6}c^{-1}d^{-1}$ in $\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}\right)^{10} = \frac{10!}{2!6!1!1!} = \frac{10\times9\times8\times7}{2} = 2520$.

4. If $(x + 1)(x + 2) \ldots\ldots (x + n) = A_0 + A_1x + \ldots. + A_nx^n$, then $A_1 + 2A_2 + \ldots. + nA_n$ is equal to

(A) $(n+1)!.\left(\frac{1}{2}+\frac{1}{3}+\cdots\frac{1}{n+1}\right)$

(B) $n!.\left(\frac{1}{2}+\frac{1}{3}+\cdots\frac{1}{n+1}\right)$

(C) $(n+1)!.\left(1+\frac{1}{2}+\frac{1}{3}+\cdots\frac{1}{n+1}\right)$

(D) none of these

Sol. **(A).** $(x + 1)(x + 2) \ldots. (x + n) = A_0 + A_1x + \ldots. + A_nx^n$ ... (1)

Put $x = 1$,

$(n + 1)! = A_0 + A_1 + \ldots.. + A_n$ ... (2)

Take log,

$\log(x + 1) + \ldots. + \log(x + n) = \log(A_0 + A_1x + \ldots. + A_nx^n)$

Differentiate with respect to 'x',

$$\frac{1}{x+1}+\frac{1}{x+2}+\cdots+\frac{1}{x+n}=\frac{A_1+2A_2x+3A_3x^2+\cdots}{A_0+A_1x+\cdots+A_nx^n}$$

$x = 1$

$$\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{n+1}=\frac{A_1+2A_2+\cdots+nA_n}{A_0+A_1+\cdots+A_n}.$$

5. The interval in which x must lie so that the numerically greatest term in the expansion of $(1-x)^{21}$ has the greatest coefficient is, $(x > 0)$

(A) $\left[\frac{5}{6}, \frac{6}{5}\right]$ (B) $\left(\frac{5}{6}, \frac{6}{5}\right)$

(C) $\left(\frac{6}{7}, \frac{7}{6}\right)$ (D) $\left[\frac{6}{7}, \frac{7}{6}\right]$

Sol. **(B).** Numerically greatest coefficient in the expansion of $(1-x)^n$ is ${}^nC_{\frac{n-1}{2}}$ or ${}^nC_{\frac{n+1}{2}}$.

Numerically greatest coefficient is
${}^{21}C_{10}$ or ${}^{21}C_{11}$.
$|{}^{21}C_{10}x^{10}| > |{}^{21}C_9x^9| \Rightarrow x > 5/6$

Similarly, $x < \frac{6}{5}$

$\Rightarrow x \in \left(\frac{5}{6}, \frac{6}{5}\right)$.

6. The number of real negative terms in the binomial expansion of $(1+ix)^{4n-2}$, $n \in N$, $x > 0$ is

(A) n (B) n + 1

(C) n − 1 (D) 2n

Sol. **(A).** $T_{r+1} = {}^{4n-2}C_r\,(ix)^r$, (Since $x > 0$)

T_{r+1} is negative, if i^r is negative and real.
$i^r = -1 \Rightarrow r = 2, 6, 10, \ldots\ldots$ which forms an A.P.
$0 \le r \le 4n-2$
$4n - 2 = 2 + (r-1)4 \Rightarrow r = n$.
Required number of terms is n.

7. If $\left(3+a\sqrt{2}\right)^{100} + \left(3+b\sqrt{2}\right)^{100} = 7+5\sqrt{2}$, number of pairs (a, b) for which the equation is true is, (a, b are rational numbers)

(A) 1 (B) 6

(C) 0 (D) infinite

Sol. **(C).** $\left(3-a\sqrt{2}\right)^{100} + \left(3-b\sqrt{2}\right)^{100} = 7-5\sqrt{2} < 0$

But L.H.S. > 0.
So there are no such pairs (a, b).

8. Let $U_i = \frac{{}^nC_i}{{}^nC_i + {}^nC_{i+1}}$. If $\sqrt[3]{\sum_{i=1}^{n-1} U_i} = \frac{3}{2}$, then n is equal to

(A) 4 (B) 5

(C) 6 (D) 7

Sol. **(D).** $n = 7$

$U_i = \frac{i+1}{n+1}$.

On solving $n = 7$.

9. The coefficient of x^5 in the expansion of $(x^2 - x - 2)^5$ is

(A) −83 (B) −82

(C) −86 (D) −81

Sol. **(D).** $(x-2)^5(x+1)^5$
$\Rightarrow [{}^5C_0x^5 - {}^5C_1x^4 \times 2 + \ldots.][{}^5C_0 + {}^5C_1x + \ldots.]$.
Coefficient of x^5
$= 1 - 5 \cdot 5 \cdot 2 + 10 \cdot 10 \cdot 4 - 10 \cdot 10 \cdot 8 + 5 \cdot 5 \cdot 16 - 32$
$= -81$.

10. The number of values of r satisfying the equation ${}^{69}C_{3r-1} - {}^{69}C_{r^2} = {}^{69}C_{r^2-1} - {}^{69}C_{3r}$ is
(A) 1
(B) 2
(C) 3
(D) 7

Sol. **(C, D).** ${}^{69}C_{3r-1} + {}^{69}C_{3r} = {}^{69}C_{r^2-1} + {}^{69}C_{r^2}$
$\Rightarrow {}^{70}C_{3r} = {}^{70}C_{r^2}$

Thus $r^2 = 3r$ or $70 - 3r = r^2$ so that $r = 0, 3$ or $7, -10$.
Hence $r = 3$ and 7 (as the given is not defined for $r = 0$ and -10).

11. If $ac > b^2$ then the sum of the coefficients in the expansion of $(a\alpha^2x^2 + 2b\alpha x + c)^n$; (a, b, c, $\alpha \in R$, $n \in N$) is
(A) positive if $a > 0$
(B) positive if $c > 0$
(C) negative if $a < 0$, n is odd
(D) positive if $c < 0$, n is even.

Sol. **(A), (B), (C), (D)**
In the expansion of $(a\alpha^2x^2 + 2b\alpha x + c)^n$
the sum of the coefficients $= (a\alpha^2 + 2b\alpha + c)^n$
Let $f(\alpha) = a\alpha^2 + 2b\alpha + c$
Its discriminant $= 4b^2 - 4ac = 4(b^2 - ac) < 0$
Hence, $f(\alpha) < 0$ or $f(\alpha) > 0$ for all $\alpha \in R$
If $a > 0$ then $f(\alpha) > 0 \Rightarrow (a\alpha^2 + 2b\alpha + c)^n > 0$
If $c > 0$ i.e. $f(0) > 0 \Rightarrow f(\alpha) > 0 \Rightarrow (a\alpha^2 + 2b\alpha + c)^n > 0$
If $a < 0$ then $f(\alpha) < 0 \Rightarrow (a\alpha^2 + 2b\alpha + c)^n < 0$ if n is odd
If $c < 0$ i.e. $f(0) < 0 \Rightarrow f(\alpha) < 0 \Rightarrow (a\alpha^2 + 2b\alpha + c)^n > 0$ if n is even.

12. If $(1 + x)^{2n} = a_0 + a_1x + a_2x^2 + \ldots + a_{2n}x^{2n}$, then
(A) $a_0 + a_2 + a_4 + \ldots = \frac{1}{2}(a_0 + a_1 + a_2 + a_3 + \ldots..)$
(B) $a_{n+1} = a_n$
(C) $a_{n-3} = a_{n+3}$
(D) $a_{n-3} > a_{n+3}$

Sol. **(A), (B), (C)**
$0 = a_0 - a_1 + a_2 - a_3 + a_4 - a_5 + \ldots.$
$\therefore a_0 + a_2 + a_4 + \ldots. = a_1 + a_3 + a_5 + \ldots.$
$a_{n-1} = {}^{2n}C_{n+1}$ and $a_n = {}^{2n}C_n$
so $a_{n+1} = a_n$
Clearly ${}^{2n}C_{n+3} = {}^{2n}C_{n-3}$.

Comprehension

Let n be a positive integer, such that $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots a_{2n}x^{2n}$, then

Answer the following question:

13. The value of a_r when $(0 \le r \le 2n)$ is
(A) a_{2n-r}
(B) a_{n-r}
(C) a_{2n}
(D) $n.a_{2n-1}$

Sol. (A)

$(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \ldots + a_{2n}x^{2n}$, put, $x = \frac{1}{x}$

$\Rightarrow \frac{(x^2+x+1)^n}{x^{2n}} = \frac{(a_0x^{2n} + a_1x^{2n-1} + a_2x^{2n-2} + \ldots + a_{2n})}{x^{2n}}$ ...(2)

$\Rightarrow$ comparing (1) and (2)

$a_0 = a_{2n}$

$a_1 = a_{2n-2}$

$a_2 = a_{2n-2}$

.....

....

$a_{n-1} = a_{n+1}$

$a_n = a_n \Rightarrow a_r = a_{2n-r}$

14. The value of $a_0 + a_1 + a_2 + \ldots a_{n-1}$ is

(A) $\frac{1}{2}(3^n)$

(B) $\frac{1}{2}(3^n - a_n)$

(C) $\frac{a_n}{2}$

(D) $3 \cdot a_n^n$

Sol. (B)

$a_0 = a_{2n}, a_1 = a_{2n-1}, \ldots, a_{n-1} = a_{n+1} \Rightarrow (a_0 + a_1 + a_2 + \ldots + a_{n-1}) + a_n + (a_{n+1} + a_{n+2} + \ldots + a_{2n}) = 3^n$

$\Rightarrow 2(a_0 + a_1 + a_2 + \ldots + a_{n-1}) + a_n = 3^n \Rightarrow a_0 + a_1 + a_2 + \ldots + a_{n-1} = \frac{1}{2}(3^n - a_n)$.

15. The value of $a_0^2 - a_1^2 + a_2^2 - a_3^2 \cdots a_{2n}^2$ is

(A) a_n

(B) $\frac{n+1}{2}(a_0 + a_1 + a_2 + \cdots a_{2n})$

(C) $2a_n$

(D) $3a_n$

Sol. (A)

$(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \ldots + a_{2n}x^{2n}$ ....(1)

put $x = -x$

$\Rightarrow (1 - x + x^2)^n = a_0 - a_1x + a_2x^2 - a_3x^2 - a_3x^3 + \ldots + a_{2n}x^{2n}$ ....(2)

(1) × (2); $\{(1+x^2)^2 - x^2\}^n = (a_0 + a_1x + a_2x^2 + \ldots + a_{2n}x^{2n}) \times (a_9 - a_1x + a_2x^2 - \ldots + a_{2n}x^{2n})$

$\therefore a_0^2 - a_1^2 + a_2^2 - a_3^2 + \ldots + a_{2n}^2$

= coefficient of x^0 + coefficient of x^2 + coefficient of x^4 + ... + coefficient of x^{2n} in RHS

$\therefore \{(1+x^2)^2 - x^2\}^n = (1 + x^2 + x^4)^n = a_0 + a_1x^2 + a_2x^4 + \ldots + a_{2n}x^{4n}$ (putting $x = x^2$ in (1))

$= a_0 + a_1 + a_2 + \ldots + a_{2n}$.

Assertion Reason

16. Statement –1: The coefficient of $x^{3\lambda+2}$ in the expansion of $(a + x)^\lambda(b + x)^{\lambda+1}(c + x)^{\lambda+2}$ $\forall \lambda \in N$ is $\lambda(a + b + c)$.

because

Statement –2: The coefficient of x^m in the expansion of $(a + x)^n$ is ${}^nC_m a^{n-m}$.

(A) Statement -1 is True, Statement -2 is True; Statement-2 is a correct explanation for Statement-1

(B) Statement -1 is True, Statement -2 is True; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement -1 is True, Statement -2 is False

(D) Statement -1 is False, Statement -2 is True

Sol. **(D).** We have $(a+x)^{\lambda}(b+x)^{\lambda+1}(c+x)^{\lambda+2} = \{(x+a)(x+a)\ldots \lambda \text{ times}\}\{(x+b)(x+b)\ldots(\lambda+1) \text{ times}\}\{(x+c)(x+c)\ldots(\lambda+2) \text{ times}\} = x^{3\lambda+3} + \{a\lambda + b(\lambda+1) + c(\lambda+2)\}x^{3\lambda+2} + \ldots$
$\therefore$ coefficient of $x^{3\lambda+2}$ is $\lambda(a+b+c) + b + 2c$.

17. Match the following Column–I with Column–II.

	Column – I		Column – II
(A)	The number of integer values of x for which 4^{th} term is numerically greatest term in the expansion of $\left(2+\frac{3x}{8}\right)^{10}$ is	(p)	197
(B)	If [x] = greatest integer less than or equal to x, then $\left[\left(\sqrt{2}+1\right)^6\right]$ equals	(q)	1
(C)	The number of prime factors of the coefficient of x^8 in $\left(1-2x+3x^2-4x^3+5x^4-6x^5+7x^6\right)^6$ is	(r)	0
(D)	The number of real roots of the equation $x + \sin(x) = 1$ is	(s)	4
		(t)	8

Sol. **(A) → (s); (B) → (p); (C) → (s); (D) → (q)**

(A) $3 \le \frac{(n+1)\left|\frac{3x}{16}\right|}{1+\left|\frac{3x}{16}\right|} = \frac{11\left|\frac{3x}{16}\right|}{1+\left|\frac{3x}{16}\right|} < 4, \left]\frac{-64}{21}, -2\right] \cup \left[2, \frac{64}{21}\right[$. The required values of x are –2, –3, 2, 3.

(B) $\left(\sqrt{2}+1\right)^6 = I + F,\ 0 < F < 1,\ 0 < f = \left(\sqrt{2}-1\right)^6 < 1,$

$I + F + f = 2\left({}^6c_0(8) + {}^6c_2(4) + {}^6c_4(2) + 1\right) = 198;\ 0 < F + f < 2 \quad F + f = 198 - i = \text{integer} = 1$

$i = 197$

(C) Coefficient of x^8 in $\left(1+8x^7+7x^8\right)^6(1+x)^{-12} = 75048 = 2^3\,3^1(59)(53)$

(D) $x + \sin(x)$ is a bijection, hence $x + \sin x = 1$ has only one solution.

Answer Q.18, Q.19 and Q.20 by appropriately matching the information given in the three columns of the following table.

Column–I	Column–II	Column–III
(I) Let $f(x) = (1 + x + x^2)^n$, then $f(x) = a_0 + a_1x + a_2x^2 \ldots\ldots a_{2n}x^{2n}$	(i) for n = 6, $a_0 + a_2 + a_4 + a_6 \ldots\ldots a_{2n} = 364$	(P) for n = 6, $a_0 + a_3 + a_6 \ldots\ldots = 486$
(II) Let $f(x) = (1 + x + 2x^2)^n$, then $f(x) = a_0 + a_1x + a_2x^2 \ldots\ldots a_{2n}x^{2n}$	(ii) for n = 6, $a_0 + a_2 + a_4 + a_6 \ldots\ldots a_{2n} = 365$	(Q) for n = 6, $a_0 + a_3 + a_6 \ldots\ldots = 1366$
(III) Let $f(x) = (1 + x - 2x^2)^n$, then $f(x) = a_0 + a_1x + a_2x^2 \ldots\ldots a_{2n}x^{2n}$	(iii) for n = 6, $a_0 + a_2 + a_4 + a_6 \ldots\ldots a_{2n} = 32$	(R) for n = 6, $a_0 + a_3 + a_6 \ldots\ldots = 0$
(IV) Let $f(x) = (1 + 2x - 2x^2)^n$, then $f(x) = a_0 + a_1x + a_2x^2 \ldots\ldots a_{2n}x^{2n}$	(iv) for n = 6, $a_0 + a_2 + a_4 + a_6 \ldots\ldots a_{2n} = 2080$	(S) for n = 6, $a_0 + a_3 + a_6 + a_9 \ldots\ldots = 243$

18. Which of the following combination is **CORRECT**?

(A) (III) (i) (P) (B) (I) (ii) (P)

(C) (III) (iii) (P) (D) (II) (iv) (Q)

Sol. **(C)**

19. Which of the following combination is **CORRECT**?
(A) (I) (i) (S) (B) (III) (ii) (Q)
(C) (IV) (ii) (R) (D) (I) (ii) (S)

Sol. **(D)**

20. Which of the following combination is **CORRECT**?
(A) (II) (iv) (S) (B) (II) (iv) (Q)
(C) (I) (ii) (R) (D) (III) (i) (Q)

Sol. **(B)**

18.-20. If $f(x)=\left(a+bx+cx^2\right)^n=\left(a_0+a_1x+a_2x^2+a_3x^3\ldots..a_{2n}x^{2n}\right)$,

then $a_0+a_2+a_4\ldots..=\dfrac{f(1)+f(-1)}{2}$

and $a_0+a_3+a_6\ldots..=\dfrac{f(1)+f(\omega)+f\left(\omega^2\right)}{3}$

21. The value of $S=\sum_{m=0}^{\infty}\sum_{n=0}^{m}\frac{1}{3^m}\left({}^mC_n\right)$ is equal to _____

Sol. **(3.00)**

$$S=\sum_{m=0}^{\infty}\left(\frac{1}{3^m}\right)\sum_{n=0}^{m}\left({}^mC_n\right)$$

$$\Rightarrow\sum_{m=0}^{\infty}\left(\frac{1}{3^m}\right)\cdot 2^m$$

$$=\sum_{m=0}^{\infty}\left(\frac{2}{3}\right)^m$$

$$=\frac{1}{1-\frac{2}{3}}=3$$

22. $10!+\dfrac{11!}{1!}+\dfrac{12!}{2!}\ldots..$ upto 12 terms $=\dfrac{(2k)!}{k\cdot(k!)}$, then $\dfrac{k-1}{2}$ is equal to _____

Sol. **(5.00)**

$$\frac{10!}{0!}+\frac{11!}{1!}+\frac{12!}{2!}\ldots..\text{ upto 12 terms}$$

$$(10!)\left\{\frac{10!}{10!0!}+\frac{11!}{10!1!}+\frac{12!}{10!2!}\ldots..\frac{(21)!}{10!(11)!}\right\}$$

$$(10!)\left\{{}^{10}C_0+{}^{11}C_1+{}^{12}C_2\ldots..{}^{21}C_2\right\}$$

$$(10!)\cdot\left\{{}^{22}C_{11}\right\}=\frac{22!}{11(11)!}$$

$$\Rightarrow k=11$$

23. If the value of $\sum_{r=0}^{n}(-1)^r \frac{{}^nC_r}{{}^{r+3}C_r} = \frac{a!}{b(n+c)}$, then $\frac{a+b+c}{5}$ is equal to ____

Sol. **(1.60)**

$$\sum_{r=0}^{n}(-1)^r \frac{{}^nC_r}{{}^{r+3}C_r} = \sum_{r=0}^{n}(-1)^r \frac{n!}{r!(n-r)!}\cdot\frac{r!3!}{(r+3)!}\cdot\frac{(n+3)!}{(n+3)!}$$

$$= \sum_{r=0}^{n}(-1)^r \frac{3!n!}{(n+3)!}\,{}^{(n+3)}C_{r+3}$$

$$= \frac{3!n!}{(n+3)!}\left[1-(n+3)+\frac{(n+3)(n+2)}{2}\right]$$

$$= \frac{3!}{2(n+3)}$$

24. If the sum of $\sum_{r=0}^{10} {}^{10+r}C_r = {}^mC_{10}$, then value of $\frac{m}{5}$ is ____

Sol. **(4.20)**

Let $S = \sum_{r=0}^{10} {}^{10+r}C_r = \sum_{r=0}^{10} {}^{10+r}C_{10}$

$\Rightarrow S = {}^{10}C_{10} + {}^{11}C_{10} + {}^{12}C_{10} \ldots\ldots {}^{20}C_{10}$

Coefficient of x^{10} in $(1+x)^{10} + (1+x)^{11} + (1+x)^{12} \ldots\ldots (1+x)^{20}$

Coefficient of x^{10} in $(1+x)^{10}\left\{\frac{(1+x)^{11}-1}{1+x-1}\right\} = \frac{(1+x)^{21}-(1+x)^{10}}{x}$

$\Rightarrow$ Coefficient of x^{11} in $(1+x)^{21} - (1+x)^{10}$

$\Rightarrow {}^{21}C_{11}$ or ${}^{21}C_{10}$

PERMUTATION & COMBINATION 17

COUNTING PRINCIPLES

There are two fundamental counting principles viz. Multiplication principle and Addition principle.

Multiplication Principle

If one experiment has n possible outcomes and another experiment has m possible outcomes, then there are $m \times n$ possible outcomes when both of these experiments are performed.

In other words, if a job has n parts and the job will be completed only when each part is completed and the first part can be completed in a_1 ways, the second part can be completed in a_2 ways and so on . . . the nth part can be completed in a_n ways, then the total number of ways of doing the job is $a_1a_2a_3...a_n$. This is known as the Multiplication principle.

Addition Principle

If one experiment has n possible outcomes and another has m possible outcomes, then there are (m + n) possible outcomes when exactly one of these experiments is performed.

In other words, if a job can be done by n methods and by using the first method, can be done in a_1 ways or by second method in a_2 ways and so on … by the nth method in a_n ways, then the number of ways to get the job done is $(a_1 + a + ... + a_n)$.

PERMUTATIONS (ARRANGEMENT OF OBJECTS)

The number of permutations of n objects, taken r at a time, is the total number of arrangements of r objects, selected from n objects where the order of the arrangement is important.

- **Without Repetition:** arranging n distinct objects, taken r at a time is equivalent to filling r places from n difference things.

r–Places:

1	2	3	4	.....	r
n	n-1	n-2	n-3		n -(r-1)

Number of Choices:

The number of ways of arranging = The number of ways of filling r places

$= n(n-1)(n-2)\ldots(n-r+1)$

$= \dfrac{n(n-1)(n-2)\ldots(n-r+1)((n-r)!)}{(n-r)!} = \dfrac{n!}{(n-r)!} = {}^nP_r$

- The number of arrangements of n different objects taken all at a time $= {}^np_n = n!$
- **With Repetition:** The number of permutations (arrangements) of n different objects, taken r at a time, when each object may occur once, twice, thrice …. upto r times in any arrangement

= The number of ways of filling r places where each place can be filled by any one of n objects .

r–Places:

1	2	3	4	.....	r
n	n	n	n		n

Number of Choices:

The number of permutations = The number of ways of filling r places

$= (n)^r$

- The number of arrangements that can be formed using n objects out of which p are identical (and of one kind), q are identical (and of another kind), r are identical (and of another kind) and the rest are distinct is $\frac{n!}{p!q!r!}$

Circular Permutations

The arrangements we have considered so far were linear. There are arrangements in closed loops also, called as circular arrangements.

Suppose n persons $(a_1, a_2, a_3, \ldots, a_n)$ are to be arranged around a circular table. There are n! ways in which they can be arranged in a row. On the other hand, all the linear arrangements depicted by

$a_1, a_2, a_3, \ldots, a_n$

$a_n, a_1, a_2, \ldots, a_{n-1}$

$a_{n-1}, a_n, a_1, a_2, \ldots, a_{n-2}$

.....

.....

.....

$a_2, a_3, a_4, \ldots, a_1$

will lead to the same arrangement for a circular table. Hence each circular arrangement corresponds to n linear arrangements (i.e. in a row). Hence the total number of circular arrangements of n persons is $\frac{n!}{n} = (n-1)!$.

Note:

- If no distinction is made between the clockwise and the anti-clockwise arrangements of n different objects around a circle, then the number of arrangements is $\frac{1}{2}(n-1)!$

COMBINATIONS

Meaning of combination is selection of objects.

Selection of Objects without Repetition

The number of selections (combinations or groups) that can be formed from n different objects taken r $(0 \le r \le n)$ at a time is ${}^{n}C_{r} = \frac{n!}{r!(n-r)!}$.

Restricted Selection / Arrangement

(a) The number of ways in which r objects can be selected from n different objects if k particular objects are
(i) always included $= {}^{n-k}C_{r-k}$
(ii) never included $= {}^{n-k}C_{r}$

(b) The number of arrangements of n distinct objects taken r at a time so that k particular objects are
(i) always included $= {}^{n-k}C_{r-k} \cdot r!$
(ii) never included $= {}^{n-k}C_{r} \cdot r!$

(c) The number of combinations of n objects, of which p are identical, taken r at a time is
$= {}^{n-p}C_{r} + {}^{n-p}C_{r-1} + {}^{n-p}C_{r-2} + \ldots + {}^{n-p}C_{0}$ if $r \le p$ and it is:
$= {}^{n-p}C_{r} + {}^{n-p}C_{r-1} + {}^{n-p}C_{r-2} + \ldots + {}^{n-p}C_{r-p}$ if $r > p$

ALL POSSIBLE SELECTIONS

Selection from Distinct Objects

The number of selections from n different objects, taken at least one
$= {}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots + {}^nC_n = 2^n - 1.$

Selection from Identical Objects

(a) The number of selections of r objects out of n identical objects is 1.

(b) Total number of selections of zero or more objects from n identical objects is n + 1.

(c) The total number of selections of at least one out of $a_1+a_2+a_3+\ldots+a_n$ objects, where a_1 are alike (of one kind), a_2 are alike (of second kind) and so on .... a_n are alike (of nth kind), is $[(a_1+1)(a_2+1)(a_3+1)\ldots(a_n+1)] - 1$.

Selection when both Identical and Distinct Objects are Present

The number of selections taking at least one out of $a_1+a_2+a_3+\ldots+a_n+k$ objects, where a_1 are alike (of one kind), a_2 are alike (of second kind) and so on ... a_n are alike (of nth kind), and k are distinct $= [(a_1+1)(a_2+1)(a_3+1)\ldots(a_n+1)]$ $2^k - 1$.

Division and Distribution of Objects (with fixed number of objects in each group)

Into groups of unequal size (different number of objects in each group)

- The number of ways in which n distinct objects can be divided into r unequal groups containing a_1 objects in the 1st group, a_2 objects in the second group and so on

$$= {}^nC_{a_1} \cdot {}^{n-a_1}C_{a_2} \cdot {}^{n-a_1-a_2}C_{a_3} \ldots {}^{a_r}C_{a_r} = \frac{n!}{a_1!a_2!a_3!\cdots a_r!}$$

Here $a_1+a_2+a_3+\text{----}+a_r = n$.

- The number of ways in which n distinct objects can be distributed among r persons such that first person gets a_1 objects, 2nd person gets a_2 objects ... , r^{th} person gets a_r objects $= \frac{n!r!}{a_1!a_2!a_3!\cdots a_r!}$.

EXPLANATION

Let us divide the task into two parts . In the first part, we divide the objects into groups. In the second part, these r groups can be assigned to r persons in r! ways.

Into groups of equal size (each group containing same number of objects)

- The number of ways in which m×n distinct objects can be divided equally into n groups (unmarked) $= \frac{(mn)!}{(m!)^n n!}$

- The number of ways in which m× n different objects can be distributed equally among n persons (or numbered groups) = (number of ways of dividing into groups) × (number of groups)! $= \frac{(mn)!n!}{(m!)^n n!} = \frac{(mn)!}{(m!)^n}$

For example if out of 50 players, 5 teams of 10 players each have to be formed this becomes a question on grouping and thus the required number of ways to form such teams is $\frac{(50)!}{(10!)^5(5!)}$.

MULTINOMIAL THEOREM

Let $x_1, x_2, \ldots, x_m$ be integers. Then the number of solutions to the equation

$$x_1 + x_2 + \ldots + x_m = n \quad \ldots(1)$$

subject to the condition $a_1 \le x_1 \le b_1, a_2 \le x_2 \le b_2, \ldots, a_m \le x_m \le b_m \quad \ldots(2)$

is equal to the coefficient of x^n in

$$\left(x^{a_1} + x^{a_1+1} + \ldots + x^{b_1}\right)\left(x^{a_2} + x^{a_2+1} + \ldots + x^{b_2}\right)\ldots\left(x^{a_m} + x^{a_{m+1}} + \ldots + x^{b_m}\right) \quad \ldots(3)$$

This is because the number of ways, in which the sum of m integers in (1) equals n, is the same as the number of times x^n comes in (3).

Note:

- We use the result that the coefficient of x^r in the expansion of $(1-x)^{-n} = {}^{n+r-1}C_r$

Different Cases of Multinomial Theorem

Case I: If the upper limit of a variable is more than or equal to the sum required and lower limit of all the variables are non-negative, then the upper limit of that variable can be taken as infinite.

Case II: When coefficients of variables are not unity

The number of solutions of $ax_1 + bx_2 + cx_3 = n$

Subject to the condition $a_1 \le x_1 \le b_1$, $a_2 \le x_2 \le b_2$, $a_3 \le x_3 \le b_3$

Is coefficient of t^n in

$$\left(\left(t^a\right)^{a_1} + \left(t^a\right)^{a_1+1} + \ldots\ldots + \left(t^a\right)^{b_1}\right) \times \left(\left(t^b\right)^{a_2} + \left(t^b\right)^{a_2+1} + \ldots\ldots + \left(t^b\right)^{b_2}\right) \times \left(\left(t^c\right)^{a_3} + \left(t^c\right)^{a_3+1} + \ldots\ldots + \left(t^c\right)^{b_3}\right)$$

Case III: When variables are distinct

In such a case, we introduce some new variables to remove the condition of distinctness

Case – IV: When the required sum is not fixed

To find the number of solutions of the equation

$x_1 + x_2 + \ldots + x_m \le n \quad \ldots \quad (1)$

We introduce a dummy variable x_{m+1} so that

$x_1 + x_2 + \ldots + x_{m+1} = n, \; x_{m+1} \ge 0 \quad \ldots. \; (2)$

The number of solutions of (1) and (2) will be same

ILLUSTRATIVE PROBLEMS

1. Let $n_1 = x_1 x_2 x_3$ and $n_2 = y_1 y_2 y_3$ be two 3-digit numbers. How many pairs of n_1 and n_2 can be formed so that n_1 can be subtracted from n_2 without borrowing?

Sol. Clearly n_1 can be subtracted from n_2 without borrowing if $y_i \ge x_i$ for $i = 1, 2, 3$. Let $x_i = r$, where $r = 0$ to 9 for $i = 2$ and 3

and $r = 1$ to 9 for $i = 1$.

Now as per our requirement $y_i = r, r+1, \ldots, 9$. Thus we have $(10 - r)$ choices for y_i.

Hence the total ways of choosing y_i and x_i

$$= \left(\sum_{r=1}^{9}(10-r)\right)\left(\sum_{r=0}^{9}(10-r)\right)^2 = 45.(55)^2.$$

2. A delegation of four students is to be selected from a total of 12 students. In how many ways can the delegation be selected

(a) if all the students are equally willing?

(b) if two particular students have to be included in the delegation?

(c) if two particular students do not wish to be together in the delegation?

(d) if two particular students wish to be included together only?

(e) if two particular students refuse to be together and two other particular students wish to be together only in the delegation?

Sol. (a) Formation of delegation means selection of 4 out of 12. Hence the number of ways $= {}^{12}C_4 = 495$

(b) If two particular students are already selected. Here we need to select only 2 out of the remaining 10. Hence the number of ways $= {}^{10}C_2 = 45$.

(c) The number of ways in which both are selected = 45. Hence the number of ways in which the two are not included together = 495 − 45 = 450.

(d) There are two possible cases

(i) Either both are selected. In this case, the number of ways in which the selection can be made = 45

(ii) Or both are not selected. In this case all the four students are selected from the remaining ten students.

This can be done in ${}^{10}C_4 = 210$ ways.

Hence the total number of ways of selection = 45 + 210 = 255.

(e) We assume that students A and B wish to be selected together and students C and D do not wish to be together. Now there are following 6 cases.

(i) (A, B, C) selected, (D) not selected

(ii) (A, B, D) selected (C) not selected

(iii) (A, B) selected (C, D) not selected

(iv) (C) selected (A, B, D) not selected

(v) (D) selected (A, B, C) not selected

(vi) A, B, C, D not selected

For (i) the number of ways of selection $= {}^{8}C_1 = 8$

For (ii) the number of ways of selection $= {}^{8}C_1 = 8$

For (iii) the number of ways of selection $= {}^{8}C_2 = 28$

For (iv) the number of ways of selection $= {}^{8}C_3 = 56$

For (v) the number of ways of selection $= {}^{8}C_3 = 56$

For (vi) the number of ways of selection $= {}^{8}C_4 = 70$

Hence total number of ways = 8 + 8 + 28 + 56 + 56 + 70 = 226.

3. In how many ways the sum of upper faces of four distinct dies can be six.

Sol. Here the number of required ways will be equal to the number of solutions of $x_1 + x_2 + x_3 + x_4 = 6$, $1 \le x_1, x_2, x_3, x_4 \le 6$

Since the upper limit is six which is equal to the sum required, so upper limit can be taken as infinite.

So number of solutions

= coefficient of t^6 in $(t + t^2 + \ldots\ldots)^4$

= coefficient of t^2 in $(1 - t)^{-4} = {}^{4+2-1}C_2 = 10$.

4. In how many ways can 15 identical blankets be distributed among six beggars such that everyone gets at least one blanket and two particular beggars get equal blankets and another three particular beggars get equal blankets.

Sol. The number of ways of distributing blankets is equal to the number of solutions of the equation $3a + 2b + c = 15$

$a, b, c \ge 1$, which is equal to coefficient of t^{15} in

$(t^3 + t^6 + t^9 \ldots)(t^2 + t^4 + \ldots)(t + t^2 + \ldots)$
= coefficient of t^9 in $(1 + t^3 + t^6 + t^9)(1 + t^2 + t^4 + t^6 + t^8)(1 + t + t^2 + \ldots + t^9)$ (neglecting higher powers)
= coefficient of t^9 in $(1 + t^2 + t^3 + t^4 + t^5 + 2t^6 + t^7 + 2t^8 + 2t^9)(1 + t + t^2 + \ldots + t^9)$
$= 1 + 1 + 1 + 1 + 1 + 2 + 1 + 2 + 2 = 12$

5. In how many ways can 14 identical toys be distributed among three boys so that each one gets at least one toy and no two boys get equal number of toys?

Sol. Let the boys get a, b and c toys respectively
$a + b + c = 14$, a , b, $c \geq 1$ and a, b and c are distinct
let $a < b < c$ and $x_1 = a$, $x_2 = b - a$, $x_3 = c - b$
So, $3x_1 + 2x_2 + x_3 = 14$, $x_1, x_2, x_3 \geq 1$
The number of solutions is equal to the coefficient of t^{14} in
$(t^3 + t^6 + t^9 \ldots)(t^2 + t^4 + \ldots)(t + t^2 + \ldots)$
= coefficient of t^8 in $(1 + t^3 + t^6)(1 + t^2 + t^4 + t^6 + t^8)(1 + t + t^2 + \ldots + t^8)$ (neglecting higher powers)
=coefficient of t^8 in $(1 + t^2 + t^3 + t^4 + t^5 + 2t^6 + t^7 + 2t^8)(1 + t + t^2 + \ldots + t^8)$
$= 1 + 1 + 1 + 1 + 1 + 2 + 1 + 2 = 10$
Since the three distinct numbers can be assigned to three boys in 3! ways. So, corresponding to each solution, we have six ways of distribution. So, the total numbers of ways $= 10 \times 6 = 60$.

SOLVED PROBLEMS

1. n-similar balls each of weight w when weighed in pairs the sum of the weights of all the possible pairs is 120 when they are weighed in triplets the sum of the weights comes out to be 480 for all possible triplets, then n is
(A) 5 (B) 10
(C) 15 (D) 20

Sol. **(B).** ${}^{n}C_{2} = \frac{n(n-1)}{2}$ number of possible pairs of n objects.

Total weight of $\frac{n(n-1)}{2}$ pairs are $\frac{n(n-1)}{2} \times 2 \times w = n(n-1)w$ units

similarly total weight of all triplets = 480 ...(1)

$\Rightarrow \frac{n(n-1)(n-2)w}{2} = 480$ ...(2)

dividing (1) and (2), we get

$\frac{n-2}{2} = 4 \Rightarrow n = 10$

2. A seven digit number without repetition and divisible by 9 is to be formed by using 7 digits out of 1, 2, 3, 4, 5, 6, 7, 8, 9. The number of ways in which this can be done is;
(A) 9! (B) 2 (7!)
(C) 4 (7!) (D) none of these

Sol. **(C).** Sum of 7 digits = a multiple of 9
Sum of number 1, 2, 3, 4, 5, 6, 7, 8, 9 = 45 so two left should also have sum of 9.
The pair of left numbers is (1, 8), (2, 7), (3, 6), (4, 5).
with each pair left number of 7 digit number = 7!
So with all 4 pairs = $4 \times 7!$.

3. A seven-digit number made up of all distinct digits 8, 7, 6, 4, 2, x and y is divisible by 3. Then possible number of order pair (x, y) is
(A) 4 (B) 8
(C) 2 (D) none of these

Sol. **(B).** 8, 7, 6, 4, 2, x and y
Any number is divisible by 3 if sum of digits is divisible by 3
i.e. x + y + 27 is divisible by 3.
x and y can take values from 0, 1, 3, 5, 9
possible pairs (5, 1) (3, 0) (9, 0) (9, 3) & (1, 5), (0, 3) (0, 9) (3, 9).

4. Total number of divisors of $n = 3^5 . 5^7 . 7^9$ that are of the form $4\lambda + 1, \lambda \geq 0$, is equal to
(A) 240 (B) 30
(C) 120 (D) 15.

Sol. **(A).** $3^{n_1} = (4-1)^{n_1} = 4\lambda_1 + (-1)^{n_1}$

$5^{n_2} = (4+1)^{n_2} = 4\lambda_2 + 1$

$7^{n_3} = (8-1)^{n_3} = 4\lambda_3 + (-1)^{n_3}$.

Hence, any positive integer power of 5 will be in the form of $4\lambda_2 + 1$. Even power of 3 and 7 will be in the form of $4\lambda + 1$ and odd power of 3 and 7 will be in the form of $4\lambda - 1$ thus required no. of divisors
$= 8 (3.5 + 3.5) = 240$

5. A delegation of five students is to be formed from a group of 10 students. If three particular students want to remain together whereas two particular students do not want to remain together then the number of selections is
(A) 10
(B) 20
(C) 30
(D) none of these

Sol. **(B).** A, B, C want to remain together
D, E do not want to remain together

IN	OUT	NUMBER OF SELECTIONS
A, B, C, D	E	$^5C_1 = 5$
A, B, C, E	D	$^5C_1 = 5$
D	A, B, C, E	$^5C_4 = 5$
E	A, B, C, D	$^5C_4 = 5$

Required number of selections is
$^5C_1 + {}^5C_1 + {}^5C_4 + {}^5C_4 = 20$.

6. In a shooting competition a man can score 5, 4, 3, 2, 1 or 0 points for each shot. Then the number of different ways in which he can score 30 in seven shots is
(A) 6538
(B) 6539
(C) 6537
(D) none of these

Sol. **(D).** The number of ways of making 30 in 7 shots is the coefficient of x^{30} in $(x^0 + x^2 + x^3 + x^4 + x^5)^7$
$\Rightarrow (x^0 + x^2 + x^3 + x^4 + x^5)^7 = [x^4(x + 1) + (x^3 + x^2 + 1)]^7$
$\Rightarrow x^{28}(x + 1)^7 + 7C_1 x^{24}(x + 1)^6 (x^3 + x^2 + 1) + 7C_2 x^{20}(x + 1)^5 (x^3 + x^2 + 1)^2 + \ldots$
so, coefficient of x^{30} is $7C_5 + 7C_1 ({}^6C_3 + {}^6C_2 + 1) + 7C_2 ({}^5C_1 + 2)$
$= 21 + 7(20 + 15 + 1) + 21(5 + 2) = 420$.

7. If m and n are positive integers more than or equal to 2, $m > n$, then $(mn)!$ is divisible by
(A) $(m!)^n$
(B) $(n!)^m$
(C) $(m + n)!$
(D) $(m - n)!$

Sol. **(A, B, C, D).** $\dfrac{(mn)!}{(m!)^n}$ is the number of ways of distributing mn distinct objects in n persons equally.

Hence $\dfrac{(mn)!}{(m!)^n}$ is an integer $\Rightarrow (m!)^n \mid (mn)!$. Similarly $(n!)^m \mid (mn)!$

Further $m + n < 2m \le mn \Rightarrow (m + n)! \mid (mn)!$ and $m - n < m < mn$
$\Rightarrow (m - n)! \mid (mn)!$

8. Let n be a positive integer with $f(n) = 1! + 2! + 3! + \ldots + n!$ and P(x), Q(x) be polynomials in x such that $f(n+2) = P(n)f(n+1) + Q(n)f(n)$ for all $n \ge 1$. Then
(A) $P(x) = x + 3$
(B) $Q(x) = -x - 2$
(C) $P(x) = -x - 2$
(D) $Q(x) = x + 3$

Sol. **(A, B).** $f(n) = 1! + 2! + 3! + \ldots + n!$
$f(n + 1) = 1! + 2! + 3! + \ldots + (n + 1)!$
$f(n + 2) = 1! + 2! + 3! + \ldots + (n + 2)!$
$f(n + 2) - f(n + 1) = (n + 2)! = (n + 2)(n + 1)! = (n + 2)[f(n + 1) - f(n)]$
$\Rightarrow f(n+2) = (n + 3)f(n + 1) - (n + 2)f(n) \Rightarrow P(x) = x + 3, Q(x) = -x - 2$

9. The number of ways in which we can choose 2 distinct integers from 1 to 100 such that the difference between them is at most 10 is

(A) ${}^{100}C_2 - {}^{90}C_2$ (B) ${}^{100}C_{98} - {}^{90}C_{88}$

(C) ${}^{100}C_2 - {}^{90}C_{88}$ (D) None of these

Sol. **(A, B, C).** Let the chosen integers be x_1 and x_2.

Let there be a integer before x_1, b integer between x_1 and x_2 and c integer after x_2

$\therefore$ $a + b + c = 98$. Where $a \geq 0, b \geq 10, c \geq 0$

Now if we consider the choices where difference is at least 11, then the number of solution is ${}^{88-3+1}C_{3-1} = {}^{90}C_2$

$\therefore$ The number of ways in which b is less than 10 is ${}^{100}C_2 - {}^{90}C_2$ which is equal to (A), (B) and (C) option.

10. 'n' locks and 'n' corresponding keys are available. But the actual combination is not known. The maximum numbers of trials that are needed to assign the keys to their corresponding locks are

(A) ${}^{n}C_2$ (B) $\sum_{k=2}^{n}(k-1)$

(C) n! (D) ${}^{n+1}C_2$

Sol. **(A), (B).** First key will be tried for at the most (n-1) locks. Second key will be tried for at the most (n - 2) locks and so on. Thus the maximum number of trials needed

$= (n-1) + (n-2) + \ldots 1 = \frac{n(n-1)}{2} = {}^{n}C_2$.

Comprehension

Let A be the set of all 2×2 matrices whose entries are binomial coefficients in the expansion of $(1 + x)^n$. (n is prime > 2)

11. Number of matrices in A are

(A) $(n+1)^4$ (B) $\frac{(n+1)^4}{16}$

(C) $(n-1)^4$ (D) $(n+1)!$

Sol. **(B)** Binomial coefficients are ${}^{n}C_0, {}^{n}C_1, \ldots, {}^{n}C_n$.

$\frac{n+1}{2}$ coefficients are distinct starting from ${}^{n}C_0$ to $\frac{{}^{n}C_{n-1}}{2}$.

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

For every linear arrangement there is a matrix.

$\therefore$ number of matrices in $A = \left(\frac{n+1}{2}\right)^4$.

12. Number of symmetric matrices in A are

(A) $(n+1)^3$ (B) $\frac{3(n+1)^3}{8}$

(C) $\frac{(n^2-1)(n-3)}{8}$ (D) $\frac{(n+1)^3}{8}$

Sol. **(D)** $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ (symmetric matrix) $\leftarrow$ (a b b c) or (a b c)

Number of symmetric matrix = number of linear arrangements of type (a b c) $= \frac{(n+1)^3}{8}$.

13. Number of matrices in A for which trace is divisible by 'n'

(A) $\frac{(n^2-1)^2}{16}$ (B) $\frac{(n+1)^4}{16}$

(C) $\frac{(n+1)^3(n-1)}{16}$ (D) $(n^2-1)^2$

Sol. **(A)** Note that ${}^nC_r$ is divisible by n if n is prime for all r except r = 0.

$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \leftarrow$ (a b c d) total number of such matrices $= \left(\frac{n-1}{2}\right)\left(\frac{n+1}{2}\right)^2\left(\frac{n-1}{2}\right) = \frac{(n^2-1)^2}{16}$.

Assertion Reason

14. Statement –1: Number of rectangle in a chess board is ${}^8C_2 \times {}^8C_2$.
because
Statement –2: To form a rectangle we have to select any two of the horizontal line and any two vertical line.

(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1
(B) Statement -1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement -1 is True, Statement -2 is False
(D) Statement -1 is False, Statement -2 is True

Sol. **(D)**

15. Match the following

	Column – I		Column – II
(A)	The number of permutations of the letters of the word HINDUSTAN such that neither the pattern 'HIN' nor 'DUS' nor 'TAN' appears, are	(p)	169194
(B)	Taking all the letters of word "MATHEMATICS" how many words can be formed in which either M or T are together?	(q)	$\frac{9\,9!}{2!}$
(C)	The number of ways of distributing 15 identical toys among some or all of the students in a classroom consisting of 15 students such that no student gets all the toys	(r)	${}^{29}C_{15} - 15$
(D)	The total number of eight-digit numbers, the sum of whose digits is odd, is	(s)	45×10^6
		(t)	0

Sol. **(A).** Total number of permutations $= \frac{9\,!}{2\,!}$; Number of those containing 'HIN' = 7!

Number of those containing 'DUS' $= \frac{7!}{2!}$; Number of those containing 'TAN' = 7!

Number of those containing 'HIN' and 'DUS' = 5!

Number of those containing 'HIN' and 'TAN' = 5!

Number of those containing 'TAN' and 'DUS' = 5!

Number of those containing 'HIN', 'DUS' and 'TAN' = 3!

Required number = $\frac{9!}{2!}-\left(7!+7!+\frac{7!}{2}\right)+3\times5!-3!=169194$

(B) M – 2, T – 2, A – 2, H – 1, E – 1, I – 1, C – 1, S – 1.

Number of words in which both M are together + Number of words in which both T are together – Number of words in which both T and both M are together = required number of words

Required number of words = $\frac{10!}{2!\,2!}+\frac{10!}{2!\,2!}-\frac{9!}{2!}=\frac{5\cdot9!+5\cdot9!-9!}{2!}=\frac{9\cdot9!}{2!}$.

(C) Let the toys be distributed to 'r' students where $2\le r\le15$

$x_1+x_2+\ldots+x_r=15$

$^{15-1}C_{r-1}$ is the required number of ways of distributing selecting r students is

$^{15}C_r=\frac{15}{r}C_r\times{}^{14}C_{r-1}$

$\sum_{r=2}^{15}{}^{15}C_r\times{}^{14}C_{r-1}$ = coefficient of x in $(1+x)^{15}\left(1+\frac{1}{x}\right)^{14}-{}^{15}C_1\times{}^{14}C_0$

= coefficient of x^{15} in $(1+x)^{29}-15={}^{29}C_{15}-15$.

(D) The numbers will vary from 10000000 to 99999999. If sum of digits of a particular number is even, then the sum of digits of its next consecutive number will be odd and the sum of digits of next consecutive number will be even and so on.

As sum of digits of first number is odd and sum of digits of last number is even,

So number of numbers with sum of digits as odd

$=\frac{\text{total number of 8-digit numbers}}{2}=\frac{90000000}{2}=45\times10^6$

Answer Q.16, Q.17 and Q.18 by appropriately matching the information given in the three columns of the following table.

For the equation $x_1+x_2+x_3\,x_4\,x_5\,x_6=n$ where $n\ge x_1, x_2, x_3\ge0$. Column-I: lists conditions on x_i, Column-II: lists conditions on n and Column-III: lists integral solutions for the equation

Column–I	Column–II	Column–III
(I) $x_4=x_5=x_6=1$	(i) 20	(P) 435
(II) $x_4=2, x_5=x_6=1$	(ii) 24	(Q) 651
(III) $x_4=x_5=x_6=-1$	(iii) 28	(R) 320
(IV) $x_1=x_2=0$	(iv) 30	(S) 512

16. Which of the following combination is **CORRECT**?

(A) (I) (ii) (Q) (B) (II) (iii) (S)

(C) (I) (iii) (P) (D) (II) (iv) (R)

Sol. **(C)**

17. Which of the following combination is **CORRECT**?

(A) (III) (i) (Q) (B) (III) (ii) (R)

(C) (I) (ii) (P) (D) (II) (iii) (S)

Sol. **(A)**

18. Which of the following combination is **INCORRECT**?
(A) (II) (ii) (P)
(B) (IV) (ii) (R)
(C) (III) (ii) (P)
(D) (IV) (iv) (S)

Sol. **(B)**

16.-18. If $x_4 = x_5 = x_6 = 1 \Rightarrow x_1 + x_2 + x_3 = n$

Number of solution $= {}^{n+2}C_2$. If $x_4 = 2, x_5 = x_6 = 1 \Rightarrow x_1 + x_2 + 2x_3 = n$

Number of solution $= {}^{n+1}C_1 + {}^{n-1}C_1 + \ldots + {}^{1}C_1 = \dfrac{(n+2)^2}{2}$ (n is even)

If $x_4 = x_5 = x_6 = -1 \Rightarrow x_1 + x_2 - x_3 = n$

Number of solution $= {}^{n+1}C_1 + {}^{n+2}C_1 + \ldots + {}^{2n+1}C_1 = \dfrac{(n+1)(3n+2)}{2}$

If $x_1 = x_2 = 0$. If $n = 24$

$\Rightarrow$ Number of solution $= {}^{6}C_3 {}^{4}C_3 \cdot \left({}^{1+3}C_2\right) = 320$

19. A 8 × 8 square grid allows movement only along the left-right or up-down direction with no back tracking. If A is the total number of paths from lower left corner to upper right corner and B is the number of paths from lower left corner to upper right corner which do not cut the diagonal joining the two end points, then $\dfrac{A}{B}$ is _____

Sol. **(9.00)**

$A = {}^{16}C_8$ (16 steps need to choose 8 up steps)

$B = \dfrac{1}{9} {}^{16}C_8$. If any path from the left bottom corner (0, 0) to right upper corner (8, 8) cuts the diagonal joining the end points we reflect the part of this path beyond the first point of intersection in the diagonal joining (0, 1) and (7, 8). This path then ends at (7, 9) and has 9 upward steps one can see that this is a 1–1 mapping so $B = {}^{16}C_8 - {}^{16}C_9 = \dfrac{1}{9} {}^{16}C_8$

20. If the number of integral solutions to the system of equation $x_1 + x_2 + x_3 + x_4 + x_5 = 20$ and $x_1 + x_2 + x_3 = 5$, when $x_k \geq 0$, is M and M can be written as p^2q, $(p, q \in N)$, then the largest possible value of p is _____

Sol. **(4.00)**

Here, $M = {}^{7}C_2 \times {}^{16}C_1 = 336 = 4^2 \cdot 21$

$\therefore p = 4$

21. Eight boxes are arranged in a row. In how many ways can five distinct balls be put into them such that each box contains at most one ball and no two empty boxes are adjacent? _____

Sol. **(2400.00)**

Consider 5 boxes with 1 ball each, now there are 6 spaces in which to put 3 empty boxes

$\therefore N = {}^{6}C_3 5! = 5 \times 4 \times 5! = 2400$

22. If minimum number of points of intersection of 8 straight lines and 8 circles be 'a' and maximum number of points of intersection be 'b', then a + b equals _____

Sol. **(212.00)**

$a = 0$

$b = {}^{8}C_2 + 2^{8}C_2 + 2 \cdot {}^{8}C_1 {}^{8}C_1 = 212$

PROBABILITY

18

RANDOM EXPERIMENT

An experiment, whose all possible outcomes are known in advance but the outcome of any specific performance can not predicted before the completion of the experiment, is known as random experiment.

Sample-space and sample point

A set whose elements represent all possible outcomes of a random experiment is called the sample – space and is usually represented by 'S'.
An element of a sample – space is called a sample point.

Event

An event is a subset of sample – space.

Simple Event and Compound Event:

If an event is a set containing only one element of the sample-space, then it is called a simple event. A compound event is one that can be represented as a union of sample points.

Definition

To find the probability of any event 'A' we sum all weights assigned to sample points in A. This sum is called measure or probability of A and is denoted by P(A).
The probability of any event A is the sum of the weights of sample points in A. Therefore, $0 \le P(A) \le 1$, $P(\phi) = 0$ and $P(S) = 1$.
If a random experiment can result in any one of N different equally likely outcomes, and if exactly n of these outcomes correspond to A, then the probability of event A, $P(A) = \frac{n}{N}$.

PROBABILITY LAWS

Mutually Exclusive Events

A set of events is said to be mutually exclusive if the occurrence of one of them precludes the occurrence of any of the other events. Using set theoretic notation, if $A_1, A_2, \cdots A_n$ be the set of mutually exclusive events then $A_i \cap A_j = \phi$ for $i \neq j$ and $1 \le i, j \le n$.
Also, $P(A_i \cup A_j) = P(A_i) + P(A_j)$

Exhaustive Event

A set of events is said to be exhaustive if the performance of random experiment always result in the occurrence of atleast one of them. In other words all sample points put together (i.e. sample space itself) would give us an exhaustive event.
If 'E' be an exhaustive event then $P(E) = 1$.

Set Theoretic Principles

Let us introduce a few notations, which would be frequently used:

If 'A' and 'B' be any two events of the sample space then

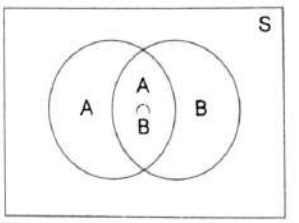
- $A \cup B$ would stand for occurrence of atleast one of them.
- $A \cap B$ stands for simultaneous occurrence of A and B.
- $\bar{A}$ (or A') stands for non-occurrence of A
- $\bar{A} \cap \bar{B}$ (or $A' \cap B'$) stands for non-occurrence of both A and B.
- $A \subseteq B$ stands for 'the occurrence of A implies the occurrence of B'.
- If A and B are any two events, then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- If A and B are mutually exclusive, $P(A \cup B) = P(A) + P(B)$.
- $P(A') = 1 - P(A)$
- $P(A \cap B') = P(A) - P(A \cap B)$
- $P(A' \cup B') = 1 - P(A \cap B)$
- P(exactly one of A, B occurs) $= P(A) + P(B) - 2P(A \cap B)$
- $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$
- P (Exactly one of A, B, C occurs)
  $= P(A) + P(B) + P(C) - 2P(A \cap B) - 2P(A \cap C) - 2P(B \cap C) + 3P(A \cap B \cap C)$
- P(Exactly two of A, B, C occur) $= P(A \cap B) + P(B \cap C) + P(A \cap C) - 3P(A \cap B \cap C)$
- P (at least two of A, B, C occur) $= P(A \cap B) + P(B \cap C) + P(A \cap C) - 2P(A \cap B \cap C)$
- If $A_1, A_2 \cdots, A_n$ are 'n' events, then $P(A_1 \cup A_2 \cdots \cup A_n)$

$$= \sum_{r=1}^{n} P(A_r) - \sum_{1 \le r_1 < r_2 \le n} (P(A_{r_1} \cup A_{r_2})) + \sum_{1 \le r_1 < r_2 < r_3 \le n} P(A_{r_1} \cap A_{r_2} \cap A_{r_3}) - \cdots (-1)^{n-1} P(A_1 \cap A_2 .. \cap A_n)$$

- $P(A \cup B) \ge \max(P(A), P(B), P(A) + P(B) - 1)$
- If out of m + n equally likely, mutually exclusive and exhaustive cases, m cases are favourable to an A event and n are not favourable to the an event A, m : n is called odds in favour of A, n : m is called odds against the event A.

Conditional Probability

The probability of occurrence of an event B when it is known that some event A has occurred is called a condition probability and is denoted by P(B/A). The symbol P(B/A) is usually read "the probability that B occurs given that A has occurred" or "simply probability of B, given A".



$$\Rightarrow P(B/A) = \frac{n(A \cap B)}{n(A)} = \frac{\frac{n(A \cap B)}{n(S)}}{\frac{n(A)}{n(S)}} = \frac{P(A \cap B)}{P(A)},$$

where $0 < P(A) \le 1$

Similarly, $P(A/B) = \frac{P(A \cap B)}{P(B)}, 0 < P(B) \le 1$

Hence, $P(A \cap B) = \begin{cases} P(A) \cdot P(B/A), & P(A) > 0 \\ P(B) \cdot P(A/B), & P(B) > 0 \end{cases}$

Independent Events

Two events A and B are said to be independent if occurrence or non-occurrence of one does not affect the occurrence or non-occurrence of the other.

i.e. $P(B/A) = P(B)$, $P(A) \neq 0$ similarly $P(A/B) = P(A)$, $P(B) \neq 0$

$$\Rightarrow P(B/A) = \frac{P(A \cap B)}{P(A)} = P(B)$$

$$\Rightarrow P(A \cap B) = P(A) \cdot P(B)$$

If the events are not independent, they are said to be dependent.

Remarks:

- If $P(A) = 0 \Rightarrow$ for event 'B', $0 \leq P(A \cap B) \leq P(A)$.
  $\Rightarrow P(A \cap B) = 0$, thus $P(A \cap B) = P(A) \cdot P(B) = 0$. Hence an impossible event would be independent of any other event.
- Distinction between independent and mutually exclusive events must be carefully made. Independence is a property of probability whereas the mutual exclusion is a set-theoretic concept. If A and B are two mutually exclusive and possible events of sample space 'S' then $P(A) > 0$, $P(B) > 0$ and $P(A \cap B) = 0 \neq P(A) \cdot P(B)$ so that A and B can't be independent. Infact $P(A/B) = 0$, similarly $P(B/A) = 0$. Consequently, mutually exclusive events are strongly dependent.
- Two events A and B are independent if and only if A and B′ are independent or A′ and B are independent or A′ and B′ are independent.
  We have $P(A \cap B) = P(A) \cdot P(B)$. Now $P(A \cap B') = P(A) \cdot P(B')$
  Thus A and B′ are independent.
  Similarly $P(A' \cap B) = P(B) - P(A \cap B) = P(B) \cdot P(A')$
  Finally $P(A' \cap B') = P(A') \cdot P(B')$
  Thus A′ and B′ are also independent.

Mutual Independence and Pairwise Independence

Three events A, B, C are said to be mutually independent if,

$P(A \cap B) = P(A) \cdot P(B)$, $P(A \cap C) = P(A) \cdot P(C)$, $P(B \cap C) = P(B) \cdot P(C)$

and $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$

These events would be said to be pairwise independent if,

$P(A \cap B) = P(A) \cdot P(B)$, $P(B \cap C) = P(B) \cdot P(C)$ and $P(A \cap C) = P(A) \cdot P(C)$.

Thus mutually independent events are pairwise independent but the converse may not be true.

Binomial Trials and Binomial Distribution

Consider a random experiment whose outcomes can be classified as success or failure. It means that experiment results in only two outcomes E_1(success) or E_2 (failure). Further assume that experiment can be repeated several times, probability of success or failure in any trial are p and q ($p + q = 1$) and don't vary from trial to trial and finally different trials are independent. Such a experiment is called Binomial experiment and trials are said to be binomial trials.

A probability distribution representing the binomial trials is said to binomial distribution.

Let us consider a Binomial experiment which has been repeated 'n' times. Let the probability of success and failure in any trial be p and q respectively and we are interested in the probability of occurrence of exactly 'r' successes in these n trials. Now number of ways of choosing 'r' success in 'n' trials = ${}^nC_r$. Probability of 'r' successes and $(n-r)$ failures is $p^r \cdot q^{n-r}$. Thus probability of having exactly r successes = ${}^nC_r \cdot p^r \cdot q^{n-r}$

Let 'X' be random variable representing the number of successes, then
$P(X = r) = {}^nC_r \cdot p^r \cdot q^{n-r}$ $(r = 0, 1, 2, \cdots, n)$

Remarks:

- Probability of utmost 'r' successes in n trials = $\sum_{\lambda=0}^{r} {}^nC_\lambda p^\lambda q^{n-\lambda}$
- Probability of atleast 'r' successes in n trials = $\sum_{\lambda=r}^{n} {}^nC_\lambda p^\lambda \cdot q^{n-\lambda}$
- Probability of having 1st success at the r^{th} trial = $p \cdot q^{r-1}$

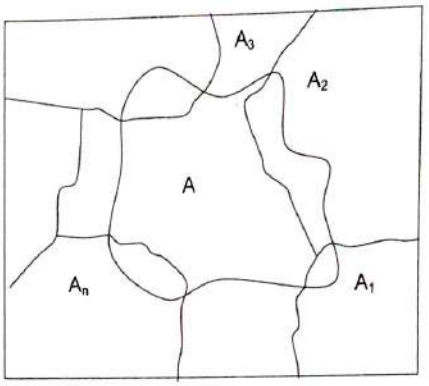
TOTAL PROBABILITY THEOREM AND BAYES' THEOREM

Partition of Sample-Space:

Consider a sample space 'S'. Let $A_1, A_2, \cdots A_n$ be the set of mutually exclusive and exhaustive set of sample-space S. These $A_1, A_2, \cdots, A_n$ events are said to partition the sample-space in to 'n'-parts.

We have $A_i \cap A_j \in \phi$ for $i \neq j$, $1 \leq i, j \leq n$

And $\sum_{i=1}^{n} P(A_i) = 1$



Total Probability Theorem

Let 'A' be any event of S. We can write $A = (A_1 \cap A) \cup (A_2 \cap A) \cdots \cup (A_n \cap A)$

As $A_1, A_2 \cdots A_n$ are mutually exclusive, $(A_1 \cap A), (A_2 \cap A), \cdots, (A_n \cap A)$ would also be mutually exclusive.

$\Rightarrow P(A) = P(A_1 \cap A) + P(A_2 \cap A) + \cdots + P(A_n \cap A)$

$= P(A_1) \cdot P(A/A_1) + P(A_2) \cdot P(A/A_2) + \cdots + P(A_n) \cdot P(A/A_n)$

$\Rightarrow P(A) = \sum_{i=1}^{n} P(A_i) \cdot P(A/A_i)$. This is known as the total probability of the event A.

Bayes' Theorem

This theorem at times is also called inverse probability theorem.

Let us consider any event 'A' of sample space 'S' (as in the previous section). Let us say that event A is found to have occurred and we have to find the probability that it has occurred to the occurrence of cause, say A_i.

From total probability theorem we get $P(A) = \sum_{i=1}^{n} P(A_i) \cdot P(A \ / \ A_i)$

Also, $P(A_i/A) = \dfrac{P(A_i \cap A)}{P(A)} = \dfrac{P(A_i) \cdot P(A \ / \ A_i)}{P(A)}$

$\Rightarrow P(A_i/A) = \dfrac{P(A_i) \cdot P(A \ / \ A_i)}{\sum_{i=1}^{n} P(A_i) \cdot P(A \ / \ A_i)}$. This result is known as Bayes' theorem.

Value of Testimony

The theory of probability can be used to estimate the value of testimony of witnesses. Such an application rests on two basic assumptions,

(a) That to each witness there pertains a constant P(his credibility), which measures the average frequency with which he speaks the truth.

(b) That the statements of witnesses are independent of one another in the sense required in the theory of probability.

ILLUSTRATIVE PROBLEMS

1. Find the probability that an year chosen at random has 53 Sundays.

Sol. Let P(L) → be the probability that an year chosen at random is leap year

$P(L) = 1/4$, $\therefore P(\bar{L}) = 3/4$

Let P(S) → be the probability that an year chosen at random has 53 Sunday.

$\therefore P(S) = P(L).P(S/L) + P(\bar{L})P(S/\bar{L})$

Now P(S/L) prob. that a leap year has 53 Sundays.

A leap year has 366 days, 52 weeks + the remaining 2 days may be Sunday-Monday, M-T, T-W, W-Th, Th-F, F-Sat or Sat - S.

Out of the 7 possibilities 2 are favourable

$\therefore P(S/L) = \frac{2}{7}$. Similarly $P(S/\bar{L}) = \frac{1}{7}$

$\therefore P(S) = \frac{1}{4}.\frac{2}{7} + \frac{3}{4}.\frac{1}{7} = \frac{5}{28}$

2. A person draws a card from a pack of 52, replaces it and shuffles it. He continues doing it until he draws a spade. What is the chance that he has to make (i) atleast 3 trials (ii) exactly 3 trials :

Sol. (i) For atleast 3 trials, he has to fail at the first 2 attempts and then after that it doesn't make a difference if he fails or wins at the 3rd or the subsequent attempts. Chance of success at any attempt $= 1/4$ $\therefore$ chance of failure $= 3/4$

$\therefore$ chance of failing in first 2 attempt $= \left(\frac{3}{4}\right)^2 = \frac{9}{16}$

(ii) For exactly 3 attempts, he has to fail in the first two attempts and succeed in the 3rd attempt.

$\therefore$ required probability $= \left(\frac{3}{4}\right)^2.\frac{1}{4} = \frac{9}{64}$.

3. In a game of tickets, 8 tickets numbered 1, 2, 3, ……., 8 are grouped in pairs at random and observed pair-wise, the lower numbered tickets of pairs are kept and others are thrown and this process is repeated once more. Find the probability that out of remaining two tickets, one is numbered 4.

Sol. Total number of ways in first process $= \frac{8!}{(2!)^4(4!)}$

and in second process $= \frac{4!}{(2!)^2(2!)}$ so total number of ways $= \frac{8!\,4!}{(2!)^7(4!)} = \frac{8!}{(2!)^7}$.

For favourable ways lets divide tickets into two groups

1, 2, 3, 4

5, 6, 7, 8

Ticket of number 4 has to be selected so that its grouping must be with a higher number and there also should be an higher number of 4 at the end of first process so favourable numbers are ${}^4C_1\ {}^1C_1 \frac{4!}{(2!)^2(2!)}$

So required probability $= \frac{4 \cdot 3 \cdot \frac{4!}{(2!)^2(2!)}}{\frac{8!}{(2!)^7}} = \frac{4 \cdot 3 \cdot (2!)^4}{8 \cdot 7 \cdot 6 \cdot 5} = \frac{4}{35}$.

4. There are four players A_1, A_2, A_3 and A_4 in pool A and four players B_1, B_2, B_3 and B_4 in pool B. They are arbitrarily paired in their respective pools to play against each other and one winner is decided in each pair to play in semifinal. It is known that whenever A_1 plays with A_2, A_1 wins the game and whenever B_1 plays with B_2, B_1 wins. Find the probability that

(i) A_1 and A_2 reach in the semifinal

(ii) A_1, A_2, B_2 and B_3 reach semifinal.

Sol. In pool a pairing may be done in following ways:

Way no 1:- A_1A_2 and A_3A_4

Way no 2:- A_1A_3 and A_2A_4

Way no 3:- A_1A_4 and A_2A_3

outcomes of different ways are :

W – 1	A_1A_3	A_1A_4		
W – 2	A_1A_2	A_1A_4	A_2A_3	A_3A_4
W – 3	A_1A_2	A_1A_3	A_2A_4	A_3A_4

(i) The probability that A_1A_2 reach the semifinal $= \frac{1}{3}\left(0 + \frac{1}{4} + \frac{1}{4}\right) = \frac{1}{6}$.

(ii) The probability that A_1A_2, B_2B_3 reach in the semifinal $= \frac{1}{6} \times \frac{1}{12} = \frac{1}{72}$

Since probability of B_2B_3 is same as that of A_2A_3 which is $\frac{1}{12}$.

5. A bag contains 'n' balls, one of which is white. The probability that A and B speak truth are P_1 and P_2 respectively. One ball is drawn from the bag and A and B both assert that it is white. Find the probability that drawn ball is actually white.

Sol. Let $A_1(A_2)$ be the event that drawn ball is white (non-white) and 'E' be the event that A and B claim that drawn ball is white. Clearly,

$P(A_1) = \frac{1}{n}$, $P(A_2) = \frac{n-1}{n}$,

$P(E/A_1) = P_1 P_2$, $P(E/A_2) = (1 - P_1)(1 - P_2) \Rightarrow P(E) = P(A_1) \cdot P(E/A_1) + P(A_2) \cdot P(E/A_2)$

$= \frac{P_1 P_2}{n} + \frac{(n-1)}{n} \cdot (1 - P_1)(1 - P_2) = \frac{(n-1) P_1 P_2 + (1 - P_1)(1 - P_2)}{n}$

$\Rightarrow P(A_1/E) = \frac{P(A_1) \cdot P(E/A_1)}{P(E)} = \frac{(n-1) P_1 P_2}{P_1 P_2 + (1 - P_1)(1 - P_2)}$.

SOLVED PROBLEMS

1. Team A plays with 5 other teams exactly once. Assuming that for each match the probabilities of a win, draw and loss are equal then

(A) the probability that A wins and losses equal number of matches is $\frac{34}{81}$

(B) the probability that A wins and losses equal number of matches is $\frac{17}{81}$

(C) the probability that A wins more number of matches than it losses is $\frac{17}{81}$

(D) the probability that A losses more number of matches than it wins is $\frac{16}{81}$

Sol. **(B).** Probability of same number of wins and losses = no wins no losses + 1 win, 1 loss + 2 win, 2 loss

$$= \left(\frac{1}{3}\right)^5 + {}^5C_1\,{}^4C_1\left(\frac{1}{3}\right)^5 + {}^5C_2\,{}^3C_2\left(\frac{1}{3}\right)^5 = \left(\frac{1}{3}\right)^5(1+20+30) = \frac{17}{81}$$

$\Rightarrow$ Probability that A wins more matches than its losses $= \frac{1}{2}\left(1-\frac{17}{81}\right) = \frac{32}{81}$.

2. 64 players play in a knock out tournament assuming all the players are of equal strength the probability that P_1 losses to P_2 and P_2 becomes the eventual champion is

(A) $\frac{1}{612}$ (B) $\frac{1}{672}$

(C) $\frac{1}{512}$ (D) $\frac{1}{63\cdot 2^6}$

Sol. **(B).** The probability that P_2 is champion is $\frac{1}{64}$. P_2 plays against 6 other players which can be selected in ${}^{63}C_6$ ways. If P_2 losses to P_1 then P_1 is included in 6 players in ${}^{62}C_5$ ways.

Hence probability $= \frac{{}^{62}C_5}{{}^{63}C_6}\cdot\frac{1}{64} = \frac{1}{672}$.

3. If the probability of choosing an integer k out of '2m' integers 1, 2, 3, ..., 2m is inversely proportional to k^4 $(1 \le k \le 2m)$, then the probability that chosen number is odd, is

(A) > 1 (B) > 1/2

(C) = 1/2 (D) < 1/3

Sol. **(B).** Let the probability of choosing integer k; $p(k) = \frac{\alpha}{k^4}$, α is constant of proportionality

$$\sum_{k=1}^{2m} p(k) = 1 \Rightarrow \alpha = \left(\sum_{k=1}^{2m}\frac{1}{k^4}\right)^{-1}\ ;\ a = p \text{ (choosing an odd number)},$$

$$a = \sum_{k=1}^{m} p(2k-1) = \alpha\cdot\sum_{k=1}^{m}\frac{1}{(2k-1)^4}$$

$$1 - a = \alpha\cdot\sum_{k=1}^{m}\left(\frac{1}{2k}\right)^4 < \alpha\cdot\sum_{k=1}^{m}\frac{1}{(2k-1)^4} = a \Rightarrow a > 1/2.$$

4. 40 teams play a tournament. Each team plays every other team just once. Each game results in a win for one team. If each team has a 50% chance of winning each game, the probability that at the end of the tournament every team has won a different number of games is

(A) $\frac{1}{780}$ (B) $\frac{40!}{2^{780}}$

(C) $\frac{40!}{3^{780}}$ (D) none of these

Sol. **(B).** Team totals must be 0, 1, 2, ..., 39.
Let the teams be $T_1, T_2, \ldots, T_{40}$, so that T_i losses to T_j for $i < j$.
In other words, this order uniquely determines the result of every game.
There are 40! such orders and 780 games, so 2^{780} possible outcomes for the games.
Hence the probability is $\frac{40!}{2^{780}}$.

5. In a game a coin is tossed (2n + m) times and a player wins if he does not get any two consecutive outcomes same for atleast 2n times in a row. The probability that player wins the game is

(A) $\frac{m+2}{2^{2n}+1}$ (B) $\frac{2n+2}{2^{2n}}$

(C) $\frac{2n+2}{2^{2n+1}}$ (D) $\frac{m+2}{2^{2n}}$

Sol. **(D).** Player should get (HT, HT, HT, ...) or (TH, TH, ...) atleast 2n times. If the sequence start from first place then probability $\frac{1}{2^{2n}}$ and if any other time probability is $\frac{1}{2^{2n+1}}$.

Hence probability $= 2\left(\frac{1}{2^{2n}} + \frac{m}{2^{2n+1}}\right) = \frac{m+2}{2^{2n}}$.

6. A 2n digit number starts with 2 and all its digits are prime, then the probability that the sum of all 2 consecutive digits of the number is prime, is

(A) 4×2^{3n} (B) 4×2^{-3n}

(C) 2^{3n} (D) none of these

Sol. **(B).**

2							

The prime digits are (2, 3, 5, 7)
If we fix 2 at first place, then other (2n – 1) places are filled by all four digits,
so total number of cases $= 4^{2n-1}$
Now, sum of 2 consecutive digits is prime when consecutive digits are (2, 3) or (2, 5) then 2 will be fixed at all alternative places

2		2		2		2	

So favourable cases $= 2^n$. Therefore probability $= \frac{2^n}{4^{2n-1}} = 2^n \cdot 2^{-4n+2} = 2^2 \cdot 2^{-3n} = \frac{4}{2^{3n}}$.

7. Two numbers a, b are chosen from the set of integers 1, 2, 3, ..., 39. Then probability that the equation $7a - 9b = 0$ is satisfied is

(A) $\frac{1}{247}$ (B) $\frac{2}{247}$

(C) $\frac{4}{741}$ (D) $\frac{5}{741}$

Sol. **(C).** Given $7a - 9b = 0 \Rightarrow b = \frac{7}{9}a$.

Hence number of pairs (a, b) can be (9, 7); (18, 14); (27, 21); (36, 28).

Hence required probability $= \frac{4}{{}^{39}C_2} = \frac{4}{741}$.

8. Four whole numbers, taken at random, are multiplied together. The probability that the last digit in the product is 1, 3, 7 or 9 is

(A) $\frac{1}{125}$ (B) $\frac{16}{625}$

(C) $\frac{1}{25}$ (D) $\frac{3}{25}$

Sol. **(B).** Each number in the product should have the last digit as one of the digits 1, 3, 7, 9.

$\therefore$ Required probability $= \frac{4^4}{10^4} = \frac{16}{625}$.

9. Two numbers x and y are choose at random (without replacement) from amongst the numbers 1, 2, 3,.....2004. The probability that $x^3 + y^3$ is divisible by 3 is

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $\frac{1}{6}$ (D) $\frac{1}{4}$

Sol. **(A).** Let $E_1 = 1, 4, 7, \ldots\ldots$ (n each) $3n = 2004$

$E_2 = 2, 5, 8, \ldots\ldots$ (n each)

$E_3 = 3, 6, 9, \ldots\ldots$(n each)

x and y belong to (E_1, E_2), (E_2, E_1) or (E_3, E_3)

so required probability $\frac{{}^{3n}C_2}{n^2 + {}^nC_2} = \frac{1}{3}$.

10. One mapping is selected at random from all mappings of the set $S = \{1, 2, 3, \ldots, n\}$ into itself. If the probability that the mapping is one–one is $\frac{3}{32}$, then the value of n is

(A) 2 (B) 3

(C) 4 (D) none of these

Sol. **(C).** Total number of mapping $= n^n$.

Number of one–one mapping $= {}^nC_1 \cdot {}^{n-1}C_1 \ldots\ldots\ldots {}^1C_1 = n!$

Hence the probability $= \frac{n!}{n^n} = \frac{3}{32} = \frac{4!}{4^4}$. Comparing, we get $n = 4$.

11. A dice is thrown $2n + 1$ times, $n \in N$. The probability that faces with even numbers show up odd numbers of times is

(A) $\frac{2n+1}{4n+3}$ (B) less than $\frac{1}{2}$

(C) greater than $\frac{1}{2}$ (D) none of these

Sol. **(D).** Let probability of showing even and odd number be p and q respectively.

$p = q = \frac{1}{2}$.

Probability that faces with even number show odd number of times

$= {}^{2n+1}C_1 p^{2n} q + {}^{2n+1}C_3 p^{2n-2} q^3 + \ldots\ldots + {}^{2n+1}C_{2n+1} q^{2n+1} = \frac{1}{2}$.

12. A and B are two events such that P (A) = 0.3 and P (B) = 0.25 and P (A ∩ B) = 0.2 then $P(\bar{A} / \bar{B})$ is equal to

(A) $\frac{12}{15}$ (B) $\frac{11}{15}$

(C) $\frac{14}{15}$ (D) $\frac{13}{15}$

Sol. **(D).** $P(\bar{A} / \bar{B}) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} = \frac{1 - P(A \cup B)}{1 - P(B)} = \frac{0.65}{0.75}$.

13. Entries of a 2 × 2 determinant are chosen from the set {–1, 1}. The probability that determinant has zero value is

(A) 1/4 (B) 1/3

(C) 1/2 (D) none of these

Sol. **(C).** $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ = ad – bc = 0 ⇒ whether ad = 1, bc = 1 or ad = – 1, bc = – 1

which occur in eight ways. Total number of 2 × 2 determinants from {–1, 1} is 16. Thus required probability is $\frac{8}{16} = \frac{1}{2}$.

14. A and B are two independent events. The probability that A and B happen simultaneously is $\frac{1}{12}$ and neither A nor B happens is $\frac{1}{2}$, the

(A) $P(A) = \frac{1}{3}, P(B) = \frac{1}{4}$ (B) $P(A) = \frac{1}{2}, P(B) = \frac{1}{6}$

(C) $P(A) = \frac{1}{4}, P(B) = \frac{1}{3}$ (D) $P(A) = \frac{1}{6}, P(B) = \frac{1}{2}$

Sol. **(A), (C)**

$P(A \cap B) = P(A) . P(B) = 1/12$ .... (1)

$P(\bar{A} \cap \bar{B}) = P(\bar{A}) . P(\bar{B}) = 1/2$

$\Rightarrow (1 - P(A)) (1 - P(B) = 1/2$

$\Rightarrow 1 - P(A) - P(B) + P(A) P(B) = 1/2$

$\Rightarrow 1 - P(A) - P(B) + 1/12 = 1/2$

$\Rightarrow P(A) + P(B) = 7/12$ .... (2)

Solving (1) and (2) gives P(A) = 1/3 or 1/4

Accordingly, P(B) = 1/ 4 or 1/3.

Comprehension

Ajay plays a game against an electronic machine. At each round he deposits one rupee in a slot and then flips a coin which has probability p of sowing a head. If head comes, he gets back the rupee he deposited and one more rupee from the machine. If tail shows he loses his coin. Suppose he started with 10 one rupee coin. Assume q as the probable event of getting a tail.

Answer the following questions

15. The probability that Ajay is left with no money by the tenth round or earlier is

(A) q^{10} (B) $\sum_{r=1}^{10} {}^{10}C_r p^r q^{10+r}$

(C) pq^9 (D) $(1 - p^{10})$

Sol. **(A).** Ajay will left with no money at the end of the tenth round or earlier if and only if he gets a tail on each of them.
Hence required probability = q^{10}.

16. The probability that his money will end up exactly in the twelfth round is

(A) q^{12} (B) $1 - p^{12}$

(C) ${}^{10}C_1 pq^{11}$ (D) ${}^{12}C_2 p^2 q^{10}$

Sol. **(C).** In order to finish up at the twelfth chance he must have exactly one head in first ten chances and tail at the eleventh and twelfth round then apparently we have required probability = ${}^{10}C_1 pq^{11}$.
Hence (C) is correct option.

17. The probability that he is left with no money by the 14^{th} round or earlier is

(A) $q^{10}(1 + 10pq + 45p^2q^2)$ (B) $q^{14}(p^2q + 36pq + 7)$

(C) $q^{12} + 3pq^{12} + 3p^{13}q + p^{12}$ (D) $1 - {}^{10}C_1 pq^{11} - {}^{10}C_2 p^2 q^{12}$

Sol. **(A).** Apparently he cannot be drained at the 13^{th} round if Ajay gets at least 2 heads in the first 10 rounds. To finish at the 14^{th} round. Ajay must get exactly 2 heads in the first 10 rounds and a tail on all the rounds from 11^{th} to 14^{th} rounds, which have probability ${}^{10}C_2 p^2 q^{12}$.
Therefore required probability = $q^{10} + {}^{10}C_1 pq^{11} + {}^{10}C_2 p^2 q^{12} = q^{10}(1 + 10pq + 45p^2q^{12})$
Hence (A) is correct option.

Assertion Reason

18. Statement -1: $P(A/B) \geq P(A)$, then $P(B/A) \geq P(B)$.
because

Statement - 2: $P(A/B) = \dfrac{P(A \cap B)}{P(B)}$.

(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1
(B) Statement -1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement -1 is True, Statement -2 is False
(D) Statement -1 is False, Statement -2 is True

Sol. (A). $P(A/B) \geq P(A) \Rightarrow \frac{P(A \cap B)}{P(B)} \geq P(A)$, or $\frac{P(A \cap B)}{P(A)} \geq P(B) \Rightarrow P(B/A) \geq P(B)$.

19. In a tournament there are twelve players $S_1, S_2, \ldots, S_{12}$ and divided into six pairs at random. From each game a winner is decided on the basis of a game played between the two players of the pair. Assuming all the pairs are of equal strength, then match the following:

	Column – I		Column – II
(A)	Probability that S_2 is among the losers is	(p)	$\frac{5}{22}$
(B)	Probability that exactly one S_3 and S_4 is among the losers, is	(q)	$\frac{10}{11}$
(C)	Probability that both S_2 and S_4 are among the winners is	(r)	$\frac{1}{2}$
(D)	Probability of S_4 and S_5 not playing against each other is	(s)	$\frac{6}{11}$
		(t)	0

Sol. **(A) → (r); (B) → (s); (C) → (p); (D) → (q)**

(A) $\frac{{}^{11}C_5}{{}^{12}C_6} = \frac{1}{2}$

(B) Let E_1 be the event that S_3 and S_4 are in same group

Let E_2 be the event that S_3 and S_4 are in different group $P(E_1) = \frac{1}{11} \Rightarrow P(E_2) = \frac{10}{11}$

Let E be the event that exactly one of S_3 and S_4 is among the losers, then

$P(E) = P(E_1)P(E/E_1) + P(E_2)P(E/E_2) = \frac{1}{11} \times 1 + \frac{10}{11} \times \left(\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2}\right) = \frac{6}{11}$

(C) S_2 and S_4 should be in different groups for both being winner

Required probability $= \frac{10}{11}\left(\frac{1}{2} \cdot \frac{1}{2}\right) = \frac{5}{22}$ or $\frac{{}^{10}C_4}{{}^{12}C_6} = \frac{5}{22}$

Answer Q.20, Q.21 and Q.22 by appropriately matching the information given in the three columns of the following table.

Cards are drawn from a pack of 52 well-shuffled cards

Column–I	Column–II (of the two cards drawn)	Column–III (probability belongs to interval)
(I) Two cards are drawn without replacement	(i) atleast one king is drawn	(P) $\left(0, \frac{1}{13}\right)$
(II) Two cards are drawn with replacement	(ii) atmost one of them is king	(Q) $\left(\frac{1}{13}, \frac{1}{2}\right)$
(III) Three cards are drawn and kept aside. Now, two cards are drawn without replacement	(iii) no king is drawn	(R) $\left(\frac{1}{2}, \frac{12}{13}\right)$
(IV) One red card is added from another deck of cards. Now, two cards are drawn	(iv) both cards drawn are kings	(S) $\left(\frac{12}{13}, 1\right)$

20. Which of the following are correct?
(A) I (ii) (R) (B) (II) (iii) (Q)
(C) (III) (iv) (P) (D) (IV) (ii) (Q)

Sol. **(C)**

$$\frac{{}^{4}C_{2}}{{}^{52}C_{2}} = \frac{1}{13}\cdot\frac{1}{13} < \frac{1}{13}$$

21. Which of the following are NOT correct?
(A) (I) (iii) R (B) (II) (iv) (P)
(C) (III) (ii) (S) (D) (IV) (i) (R)

Sol. **(D)**

$$\frac{12}{13} > 1 - \frac{{}^{48}C_{2}}{{}^{52}C_{2}} > \frac{1}{12}$$

22. Which of the following are correct?
(A) (I) (iv) P (B) (II) (i) (R)
(C) (III) (iii) (Q) (D) (IV) (iv) (Q)

Sol. **(A)**

$$\frac{{}^{4}C_{2}}{{}^{52}C_{2}} = \left(\frac{1}{13}\right)^{2} < \frac{1}{13}$$

23. Number 1, 2, 3, ..., 9 are put into 3 × 3 matrix such that all elements of matrix are distinct. If probability that sum of numbers in atleast one row is greater than 21 can be expressed in $\frac{m}{n}$ (where m, n are coprime), then n – m is equal to _____

Sol. **(6.00)**

$$\text{Probability} = \frac{3\times 4\times 3!\times 6!}{9!} = \frac{1}{7}$$

$n - m = 7 - 1 = 6$

24. A bag contains red and black balls, at most 17 in total, when two balls are randomly selected without replacement, probability that both are black or both are red is $\frac{1}{2}$. If largest possible number of red balls that can be present in the bag be 'm', then sum of digits of m is _____

Sol. **(1.00)**

Let r be red balls b are black balls

$$\Rightarrow \frac{{}^{b}C_{2} + {}^{r}C_{2}}{{}^{b+r}C_{2}} = \frac{1}{2}$$

$\Rightarrow (b - r)^2$

$= b + r \le 17$

$\Rightarrow r - b \le \sqrt{17}$

$\Rightarrow r - b \le 4$

and $r + b \le 17$

$\Rightarrow r \le \frac{21}{2}, r = 10$

25. Two integers n_1 and n_2 are drawn out of the set $\{1, 2, 3, ..., 9\}$ with replacement and the equation, $x^2 + \sqrt{2}|n_1 - n_2|x + n_1 = 0$ is formed, then the probability that the equation has real roots is _____

(A) $\frac{4}{9}$

(B) $\frac{7}{11}$

(C) $\frac{8}{15}$

(D) $\frac{2}{5}$

Sol. **(0.44)**

$(n_1 - n_2)^2 \geq 2n_1$

n_1	n_2	Total
1	3, 4, 5, 6, 7, 8, 9	7
2	4, 5, 6, 7, 8, 9	6
3	6, 7, 8, 9	4
4	1, 7, 8, 9	4
5	1, 9	2
6	1, 2	2
7	1, 2, 3	3
8	1, 2, 3, 4	4
9	1, 2, 3, 4	4

$P(E) = \frac{36}{9.9} = \frac{4}{9}$

26. A is a set containing 'n' elements. A subset P of A is chosen at random. The set A is reconstructed by replacing the elements of the subset P. A subset Q of A is again chosen at random the probability such that $P \cap Q$ contains 2 elements $\frac{{}^nC_a \cdot 3^{n-b}}{4^n}$, then $\frac{(a+b)}{3}$ is _____

Sol. **(1.33)**

Let $A = \{a_1, a_2, ... a_n\}$

For each $a_i \in A (1 \leq i \leq n)$ we have the following 4 cases:

(i) $a_i \in P$ and $a_i \in Q$

(ii) $a_i \notin P$ and $a_i \in Q$

(iii) $a_i \in P$ and $a_i \notin Q$

(iv) $a_i \in P$ and $a_i \notin Q$

Thus the total number of ways of choosing P and Q is 4^n

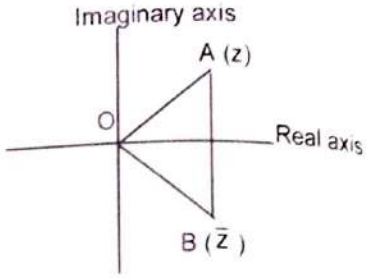
$P \cap Q$ contains exactly two element is ${}^nC_2 (3^{n-2})$

So, probability of $P \cap Q$ contains two elements $= \frac{{}^nC_2 \cdot 3^{n-2}}{4^n}$

COMPLEX NUMBERS

PROPERTIES OF CONJUGATE, MODULUS AND ARGUMENT

Conjugate



$\bar{z}$ is the mirror image of z in the real axis.

- $|z| = |\bar{z}|$
- $z + \bar{z} = 2\mathrm{Re}(z)$
- $z - \bar{z} = 2i\,\mathrm{Im}(z)$
- $z\,\bar{z} = |z|^2$
- $\overline{z_1 + z_2 + \cdots + z_n} = \bar{z}_1 + \bar{z}_2 + \cdots + \bar{z}_n$
- $\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$
- $\overline{z_1 z_2} = \bar{z}_1 . \bar{z}_2$
- $\overline{\left(\frac{z_1}{z_2}\right)} = \left(\frac{\bar{z}_1}{\bar{z}_2}\right)$ (provided $z_2 \neq 0$)
- $\overline{z^n} = (\bar{z})^n$
- $\overline{(\bar{z})} = z$
- If $\alpha = f(z)$, then $\bar{\alpha} = f(\bar{z})$

  where $\alpha = f(z)$ is a function in complex variable with real coefficients.

Modulus

- $|z| = 0 \Rightarrow z = 0$
- $-|z| \leq \mathrm{Re}\, z \leq |z|$,
- $-|z| \leq \mathrm{Im}\, z \leq |z|$
- $|z| \leq |\mathrm{Re}(z)| + |\mathrm{Im}(z)| \leq \sqrt{2}\,|z|$
- $|z|^2 = z\,\bar{z}$
- $|z_1 z_2 \ldots z_n| = |z_1|\,|z_2| \ldots |z_n|$
- $|z^n| = |z|^n$, $n \in I$
- $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$
- $|z_1 + z_2 + \ldots + z_n| \leq |z_1| + |z_2| + \ldots + |z_n|$
- $|z_1 - z_2| \geq ||z_1| - |z_2||$
- $|z_1 + z_2|^2 = (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) = |z_1|^2 + |z_2|^2 + z_1\bar{z}_2 + z_2\bar{z}_1$

Argument

- $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$
- $\arg(z_1/z_2) = \arg(z_1) - \arg(z_2)$
- $\arg(z^n) = n\arg(z)$, $n \in I$

- If $\arg(z) = 0, \pi \Rightarrow z$ is a purely real number $\Rightarrow z = \bar{z}$.
- If $\arg(z) = \pi/2, -\pi/2 \Rightarrow z$ is a purely imaginary number $\Rightarrow z = -\bar{z}$.

Note that argument function behaves like the log function.

De Moivre's Theorem

If n is any integer, then $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.

If n is a rational number, then one of the values of $(\cos\theta + i\sin\theta)^n$ is $\cos n\theta + i\sin n\theta$. If $n = p/q$, where p and q are integers $(q > 0)$ and p, q have no common factor, then $(\cos\theta + i\sin\theta)^n$ has q distinct values, one of which is $\cos n\theta + i\sin n\theta$.

If $z = r(\cos\theta + i\sin\theta)$, and n is a positive integer, then

$$z^{1/n} = r^{1/n}\left[\cos\frac{2k\pi+\theta}{n} + i\sin\frac{2k\pi+\theta}{n}\right], \; k = 0,1,2,.....n-1.$$

The nth Root of Unity

Let x be the nth root of unity. Then $x^n = 1 = \cos 2k\pi + i\sin 2k\pi$ (where k is an integer)

$$\Rightarrow x = \cos\frac{2k\pi}{n} + i\sin\frac{2k\pi}{n} \quad k = 0,1,2,.....,n-1.$$

Let $\alpha = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$.

Then the nth roots of unity are α^t $(t = 0,1,2,.....,n-1)$, i.e. the nth roots of unity are $1,\alpha,\alpha^2,.....\alpha^{n-1}$.

Sum of the Roots

$$1 + \alpha + \alpha^2 + ... + \alpha^{n-1} = \frac{1-\alpha^n}{1-\alpha} = 0 \Rightarrow \sum_{k=0}^{n-1}\cos\frac{2k\pi}{n} = 0 \text{ and } \sum_{k=0}^{n-1}\sin\frac{2k\pi}{n} = 0$$

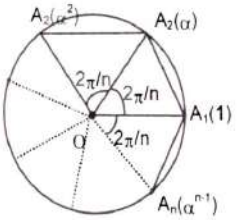
Thus the sum of the roots of unity is zero.

Product of the Roots

$$1.\alpha.\alpha^2. ..... \alpha^{n-1} = (-1)^n(-1) = (-1)^{n+1}$$

Remarks

- The points represented by the n nth roots of unity are located at the vertices of a regular polygon of n sides inscribed in a unit circle having centre at the origin, one vertex being on the positive real axis (Geometrically represented as shown).

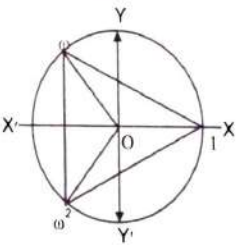


- **Cube roots of unity**

  For n = 3, we get the cube roots of unity and they are 1,

  $\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}$ and $\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}$

  i.e.1, $\frac{-1+\sqrt{3}i}{2}$ and $\frac{-1-\sqrt{3}i}{2}$.



They are generally denoted by 1, ω and ω^2 and geometrically represented by the vertices of an equilateral triangle whose circumcentre is the origin and circumradius is unity.

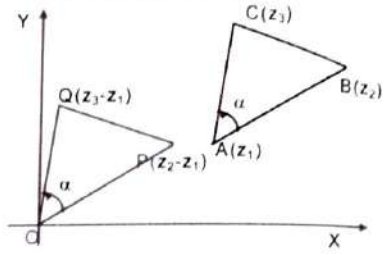
- $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$
- $1 + \omega^n + \omega^{2n} = 3$ (n is a multiple of 3)
- $1 + \omega^n + \omega^{2n} = 0$ (n is an integer, not a multiple of 3)

Concept of Rotation

If z_1, z_2, z_3 are the three vertices of a triangle ABC described in the counter-clock wise sense, then

$$\frac{z_3 - z_1}{z_2 - z_1} = \frac{OQ}{OP}(\cos\alpha + i\sin\alpha) = \frac{CA}{BA}\cdot e^{i\alpha} = \frac{|z_3 - z_1|}{|z_2 - z_1|}\cdot e^{i\alpha}$$

Note that $\arg(z_3 - z_1) - \arg(z_2 - z_1) = \alpha$ is the angle through which OP must be rotated in the anti-clock-wise direction so that it becomes parallel to OQ.



GEOMETRICAL APPLICATIONS

Section Formula

If z_1 and z_2 be any two complex numbers representing the points A and B respectively in the argand plane, then the point C(z) dividing the line segment AB internally in the ratio m : n, i.e., $\frac{AC}{BC} = \frac{m}{n}$, is given by $\frac{nz_1 + mz_2}{m+n}$.

Similarly if C(z) divides the segment AB externally in the ration of m:n, then $z = \frac{nz_1 - mz_2}{n-m}$.

In the specific case if C(z) is the mid–point of AB then $z = \frac{z_1 + z_2}{2}$.

Condition for Collinearity

If there are three real numbers (other than 0) l, m and n such that $lz_1 + mz_2 + nz_3 = 0$ and $l + m + n = 0$ then complex numbers z_1, z_2 and z_3 will be collinear.

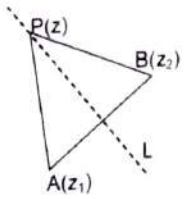
Equation of a Straight Line

- **Equation of a straight line with the help of rotation formula :**
  Let $A(z_1)$ and $B(z_2)$ be any two points lying on any line and we have to obtain the equation of this line. For this purpose let us take any point C(z) on this line

  since $\arg\left(\frac{z - z_1}{z_2 - z_1}\right) = 0$ or $\pi \Rightarrow \frac{z - z_1}{z_2 - z_1} = \frac{\bar{z} - \bar{z}_1}{\bar{z}_2 - \bar{z}_1}$ ... (1)

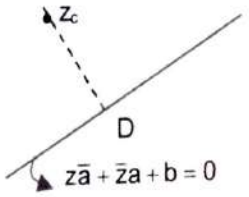
- **General equation of the line:**
  From equation(1) we get $z\bar{a} + \bar{z}a + b = 0$,
  where $\bar{a} = i(\bar{z}_1 - \bar{z}_2)$ and $ib = \bar{z}_1 z_2 - z_1\bar{z}_2$, $b \in R$
  This is the general equation of a line in the complex plane.
- **Slope of a given line:**
  If $z\bar{a} + \bar{z}a + a = 0$ is the given line then its slope $= -\frac{Re(a)}{Im(a)}$.
- Equation of a line parallel to the line $z\bar{a} + \bar{z}a + b = 0$ is $z\bar{a} + \bar{z}a + \lambda = 0$ (where λ is a real number).
- Equation of a line perpendicular to the line $z\bar{a} + \bar{z}a + b = 0$ is $z\bar{a} - \bar{z}a + i\lambda = 0$ (where λ is a real number).

- **Equation of Perpendicular Bisector:**
  Consider a line segment joining $A(z_1)$ and $B(z_2)$. Let the line 'L' be it's perpendicular bisector. If $P(z)$ be any point on the 'L', we have PA = PB
  $\Rightarrow |z - z_1| = |z - z_2|$
  $\Rightarrow z(\bar{z}_2 - \bar{z}_1) + \bar{z}(z_2 - z_1) + z_1\bar{z}_1 - z_2\bar{z}_2 = 0$



- **Distance of a given point from a given line:**

  Let the given line be $z\bar{a} + \bar{z}a + b = 0$, and the given point be z_c then distance of z_c from this line is $\dfrac{|z_c\bar{a} + \bar{z}_c a + b|}{2|a|}$.



- $\arg(z - z_0) = \theta$ represents a line passing through z_0 with slope $\tan\theta$, (making angle θ with the positive direction of x-axis).

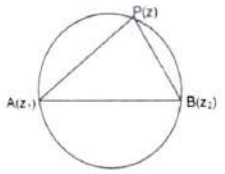
Equation of a Circle

Equation of a circle of radius r and having its centre at z_0 is $|z - z_0| = r$.
$\Rightarrow |z - z_0|^2 = r^2 \Rightarrow (z - z_0)(\bar{z} - \bar{z}_0) = r^2 \Rightarrow z\bar{z} + a\bar{z} + \bar{a}z + b = 0$, where $-a = z_0$ and $b = z_0\bar{z}_0 - r^2$.

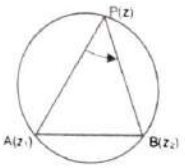
It represents the general equation of a circle in the complex plane.

- $z\bar{z} + a\bar{z} + \bar{a}z + b = 0$ represents a circle whose centre is $-a$ and radius is $\sqrt{a\bar{a} - b}$. Thus $z\bar{z} + a\bar{z} + \bar{a}z + b = 0$, $(b \in R)$ represents a real circle if and only if $a\bar{a} - b \geq 0$.

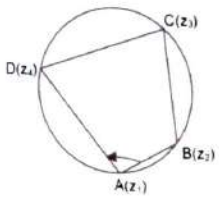
- Equation of a circle described on a line segment AB, $(A(z_1), B(z_2))$ as diameter is
  $(z - z_1)(\bar{z} - \bar{z}_2) + (z - z_2)(\bar{z} - \bar{z}_1) = 0$.



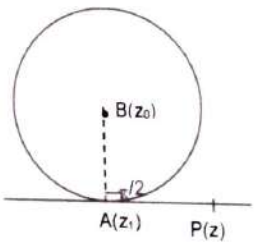
- Let z_1 and z_2 be two given complex numbers and z be a any complex number such that, $\arg\left(\dfrac{z - z_1}{z - z_2}\right) = \alpha$, where $\alpha \in (0, \pi)$.
  Then 'z' will lie on the arc of a circle.



- Let ABCD be a cyclic quadrilateral such that $A(z_1)$, $B(z_2)$, $C(z_3)$ and $D(z_4)$ lie on a circle.
  Then $\dfrac{(z_4 - z_1)(z_2 - z_3)}{(z_2 - z_1)(z_4 - z_3)}$ is purely real.



- **Equation of tangent to a given circle at a given point :**
  Let $|z| = r$ be the given circle then equation of tangent at the point z_1 is;
  $z\bar{z}_1 + \bar{z}z_1 = 2|z_1|^2 = 2r^2$
  or $\dfrac{z}{z_1} + \dfrac{\bar{z}}{\bar{z}_1} = 2$ or $\text{Re}\left(\dfrac{z}{z_1}\right) = 1$



- If $\left|\dfrac{z - z_1}{z - z_2}\right| = \lambda$, $(\lambda \in R^+, \lambda \neq 1)$, then 'z' will lie on a circle.

Some Important Result to Remember

- The triangle whose vertices are the points represented by complex numbers z_1, z_2, z_3 is equilateral if $\frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} + \frac{1}{z_1 - z_2} = 0$ i.e., if $z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$.
- $|z - z_1| + |z - z_2| = \lambda$, represents an ellipse if $|z_1 - z_2| < \lambda$, having the points z_1 and z_2 as its foci. And if $|z_1 - z_2| = \lambda$, then z lies on a line segment connecting z_1 and z_2.
- $|z - z_1| \sim |z - z_2| = \lambda$, represents a hyperbola if $|z_1 - z_2| > \lambda$, having the points z_1 and z_2 as its foci. And if $|z_1 - z_2| = \lambda$, the z lies on the line passing through z_1 and z_2 excluding the points between z_1 and z_2.

ILLUSTRATIVE PROBLEMS

1. Solve for z i.e., find all complex numbers z which satisfy $|z|^2 - 2iz + 2c(1+i) = 0$ where c is real.

Sol. Put $z = a + ib$ then $a^2 + b^2 - 2ai + 2b + 2c + 2ci = 0$

$\Rightarrow (a^2 + b^2 + 2b + 2c) + (2c - 2a)i = 0 \Rightarrow a^2 + b^2 + 2b + 2c = 0$ and $2c - 2a = 0 \Rightarrow a = c$.

Now $b^2 + 2b + (c^2 + 2c) = 0 \Rightarrow b = -1 \pm\sqrt{1 - 2c - c^2}$

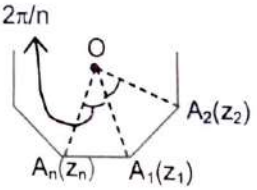
Since b is real, $1 - 2c - c^2 \geq 0 \Rightarrow c \in [-1 - \sqrt{2}, -1 + \sqrt{2}]$.

$\Rightarrow z = c + i(-1 \pm\sqrt{1 - 2c - c^2})$

z does not exist for $c \notin [-1 - \sqrt{2}, -1 + \sqrt{2}]$.

2. Consider an 'n' sided regular polygon with origin as its centre. If z_1 be the complex number representing a vertex A_1, of the polygon, find the complex number associated with the vertex that is adjacent to A_1.

Sol. Vertex adjacent to A_1 is either A_2 or A_n. Let z_2 and z_n represent the vertices A_2 and A_n respectively. Considering the rotation about the origin, we get



$$\frac{z_2 - 0}{z_1 - 0} = \frac{|z_2|}{|z_1|}e^{i2\pi/n} = e^{i2\pi/n} \Rightarrow z_2 = z_1 e^{i2\pi/n}$$

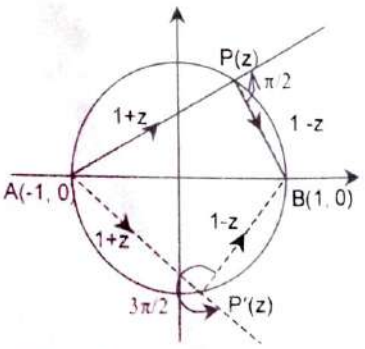
Similarly, $\frac{z_n}{z_1} = e^{-i2\pi/n} \Rightarrow z_n = z_1 e^{-i2\pi/n}$.

3. If $|z| = 1$, then prove that the points represented by $\sqrt{\frac{1+z}{1-z}}$ lie on one or other of two fixed perpendicular straight lines.

Sol. Since $|z| = 1$, z lies on a unit circle having centre at the origin.

$$\text{Arg}\left(\frac{1+z}{1-z}\right) = +\frac{\pi}{2} \text{ or } +\frac{3\pi}{2}$$

$\Rightarrow \frac{1+z}{1-z} = ke^{i\pi/2}$ or $ke^{i3\pi/2}$, where k is a real parameter and its value depends upon the position of z.

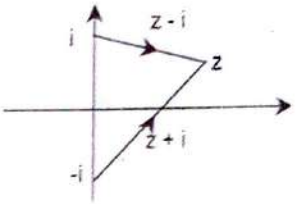


Let $\alpha = \sqrt{\frac{1+z}{1-z}}$

$\Rightarrow \alpha = \sqrt{k}\ e^{i\pi/4}$ or $\sqrt{k}\ e^{i3\pi/4} \Rightarrow \alpha$ lies on one or other of the two perpendicular lines.

4. If $\alpha = \frac{z-i}{z+i}$, show that, when z lies above the real axis, α will lie within the unit circle which has centre at the origin. Find the locus of α as z travels on the real axis from $-\infty$ to $+\infty$.

Sol. From figure, it is clear that $|z-i| < |z+i|$ (as z lies above the real axis).



$$\Rightarrow |\alpha| = \frac{|z-i|}{|z+i|} < 1$$

$\Rightarrow$ α lies within the unit circle which has centre at the origin.

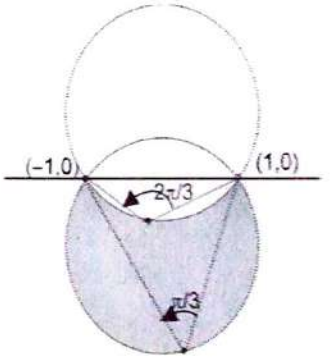
Now if z is travelling on the real axis Im(z)=0 and Re(z) varies from $-\infty$ to $+\infty$.

Let $z = x + i\,0 \Rightarrow \alpha = \frac{x-i}{x+i} \Rightarrow |\alpha| = \frac{|x-i|}{|x+i|} = 1$ (as $|x-i| = |x+i|\ \forall\, x \in R$

$\Rightarrow$ α moves on the unit circle which has centre at the origin.

5. Plot the region represented by $\frac{\pi}{3} \le \arg\left(\frac{z+1}{z-1}\right) \le \frac{2\pi}{3}$ in the argand plane.

Sol. Let us take $\arg\left(\frac{z+1}{z-1}\right) = 2\pi/3$, Clearly z lies on the minor arc of circle passing through (1, 0) and (-1, 0).



Similarly, $\text{Arg}\left(\frac{z+1}{z-1}\right) = \pi/3$.

Means that 'z' is lying on the major arc of the circle passing through (1, 0) and (-1, 0).

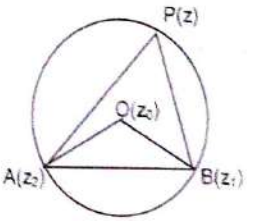
Now if we take any point in the region included between the two arcs, say $P_1(z_1)$,

We get $\frac{\pi}{3} \le \arg\left(\frac{z+1}{z-1}\right) \le \frac{2\pi}{3}$.

Thus $\frac{\pi}{3} \le \arg\left(\frac{z+1}{z-1}\right) \le \frac{2\pi}{3}$ represents the shaded region (excluding the points (1, 0) and (-1, 0)).

6. Consider the complex numbers $z_1 = 10 + 6i$ and $z_2 = 4 + 2i$. If $\arg\left(\frac{z-z_1}{z-z_2}\right) = \frac{\pi}{4}$, find the centre and radius of the locus of complex number z.

Sol. Let $O(z_0)$ be the centre of the circle.



We have $\angle AOB = \frac{\pi}{2}$. $AB = |z_1 - z_2|$

$= |6 + 4i| = \sqrt{52}$

Let $OA = OB = r \Rightarrow AB = r\sqrt{2} \Rightarrow r = \sqrt{26}$

Also $\frac{z_1 - z_0}{z_2 - z_0} = e^{i\pi/2} = i \Rightarrow z_1 - z_0 = i(z_2 - z_0)$

$\Rightarrow z_0 = \frac{1}{2}(z_2 + iz_1 - iz_2 + z_1)$

$= 5 + 7i$

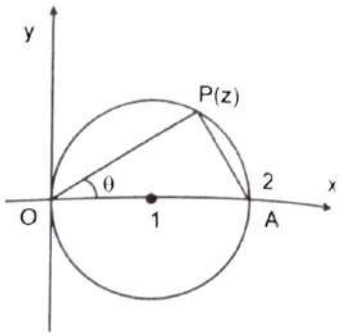
SOLVED PROBLEMS

1. If $|z-1| = 1$ and $\arg(z) = \theta$, where $z \neq 0$ and θ is acute, then $1 - \frac{2}{z}$ is equal to

(A) $\tan\theta$ (B) $i\tan\theta$

(C) $\tan\frac{\theta}{2}$ (D) $\tan\frac{\theta}{2}$

Sol. **(B).** $|z-1| = 1$ represents a circle with centre at 1 and radius equal to 1.



We have $\angle OPA = \frac{\pi}{2}$

$\Rightarrow \arg\left(\frac{2-z}{0-z}\right) = \frac{\pi}{2} \Rightarrow \frac{z-2}{z} = \frac{AP}{OP}\,i.$

Now in ΔOAP

$\tan\theta = \frac{AP}{OP}.$

Thus $\frac{z-2}{z} = i\tan\theta.$

2. If z be a complex number satisfying $|z|^2 + 2(z+\bar{z}) + 3i(z-\bar{z}) + 4 = 0$, then complex number $z + 3 + 2i$ will lie on

(A) circle with centre $1 - 5i$ radius 4 (B) circle with centre $1 + 5i$ radius 4

(C) circle with centre $1 + 5i$ radius 3 (D) circle with centre $1 - 5i$ radius 3

Sol. **(C).** Given equation of circle

$|z|^2 + z(2 + 3i) + \bar{z}(2 - 3i) + 4 = 0$

centre $-(2 - 3i)$, radius $= \sqrt{a\bar{a} - b} = 3 \Rightarrow |z + 2 - 3i| = 3.$

Let $\omega = z + 3 + 2i = z + 2 - 3i + 1 + 5i$

$|\omega - 1 - 5i| = |z + 2 - 3i| = 3$

so, ω lies on circle whose centre is $1 + 5i$, radius $= 3$.

3. If $|z-2-i| = |z| \left|\sin\left(\frac{\pi}{4} - \arg z\right)\right|$, then locus of z is

(A) a pair of straight lines (B) circle

(C) parabola (D) ellipse

Sol. **(C).** We have $|(x-2) + i(y-1)| = |z|\left|\frac{1}{\sqrt{2}}\cos\theta - \frac{1}{\sqrt{2}}\sin\theta\right|$, where $\theta = \arg z$

$\sqrt{(x-2)^2 + (y-1)^2} = \frac{1}{\sqrt{2}}|x-y|$, which is a parabola.

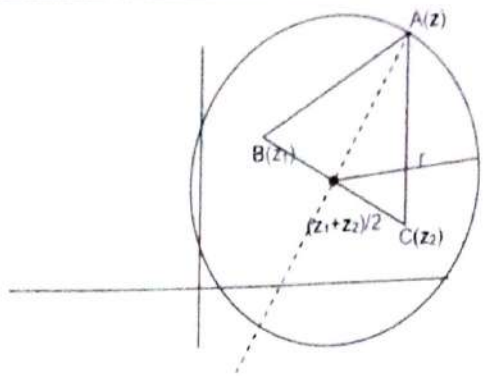
4. The maximum area of the triangle formed by the complex co–ordinates z, z_1, z_2 which satisfy the relations $|z - z_1| = |z - z_2|$, $\left|z - \left(\frac{z_1 + z_2}{2}\right)\right| \leq r$, where $r > |z_1 - z_2|$ is

(A) $\frac{1}{2}|z_1 - z_2|^2$ (B) $\frac{1}{2}|z_1 - z_2|r$

(C) $\frac{1}{2}|z_1 - z_2|^2 r^2$ (D) $\frac{1}{2}|z_1 - z_2|r^2$

Sol. **(B).** By the given conditions, the area of the triangle ABC is

$= \frac{1}{2}|z_1 - z_2|r$



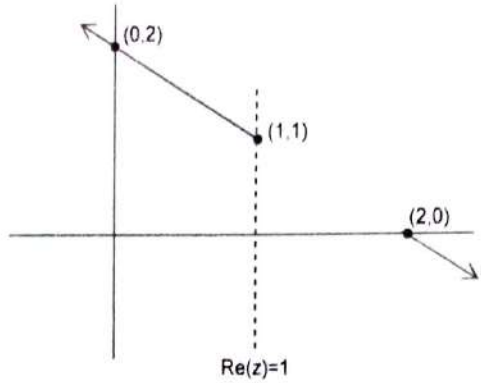
5. Locus of z if $\arg[z-(1+i)] = \begin{cases} \frac{3\pi}{4} & \text{when } |z| \le |z-2| \\ \frac{-\pi}{4} & \text{when } |z| > |z-2| \end{cases}$ is

(A) straight lines passing through (2, 0)
(B) straight lines passing through (2, 0), (1, 1)
(C) a line segment
(D) a set of two rays

Sol. **(D).** The given equation is written as

$$\arg(z-(1+i)) = \begin{cases} \frac{3\pi}{4} & \text{when } x \le 1 \\ \frac{-\pi}{4} & \text{when } x > 1 \end{cases}$$

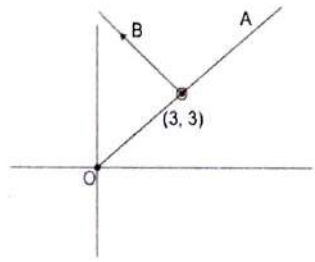
$\Rightarrow$ the locus is a set of two rays.



6. Let $z \in C$ and if $A = \left\{z, \arg(z) = \frac{\pi}{4}\right\}$ and $B = \left\{z, \arg(z-3-3i) = \frac{2\pi}{3}\right\}$. Then $n(A \cap B)$ is equal to

(A) 1
(B) 2
(C) 3
(D) 0

Sol. **(D).** We can observe that $3 + 3i \in A$ but $\notin B$
$\therefore n(A \cap B) = 0$.



7. If z_1, z_2, z_3 are complex numbers such that $|z_1| = |z_2| = |z_3| = 1$, then $|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_1|^2$ can not exceed

(A) 6
(B) 9
(C) 12
(D) none of these

Sol. **(B).** Let $y = |z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_1|^2$
$= (z_1 - z_2)(\bar{z}_1 - \bar{z}_2) + (z_2 - z_3)(\bar{z}_2 - \bar{z}_3) + (z_3 - z_1)(\bar{z}_3 - \bar{z}_1)$

$6 \ge (z_1\bar{z}_2 + z_2\bar{z}_1 + z_2\bar{z}_3 + z_3\bar{z}_2 + z_3\bar{z}_1 + z_1\bar{z}_3)$ ...(1)

Now we know $|z_1 + z_2 + z_3|^2 \ge 0$

$\Rightarrow 3 + (z_1\bar{z}_2 + z_2\bar{z}_1 + z_1\bar{z}_3 + z_3\bar{z}_1 + z_2\bar{z}_3 + z_3\bar{z}_2) \ge 0$ ...(2)

From (1) and (2)

$y \le 9$

8. Dividing $f(z)$ by $z - i$, we obtain the remainder i and dividing it by $z + i$, we get the remainder $1 + i$. The remainder upon the division of $f(z)$ by $z^2 + 1$ is

(A) $\frac{1}{2}(z+1)+i$ (B) $\frac{1}{2}(iz+1)+i$

(C) $\frac{1}{2}(iz-1)+i$ (D) $\frac{1}{2}(z+i)+1$

Sol. **(B).** $f(z) = g(z)(z - i)(z + i) + az + b$; $a, b \in C$

$f(i) = i \Rightarrow ai + b = i$ ...(1)

$f(-i) = 1 + i \Rightarrow a(-i) + b = 1 + i$ ...(2)

From (1) and (2)

$a = \frac{i}{2}, b = \frac{1}{2} + i$.

Hence required remainder $= az + b = \frac{1}{2}iz + \frac{1}{2} + i$.

9. If $z = \sqrt{20i - 21} + \sqrt{21 - 20i}$, then the principal value of arg z can be

(A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$

(C) $-\frac{3\pi}{4}$ (D) $-\frac{\pi}{4}$

Sol. **(A, B, C, D).** $\sqrt{20i - 21} = \sqrt{(-25) + 4 + 2(5)(2i)}$

$= \sqrt{(5i)^2 + 2^2 + 2 \times 2 \times 5i} = \pm(2 + 5i)$

and $\sqrt{21 + 20i} = \sqrt{25 + (2i)^2 + 2(5)(2i)} = \sqrt{(5 + 2i)^2} = \pm(5 + 2i)$

$\Rightarrow z = 7 + 7i, -3 + 3i, -7 - 7i, 3 - 3i$

10. If $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ are complex numbers such that $|z_1| = 1$, $|z_2| = 2$ and $\text{Re}(z_1z_2) = 0$, then the pair of complex numbers $\omega_1 = a_1 + \frac{ia_2}{2}$ and $\omega_2 = 2b_1 + ib_2$ satisfy

(A) $|\omega_1| = 1$ (B) $|\omega_2| = 2$

(C) $\text{Re}(\omega_1\omega_2) = 0$ (D) $\text{Im}(\omega_1\omega_2) = 2$

Sol. **(A, B, C, D).** $a_1^2 + b_1^2 = 1$

$a_2^2 + b_2^2 = 4$ and $a_1a_2 = b_1b_2$

$a_2^2 + b_2^2 = 4a_1^2 + 4b_1^2$

$(a_2 + 2ia_1)^2 = (2b_1 + ib_2)^2$

$\Rightarrow a_2 = \pm 2b_1$

$2a_1 = \pm b_2$

$|\omega_1|^2 = a_1^2 + \frac{a_2^2}{4} = a_1^2 + b_1^2 = 1$

$\Rightarrow |\omega_1| = 1$

and $|\omega_2|^2 = 4b_1^2 + b_2^2 = 4$

$\Rightarrow |\omega_2| = 2$

$\text{Re}(\omega_1\omega_2) = 2a_1b_1 - \frac{a_2b_2}{2} = 0$

$\text{Im}(\omega_1\omega_2) = a_1b_2 + a_2b_1 = 2a_1^2 + 2b_1^2 = 2$.

11. If from a point P representing the complex number z_1 on the curve $|z| = 2$, pair of tangents are drawn to the curve $|z| = 1$, meeting at point $Q(z_2)$ and $R(z_3)$, then

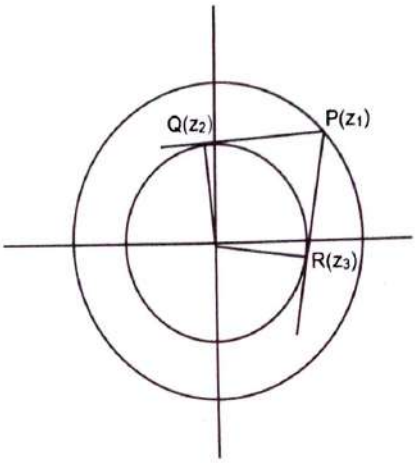
(A) complex number $\frac{z_1 + z_2 + z_3}{3}$ will lie on the curve $|z| = 1$

(B) $\left(\frac{4}{\bar{z}_1} + \frac{1}{\bar{z}_2} + \frac{1}{\bar{z}_3}\right)\left(\frac{4}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right) = 9$

(C) $\arg\left(\frac{z_2}{z_3}\right) = \frac{2\pi}{3}$

(D) orthocentre and circumcentre of ΔPQR will coincide

Sol. **(A, B, C, D).** A, D is true as PQR is equilateral triangle so orthocentre, circumcentre and centroid will coincide.



(B) $\left|\frac{z_1 + z_2 + z_3}{3}\right| = 1$

$\Rightarrow |z_1 + z_2 + z_3|^2 = 9$

$(z_1 + z_2 + z_3)(\bar{z}_1 + \bar{z}_2 + \bar{z}_3) = 9$

$\left(\frac{4}{\bar{z}_1} + \frac{1}{\bar{z}_2} + \frac{1}{\bar{z}_3}\right)\left(\frac{4}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right) = 9.$

(C) $\angle QOR = 120°$.

12. One vertex of the triangle of maximum area that can be inscribed in the curve $|z - 2i| = 2$, is $2 + 2i$, remaining vertices is / are

(A) $-1 + i(2 + \sqrt{3})$ (B) $-1 - i(2 + \sqrt{3})$

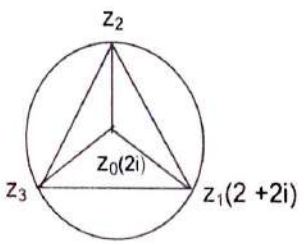
(C) $-1 + i(2 - \sqrt{3})$ (D) $-1 - i(2 - \sqrt{3})$

Sol. **(A), (C)**

Clearly the inscribed triangle is equilateral.

$\Rightarrow \frac{z_2 - z_0}{z_1 - z_0} = e^{i\frac{2\pi}{3}}, \frac{z_3 - z_0}{z_1 - z_0} = e^{-i\frac{2\pi}{3}}$

$\Rightarrow z_2 = -1 + i(2 + \sqrt{3})$ and $z_3 = -1 + i(2 - \sqrt{3})$



13. Let 'z' be a complex number and 'a' be a real parameter such that $z^2 + az + a^2 = 0$, then
(A) locus of z is a pair of straight lines
(B) locus of z is a circle
(C) $\arg(z) = \pm\frac{2\pi}{3}$
(D) $|z| = |a|$.

Sol. **(A), (C), (D)**

$z^2 + az + a^2 = 0$

$\Rightarrow$ $z = a\omega,\ a\omega^2$ (where 'ω' is non real root of cube unity)

$\Rightarrow$ locus of z is a pair of straight lines

and $\arg(z) = \arg(a) + \arg(\omega)$

or $\arg(a) + \arg(\omega^2)$

$\Rightarrow \arg(z) = \pm\frac{2\pi}{3}$

also, $|z| = |a||\omega|$ or $|a|\,|\omega^2|$

$\Rightarrow |z| = |a|$.

Comprehension

In argand plane $|z|$ represent the distance of a point z from the origin. In general $|z_1 - z_2|$ represent the distance between two points z_1 and z_2. Also for a general moving point z in argand plane, if $\arg(z) = \theta$, then $z = |z|e^{i\theta}$, where $e^{i\theta} = \cos\theta + i\sin\theta$.

Answer the following questions

14. The equation $|z - z_1| + |z - z_2| = 10$ if $z_1 = 3 + 4i$ and $z_2 = -3 - 4i$ represents
(A) point circle
(B) ordered pair (0, 0)
(C) ellipse
(D) none of these

Sol. **(D)**

15. $||z - z_1| - |z - z_2|| = t$, where t is a real parameter always represents
(A) ellipse
(B) hyperbola
(C) circle
(D) none of these

Sol. **(D).** Given equation can represent a hyperbola if $|z_1 - z_2| > t$.

16. If $|z - (3 + 2i)| = \left|z\cos\left(\frac{\pi}{4} - \arg z\right)\right|$, then locus of z is
(A) circle
(B) parabola
(C) ellipse
(D) hyperbola

Sol. **(B).** $|(x - 3) + i(y - 2)|$

$= |z|\left|\frac{1}{\sqrt{2}}\cos\theta + \frac{1}{\sqrt{2}}\sin\theta\right|$, where $\theta = \arg z$

$\Rightarrow \sqrt{(x-3)^2 + (y-2)^2}$

$= \frac{1}{\sqrt{2}}|x + iy|$, which is a parabola.

Assertion Reason

17. Statement –1: The points $z = x + iy$ on the complex plane satisfying $|z + 1| < |z - x| < |z - 1|$ lie on the same side of the origin of a parabola with vertex on imaginary axis.
because
Statement –2: $|z - x| < |z - 1|$ is always true.
(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1
(B) Statement-1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement -1 is True, Statement -2 is False
(D) Statement -1 is False, Statement -2 is True

Sol. **(D).** The first statement

$\Rightarrow x^2 < -2\left(y + \frac{1}{2}\right)$ and $|z - x| < |z - 1|$

2nd statement

$\Rightarrow y^2 < (x + 1)^2 + y^2$

so first statement is wrong because z lies on the side opposite to the side of the origin.

18. Match the following:

	Column – I		Column – II
(A)	$\arg\left(\frac{z^2-1}{z^2+1}\right) = 0 ; z \neq \pm i, \pm 1$	(p)	portions of a line
(B)	$\left\|\|z - \cos^{-1}\cos 12\| - \|z - \sin^{-1}\sin 12\|\right\| = 8(\pi - 3)$	(q)	point of intersection of hyperbolae
(C)	$z^2 + k_1 = i\|z_1\|^2 + k_2; k_1 \neq k_2 \in R - \{0\}$ and z_1 is fixed $\neq 0$	(r)	pair of open rays
(D)	$\left\|z - 1 - \sin^{-1}\frac{1}{\sqrt{3}}\right\| + \left\|z + \cos^{-1}\frac{1}{\sqrt{3}} - \frac{\pi}{2}\right\| = 1$	(s)	line segment
		(t)	

Sol. **(A) → (r); (B) → (p); (C) → (q); (D) → (s)**

(A) $\arg\left(\frac{z^2-1}{z^2+1}\right) = 0 ; z \neq \pm i$

$\frac{z^2-1}{z^2+1} = \frac{\bar{z}^2-1}{\bar{z}^2+1} \Rightarrow z - \bar{z} = 0, z + \bar{z} = 0$

$y = 0, x = 0$

locus of z is portion of pair of lines $xy = 0$. $\left[\because \left(\frac{z^2-1}{z^2+1}\right) > 0\right]$.

(B) $\left\|z - \cos^{-1}\cos 12\right| - \left|z - \sin^{-1}\sin 12\right\| = 8(\pi - 3)$

Since $\left|\cos^{-1}\cos 12 - \sin^{-1}\sin 12\right| = 8(\pi - 3)$

$\therefore$ locus of z is portion of a line joining z_1 and z_2 except the segment between z_1 and z_2.

(C) $z^2 - i|z_1|^2 = k_2 - k_1$

$x^2 - y^2 + 2ixy - i\lambda_1 = \lambda_2$

$\therefore x^2 - y^2 = \lambda_2$ and $xy = \frac{\lambda_1}{2}$

$\therefore$ locus of z is point of intersection of hyperbolae.

(D) $\left|z-1-\sin^{-1}\frac{1}{\sqrt{3}}\right|+\left|z+\cos^{-1}\frac{1}{\sqrt{3}}-\frac{\pi}{2}\right|=1$

Since $1+\sin^{-1}\frac{1}{\sqrt{3}}+\cos^{-1}\frac{1}{\sqrt{3}}-\frac{\pi}{2}=1$

$\therefore |z-z_1|+|z-z_2|=|z_1+z_2|$

$\Rightarrow$ locus of z is line segment joining z_1 and z_2.

Answer Q.19, Q.20 and Q.21 by appropriately matching the information given in the three columns of the following table.

Consider $z_1 = 1+4i$, $z_2 = 3+5i$ and z satisfying $L \equiv (1+i)z + (1-i)\bar{z} + 2 = 0$
Column-I contains statements about relation between z, z_1 and z_2
Column-II contains statements about value of z and
Column-III contains statements about z_1, z_2 and L

Column–I	Column–II	Column–III
(I) $\|z-z_1\|+\|z-z_2\|$ is minimum	(i) $\frac{5}{2}+\frac{7}{2i}$	(P) Reflection of z_1 in L is $3+2i$
(II) $\|\|z-z_1\|-\|z-z_2\|\|$ is minimum	(ii) $2+3i$	(Q) Foot of perpendicular form z_2 to L is $5+4i$
(III) $\|z-z_1\|-\|z-z_2\|$ is maximum	(iii) $3+4i$	(R) Reflection of z_2 in L is $4+3i$
(IV) $\text{Re}\left(\frac{z-z_1}{z_1-z_2}\right)=0$	(iv) $5+6i$	(S) Foot of perpendicular from z_1 to L is $2+3i$

19. Which of the following combination is **CORRECT**?
(A) (I) (i) (P)
(B) (I) (iii) (R)
(C) (IV) (ii) (Q)
(D) (II) (i) (P)

Sol. **(D)**

20. Which of the following combination is **CORRECT**?
(A) (II) (i) (Q)
(B) (III) (ii) (P)
(C) (I) (iii) (S)
(D) (IV) (iii) (Q)

Sol. **(C)**

21. Which of the following combination is **CORRECT**?
(A) (I) (iii) (Q)
(B) (IV) (ii) (P)
(C) (II) (i) (R)
(D) (I) (iii) (P)

Sol. **D**

19.-21. (I) z is collinear with reflection of z_1 and z_2
(II) z is on perpendicular bisector of line joining z_1 and z_2
(III) z point of intersection of line joining z_1 and z_2 and L

22. If z is a complex number lying in the first quadrant such that $\text{Re}(z) + \text{Im}(z) = 3$, then the maximum value of $\{\text{Re}(z)\}^2\,\text{Im}(z)$ is _____

Sol. **(4.00)**

Let $z = a + ib$ where $a > 0$, $b > 0$ (given)

Since, $\text{Re}(z) + \text{Im}(z) = 3$ (given)

$\Rightarrow a + b = 3$ ... (1)

Now, let $E = (\text{Re}(z))^2\,\text{Im}(z)$

$\Rightarrow E = a^2b = a^2(3-a)$ ... (2)

Now, E is maximum or minimum if $\frac{dE}{da} = 0$

$\Rightarrow 6a - 3a^2 = 0$

$\Rightarrow 3a(2-a) = 0$ either $a = 0$ or $a = 2$

Again, $\frac{d^2E}{da^2} < 0$, when $a = 2$.

Hence, E will attain its maximum if $a = 2$

$\therefore$ maximum value $= (2)^2(3-2) = 4$.

23. $\alpha_1, \alpha_2, \alpha_3, ..., \alpha_{100}$ are all the 100^{th} roots of unity. The numerical value of $\sum\sum_{1\le i<j\le 100}(\alpha_i\alpha_j)^5$ is _____

Sol. **(0.00)**

$$2\sum\sum_{1\le i<j\le 100}\alpha_i^5\alpha_j^5 = (\alpha_1^5+\alpha_2^5+\alpha_3^5+\alpha_4^5+\alpha_5^5+...)^2 - (\alpha_1^{10}+\alpha_2^{10}+\alpha_3^{10}+\alpha_4^{10}+\alpha_5^{10}+...) = 0-0=0$$

because $(\alpha_1^r+\alpha_2^r+...+\alpha_{100}^r) = \begin{cases}100 & \text{if} \quad r = 100k \\ 0. & \text{if} \quad r \neq 100k\end{cases}$

24. If z_1 satisfies $|z_1 + 1 - 2i| + |z_1 - 2i| = 9$ and z_2 satisfies $|z_2 - 2i| = 1$, then the minimum distance between z_1 and z_2 is _____

Sol. **(3.00)**

z_1 lies on ellipse with foci $-1 + 2i$ and $2i$ while z_2 lies on circle now find common normal

25. Let $M = \max\{(Re(z))^2 + 5(Im(2))^2 : |z| = 1\}$ and $m = \min\{2(Re(z))^2 + (Im(z))^2 : |z| = 1\}$, then $\frac{M^2}{m^2}$ is _____

Sol. **(25.00)**

$M = 5$ and $m = 1$

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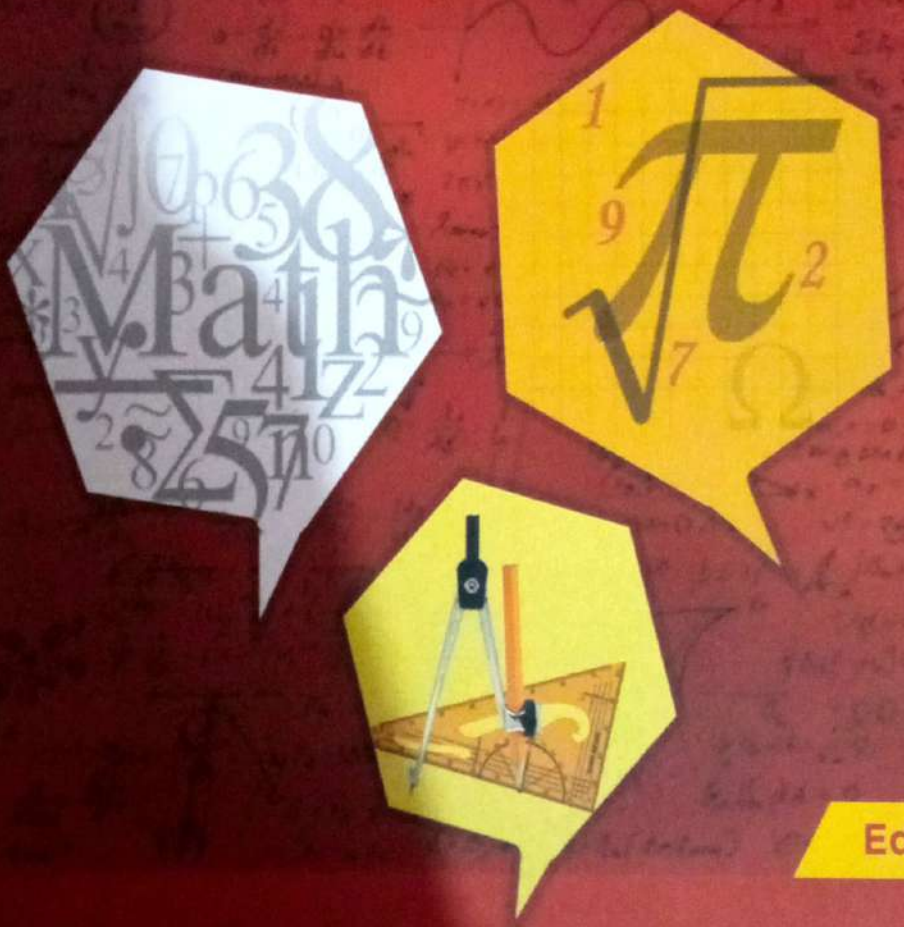


Edition 2020

MATHEMATICS

MATRICES

20

DEFINITION

A rectangular array of symbols (which could be real or complex numbers) along rows and columns is called a matrix.

Thus, $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$ is a matrix of order $m \times n$.

In a compact form the above matrix is represented by $A = [a_{ij}]$, $1 \le i \le m$, $1 \le j \le n$ or simply $[a_{ij}]_{m\times n}$

The numbers $a_{11}, a_{12}, \ldots$ etc of this rectangular array are called the elements of the matrix. The element a_{ij} belongs to the i^{th} row and j^{th} column and is called the $(i, j)^{th}$ element of a matrix.

Note

- The diagonal from left hand side upper corner to right hand side lower side lower corner is known as leading diagonal or principal diagonal.

Trace of a Matrix:

The sum of the elements of a square matrix A lying along the principal diagonal is called the trace of A and briefly written as tr(A).

Thus if $A = [a_{ij}]_{n\times n}$

Then $tr(A) = \sum_{i=1}^{n} a_{ii} = a_{11} + a_{22} + \ldots\ldots + a_{nn}$

Properties of Trace of a Matrix:

Let $A = [a_{ij}]_{n\times n}$ and $B = [b_{ij}]_{n\times n}$ and λ be a scalar

(i) $tr(\lambda A) = \lambda\, tr(A)$

(ii) $tr(A + B) = tr(A) + tr(B)$

(iii) $tr(AB) = tr(BA)$

TRANSPOSE OF A MATRIX

The matrix obtained from any given matrix A, by interchanging rows and columns, is called the transpose of A and is denoted by A'.

If $A = [a_{ij}]_{m\times n}$ and $A' = [b_{ij}]_{n\times m}$ then $b_{ij} = a_{ji}$, $\forall$ i, j.

Properties of Transposes

(i) $(A')' = A$

(ii) $(A + B)' = A' + B'$, A and B being conformable matrices for addition

(iii) $(\alpha A)' = \alpha A'$, α being scalar

(iv) $(AB)' = B'A'$, A and B being conformable for multiplication

Conjugate of a Matrix

The matrix obtained from any given matrix A containing complex number as its elements, on replacing its elements by the corresponding conjugate complex numbers is called conjugate of A and is denoted by $\bar{A}$.

Properties of Conjugates

(i) $\overline{(\bar{A})} = A$

(ii) $\overline{(A+B)} = \bar{A} + \bar{B}$

(iii) $\overline{(\alpha A)} = \bar{\alpha}\bar{A}$, α being any number

(iv) $\overline{(AB)} = \bar{A}\bar{B}$, A and B being conformable for multiplication.

Transpose Conjugate of a Matrix

$\overline{(A')} = (\bar{A})' = A^{\theta}$.

If $A = [a_{ij}]_{m\times n}$, then $A^{\theta} = [b_{ji}]_{n\times m}$ where $b_{ji} = \bar{a}_{ij}$

Properties of Transpose Conjugate

(i) $(A^{\theta})^{\theta} = A$

(ii) $(A+B)^{\theta} = A^{\theta} + B^{\theta}$

(iii) $(kA)^{\theta} = \bar{k}\ A^{\theta}$, k being any number

(iv) $(AB)^{\theta} = B^{\theta}A^{\theta}$

ALGEBRA OF MATRICES

Notes

- Only matrices of the same order can be added or subtracted.
- Addition of matrices is commutative as well as associative.
- Cancellation laws hold well in case of addition.
- The equation A + X = 0 has a unique solution in the set of all m×n matrices.
- All the laws of ordinary algebra hold for the addition or subtraction of matrices and their multiplication by scalars.

Multiplication of Matrices

Two matrices can be multiplied only when the number of columns in the first is equal to the number of rows in the second. Such matrices are said to be **conformable** for multiplication.

Notes

- Commutative law does not necessarily hold for matrices.
- If AB = BA, then matrices A and B are called ***commutative*** matrices.
- If AB = –BA, then matrices A and B are called ***anti-commutative*** matrices.
- Matrix multiplication is associative.
- Matrix multiplication is distributive with respect to addition.
- The matrices possesses divisors of zero, i.e. if the product AB = 0, it is not necessary that atleast one of the matrix should be zero matrix.
- Cancellation law does not necessarily hold, i.e. if AB = AC then in general $B \neq C$, even if $A \neq O$.

SPECIAL MATRICES

Hermitian and Skew-Hermitian Matrix

A square matrix $A = [a_{ij}]$ is said to be Hermitian matrix if $a_{ij} = \bar{a}_{ji} \quad \forall\, i, j$ i.e. $A = A^{\theta}$

and a square matrix, $A = [a_{ij}]$ is said to be a skew-Hermitian if $a_{ij} = -\bar{a}_{ji}$, $\forall\, i, j$ i.e. $A^{\theta} = -A$.

Notes:

- If A is a Hermitian matrix then $a_{ii} = \bar{a}_{ii} \Rightarrow a_{ii}$ is real $\forall$ i. thus every diagonal element of a Hermitian Matrix must be real.
- A Hermitian Matrix over the set of real numbers is actually a real symmetric matrix
- If A is a skew-Hermitian matrix then $a_{ii} = -\bar{a}_{ii} \Rightarrow a_{ii} + \bar{a}_{ii} = 0$ i.e. a_{ii} must be purely imaginary or zero.
- A skew-Hermitian Matrix over the set of real numbers is actually a real skew-symmetric matrix

Orthogonal Matrix

Any square matrix A of order n is said to be orthogonal if $AA' = A'A = I_n$.

Idempotent Matrix

A square matrix A is called idempotent provided it satisfies the relation $A^2 = A$.

Involutary Matrix

A square matrix such that $A^2 = I$ is called **involutary matrix.**

Nilpotent Matrix

A square matrix A is called a **nilpotent matrix** if there exists a positive integer m such that $A^m = O$. If m is the least positive integer such that $A^m = O$, then m is called the **index** of the nilpotent matrix A.

Adjoint of a Square Matrix

Let $A = [a_{ij}]$ be a square matrix of order n and let C_{ij} be cofactor of a_{ij} in A. Then
$\text{adj}A = [C_{ij}]^T \Rightarrow (\text{adj } A)_{ij} = C_{ji}$.

Theorem:
Let A be square matrix of order n. Then $A(\text{adj } A) = |A|\, I_n = (\text{adj } A)A$.

Note:

- The adjoint of a square matrix of order 2 can be easily obtained by interchanging the diagonal elements and changing the signs of off–diagonal (left hand side lower corner to right hand side upper corner) elements.

Inverse of a Matrix

The inverse of A is given by $A^{-1} = \frac{1}{|A|} \cdot \text{adj } A$

Properties of Inverse of a Matrix

(i) Every invertible matrix possesses a unique inverse.

(ii) (Reversal Law) If A and B are invertible matrices of the same order, then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

In general, if A, B, C, ... are invertible matrices then
$(ABC.....)^{-1} = ...C^{-1}B^{-1}A^{-1}$.

(iii) If A is an invertible square matrix, then A^T is also invertible and $(A^T)^{-1} = (A^{-1})^T$.

(iv) If A is a non-singular square matrix of order n. Then $|adjA| = |A|^{n-1}$

(v) If A and B are non-singular square matrices of the same order, then
adj (AB) = (adj B) (adj A).

(vi) If A is an invertible square matrix, then $adj(A^T) = (adj\ A)^T$

(vii) If A is a non-singular square matrix, then $adj(adjA) = |A|^{n-2} A$

Matrix Method of Solving non-homogeneous system of linear equations

Consider the equations

$$\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned} \quad \ldots (i)$$

$$\text{If } A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix},$$

then the equation (i) is equivalent to the matrix equation

$$AX = D \quad \ldots (ii)$$

Multiplying both sides of (ii) by the reciprocal matrix A^{-1}, we get

$$A^{-1}(AX) = A^{-1}D \Rightarrow IX = A^{-1}D \qquad [\because A^{-1}A = I]$$

$$\Rightarrow X = A^{-1}D \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{\Delta} \begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix} \times \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \quad \ldots (iii)$$

where A_1, B_1 etc. are the cofactors of a_1, b_1 etc. in the determinant

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \qquad (\Delta \neq 0)$$

(i) If A is a non-singular matrix, then the system of equations given by AX = B has a unique solution given by $X = A^{-1}B$.

(ii) If A is singular matrix, and (adjA)D = 0, then the system of equations given by AX = D is consistent with infinitely many solutions.

(iii) If A is singular matrix, and $(adjA)D \neq 0$, then the system of equation given by AX = D is inconsistent.

Solution of Homogeneous System of Linear Equations

Let AX = 0 be a homogeneous system of n linear equation with n unknowns. Now if A is non-singular then the system of equations will have a unique solution i.e. trivial solution and if A is a singular, then the system of equations will infinitely many solutions.

ILLUSTRATIVE PROBLEMS

1. Let $A=\begin{bmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{bmatrix}$, then find $\lim\limits_{n\to\infty}\frac{A_n}{n}$, where $\theta\in R$

Sol. $A^n=\begin{bmatrix}\cos n\theta & \sin n\theta\\ -\sin n\theta & \cos n\theta\end{bmatrix}$

$$\frac{A^n}{n}=\begin{bmatrix}\frac{\cos n\theta}{n} & \frac{\sin n\theta}{n}\\ \frac{-\sin n\theta}{n} & \frac{\cos n\theta}{n}\end{bmatrix}$$

$$\lim_{n\to\infty}\frac{A^n}{n}=\begin{bmatrix}\lim\limits_{n\to\infty}\frac{\cos n\theta}{n} & \lim\limits_{n\to\infty}\frac{\sin n\theta}{n}\\ -\lim\limits_{n\to\infty}\frac{\sin n\theta}{n} & \lim\limits_{n\to\infty}\frac{\cos n\theta}{n}\end{bmatrix}$$

$=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$ = **Zero Matrix**

2. **If A is a square matrix of order 3, then find $\left|\left(A-A^T\right)^{2011}\right|$.**

Sol. $A - A^T$ is skew-symmetric of order 3.
$\Rightarrow (A - A^T)^{2011}$ is skew-symmetric of order 3.
$\Rightarrow |(A - A^T)^{2011}| = 0$.

3. **If A is skew symmetric matrix, then I – A is (where I is identity matrix of the order equal to that of A)**

Sol. Consider a homogeneous system of equation $(I - A)X = 0 \Rightarrow AX = IX = X$ ...(1)
$\Rightarrow (AX)^T = X^T \Rightarrow X^TA^T = X^T \Rightarrow -X^TA = X^T$ ($\because$ A is skew symmetric)
$\Rightarrow -X^TAX = X^TX \Rightarrow -X^TX = X^TX$ (using (1))
$\Rightarrow X^TX = 0 \Rightarrow X = 0$
$\therefore (I - A)X = 0$ has only trivial solution
$\therefore \det(I - A) \neq 0$ or $I - A$ is non-singular.

4. **If $\omega^{x+1}+\omega^{y+2}+\omega^{z+3}=0$ where 'ω' being the cube root of unity. Then solve the system of equations**
$\lambda x+y-\lambda z=-\lambda$
$x-\lambda y+z=-\lambda$
$-\lambda x+y-\lambda z=\lambda$

Sol. Given systems are
$\lambda x + y - \lambda z = -\lambda$
$x - \lambda y + z = -\lambda$
$-\lambda x + y - \lambda z = \lambda$
$AX = B$

When $A=\begin{bmatrix}\lambda & 1 & -\lambda\\ 1 & -\lambda & 1\\ -\lambda & 1 & -\lambda\end{bmatrix}$, $X=\begin{bmatrix}x\\ y\\ z\end{bmatrix}$, $B=\begin{bmatrix}-\lambda\\ -\lambda\\ \lambda\end{bmatrix}$

$$\Delta = \begin{vmatrix} \lambda & 1 & -\lambda \\ 1 & -\lambda & 1 \\ -\lambda & 1 & -\lambda \end{vmatrix} = 2(\lambda+1)\lambda(\lambda-1)$$

$$\Delta_1 = \begin{vmatrix} -\lambda & 1 & -\lambda \\ -\lambda & -\lambda & 1 \\ \lambda & 1 & -\lambda \end{vmatrix} = -2\lambda(\lambda-1)(\lambda+1)$$

$$\Delta_2 = \begin{vmatrix} \lambda & -\lambda & -\lambda \\ 1 & -\lambda & 1 \\ -\lambda & \lambda & -\lambda \end{vmatrix} = 2\lambda^2(\lambda-1)$$

$$\Delta_3 = \begin{vmatrix} \lambda & 1 & -\lambda \\ 1 & -\lambda & -\lambda \\ -\lambda & 1 & \lambda \end{vmatrix} = 2\lambda(\lambda-1)$$

$\therefore x = -1$

$y = \frac{\lambda}{\lambda+1}$, $z = \frac{1}{\lambda+1}$

Now, $\omega^{x+1} + \omega^{y+2} + \omega^{z+3} = 0$

$\Rightarrow x + 1 = 0$ which is true as $x = -1$

Then $y + 2 = 3n + 1$

$\frac{\lambda}{\lambda+1} + 2 = 3n+1$, $\frac{1}{\lambda+1} + 3 = 3n+2$ $\left\{x = -1, y = z = \frac{1}{2}\right\}$

$\Rightarrow \lambda = 1$

$\therefore y = \frac{1}{2}$, $z = \frac{1}{2}$

5. **$A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$ and f(x) is defined as $f(x) = \det(A^T A^{-1})$ then the value of f(f(f(f ..... f(x)))) is ($x \geq$ 2)**

Sol. $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$

Hence, $\det A = \sec^2 x$

$\therefore \det A^T = \sec^2 x$

Now, $f(x) = \det(A^T A^{-1}) = (\det A^T)(\det A^{-1}) = (\det A^T)(\det A)^{-1} = \frac{\det(A^T)}{\det(A)} = 1$

Hence, $f(x) = 1$

SOLVED PROBLEMS

1. If $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, then $(BB^TA)^5$ is equal to

(A) $\begin{bmatrix} 2+\sqrt{3} & 1 \\ -1 & 2-\sqrt{3} \end{bmatrix}$ (B) $\frac{1}{2}\begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 5 & 1 \\ 0 & 1 \end{bmatrix}$

Sol. **(C).** $BB^T = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}\begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} \frac{3}{4}+\frac{1}{4} & -\frac{\sqrt{3}}{4}+\frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4}+\frac{\sqrt{3}}{4} & \frac{1}{4}+\frac{3}{4} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$(BB^TA)^5 = (A)^5 = \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right)^5 = {}^5C_0\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right)^5 + {}^5C_1\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^4\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + O$

$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + 5\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$.

2. In a 4×4 matrix the sum of each row, column and both the main diagonals is α. Then the sum of the four corner elements

(A) is also α (B) may not be α

(C) is never equal to α (D) none of these

Sol. **(A).** Let the numbers be a_{ij}

$\Rightarrow 2\alpha = (a_{11} + a_{12} + a_{13} + a_{14}) + (a_{41} + a_{42} + a_{43} + a_{44}) + (a_{11} + a_{22} + a_{33} + a_{44}) + (a_{14} + a_{23} + a_{32} + a_{41})$
$- (a_{12} + a_{22} + a_{32} + a_{42}) - (a_{13} + a_{23} + a_{33} + a_{43}) = 2(a_{11} + a_{14} + a_{41} + a_{44})$.

3. If A and B are square matrices of same order and A is non-singular, then for a positive integer n, then $(A^{-1}BA)^n$ is equal to

(A) $A^{-n}B^nA^n$ (B) $A^nB^nA^{-n}$

(C) $A^{-1}B^nA$ (D) $n(A^{-1}BA)$

Sol. **(C).** $(A^{-1}BA)^2 = (A^{-1}BA)(A^{-1}BA) = A^{-1}B^2A$

Similarly $(A^{-1}BA)^n = A^{-1}B^nA$.

4. $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $G(x) = \begin{bmatrix} \cos x & 0 & \sin x \\ 0 & 1 & 0 \\ -\sin x & 0 & \cos x \end{bmatrix}$, then $[F(x)\,G(y)]^{-1}$ is equal to

(A) $F(-x)\,G(-y)$ (B) $F(x-1)\,G(y-1)$

(C) $G(-y)\,F(-x)$ (D) $G(y^{-1})F(x^{-1})$

Sol. **(C).** $[F(x)\,G(y)]^{-1} = [G(y)]^{-1}\,[F(x)]^{-1}$

$[F(x)]^{-1} = \begin{bmatrix} \cos x & \sin x & 0 \\ -\sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix} = F(-x) \Rightarrow G(y)^{-1} = G(-y)$. Similarly, $[F(x)\,G(y)]^{-1} = G(-y)\,F(-x)$.

5. If $D = \text{diag}[d_1, d_2 \ldots d_n]$ is a diagonal matrix with $d_i \neq 0$, $i = 1, 2, 3, \ldots, n$, then D^{-1} is
(A) $d_1^{-1}d_2^{-1}\ldots d_n^{-1}I_n$
(B) $\text{diag}[d_n^{-1}, d_{n-1}^{-1}\ldots d_1^{-1}]$
(C) $\text{diag}[d_1^{-1}, d_2^{-1}\ldots d_n^{-1}]$
(D) none of these

Sol. **(C).** Here $|D| = d_1d_2d_3 \ldots\ldots\ldots d_n$ and
$\text{adj } D = [d_2d_3 \ldots\ldots\ldots d_n, d_1d_3 \ldots\ldots\ldots d_n, \ldots, d_1d_2 \ldots d_{n-1}]$
$\Rightarrow D^{-1} = \text{diag}[d_1^{-1}, d_2^{-1}, \ldots, d_n^{-1}]$.

6. If A is a matrix order $n \times n$, then $A + A'$ is a matrix of type,
(A) null matrix
(B) symmetric matrix
(C) skew-symmetric matrix
(D) skew-Hermition

Sol. **(B).** Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ and $A' = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}$

Therefore $A + A' = \begin{bmatrix} 2a & b+d & g+c \\ b+d & 2e & f+h \\ g+c & h+f & 2i \end{bmatrix}$, which is a symmetric matrix.

7. $A^3 - 2A^2 - A + 2I = 0$ if A is
(A) I
(B) 2I
(C) $\begin{bmatrix} 2 & -1 & 2 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$
(D) $\begin{bmatrix} 2 & 1 & -2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

Sol. **(A, B, C, D)**
$A(A^2 - I) - 2(A^2 - I) = 0$
$\Rightarrow (A - 2I)(A^2 - I) = 0$

8. If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ (where $bc \neq 0$) satisfies the equations $x^2 + k = 0$, then
(A) $a + d = 0$
(B) $k = -|A|$
(C) $k = |A|$
(D) none of these

Sol. **(A, C)**
We have $A^2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2+bc & ad+db \\ ac+cd & bc+d^2 \end{bmatrix}$
As A satisfies, $x^2 + k = 0$, $A^2 + kI = 0$
$\Rightarrow \begin{bmatrix} a^2+bc+k & (a+d)b \\ (a+d)c & bc+d^2+k \end{bmatrix} = 0$
$\Rightarrow a^2 + bc + k = 0$
and $bc + d^2 + k = 0$
and $(a + d)b = (a + d)c = 0$
As $bc \neq 0$, $b \neq 0$, $c \neq 0$
$\Rightarrow a + d = 0 \Rightarrow a = -d$
Also, $k = -(a^2 + bc) = -(d^2 + bc)$
$= -((-ad) + bc) = |A|$

9. Let $A = a_{ij}$ be a matrix of order 3 where $a_{ij} = \begin{cases} x & \text{if } i = j, x \in R \\ 1 & \text{if } |i-j| = 1 \\ 0 & \text{otherwise} \end{cases}$, then which of the following hold(s) good?

(A) for $x = 2$, A is a diagonal matrix
(B) A is a symmetric matrix
(C) for $x = 2$, det A has the value equal to 6
(D) let $f(x) = \det A$, then the function $f(x)$ has both the maxima and minima

Sol. **(B, D)**

$$A = \begin{bmatrix} x & 1 & 0 \\ 1 & x & 1 \\ 0 & 1 & x \end{bmatrix} \Rightarrow |A| = x^3 - 2x = f(x)$$

Comprehension

$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$, if U_1, U_2 and U_3 are columns matrices satisfying.

$AU_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $AU_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$, $AU_3 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ and U is 3×3 matrix whose columns are U_1, U_2, U_3 then

Answer the following questions:

10. The value of $|U|$ is
(A) 3 (B) -3
(C) 3/2 (D) 2

Sol. **(A).** Let U_1 be $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ so that

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

Similarly $U_2 = \begin{bmatrix} 2 \\ -1 \\ -4 \end{bmatrix}$, $U_3 = \begin{bmatrix} 2 \\ -1 \\ -3 \end{bmatrix}$.

Hence $U = \begin{bmatrix} 1 & 2 & 2 \\ -2 & -1 & -1 \\ 1 & -4 & -3 \end{bmatrix}$ and $|U| = 3$.

11. The sum of the elements of U^{-1} is
(A) -1 (B) 0
(C) 1 (D) 3

Sol. **(B).** Moreover adj $U = \begin{bmatrix} -1 & -2 & 0 \\ -7 & -5 & -3 \\ 9 & 6 & 3 \end{bmatrix}$.

Hence $U^{-1} = \frac{\text{adj}U}{3}$

and sum of the elements of $U^{-1} = 0$.

12. The value of $\begin{bmatrix} 3 & 2 & 0 \end{bmatrix} U \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$ is

(A) 5 (B) 5/2
(C) 4 (D) 3/2

Sol. **(A).** The value of $\begin{bmatrix} 3 & 2 & 0 \end{bmatrix} U \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$

$= \begin{bmatrix} 3 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ -2 & -1 & -1 \\ 1 & -4 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$

$= \begin{bmatrix} -1 & 4 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$

$= -3 + 8 = 5.$

Assertion Reason

13. Statement 1: If $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$, then the matrix represented by

$A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$ is $A^2 + A + I$.

because

Statement 2: The chareateristic equation of A is $\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$.

(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1
(B) Statement -1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement -1 is True, Statement -2 is False
(D) Statement -1 is False, Statement -2 is True

Sol. **(A).** The chareateristic equation of A is $|A - \lambda I| = 0$

$\Rightarrow \lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$

$\therefore A^3 - 5A^2 + 7A - 3I = 0$

Now $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I = A^5(A^3 - 5A^2 + 7A - 3I) + A(A^3 - 5A^2 + 7A - 3I) + A^2 + A + I$

$= A^5 . O + A . O + A^2 + A + I$

$= A^2 + A + I.$

14. Match the following

Column–I		Column–II	
(A)	If a, b, c $\in$ R – {0} such that $a \neq b \neq c$ and $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=0$ then $\begin{bmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{bmatrix}$ is	(p)	symmetric
(B)	If $\alpha, \beta, \gamma \in R$ then $\begin{bmatrix} 1 & \cos(\alpha-\beta) & \cos(\alpha-\gamma) \\ \cos(\beta-\alpha) & 1 & \cos(\beta-\gamma) \\ \cos(\gamma-\alpha) & \cos(\gamma-\beta) & 1 \end{bmatrix}$ is	(q)	singular
(C)	If $\omega \neq 1$, be cube root of unity then $\begin{bmatrix} 1+2\omega^{100}+\omega^{200} & \omega^2 & 1 \\ 1 & 1+\omega^{101}+2\omega^{202} & \omega \\ \omega & \omega^2 & 2+\omega^{100}+2\omega^{200} \end{bmatrix}$ is	(r)	non-singular
(D)	If a, b, c $\in$ R – {0} such that $a \neq b \neq c$ $\begin{bmatrix} 0 & (a-b)^3 & (a-c)^3 \\ (b-a)^3 & 0 & (b-c)^3 \\ (c-a)^3 & (c-b)^3 & 0 \end{bmatrix}$ is	(s)	invertible
		(t)	

Sol. **(A) → (p, r, s); (B) → (p, q); (C) → (q); (D) →(q)**

(A) It is symmetric

$$\text{Let } w=\begin{bmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{bmatrix}$$

$$|w| = abc\begin{vmatrix} 1+\frac{1}{a} & \frac{1}{b} & \frac{1}{c} \\ \frac{1}{a} & 1+\frac{1}{b} & \frac{1}{c} \\ \frac{1}{a} & \frac{1}{b} & 1+\frac{1}{c} \end{vmatrix}$$

Applying $C_1 \to C_1 + C_2 + C_3$

$$|w| = abc\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)\begin{vmatrix} 1 & \frac{1}{b} & \frac{1}{c} \\ 1 & 1+\frac{1}{b} & \frac{1}{c} \\ 1 & \frac{1}{b} & 1+\frac{1}{c} \end{vmatrix}$$

Applying $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$, we get $|w| \neq 0$.

Hence it is non-singular and invertible.

(B) Clearly It is symmetric matrix.

$$\text{Let } |w| = \begin{vmatrix} \cos\alpha & \sin\alpha & 0 \\ \cos\beta & \sin\beta & 0 \\ \cos\gamma & \sin\gamma & 0 \end{vmatrix} \times \begin{vmatrix} \cos\alpha & \sin\alpha & 0 \\ \cos\beta & \sin\beta & 0 \\ \cos\gamma & \sin\gamma & 0 \end{vmatrix} = 0 \times 0$$

$\therefore$ Hence $|w| = 0$

$\therefore$ it is singular matrix.

(C) $A = \begin{bmatrix} 1+2\omega^{100}+\omega^{200} & \omega^2 & 1 \\ 1 & 1+\omega^{101}+2\omega^{202} & \omega \\ \omega & \omega^2 & 2+\omega^{100}+2\omega^{200} \end{bmatrix}$

$$|A| = \begin{vmatrix} \omega & \omega^2 & 1 \\ 1 & \omega & \omega \\ \omega & \omega^2 & -\omega \end{vmatrix} = 0$$

(D) Since given matrix is skew-symmetric matrix. So its determinant is zero. Hence it is singular.

Answer Q.15, Q.16 and Q.17 by appropriately matching the information given in the three columns of the following table.

Match the following Column(s)

Column–I	Column–II	Column–III
(I) $A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ -2 & 0 & 1 \end{bmatrix}$	(i) $\text{Adj } A = \begin{bmatrix} -10 & 5 & -5 \\ -5 & 0 & -5 \\ -15 & -20 & -10 \end{bmatrix}$	(P) 625
(II) $A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix}$	(ii) $\text{Adj } A = \begin{bmatrix} 3 & 1 & -11 \\ -12 & 5 & -1 \\ 6 & 2 & 5 \end{bmatrix}$	(Q) 121
(III) $A = \begin{bmatrix} 4 & -6 & 1 \\ -1 & -1 & 1 \\ -4 & 11 & -1 \end{bmatrix}$	(iii) $\text{Adj } A = \begin{bmatrix} 30 & 12 & -3 \\ -20 & -8 & 2 \\ -50 & -20 & 5 \end{bmatrix}$	(R) 0
(IV) $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 3 & 0 \\ 18 & 2 & 10 \end{bmatrix}$	(iv) $\text{Adj } A = \begin{bmatrix} 0 & 3 & 2 \\ -11 & 1 & 8 \\ 0 & -1 & 3 \end{bmatrix}$	(S) 729

15. Which of the following combination is **CORRECT**?

(A) (I) (ii) (S) (B) (I) (iii) (S)

(C) (II) (ii) (S) (D) (II) (iii) (S)

Sol. **(A)**

16. Which of the following combination is **CORRECT**?

(A) (III) (i) (P) (B) (IV) (iii) (P)

(C) (III) (iii) (R) (D) (IV) (i) (P)

Sol. **(A)**

17. Which of the following combination is **INCORRECT**?

(A) (III) (i) (P) (B) (IV) (iii) (P)

(C) (IV) (iii) (R) (D) (II) (iv) (Q)

Sol. **(B)**

15-17. $|\text{Adj } A| = |A|^{n-1}$

18. The index of the matrix $\begin{pmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{pmatrix}$ is

(A) 2 (B) 3
(C) 4 (D) none of these

Sol. **(3.00)**

$A^2 = \begin{pmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{pmatrix}$; $A^3 = \begin{pmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{pmatrix}\begin{pmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. So index is 3.

19. Let M be a square matrix of order 3 such that $MM^T = I$ and $M^2 = I$. Also $M^{-1} + \text{adj}(M) = O$. If P is another matrix such that $P + 2M = O$, then find the value of $\det.(PP^T P^{-1})$ _____

Sol. **(8.00)**

$MM^T = I$; $M^2 = I$; $M^{-1} + \text{adj}(M) = 0$; $M^{-1} = -\text{adj } M$;

Take determinant of both sides, $|M^{-1}| = |-\text{adj } M|$

$$\frac{1}{|M|} = (-1)^3 |\text{adj } M| \Rightarrow \frac{1}{|M|} = -|M|^2 ; |M| = -1$$

$$\Rightarrow \det.\left(PP^T P^{-1}\right) = |P|\left|P^T\right|\left|P^{-1}\right| = |P||P|\frac{1}{|P|} = |P|$$

As $P = -2M$; $|P| = |-2M| = (-2)^3 |M| = (-8)(-1) = 8$; $\det.(PP^T P^{-1}) = 8$

20. If $3A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ x & 2 & y \end{bmatrix}$ and $A \cdot A^T = I$, then $x + y$ is p, then $\frac{|p|}{2}$ is equals to _____

Sol. **(1.50)**

$$AA^T = \frac{1}{9}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ x & 2 & y \end{bmatrix}\begin{bmatrix} 1 & 2 & x \\ 2 & 1 & 2 \\ 2 & -2 & y \end{bmatrix} = \frac{1}{9}\begin{bmatrix} 9 & 0 & x+4+2y \\ 0 & 9 & 2x+2-2y \\ x+4+2y & 2x+2-2y & x^2+4+y^2 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\Rightarrow x = -2, y = -1$

$\Rightarrow x + y = -3$

21. If A and B are two square matrices such that $A^2B = BA$ and $(AB)^{10} = A^k \cdot B^{10}$ then the value of $(k - 1020)$ is k, then $\frac{k}{2}$ is equal to _____

Sol. **(1.50)**

$(AB)^2 = AB \cdot AB = A^3B^2$

$(AB)^3 = (AB)^2 AB = A^3B^2AB = A^7B^3$

$(AB)^4 = (AB)^3 \cdot AB = A^7B^3 \cdot AB = A^{15}B^4$

$\Rightarrow (AB)^{10} = A^{1023}B^{10} = A^kB^{10}$

$\Rightarrow k - 1020 = 3$

21

DETERMINANTS AND LINEAR EQUATIONS

BASIC CONCEPTS

- A determinant of order three consisting of 3 rows and 3 columns is written as $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and is equal to $a_1\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1\begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1\begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$

  $= a_1(b_2c_3 - c_2b_3) - b_1(a_2c_3 - c_2a_3) + c_1(a_2b_3 - b_2a_3)$
- The numbers a_i, b_i, c_i ($i = 1, 2, 3$) are called the elements of the determinant.
- The determinant obtained by deleting the i^{th} row and j^{th} column is called the minor of element at the i^{th} row and the j^{th} column. The cofactor of this element is $(-1)^{i+j}$ (minor). Note that:

  $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1A_1 + b_1B_1 + c_1C_1$

  where A_1, B_1 and C_1 are the cofactors of a_1, b_1 and c_1 respectively.
- We can expand the determinant through any row or column. It means that we can write

  $\Delta = a_2A_2 + b_2B_2 + c_2C_2 = a_1A_1 + a_2A_2 + a_3A_3$ etc.

  Also $a_1A_2 + b_1B_2 + c_1C_2 = 0$

  $\Rightarrow a_iA_j + b_iB_j + c_iC_j = \Delta$ if $i = j$

  $= 0$ if $i \neq j$

  These results are true for determinants of any order.

Properties of Determinants

- The determinant remains unaltered if its rows are changed into columns and the columns into rows.
- If all the elements of a row (or column) are zero, then the determinant is zero.
- If the elements of a row (column) are proportional (or identical) to the elements of any other row (column), then the determinant is zero.
- The interchange of any two rows (columns) of the determinant changes its sign.
- If all the elements of a row (column) of a determinant are multiplied by a non-zero constant then the determinant gets multiplied by the same constant.
- A determinant remains unaltered under a column (C_i) operation of the form $C_i + \alpha C_j + \beta C_k$ ($j,k \neq i$) or a row (R_i) operation of the form $R_i + \alpha R_j + \beta R_k$ ($j,k \neq i$).
- If each element in any row (column) is the sum of r terms, then the determinant can be expressed as the sum of r determinants.
- If the determinant $\Delta = f(x)$ and $f(a) = 0$, then $(x - a)$ is a factor of the determinant. In other word, if two rows (or two columns) become proportional (identical) for $x = a$ then $(x - a)$ is a factor of determinant. In general, if r rows become identical for $x = a$ then $(x - a)^{r-1}$ is a factor of the determinant.

- If in a determinant (of order three or more) the elements in all the rows (columns) are in A.P. with same or different common difference, the value of the determinant is zero.
- The determinant value of an odd order skew symmetric determinant is always zero.

Product of two Determinants

We can write

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \begin{vmatrix} \alpha_1 & \beta_1 & \gamma_1 \\ \alpha_2 & \beta_2 & \gamma_2 \\ \alpha_3 & \beta_3 & \gamma_3 \end{vmatrix} = \begin{vmatrix} a_1\alpha_1 + b_1\beta_1 + c_1\gamma_1 & a_1\alpha_2 + b_1\beta_2 + c_1\gamma_2 & a_1\alpha_3 + b_1\beta_3 + c_1\gamma_3 \\ a_2\alpha_1 + b_2\beta_1 + c_2\gamma_1 & a_2\alpha_2 + b_2\beta_2 + c_2\gamma_2 & a_2\alpha_3 + b_2\beta_3 + c_2\gamma_3 \\ a_3\alpha_1 + b_3\beta_1 + c_3\gamma_1 & a_3\alpha_2 + b_3\beta_2 + c_3\gamma_2 & a_3\alpha_3 + b_3\beta_3 + c_3\gamma_3 \end{vmatrix}$$

Here we have multiplied rows by rows. We can also multiply rows by columns or columns by rows, or columns by columns.

Note:

- If $\Delta = |a_{ij}|$ is a determinant of order n, then the value of the determinant $|A_{ij}|$, where A_{ij} is the cofactor of a_{ij} is Δ^{n-1}. This is known as **power cofactor formula**.

Differentiation of a Determinant

- Let $\Delta(x) = \begin{vmatrix} a_1(x) & b_1(x) \\ a_2(x) & b_2(x) \end{vmatrix}$. Then $\Delta'(x) = \begin{vmatrix} a_1'(x) & b_1'(x) \\ a_2(x) & b_2(x) \end{vmatrix} + \begin{vmatrix} a_1(x) & b_1(x) \\ a_2'(x) & b_2'(x) \end{vmatrix}$ where a dash denotes derivative with respect to x.
- If we write $\Delta(x) = |C_1C_2C_3|$, then $\Delta'(x) = |C_1'C_2C_3| + |C_1C_2'C_3| + |C_1C_2C_3'|$
- Similarly if $\Delta(x) = \begin{vmatrix} R_1 \\ R_2 \\ R_3 \end{vmatrix}$, then $\Delta'(x) = \begin{vmatrix} R_1' \\ R_2 \\ R_3 \end{vmatrix} + \begin{vmatrix} R_1 \\ R_2' \\ R_3 \end{vmatrix} + \begin{vmatrix} R_1 \\ R_2 \\ R_3' \end{vmatrix}$

Summation of Determinants

Let $\Delta_r = \begin{vmatrix} f(r) & a & l \\ g(r) & b & m \\ h(r) & c & n \end{vmatrix}$ where a, b, c, l, m and n are constants, independent of r.

Then $\sum_{r=1}^{n} \Delta_r = \begin{vmatrix} \sum_{r=1}^{n} f(r) & a & l \\ \sum_{r=1}^{n} g(r) & b & m \\ \sum_{r=1}^{n} h(r) & c & n \end{vmatrix}$. Here function of r can be the elements of only one row or one column.

Determinants Involving Integrations

Let $\Delta(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ a & b & c \\ l & m & n \end{vmatrix}$ where a, b, c, l, m and n are constants.

$$\Rightarrow \int_a^b \Delta(x)dx = \begin{vmatrix} \int_a^b f(x)dx & \int_a^b g(x)dx & \int_a^b h(x)dx \\ a & b & c \\ l & m & n \end{vmatrix}$$

Note:

- If the elements of more than one column or rows are functions of x then the integration can be done only after evaluation / expansion of the determinant.

Cramer's Rule

If $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \neq 0,$

then the solution of the system of non-homogeneous simultaneous linear equations

$a_1x + b_1y + c_1z = d_1,$

$a_2x + b_2y + c_2z = d_2,$

$a_3x + b_3y + c_3z = d_3$ is given by {where $(d_1, d_2, d_3) \neq (0, 0, 0)$}

$x = \frac{\Delta_x}{\Delta}, \quad y = \frac{\Delta_y}{\Delta}, \quad z = \frac{\Delta_z}{\Delta}$ where

$$\Delta_x = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}, \quad \Delta_y = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}, \quad \Delta_z = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}.$$

- If any of $\Delta_x, \Delta_y, \Delta_z \in R$ and $\Delta \neq 0$, system of equation will have unique solution and is said to be consistence independent.
- If $\Delta_x = \Delta_y = \Delta_z = 0$ and Δ is also zero then the system of equation will have infinitely many solutions and is said to be consistent dependent.
- If $\Delta_x, \Delta_y, \Delta_z$ is non zero and Δ is zero, then the system of equations will have no solution and is said to be inconsistent.

Linear Equations

The system of homogeneous simultaneous linear equations

$a_1x + b_1y + c_1z = 0, \qquad a_2x + b_2y + c_2z = 0, \qquad a_3x + b_3y + c_3z = 0$

Here notice that $\Delta_1 = \Delta_2 = \Delta_3 = 0$ has a non-trivial solution (i.e. at least one of x,y,z,is different from zero) if

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

If $\Delta \neq 0$, then the only solution of the above system of equations is $x = 0, y = 0, z = 0$ (called trivial solution).

ILLUSTRATIVE PROBLEMS

1. **Let a, b, c be real numbers with $a^2 + b^2 + c^2 = 1$. Show that the equation** $\begin{vmatrix} ax-by-c & bx+ay & cx+a \\ bx+ay & -ax+by-c & cy+b \\ cx+a & cy+b & -ax-by+c \end{vmatrix} = 0$ **represents a straight line.**

Sol. Given, $\begin{vmatrix} ax-by-c & bx+ay & cx+a \\ bx+ay & -ax+by-c & cy+b \\ cx+a & cy+b & -ax-by+c \end{vmatrix} = 0$

$C_1 \to aC_1 + bC_2 + cC_3$

$$\Rightarrow \frac{1}{a}\begin{vmatrix} (a^2+b^2+c^2)x & bx+ay & cx+a \\ (a^2+b^2+c^2)y & -ax+by-c & cy+b \\ (a^2+b^2+c^2)z & cy+b & -ax-by+c \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x & bx+ay & cx+a \\ y & -ax+by-c & cy+b \\ 1 & cy+b & -ax-by+c \end{vmatrix} = 0$$

Applying $C_2 \to C_2 - bC_1$ and $C_3 \to C_3 - cC_1$

$$\Delta = \begin{vmatrix} x & ay & a \\ y & -ax-c & b \\ 1 & cy & -ax-by \end{vmatrix} = 0$$ applying $R_3 \to R_3 + xR_1 + yR_2$, we get

$$\Delta = \begin{vmatrix} x & ay & a \\ y & -ax-c & b \\ x^2+y^2+1 & 0 & 0 \end{vmatrix} = 0$$

$\Rightarrow (x^2 + y^2 + 1)(aby + a^2x + ac) = 0$

$\Rightarrow ax + by + c = 0$, which represents a straight line.

2. **Express** $\Delta = \begin{vmatrix} (a-x)^2 & (a-y)^2 & (a-z)^2 \\ (b-x)^2 & (b-y)^2 & (b-z)^2 \\ (c-x)^2 & (c-y)^2 & (c-z)^2 \end{vmatrix}$ **as the product of two determinants and evaluate it.**

Sol. $$\Delta = \begin{vmatrix} (a^2-2ax+x^2) & (a^2-2ay+y^2) & (a^2-2az+z^2) \\ (b^2-2bx+x^2) & (b^2-2by+y^2) & (b^2-2bz+z^2) \\ (c^2-2cx+x^2) & (c^2-2cy+y^2) & (c^2-2cz+z^2) \end{vmatrix}$$

$$= \begin{vmatrix} a^2 & -2a & 1 \\ b^2 & -2b & 1 \\ c^2 & -2c & 1 \end{vmatrix} \times \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

$= 2(a-b)(b-c)(c-a)(x-y)(y-z)(z-x)$.

3. **Let** $\Delta_a = \begin{vmatrix} a-1 & n & 6 \\ (a-1)^2 & 2n^2 & 4n-2 \\ (a-1)^3 & 3n^3 & 3n^2-3n \end{vmatrix}$. **Show that** $\sum_{a=1}^{n} \Delta_a = 0$.

Sol. By adding all the corresponding elements of C_1 of all determinants Δ_a we have

$$\sum_{a=1}^{n} \Delta_a = \begin{vmatrix} \sum_{a=1}^{n}(a-1) & n & 6 \\ \sum_{a=1}^{n}(a-1)^2 & 2n^2 & 4n-2 \\ \sum_{a=1}^{n}(a-1)^3 & 3n^3 & 3n^2-3n \end{vmatrix}$$

$$= \begin{vmatrix} \frac{(n-1)n}{2} & n & 6 \\ \frac{(n-1)n(2n-1)}{6} & 2n^2 & 4n-2 \\ \frac{(n-1)^2n^2}{4} & 3n^3 & 3n^2-3n \end{vmatrix}.$$

By taking $\frac{(n-1)n}{2}$ as common factor from C_1 and 6 as common factor from C_3, we get

$$\sum_{a=1}^{n} \Delta_a = 3n(n-1).\begin{vmatrix} 1 & n & 1 \\ \frac{2n-1}{3} & 2n^2 & \frac{2n-1}{3} \\ \frac{n(n-1)}{2} & 3n^3 & \frac{n(n-1)}{2} \end{vmatrix}$$

Since C_1 and C_3 are identical $\sum_{a=1}^{n} \Delta_a = 0$.

4. **Let** $f(x) = \begin{vmatrix} \sec x & \cos x & \sec^2 x + \cot x \operatorname{cosec} x \\ \cos^2 x & \cos^2 x & \operatorname{cosec}^2 x \\ 1 & \cos^2 x & \cos^2 x \end{vmatrix}$, **prove that at** $\int_0^{\pi/2} f(x)dx = -\left(\frac{\pi}{4} + \frac{8}{15}\right)$.

Sol. Operate $R_1 \to R_1 - \sec x\, R_3$

$$\Rightarrow f(x) = \begin{vmatrix} 0 & 0 & \sec^2 x + \cot x \operatorname{cosec} x - \cos x \\ \cos^2 x & \cos^2 x & \operatorname{cosec}^2 x \\ 1 & \cos^2 x & \cos^2 x \end{vmatrix}$$

$$= \left(\sec^2 x + \cot x \operatorname{cosec} x - \cos x\right)\left(\cos^4 x - \cos^2 x\right) = \left(1 + \frac{\cos^3 x}{\sin^2 x} - \cos^3 x\right)\left(\cos^2 x - 1\right)$$

$$= -\sin^2 x - \cos^3 x + \cos^3 x \sin^2 x = -\left(\sin^2 x + \cos^5 x\right)$$

$$\Rightarrow \int_0^{\pi/2} f(x)dx = -\int_0^{\pi/2} (\sin^2 x + \cos^5 x)dx = -\left(\frac{1}{2}.\frac{\pi}{2} + \frac{4.2}{5.3.1}\right) = -\left(\frac{\pi}{4} + \frac{8}{15}\right).$$

5. For what values of p and q, the system of equations $2x + py + 6z = 8, x + 2y + qz = 5, x + y + 3z = 4$ has (i) no solution (ii) a unique solution (iii) infinitely many solutions?

Sol. $\Delta = \begin{vmatrix} 2 & p & 6 \\ 1 & 2 & q \\ 1 & 1 & 3 \end{vmatrix} = 2(6 - q) - p(3 - q) + 6(1 - 2)$

$= 12 - 2q - 3p + pq - 6$

$= pq - 2q - 3p + 6 = (p - 2)(q - 3)$

$\Delta_1 = \begin{vmatrix} 8 & p & 6 \\ 5 & 2 & q \\ 4 & 1 & 3 \end{vmatrix} = 8(6 - q) - p(15 - 4q) + 6(5 - 8)$

$= 48 - 8q - 15p + 4pq - 18 = 4pq - 8q - 15p + 30$

$= 4q(p - 2) - 15(p - 2) = (4q - 15)(p - 2)$

$\Delta_2 = \begin{vmatrix} 2 & 8 & 6 \\ 1 & 5 & q \\ 1 & 4 & 3 \end{vmatrix} = 2(15 - 4q) - 8(3 - q) + 6(4 - 5) = 0$

$\Delta_3 = \begin{vmatrix} 2 & p & 8 \\ 1 & 2 & 5 \\ 1 & 1 & 4 \end{vmatrix} = 2(8 - 5) - p(4 - 5) + 8(1 - 2) = p - 2$

Case –I

When $q = 3, p \neq 2, \Delta = 0, \Delta_1 \neq 0$.

$\therefore$ The given system of equations will have no solution.

Case–II

When $\Delta \neq 0$, i.e. $p \neq 2, q \neq 3$, the given system of equation has unique solution.

Case–III

When $\Delta = 0$, i.e. $p = 2$, or $q = 3$

When $p = 2, \Delta = 0, \Delta_1 = 0, \Delta_2 = 0, \Delta_3 = 0$

$\therefore$ given system of equation has infinitely many solutions

SOLVED PROBLEMS

1. If α, β and γ are the roots of the equation $x^2(px + q) = r(x + 1)$. Then the value of determinant $\begin{vmatrix} 1+\alpha & 1 & 1 \\ 1 & 1+\beta & 1 \\ 1 & 1 & 1+\gamma \end{vmatrix}$ is

(A) $\alpha\beta\gamma$

(B) $1+\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}$

(C) 0

(D) none of these

Sol. **(C).** The given equation is $px^3 + qx^2 - rx - r = 0$. Its roots are given α, β, γ so we have

$\alpha+\beta+\gamma = \frac{-q}{p}$; $\alpha\beta+\beta\gamma+\gamma\alpha = -\frac{r}{p}$; $\alpha\beta\gamma = \frac{r}{p}$.

$D = \alpha\beta\gamma\left(1+\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}\right) = \alpha\beta\gamma\left(\frac{\alpha\beta\gamma+\alpha\beta+\beta\gamma+\gamma\alpha}{\alpha\beta\gamma}\right) = 0.$

2. One factor of $\begin{vmatrix} a^2+\lambda & ab & ac \\ ab & b^2+\lambda & cb \\ ca & cb & c^2+\lambda \end{vmatrix}$ is

(A) λ^2

(B) $(a^2 + \lambda)(b^2 + \lambda)(c^2 + \lambda)$

(C) $\frac{1}{\lambda}$

(D) none of these

Sol. **(A).** Put $\lambda = 0$

The given determinant becomes $\begin{vmatrix} a^2 & ab & ac \\ ab & b^2 & cb \\ ca & cb & c^2 \end{vmatrix} = abc\begin{vmatrix} a & b & c \\ a & b & c \\ a & b & c \end{vmatrix} = 0 \Rightarrow \lambda^2 = 0$

3. If $\begin{vmatrix} x^2+1 & 1 & x+1 \\ 2x^2-1 & 1 & x+2 \\ 3x^2-2 & 1 & x+3 \end{vmatrix} = a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots.$, then a_1 is equal to

(A) 1

(B) 2

(C) 3

(D) 0

Sol. **(D).** $a_1 = f'(0)$ where $f(x) = \begin{vmatrix} x^2+1 & 1 & x+1 \\ 2x^2-1 & 1 & x+2 \\ 3x^2-2 & 1 & x+3 \end{vmatrix}$

$f'(x) = \begin{vmatrix} 2x & 1 & x+1 \\ 4x & 1 & x+2 \\ 6x & 1 & x+3 \end{vmatrix} + 0 + 0 = 0 \ (R_3 - R_2,\ R_2 - R_1)$

$\therefore f'(0) = 0$

$\Rightarrow a_1 = 0.$

4. The system of equations $6x + 5y + \lambda z = 0,\ 3x - y + 4z = 0,\ x + 2y - 3z = 0$ has non-trivial solutions for

(A) $\lambda = 0$ (B) $\lambda = 1$

(C) $\lambda = -5$ (D) none of these

Sol. **(C).** For non-trivial solution $\Delta = 0$

$$\begin{vmatrix} 6 & 5 & \lambda \\ 3 & -1 & 4 \\ 1 & 2 & -3 \end{vmatrix} = 0$$

$\Rightarrow -30 + 65 + 7\lambda = 0$

$\Rightarrow \lambda = -5.$

5. Consider an identity matrix $(n \times n)\ I_n$; $\lambda \in R^+$ then $|\text{Adj}(\lambda\, I_n)|$ is

(A) λ^{n-1} (B) $\lambda^{n(n-1)}$

(C) λ^n (D) $\lambda^{2(n-1)}$

Sol. **(B)**

$(\lambda\, I_n)\ \text{Adj}(\lambda\, I_n) = |\lambda\, I_n|\, I$

$\Rightarrow \text{Adj}(\lambda\, I_n) = \lambda^{n-1} I_n$

$\Rightarrow |\text{Adj}(\lambda\, I_n)| = \lambda^{n(n-1)}$

6. If $\Delta(x) = \begin{vmatrix} e^x & \sin 2x & \tan x^2 \\ \ln(1+x) & \cos x & \sin x \\ \cos x^2 & e^x - 1 & \sin x^2 \end{vmatrix} = A + Bx + Cx^2 + \ldots\ldots$, then B is equal to

(A) 0 (B) 1

(C) 2 (D) none of these

Sol. **(A)**

$$\Delta'(x) = \begin{vmatrix} e^x & 2\cos 2x & 2x\sec^2 x^2 \\ \ln(1+x) & \cos x & \sin x \\ \cos x^2 & e^x - 1 & \sin x^2 \end{vmatrix} + \begin{vmatrix} e^x & \sin 2x & \tan x^2 \\ \frac{1}{(1+x)} & -\sin x & \cos x \\ \cos x^2 & e^x - 1 & \sin x^2 \end{vmatrix} + \begin{vmatrix} e^x & \sin 2x & \tan x^2 \\ \ln(1+x) & \cos x & \sin x \\ -2x\sin x^2 & e^x & 2x\cos x^2 \end{vmatrix}$$

$= B + 2Cx + \ldots.$

$$\text{Put } x = 0,\ B = \begin{vmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 0 & 0 \\ 1 & 0 & -1 \\ 1 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{vmatrix} = 0$$

7. If $f(x) = \begin{vmatrix} x+\alpha & x+a & x+a \\ x+b & x+\beta & x+a \\ x+b & x+b & x+\gamma \end{vmatrix}$, $g(x) = (\alpha - x)(\beta - x)(\gamma - x)$, then the value of $\dfrac{b\,g(a) - a\,g(b)}{b-a}$ is

(A) $f(1)$ (B) (-1)

(C) $f(0)$ (D) none of these

Sol. **(C).** $f(x) = \begin{vmatrix} x+\alpha & a-\alpha & 0 \\ x+b & \beta-b & a-\beta \\ x+b & 0 & \gamma-b \end{vmatrix} \begin{pmatrix} C_3 \to C_3 - C_2 \\ C_2 \to C_2 - C_1 \end{pmatrix}$ = a linear expression in $x = px + q$ (say)

$f(-a) = -pa + q;\ f(-b) = -pb + q$

$$f(-a)=\begin{vmatrix}\alpha-a & a-\alpha & 0\\ b-a & \beta-b & a-\beta\\ b-a & 0 & \gamma-b\end{vmatrix}=\begin{vmatrix}\alpha-a & 0 & 0\\ b-a & \beta-a & a-\beta\\ b-a & b-a & \gamma-b\end{vmatrix}(C_2\to C_2+C_1)=(\alpha-a)(\beta-a)(\gamma-a)=g(a)$$

$$f(-b)=\begin{vmatrix}\alpha-b & \alpha-b & a-b\\ 0 & \beta-b & a-b\\ 0 & 0 & \gamma-b\end{vmatrix}=(\alpha-b)(\beta-b)(\gamma-b)=g(b)$$

$\therefore g(a)=pa+q,$

$g(b)=pb+q$

$\because bg(a)-ag(b)=(b-a)q=(b-a)f(0).$

8. If p, q, r be real, then $f(x)=\begin{vmatrix}x+p^2 & pq & pr\\ pq & x+q^2 & qr\\ pr & qr & x+r^2\end{vmatrix}$

(A) increases in interval $(0,\infty)$ (B) decreases in interval $\left(\frac{-2(p^2+q^2+r^2)}{3},\ 0\right)$

(C) increases in interval $\left(-\infty,\ \frac{-2(p^2+q^2+r^2)}{3}\right)$ (D) none of these

Sol. **(A, B, C).** $f'(x)=\begin{vmatrix}1 & 0 & 0\\ pq & x+q^2 & qr\\ pr & qr & x+r^2\end{vmatrix}+\begin{vmatrix}x+p^2 & pq & pr\\ 0 & 1 & 0\\ pr & qr & x+r^2\end{vmatrix}+\begin{vmatrix}x+p^2 & pq & pr\\ pq & x+q^2 & qr\\ 0 & 0 & 1\end{vmatrix}$

$=3x^2+2(p^2+q^2+r^2)x.$

9. If $a_i, b_i, c_i \in R$ $(i=1,2,3)$ and $x \in B$ and $\begin{vmatrix}a_1+b_1x & a_1x+b_1 & c_1\\ a_2+b_2x & a_2x+b_2 & c_2\\ a_3+b_3x & a_3x+b_3 & c_3\end{vmatrix}=0$, then

(A) $x=1$ (B) $x=-1$

(C) $\begin{vmatrix}a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{vmatrix}=0$ (D) 0

Sol. **(A, B, C)**

Applying $C_1\to C_1-xC_2$

$$\begin{vmatrix}a_1(1-x^2) & a_1x+b_1 & c_1\\ a_2(1-x^2) & a_2x+b_2 & c_2\\ a_3(1-x^2) & a_3x+b_3 & c_3\end{vmatrix}=0\Rightarrow(1-x^2)\begin{vmatrix}a_1 & a_1x+b_1 & c_1\\ a_2 & a_2x+b_2 & c_2\\ a_3 & a_3x+b_3 & c_3\end{vmatrix}=0$$

Applying $C_2\to C_2-xC_1$

$$\Rightarrow(1-x^2)\begin{vmatrix}a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{vmatrix}=0;\ \therefore x=\pm1 \text{ and } \begin{vmatrix}a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{vmatrix}=0$$

10. If maximum and minimum values of the determinant $\begin{vmatrix} 1+\sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1+\cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1+\sin 2x \end{vmatrix}$ are α and β, then

(A) $\alpha + \beta^{99} = 4$
(B) $\alpha^3 - \beta^{17} = 26$
(C) $(\alpha^{2n} - \beta^{2n})$ is always an even integer for $n \in N$
(D) a triangle can be constructed having it's sides as α, β and $\alpha - \beta$.

Sol. **(A), (B), (C)**

Applying $C_1 \to C_1 + C_2$, we get

$$\begin{vmatrix} 2 & \cos^2 x & \sin 2x \\ 2 & 1+\cos^2 x & \sin 2x \\ 1 & \cos^2 x & 1+\sin 2x \end{vmatrix}$$

Applying $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$, we get

$$\begin{vmatrix} 2 & \cos^2 x & \sin 2x \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = 2 + \sin 2x.$$

Since the maximum value of sin 2x is 1, and min value of sin 2x is (–1).
Therefore $\alpha = 3$, $\beta = 1$.

11. If a_i, $i = 1, 2, \ldots, 9$ are perfect odd squares, then $\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$ is always a multiple of

(A) 4
(B) 7
(C) 16
(D) 5

Sol. **(A), (C)**

Let $a_1 = (2m+1)^2$, $a_2 = (2n+1)^2$
$\Rightarrow a_1 - a_2 = 4(m(m+1) - n(n+1)) = 8k$
so, difference of any two odd square is always a multiple of 8
Now apply $C_1 - C_3$ and $C_2 - C_3$
then C_1 and C_2 both become multiple of 8 so Δ always a multiple of 64.

12. If $g(x) = \begin{vmatrix} a & -1 & 0 \\ ax & a & -1 \\ ax^2 & ax & a \end{vmatrix}$ then $g(2x) - g(x)$ is divisible by

(A) x
(B) a
(C) 2a + 3x
(D) x^2

Sol. **A, B, C**

$g(x) = a(a + x)^2$
$\therefore g(2x) - g(x)$
$= a(a + 2x)^2 - a(a + x)^2$
$= a((2a + 3x)x)$
$= ax(2a + 3x)$

Comprehension

Consider the determinant, $\Delta = \begin{vmatrix} p & q & r \\ x & y & z \\ l & m & n \end{vmatrix}$.

M_{ij} denotes the minor of an element in i^{th} row and j^{th} column.
C_{ij} denotes the cofactor of an element in i^{th} row and j^{th} column.

Now answer the following questions based on the above comprehension:

13. The value of $p \cdot C_{21} + q \cdot C_{22} + r \cdot C_{23}$ is equal to
(A) 0 (B) $-\Delta$
(C) Δ (D) Δ^2

Sol. **(A)**
p, q, r are the entries of first row and C_{21}, C_{22}, C_{23} are the cofactors of second row.
$\therefore p \cdot C_{21} + q \cdot C_{22} + r \cdot C_{23} = 0$.

14. The value of $x \cdot C_{21} + y \cdot C_{22} + z \cdot C_{23}$ is equal to
(A) 0 (B) $-\Delta$
(C) Δ (D) Δ^2

Sol. **(C)**
x, y, z are the entries of second row and C_{21}, C_{22}, C_{23} are the cofactors of second row.
$\therefore x \cdot C_{21} + y \cdot C_{22} + z \cdot C_{23} = \Delta$

15. The value of $q \cdot M_{12} - y \cdot M_{22} + m \cdot M_{32}$ is equal to
(A) 0 (B) $-\Delta$
(C) Δ (D) Δ^2

Sol. **(B)**
$q \cdot M_{12} - y \cdot M_{22} + m \cdot M_{32}$
$= -q \cdot C_{12} - y \cdot C_{22} - m \cdot C_{32}$
$= -(-q \cdot C_{12} - y \cdot C_{22} - m \cdot C_{32}) = -\Delta$
($\because$ q, y, m are entries of second column and C_{12}, C_{22}, C_{32} are cofactor of second column)

Assertion Reason

16. Statement –I: The number of possible dimensions of a matrix containing 32 elements is 6
Because
Statement –II: The number of ways of expressing 32 as the product of two possible integers is 6
(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1
(B) Statement -1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement -1 is True, Statement -2 is False
(D) Statement -1 is False, Statement -2 is True

Sol. **(C)**
The possible types of matrix are 1×32, 2×16, 4×8, 32×1, 16×2, 8×4
So A is true
But 32 can only be expressed as the product of two positive integers in 3 ways.
So A is true but R is false.

17. Match the following Column-I with Column-II

Column – I		Column – II	
(A)	If $\Delta(x) = \begin{vmatrix} x^2 & 5x+3 & 2x \\ 3x^2+x+4 & 6x+1 & 9 \\ 7x^2-6x+9 & 14x-6 & 21 \end{vmatrix} = ax^3+bx^2+cx+d$, then	(p)	$3a+4b+5c+d$ 141
(B)	If $\Delta(x) = \begin{vmatrix} x-1 & 5x & 7 \\ x^2 & 1 & x \\ 2x & 3x & 0 \end{vmatrix} = ax^3+bx^2+cx+d$, then	(q)	$a+2b+3c+5d$ 156
(C)	If $\Delta(x) = \begin{vmatrix} 2x^3-3x^2 & 5x+7 & 2 \\ 4x^3 & 7x & 3x+2 \\ 7x^3-8x^2 & x-1 & 3 \end{vmatrix} = a+bx+cx^2+dx^3+ex^4$, then	(r)	$c-d=119$
(D)	If $\Delta(x) = \begin{vmatrix} e^x & \sin x \\ \cos x & \ln(1+x^2) \end{vmatrix} = a+bx+cx^2+\ldots..$ then	(s)	a 42b 42
		(t)	0

Sol. **(A) → (p); (B) → (q); (C) → (r); (D) → (s)**

(P) $\Delta'(x) = 0 \Rightarrow a = b = c = 0$

$$\therefore \Delta(0) = \begin{vmatrix} 3 & 5 & 3 \\ 4 & 1 & 9 \\ 9 & -6 & 21 \end{vmatrix} \Rightarrow d = 141$$

(Q) At $x = 0$

$$\Delta(0) = \begin{vmatrix} -1 & 0 & 7 \\ -1 & -1 & 8 \\ 0 & 0 & 0 \end{vmatrix} \Rightarrow d = 0$$

$$\Delta'(x) = \begin{vmatrix} 1 & 5x & 7 \\ 2x & x-1 & 8 \\ 2 & 3x & 0 \end{vmatrix} + \begin{vmatrix} x-1 & 5 & 7 \\ x^2-1 & 1 & 8 \\ 2x & 3 & 0 \end{vmatrix} = 3ax^2+2b+c$$

Put $x = 0 \Rightarrow c = 17$

$$\Delta''(x) = \begin{vmatrix} 0 & 5x & 7 \\ 2 & x-1 & 8 \\ 0 & 3x & 0 \end{vmatrix} + \begin{vmatrix} 1 & 5 & 7 \\ 2x & 1 & 8 \\ 2 & 3 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 5 & 7 \\ 2x & 1 & 8 \\ 2 & 3 & 0 \end{vmatrix} = 6ax+2b$$

Put $x = 0$

$2b = 84 \Rightarrow b = 42$

$$\Delta'''(x) = \begin{vmatrix} 0 & 5 & 7 \\ 2 & 1 & 8 \\ 0 & 3 & 0 \end{vmatrix} + 2\begin{vmatrix} 0 & 5 & 7 \\ 2 & 1 & 8 \\ 0 & 3 & 0 \end{vmatrix} = 6a \Rightarrow 6a = 42 + 2 \times 42 \Rightarrow a = 21$$

(R) $a = 0, b = 161, c = 46, d = -73, e = -43$

(S) Put $x = 0$ then $a = 0$

$$\begin{vmatrix} e^x & \cos x \\ \cos x & \ln(1+x^2) \end{vmatrix} + \begin{vmatrix} e^x & \sin x \\ -\sin x & \dfrac{2x}{1+x^2} \end{vmatrix} = B + 2Cx + \ldots.. \Rightarrow b = -1$$

18. If $f(x) = \begin{vmatrix} \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ \sin(x+\alpha) & \sin(x+\beta) & \sin(x+\gamma) \\ \sin(\beta-\gamma) & \sin(\gamma-\alpha) & \sin(\alpha-\beta) \end{vmatrix}$, then $f(\theta) - 2f(\phi) + f(\psi)$ is equal to _____

Sol. **(0.00)**

Now, $f'(x) = -\begin{vmatrix} \sin(x+\alpha) & \sin(x+\beta) & \sin(x+\gamma) \\ \sin(x+\alpha) & \sin(x+\beta) & \sin(x+\gamma) \\ \sin(\beta-\gamma) & \sin(\gamma-\alpha) & \sin(\alpha-\beta) \end{vmatrix} + \begin{vmatrix} \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ \sin(\beta-\gamma) & \sin(\gamma-\alpha) & \sin(\alpha-\beta) \end{vmatrix}$

$= 0 + 0 = 0 \Rightarrow f(x)$ is a constant function, let $f(x) = c \therefore f(\theta) - 2f(\phi) + f(\psi) = c - 2c + c = 0$.

19. If $a_1, a_2, a_3, \ldots, a_{12}$ are in A.P. and $\Delta_1 = \begin{vmatrix} a_1a_5 & a_1 & a_2 \\ a_2a_6 & a_2 & a_3 \\ a_3a_7 & a_3 & a_4 \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} a_2a_{10} & a_2 & a_3 \\ a_3a_{11} & a_3 & a_4 \\ a_3a_{12} & a_4 & a_5 \end{vmatrix}$ then $\Delta_1 : \Delta_2$ is equal to _____

Sol. **(1.00)**

Here, $\Delta_1 = d(a_1 - a_2)(a_2 - a_3)(a_3 - a_1) = -2d^4$. Also, $\Delta_2 = -2d^4$

20. If $\begin{vmatrix} x & 2 & x \\ x^2+1 & x & 5 \\ 2x-1 & 1 & x \end{vmatrix} = Ax^5 + Bx^4 + Cx^3 + Dx^2 + Ex + F$ then the value of $5A + 4B + 3C + 2D + E$ is k, then $\frac{k}{5}$ is equal to _____

Sol. **(2.60)**

$Ax^5 + Bx^4 + Cx^3 + Dx^2 + Ex + F = \begin{vmatrix} x & 2x & x \\ x^2+1 & x^2 & 5 \\ 2x-1 & x & x \end{vmatrix}$. Diff. $F'(x) = 5Ax^4 + 4Bx^3 + 3Cx^2 + 2Dx + E$

$= \begin{vmatrix} 1 & 2x & x \\ 2x & x^2 & 5 \\ 2 & x & x \end{vmatrix} + \begin{vmatrix} x & 2 & x \\ x^2+1 & 2x & 5 \\ 2x-1 & 1 & x \end{vmatrix} + \begin{vmatrix} x & 2x & 1 \\ x^2+1 & x^2 & 0 \\ 2x-1 & x & 1 \end{vmatrix}$. $F'(1) = 5A + 4B + 3C + 2D + E = 12 + 3 - 2 = 13$

21. If $a_1, a_2, a_3, 5, 4, a_6, a_7, a_8, a_9$ are in H.P. and $\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ 5 & 4 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$, then $[3\Delta]$ is k, find $\frac{k}{2}$ is equal to _____

(where [.] denote the greatest integer function)

Sol. **(3.50)**

By applying $\begin{matrix} a+3d = \frac{1}{5} \\ a+4d = \frac{1}{4} \\ \hline a = d = \frac{1}{20} \end{matrix}$, r^{th} term of H.P. $= \frac{20}{r}$; $\Delta = \begin{vmatrix} 20 & 10 & \frac{20}{3} \\ 5 & 4 & \frac{20}{6} \\ \frac{20}{7} & \frac{20}{8} & \frac{20}{9} \end{vmatrix} = \begin{vmatrix} 20 & 10 & \frac{20}{3} \\ 5 & 4 & \frac{10}{3} \\ \frac{20}{7} & \frac{5}{2} & \frac{20}{9} \end{vmatrix} = \frac{50}{21}$

$\Rightarrow [3\Delta] = \left[\frac{50}{7}\right] = 7$

VECTORS

22

LINEAR COMBINATION OF VECTORS

The linear combination of a finite set of vectors $\bar{a}_1, \bar{a}_2, \ldots, \bar{a}_n$ is defined as a vector $\bar{r}$ such that $\bar{r} = k_1\bar{a}_1 + k_2\bar{a}_2 + \ldots\ldots + k_n\bar{a}_n$, where $k_1, k_2, \ldots, k_n$ are any scalars (real numbers).

Fundamental Theorem in Two Dimensions

Let $\bar{a}$ and $\bar{b}$ be two non-zero, non-collinear vectors. Then any vector $\bar{r}$ in the plane of $\bar{a}$ and $\bar{b}$ can be expressed uniquely as a linear combination of $\bar{a}$ and $\bar{b}$, i.e. there exist unique $l, m \in R$ such that $l\bar{a} + m\bar{b} = \bar{r}$.

This also means that if $l_1\bar{a} + m_1\bar{b} = l_2\bar{a} + m_2\bar{b}$ then $l_1 = l_2$ and $m_1 = m_2$.

Fundamental Theorem in Three Dimensions

Let $\bar{a}$, $\bar{b}$ and $\bar{c}$ be the three non-zero non-coplanar vectors in space. Then any vector $\bar{r}$ in space can be expressed uniquely as a linear combination of $\bar{a}$, $\bar{b}$ and $\bar{c}$, i.e. there exist unique $l, m, n \in R$ such that

$$l\bar{a} + m\bar{b} + n\bar{c} = \bar{r}.$$

This also means that if $l_1\bar{a} + m_1\bar{b} + n_1\bar{c} = l_2\bar{a} + m_2\bar{b} + n_2\bar{c}$. then $l_1 = l_2$, $m_1 = m_2$ and $n_1 = n_2$

Linearly Dependent and Independent Vectors

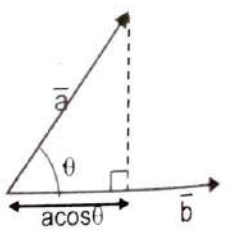
A system of vectors $\bar{a}_1, \bar{a}_2, \ldots, \bar{a}_n$ is said to be linearly dependent if there exists a system of scalars $k_1, k_2 \ldots, k_n$ (not all zero) such that $k_1\bar{a}_1 + k_2\bar{a}_2 + \ldots + k_n\bar{a}_n = \bar{0}$.

They are said to be linearly independent if every relation of the type $k_1\bar{a}_1 + k_2\bar{a}_2 + \ldots + k_n\bar{a}_n = \bar{0}$ implies that $k_1 = k_2 = \ldots = k_n = 0$

- Two collinear vectors are always linearly dependent.
- Two non-collinear non-zero vectors are always linearly independent.
- Three coplanar vectors are always linearly dependent.
- Three non-coplanar non-zero vectors are always linearly independent.
- More than 3 vectors are always linearly dependent.
- Three points with position vectors $\bar{a}, \bar{b}, \bar{c}$ are collinear if $\lambda_1\bar{a} + \lambda_2\bar{b} + \lambda_3\bar{c} = 0$ with $\lambda_1 + \lambda_2 + \lambda_3 = 0$.
- Four points with position vectors $\bar{a}, \bar{b}, \bar{c}, \bar{d}$ are coplanar if $\lambda_1\bar{a} + \lambda_2\bar{b} + \lambda_3\bar{c} + \lambda_4\bar{d} = 0$ with $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 0$.

Scalar (or Dot) Product of Two Vectors

The scalar product of $\bar{a}$ and $\bar{b}$ written as $\bar{a}\cdot\bar{b}$ is defined to be the number $|\bar{a}||\bar{b}|\cos\theta$ where θ is the angle between the vectors $\bar{a}$ and $\bar{b}$ i.e. $\bar{a}\cdot\bar{b} = ab\cos\theta$.



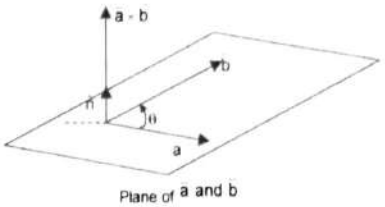
Properties

- $\bar{a}\cdot\bar{a} = |\bar{a}|^2 = a^2 \Rightarrow \hat{i}\cdot\hat{i} = \hat{j}\cdot\hat{j} = \hat{k}\cdot\hat{k} = 1$
- $\bar{a}\cdot\bar{b} = \bar{b}\cdot\bar{a}$
- $\bar{a}\cdot\bar{0} = 0$
- $\bar{a}\cdot\bar{b} =$ $(a\cos\theta)b$ = (projection of $\bar{a}$ on $\bar{b}$) b = (projection of $\bar{b}$ on $\bar{a}$) a
- $\bar{a}\cdot(\bar{b}+\bar{c}) = \bar{a}\cdot\bar{b} + \bar{a}\cdot\bar{c}$ (Distributive law)
- $(l\bar{a})\cdot(m\bar{b}) = lm(\bar{a}\cdot\bar{b})$; l, m are scalars
- $\bar{a}\cdot\bar{b} = 0 \Leftrightarrow \bar{a}$, $\bar{b}$ are perpendicular to each other
  $\Rightarrow \hat{i}\cdot\hat{j} = \hat{j}\cdot\hat{k} = \hat{k}\cdot\hat{i} = 0$
- $(\bar{a}\pm\bar{b})^2 = (\bar{a}\pm\bar{b})\cdot(\bar{a}\pm\bar{b}) = \bar{a}^2 + \bar{b}^2 \pm 2\bar{a}\cdot\bar{b}$
- $(\bar{a}+\bar{b})\cdot(\bar{a}-\bar{b}) = \bar{a}^2 - \bar{b}^2 = a^2 - b^2$
- If $\bar{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\bar{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, then $\bar{a}\cdot\bar{b} = a_1b_1 + a_2b_2 + a_3b_3$
- If $\bar{a}$, $\bar{b}$ are non-zero, the angle between them is given by $\cos\theta = \dfrac{\bar{a}\cdot\bar{b}}{|\bar{a}||\bar{b}|}$

$$= \frac{a_1b_1 + a_2b_2 + a_3b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2}\sqrt{b_1^2 + b_2^2 + b_3^2}}.$$

Vector (or Cross) Product of Two Vectors:

The vector product of two vectors $\bar{a}$ and $\bar{b}$ denoted by $\bar{a}\times\bar{b}$ is defined as the vector $|\bar{a}||\bar{b}|\sin\theta\hat{n}$, where θ is the angle between the vectors $\bar{a}$, $\bar{b}$ and $\hat{n}$ is a unit vector perpendicular to both $\bar{a}$ and $\bar{b}$ (i.e., perpendicular to the plane of $\bar{a}$ and $\bar{b}$). The sense of $\hat{n}$ is obtained by the right hand thumb rule i.e. $\bar{a}$, $\bar{b}$ and $\hat{n}$ form a right-handed screw.



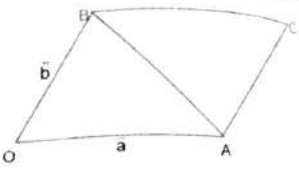
It is evident that $|\bar{a}\times\bar{b}| = ab\sin\theta$.

Properties

- $\bar{a}\times\bar{a} = \bar{0} \Rightarrow \hat{i}\times\hat{i} = \hat{j}\times\hat{j} = \hat{k}\times\hat{k} = \bar{0}$
- $\bar{a}\times\bar{b} = -(\bar{b}\times\bar{a})$ (non-commutative)
- $\bar{a}\times(\bar{b}+\bar{c}) = \bar{a}\times\bar{b} + \bar{a}\times\bar{c}$ (Distributive)
- $(l\bar{a})\times(m\bar{b}) = lm(\bar{a}\times\bar{b})$; l, m are scalars
- $\bar{a}\times\bar{b} = \bar{0} \Leftrightarrow \bar{a}$ and $\bar{b}$ are collinear (if none of $\bar{a}$, or $\bar{b}$ is a zero vector)
- $\hat{i}\times\hat{j} = \hat{k}$, $\hat{j}\times\hat{k} = \hat{i}$, $\hat{k}\times\hat{i} = \hat{j}$
  $\hat{j}\times\hat{i} = -\hat{k}$, $\hat{k}\times\hat{j} = -\hat{i}$, $\hat{i}\times\hat{k} = -\hat{j}$
- If $a = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then $\bar{a}\times\bar{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$.
- Any vector perpendicular to the plane of $\bar{a}$ and $\bar{b}$ is $\lambda(\bar{a}\times\bar{b})$ where λ is a real number. Unit vector perpendicular to $\bar{a}$ and $\bar{b}$ is $\pm\dfrac{\bar{a}\times\bar{b}}{|\bar{a}\times\bar{b}|}$.

- $|\bar{a}\times\bar{b}|$ denotes the area of the parallelogram OACB, whereas the area of $\Delta OAB = \frac{1}{2}|\bar{a}\times\bar{b}|$.

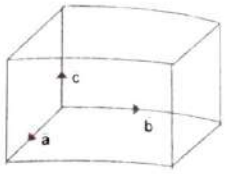
  Area is also treated as a vector with its direction in the proper sense.



Scalar Triple Product

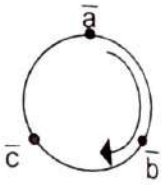
It is defined for three vectors $\bar{a}$, $\bar{b}$, $\bar{c}$, in that order as the scalar $(\bar{a}\times\bar{b})\cdot\bar{c} = [\bar{a}\ \ \bar{b}\ \ \bar{c}]$, which can also be written simply as $\bar{a}\times\bar{b}\cdot\bar{c}$. It denotes the volume of the parallelopiped formed by taking $\bar{a}$, $\bar{b}$, $\bar{c}$ as the co-terminus edges.

i.e. V = magnitude of $\bar{a}\times\bar{b}\cdot\bar{c} = |\bar{a}\times\bar{b}\cdot\bar{c}|$.



Properties

- If $\bar{a}$, $\bar{b}$, $\bar{c}$ are given as $\bar{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, etc., then $\bar{a}\times\bar{b}\cdot\bar{c} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$.
- $\bar{a}\times\bar{b}\cdot\bar{c} = \bar{a}\cdot\bar{b}\times\bar{c}$, i.e. position of dot and cross can be interchanged without altering the product. Hence it is also represented by $[\bar{a}\ \ \bar{b}\ \ \bar{c}]$.
- $[\bar{a}\ \ \bar{b}\ \ \bar{c}] = [\bar{b}\ \ \bar{c}\ \ \bar{a}] = [\bar{c}\ \ \bar{a}\ \ \bar{b}]$
- $[\bar{a}\ \ \bar{b}\ \ \bar{c}] = -[\bar{b}\ \ \bar{a}\ \ \bar{c}]$
- $[k\bar{a}\ \ \bar{b}\ \ \bar{c}] = k[\bar{a}\ \ \bar{b}\ \ \bar{c}]$
- $[\bar{a}+\bar{b}\ \ \bar{c}\ \ \bar{d}] = [\bar{a}\ \ \bar{c}\ \ \bar{d}] + [\bar{b}\ \ \bar{c}\ \ \bar{d}]$
- $\bar{a}$, $\bar{b}$, $\bar{c}$ in that order form a right handed system if $[\bar{a}\ \ \bar{b}\ \ \bar{c}] > 0$.



Vector Triple Product

It is defined for three vectors $\bar{a},\bar{b},\bar{c}$ as the vector $\bar{a}\times(\bar{b}\times\bar{c}) = (\bar{a}\cdot\bar{c})\bar{b} - (\bar{a}\cdot\bar{b})\bar{c}$

In general, $\bar{a}\times(\bar{b}\times\bar{c}) \neq (\bar{a}\times\bar{b})\times\bar{c}$.

$\bar{a}\times(\bar{b}\times\bar{c}) = (\bar{a}\times\bar{b})\times\bar{c}$ if some or all of $\bar{a}$, $\bar{b}$, $\bar{c}$ are zero vectors or $\bar{a}$ and $\bar{c}$ are collinear.

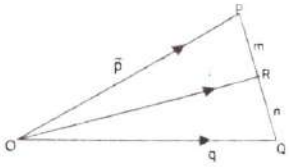
APPLICATIONS OF VECTORS

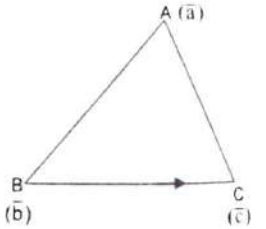
Section Formula

Let the position vector of a point P be $\bar{p}$ and that of another point Q be $\bar{q}$. If the line joining P and Q is divided by a point R in the ratio of m:n (internally or externally), then $\bar{r} = \frac{m\bar{q}+n\bar{p}}{m+n}$.

For internal division take m : n as positive and for external division take m : n as negative (i.e., either of m or n as negative).

If R is the mid-point of PQ, then $\bar{r} = \frac{\bar{p}+\bar{q}}{2}$. If A, B, C are the vertices of a triangle with position vectors $\bar{a}$, $\bar{b}$ $\bar{c}$ respectively, then the position vector of the centroid $G(\bar{g})$ of the triangle is given by $\bar{g} = \frac{\bar{a}+\bar{b}+\bar{c}}{3}$.

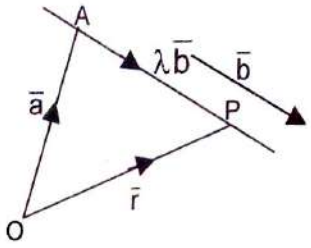




Equation of a Line in Vector Form

Following are the two most useful forms of the equation of a line.

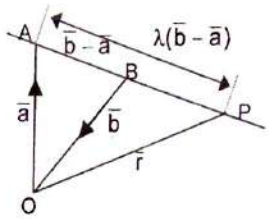
(i) Equation of the line passing through a given point A $(\bar{a})$ and parallel to a vector B $(\bar{b})$ is given by $\bar{r} = \bar{a} + \lambda\bar{b}$, where $\bar{r}$ is the position vector of any general point P on the line and λ is any real number.



(ii) Equation of the line passing through two given points A $(\bar{a})$ and B $(\bar{a})$ is given by $\bar{r} = \bar{a} + \lambda(\bar{b} - \bar{a})$.

For each particular value of λ, we get a particular point on the line. Each of the above equations can be written easily in Cartesian form also.

For example, in case (i), writing



$\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $\bar{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

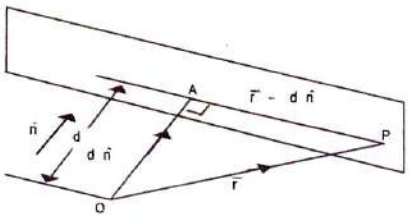
we get $\dfrac{x - a_1}{b_1} = \dfrac{y - a_2}{b_2} = \dfrac{z - a_3}{b_3} = \lambda$

$\Rightarrow x = a_1 + \lambda b_1,\ y = a_2 + \lambda b_2,\ z = a_3 + \lambda b_3.$

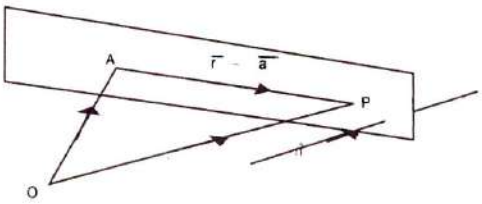
Equation of a Plane in Vector Form

Following are the four useful ways of specifying a plane.

(i) A plane at a perpendicular distance d from the origin and normal to a given direction $(\hat{n})$ has the equation $(\bar{r} - d\hat{n}) \cdot d\hat{n} = 0$ or $\bar{r} \cdot \hat{n} = d$ ($\hat{n}$ is a unit vector).



(ii) A plane passing through the point A $(\bar{a})$ and normal to $\hat{n}$ has the equation $(\bar{r} - \bar{a}) \cdot \hat{n} = 0$.



(iii) Parametric equation of the plane passing through A $(\bar{a})$ and parallel to the plane of vectors $(\bar{b})$ and $(\bar{c})$ is given by $\bar{r} = \bar{a} + \lambda\bar{b} + \mu\bar{c} \Rightarrow \bar{r} \cdot (\bar{b} \times \bar{c}) = [\bar{a}\ \bar{b}\ \bar{c}]$

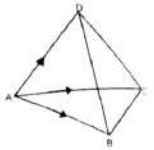
(iv) Parametric equation of the plane passing through A $(\bar{a})$, B $(\bar{b})$, C $(\bar{c})$ (A, B, C non-collinear) is given by $\bar{r} = (1 - \lambda - \mu)\bar{a} + \lambda\bar{b} + \mu\bar{c} \Rightarrow \bar{r} \cdot [\bar{b} \times \bar{c} + \bar{c} \times \bar{a} + \bar{a} \times \bar{b}] = [\bar{a}\ \ \bar{b}\ \ \bar{c}]$.

In Cartesian form, the equation of the plane assumes the form $Ax + By + Cz = D$. The vector normal to this plane is $A\hat{i} + B\hat{j} + C\hat{k}$ and the perpendicular distance of the plane from the origin is $\frac{|D|}{\sqrt{A^2 + B^2 + C^2}}$.

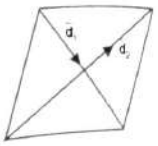
Angle between Two Planes

It is equal to the angle between their normal unit vectors $\hat{n}_1$ and $\hat{n}_2$, i.e. $\cos\theta = \hat{n}_1 \cdot \hat{n}_2$

Some Miscellaneous Results:



(i) Volume of the tetrahedron ABCD $= \frac{1}{6}[\overline{AB} \quad \overline{AC} \quad \overline{AD}]$.



(ii) Area of the quadrilateral with diagonals $\bar{d}_1$ and $\bar{d}_2 = \frac{1}{2}|\bar{d}_1 \times \bar{d}_2|$.

(iii) The internal bisector of angle between unit vectors $\hat{a}$ and $\hat{b}$ is along the vector $\hat{a} + \hat{b}$. The external bisector is along $\hat{a} - \hat{b}$. Equation of internal and external bisectors of the lines $\bar{r} = \bar{a} + \lambda\bar{b}_1$ and $\bar{r} = \bar{a} + \mu\bar{b}_2$ (intersecting at A($\bar{a}$)) are given by $\bar{r} = \bar{a} + t\left(\frac{\bar{b}_1}{|\bar{b}_1|} \pm \frac{\bar{b}}{|\bar{b}_2|}\right)$.

(iv) Reciprocal system of vectors: If $\bar{a}$, $\bar{b}$, $\bar{c}$ are three non-coplanar vectors, then a system of vectors $\bar{a}'$, $\bar{b}'$, $\bar{c}'$ defined by $\bar{a}' = \frac{\bar{b} \times \bar{c}}{[\bar{a} \quad \bar{b} \quad \bar{c}]}$, $\bar{b}' = \frac{\bar{c} \times \bar{a}}{[\bar{a} \quad \bar{b} \quad \bar{c}]}$, $\bar{c}' = \frac{\bar{a} \times \bar{b}}{[\bar{a} \quad \bar{b} \quad \bar{c}]}$ is called the reciprocal system of vectors because $\bar{a} \cdot \bar{a}' = \bar{b} \cdot \bar{b}' = \bar{c} \cdot \bar{c}' = 1$.

Further $[\bar{a}' \quad \bar{b}' \quad \bar{c}'] = \frac{1}{[a \quad b \quad c]}$.

The scalar product of any vector of one system with a vector of other system which does not correspond to it is zero i.e. $\bar{a} \cdot \bar{b}' = \bar{a} \cdot \bar{c}' = \bar{b} \cdot \bar{c}' = \bar{c} \cdot \bar{a}' = \cdots = 0$.

If $\bar{a}'$, $\bar{b}'$, $\bar{c}'$ is a reciprocal system to $\bar{a}$, $\bar{b}$, $\bar{c}$, then $\bar{a}$, $\bar{b}$, $\bar{c}$ is also reciprocal system to $\bar{a}'$, $\bar{b}'$, $\bar{c}'$.

ILLUSTRATIVE PROBLEMS

1. Show that the sum of the vectors represented by the sides $\overline{AB}$, $\overline{DC}$ of a quadrilateral ABCD is equivalent to the sum of the vectors represented by the diagonals $\overline{AC}$ and $\overline{DB}$.

Sol. From the triangular law of addition

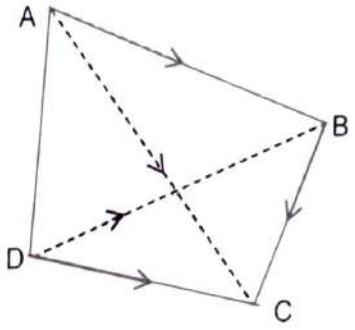
$\overline{AB} + \overline{BC} = \overline{AC}$ and

$\overline{DC} + \overline{CB} = \overline{DB}$

Since $\overline{BC} = -\overline{CB}$,

addition of these two equation gives,

$\overline{AB} + \overline{DC} = \overline{AC} + \overline{DB}$



2. Show that any two opposite edges of a regular tetrahedron are perpendicular.
Note: (A tetrahedron whose edges are all equal in length is called a regular tetrahedron).

Sol. Take one of the vertices of the tetrahedron as the initial point. Let it be O. Then the position vectors of the points A, B, C of the tetrahedron OABC are $\bar{a}$, $\bar{b}$, $\bar{c}$, where $|\bar{a}| = |\bar{b}| = |\bar{c}|$. Since all the edges are of equal length, all the four faces of the tetrahedron are equilateral triangles.

$\Rightarrow \bar{a}\cdot\bar{b} = \bar{a}\cdot\bar{c} = \bar{b}\cdot\bar{c}$

$\Rightarrow \bar{a}\cdot(\bar{b}-\bar{c}) = 0.$

Hence the opposite edges are perpendicular to each other.

3. ABC is a triangle. A line is drawn parallel to BC to meet AB and AC in D and E respectively. Prove that the median through A bisects DE.

Sol. Take the vertex A of the triangle ABC as the origin. Let $\bar{b}$ and $\bar{c}$ be the position vector of B and C. The mid point of BC has the position vector $= \dfrac{\bar{b}+\bar{c}}{2}$.

The equation of the median is $r = t\dfrac{\bar{b}+\bar{c}}{2}$.

Let D divide AB in the ratio $1 : \mu$

$\Rightarrow$ position vector of $D = \dfrac{\bar{b}}{1+\mu}$.

Let E divide AC in the ratio $1 : \lambda$

$\Rightarrow$ position vector $E = \dfrac{\bar{c}}{1+\lambda}$.

$\Rightarrow \lambda = \mu$.

Position vector of the mid-point of

$DE = \dfrac{\bar{b}+\bar{c}}{2(1+\mu)}$ which lies on the median.

Hence the median bisects DE.

4. **Show that the line of intersection of the planes $\bar{r} \cdot (\hat{i}+2\hat{j}+3\hat{k}) = 0$ and $\bar{r} \cdot (3\hat{i}+2\hat{j}+\hat{k}) = 0$ is equally inclined to $\hat{i}$ and $\hat{k}$. Also find the angle it makes with $\hat{j}$.**

Sol. The line of intersection of the two planes will be perpendicular to the normals to the planes. Hence it is parallel to the vector

$(\hat{i}+2\hat{j}+3\hat{k})\times(3\hat{i}+2\hat{j}+\hat{k}) = (-4\hat{i}+8\hat{j}-4\hat{k})$

Now $(-4\hat{i}+8\hat{j}-4\hat{k})\cdot\hat{i} = -4$ and $(-4\hat{i}+8\hat{j}-4\hat{k})\cdot\hat{k} = -4$

Hence the line is equally inclined to $\hat{i}$ and $\hat{k}$.

Also $\dfrac{(-4\hat{i}+8\hat{j}-4\hat{k})}{\sqrt{16+64+16}}\cdot\hat{j} = \dfrac{8}{\sqrt{96}} = \sqrt{\dfrac{2}{3}}$.

If θ is the required angle, then

$\cos\theta = \sqrt{\dfrac{2}{3}} \Rightarrow \theta = \cos^{-1}\sqrt{\dfrac{2}{3}}$.

5. **Show that if $\bar{a}$ and $\bar{b}$ are two non-zero vectors, then $\bar{a}$ and $\bar{b}$ make equal angles with $\bar{c}$ if $\bar{c} = \dfrac{|\bar{b}|}{|\bar{a}|+|\bar{b}|}\bar{a} + \dfrac{|\bar{a}|}{|\bar{a}|+|\bar{b}|}\bar{b}$.**

Sol. It is evident that $\bar{c}$ is along the bisector of the angle between the vectors $\bar{a}$ and $\bar{b}$

$\Rightarrow \bar{c} = \lambda\left(\dfrac{\bar{a}}{|\bar{a}|}+\dfrac{\bar{b}}{|\bar{b}|}\right) = \lambda\left(\dfrac{|\bar{b}|\bar{a}+|a|\bar{b}}{|\bar{a}||\bar{b}|}\right)$.

If we take $\lambda = \dfrac{|\bar{a}||\bar{b}|}{|\bar{a}|+|\bar{b}|}$,

Then $\bar{c} = \dfrac{|\bar{a}||\bar{b}|}{|\bar{a}|+|\bar{b}|}\left(\dfrac{|\bar{b}|\bar{a}+|\bar{a}|\bar{b}}{|\bar{a}||\bar{b}|}\right)$

$= \dfrac{|\bar{b}|}{|\bar{a}|+|\bar{b}|}\bar{a} + \dfrac{|\bar{a}|}{|\bar{a}|+|\bar{b}|}\bar{b}$.

SOLVED PROBLEMS

1. If the vectors $2a\hat{i}+b\hat{j}+c\hat{k}$, $b\hat{i}+c\hat{j}+2a\hat{k}$ and $c\hat{i}+2a\hat{j}+b\hat{k}$ are coplanar vectors, then the straight lines $ax + by + c = 0$ will always pass through
(A) (1, 2)
(B) (2, –1)
(C) (2, 1)
(D) (1, – 2)

Sol. **(C).** Scaler triple product = 0
$\Rightarrow 2a + b + c = 0$, hence pass through (2, 1).

2. Plane is parallel to the vectors $\hat{i}+\hat{j}+\hat{k}$ and $2\hat{k}$ and another plane is parallel to $\hat{i}+\hat{j}$ and $\hat{i}-\hat{k}$ then the acute angle between $4\hat{i}-\hat{j}$ and the line of intersection of the two plane is

(A) $\cos^{-1}\frac{2}{\sqrt{34}}$
(B) $\cos^{-1}\frac{3}{\sqrt{34}}$
(C) $\cos^{-1}\frac{4}{\sqrt{34}}$
(D) $\cos^{-1}\frac{5}{\sqrt{34}}$

Sol. **(B).** Normal vector to plane I and II are $2\hat{i}-2\hat{j}$, $-\hat{i}+\hat{j}-\hat{k}$ and vector parallel to line of intersection of the planes $= (2\hat{i}-2\hat{j})\times(-\hat{i}+\hat{j}+\hat{k}) = 2\hat{i}+2\hat{j}$

Required angle $= \cos^{-1}\left(\frac{(4\hat{i}-\hat{j})(2\hat{i}+2\hat{j})}{\sqrt{17}\sqrt{8}}\right) = \cos^{-1}\left(\frac{3}{\sqrt{34}}\right)$

3. If the p^{th}, q^{th}, r^{th} terms of a G.P. are the positive numbers a, b, c, then the angle between the vectors. $\log a^2\hat{i}+\log b^2\hat{j}+\log c^2\hat{k}$ and $(q-r)\hat{i}+(r-p)\hat{j}+(p-q)\hat{k}$ is

(A) $\frac{\pi}{3}$
(B) $\frac{\pi}{2}$
(C) $\sin^{-1}\left(\frac{1}{\sqrt{a^2+b^2+c^2}}\right)$
(D) none of these

Sol. **(B).** Let A be the 1st term and R the common ratio
$\therefore a = AR^{p-1}$, $b = AR^{q-1}$, $c = AR^{r-1} \Rightarrow \log a = \log A + (p-1)\log R$
$\log b = \log A + (q-1)\log R$
$\log c = \log A + (r-1)\log R$
If $\vec{x} = \log a^2\hat{i}+\log b^2\hat{j}+\log c^2\hat{k}$ and $\vec{y} = (q-r)\hat{i}+(r-p)\hat{j}+(p-q)\hat{k}$
$\therefore \vec{x}\cdot\vec{y} = \sum 2(\log a)(q-r) = 2\sum(\log A+(p-1)\log R)(q-r)$
$= 2\left[\log A(q-r+r-p+p-q)+\log R\{(p-1)(q-r)+(q-1)(r-p)+(r-1)(p-q)\}\right]$
$= 2[\log A\cdot 0+\log R\cdot 0] = 0$
$\therefore$ Angle between $\vec{x}$ and $\vec{y} = \frac{\pi}{2}$.

4. If in a triangle ABC, $\overrightarrow{AB} = \frac{\vec{u}}{|\vec{u}|}-\frac{\vec{v}}{|\vec{v}|}$ and $\overrightarrow{AC} = \frac{2\vec{u}}{|\vec{u}|}$, where $|\vec{u}| \neq |\vec{v}|$, then

(A) $1 + \cos 2A + \cos 2B + \cos 2C = 0$
(B) $1 + \cos 2A + \cos 2B + \cos 2C = 2$
(C) both A and B correct
(D) none of these

Sol. **(A).** $\overline{AB}+\overline{BC}=\overline{AC}$

$$\overline{BC}=\frac{2\bar{u}}{|\bar{u}|}-\frac{\bar{u}}{|\bar{u}|}+\frac{\bar{v}}{|\bar{v}|}=\frac{\bar{u}}{|\bar{u}|}+\frac{\bar{v}}{|\bar{v}|}.$$

$$\overline{AB}\cdot\overline{BC}=\left(\frac{\bar{u}}{|\bar{u}|}-\frac{\bar{v}}{|\bar{v}|}\right)\left(\frac{\bar{u}}{|\bar{u}|}+\frac{\bar{v}}{|\bar{v}|}\right)=(\hat{u}-\hat{v})\cdot(\hat{u}+\hat{v})=1-1=0.$$

$\angle B=90°$.

$\Rightarrow 1+\cos 2A+\cos 2B+\cos 2C=0.$

5. If A, B and C are three points with position vectors $\bar{a}$, $\bar{b}$, $\bar{c}$ respectively, then perpendicular distance of A from the line joining B and C is

(A) $\dfrac{\left|\bar{a}\times(\bar{b}\times\bar{c})\right|}{2\left|(\bar{b}-\bar{c})\right|}$

(B) $\dfrac{\left|\bar{a}\times\bar{b}+\bar{b}\times\bar{c}+\bar{c}\times\bar{a}\right|}{2\left|(\bar{b}-\bar{c})\right|}$

(C) $\dfrac{\left|\bar{a}\times\bar{b}+\bar{c}\times\bar{a}-\bar{b}\times\bar{c}\right|}{2\left|(\bar{b}-\bar{c})\right|}$

(D) none of these

Sol. **(D).** Area of $\Delta ABC=\dfrac{1}{2}BC\cdot AD=\dfrac{1}{2}\left|\overline{AB}\times\overline{AC}\right|$

$$AD=\frac{\left|\bar{a}\times\bar{b}+\bar{b}\times\bar{c}+\bar{c}\times\bar{a}\right|}{\left|(\bar{b}-\bar{c})\right|}.$$

6. Let $\vec{a}$ and $\vec{c}$ are unit vectors and $|\vec{b}|=4$ with $\vec{a}\times\vec{b}=2\vec{a}\times\vec{c}$. The angle between $\vec{a}$ and $\vec{c}$ is $\cos^{-1}\left(\dfrac{1}{4}\right)$. If $\vec{b}-2\vec{c}=\lambda\vec{a}$, then λ equals

(A) $\dfrac{1}{3}$

(B) $\dfrac{1}{4}$

(C) -4

(D) 3

Sol. **(C, D).** $\vec{a}\cdot\vec{c}=|\vec{a}|\ |\vec{c}|\cos\left(\cos^{-1}\dfrac{1}{4}\right)$

$\Rightarrow \vec{a}\cdot\vec{c}=\dfrac{1}{4}$ ... (1)

Taking dot product by $\vec{a}$, $\vec{b}$, $\vec{c}$, we have

$\vec{a}\cdot\vec{b}-2(\vec{a}\cdot\vec{c})=\lambda(\vec{a}\cdot\vec{a})$

$\Rightarrow \vec{a}\cdot\vec{b}-\dfrac{1}{2}=\lambda \Rightarrow \vec{a}\cdot\vec{b}=\lambda+\dfrac{1}{2}$ ... (2)

Similarly, $\vec{b}\cdot\vec{c}=8-\dfrac{\lambda^2}{2}-\dfrac{\lambda}{4}$ ... (3)

And $\vec{b}\cdot\vec{c}-2=\lambda(\vec{a}\cdot\vec{c})$ ... (4)

From (2), (3) and (4), we get $8-\dfrac{\lambda^2}{2}-\dfrac{\lambda}{4}-2=\lambda\left(\dfrac{1}{4}\right) \Rightarrow \lambda=3,-4.$

7. In ΔABC perpendiculars are drawn from A, B, C to meet BC, CA, AB at D, E, F. If the position vectors of D, E, F are $(\bar{x}, \bar{y}, \bar{z})$, then

(A) position vector of orthocentre of $\Delta ABC = \dfrac{\bar{x}EF + \bar{y}DF + \bar{z}DE}{EF + DF + DE}$

(B) $\left|DF(\bar{y} - \bar{x}) + DE(\bar{z} - \bar{x})\right| = 2R\cos B\cos C$, where R is the circumradius of ΔABC

(C) $\left|DF(\bar{y} - \bar{x}) + DE(\bar{z} - \bar{x})\right| = 2R\sin B\sin C$

(D) none of these

Sol. **(A, B).** DEF is the pedal triangle of ABC.

Orthocentre of ΔABC is incentre of ΔDEF.

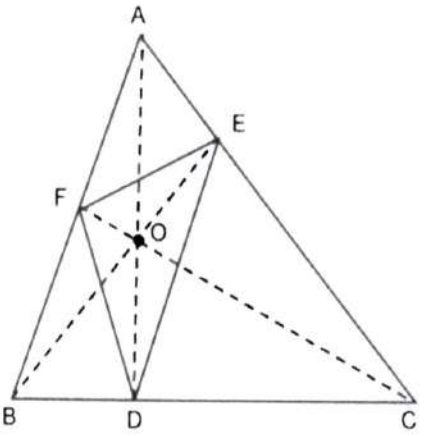
Position vector of orthocentre

$= \dfrac{\bar{x}EF + \bar{y}DF + \bar{z}DE}{EF + DF + DE}$

$OD = 2R\cos B\cos C$

$\left|\dfrac{\bar{x}EF + \bar{y}DF + \bar{z}DE}{EF + DF + DE} - \bar{x}\right| = 2R\cos B\cos C$

$\left|DF(\bar{y} - \bar{x}) + DE(\bar{z} - \bar{x})\right| = 2R\cos B\cos C$.



8. Let $\bar{r}$ be a unit vector satisfying $\bar{r} \times \bar{a} = \bar{b}$, where $|\bar{a}| = \sqrt{3}$ and $|\bar{b}| = \sqrt{2}$, then

(A) $\bar{r} = \dfrac{2}{3}\left(\bar{a} + \bar{a} \times \bar{b}\right)$ (B) $\bar{r} = \dfrac{1}{3}\left(\bar{a} + \bar{a} \times \bar{b}\right)$

(C) $\bar{r} = \dfrac{2}{3}\left(\bar{a} - \bar{a} \times \bar{b}\right)$ (D) $\bar{r} = \dfrac{1}{3}\left(-\bar{a} + \bar{a} \times \bar{b}\right)$

Sol. **(B, D).** $\bar{a} \times (\bar{r} \times \bar{a}) = \bar{a} \times \bar{b}$

$3\bar{r} - (\bar{a} \cdot \bar{r})\bar{a} = \bar{a} \times \bar{b}$

$|\bar{r} \times \bar{a}| = |\bar{b}|$

$|\bar{r}| \times |\bar{a}| \sin\theta = |\bar{b}|$

$|\bar{r}|^2 3\sin^2\theta = 2$

$|\bar{r}|^2 (1 - \cos^2\theta) = \dfrac{2}{3} \Rightarrow |\bar{r}|^2 - \dfrac{2}{3} = |\bar{r}|^2 \cos^2\theta$

$\dfrac{1}{3} = \cos^2\theta \Rightarrow 3\bar{r} - \bar{a} = \bar{a} \times \bar{b}$

$\bar{r} = \dfrac{1}{3}\left(\bar{a} \times \bar{b} \pm \bar{a}\right)$.

9. Let $\vec{\alpha} = a\hat{i} + b\hat{j} + c\hat{k}$, $\vec{\beta} = b\hat{i} + c\hat{j} + a\hat{k}$ and $\vec{\gamma} = c\hat{i} + a\hat{j} + b\hat{k}$ be three coplaner vectors with $a \neq b$, and $\vec{v} = \hat{i} + \hat{j} + \hat{k}$. Then $\vec{v}$ is perpendicular to

(A) $\vec{\alpha}$ B) $\vec{\beta}$

(C) $\vec{\gamma}$ (D) none of these.

Sol. **(A, B, C).** Given vectors are coplanar $\Rightarrow \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$

$\Rightarrow a^3 + b^3 + c^3 - 3abc = 0 = (a+b+c)\left[(a-b)^2 + (b-c)^2 + (c-a)^2\right] = 0$

$\Rightarrow a + b + c = 0$ as $a \neq b$

$\Rightarrow \vec{v}\cdot\left(a\hat{i} + b\hat{j} + c\hat{k}\right) = \vec{v}\cdot\left(b\hat{i} + c\hat{j} + a\hat{k}\right) = \vec{v}\cdot\left(c\hat{i} + a\hat{j} + b\hat{k}\right) = 0.$

10. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non-coplanar vectors and $\vec{d}$ be a non-zero vector which is perpendicular to $\left(\vec{a} + \vec{b} + \vec{c}\right)$ and is represented as $\vec{d} = x\left(\vec{a} \times \vec{b}\right) + y\left(\vec{b} \times \vec{c}\right) + z\left(\vec{c} \times \vec{a}\right)$. Then

(A) $x^3 + y^3 + z^3 = 3xyz$ (B) $xy + yz + xz \leq 0$

(C) $x = y = z$ (D) $x^2 + y^2 + z^2 = xy + yz + zx$

Sol. **(A, B).** $\vec{d} \cdot (\vec{a} + \vec{b} + \vec{c}) = \left[x\left(\vec{a} \times \vec{b}\right) + y\left(\vec{b} \times \vec{c}\right) + z\left(\vec{c} \times \vec{a}\right)\right] \cdot (\vec{a} + \vec{b} + \vec{c}) = 0$

$\Rightarrow [abc](x + y + z) = 0$

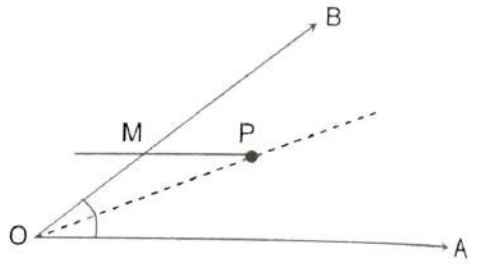
$\Rightarrow x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx).$

Here $x + y + z = 0$, $x^3 + y^3 + z^3 = 3xyz$ and $xy + yz + xz \leq 0$.

Comprehension

Equations of Bisectors of the Angle between Two Vectors:

Let two unit vectors along two lines $\overrightarrow{OA}$ and $\overrightarrow{OB}$ be $\hat{a}$ and $\hat{b}$ respectively. Take their point of intersection as the origin and let P be any point on the bisector of angle between the lines OA and OB. Draw PM parallel to AO cutting OB at M.



$\angle AOP = \angle POM = \angle OPM$ and hence OM = PM.

But $\overrightarrow{OM} = t\hat{b}$ and $\overrightarrow{MP} = t\hat{a}$

(since $\overrightarrow{OM} \parallel \hat{b}$ and $\overrightarrow{MP} \parallel \hat{a}$ and their magnitudes are same).

Then $\overrightarrow{OP} = \vec{r} = \overrightarrow{OM} + \overrightarrow{MP} = t\left(\hat{b} + \hat{a}\right)$ ...(i)

For external bisector OP', the angle between OB and OA is the same as the internal bisector of the angle between the unit vectors along them being $-\hat{b}$ and $\hat{a}$ and hence the equation of $\overrightarrow{OP'}$ be

$\overrightarrow{OP'} = \vec{r}' = t\left(\hat{a} - \hat{b}\right)$ ...(ii)

For any two vectors $\vec{a}$ and $\vec{b}$ the equations (i) and (ii) reduce to $\vec{r} = t\left(\frac{\vec{a}}{|\vec{a}|} \pm \frac{\vec{b}}{|\vec{b}|}\right)$.

Answer the following questions:

11. A vector $\vec{c}$, directed along the internal bisector of the angle between the vectors $\vec{a} = 7\hat{i} - 4\hat{j} - 4\hat{k}$ and $\vec{b} = -2\hat{i} - \hat{j} + 2\hat{k}$ with $|\vec{c}| = 5\sqrt{6}$, is

(A) $\frac{5}{3}\left(\hat{i} - 7\hat{j} + 2\hat{k}\right)$ (B) $\frac{5}{3}\left(5\hat{i} + 5\hat{j} + 2\hat{k}\right)$

(C) $\frac{5}{3}\left(\hat{i} + 7\hat{j} + 2\hat{k}\right)$ (D) $\frac{5}{3}\left(-5\hat{i} + 5\hat{j} + 2\hat{k}\right)$

Sol. **(A).** The vector along the internal bisector of the given angle is

$$\vec{c} = t\left(\frac{7\hat{i}-4\hat{j}-4\hat{k}}{\sqrt{81}} + \frac{-2\hat{i}-\hat{j}+2\hat{k}}{\sqrt{9}}\right) = t\left(\frac{\hat{i}-7\hat{j}+2\hat{k}}{9}\right)$$

Since $|\vec{c}| = 5\sqrt{6}, 5\sqrt{6} = t\frac{\sqrt{54}}{9}$

$\Rightarrow t = 15.$

12. Let ABC be a triangle and $\vec{a}$, $\vec{b}$, $\vec{c}$ be the position vectors of the point A, B, C respectively. External bisectors of $\angle B$ and $\angle C$ meet at P with the sides of the triangle as a, b, c, the position vectors of P becomes

(A) $\frac{(-b)\vec{b}+(-c)\vec{c}}{(b+c)}$

(B) $\frac{a\vec{a}+(-b)\vec{b}+(-c)\vec{c}}{(a-b-c)}$

(C) $\left(\frac{\vec{a}+\vec{b}+\vec{c}}{3}\right)(abc)$

(D) $\frac{a\vec{a}+b\vec{b}+c\vec{c}}{(\vec{a}+\vec{b}+\vec{c})}$

Sol. **(B).** Vectors along AB, BC and CA are $\vec{b}-\vec{a}$, $\vec{c}-\vec{b}$ and $\vec{a}-\vec{c}$ respectively. Hence the bisectors of the angular B and C respectively are

$\vec{r} = \vec{b} + \mu\left(\frac{\vec{b}-\vec{a}}{c} + \frac{\vec{c}-\vec{b}}{a}\right)$ and $\vec{r} = \vec{c} + t\left(\frac{\vec{b}-\vec{c}}{a} + \frac{\vec{c}-\vec{a}}{b}\right)$ so that for the point P

$$\vec{b} + \mu\left(\frac{\vec{b}-\vec{a}}{c} + \frac{\vec{c}-\vec{b}}{a}\right) = \vec{c} + t\left(\frac{\vec{b}-\vec{c}}{a} + \frac{\vec{c}-\vec{a}}{b}\right)$$

$\Rightarrow t = \frac{b\mu}{c}$ and $\mu = \frac{ac}{b+c-a}$ $\Rightarrow$ the p.v. of P is $\frac{b\vec{b}+c\vec{c}-a\vec{a}}{b+c-a}$.

13. If the interior and exterior bisectors of the angle A of a triangle ABC meet the base BC at D and E, then

(A) $2BC = BD + BE$

(B) $BC^2 = BD \times BE$

(C) $\frac{2}{BC} = \frac{1}{BD} + \frac{1}{BE}$

(D) None of these.

Sol. **(C).** Let A be the initial point and the p.v. of B and C be $\vec{b}$ and $\vec{c}$ respectively. Hence, let AB = c and AC = b. The internal bisector is $\vec{r} = t\left(\frac{\vec{b}}{c} + \frac{\vec{c}}{b}\right)$ and the equation of the line BC is $\vec{r} = \vec{b} + k(\vec{c}-\vec{b})$ and hence the p.v. of D is $\frac{b\vec{b}+c\vec{c}}{b+c}$

$$\Rightarrow \overrightarrow{BD} = \frac{b\vec{b}+c\vec{c}}{b+c} - \vec{b} = \frac{c(\vec{c}-\vec{b})}{b+c}$$

$$\Rightarrow BD = \frac{c|\vec{c}-\vec{b}|}{b+c} = \frac{ca}{b+c}.$$

Similarly $BE = \frac{c|\vec{c}-\vec{b}|}{c-b} = \frac{ca}{c-b}$

and hence $\frac{1}{BE} + \frac{1}{BD} = \frac{c-b}{ac} + \frac{c+b}{ac} = \frac{2c}{ac} = \frac{2}{a} = \frac{2}{BC}$.

Assertion Reason

14. Let B_1, C_1 and D_1 are points on AB, AC and AD of the parallelogram ABCD, such that $\overrightarrow{AB_1} = k_1\overrightarrow{AB}$, $\overrightarrow{AC_1} = k_2\overrightarrow{AC}$ and $\overrightarrow{AD_1} = k_3\overrightarrow{AD}$, where k_1, k_2 and k_3 are scalar.

Statement 1: k_1, $2k_2$ and k_3 are in harmonic progression. If B_1, C_1 and D_1 are collinear.

because

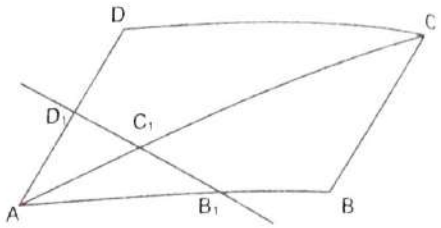
Statement 2: $\dfrac{\overrightarrow{AB_1}}{k_1} + \dfrac{\overrightarrow{AD_1}}{k_3} = \dfrac{\overrightarrow{AC_1}}{k_2}$.

(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement -1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement -1 is True, Statement -2 is False

(D) Statement -1 is False, Statement -2 is True



Sol. **(A).** Equation of line B_1D_1

$\vec{r} = \overrightarrow{AB_1} + \lambda\left(\overrightarrow{B_1D_1}\right)$

$\Rightarrow \vec{r} = \overrightarrow{AB_1} + \lambda\left(\overrightarrow{AD_1} - \overrightarrow{AB_1}\right)$.

If points B_1, C_1 and D_1 are collinear, then

$\overrightarrow{AC_1} = \overrightarrow{AB_1} + \lambda\left(\overrightarrow{AD_1} - \overrightarrow{AB_1}\right)$

Since, $\dfrac{\overrightarrow{AB_1}}{k_1} + \dfrac{\overrightarrow{AD_1}}{k_3} = \dfrac{\overrightarrow{AC_1}}{k_2}$ $\left\{\because \overrightarrow{AB} + \overrightarrow{AD} = \overrightarrow{AC}\right\}$

$\Rightarrow \dfrac{k_2}{k_1}\overrightarrow{AB_1} + \dfrac{k_2}{k_3}\overrightarrow{AD_1} = (1-\lambda)\overrightarrow{AB_1} + \lambda\overrightarrow{AD_1}$

Since, $\overrightarrow{AB}$ and $\overrightarrow{AD}$ form linearly independent system of vectors

$\Rightarrow \lambda = \dfrac{k_2}{k_3}$ and $1-\lambda = \dfrac{k_2}{k_1}$

$\Rightarrow 1 - \dfrac{k_2}{k_3} = \dfrac{k_2}{k_1} \Rightarrow \dfrac{1}{k_2} = \dfrac{1}{k_1} + \dfrac{1}{k_3}$.

15. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be the position vectors of the vertices A, B, C of a triangle ABC respectively

Statement –I: If for some non-zero vector $\vec{r}$, $\vec{r}\cdot\vec{a} = \vec{r}\cdot\vec{b} = \vec{r}\cdot\vec{c} = \dfrac{1}{2}$ then the area of the triangle ABC is $\left[\vec{a}\ \vec{b}\ \vec{c}\right]|\vec{r}|$.

because

Statement –II: The area of the triangle ABC is given by $\dfrac{1}{2}\left|\vec{a}\times\vec{b} + \vec{b}\times\vec{c} + \vec{c}\times\vec{a}\right|$.

Sol. **(A)**. Let $\vec{r} = x(\vec{a}\times\vec{b}) + y(\vec{b}\times\vec{c}) + z(\vec{c}\times\vec{a})$

Then $x = \dfrac{\vec{r}\cdot\vec{c}}{\left[\vec{a}\ \vec{b}\ \vec{c}\right]}$, $y = \dfrac{\vec{r}\cdot\vec{a}}{\left[\vec{a}\ \vec{b}\ \vec{c}\right]}$ and $z = \dfrac{\vec{r}\cdot\vec{b}}{\left[\vec{a}\ \vec{b}\ \vec{c}\right]}$

$\therefore \left[\vec{a}\ \vec{b}\ \vec{c}\right]\vec{r} = \dfrac{1}{2}\left[\vec{a}\times\vec{b} + \vec{b}\times\vec{c} + \vec{c}\times\vec{a}\right]$

$\therefore$ Area of $\Delta ABC = \dfrac{1}{2}\left|\vec{a}\times\vec{b} + \vec{b}\times\vec{c} + \vec{c}\times\vec{a}\right| = \left|\left[\vec{a}\ \vec{b}\ \vec{c}\right]\right||\vec{r}|$.

16. Let $\vec{a}$, $\vec{b}$, $\vec{c}$ are the three vectors such that $|\vec{a}| = |\vec{b}| = |\vec{c}| = 2$ and angle between $\vec{a}$ and $\vec{b}$ is $\pi/3$, $\vec{b}$ and $\vec{c}$ is $\pi/3$ and $\vec{a}$ and $\vec{c}$ $\pi/3$.

	Column – I		Column – II
(A)	If $\vec{a}$, $\vec{b}$, $\vec{c}$ represents adjacent edges of parallelepiped then its volume is	(p)	$\frac{2\sqrt{2}}{\sqrt{3}}$
(B)	If $\vec{a}$, $\vec{b}$, $\vec{c}$ represents adjacent edges of parallelepiped then its height is	(q)	$\frac{2\sqrt{2}}{3}$
(C)	If $\vec{a}$, $\vec{b}$, $\vec{c}$ represents adjacent edges of tetrahedron then its volume is	(r)	$4\sqrt{2}$
(D)	If $\vec{a}$, $\vec{b}$, $\vec{c}$ represents adjacent edges of tetrahedron then its height is	(s)	$\sqrt{\frac{2}{3}}$
		(t)	$2\sqrt{2}$

Sol. **(A) → (r); (B) → (p); (C) → (q); (D) → (p)**

Let $\vec{c} = \lambda(\vec{a} + \vec{b}) + \mu(\vec{a} + \vec{b})$

Taking dot product with $\vec{a}$, we get $2 = \lambda[4 + 2] \Rightarrow \lambda = \frac{1}{3}$

Squaring $4 = \lambda^2[4 + 4 + 4] = \mu^2 \cdot 12 \Rightarrow \mu^2 = \frac{2}{9} \Rightarrow$ taking dot product with $(\vec{a} \times \vec{b})$

$\left[(\vec{a} \times \vec{b}) \cdot \vec{c}\right] = \mu(12) = \frac{\sqrt{2}}{3} \times 12 = 4\sqrt{2}$.

Answer Q.17, Q.18 and Q.19 by appropriately matching the information given in the three columns of the following table.

Column-I: contains three point with p.v. $A(\vec{a}), B(\vec{b}), C(\vec{c})$, $\vec{a}, \vec{b}, \vec{c}$ and fourth point origin forms a tetrahedron.

Column-II: contains volume of tetrahedron and Column-III: contains position vector of foot of perpendicular from origin on face ABC

Column–I	Column–II	Column–III
(I) $\vec{a} = \hat{i} + \hat{j}$ $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ $\vec{c} = 2\hat{i} + \hat{j}$	(i) $\frac{5}{6}$	(P) $\frac{2\hat{i} - \hat{j} + 2\hat{k}}{9}$
(II) $\vec{a} = \hat{i} - \hat{j}$ $\vec{b} = \hat{i} + \hat{j} - \hat{k}$ $\vec{c} = 2\hat{i} - \hat{j} + \hat{k}$	(ii) $\frac{1}{3}$	(Q) $\frac{\hat{j} + 2\hat{k}}{5}$
(III) $\vec{a} = \hat{i} + \hat{j}$ $\vec{b} = \hat{i} - \hat{j} - \hat{k}$ $\vec{c} = 2\hat{i} + \hat{j} + \hat{k}$	(iii) $\frac{1}{6}$	(R) $\frac{5}{21}(4\hat{i} - \hat{j} - 2\hat{k})$
(IV) $\vec{a} = \hat{i} - \hat{j}$ $\vec{b} = \hat{i} + \hat{j} - \hat{k}$ $\vec{c} = 2\hat{i} - \hat{j} + \hat{k}$	(iv) $\frac{1}{2}$	(S) $\frac{2\hat{i} - \hat{j} - 2\hat{k}}{3}$

17. Which of the following combination is **CORRECT**?
(A) (I) (ii) (P) (B) (III) (iii) (S)
(C) (I) (iii) (Q) (D) (III) (ii) (R)

Sol. **(C)**

18 Which of the following combination is **CORRECT**?
(A) (II) (iv) (P) (B) (III) (ii) (P)
(C) (III) (iii) (R) (D) (II) (iv) (S)

Sol. **(D)**

19. Which of the following combination is **CORRECT**?
(A) (IV) (i) (R) (B) (IV) (i) (P)
(C) (II) (iv) (R) (D) (II) (iii) (R)

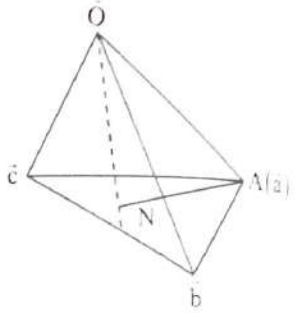
Sol. **(A)**

17 -19. Volume of tetrahedron $= \frac{1}{6}\left[\vec{a}\ \vec{b}\ \vec{c}\right]$

Equation of line passes through origin and perpendicular to plane ABC is $\vec{r} = \lambda\vec{n}$ where $\left[\vec{n} = \vec{a}\times\vec{b} + \vec{b}\times\vec{c} + \vec{c}\times\vec{a}\right]$

Then vector $\overrightarrow{AN} \perp \vec{n} \Rightarrow (\lambda\vec{n} - \vec{a})\cdot\vec{n} = 0$

$\therefore \lambda = \frac{\vec{a}\cdot\vec{n}}{|\vec{n}|^2}$



20. If A, B, C and D are four points in space, then $\left|\overline{AB}\times\overline{CD} + \overline{BC}\times\overline{AD} + \overline{CA}\times\overline{BD}\right| = k \cdot$ area of ΔABC, where k is equal to _____

Sol. **(4.00)**

$\overline{OA} = \bar{a},\ \overline{OB} = \bar{b},\ \overline{OC} = \bar{c}$

and $\overline{OD} = \bar{d}$

$\left|\overline{AB}\times\overline{CD} + \overline{BC}\times\overline{AD} + \overline{CA}\times\overline{BD}\right|$

$= 2\left|\bar{a}\times\bar{b} + \bar{b}\times\bar{c} + \bar{c}\times\bar{a}\right|$

$= 4\cdot\frac{1}{2}\left|\bar{a}\times\bar{b} + \bar{b}\times\bar{c} + \bar{c}\times\bar{a}\right|$

$= 4 \cdot$ area of ΔABC

$\Rightarrow k = 4$

21. If $4\bar{a} + 5\bar{b} + 9\bar{c} = 0$, then $(\bar{a}\times\bar{b})\times\left[(\bar{b}\times\bar{c})\times(\bar{c}\times\bar{a})\right]$ is equal to _____

Sol. $(\bar{0})$

$4\bar{a} + 5\bar{b} + 9\bar{c} = 0$

$\Rightarrow$ vectors $\bar{a}$, $\bar{b}$ and $\bar{c}$ are coplanar

$\Rightarrow \bar{b}\times\bar{c}$ and $\bar{c}\times\bar{a}$ are collinear

$\Rightarrow (\bar{b}\times\bar{c})\times(\bar{c}\times\bar{a}) = \bar{0}$

22. Let $A\left(2\hat{i}+3\hat{j}+5\hat{k}\right)$, $B\left(-\hat{i}+3\hat{j}+2\hat{k}\right)$ and $C\left(\lambda\hat{i}+5\hat{j}+\mu\hat{k}\right)$ are vertices of a triangle and its median through A is equally inclined to the positive directions of the axes. The value of $3\lambda - \mu$ is k, then $\frac{k}{5}$ is equal to _____

Sol. **(2.20)**

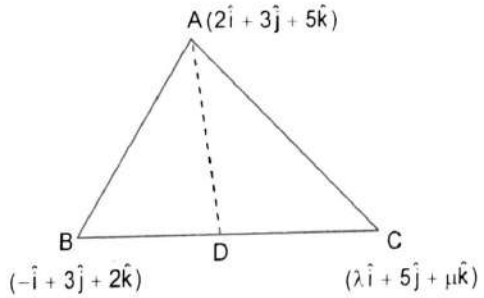
P.V. of D $= \frac{\lambda-1}{2}\hat{i}+4\hat{j}+\frac{\mu+2}{2}\hat{k}$

D.R. of AD $= \frac{\lambda-5}{2}, 1, \frac{\mu-8}{2}$.

But direction cosines of AD should be

$\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \Rightarrow \frac{\lambda-5}{2} = 1 = \frac{\mu-8}{2}$

$\lambda = 7, \mu = 10 \Rightarrow 3\lambda - \mu = 11$



23. The position vectors of the vertices of a triangle are $3\hat{i}+4\hat{j}+5\hat{k}$, $\hat{i}+7\hat{k}$ and $5\hat{i}+5\hat{j}$. The distance between the circumcentre and orthocentre is $\sqrt{k}$. Then k is equal to _____

Sol. **(306.00)**

Let $a = 3\hat{i}+4\hat{j}+5\hat{k}$; $b = \hat{i}+7\hat{k}$

and $c = 5\hat{i}+5\hat{j}$

then $|a| = |b| = |c|$

So, origin is the circumcentre and the orthocentre is

$a + b + c = 9\hat{i}+9\hat{j}+12\hat{k}$

$\therefore$ The distance between the circumcentre and orthocentre is

$|a + b + c| = \sqrt{9^2+9^2+12^2} = \sqrt{306}$

So, $k = 306$

TRIGONOMETRIC RATIOS AND IDENTITIES

BASIC FORMULAE

- $\cos^2 A + \sin^2 A = 1 \Rightarrow \cos^2 A = 1 - \sin^2 A$ or $\sin^2 A = 1 - \cos^2 A$
- $1 + \tan^2 A = \sec^2 A \Rightarrow \sec^2 A - \tan^2 A = 1$
- $\cot^2 A + 1 = \operatorname{cosec}^2 A \Rightarrow \operatorname{cosec}^2 A - \cot^2 A = 1$
- $\tan A = \dfrac{\sin A}{\cos A}$ and $\cot A = \dfrac{\cos A}{\sin A}$
- Fundamental inequalities: For $0 < A < \dfrac{\pi}{2}$, $0 < \cos A < \dfrac{\sin A}{A} < \dfrac{1}{\cos A}$.
- It is possible to express trigonometrical ratios in terms of any one of them

$$\text{e.g. } \sin\theta = \frac{1}{\sqrt{1+\cos^2\theta}}, \quad \cos\theta = \frac{\cot\theta}{\sqrt{1+\cot^2\theta}}, \quad \tan\theta = \frac{1}{\cot\theta}$$

$$\operatorname{cosec}\theta = \sqrt{1+\cot^2\theta}, \quad \sec\theta = \frac{\sqrt{1+\cot^2\theta}}{\cot\theta}$$

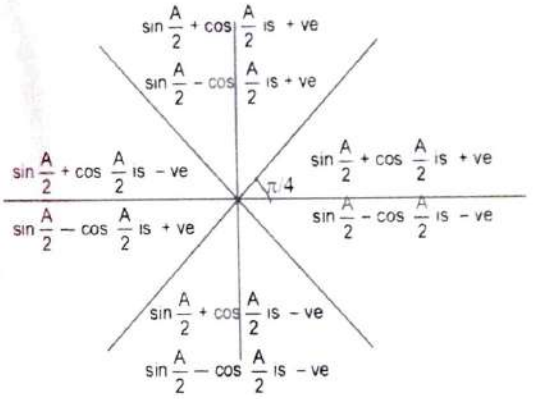
- $\sin(A + B) = \sin A \cos B + \cos A \sin B$
- $\sin(A - B) = \sin A \cos B - \cos A \sin B$
- $\cos(A + B) = \cos A \cos B - \sin A \sin B$
- $\cos(A - B) = \cos A \cos B + \sin A \sin B$
- $\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$
- $\tan(A-B) = \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$
- $\sin(A + B)\sin(A - B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A.$
- $\cos(A + B)\cos(A - B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A.$
- $\sin 2A = 2\sin A \cos A = \dfrac{2\tan A}{1 + \tan^2 A}$
- $\cos 2A = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1 = \dfrac{1 - \tan^2 A}{1 + \tan^2 A}$
- $\tan 2A = \dfrac{2\tan A}{1 - \tan^2 A}$
- $\sin 3A = 3\sin A - 4\sin^3 A = 4\sin(60^\circ - A)\sin A \sin(60^\circ + A)$
- $\cos 3A = 4\cos^3 A - 3\cos A = 4\cos(60^\circ - A)\cos A \cos(60^\circ + A)$
- $\tan 3A = \dfrac{3\tan A - \tan^3 A}{1 - 3\tan^2 A} = \tan(60^\circ - A)\tan A \tan(60^\circ + A)$
- $\sin A + \sin B = 2\sin\dfrac{A+B}{2}\cos\dfrac{A-B}{2}$
- $\sin A - \sin B = 2\sin\dfrac{A-B}{2}\cos\dfrac{A+B}{2}$

- $\cos A + \cos B = 2\cos\frac{A-B}{2}\cos\frac{A+B}{2}$
- $\cos A - \cos B = 2\sin\frac{B-A}{2}\sin\frac{A+B}{2}$ (here notice (B – A) !)
- $\tan A + \tan B = \frac{\sin(A+B)}{\cos A.\cos B}$
- $2\sin A\cos B = \sin(A+B) + \sin(A-B)$
- $2\cos A\sin B = \sin(A+B) - \sin(A-B)$
- $2\cos A\cos B = \cos(A+B) + \cos(A-B)$
- $2\sin A\sin B = \cos(A-B) - \cos(A+B)$
- $\left|\sin\frac{A}{2} + \cos\frac{A}{2}\right| = \sqrt{1+\sin A}$.

  or $\sin\frac{A}{2} + \cos\frac{A}{2} = \pm\sqrt{1+\sin A}\begin{cases} + & \text{if } 2n\pi - \frac{\pi}{4} \le \frac{A}{2} \le 2n\pi + \frac{3\pi}{4} \\ - & \text{otherwise} \end{cases}$
- $\left|\sin\frac{A}{2} - \cos\frac{A}{2}\right| = \sqrt{1-\sin A}$

  or $\sin\frac{A}{2} - \cos\frac{A}{2} = \pm\sqrt{1-\sin A}\begin{cases} + & \text{if } 2n\pi + \frac{\pi}{4} \le \frac{A}{2} \le 2n\pi + \frac{5\pi}{4} \\ - & \text{otherwise} \end{cases}$
- $\tan\frac{A}{2} = \frac{\pm\sqrt{\tan^2 A + 1} - 1}{\tan A}$

  The ambiguities of signs are removed by locating the quadrant in which $\frac{A}{2}$ lies or you can follow the given figure.



- $|a\cos A + b\sin A| \le \sqrt{a^2+b^2}$

  Also $\cos A \pm \sin A$

  $= \sqrt{2}\sin\left(\frac{\pi}{4} \pm A\right) = \sqrt{2}\cos\left(A \mp \frac{\pi}{4}\right)$.

IDENTITIES

For any Angles A, B, C

$\sin(A+B+C) = \sin A\cos B\cos C + \cos A\sin B\cos C + \cos A\cos B\sin C - \sin A\sin B\sin C$

$\cos(A+B+C) = \cos A\cos B\cos C - \cos A\sin B\sin C - \sin A\cos B\sin C - \sin A\sin B\cos C$

$$\tan(A+B+C) = \frac{\tan A + \tan B + \tan C - \tan A\tan B\tan C}{1 - \tan A\tan B - \tan B\tan C - \tan A\tan C};$$

$$\cot(A+B+C) = \frac{\cot A\cot B\cot C - \cot A - \cot B - \cot C}{\cot A\cot B + \cot B\cot C + \cot A\cot C - 1}$$

If A, B, C are Angles of a Triangle (or A + B + C = π)

- $\sin A\cos B\cos C + \cos A\sin B\cos C + \cos A\cos B\sin C = \sin A\sin B\sin C$
- $\cos A\sin B\sin C + \sin A\cos B\sin C + \sin A\sin B\cos C = 1 + \cos A\cos B\cos C$

- $\tan A + \tan B + \tan C = \tan A \tan B \tan C$
- $\cot B \cot C + \cot C \cot A + \cot A \cot B = 1$
- $\tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2} + \tan\frac{A}{2}\tan\frac{B}{2} = 1$
- $\cot\frac{A}{2} + \cot\frac{B}{2} + \cot\frac{C}{2} = \cot\frac{A}{2}\cot\frac{B}{2}\cot\frac{C}{2}$
- $\sin 2A + \sin 2B + \sin 2C = 4\sin A \sin B \sin C$
- $\cos 2A + \cos 2B + \cos 2C = -1 - 4\cos A \cos B \cos C$
- $\cos^2 A + \cos^2 B + \cos^2 C = 1 - 2\cos A \cos B \cos C$
- $\sin A + \sin B + \sin C = 4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$
- $\cos A + \cos B + \cos C = 1 + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

ILLUSTRATIVE PROBLEMS

1. If A, B and C are the angles of a triangle, show that $\tan^2\frac{A}{2} + \tan^2\frac{B}{2} + \tan^2\frac{C}{2} \geq 1$.

Sol. We have $\frac{A}{2} + \frac{B}{2} + \frac{C}{2} = \frac{\pi}{2}$, so that $\tan\left(\frac{A}{2} + \frac{B}{2}\right) = \tan\left(\frac{\pi}{2} - \frac{C}{2}\right)$

$$\Rightarrow \frac{\tan\frac{A}{2} + \tan\frac{B}{2}}{1 - \tan\frac{A}{2}\tan\frac{B}{2}} = \frac{1}{\tan\frac{C}{2}} \Rightarrow \sum \tan\frac{A}{2}\tan\frac{B}{2} = 1$$

$$\Rightarrow \tan^2\frac{A}{2} + \tan^2\frac{B}{2} + \tan^2\frac{C}{2} - 1 = \frac{1}{2}\left[\sum 2\tan^2\frac{A}{2} - \sum 2\tan\frac{A}{2}\tan\frac{B}{2}\right]$$

$$= \frac{1}{2}\left[\left(\tan\frac{A}{2} - \tan\frac{B}{2}\right)^2 + \left(\tan\frac{B}{2} - \tan\frac{C}{2}\right)^2 + \left(\tan\frac{C}{2} - \tan\frac{A}{2}\right)^2\right] \geq 0.$$

2. If a right angle is divided in three parts α, β and γ, prove that for all possible divisions $\tan\alpha + \tan\beta + \tan\gamma > 1 + \tan\alpha \tan\beta \tan\gamma$.

Sol. $(\tan\alpha - \tan\beta)^2 + (\tan\beta - \tan\gamma)^2 + (\tan\gamma - \tan\alpha)^2 \geq 0$

$\Rightarrow \tan^2\alpha + \tan^2\beta + \tan^2\gamma - \tan\alpha\tan\beta - \tan\beta\tan\gamma - \tan\gamma\tan\alpha \geq 0$

$\Rightarrow \tan^2\alpha + \tan^2\beta + \tan^2\gamma \geq 1$, since $\sum \tan\alpha\tan\beta = 1$.

$\Rightarrow \left(\sum \tan\alpha\right)^2 \geq 3 \Rightarrow \sum \tan\alpha \geq \sqrt{3}$.

If we prove that $1 + \tan\alpha\tan\beta\tan\gamma < \sqrt{3}$ the result is obvious.

Now, $(\tan\alpha\tan\beta\tan\gamma)^{1/3} \geq \dfrac{3}{\frac{1}{\tan\alpha} + \frac{1}{\tan\beta} + \frac{1}{\tan\gamma}}$ [G.M. ≥ H.M.]

$$\Rightarrow (\tan\alpha\tan\beta\tan\gamma)^{1/3} \geq \frac{3\tan\alpha\tan\beta\tan\gamma}{\tan\alpha\tan\beta + \tan\beta\tan\gamma + \tan\gamma\tan\alpha}$$

$$\Rightarrow (\tan\alpha\tan\beta\tan\gamma)^{2/3} \leq \frac{1}{3} \Rightarrow 1 + \tan\alpha\tan\beta\tan\gamma \leq 1 + \frac{1}{3\sqrt{3}} < \sqrt{3}.$$

Hence proved.

3. In a triangle ABC, prove that $\frac{b+c}{a} \le \operatorname{cosec}\frac{A}{2}$. Hence declare that $\operatorname{cosec}^n\frac{A}{2}\cdot\operatorname{cosec}^n\frac{B}{2}\cdot\operatorname{cosec}^n\frac{C}{2} \ge 2^{3n}$ for all integers $n \ge 1$.

Sol. In the triangle ABC, $\frac{b+c}{a} = \frac{\sin B+\sin C}{\sin A} = \frac{2\sin\frac{B+C}{2}\cos\frac{B-C}{2}}{2\sin\frac{A}{2}\cdot\cos\frac{A}{2}} = \frac{\cos\frac{B-C}{2}}{\sin\frac{A}{2}} \le \frac{1}{\sin\frac{A}{2}}$

$\Rightarrow \operatorname{cosec}\frac{A}{2} \ge \frac{b+c}{a}$

$\Rightarrow \operatorname{cosec}\frac{A}{2} \ge \frac{b}{a}+\frac{c}{a} = 2\left(\frac{\frac{b}{a}+\frac{c}{a}}{2}\right) \ge 2\left(\frac{b}{a}\cdot\frac{c}{a}\right)^{1/2} \Rightarrow \operatorname{cosec}^n\frac{A}{2} \ge 2^n\left(\frac{b}{a}\cdot\frac{c}{a}\right)^{n/2}$.

Also $\operatorname{cosec}^n\frac{B}{2} \ge 2^n\left(\frac{c}{b}\cdot\frac{a}{b}\right)^{n/2}$ and $\operatorname{cosec}^n\frac{C}{2} \ge 2^n\left(\frac{a}{c}\cdot\frac{b}{c}\right)^{n/2}$.

Multiplying, these inequalities, we get $\operatorname{cosec}^n\frac{A}{2}\cdot\operatorname{cosec}^n\frac{B}{2}\cdot\operatorname{cosec}^n\frac{C}{2} \ge 2^{3n}$.

4. Find the possible integral values of 'a' for which $\cos 2x + a\sin x = 2a - 7$ has solutions.

Sol. Here we have $1 - 2\sin^2 x + a\sin x = 2a - 7$

$\Rightarrow 2\sin^2 x - a\sin x + 2a - 8 = 0$

$\Rightarrow \sin x = \frac{a \pm \sqrt{a^2 - 8(2a-8)}}{4} \Rightarrow \sin x = \frac{a-4}{2}$ or 2

$\Rightarrow -1 \le \frac{a-4}{2} \le 1 \Rightarrow -2 \le a-4 \le 2 \Rightarrow 2 \le a \le 6 \Rightarrow a = 2, 3, 4, 5, 6.$

5. Prove that $\sum_{r=1}^{n}\tan^{-1}\frac{2r}{2r^4+1} = \tan^{-1}\frac{n(n+1)}{1+n(n+1)}$.

Sol. Let $t_r = \tan^{-1}\frac{2r}{2r^4+1} = \tan^{-1}\frac{4r}{4r^4+2} = \tan^{-1}\frac{4r}{1+(2r^2+2r+1)(2r^2-2r+1)}$

$= \tan^{-1}\frac{(2r^2+2r+1)-(2r^2-2r+1)}{1+(2r^2+2r+1)(2r^2-2r+1)} = \tan^{-1}(2r^2+2r+1) - \tan^{-1}(2r^2-2r+1).$

$t_1 = \tan^{-1}5 - \tan^{-1}1$

$t_2 = \tan^{-1}13 - \tan^{-1}5$

$t_3 = \tan^{-1}25 - \tan^{-1}13$

$\ldots \quad \ldots \quad \ldots$

$\ldots \quad \ldots \quad \ldots$

$t_n = \tan^{-1}(2n^2+2n+1) - \tan^{-1}(2n^2-2n+1)$

$\Sigma t_r = \tan^{-1}(2n^2+2n+1) - \tan^{-1}1$

$\therefore \Sigma t_r = \tan^{-1}\frac{2n^2+2n+1-1}{1+(2n^2+2n+1)} = \tan^{-1}\frac{2n^2+2n}{(2n^2+2n+2)} = \tan^{-1}\left(\frac{n(n+1)}{n^2+n+1}\right)$.

SOLVED PROBLEMS

1. The number of solutions of the equation $\sin^{-1}\frac{1+x^2}{2x} = \frac{\pi}{2}\sec(x-1)$ is/are

(A) 1 (B) 2
(C) 3 (D) infinite

Sol. **(A).** $\sin^{-1}\frac{1+x^2}{2x}$ is defined for $\left|\frac{1+x^2}{2x}\right| \le 1 \Rightarrow x = \pm 1$. Out of these two values of x, only $x = 1$ satisfies the given equation.

2. The maximum value of the expression $\left|\sqrt{\sin^2 x + 2a^2} - \sqrt{2a^2 - 1 - \cos^2 x}\right|$, where a and x are real numbers, is

(A) $\sqrt{2}$ (B) 1
(C) $\sqrt{3}$ (D) none of these

Sol. **(A).** Since $\left|\sqrt{A} - \sqrt{B}\right| \le \sqrt{|A - B|}$

$\Rightarrow$ Given expression $\le \sqrt{\sin^2 x + 2a^2 - 2a^2 + 1 + \cos^2 x} \le \sqrt{2}$

3. If $4n\alpha = \pi$ then the value of $\tan\alpha \tan2\alpha \tan3\alpha \ldots \tan(2n-1)\alpha$ is equal to

(A) 1 (B) 0
(C) -1 (D) none of these

Sol. **(A).** $\tan\alpha \tan2\alpha \tan3\alpha \ldots \tan(2n-1)\alpha$

$= \{\tan\alpha \tan(2n-1)\alpha\}\{\tan2\alpha \tan(2n-2)\alpha\} \ldots \{\tan(n-1)\alpha \tan(n+1)\alpha\} \tan n\alpha$,

$= \{\tan\alpha \tan(\pi/2 - \alpha)\}\{\tan2\alpha \tan(\pi/2 - 2\alpha)\} \ldots \tan\frac{\pi}{4} = 1.1.1. \ldots 1 = 1$

4. If $0 < \alpha < \beta < \gamma < \frac{\pi}{2}$, then $\frac{\sin\alpha + \sin\beta + \sin\gamma}{\cos\alpha + \cos\beta + \cos\gamma}$ lies between

(A) $\sin\alpha$ and $\sin\gamma$ (B) $\tan\alpha$ and $\tan\gamma$
(C) $\cos\alpha$ and $\cos\gamma$ (D) none of these

Sol. **(B).** $3\sin\alpha < \sin\alpha + \sin\beta + \sin\gamma < 3\sin\gamma$

Also, $\frac{1}{3\cos\alpha} < \frac{1}{\cos\alpha + \cos\beta + \cos\gamma} < \frac{1}{3\cos\gamma}$

Multiplying, $\tan\alpha < \frac{\sin\alpha + \sin\beta + \sin\gamma}{\cos\alpha + \cos\beta + \cos\gamma} < \tan\gamma$.

5. If $\cos2\theta - \cos3\theta - \cos4\theta + \cos5\theta = \lambda \sin\frac{\theta}{2} \sin\theta \cos\frac{7}{2}\theta$, then λ is equal to

(A) $\lambda = -2$ (B) $\lambda = +2$
(C) $\lambda = +4$ (D) $\lambda = -4$

Sol. **(D).** $2\cos\frac{7\theta}{2}\cos\frac{3\theta}{2} - 2\cos\frac{7\theta}{2}\cos\frac{\theta}{2} = -4\cos\frac{7\theta}{2}\left[\sin\frac{\theta}{2} \cdot \sin\theta\right]$

$\Rightarrow \lambda = -4$.

6. If $\varphi = \sin^{-1}\left(\sin\sqrt{z}\right)$, where $z \in (80, 140)$ and z is sum of cubes of first n natural numbers, then

(A) $\varphi = (n+1)\pi - 10$ (B) $\varphi = (n-1)\pi - 10$

(C) $\varphi = (n+1)\pi + 10$ (D) $\varphi = (n-1)\pi + 10$

Sol. **(B).** $\phi = \sin^{-1}(\sin\sqrt{z})$

cubes of first n natural numbers is a perfect square.

i.e. 81, 100, 121 in $Z \in (80, 140)$

i.e. $\left[\frac{n(n+1)}{2}\right]$ = either 9, 10 or 11.

i.e. $n(n+1)$ = either 18, 20 or 22

but $n(n+1) \neq 18$ or $22 \Rightarrow n(n+1) = 20 \Rightarrow n = 4$

$\Rightarrow Z = 100$

$\Rightarrow \phi = \sin^{-1}(\sin 10)$

$= 3\pi - 10.$

7. The value of $\sin^3 10^0 + \sin^3 50^0 - \sin^3 70^0$ is equal to

(A) $-\frac{3}{2}$ (B) $\frac{3}{4}$

(C) $-\frac{3}{4}$ (D) $-\frac{3}{8}$

Sol. **(D).** We have $\sin^3 10° + \sin^3 50° - \sin^3 70°$

$= \frac{1}{4}[(3\sin 10° - \sin 30°) + (3\sin 50° - \sin 150°) - (3\sin 70° - \sin 210°)]$

$= \frac{1}{4}\left[3(\sin 10° + \sin 50° - \sin 70°) - \frac{3}{2}\right]$

$= \frac{1}{4}\left[3(\sin 10° - 2\cos 60°\cdot\sin 10°) - \frac{3}{2}\right] = -\frac{3}{8}.$

8. If $\left(\cos^2 x + \frac{1}{\cos^2 x}\right)\left(1+\tan^2 2y\right)\left(3+\sin 3z\right) = 4$, then

(A) x may be an integer multiple of π (B) x can not be an even integer multiple of π

(C) z can be an integer multiple of π (D) y can be an integer multiple of $\pi/2$

Sol. **(A, D).** $\left(\cos^2 x + \frac{1}{\cos^2 x}\right)\left(1+\tan^2 2y\right)\left(3+\sin 3z\right) = 4$

Since, $\cos^2 x + \frac{1}{\cos^2 x} \geq 2$, $1 + \tan^2 2y \geq 1$, $2 \leq 3 + \sin 3z \leq 4$

So, the only possiblility is

$\cos^2 x + \frac{1}{\cos^2 x} = 2,\ 1 + \tan^2 2y = 1,\ 3 + \sin 3z = 2$

$\Rightarrow \cos x = \pm 1 \Rightarrow x = n\pi$

$\tan 2y = 0 \Rightarrow y = \frac{m\pi}{2}$

$\sin 3z = -1 \Rightarrow z = (4k-1)\frac{\pi}{6}$; $m, n, k \in I$.

9. If $\cos x + \cos y = a$, $\cos 2x + \cos 2y = b$, $\cos 3x + \cos 3y = c$, then

(A) $\cos^2 x + \cos^2 y = 1 + \frac{b}{2}$

(B) $\cos x \cos y = \frac{a^2}{2} - \left(\frac{b+2}{4}\right)$

(C) $2a^3 + c = 3a(1 + b)$

(C) $a + b + c = 3abc$

Sol. **(A, B, C).** $(\cos x + \cos y)^2 = a^2$

$\Rightarrow \cos^2 x + \cos^2 y + 2\cos x \cos y = a^2$ ....(1)

$\cos 2x + \cos 2y = b$

$2\cos^2 x - 1 + 2\cos^2 y - 1 = b$

$2[\cos^2 x + \cos^2 y] = b + 2$ ....(2)

$\Rightarrow \cos^2 x + \cos^2 y = \frac{b}{2} + 1$

From (1) and (2),

$2\cos x \cos y = a^2 - \left(\frac{b+2}{2}\right)$

$\Rightarrow \cos x \cos y = \frac{a^2}{2} - \left(\frac{b+2}{4}\right)$

$\cos 3x + \cos 3y = c$

$4\cos^3 x - 3\cos x + 4\cos^3 y - 3\cos y = c$

$4[\cos^3 x + \cos^3 y] - 3[\cos x + \cos y] = c$

$\Rightarrow 4[(\cos x + \cos y)(\cos^2 x + \cos^2 y - \cos x \cos y)] - 3(\cos x + \cos y) = c$

$\Rightarrow 4\left[a\left(\frac{b+2}{2} - \frac{1}{2}\left(a^2 - \frac{b+2}{2}\right)\right)\right] - 3a = c$

$\Rightarrow 2ab + 4a - 2a^3 + ab + 2a = 3a + c$

$\Rightarrow 2a^3 + c = 3a(1 + b)$.

10. If $x = \sin(\alpha - \beta)\cdot\sin(\gamma - \delta)$, $y = \sin(\beta - \gamma)\cdot\sin(\alpha - \delta)$, $z = \sin(\gamma - \alpha)\cdot\sin(\beta - \delta)$, then

(A) $x + y + z = 0$

(B) $x^3 + y^3 + z^3 = 3xyz$

(C) $x + y - z = 0$

(D) $x^3 + y^3 - z^3 = 3xyz$

Sol. **(A, B)**

If $x = \sin(\alpha - \beta)\sin(\gamma - \delta)$

$2x = \cos(\alpha - \beta - \gamma + \delta) - \cos(\alpha - \beta + \gamma - \delta)$

$y = \sin(\beta - \gamma)\sin(\alpha - \delta)$

$\Rightarrow 2y = \cos(\beta - \gamma - \alpha + \delta) - \cos(\beta - \gamma + \alpha - \delta)$

$z = \sin(\gamma - \alpha)\sin(\beta - \delta)$

$\Rightarrow 2z = \cos(\gamma - \alpha - \beta + \delta) - \cos(\gamma - \alpha + \beta - \delta)$

$2x + 2y + 2z = 0 \Rightarrow x + y + z = 0$

If $x + y + z = 0$ then $x^3 + y^3 + z^3 = 3xyz$

11. If $0 < x < \frac{\pi}{2}$ and $\sin^n x + \cos^n x \geq 1$, then 'n' may belong to interval

(A) [1, 2)

(B) [3, 4]

(C) $(-\infty, 2]$

(D) [–1, 1]

Sol. **(A, C, D)**

$0 < \sin x < 1$, $0 < \cos x < 1$

If $\sin^n x + \cos^n x = 1$ $n = 2$

$\sin^n x + \cos^n x > 1$ $n < 2$

$\sin^n x + \cos^n x < 1$ $n > 2$

Comprehension

Sometimes we have quadratic equation having unknown variables/parameter, in terms of some trigonometric functions.
Apparently it is seen while solving such type of equations we have to keep in consideration the domain and range of the corresponding trigonometric functions while finding out the final roots.
For example:

$f(x) = 4\sin^2 x + 3\cos^2 x + 3\sin x \ \forall\ x \in \left[0, \frac{\pi}{2}\right]$

find the minimum value of $f(x)$

As we know that sinx is increasing function in $\left[0, \frac{\pi}{2}\right]$

Let $f(x) = 4\sin^2 x + 3 - 3\sin^2 x + 3\sin x$
$\Rightarrow f(x) = \sin^2 x + 3\sin x + 3$
Now, let us take $\sin x = t$

$t \in [0, 1]$, since $0 \le \sin x \le 1\ \forall\ x \in \left[0, \frac{\pi}{2}\right]$

This reduces to $t^2 + 3t + 3$.
As coefficient of $t^2 > 0$

Hence this will be upward parabola such that minimum value of the function exists at $\left(\frac{4ac - b^2}{4a}\right)$

$\Rightarrow t_{min} = \frac{4\times 3\times 1 - 9}{4} = \frac{3}{4}$.

In such a manner we can interrelate these two concepts keeping in mind the domain and codomain of the respective trigonometric function.

Answer the following questions:

12. If $\sec\theta + \operatorname{cosec}\theta = p$ has four solutions between $(0, 2\pi)$, then
(A) $p^2 - 8 > 0$ (B) $p^2 + 8 > 0$
(C) $p^2 - 8 < 0$ (B) $p^2 + 8 < 0$

Sol **(A).** $\sec.\theta + \operatorname{cosec}\theta = p$

$$\sqrt{1+\tan^2\theta} + \sqrt{1+\cot^2\theta} = p$$

$$\sqrt{1+t^2} + \sqrt{1+\frac{1}{t^2}} = p \quad (\tan\theta = t)$$

$$\sqrt{1+t^2}\left(\frac{t+1}{t}\right) = p$$

$$(t^2 + 1)(t^2 + 2t + 1) = p^2t^2$$

$$(t^2 + t + 1)^2 = (p^2 + 1)t^2$$

$$\left(t^2 + t + 1\right) \mp t\sqrt{(p^2+1)} = 0$$

$$t^2 + t + 1 - t\sqrt{p^2+1} = 0$$

or $t^2 + t + 1 + t\sqrt{(p^2+1)} = 0$

$$D_1 = \left(1 - \sqrt{p^2+1}\right)^2 - 4$$

$D_2 = \left(1+\sqrt{p^2+1}\right)^2 - 4$

we have, $D_1 > 0$ and $D_2 > 0$

$D_1 > 0 \Rightarrow \left(1-\sqrt{p^2+1}\right)^2 > 4$

$\Rightarrow 1+p^2+1-2\sqrt{p^2+1} > 4 \Rightarrow p^2-2 > 2\sqrt{p^2+1}$

$\Rightarrow p^4 + 4 - 4p^2 > 4(p^2+1)$

$\Rightarrow p^4 - 8p^2 > 0 \Rightarrow p^2 > 8.$

$D_2 > 0 \Rightarrow \left(1+\sqrt{p^2+1}\right)^2 > 4$

$\Rightarrow 1+p^2+1+2\sqrt{p^2+1} > 4 \Rightarrow p^2+2\sqrt{p^2+1}-2 > 0$

This true always $\left(\because \sqrt{p^2+1} > 1 \quad 2\sqrt{p^2+1} > 2\right)$.

13. If in above problem the equation $\sec\theta + \operatorname{cosec}\theta = p$ has only two real solutions in $(0, 2\pi)$, then

(A) $p^2 - 8 > 0$ (B) $p^2 - 8 < 0$

(C) $p^2 = 8$ (D) none of these

Sol. **(B).** Since $D_2 > 0$ always

We should many $D_1 < 0$

$\Rightarrow \left(1-\sqrt{p^2+1}\right) < 4$

$\Rightarrow p^2 < 8$ (by previous).

14. The number of isosceles ΔABC(s) possible such that $\tan A + \tan B + \tan C = 100$, is

(A) 1 (B) 2

(C) 3 (D) 4

Sol. **(B).** $\tan A + \tan B + \tan C = 100$

Triangle ABC is isosceles triangle. Let $\angle A = \angle C$, then $B = \pi - 2C \therefore 2\tan A - \tan 2A = 100$

$2\tan A - \dfrac{2\tan A}{1-\tan^2 A} = 100$

$\tan^3 A - 50\tan^2 A + 50 = 0$

let $x = \tan A$

then $f(x) = x^3 - 50x^2 + 50$ ...(1)

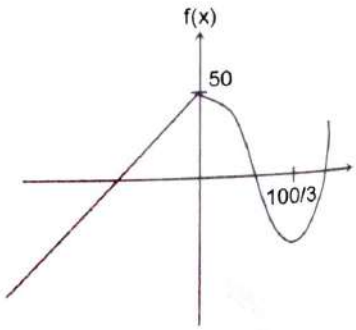
$f(x) = 3x^2 - 100x = 0$

critical points $x = 0$ and $x = \dfrac{100}{3}$

As $x \to -\infty$, $f(x) \to -\infty$

At $x = 0$, $f(x) = 50$

At $x = \dfrac{100}{3}$ $f(x)$ = negative



$\Rightarrow$ tanA takes one negative value which is not possible

($\because$ A = C and A is acute)

$\Rightarrow$ tanA has two distinct positive values

$\Rightarrow$ number of isosceles triangles = 2.

Assertion Reason

15. The angles of a triangle are given by the three values of x obtained from the equation
$\tan^3 x - 3\sqrt{3}\tan^2 x + 3\sqrt{3}\tan x - 3 = 0$
Statement 1: The triangle so obtained is an equilateral triangle.
because
Statement 2: If roots of the equation be tanA, tanB and tanC, then
$\tan A + \tan B + \tan C = 3\sqrt{3}$.
(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1
(B) Statement -1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement -1 is True, Statement -2 is False
(D) Statement -1 is False, Statement -2 is True

Sol. **(D).** $\tan A + \tan B + \tan C = 3\sqrt{3}$ and $\tan A \tan B \tan C = 3$
$\therefore \tan A + \tan B + \tan C \neq \tan A \tan B \tan C$
$\Rightarrow$ triangle does not exist.

16. If $A + B + C = \pi$, then match the following (equality must hold)

	Column – I		Column – II
(A)	$\cos A + \cos B + \cos C$	(p)	$\le 3/2$
(B)	$\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$	(q)	≥ -3
(C)	$\frac{1}{2}(\cos A\cos B + \cos B\cos C + \cos C\cos A)$	(r)	$\le \frac{3}{8}$
(D)	$\cos A \cos B \cos C$	(s)	≥ -1
		(t)	0

Sol. **(A) → (p, q)** **(B) → (s)** **(C) → (p)** **(D) → (s)**
(A) $\cos A + \cos B + \cos C$ becomes -3 for $A = B = -\pi$
and $C = 3\pi$ so $\cos A + \cos B + \cos C \ge -3$
and using $y = \cos A + \cos B + \cos C$

$$y = 2\sin\frac{C}{2}\cos\frac{A-B}{2} + 1 - 2\sin^2\frac{C}{2}$$

we get, $y \le \frac{3}{2}$.

(B) Also, $\cos A + \cos B + \cos C$

$$= 1 + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

$$\Rightarrow -1 \le \sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} \le \frac{1}{8}.$$

(C) $\frac{1}{2}(\cos A\cos B + \cos B\cos C + \cos C\cos A) \le \frac{3}{2}$
equality holds when all of cosA, cosB, cosC are -1.
(D) $\cos A \cos B \cos C \ge -1$
(when all of cosA, cosB, cosC are -1).

Answer Q.17, Q.18 and Q.19 by appropriately matching the information given in the three columns of the following table.

Match the following column(s)

Column–I	Column–II	Column–III
(I) $\left[\tan\left(\frac{\pi}{4}+\theta\right)+\tan\left(\frac{\pi}{4}-\theta\right)\right]\frac{1}{2}$	(i) $\frac{\cot\theta+\tan\theta}{\cot\theta-\tan\theta}$	(P) $\sec 2\theta$
(II) $\left[\tan\left(\frac{\pi}{4}+\theta\right)-\tan\left(\frac{\pi}{4}-\theta\right)\right]\frac{1}{2}$	(ii) $\tan 2\theta$	(Q) $2[\cot\theta-\tan\theta]^{-1}$
(III) $8\sin\frac{\pi}{14}\cdot\sin\frac{3\pi}{14}\cdot\sin\frac{5\pi}{14}\cdot\sin\frac{7\pi}{14}\cdot\sin\frac{9\pi}{14}\cdot\sin\frac{11\pi}{14}\cdot\sin\frac{13\pi}{14}$	(iii) $\frac{1}{8}$	(R) $\sin\left(\frac{\pi}{18}\right)\cdot\sin\frac{5\pi}{18}\cdot\sin\frac{7\pi}{18}$
(IV) $8\left(1+\cos\frac{\pi}{8}\right)\cdot\left(1+\cos\frac{3\pi}{8}\right)\cdot\left(1+\cos\frac{5\pi}{8}\right)\cdot\left(1+\cos\frac{7\pi}{8}\right)$	(iv) 1	(S) $16\cos\left(\frac{2\pi}{15}\right)\cdot\cos\left(\frac{4\pi}{15}\right)\cdot\cos\left(\frac{8\pi}{15}\right)\cdot\cos\left(\frac{16\pi}{15}\right)$

17. Which of the following combination is **CORRECT**?
(A) (I) (i) (R) (B) (II) (ii) (R)
(C) (I) (i) (P) (D) (II) (i) (Q)

Sol. **(C)**

18. Which of the following combination is **CORRECT**?
(A) (II) (iii) (S) (B) (II) (ii) (Q)
(C) (II) (iii) (R) (D) (I) (ii) (R)

Sol. **(B)**

19. Which of the following combination is **INCORRECT**?
(A) (II) (i) (S) (B) (III) (iii) (R)
(C) (IV) (iv) (S) (D) (I) (i) (P)

Sol. **(A)**

17.-19. $\tan\left(\frac{\pi}{4}+\theta\right)+\tan\left(\frac{\pi}{4}-\theta\right)=2\sec 2\theta$; $\tan\left(\frac{\pi}{4}+\theta\right)-\tan\left(\frac{\pi}{4}-\theta\right)=2\tan 2\theta$

$\cot\theta+\tan\theta=2\sec 2\theta$; $\cot\theta-\tan\theta=2\cot 2\theta$

$$\sin\frac{\pi}{14}\cdot\sin\frac{3\pi}{14}\cdot\sin\frac{5\pi}{14}\cdot\sin\frac{7\pi}{14}\cdot\sin\frac{9\pi}{14}\cdot\sin\frac{11\pi}{14}\cdot\sin\frac{13\pi}{14}=\left(\sin\frac{\pi}{14}\cdot\sin\frac{3\pi}{14}\cdot\sin\frac{5\pi}{14}\right)^2=\frac{1}{64}$$

$$\sin\frac{\pi}{18}\cdot\sin\frac{5\pi}{18}\cdot\sin\frac{7\pi}{18}=\cos\frac{\pi}{9}\cdot\cos\frac{2\pi}{9}\cdot\cos\frac{4\pi}{9}=\frac{1}{8};\ \cos\frac{2\pi}{15}\cdot\cos\frac{4\pi}{15}\cdot\cos\frac{8\pi}{15}\cdot\cos\frac{16\pi}{15}=\cos A\cdot\cos 2A$$

$$\cos 2^2A\cdot\cos 2^3A\cdot\ldots\ldots\cdot\cos 2^nA=\frac{1}{2^{n+1}\cdot\sin A}\cdot\sin\left(2^{n+1}A\right)=\frac{1}{16}$$

$$\left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1+\cos\frac{5\pi}{8}\right)\left(1+\cos\frac{7\pi}{8}\right)$$

$$=\sin^2\frac{\pi}{8}\cdot\sin^2\frac{3\pi}{8}=\frac{1}{8}$$

20. Let $a = \sin 10^\circ$, $b = \sin 50^\circ$, $c = \sin 70^\circ$, then $8abc\left(\frac{a+b}{c}\right)\left(\frac{1}{a}+\frac{1}{b}-\frac{1}{c}\right)$ is equal to _____

Sol. **(6.00)**

$8abc = 8 \sin 10^\circ \sin 50^\circ \sin 70^\circ = 1$

$$\frac{a+b}{c} = \frac{\sin 10^\circ + \sin 50^\circ}{\sin 70^\circ} = \frac{2 \sin 30^\circ \cos 20^\circ}{\sin 70^\circ} = 1$$

$$\frac{1}{a}+\frac{1}{b}-\frac{1}{c} = \frac{1}{\sin 10^\circ}+\frac{1}{\sin 50^\circ}-\frac{1}{\sin 70^\circ} = \frac{\sin 50^\circ \sin 70^\circ + \sin 10^\circ \sin 70^\circ - \sin 10^\circ \sin 50^\circ}{\sin 10^\circ \sin 50^\circ \sin 70^\circ} = 6$$

21. The number of real roots of the equation $\cos^7 x + \sin^4 x = 1$ in $(-\pi, \pi)$ is _____

Sol. **(3.00)**

$\cos^7 x + \sin^4 x = 1$

$\cos^7 x = (1 + \sin^2 x)\cos^2 x$

$\Rightarrow \cos x = 0$ or $\cos^5 x = 1 + \sin^2 x$

$\cos x = 0 \Rightarrow x = -\frac{\pi}{2}, \frac{\pi}{2}$; $\cos^5 x = 1 + \sin^2 x \Rightarrow x = 0$ ($\because$ LHS ≤ 1 and RHS ≥ 1)

$\therefore x = -\frac{\pi}{2}, 0, \frac{\pi}{2}$

22. If $(1 + \tan 1^\circ)(1 + \tan 2^\circ)(1 + \tan 3^\circ) \ldots\ldots (1 + \tan 45^\circ) = 2^n$, then $\frac{n}{5}$ is _____

Sol. **(4. 60)**

We know that $(1+\tan k^\circ)\{1+\tan(45-k^\circ)\} = (1+\tan k^\circ)\left(1+\frac{1-\tan k^\circ}{1+\tan k^\circ}\right) = 2$

$2^{23} = 2^n \Rightarrow n = 23$

23. Let $0 \le \alpha, \beta, \gamma, \delta \le \pi$ where β and γ are not complementary such that
$2 \cos \alpha + 6 \cos \beta + 7 \cos \gamma + 9 \cos \delta = 0$ and $2 \sin \alpha - 6 \sin \beta + 7 \sin \gamma - 9 \sin \delta = 0$.

If $\frac{\cos(\alpha+\delta)}{\cos(\beta+\gamma)} = \frac{m}{n}$ where m and n are relatively prime positive numbers, then the value of $(m + n)$ is k,

then $\frac{k}{3}$ is equal to _____

Sol. **(3.33)**

Gives equations can be written as

$2 \cos \alpha + 9 \cos \delta = -6 \cos \beta - 7 \cos \gamma$ ..... (1)

$2 \sin \alpha - 9 \sin \delta = 6 \sin \beta - 7 \sin \gamma$ ..... (2)

Square and add equation (1) and (2), we get

$\Rightarrow 4 + 36 + 36 [\cos \alpha \cos \delta - \sin \delta \sin \alpha]$

$= 36 + 49 + 84 [\cos \beta \cos \gamma - \sin \beta \sin \gamma]$

$\Rightarrow 36[\cos(\alpha + \delta)] = 84[\cos(\beta + \gamma)]$

$$\frac{\cos(\alpha+\delta)}{\cos(\beta+\gamma)} = \frac{84}{36} = \frac{7}{3} = \frac{m}{n}; \; m + n = 10$$

TRIGONOMETRIC EQUATIONS 24

BASIC FORMULAE

- $\sin\theta = 0 \Rightarrow \theta = n\pi$,
- $\cos\theta = 0 \Rightarrow \theta = (2n+1)\frac{\pi}{2}$,
- $\tan\theta = 0 \Rightarrow \theta = n\pi$,
- $\sin\theta = \sin\alpha \Rightarrow \theta = n\pi + (-1)^n\alpha$, where $\alpha \in [-\pi/2, \pi/2]$
- $\cos\theta = \cos\alpha \Rightarrow \theta = 2n\pi \pm \alpha$, where $\alpha \in [0, \pi]$
- $\tan\theta = \tan\alpha \Rightarrow \theta = n\pi + \alpha$, where $\alpha \in (-\pi/2, \pi/2)$
- $\sin^2\theta = \sin^2\alpha$, $\cos^2\theta = \cos^2\alpha$, $\tan^2\theta = \tan^2\alpha \Rightarrow \theta = n\pi \pm \alpha$,
- $\sin\theta = 1 \Rightarrow \theta = (4n+1)\frac{\pi}{2}$,
- $\sin\theta = -1 \Rightarrow \theta = (4n-1)\frac{\pi}{2}$,
- $\cos\theta = 1 \Rightarrow \theta = 2n\pi$,
- $\cos\theta = -1 \Rightarrow \theta = (2n+1)\pi$,
- $\sin\theta = \sin\alpha$ and $\cos\theta = \cos\alpha \Rightarrow \theta = 2n\pi + \alpha$.

Notes:

- Everywhere in this chapter 'n' is taken as an integer, If not stated otherwise.
- The general solution should be given unless the solution is required in a specified interval or range.
- α is taken as the principal value of the angle. Numerically least angle is called the principal value.

Equations of the Form $a\cos\theta + b\sin\theta = c$

To solve the equation $a\cos\theta + b\sin\theta = c$, put $a = r\cos\phi$, $b = r\sin\phi$ such that $r = \sqrt{a^2+b^2}$, $\phi = \tan^{-1}\frac{b}{a}$. Substituting these values in the equation we have,

$r\cos\phi\cos\theta + r\sin\phi\sin\theta = c. \Rightarrow \cos(\theta - \phi) = \frac{c}{r} \Rightarrow \cos(\theta - \phi) = \frac{c}{\sqrt{a^2+b^2}}$. If $|c| > \sqrt{a^2+b^2}$, then the equation $a\cos\theta + b\sin\theta = c$ has no solution.

If $|c| \le \sqrt{a^2+b^2}$ then put $\frac{|c|}{\sqrt{a^2+b^2}} = \cos\alpha$, so that $\cos(\theta - \phi) = \cos\alpha$

$\Rightarrow (\theta - \phi) = 2n\pi \pm \alpha \Rightarrow \theta = 2n\pi \pm \alpha + \phi$, where $n \in I$

Remarks:

- While solving a trigonometric equation, squaring the equation at any step should be avoided as far as possible. If squaring is necessary, check the solution for extraneous values.
- Never cancel the terms containing unknown terms on the two sides, which are in product. It may cause loss of genuine solution.

- The answer should not contain such values of angles which make any of the terms undefined or infinite.
- Domain should not change. If it changes, necessary corrections must be made.
- Check that denominator is not zero at any stage while solving equations.

Solving Simultaneous Equations

Here, we will discuss the problems related to the solution of two equations satisfied simultaneously. We may divide the problems in two categories:

(i) Two equations in one 'unknown' satisfied simultaneously.

(ii) Two equations in two 'unknowns' satisfied simultaneously.

TRIGONOMETRIC INEQUALITY

Inverse Circular Functions (Inverse Trigonometric Functions)

$\sin\theta = x \Rightarrow \theta = \sin^{-1}x$ where $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and $x \in [-1, 1]$.

$\operatorname{cosec}\theta = x \Rightarrow \theta = \operatorname{cosec}^{-1}x$ where $\theta \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$ and $x \in (-\infty, -1] \cup [1, \infty)$

$\tan\theta = x \Rightarrow \theta = \tan^{-1}x$ where $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and $x \in (-\infty, \infty)$

$\cos\theta = x \Rightarrow \theta = \cos^{-1}x$ where $\theta \in [0, \pi]$ and $x \in [-1, 1]$

$\sec\theta = x \Rightarrow \theta = \sec^{-1}x$ where $\theta \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$ and $x \in (-\infty, -1] \cup [1, \infty)$

$\cot\theta = x \Rightarrow \theta = \cot^{-1}x$ where $\theta \in (0, \pi)$ and $x \in (-\infty, \infty)$

Notes:

- $\sin^{-1}(\sin\theta) = \theta$ if and only if $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$
  and $\sin(\sin^{-1}x) = x$ where $-1 \le x \le 1$
- $\operatorname{cosec}^{-1}(\operatorname{cosec}\theta) = \theta$ if and only if $-\frac{\pi}{2} \le \theta < 0$ or $0 < \theta \le \frac{\pi}{2}$
  and $\operatorname{cosec}(\operatorname{cosec}^{-1}x) = x$ where $-\infty < x \le -1$ or $1 \le x < \infty$
- $\tan^{-1}(\tan\theta) = \theta$ if and only if $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$
  and $\tan(\tan^{-1}x) = x$ where $-\infty < x < \infty$
- $\cos^{-1}(\cos\theta) = \theta$ if and only if $0 \le \theta \le \pi$
  and $\cos(\cos^{-1}x) = x$ where $-1 \le x \le 1$
- $\sec^{-1}(\sec\theta) = \theta$ if and only if $0 \le \theta < \frac{\pi}{2}$ or $\frac{\pi}{2} < \theta \le \pi$.
  and $\sec(\sec^{-1}x) = x$ where $-\infty < x \le -1$ or $1 \le x < \infty$
- $\cot^{-1}(\cot\theta) = \theta$ if and only if $0 < \theta < \pi$
  and $\cot(\cot^{-1}x) = x$ where $-\infty < x < \infty$

Some Important Results

$$\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2},\qquad \tan^{-1}x+\cot^{-1}x=\frac{\pi}{2},\qquad \sec^{-1}x+\operatorname{cosec}^{-1}x=\frac{\pi}{2},$$

$\sin^{-1}(-x)=-\sin^{-1}(x)$, $\operatorname{cosec}^{-1}(-x)=-\operatorname{cosec}^{-1}(x)$, $\tan^{-1}(-x)=-\tan^{-1}(x)$

$\cos^{-1}(-x)=\pi-\cos^{-1}(x)$, $\sec^{-1}(-x)=\pi-\sec^{-1}(x)$, $\cot^{-1}(-x)=\pi-\cot^{-1}(x)$

$$\sin^{-1}x=\tan^{-1}\frac{x}{\sqrt{1-x^2}}=\operatorname{cosec}^{-1}\frac{1}{x},\ -1\le x<0 \text{ or } 0<x\le 1$$

$$\sin^{-1}x=\cos^{-1}\sqrt{1-x^2}=\cot^{-1}\frac{\sqrt{1-x^2}}{x}=\sec^{-1}\frac{1}{\sqrt{1-x^2}},\qquad x>0$$

$$\cos^{-1}x=\sin^{-1}\sqrt{1-x^2}=\tan^{-1}\frac{\sqrt{1-x^2}}{x}=\operatorname{cosec}^{-1}\frac{1}{\sqrt{1-x^2}},\qquad x>0$$

$$\tan^{-1}x=\sin^{-1}\frac{x}{\sqrt{1+x^2}}=\operatorname{cosec}^{-1}\frac{\sqrt{1+x^2}}{x}$$

$$\cot^{-1}x=\cos^{-1}\frac{x}{\sqrt{1+x^2}}=\sec^{-1}\frac{\sqrt{1+x^2}}{x}$$

$$\cos^{-1}x+\cos^{-1}y=\cos^{-1}\left[xy-\sqrt{1-x^2}\sqrt{1-y^2}\right],\qquad x\ge 0,\quad y\ge 0$$

$$\cos^{-1}x-\cos^{-1}y=\begin{cases}\cos^{-1}\left[xy+\sqrt{1-x^2}\sqrt{1-y^2}\right], & x\ge 0,\quad y\ge 0,\quad x\le y\\ -\cos^{-1}\left[xy+\sqrt{1-x^2}\sqrt{1-y^2}\right], & x\ge 0,\quad y\ge 0,\quad x>y\end{cases}$$

$$\sin^{-1}x+\sin^{-1}y=\begin{cases}\sin^{-1}\left[x\sqrt{1-y^2}+y\sqrt{1-x^2}\right], & x\ge 0,\ y\ge 0,\ x^2+y^2\le 1\\ \pi-\sin^{-1}\left[x\sqrt{1-y^2}+y\sqrt{1-x^2}\right], & x\ge 0,\ y\ge 0,\ x^2+y^2>1\end{cases}$$

$$\sin^{-1}x-\sin^{-1}y=\sin^{-1}\left[x\sqrt{1-y^2}-y\sqrt{1-x^2}\right],\ x\ge 0,\quad y\ge 0$$

$$\tan^{-1}x+\tan^{-1}y=\begin{cases}\tan^{-1}\dfrac{x+y}{1-xy}, & x\ge 0,\ y\ge 0,\ xy<1\\ \pi+\tan^{-1}\dfrac{x+y}{1-xy}, & x>0,\ y>0,\ xy>1\\ \dfrac{\pi}{2}, & x>0,y>0,xy=1\end{cases}$$

$$\tan^{-1}x-\tan^{-1}y=\tan^{-1}\frac{x-y}{1+xy}\qquad x\ge 0,\quad y\ge 0$$

$$2\cos^{-1}x=\cos^{-1}(2x^2-1),\quad x\ge 0$$

$$2\sin^{-1}x=\begin{cases}-\pi-\sin^{-1}\left[2x\sqrt{1-x^2}\right], & x<-\dfrac{1}{\sqrt{2}}\\ \sin^{-1}\left[2x\sqrt{1-x^2}\right], & -\dfrac{1}{\sqrt{2}}\le x\le\dfrac{1}{\sqrt{2}}\\ \pi-\sin^{-1}\left[2x\sqrt{1-x^2}\right], & x>\dfrac{1}{\sqrt{2}}\end{cases}$$

$$2\tan^{-1}x=\cos^{-1}\frac{1-x^2}{1+x^2},\ x\ge 0.\quad 2\tan^{-1}x=\sin^{-1}\frac{2x}{1+x^2}=\tan^{-1}\frac{2x}{1-x^2},\quad |x|<1$$

$$2\tan^{-1}x = \pi - \sin^{-1}\frac{2x}{1+x^2} = \pi + \tan^{-1}\frac{2x}{1-x^2}, \qquad x > 1.$$

ILLUSTRATIVE PROBLEMS

1. If $\tan(A - B) = 1$ and $\sec(A + B) = 2/\sqrt{3}$, find the smallest positive values of A and B and their most general values.

Sol. For the smallest positive values, find A + B and A – B between 0 and 2π from the two given equations. Since A and B are positive angles, A + B > A – B. Solve the two equations to get A and B.
For the most general values, find the general values of A – B and A + B by solving the given equations separately. Solve the two to get A and B.

$$\tan(A - B) = 1 \Rightarrow A - B = \frac{\pi}{4} \text{ or } \frac{5\pi}{4} \qquad \ldots\ldots (1)$$

$$\text{Also } \sec(A + B) = \frac{2}{\sqrt{3}} \Rightarrow A + B = \frac{\pi}{6} \text{ or } \frac{11\pi}{6}.$$

$$\text{Since } A+B > A - B,\ A + B = \frac{11\pi}{6} \qquad \ldots\ldots (2)$$

Solving (1) and (2),

$$\text{we get } A = \frac{25\pi}{24},\ B = \frac{19\pi}{24} \text{ or } A = \frac{37\pi}{24},\ B = \frac{7\pi}{24}$$

For the most general values.

$$\tan(A - B) = 1 \Rightarrow A - B = n\pi + \frac{\pi}{4} \qquad \ldots\ldots (3)$$

$$\sec(A + B) = \frac{2}{\sqrt{3}} \Rightarrow \cos(A+B) = \frac{\sqrt{3}}{2}$$

$$\Rightarrow A + B = 2m\pi \pm \frac{\pi}{6} \qquad \ldots\ldots (4)$$

Solving (3) and (4), we get $A = \frac{1}{2}\left[(2m+n)\pi + \frac{\pi}{4} \pm \frac{\pi}{6}\right]$

$B = \frac{1}{2}\left[(2m-n)\pi - \frac{\pi}{4} \pm \frac{\pi}{6}\right]$, where $m, n \in I$.

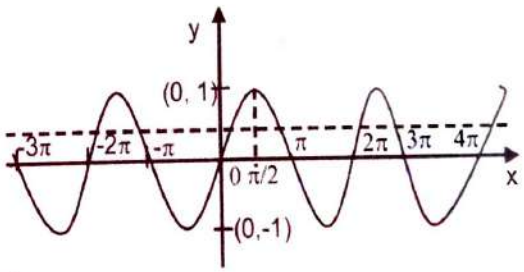
2. Find the solution set of inequality $\sin x > 1/2$.

Sol. When $\sin x = 1/2$, the two values of x between 0 and 2π are $\pi/6$ and $5\pi/6$.
From, the graph of $y = \sin x$, it is obvious that, between 0 and 2π

$$\sin x > 1/2 \text{ for } \frac{\pi}{6} < x < \frac{5\pi}{6}.$$

Hence $\sin x > 1/2 \Rightarrow 2n\pi + \pi/6 < x < 2n\pi + 5\pi/6$.

The required solution set is $\bigcup_{n\in I}\left(2n\pi + \frac{\pi}{6},\ 2n\pi + \frac{5\pi}{6}\right)$.



3. Solve $\sin^2 x + \cos^2 y = 2\operatorname{cosec}^2 z$.

Sol. L.H.S. $= \sin^2 x + \cos^2 y \le 2$. R.H.S. $= 2\operatorname{cosec}^2 z \ge 2$

Hence L.H.S. = R.H.S. only when $\sin^2 x = 1$, $\cos^2 y = 1$, $\operatorname{cosec}^2 z = 1$

$\Rightarrow \cos^2 x = 0$, $\sin^2 y = 0$, $\sin^2 z = 1 \Rightarrow \cos x = 0$, $\sin y = 0$, $\cos z = 0$

$x = (2m+1)\frac{\pi}{2}$, $y = n\pi$ and $z = (2t+1)\frac{\pi}{2}$ where m, n,t are integers.

4. Find all the values of 'θ' in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ satisfying the equation

$(1 - \tan\theta)(1 + \tan\theta)\sec^2\theta + 2^{\tan^2\theta} = 0$

Sol. $(1 - \tan\theta)(1 + \tan\theta)\sec^2\theta + 2^{\tan^2\theta} = 0$

or $(1 - \tan^2\theta)(1 + \tan^2\theta) + 2^{\tan^2\theta} = 0$

Put $\tan^2\theta = t$

$(1 - t)(1 + t) = 2^{-t}$

$t = 3$, we get

$\tan^2\theta = 3$

$\theta = \pm\frac{\pi}{3}$ is the given interval

5. The number of solution of the equation $1 + \cos x + \cos 2x + \sin x + \sin 2x + \sin 3x = 0$, which satisfy the condition $\frac{\pi}{2} < \left|3x - \frac{\pi}{2}\right| \le \pi$ is

Sol. Given $\frac{\pi}{2} < \left|3x - \frac{\pi}{2}\right| < \pi$

$\Rightarrow \frac{\pi}{2} < \left(3x - \frac{\pi}{2}\right) \le \pi$

or $-\pi \le \left(3x - \frac{\pi}{2}\right) < \frac{\pi}{2}$

$x \in \left[-\frac{\pi}{6}, 0\right) \cup \left(\frac{\pi}{3}, \frac{\pi}{2}\right]$

Now, $1 + \cos x + \cos 2x + \sin x + \sin 2x + \sin 3x = 0$

$\cos x(1 + 2\sin x)(2\cos x + 1) = 0$

$\cos x = 0$

$\sin x = -\frac{1}{2}$

$x = -\frac{\pi}{6}$

or $x = \frac{\pi}{2}$ there are '2' solution

SOLVED PROBLEMS

1. If $y = \cot^{-1}2 + \cot^{-1}8 + \cot^{-1}18 + \ldots + \infty$, then tany is

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$

(C) $\frac{\pi}{4}$ (D) none of these

Sol. **(C).** $y = \cot^{-1}2 + \cot^{-1}8 + \cot^{-1}18 + \ldots + \infty$

n^{th} term $= \cot^{-1}2n^2$

$\Rightarrow t_n = \cot^{-1}(2n-1) - \cot^{-1}(2n+1)$

$$\left(\because \cot^{-1}(2n-1) - \cot^{-1}(2n+1) = \cot^{-1}\left(\frac{(2n-1)(2n+1)+1}{2n+1-2n+1}\right) = \cot^{-1}2n^2\right)$$

$t_1 = \cot^{-1}1 - \cot^{-1}3$

$t_2 = \cot^{-1}3 - \cot^{-1}5$

$t_3 = \cot^{-1}5 - \cot^{-1}7$

$\vdots \quad \vdots \quad \vdots$

$y = \cot^{-1}1 - \cot^{-1}(2n+1)$

as $n \to \infty$, $\cot^{-1}(2n+1) \to 0$

$y = \frac{\pi}{4}$

2. Let $k = 1°$, then $2\sin2k + 4\sin4k + 6\sin6k + \ldots + 180\sin180k$ is equal to

(A) 90cosk (B) 90tan89°

(C) 90 tank (D) 90cot89°

Sol. **(B).** Let $y = 2\sin2° + 4\sin4° + \ldots + 178\sin178°$

$\Rightarrow y = 178\sin178° + 176\sin176° + \ldots + 2\sin2°$

$\Rightarrow 2y = 180(\sin2° + \sin4° + \ldots + \sin178°)$

$\Rightarrow y = 90.\frac{\sin89°}{\sin1°}\sin(90°) = 90\tan89°.$

3. If $x, y > 0$, then the range of $\sin^{-1}\left(\frac{x}{1+x^2}\right) + \sin^{-1}\left(\frac{2y}{1+y^2}\right)$ is

(A) $[0, \pi]$ (B) $(-\pi, \pi]$

(C) $\left[0, \frac{2\pi}{3}\right]$ (D) $\left(0, \frac{2\pi}{3}\right]$

Sol. **(D).** $\frac{1+x^2}{x} \geq 2 \quad (\because x > 0)$

$\Rightarrow \frac{x}{1+x^2} \leq \frac{1}{2} \Rightarrow \sin^{-1}\left(\frac{x}{1+x^2}\right) \in \left(0, \frac{\pi}{6}\right]$

similarly $\sin^{-1}\left(\frac{2y}{1+y^2}\right) \in \left(0, \frac{\pi}{2}\right]$

$\Rightarrow$ range of given expression is $\left(0, \frac{2\pi}{3}\right]$.

4. Number of solution of the equation $\tan^{-1}\left(\frac{x}{1-x^2}\right) + \tan^{-1}\left(\frac{1}{x^3}\right) = \frac{3\pi}{4}$, belonging to the interval $(0, 1)$ is

(A) 0 (B) 1

(C) 2 (D) infinite

Sol. **(A).** $\left(\frac{x}{1-x^2}\right)\times\frac{1}{x^3}=\frac{1}{\left(1-x^2\right)x^2}>1$

So $\tan^{-1}\left(\frac{x}{1-x^2}\right)+\tan^{-1}\left(\frac{1}{x^3}\right)=\pi+\tan\frac{\frac{x}{1-x^2}+\frac{1}{x^3}}{1-\frac{1}{x^2\left(1-x^2\right)}}$

$=\pi+\tan^{-1}\frac{x^4+1-x^2}{\left(x^2-x^4-1\right)x}=\pi+\tan^{-1}\left(-\frac{1}{x}\right)=\frac{3\pi}{4}\Rightarrow x=1.$

5. The number of values of α in the interval $[-\pi, 0]$ satisfying $\sin\alpha+\int_{\alpha}^{2\alpha}\cos 2x\ dx=0$

(A) 2
(B) 3
(C) 4
(D) 5

Sol. **(B).** $\sin\alpha+\left[\frac{\sin 2x}{2}\right]_{\alpha}^{2\alpha}=0\Rightarrow\sin\alpha(1+\cos 3\alpha)=0.$

6. The least positive solution of $\cot\left(\frac{\pi}{3\sqrt{3}}\sin 2x\right)=\sqrt{3}$ lies in

(A) $\left(0, \frac{\pi}{6}\right]$
(B) $\left(\frac{\pi}{9}, \frac{\pi}{6}\right]$
(C) $\left(\frac{\pi}{9}, \frac{\pi}{3}\right]$
(D) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right]$

Sol. **(A).** $\cot\left(\frac{\pi}{3\sqrt{3}}\sin 2x\right)=\sqrt{3}\Rightarrow\frac{\pi}{3\sqrt{3}}\sin 2x=n\pi+\frac{\pi}{6}\Rightarrow\sin 2x=3\sqrt{3}\,n+\frac{\sqrt{3}}{2}\Rightarrow n=0.$

7. The number of solutions of $\theta\in[0, 2\pi]$ satisfying the equation $\left(\log_{\sqrt{3}}\tan\theta\right)\left(\sqrt{\left(\log_{\tan\theta}3+\log_{\sqrt{3}}3\sqrt{3}\right)}\right)=-1$, is/are

(A) 0
(B) 1
(C) 2
(D) > 2

Sol. **(C).** $\log_{\sqrt{3}}\tan\theta\left[\sqrt{\frac{\log_{\sqrt{3}}3}{\log_{\sqrt{3}}\tan\theta}+\frac{\log\left(\sqrt{3}\right)^3}{\log\sqrt{3}}}\right]=-1$

$\Rightarrow\log_{\sqrt{3}}\tan\theta\left[\sqrt{\frac{2}{\log_{\sqrt{3}}\tan\theta}+3}\right]=-1$

Let $\log_{\sqrt{3}}\tan\theta=y$

$\Rightarrow y\sqrt{\frac{2}{y}+3}=-1\Rightarrow\sqrt{\frac{2}{y}+3}=\frac{-1}{y}$

$\Rightarrow\frac{2}{y}+3=\frac{1}{y^2}$ or $y^2(2+3y)=y\Rightarrow y[3y^2+2y-1]=0$

$\because y<0.$

$y(3y-1)(y+1)=0$

$y=-1$ ($\because$ y can not be positive)

$\Rightarrow \log_{\sqrt{3}} \tan\theta = -1$

$\tan\theta = \frac{1}{\sqrt{3}} \therefore \theta = \frac{\pi}{6}$ and $\frac{7\pi}{6}$.

$\therefore$ There are two values of θ in $[0, 2\pi]$.

8. If k times the sum of first n natural numbers is equal to sum of the squares of first n natural numbers, then $\sin^{-1}\left(\frac{9k^2-4n^2}{6k+4n}\right)$ is equal to

(A) π (B) $\frac{\pi}{2}$

(C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

Sol. **(D).** $k\frac{n(n+1)}{2} = \frac{n(n+1)(2n+1)}{6}$

$2n+1 = 3k$

$\sin^{-1}\left(\frac{9k^2-4n^2}{6k+4n}\right) = \sin^{-1}\left(\frac{(3k+2n)(3k-2n)}{2(3k+2n)}\right) = \sin^{-1}\frac{1}{2} = \frac{\pi}{6}$

9. If $2^{\sqrt{\sin^2 x - 2\sin x + 5}} \frac{1}{4^{\operatorname{cosec}^2 y}} \le 1$, then ordered pair (x, y) is equal to

(A) $x = (4n+1)\frac{\pi}{2}$, $y = (2m+1)\frac{\pi}{2}$ (B) $x = 2n\pi$, $y = 2m\pi$

(C) $x = (2n+1)\frac{\pi}{2}$, $y = (2m+1)\frac{\pi}{2}$ (D) none of these

Sol. **(A).** $\sin^2 x - 2\sin x + 5 = (\sin x - 1)^2 + 4 \ge 4$

$\therefore 2^{\sqrt{\sin^2 x - 2\sin x + 5}} \ge 2^2 = 4$ and $\operatorname{cosec}^2 y \ge 1 \Rightarrow \frac{1}{4^{\operatorname{cosec}^2 y}} \ge \frac{1}{4}$

$\therefore$ L.H.S. $\ge 4 \times \frac{1}{4} = 1$ and R.H.S. $= 1$

$\therefore \sin^2 x - 2\sin x + 5 = 4$

$(\sin x - 1)^2 = 0$

$\sin x = 1 \Rightarrow x = (4n+1)\frac{\pi}{2}$

and $\operatorname{cosec}^2 y = 1$, $\sin^2 y = 1$ or $\cos y = 0$

$y = (2m+1)\frac{\pi}{2}$.

10. The smallest positive root of the equation $\sqrt{\sin(1-x)} = \sqrt{\cos x}$ is

(A) $\frac{\pi}{4}$ (B) $\frac{1}{2} + \frac{\pi}{4}$

(C) $\frac{1}{2} + \frac{3\pi}{4}$ (D) $\frac{1}{2} + \frac{7\pi}{4}$

Sol. **(D).** The given equation holds if $\sin(1-x) \geq 0$ and $\cos x \geq 0$

Squaring, we get $\sin(1-x) = \cos x$

$\Rightarrow x = 2n\pi \pm \left\{\frac{\pi}{2} - 1 + x\right\}$ or $x = n\pi - \frac{\pi}{4} + \frac{1}{2}, n \in I$ ...(i)

If $n \leq 0$, then $x < 0$

If $n = 1$, $x = \frac{1}{2} + \frac{3\pi}{4}$.

Clearly $x \in \left(\frac{\pi}{2}, \pi\right)$, so $\cos x < 0$, hence rejected.

If $n = 2$, $x = \frac{1}{2} + \frac{7\pi}{4}$.

Then $\cos x = \cos\left\{2\pi - \frac{\pi}{4} + \frac{1}{2}\right\} = \cos\left(\frac{\pi}{4} - \frac{1}{2}\right) > 0$

Also, $\sin(1-x) = \sin\left(\frac{1}{2} - \frac{7\pi}{4}\right) = \sin\left(\frac{1}{2} + \frac{\pi}{4} - 2\pi\right) = \sin\left(\frac{1}{2} + \frac{\pi}{4}\right) > 0$

Hence $x = \frac{1}{2} + \frac{7\pi}{4}$ is the smallest positive root.

11. If $\sin^{-1}x + \sin^{-1}y = -\pi - \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$, then the area of the region or rectangular Cartesian plane in which the point P(x, y) can lie is

(A) π (B) $\frac{4-\pi}{4}$

(C) $\frac{2-\pi}{2}$ (D) $\pi - 2$

Sol. **(B).** $\sin^{-1}x + \sin^{-1}y = -\pi - \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$, if $-1 \leq x < 0$, $-1 \leq y < 0$ and $x^2 + y^2 > 1$.

12. If $k_1 = \sin x \cos^3 x$ and $k_2 = \cos x \sin^3 x$ then

(A) $k_1 - k_2 > 0, x \in \left(0, \frac{\pi}{4}\right)$ (B) $k_1 - k_2 < 0, x \in \left(0, \frac{\pi}{4}\right)$

(C) $k_1 + k_2 > 0, x \in \left(0, \frac{\pi}{2}\right)$ (D) $k_1 + k_2 < 0, x \in \left(0, \frac{\pi}{4}\right)$

Sol. **(A, C).** $k_1 - k_2 = \sin x \cos x (\cos^2 x - \sin^2 x)$

$= \frac{1}{2}\sin 2x \cos 2x$

$= \frac{1}{4}\sin 4x$

$k_1 - k_2 > 0, 0 < 4x < \pi$

$0 < x < \frac{\pi}{4}$

$k_1 + k_2 = \frac{\sin 2x}{2}$

$k_1 + k_2 > 0$

$0 < 2x < \pi$

$0 < x < \frac{\pi}{2}$.

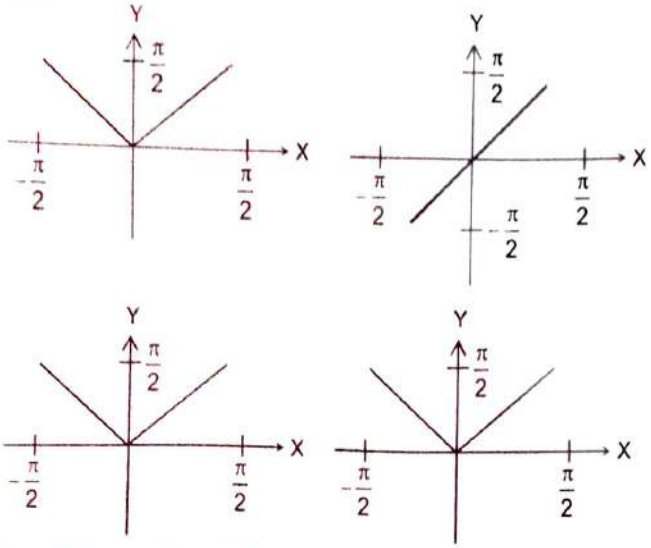
13. If $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$ then which of the following is / are correct?

(A) $\cos^{-1}\cos x = |x|$ (B) $\sin^{-1}\sin x = |x|$

(C) $\cos^{-1}\cos|x| = |x|$ (D) $\sin^{-1}\sin|x| = |x|$

Sol. **(A, C, D).**

The graph of $y = \cos^{-1}\cos x$, $y = \sin^{-1}\sin x$, $y = \cos^{-1}\cos|x|$ and $y = \sin^{-1}\sin|x|$ are shown below in that order



Except the second graph all other represent $y = |x|$ also.

Comprehension

Use boundedness for solving Trigonometric equations

Consider an equation $\sin x + \sin y = 2$ ...(1)

We know that $\sin x \le 1$ and $\sin y \le 1$ for all x, y

So, $\sin x + \sin y \le 2$ for all x and y

Therefore, $\sin x + \sin y = 2$ if and only if $\sin x = 1$ and $\sin y = 1$

$\Leftrightarrow x = 2n\pi + \frac{\pi}{2}$ and $y = 2m\pi + \frac{\pi}{2}$

Which is the required solution of equation? To solve the equation (1), we have used the boundedness of sin x rather than using conventional methods of solving equations

In general we employ one or more of the following extreme value conditions

i) $-1 \le \sin x \le 1 \Rightarrow |\sin x| \le 1$ and $\sin^2 x \le 1$

ii) $-1 \le \cos x \le 1 \Rightarrow |\cos x| \le 1$ and $\cos^2 x \le 1$

iii) $-\sqrt{a^2+b^2} \le a\sin x + b\cos x \le \sqrt{a^2+b^2} \Rightarrow |a\sin x + b\cos x| \le \sqrt{a^2+b^2}$

Answer the following question:

14. The minimum value of $27^{\cos 2x} \cdot 81^{\sin 2x}$ is

(A) 1 (B) 1/9

(C) 1/81 (D) 1/243

Sol. **(D).** Let $y = 27^{\cos 2x} \cdot 81^{\sin 2x} = 3^{3\cos 2x + 4\sin 2x}$. Now $-\sqrt{3^2+4^2} \le 3\cos 2x + 4\sin 2x \le \sqrt{3^2+4^2}$

or $-5 \le 3\cos 2x + 4\sin 2x \le 5 \therefore 3^{-5} \le 3^{3\cos 2x + 4\sin 2x} \le 3^5$

15. Number of roots of the equation $\cos^7 x + \sin^4 x = 1$ in the internal $[0, 2\pi]$ is
(A) 0 (B) 1
(C) 2 (D) 4

Sol. **(D).** $\cos^7 x \le \cos^2 x$ and $\sin^4 x = \sin^2 x$
$\Rightarrow \cos^7 x + \sin^4 x \le \cos^2 x + \sin^2 x = 1$
So, the given equation is satisfies if and only if
$\cos^7 x = 0$ and $\sin^2 = 1$ or $\cos^7 = 1$ and $\sin^2 x = 0$
$\Leftrightarrow x = (2n+1)\frac{\pi}{2}$ or $x = 2m\pi$
$\because 0 \le x \le 2\pi$, so $x = 0, \frac{\pi}{2}, \frac{3\pi}{2}, 2\pi$

16. The smallest positive number p for which the equation $\cos(p \sin x) = \sin(p \cos x)$ has a solution in $[0, 2\pi]$ is
(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$
(C) $\frac{\pi}{4\sqrt{2}}$ (D) $\frac{\pi}{2\sqrt{2}}$

Sol. **(D).** $\cos(p \sin x) = \sin(p \cos x) = \cos\left(\frac{\pi}{2} - p\cos x\right) \Rightarrow p \sin x = 2n\pi \pm \left(\frac{\pi}{2} - p\cos x\right)$, $n \in I$
consider the positive sign, we have
$p(\sin x + \cos x) = (4n+1)\frac{\pi}{2} \Rightarrow \sin x + \cos x = \frac{(4n+1)\pi}{2p}$
$\because |\cos x + \sin x| \le \sqrt{2} \Rightarrow \left|\frac{(4n+1)\pi}{2p}\right| \le \sqrt{2}$ or $p \ge \frac{|(4n+1)|\pi}{2\sqrt{2}}$ (for positive values of p)
p is smallest positive number if $n = 0$, giving $p = \frac{\pi}{2\sqrt{2}}$
Now consider the negative sign, we have
$p(\sin x - \cos x) = \frac{|(4n-1)|\pi}{2}$. So $\left|\frac{(4n-1)\pi}{2}\right| \le \sqrt{2}$ or $p \ge \frac{|(4n-1)|\pi}{2\sqrt{2}}$
p is smallest positive number if $n = 0$, giving again $p = \frac{\pi}{2\sqrt{2}}$.

Assertion Reason

17. Statement –1: $2\cos^{-1}x = \sin^{-1}\left(2x\sqrt{1-x^2}\right)$ is an identity
Because
Statement –2: $x = \cos\theta$ satisfies above $\forall\ \theta \in R$
(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1
(B) Statement -1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement -1 is True, Statement -2 is False
(D) Statement -1 is False, Statement -2 is True

Sol. **(C).**

18. Match the following:

Column – I		Column – II	
(A)	The number of solution of $\frac{x}{2}+\frac{\sin x}{\cos x}=\frac{\pi}{4}$ in $[-\pi, \pi]$	(p)	1
(B)	The number of solution of equation $\sin^{-1}\left(\lvert x^2-1\rvert\right)+\cos^{-1}\left(\lvert 2x^2-5\rvert\right)=\frac{\pi}{2}$	(q)	0
(C)	The number of solution of $x^4-2x^2\sin^2\frac{\pi}{2}x+1=0$	(r)	3
(D)	The number of solution of $x^2+2x+2\sec^2\pi x+\tan^2\pi x=0$	(s)	2
		(t)	1/2

Sol. **(A) → (r); (B) → (s); (C) → (s); (D) → (q)**

(A) $\frac{x}{2}+\tan x=\frac{\pi}{4}$

from graph, the equation has 3 solutions in $[-\pi, \pi]$.

(B) $\sin^{-1}|x^2-1|+\cos^{-1}|2x^2-5|=\frac{\pi}{2}\Rightarrow|x^2-1|=|2x^2-5|$

$\Rightarrow x^2=2\Rightarrow x=\pm\sqrt{2}$ (Two solutions)

(C) $x^4-2x^2\sin\frac{\pi x}{2}+1=0\Rightarrow\left(x^2-\sin^2\frac{\pi x}{2}\right)^2+1-\sin^4\frac{\pi x}{2}=0$

$\Rightarrow x=(2n+1), n\in I$ and $x^2=\sin^2\frac{\pi x}{2}=1$

$\Rightarrow x=\pm 1$ is the solution.

(D) Let $y=x^2+2x+2\sec^2\pi x+\tan^2\pi x$

$\Rightarrow y=(x+1)^2+(2\sec^2\pi x-1)+\tan^2\pi x>0\Rightarrow$ no solution.

Answer Q.19, Q.20 and Q.21 by appropriately matching the information given in the three columns of the following table.

Column-I: represents trigonometric equation, Column-II: represents general solution of the equation and Column-III: represents number of solutions of the equation in $[0, \pi]$

Column–I	Column–II	Column–III
(I) $\sin^2 x+\cos^2 3x=1$	(i) $n\pi$	(P) 1
(II) $e^{\cot^2\theta}+\sin^2\theta-2\cos^2 2\theta+4=4\sin\theta$	(ii) $\frac{n\pi}{4}$	(Q) 2
(III) $a^2\sin x-a\sin 2x+\sin x=0, a\in R$	(iii) $n\pi+\frac{\pi}{4}$	(R) 3
(IV) $\sqrt[3]{2\tan\theta-1}+\sqrt[3]{\tan\theta-1}=1$	(iv) $(4n+1)\frac{\pi}{2}$	(S) 5

19. Which of the following combination is **CORRECT**?

(A) (I) (i) (S) (B) (II) (iv) (P)

(C) (III) (i) (P) (D) (II) (iii) (R)

Sol. **(B)**

20. Which of the following combination is **CORRECT**?
(A) (I) (ii) (S) (B) (II) (ii) (P)
(C) (III) (i) (R) (D) (IV) (iii) (S)

Sol. **(A)**

21. Which of the following combination is **INCORRECT**?
(A) (I) (ii) (S) (B) (II) (iii) (R)
(C) (III) (i) (Q) (D) (IV) (iii) (P)

Sol. **(B)**

19.-21. (I) $\sin^2 x + \cos^2 3x = 1$

$\Rightarrow \sin^2 x = \sin^2 3x \Rightarrow 3x = n\pi \pm x$

$\Rightarrow 4x = n\pi$ or $2x = n\pi$

$\Rightarrow x = \frac{n\pi}{4}, \frac{n\pi}{2} \Rightarrow x = \frac{n\pi}{4}$

Number of solutions in $[0, \pi] = 5$

(II) $e^{\cot^2\theta} + \sin^2\theta - 2\cos^2 2\theta + 4 = 4\sin\theta \Rightarrow e^{\cot^2\theta} + (\sin\theta - 2)^2 = 2\cos^2 2\theta$

LHS ≥ 2, RHS $\leq 2 \Rightarrow$ LHS = RHS = 2

$\Rightarrow \cot^2\theta = 0, \cos^2 2\theta = 1$ and $\sin\theta = 1 \Rightarrow \theta = (4n+1)\frac{\pi}{2}$

Number of solutions in $[0, \pi] = 1$

(III) $a^2 \sin x - a \sin 2x + \sin x = 0 \Rightarrow \sin x\,(a^2 - 2a\cos x + 1) = 0$

$\Rightarrow \sin x\,((a - \cos x)^2 + \sin^2 x) = 0 \Rightarrow \sin x = 0$ and $a = \cos x$

$\Rightarrow x = n\pi$ number of solutions in $[0, \pi] = 2$

(IV) $\sqrt[3]{2\tan\theta - 1} + \sqrt[3]{\tan\theta - 1} = 1$

$\Rightarrow (2\tan\theta - 1) + (\tan\theta - 1) + 3\sqrt[3]{(2\tan\theta - 1)(\tan\theta - 1)}\left(\sqrt[3]{2\tan\theta - 1} + \sqrt[3]{\tan\theta - 1}\right) = 1$

$\Rightarrow 3\sqrt[3]{2\tan^2\theta - 3\tan\theta + 1} = 3(1 - \tan\theta)$

$\Rightarrow (2\tan\theta - 1)(\tan\theta - 1) = (1 - \tan\theta)^3$

$\Rightarrow (1 - \tan\theta)(1 + \tan^2\theta - 2\tan\theta + 2\tan\theta - 1) = 0$

$\Rightarrow (1 - \tan\theta)\tan^2\theta = 0$

$\Rightarrow \tan\theta = 1$ as $\tan\theta \neq 0$

$\Rightarrow \theta = n\pi + \frac{\pi}{4}$

Number of solutions = 1

22. The number of solution $\sum_{r=1}^{5} \cos rx = 5$ in the interval $[0, 2\pi]$ is _____

Sol. **(5.00)**

$\sum_{r=1}^{5} \cos rx = 5$

$\Rightarrow \cos x + \cos 2x + \cos 3x + \cos 4x + \cos 5x = 5$

Which is possible only when $\cos x = \cos 2x = \cos 3x = \cos 4x = \cos 5x = 1$

$\Rightarrow x = 0$ and $x = 2\pi$

23. Number of value(s) of α for which the three elements set $S = \{\sin\alpha, \sin 2\alpha, \sin 3\alpha\}$ is equal to the three element set $T = \{\cos\alpha, \cos 2\alpha, \cos 3\alpha\}$, where $\alpha \in (-\pi, \pi)$ _____

Sol. **(4.00)**

S and T can be the same set if the elements of S and T are the roots of the same cubic i.e.

(i) $\sin\alpha + \sin 2\alpha + \sin 3\alpha = \cos\alpha + \cos 2\alpha + \cos 3\alpha$

$2\sin 2\alpha\cos\alpha + \sin 2\alpha = 2\cos 2\alpha\cos\alpha + \cos 2\alpha$

$\sin 2\alpha(2\cos\alpha + 1) = \cos 2\alpha(2\cos\alpha + 1)$

$\therefore (\sin 2\alpha - \cos 2\alpha)(2\cos\alpha + 1) = 0 \Rightarrow \tan 2\alpha = 1$ or $\cos\alpha = -\frac{1}{2}$

$\therefore 2\alpha = n\pi + \frac{\pi}{4}$ or $\alpha = 2n\pi \pm \frac{2\pi}{3}$; $\alpha = \frac{n\pi}{2} + \frac{\pi}{8}$ or $\alpha = 2n\pi + \frac{2\pi}{3}$ or $2n\pi - \frac{2\pi}{3}$ (rejected)

(ii) $\sin\alpha\sin 2\alpha + \sin 2\alpha\sin 3\alpha + \sin 3\alpha\sin\alpha = \cos\alpha\cos 2\alpha + \cos 2\alpha\cos 3\alpha + \cos 3\alpha\cos\alpha$

$\cos\alpha - \cos 3\alpha + \cos\alpha - \cos 5\alpha + \cos 2\alpha - \cos 4\alpha$

$= \cos 3\alpha + \cos\alpha + \cos 5\alpha + \cos\alpha + \cos 4\alpha + \cos 2\alpha$

$2(\cos 3\alpha + \cos 4\alpha + \cos 5\alpha) = 0$

$\cos 4\alpha + 2\cos 4\alpha\cos\alpha = 0 \Rightarrow \cos 4\alpha(2\cos\alpha + 1) = 0$

$4\alpha = 2n\pi \pm \frac{\pi}{2} \Rightarrow \alpha = \frac{n\pi}{2} \pm \frac{\pi}{8}$ or $\alpha = 2n\pi + \frac{2\pi}{3}$ or $2n\pi - \frac{2\pi}{3}$ (rejected)

(iii) $\sin\alpha\sin 2\alpha\sin 3\alpha = \cos\alpha\cos 2\alpha\cos 3\alpha$ is satisfied by only by $\alpha = \frac{n\pi}{2} + \frac{\pi}{8}$

24. If the sum of all values of x satisfying the system of equations $\tan x + \tan y + \tan x\tan y = 5$ and $\sin(x + y) = 4\cos x\cos y$ is $\frac{25k\pi}{2}$, where $x \in \left(0, \frac{\pi}{2}\right)$, then k is equal to _____

Sol. **(0.04)**

$\sin(x + y) = 4\cos x\cos y \Rightarrow \sin x\cos y + \cos x\sin y = 4\cos x\cos y$

$\Rightarrow \tan x + \tan y = 4 \Rightarrow \tan x\tan y = 1$

$\Rightarrow \tan x + \frac{1}{\tan x} = 4 \Rightarrow \tan^2 x - 4\tan x + 1 = 0$

$\Rightarrow \tan x = \frac{4 \pm \sqrt{12}}{2} = 2 \pm \sqrt{3} \Rightarrow x = \frac{\pi}{12}, \frac{5\pi}{12}$

$\Rightarrow \frac{25k\pi}{2} = \frac{\pi}{2} \Rightarrow k = \frac{1}{25} = 0.04$

25. Find the number of solution of the equation $2^{\sin^2 x} + 5\cdot 2^{\cos^2 x} = 7$ in the interval [0, 315] is k, then $\frac{k}{7}$ is equal to _____

Sol. **(14.28)**

$2^{\sin^2 x} + 5\cdot 2^{\cos^2 x} = 7$

$2^{\sin^2 x} + 5\cdot 2^{1-\sin^2 x} = 7$

$2^{\sin^2 x} = t \Rightarrow t + \frac{10}{t} = 7 \Rightarrow t^2 - 7t + 10 = 0 \Rightarrow (t - 5)(t - 2) = 0 \Rightarrow t = 2$ or $5 \Rightarrow t = 2^{\sin^2 x} = 2$ or 5

But $2^{\sin^2 x} \neq 5$ (think!)

$\therefore 2^{\sin^2 x} = 2 = 2^1 \Rightarrow \sin^2 x = 1$

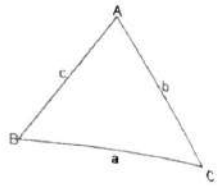
Hence, value of x in [0, 315] is $\frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots \frac{199\pi}{2}$

$\therefore$ Total number of solution in [0, 315] is 100

SOLUTION OF TRIANGLES

BASIC RULES

In a triangle ABC, the angles are denoted by capital letters A, B, and C and the lengths of the sides opposite to these angles are denoted by small letters a, b, and c respectively. Semi – perimeter of the triangle is written as $s = \frac{a+b+c}{2}$, and its area denoted by S or Δ.



Sine Rule

$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} = \frac{1}{2R}$, where R is the radius of the circumcircle of the Δ ABC.

Cosine Rule

$\cos A = \frac{b^2+c^2-a^2}{2bc}$, $\cos B = \frac{a^2+c^2-b^2}{2ac}$, $\cos C = \frac{a^2+b^2-c^2}{2ab}$.

Projection Rule

$a = b\cos C + c\cos B$, $b = c\cos A + a\cos C$, $c = a\cos B + b\cos A$.

Napier's Analogy

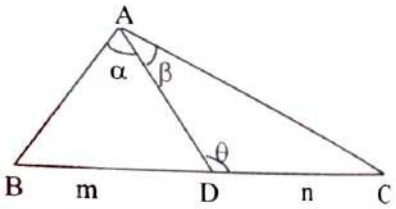
$\tan\frac{B-C}{2} = \frac{b-c}{b+c}\cot\frac{A}{2}$, $\tan\frac{C-A}{2} = \frac{c-a}{c+a}\cot\frac{B}{2}$, $\tan\frac{A-B}{2} = \frac{a-b}{a+b}\cot\frac{C}{2}$.

m-n Theorem

If in a triangle ABC, D is a point on the line BC such that BD:DC = m:n and ∠ADC = θ, ∠BAD = α, ∠DAC = β, then

(a) $(m+n)\cot\theta = m\cot\alpha - n\cot\beta$

(b) $(m+n)\cot\theta = n\cot B - m\cot C$



AUXILIARY FORMULAE

Trigonometric Ratios of Half-Angles

$\sin\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$, $\sin\frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{ca}}$, $\sin\frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$,

$\cos\frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$, $\cos\frac{B}{2} = \sqrt{\frac{s(s-b)}{ca}}$, $\cos\frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$,

$$\tan\frac{A}{2}=\sqrt{\frac{(s-b)(s-c)}{s(s-a)}},\ \tan\frac{B}{2}=\sqrt{\frac{(s-c)(s-a)}{s(s-b)}},\ \tan\frac{C}{2}=\sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$$

Area of a Triangle

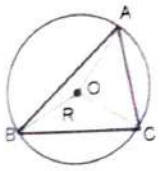
$$\Delta=\frac{1}{2}bc\sin A=\frac{1}{2}ca\sin B=\frac{1}{2}ab\sin C=\sqrt{s(s-a)(s-b)(s-c)}=\frac{abc}{4R}=rs$$

Where R and r are the radii of the circumcircle and the incircle of the Δ ABC respectively.

CIRCLES CONNECTED WITH TRIANGLE

Circumcircle

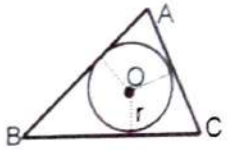
The circle passing through the vertices of the triangle ABC is called the circum-circle. Its radius R is called the circum-radius. In the triangle ABC,



$$R=\frac{a}{2\sin A}=\frac{b}{2\sin B}=\frac{c}{2\sin C}=\frac{abc}{4\Delta}.$$

In-Circle

The circle touching the three sides of the triangle internally is called the inscribed or the in-circle of the triangle. Its radius r is called the in-radius of the circle. In the triangle ABC,



$$r=\frac{\Delta}{s}=(s-a)\tan\frac{A}{2}=(s-b)\tan\frac{B}{2}=(s-c)\tan\frac{C}{2}=4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}.$$

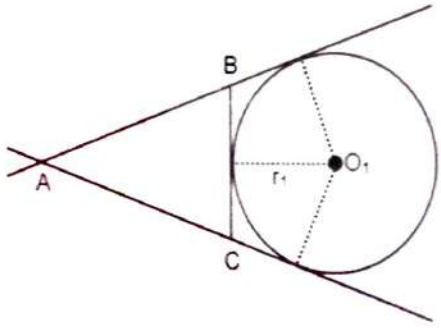
Remark:

- From $r=4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$, we find that $r\le 4R.\frac{1}{8}$

  $\Rightarrow 2r\le R$. Here equality holds for the equilateral triangle.

Escribed Circles

The circle touching BC and the two sides AB and AC produced of Δ ABC externally is called the escribed circle opposite A. Its radius is denoted by r_1. Similarly r_2, and r_3 denote the radii of the escribed circles opposite angles B and C respectively. r_1, r_2, r_3 are called the ex-radii of Δ ABC. Here



$$r_1=\frac{\Delta}{s-a}=s\tan\frac{A}{2}=4R\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2},$$

$$r_2=\frac{\Delta}{s-b}=s\tan\frac{B}{2}=4R\sin\frac{B}{2}\cos\frac{C}{2}\cos\frac{A}{2},$$

$$r_3=\frac{\Delta}{s-c}=s\tan\frac{C}{2}=4R\sin\frac{C}{2}\cos\frac{A}{2}\cos\frac{B}{2}.$$

SOLUTION OF TRIANGLES

The three sides a, b, c and the three angles A, B, C are called the elements of the triangle ABC. When any three of these six elements (except all the three angles) of a triangle are given, the triangle is known completely; that is the other three elements can be expressed in terms of the given elements and can be evaluated. This process is called the solution of triangles.

- If the three sides a,b,c are given, angle A is obtained from $\tan\frac{A}{2}=\sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$

  or $\cos A=\frac{b^2+c^2-a^2}{2bc}$. B and C can be obtained in a similar way.

- If two sides b and c and the included angle A are given, then $\tan\frac{B-C}{2}=\frac{b-c}{b+c}\cot\frac{A}{2}$ gives $\frac{B-C}{2}$. Also $\frac{B+C}{2}=90^\circ-\frac{A}{2}$, so that B and C can be evaluated. The third side is given by $a=b\frac{\sin A}{\sin B}$ or $a^2=b^2+c^2-2bc\cos A$.

- If two sides b and c and the angle B (opposite to side b) are given, then $\sin C=\frac{c}{b}\sin B$, $A=180^\circ-(B+C)$ and $a=\frac{b\sin A}{\sin B}$ give the remaining elements. If $b<c\sin B$, there is no triangle possible (fig1). If $b=c\sin B$ and B is an acute angle, then there is only one triangle possible (fig 2).If $c\sin B<b<c$ and B is an acute angle, then there are two value of angle C (fig 3). If $c<b$ and B is an acute angle, then there is only one triangle (fig 4).

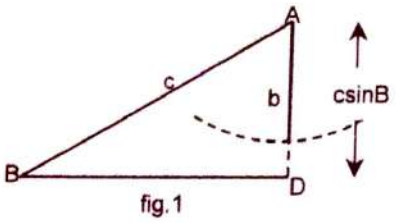


fig.1

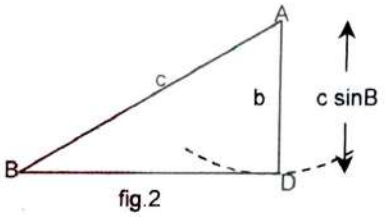


fig.2

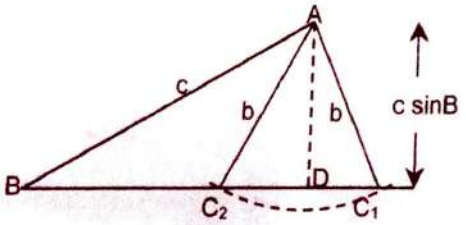


fig.3

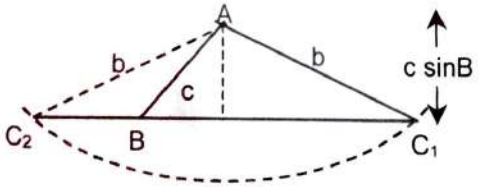


fig. 4

This is, sometimes, called an ambiguous case.

Alternative Method:

By applying cosine rule, we have $\cos B=\frac{a^2+c^2-b^2}{2ac}$

$\Rightarrow a^2-(2c\cos B)a+(c^2-b^2)=0 \Rightarrow a=c\cos B\pm\sqrt{(c\cos B)^2-(c^2-b^2)}$

$\Rightarrow a=c\cos B\pm\sqrt{b^2-(c\sin B)^2}$

This equation leads to following cases:

Case I: If $b<c\sin B$, no such triangle is possible .

Case II: Let $b = c \sin B$. There are further following case:

(a) B is an obtuse angle $\Rightarrow$ cosB is negative. There exists no such triangle.

(b) B is an acute angle$\Rightarrow$cosB is positive. There exists only one such triangle.

Case III: Let $b > c \sin B$. There are further following cases:

(a) B is an acute angle $\Rightarrow$ cosB is positive. In this case two values of a will exists if and only if $c \cos B > \sqrt{b^2 - (c \sin B)^2}$ or $c > b \Rightarrow$ Two such triangle is possible. If $c < b$, only one such triangle is possible.

(b) B is an obtuse angle $\Rightarrow$ cos B is negative. In this case triangle will exist if and only if $\sqrt{b^2 - (c \sin B)^2} > |c \cos B| \Rightarrow b > c$. So in this case only one such triangle is possible. If $b < c$ there exists no such triangle.

- If one side a and angles B and C are given, then $A = 180° - (B + C)$, and $b = \frac{a \sin B}{\sin A}$, $c = \frac{a \sin C}{\sin A}$.
- If the three angles A, B, C are given, we can only find the ratios of the sides a, b, c by using sine rule (since there are infinite similar triangles possible).

ILLUSTRATIVE PROBLEMS

1. A train travelling on one of two intersecting railway lines, subtends at a certain station on the other line, an angle α when the front of the carriage reaches the junction, and an angle β when the end of the carriage reaches it. Prove that the two lines are inclined to each other at an angle θ determined by $2\cot\theta = \cot\alpha - \cot\beta$.

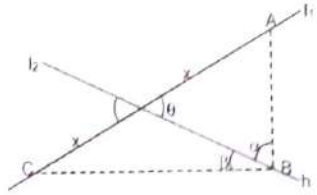
Sol. L_1 and L_2 are two intersecting lines

In ΔABC,

by m-n theorem, we have

$$(x+x)\cot\theta = x\cot\alpha - x\cot\beta$$

$$\Rightarrow 2\cot\theta = \cot\alpha - \cot\beta$$



2. Let O be a point inside a triangle ABC, such that $\angle OAB = \angle OBC = \angle OCA = w$. Then show that $\cot w = \cot A + \cot B + \cot C$.

Sol. In ΔOAC, we have from sine rule

$$\frac{\sin(A-w)}{OC} = \frac{\sin(180-A)}{b} = \frac{\sin A}{b} \quad \ldots(1)$$

Also in ΔOBC

$$\frac{\sin w}{OC} = \frac{\sin(180-C)}{a} = \frac{\sin C}{a} \quad \ldots(2)$$

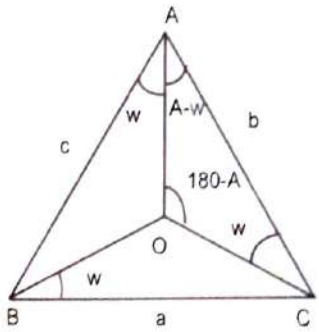
Dividing (1) by (2), we get

$$\frac{\sin(A-w)}{\sin w} = \frac{a\sin A}{b\sin C} = \frac{2R\sin A\sin A}{2R\sin B\sin C}$$

$$= \frac{\sin A\sin(180-B-C)}{\sin B\sin C}$$

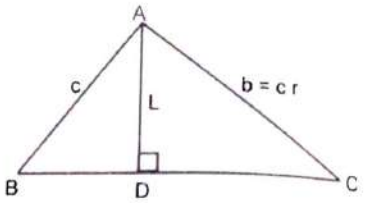
$$\Rightarrow \frac{\sin A\cos w - \sin w\cos A}{\sin w} = \sin A\frac{(\sin B\cos C + \cos B\sin C)}{\sin B\sin C} = \sin A\frac{(\sin B\cos C + \cos B\sin C)}{\sin B\sin C}$$

$\Rightarrow \sin A(\cot w) - \cos A = \sin A(\cot C + \cot B) \Rightarrow \cot w = \cot A + \cot B + \cot C.$



3. In a triangle of base a, the ratio of the other sides is r (< 1). Show that the altitude of the triangle is less than or equal to $\frac{ar}{1-r^2}$.

Sol. Let D be the foot of the altitude from A, and BC = a,
AB = c, AC = cr, AD = L = c sinB



$$\text{Now } \frac{ar}{1-r^2} = \frac{ac^2r}{c^2-r^2c^2}$$

$$= \frac{abc}{c^2-b^2}$$

$$= \frac{a\sin B\sin C}{\sin^2 C-\sin^2 B}$$

$$= \frac{a\sin B\sin C}{\sin(C-B)\sin(C+B)}$$

$$= \frac{a\sin B\sin C}{\sin A\sin(C-B)}$$

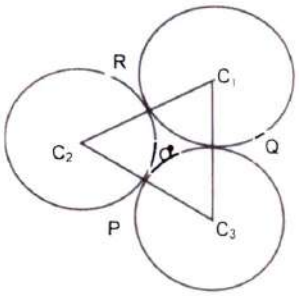
$$= \frac{c\sin B}{\sin(C-B)}$$

$$= \frac{L}{\sin(C-B)}$$

$$\Rightarrow L \le \frac{ar}{1-r^2}.$$

4. Three circles touch one - another externally. The tangents at their points of contact meet at a point whose distance from a point of contact is 4. Find the ratio of the product of the radii to the sum of the radii of circles.

Sol. Let r_1, r_2, and r_3 be the radii of the three circles with centres at C_1, C_2 and C_3. Let the circles touch at P, Q, and R.



Also $C_1C_2 = r_1 + r_2$, $C_2C_3 = r_2 + r_3$, $C_3C_1 = r_3 + r_1$

Let O be the point whose distance from the points of contact is 4.

Then O is the in - centre of the $\Delta C_1C_2C_3$ with OP = OQ = OR = 4 being the radius of the in - circle.

$$\text{Hence } 4 = \frac{\Delta C_1C_2C_3}{\frac{1}{2}[C_1C_2 + C_2C_3 + C_3C_1]} = \frac{S}{s} \qquad \ldots(1)$$

where $s = r_1 + r_2 + r_3$, S^2

$= s(s - C_1C_2)(s - C_2C_3)(s - C_3C_1)$

$= s(r_1)(r_2)(r_3)$

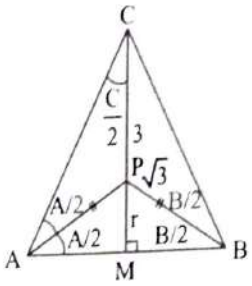
(1) gives $16 = \frac{S^2}{s^2}$

$$= \frac{s(r_3)(r_1)(r_2)}{s^2} = \frac{r_1r_2r_3}{r_1+r_2+r_3}$$

Hence the ratio of the product of the radii to the sum of the radii = 16 : 1.

5. In ΔABC, P is the incentre with $AP = BP = \sqrt{3}$ and $CP = 3$. If the area of ΔABC is $\sqrt{N}$, then find N.

Sol. Given PA = PB



$\Rightarrow \frac{A}{2} = \frac{B}{2} \therefore A = B \Rightarrow \Delta$ is isosceles

$\therefore$ CM is also altitude

$\Rightarrow$ Area of $\Delta = \frac{1}{2} \times AB \times CM$ ..... (1)

Now, $PA = PB = \sqrt{3}$

$\Rightarrow r \operatorname{cosec} \frac{A}{2} = r \operatorname{cosec} \frac{B}{2} = \sqrt{3}$ ..... (2)

Also $PC = 3$

$\therefore r \operatorname{cosec} \frac{C}{2} = 3$ ..... (3)

$$\therefore \frac{\operatorname{cosec} \frac{A}{2}}{\operatorname{cosec} \frac{C}{2}} = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \sqrt{3} \sin \frac{C}{2} = \sin \frac{A}{2}$$

$C = \pi - 2A$

$$\Rightarrow \frac{C}{2} = \frac{\pi}{2} - A$$

$$\Rightarrow \sin \frac{C}{2} = \cos A$$

$$\sqrt{3} \cos A = \sin \frac{A}{2} \quad \text{..... (4)}$$

$$\Rightarrow \sqrt{3}\left(1 - 2\sin^2 \frac{A}{2}\right) = \sin \frac{A}{2}$$

$$\Rightarrow 2\sqrt{3} \sin^2 \frac{A}{2} + \sin \frac{A}{2} - \sqrt{3} = 0$$

$$\Rightarrow 2\sqrt{3} \sin^2 \frac{A}{2} + 3\sin \frac{A}{2} - 2\sin \frac{A}{2} - \sqrt{3} = 0$$

$$\Rightarrow \left(\sqrt{3} \sin \frac{A}{2} - 1\right)\left(2\sin \frac{A}{2} + \sqrt{3}\right) = 0$$

SOLVED PROBLEMS

1. If in a triangle ABC, $A \geq B \geq C$ and $\sin A \geq \sin B \geq \sin C$ then
(A) the triangle ABC must be acute angled
(B) the triangle ABC must be obtuse angled
(C) the triangle ABC must be equilateral
(D) ABC can be any triangle

Sol. **(D).** The result is true for any triangle.

2. In a triangle ABC the altitude from A is not less than BC and altitude from B is not less than AC. The triangle is
(A) right angled
(B) isosceles
(C) obtuse angled
(D) equilateral

Sol. **(A).** $c \sin B \geq a$
$\Rightarrow \sin C \sin B \geq \sin A$ and $a \sin C \geq b$
$\Rightarrow \sin C \sin A \geq \sin B$
$\Rightarrow \sin C \geq 1 \Rightarrow \angle C$ is $\frac{\pi}{2}$.

3. If a, b, c are sides of the triangle ABC, such that $\left(1+\frac{b-c}{a}\right)^a \cdot \left(1+\frac{c-a}{b}\right)^b \cdot \left(1+\frac{a-b}{c}\right)^c \geq 1$ then triangle ABC must be
(A) right angled
(B) isosceles
(C) obtuse angled
(D) equilateral

Sol. **(D).** We have,

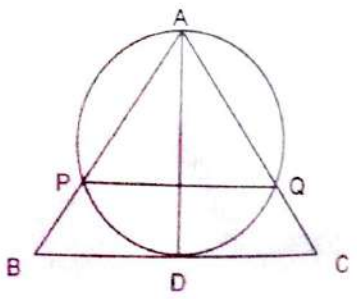
$$\frac{a\left(1+\frac{b-c}{a}\right)+b\left(1+\frac{c-a}{b}\right)+c\left(1+\frac{a-b}{c}\right)}{a+b+c} \geq \left(\left(1+\frac{b-c}{a}\right)^a \cdot \left(1+\frac{c-a}{b}\right)^b \cdot \left(1+\frac{a-b}{c}\right)^c\right)^{\frac{1}{a+b+c}}$$

$$\Rightarrow \left(1+\frac{b-c}{a}\right)^a \cdot \left(1+\frac{c-a}{b}\right)^b \cdot \left(1+\frac{a-b}{c}\right)^c \leq 1$$

$\Rightarrow a = b = c.$

4. In a triangle ABC, $a = 5$, $\angle B = 45°$, $\angle C = 60°$, AD is altitude. Assuming AD as diameter a circle is drawn which cuts side AB and AC at P and Q respectively. PQ is equal to
(A) $\frac{5\sqrt{3}}{4\sqrt{2}}$
(B) $\frac{5\sqrt{3}}{2}$
(C) $\frac{5\sqrt{3}}{2\sqrt{2}}$
(D) none of these

Sol. **(C).** $AD = c \sin B$.
Also, applying sine rule in Δ APQ
$\frac{PQ}{\sin A} = AD \Rightarrow PQ = AD \sin A \Rightarrow PQ = \sin A \, c \sin B$
$= 2R \sin A \sin B \sin C = a \sin B \sin C$
$= 5 \times \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} = \frac{5\sqrt{3}}{2\sqrt{2}}$.



5. If $x \in \left(-\frac{\pi}{2}, \frac{3\pi}{2}\right)$, then $\tan^{-1}\left(\frac{\cos x}{1+\sin x}\right)$ equals

(A) $\frac{\pi}{4}+\frac{x}{2}$ (B) $\frac{\pi}{4}-\frac{x}{2}$

(C) $\frac{x}{2}-\frac{\pi}{4}$ (D) none of these

Sol. **(B).** $\frac{\cos x}{1+\sin x} = \frac{\sin\left(\frac{\pi}{2}-x\right)}{1+\cos\left(\frac{\pi}{2}-x\right)} = \frac{2\sin\left(\frac{\pi}{4}-\frac{x}{2}\right)\cos\left(\frac{\pi}{4}-\frac{x}{2}\right)}{2\cos^2\left(\frac{\pi}{4}-\frac{x}{2}\right)} = \tan\left(\frac{\pi}{4}-\frac{x}{2}\right)$

$\therefore \tan^{-1}\left(\tan\left(\frac{\pi}{4}-\frac{x}{2}\right)\right) = \frac{\pi}{4}-\frac{x}{2}$ since $-\frac{\pi}{2} < \frac{\pi}{4}-\frac{x}{2} < \frac{\pi}{2}$

6. In any triangle ABC, if $\cot A = \sqrt{ac}$, $\cot B = \sqrt{\frac{c}{a}}$, $\cot C = \sqrt{\frac{a^3}{c}}$, then which of the following is true

(A) $a + a^2 = 1 - c$ (B) $a + a^2 = 1 + c$

(C) $a + a^2 = 2 - c$ (D) none of these

Sol. **(A).** In any triangle ABC

$\cot A \cot B + \cot B \cot C + \cot C \cot A = 1 \Rightarrow c + a + a^2 = 1 \Rightarrow a + a^2 = 1 - c.$

7. If in a triangle $\sin^4 A + \sin^4 B + \sin^4 C = \sin^2 B \sin^2 C + 2\sin^2 C \sin^2 A + 2\sin^2 A \sin^2 B$, then its angle A is equal to

(A) $45^0, 135^0$ (B) $30^0, 150^0$

(C) $60^0, 120^0$ (D) none of these

Sol. **(B).** Adding $2\sin^2 B \sin^2 C$ to both sides to make perfect square and using sine rule, we get

$a^4 + b^4 + c^4 + 2b^2c^2 - 2a^2c^2 - 2c^2b^2 = 3b^2c^2$

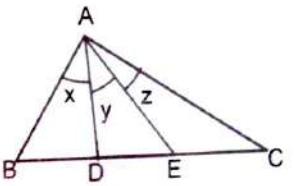
$\Rightarrow (b^2 + c^2 - a^2)^2 = (\sqrt{3}\, bc)^2 \Rightarrow (2bc\cos A)^2 = 3b^2c^2 \Rightarrow 4\cos^2 A = 3 \Rightarrow \cos A = \pm\frac{\sqrt{3}}{2}.$

Hence $A = 30^0, 150^0$.

8. Points D, E are taken on the side BC of a triangle ABC such that BD = DE = EC. If $\angle BAD = x$, $\angle DAE = y$, $\angle EAC = z$; then the value of $\frac{\sin(x+y)\sin(y+z)}{\sin x \cdot \sin z}$ is equal to

(A) 4 (B) 1

(C) 2 (D) none of these



Sol. **(C).** In triangle ADC,

$\frac{\sin(y+z)}{DC} = \frac{\sin C}{AD}$ ... (1)

In triangle ABD,

$\frac{\sin x}{BD} = \frac{\sin B}{AD}$ ... (2)

and from triangle AEC,

$$\frac{\sin z}{EC} = \frac{\sin C}{AE} \quad \ldots (3)$$

and from triangle ABE,

$$\frac{\sin(x+y)}{BE} = \frac{\sin B}{AE} \quad \ldots (4)$$

From (1), (2), (3) and (4),

$$\frac{\sin(x+y)\sin(y+z)}{\sin x \sin z} = \frac{BE}{AE} \times \frac{DC}{AD} \times \frac{AD}{BD} \times \frac{AE}{EC} = \frac{BE}{BD} \times \frac{DC}{EC} = 2 \times 2 = 4.$$

9. If in an obtuse angled triangle the obtuse angle is $\frac{3\pi}{4}$ and the other two angles are equal to two values of θ satisfying $a \tan\theta + b \sec\theta = c$, where $|b| \le \sqrt{a^2 + c^2}$; then $a^2 - c^2$ is equal to

(A) ac (B) 2ac

(C) a/c (D) none of these

Sol. **(B).** As, $A + B + C = \pi \Rightarrow A + C = \frac{\pi}{4}$; let $\angle B = \frac{3\pi}{4}$

$\Rightarrow \tan(A + C) = 1 \Rightarrow \tan A + \tan C = 1 - \tan A \tan C \quad \ldots (1)$

$\because a \tan\theta + b \sec\theta = c \Rightarrow (a \tan\theta - c)^2 = b^2 \sec^2\theta = b^2 + b^2 \tan^2\theta$

$\Rightarrow (a^2 - b^2)\tan^2\theta - 2ac \tan\theta + c^2 - b^2 = 0$

$\because \tan A + \tan C = \frac{2ac}{a^2 - b^2}$; $\tan A \tan C = \frac{c^2 - b^2}{a^2 - b^2} \Rightarrow \frac{2ac}{a^2 - b^2} = 1 - \frac{c^2 - b^2}{a^2 - b^2}$ (from (1))

$\Rightarrow 2ac = a^2 - c^2.$

10. Three equal circles, each of radius r touch one another. The radius of the circle touching all the three given circle, internally is

(A) $\left(2 + \sqrt{3}\right)r$ (B) $\frac{\left(2 + \sqrt{3}\right)}{\sqrt{3}}r$

(C) $\frac{\left(2 - \sqrt{3}\right)}{\sqrt{3}}r$ (D) $\left(2 - \sqrt{3}\right)r$

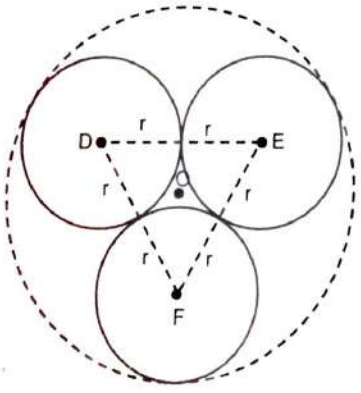
Sol. **(B).** Since DEF is equilateral with side 2r if radius of circumcircle DEF is R_1, then area of triangle

$$DEF = \frac{\sqrt{3}}{4}(2r)^2 = \sqrt{3}\, r^2$$

$$\Rightarrow \sqrt{3}\, r^2 = \frac{2r \cdot 2r \cdot 2r}{4R_1} \Rightarrow R_1 = \frac{2r}{\sqrt{3}}$$

$\therefore$ Radius of the circle touching all the three given circles

$$= r + R_1 = r + R_1 = r + \frac{2r}{\sqrt{3}} = \frac{\left(2 + \sqrt{3}\right)r}{\sqrt{3}}.$$



11. If $\cos B \cos C + \sin B \sin C \sin^2 A = 1$; then triangle ABC is

(A) isosceles (B) right angled

(C) equilateral (D) none of these

Sol. **(A, B).** We are given $\cos B \cos C + \sin B \sin C \sin^2 A = 1$ ... (1)

and $\sin^2 A \le 1$ ... (2)

Now, sin B, sin C are positive in a triangle

$\therefore \sin B \sin C \sin^2 A \le \sin B \sin C$ ... (3)

or, $1 - \cos B \cos C \le \sin B \sin C$; by (1)

or, $1 \le \cos (B - C)$; or $\cos (B - C) \ge 1$.

But $\cos \theta$ cannot be > 1,

hence we must have sign of equality throughout

$\therefore \cos (B - C) = 1 \therefore B - C = 0$ or $B = C$.

Also, $\sin^2 A = 1$, i.e. $A = \frac{\pi}{2}$;

hence the triangle both right angled and isosceles.

Comprehension

Let ABC be a triangle, from vertices A, B and C altitudes AD, BE and CF are drawn to opposite sides BC, CA and AB respectively which meet at a point O. Now triangle DEF is completed. also length OA, OB and OC respectively are p, q and r.

Answer the following questions

12. Ratio of area of ΔDEF to ΔABC is

(A) 2cosA cosB cosC

(B) 2sinA sinB sinC

(C) 2cosA cosB sinC

(D) none of these

Sol. **(A).** $\Delta_{DEF} = \frac{1}{2} a \cos A \, b \cos B \sin(\pi - 2C)$

$= \frac{ab \cos A \cos B \sin 2C}{2} = 2 \cos A \cos B \cos C \cdot \frac{ab \sin C}{2}$

$\Rightarrow \frac{\Delta_{DEF}}{\Delta_{ABC}} = 2 \cos A \cos B \cos C$ (since $\Delta_{ABC} = ab \frac{\sin C}{2}$).

13. Radius of the circumcircle of ΔDEF is

(A) $\frac{R}{2}$

(B) R

(C) $\frac{R}{4}$

(D) none of these

Sol. **(A).** Let R' be the radius of the circumcircle of ΔDEF, then

$R' = \frac{(a \cos A)(b \cos B)(c \cos C)}{4\Delta_{DEF}} = \frac{(abc)(\cos A \cos B \text{codC})}{4 \cdot (2 \cos A \cos B \cos C)\Delta_{ABC}}$

$= \frac{1}{2} \frac{abc}{4\Delta_{ABC}} = \frac{1}{2} R$ (here R is circumradius of ΔABC).

14. Radius of the incircle of ΔDEF is

(A) 2R cosA cosB cosC

(B) R cosA cosB cosC

(C) $\frac{abc \cos A \cos B \cos C}{\Delta_{ABC}}$

(D) none of these

Sol. **(A).** Radius of ΔDEF

$$= \frac{\Delta_{DEF}}{s_{DEF}} = \frac{(2\cos A\cos B\cos C)\frac{1}{2}ab\sin C}{\left(\frac{a\cos A + b\cos B + c\cos C}{2}\right)}$$

$$= 2R\cos A\cos B\cos C.$$

Assertion Reason

15. Consider a circle of radius $\sqrt{8}$ inscribed in the ΔABC. The circle touches AC and BC at M and N points respectively.

Statement 1: If $\frac{AM}{MC} = \frac{2}{3}$ and $\frac{BN}{NC} = \frac{5}{6}$, then BC = 11.

because

Statement 2: Length of the tangents drawn to a circle from a point are equal.

(A) Statement-1 is True, Statement-2 is True; Statement -2 is a correct explanation for Statement-1.

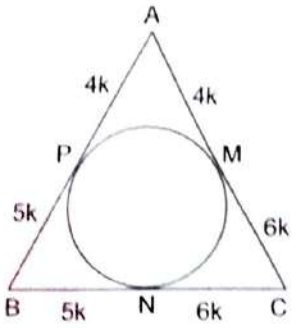
(B) Statement-1 is True, Statement-2 is True; Statement -2 is NOT a correct explanation for Statement-1.

(C) Statement -1 is True, Statement-2 is False.

(D) Statement -1 is False, Statement-2 is True.

Sol. **(A).** $r = \frac{\Delta}{s}$

$\Rightarrow k = 1 \Rightarrow BC = 11.$



16. Match the following

	Column – I		Column – II
(A)	If in a triangle ABC, $\sin^2A + \sin^2B = \sin(A + B)$, then the triangle must be	(p)	right angled
(B)	If in a triangle ABC, $\frac{bc}{2\cos A} = b^2 + c^2 - 2bc\cos A$, then the triangle must be	(q)	equilateral
(C)	If in a triangle ABC, $\tan\frac{A}{2} + \tan\frac{B}{2} + \tan\frac{C}{2} = \sqrt{3}$, then the triangle must be	(r)	isosceles
(D)	If in a triangle the sides and the altitudes are in A.P., then the triangle must be	(s)	obtuse angled
		(t)	acute angled

Sol. **(A) → (p); (B) → (r); (C) → (q); (D) → (q)**

(A) $\sin^2(A) + \sin^2(B) = \sin(A + B)$

$\Rightarrow 1 - \dfrac{\cos 2A + \cos 2B}{2} = \sin C$

$\Rightarrow 1 - \cos(A + B)\cos(A - B) = \sin C$

$\Rightarrow \cos^2 C \cos^2(A - B) + 2\cos(C)\cos(A - B) = -\cos^2 C$

$\Rightarrow \cos C = 0 \Rightarrow$ triangle is right angled.

(B) $\dfrac{bc}{2\cos A} = b^2 + c^2 - 2bc\cos A$

$\Rightarrow \cos A = \dfrac{bc}{2a^2} \Rightarrow b^2c^2 = a^2(b^2 + c^2 - a^2)$

$\Rightarrow (a^2 - b^2)(a^2 - c^2) = 0 \Rightarrow a = b$ or $a = c$.

Hence triangle is isosceles.

(C) $\tan\dfrac{A}{2} + \tan\dfrac{B}{2} + \tan\dfrac{C}{2} = \sqrt{3}$

$\Rightarrow \left(\tan\dfrac{A}{2} - \tan\dfrac{B}{2}\right)^2 + \left(\tan\dfrac{B}{2} - \tan\dfrac{C}{2}\right)^2 + \left(\tan\dfrac{C}{2} - \tan\dfrac{A}{2}\right)^2 = 0 \Rightarrow$ triangle is equilateral.

(D) If a, b, c are in A.P. and h_a, h_b, h_c are in A.P., where h_a, h_b, h_c are the altitudes, then $a = b = c$

$\Rightarrow$ the triangle is equilateral.

Answer Q.17, Q.18 and Q.19 by appropriately matching the information given in the three columns of the following table.

Column-I: contains the relation between angles and sides of a triangle, Column-II: contains the type of triangle and Column-III: contains the radius of circumcircle of triangle.

	Column–I	Column–II	Column–III
(I)	The sines of the angles A and B of a triangle ABC satisfy the equation $c^2x^2 - c(a + b)x + ab = 0$	(i) right angled	(P) $R = \dfrac{b}{\sqrt{3}}$
(II)	One angle of a Δ is 30° and the lengths of the sides adjacent to it are 40 and $40\sqrt{3}$	(ii) isosceles	(Q) $R = \dfrac{c}{2}$
(III)	Two angles of a triangle ABC satisfy the equation $81^{\sin^2 x} + 81^{\cos^2 x} = 30$	(iii) equilateral	(R) $R = a$
(IV)	In a ΔABC, $\cos A \cos B + \sin A \sin B \sin C = 1$	(iv) obtuse angled	(S) $R = \dfrac{a}{\sqrt{2}}$

17. Which of the following combination is **CORRECT**?

(A) (I) (i) (P) (B) (III) (ii) (S)

(C) (I) (i) (Q) (D) (II) (iii) (R)

Sol. **(C)**

$\sin A$ and $\sin B$ are roots of $c^2x^2 - c(1 + b) + ab = 0$

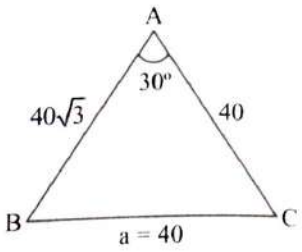
$\sin A + \sin B = \dfrac{c(a+b)}{c^2}$

$\dfrac{a+b}{2R} = \dfrac{c(a+b)}{c^2} \Rightarrow 2R = c$

By sine formula $\dfrac{c}{\sin C} = 2R \Rightarrow R = \dfrac{c}{2}, c = 90^\circ$

18. Which of the following combination is **CORRECT**?
(A) (II) (iv) (R) (B) (II) (iii) (P)
(C) (IV) (iii) (S) (D) (III) (ii) (Q)

Sol. **(A)**



Using cosine formula. Let $\angle A = 30^\circ$

$$\cos 30^\circ = \frac{40^2 + \left(40\sqrt{3}\right)^2 - a^2}{2 \times 40 \times 40\sqrt{3}}$$

$a = 40^\circ$

So $\angle A = \angle C = 30^\circ$

$\angle B = 120^\circ$

So, ABC is obtuse isosceles triangle

$$\frac{a}{\sin 30^\circ} = \frac{b}{\sin 120^\circ} = \frac{c}{\sin 30^\circ} = 2R$$

$$R = a,\ R = \frac{b}{\sqrt{3}},\ R = c$$

19. Which of the following combination is **INCORRECT**?
(A) (III) (i) (P) (B) (IV) (ii) (S)
(C) (III) (i) (R) (D) (I) (iii) (R)

Sol. **(D)**

$$81^{\sin^2 x} + 81^{\cos^2 x} = 30$$

$$81^{\sin^2 x} = t$$

Solving $t + \frac{81}{t} = 30$

$t = 27, t = 3$

$x = 30^\circ$ or $x = 60^\circ$

So two angles are 30° and 60°

Remaining angle is 90°

$$\frac{a}{\sin 30} = \frac{b}{\sin 60} = \frac{c}{\sin 90} = 2R$$

$$R = a,\ R = \frac{b}{\sqrt{3}},\ R = c$$

Also, in ΔABC

$\cos A \cos B + \sin A \sin B \sin C = 1$

$$\sin C = \frac{1 - \cos A \cos B}{\sin A \sin B}$$

As $0 < \sin C \le 1$

So, $0 < \frac{1 - \cos A \cos B}{\sin A \sin B} \le 1$ or $1 - \cos A \cos B \le \sin A \sin B$

$1 \le \sin A \sin B + \cos A \cos B$; $\cos(A - B) \ge 1 \Rightarrow \cos(A - B) = 1$; $A = B$

Also, $\sin C = \frac{1 - \cos^2 A}{\sin^2 A} = 1 \Rightarrow c = 90^\circ$

So, ABC is right angle isosceles triangle $\frac{a}{\sin 45} = \frac{b}{\sin 45} = \frac{c}{\sin 90} = 2R$

$$R = \frac{a}{\sqrt{2}},\ R = \frac{b}{\sqrt{2}},\ R = \frac{c}{2}$$

20. In ΔABC, if $\cos(2A - B) + \sin(A + B) = 2$ and $AB = 4$, then the value of BC is _____

Sol. **(2.00)**

$\cos(2A - B) = 1$ and $\sin(A + B) = 1$ to get $C = 90°$, $A = \frac{\pi}{6}$, $B = \frac{\pi}{3}$

21. Let a, b and c be the side lengths of a triangle ABC and assume that $a \le b$ and $a \le c$. If $x = \frac{b+c-a}{2}$, then find the minimum value of $\frac{ax}{rR}$, where r and R denotes Inradius and circumradius of triangle ABC _____

Sol. **(3.00)**

$$\text{Let } y = \frac{ax}{rR} = \frac{a(b+c-a)}{2\cdot\frac{\Delta}{S}\cdot\frac{abc}{4\Delta}} = \frac{a(b+c-a)(b+c+a)}{abc} = \frac{(b+c)^2 - a^2}{bc}$$

$$y = \left(\frac{b}{c} + \frac{c}{b} + 2\right) - \frac{a^2}{bc} \qquad \ldots\ldots (1)$$

as $a \le b$ and $a \le c$

$\frac{a}{b} \le 1$ and $\frac{a}{c} \le 1$

$\therefore \frac{a^2}{bc} \le 1$

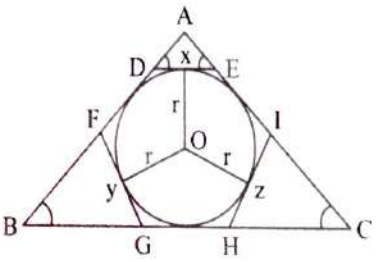
$\therefore y \ge \left(\frac{b}{c} + \frac{c}{b} + 2\right) - 1$

or $y \ge 2 + 2 - 1$ or $y \ge 3$

$\Rightarrow y_{min} = 3$

22. Tangents are drawn to the incircle of ΔABC which are parallel to its sides. If x, y, z are length of tangents intercepted between the sides of triangle and a, b, c that of sides BC, CA and AB respectively, then find $\frac{2x}{a} + \frac{2y}{b} + \frac{2z}{c}$ _____

Sol. **(2.00)**



Applying sine rule to ΔADE, we have $\frac{x}{\sin A} = \frac{AE}{\sin B} = \frac{AD}{\sin C}$

$AE = \frac{x \sin B}{\sin A}$ and $AD = \frac{x \sin C}{\sin A}$

Perimeter of ΔADE

$\Rightarrow AD + AE + DE = \frac{x \sin C}{\sin A} + \frac{x \sin B}{\sin A} + x$

$= x\frac{c}{a} + x\frac{b}{a} + x = \frac{x}{a}(a+b+c)$

Semi-perimeter of $\Delta ADE = \frac{x}{2a}(a+b+c)$

$s' = s\frac{x}{a}$

Clearly incircle of ΔABC is excircle of ΔADE

$$r = s'\tan\frac{A}{2} \Rightarrow r = s\frac{x}{a}\tan\frac{A}{2}$$

$$(s-a)\tan\frac{A}{2} = s\frac{x}{a}\tan\frac{A}{2}$$

$$s-a = s\frac{x}{a} \Rightarrow \frac{x}{a} = \frac{s-a}{s}$$

Similarly, $\frac{y}{b} = \frac{s-b}{s}$

and $\frac{z}{c} = \frac{s-c}{s}$

So, $\frac{x}{a}+\frac{y}{b}+\frac{z}{c} = \frac{s-a}{s}+\frac{s-b}{s}+\frac{s-c}{s}$

$$= \frac{3s-(a+b+c)}{s} = \frac{3s-2s}{s} = 1$$

So, $\frac{2x}{a}+\frac{2y}{b}+\frac{2z}{c} = 2$

23. In ΔABC if the maximum of $\dfrac{a\cos^2\frac{A}{2}+b\cos^2\frac{B}{2}+c\cos^2\frac{C}{2}}{a+b+c}$ is λ ____

Sol. **(0.75)**

$$\frac{a\cos^2\frac{A}{2}+b\cos^2\frac{B}{2}+c\cos^2\frac{C}{2}}{a+b+c}$$

$$\Rightarrow \frac{\frac{a(\cos A+1)}{2}+\frac{b(\cos B+1)}{2}+\frac{c(\cos C+1)}{2}}{a+b+c}$$

$$\Rightarrow \frac{1}{2}+\frac{1}{2}\left(\frac{a\cos A+b\cos B+c\cos C}{a+b+c}\right)$$

$$= \frac{1}{2}+\frac{1}{4s}R(\sin 2A+\sin 2B+\sin 2C)$$

$$= \frac{1}{2}+\frac{1}{4s}R(4\sin A\sin B\sin C)$$

$$= \frac{1}{2}+\frac{R}{s}\left(\frac{a}{2R}\times\frac{b}{2R}\times\frac{c}{2R}\right)$$

$$\frac{1}{2}+\frac{abc}{8sR^2} = \frac{1}{2}+\frac{4R\Delta}{8sR^2}$$

$$= \frac{1}{2}+\frac{\Delta}{2sR} = \frac{1}{2}+r\left(\frac{1}{2R}\right)$$

$$= \frac{1}{2}+\frac{r}{2R} \le \frac{3}{4} \text{ (We known } \frac{r}{R}\le\frac{1}{2}\text{)}$$

$$\lambda = \frac{3}{4} = 0.75$$

TEST YOURSELF

SUBJECTIVE TYPE

1. Let $f(x) = x^2 - 2ax + a - 2$ and $g(x) = \left[2 + \sin^{-1}\left(\frac{2x}{1+x^2}\right)\right]$. If the set of real values of 'a' for which $f[g(x)] < 0\ \forall\ x \in R$ is (k_1, k_2), then the equation $h(x) = [5k_1 - 3k_2]$ has 6 distinct solution where $f(x) = |x - 2|$ and $h(x) = \underbrace{f\left(f\left(.....\left(f(x)\right)\right).....\right)}_{\text{n times}}$, then find the value of 'n' (where [.] denotes the greatest integer function)

2. Find the range of function $f(x) = \cos^2(\sin x) + \sin^2(\cos x)$

3. Find all functions f: $R \to R$ such that $x^2 f(x) + f(1 - x) = 2x - x^4$

4. Let f: $R \to R$, satisfy the equation $(x - y)f(x + y) - (x + y)f(x - y) = 4xy(x^2 - y^2)\ \forall\ x, y \in R$. If $f(1) = -2$, then the absolute value of difference between maximum and minimum value of $f(x)$ on the interval $x \in \left[-\sqrt{3}, \sqrt{3}\right]$ is ____

5. Let $f(x) = -x^3 + x^2 - x + 1$

$$g(x) = \begin{cases} \min(f(t)), & 0 \le t \le x, \quad 0 \le x \le 1 \\ x - 1, & 1 < x \le 2 \end{cases}$$

. Then find $\lim_{x \to 1} g(g(x))$

6. Find the minimum value of the expression $f(x) = Ax^2 + Bx + C$ whose zeroes are β and $\beta + 2$ which can be obtained by the relation $\beta = [2\alpha - 1]$ (where [.] denote the greatest integer function) and $\alpha = \lim_{x \to 0} \frac{1 - \cos(1 - \cos x)}{x^4}$

7. If f: $(0, \infty) \to N$ and $f(x) = \left[\frac{x^2 + x + 1}{x^2 + 1}\right] + \left[\frac{4x^2 + x + 2}{2x^2 + 1}\right] + \left[\frac{9x^2 + x + 3}{3x^2 + 1}\right] + ... + \left[\frac{n^2x^2 + x + n}{nx^2 + 1}\right]$, $n \in N$, then find the value of $\lim_{n \to \infty} \left[\frac{f(x) - n}{(f(x))^2 - \frac{n^3(n+2)}{4}}\right]$ (where [y] denotes the greatest integer less than or equal to y)

8. Let $\alpha(p)$ and $\beta(p)$ be the roots of the equation $\left(\sqrt[6]{1+p} - 1\right)x^2 + \left(\sqrt[3]{1+p} - 1\right)x + \left(\sqrt[9]{1+p} - 1\right) = 0$ where $p > -1$ then $\lim_{p \to 0^+} \left|\left[\alpha(p) + \beta(p)\right]\right|$ is equal to ____

9. If $f(x)$ is a double differentiable function such that $|f''(x)| \le 5$ for each x in the interval [0, 4] and f takes its largest value at an interior point of this interval, find the maximum possible value of $|f'(0)| + |f'(4)|$

10. A container is to be built consisting of a rectangular box of length 'ℓ', width 'b', height 'h' and a semi-cylindrical lid on it whose diameter is equal to the width of the box. If V be the volume, 'S' the surface area and 'S_1' the area of the base of the container. Express S in terms of V, S_1 and ℓ. If V and S_1 are fixed ($S_1 > 1$) and $1 \le \ell \le S_1$, find the maximum and minimum values of S

11. Let $y = f(x)$ be a parametrically defined expression such that $x = 3t^2 - 18t + 7$ and $y = 2t^3 - 15t^2 + 24t + 10\ \forall\ t \in [0, 6]$. Determine the minimum and maximum values of $y = f(x)$

12. Let $f(x) = 1 - x - 4x^3$, $f: R \to R$ then find the number of integral values of $x \in [0, 4]$ satisfying the inequality $4f(x)^3 + f(1 - 2x) + f(x) < 1$

13. Evaluate $\int \sqrt{\frac{(1-\sin x)(2-\sin x)}{(1+\sin x)(2+\sin x)}}dx$

14. Evaluate $\int \frac{dx}{x^{5/6}\left(1+5x^{1/3}\right)\sqrt{1+4x^{1/3}}}$

15. Integrate $I = \int \frac{\left(x^{-7/6} - x^{5/6}\right)dx}{x^{1/3}\left(x^2+x+1\right)^{1/2} - x^{1/2}\left(x^2+x+1\right)^{1/3}}$

16. Integrate $I = \int \frac{(x-1)\sqrt{x^4+2x^3-x^2+2x+1}}{x^2(x+1)}dx$

17. Evaluate $\int_0^1 \frac{e^x(1+x)\sin^2\left(xe^x\right)dx}{\sin^2\left(xe^x\right)+\sin^2\left(e-e^x x\right)}$

18. If $\lim_{r\to\infty} \frac{r^C \int_0^{\pi/2} x^r \sin x dx}{\int_0^{\pi/2} x^r \cos x dx} = L$, then find the value of $\pi L - C$; $(C \in R, L > 0)$

19. Evaluate $\frac{\int_0^{\pi} x^3 \log_e(\sin x)dx}{\int_0^{\pi} x^2 \log_e\left(\sqrt{2}\sin x\right)dx}$

20. Evaluate $\int_0^{\infty} \frac{\tan^{-1}(ax) - \tan^{-1}(x)}{x}dx$; $a > 0$

21. Let A_r be the area of the region bounded between the curve $y^2 = (e^{-kr})x$, $k > 0$, $r \in N$ and the line $y = mx$, $m \neq 0$, k and m are some constants. Show that $A_1, A_2, A_3, \ldots$ are in geometric progression. Also find the sum to infinite terms of this G.P.

22. Let A_n denotes the area of rectangle formed by taking two of its vertices on x-axis and remaining two on curve $f(x) = e^{-2^{4n-2}x^4}$ and B_n denotes the maximum value of A_n for $x \in R$. If $\sum_{n=1}^{\infty} B_n = P$, then find the value of $16\log_e\left(\frac{2}{p}\right)$.

23. Consider the two lines $L_1 \equiv x - y = 0$ and $L_2 \equiv x + y = 0$ and a moving point P(x, y). Let D(P, L) represents distance of point P from line L. If P moves in a region such that $2 \le D(P, L_1) + D(P, L_2) \le 4$, then prove that the area is 24

24. If the solution of the differential equation
$y(x^2 + y^2 + 1)dy - x(1 - 2y^2 - 2x^2)dx = 0$ is $ax^2 + y^2 + b\log(2 - x^2 - y^2) = c$ then find the value of $a + b$

25. Find the continuous function which satisfies the relation, $\int_0^x t f(x-t)dt = \int_0^x f(t)dt + \sin x + \cos x - x - 1$ for all real number 'x'.

26. Let I be the purchase value of an equipment and V(t) be the value after it has been used for t years. The value V(t) depreciates at a rate given by differential equation $\frac{dV(t)}{dt} = -K(T-t)$, where $K > 0$ is a constant and T is the total life in years of the equipment. Then the scrap value V(T) of the equipment is

27. In a acute ΔABC, the base BC has the equation $4x - 3y + 3 = 0$. If the coordinates of orthocenter H and circumcentre (P) of Δ's are (1, 2) and (2, 3) respectively, then radius of the circle circumscribing the Δ is $\frac{\sqrt{m}}{n}$ where m and n are relatively prime. Find the value of $(m + n)$, (you may use fact that distance between orthocenter and circumcentre of Δ is given by $R\sqrt{1 - 8\cos A\cos B\cos C}$

28. A line 'L' is drawn from P(4, 3) to meet the lines L_1: $3x + 4y + 5 = 0$ and L_2: $3x + 4y + 15 = 0$ at point A and B respectively. From point 'A', a line perpendicular to L, is drawn meeting the line L_2 at A_1. Similarly, from point 'B', a line perpendicular to L, is drawn meeting the line L_1 at B_1. Thus a parallelogram AA_1BB_1 is formed. Find the equation(s) of 'L' so that the area of the parallelogram AA_1BB_1 is least

29. The triangle ABC, right angled at C, has median AD, BE and CF. AD lies along the line $y = x + 3$, BE lies along the line $y = 2x + 4$. If length of hypotenuse is 60, find area of triangle ABC

30. A variable line through P(1, 2) meets the line $3x^2 + 7xy - 6y^2 = 0$ is A and B. Then find the locus of mid-point of AB

31. Three circles C_1, C_2 and C_3 with radii r_1, r_2 and $r_1 + r_2$ respectively are such that C_1 and C_2 touch each other externally and C_3 internally. Another circle with radius r_3 touches all the three circles. If $r_1 > r_2 > r_3$ and r_1, r_2, r_3 are in A.P., then find the value of $\left(\frac{r_1}{r_2}\right)^3$

32. Let L_1: $2x + 3y + \lambda - 3 = 0$ and L_2: $2x + 3y + \lambda + 3 = 0$ be two lines where λ is an integer and
C: $x^2 + y^2 + 6x + 10y + 30 = 0$ is a circle, then find all possible values of λ for which both the lines are chords of the given circle

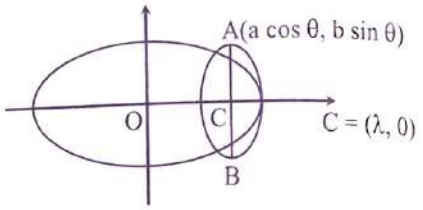
33. Let P be any point on the parabola $y^2 = 4ax$ between its vertex and extremity of latus rectum (with positive y coordinate). If M is the foot of the perpendicular from focuses S to tangent at P, then find the maximum value of area of triangle SPM

34. A variable parabola is drawn to pass through A and B ends of diameters of a given circle centred at origin and radius c and to have as directrix a tangent to concentric circle of a radius 'a' (a > c). The coordinate axis being AB and perpendicular diameter. Find the locus of focus of the parabola

35. Points A and B lie on the parabola $y = 2x^2 + 4x - 2$, such that origin is the mid-point of the line segment AB. If 'l' be the length of the line segment AB, then find the unit digit of l^2.

36. From any point P on the ellipse PN is drawn perpendicular to the major axis and produced at Q so that NQ equals to PS, where S is a focus. Prove that the locus of Q is the two straight lines $y \pm ex + a = 0$

37. A circle intersects an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ precisely cut at three points A, B, C as shown in figure. AB is a diameter of the circle and is perpendicular to the major axis of the ellipse. If the eccentricity of the ellipse is $\frac{4}{5}$, find the length of the diameter AB in terms of 'a'



38. Consider two concentric circles $S_1 : |z| = 1$ and $S_2 : |z| = 2$ on the Argand plane. A parabola is drawn through the points where 'S_1' meets the real axis and having arbitrary tangent of 'S_2' as its directrix. If the locus of the focus of drawn parabola is a conic C, then find the area of the quadrilateral formed by the tangents at the ends of the latus-rectum of conic C.

39. Let points S_1 and S_2 (lying an positive x–axis) be the foci of the curves $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and $\frac{x^2}{a^2}-\frac{y^2}{c^2}=\frac{1}{4}$. If a point of intersection of the two curves is equidistant from S_1 and S_2 and $a : b = \sqrt{7} : \sqrt{6}$ then the value of a : c is

40. Normal to a rectangular hyperbola at P meets the transverse axis at N. Foci of hyperbola are S and S′ then find the value of $6\left(\frac{SP}{SN}\right)^2$

41. A circle of variable radius cuts the rectangular hyperbola $x^2 - y^2 = \frac{16}{9}b^2$ in points A, B, C and D. Determine the equation of the locus of the centroid of triangle ABC

42. Show that the equation to the plane containing the line $\frac{y}{b}+\frac{z}{c}=1$, $x = 0$ and parallel to the line $\frac{x}{a}-\frac{z}{c}=1$, $y = 0$ is $\frac{x}{a}-\frac{y}{b}-\frac{z}{c}+1=0$. Hence show that, if 2d be the shortest distance between the lines, then $\frac{1}{d^2}=\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}$

43. A point P moves on a fixed plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$. The plane through P perpendicular to OP meets the axes in A, B, C. The planes through A, B, C parallel to co-ordinate planes intersect in Q. Show that the locus of Q is $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{ax}+\frac{1}{by}+\frac{1}{cz}$

44. Consider a 3-D figures represented as $xyz^2 = 2$, then find its minimum distance from (0, 0, 0).

45. Find the number of irrational solutions of the equation $\sqrt{x^2+\sqrt{x^2+11}}+\sqrt{x^2-\sqrt{x^2+11}}=4$

46. Let $P(x)=\frac{5}{3}-6x-9x^2$ and $Q(y)=-4y^2+4y+\frac{13}{2}$. If the unique pair of real numbers (x, y) so that P(x) Q(y) = 20, then find the value of (6x + 10y).

47. If the equation $a_8x^8 + a_7x^7 + a_6x^6 + \ldots + a_0 = 0$ has all its roots positive and real (where $a_8 = 1$, $a_7 = -4$, $a_0 = 1/2^8$), then find the value of a_1, a_2 and a_3

48. Find the number of terms in the longest geometric progression that can be obtained from the set (100, 101, ..., 1000)

49. Find $\sum_{r=0}^{n}(-1)^r\ {}^nC_r\left\{\frac{1}{3^r}+\frac{5^r}{3^{2r}}+\frac{19^r}{3^{3r}}+\frac{65^r}{3^{4r}}+\ldots..\text{ upto 10 terms}\right\}$

50. Evaluate $\sum_{r=1}^{n}\left[\sum_{k=1}^{r}k\right]\left[\log_{1/2}\sqrt{4x-x^2}\right]^r$

51. The value of the expression $\sum\sum_{0\le i<j\le n}\left((-1)^{i+j-1}\ {}^nC_i.{}^nC_j\right)$ is

52. Let the numbers tan x, cos x and sec x are the first three terms of an A.P. If the nth terms of the progression is cot x, find the value of n

53. a, b, c are positive integers forming an increasing G.P and b – a is a perfect cube and $\log_6 a + \log_6 b + \log_6 c = 6$, then a + b + c is

54. John has x children by his first wife. Mary has (x + 1) children by her first husband. They marry and have children of their own, the whole has 24 children. Assuming that two children of the same parents do not fight. Prove that the maximum number of fights that can be take place in 191

55. There are 15 seats in the first row of a theatre of which 6 are to be occupied. Find the number of ways of arranging 6 person so that no two persons are side by side

56. A bag contains 6n tickets numbered 0, 1, 2, ..., 6n – 1. In how many ways, 3 tickets can be selected so that the sum of the numbers shown on them is equal to 6n?

57. Pal's gardener is not dependable, the probability that he will forget to water the rose bush is 2/3. The rose bush is in questionable condition. Any how if watered, the probability of its withering is 1/2, if not watered, the probability of its withering is 3/4. Pal went out of station and upon returning, he finds that the rose bush has withered. What is the chance that the gardener did not water the rose bush?

58. A missile target may be at a point P with probability .. or at a point Q with probability $\frac{1}{10}$ we have 20 shells each of which can be fired either at point P or Q. Each shell may hit the target independently of the other shoot with probability $\frac{2}{3}$. Then number of shells must be fired at point P to hit any target with maximum probability is

59. A coin is tossed (m + n) times (m > n). Show that the probability of atleast m consecutive heads is $\frac{(n+2)}{2^{m+1}}$

60. $1, z_1, z_2, z_3, \ldots\ldots, z_{n-1}$ are the nth roots of unity, find the value of $\frac{1}{3-z_1}+\frac{1}{3-z_2}+\ldots..+\frac{1}{3-z_{n-1}}$

61. If $A(z_1)$ and $B(z_2)$ be two complex numbers satisfying the equation $z + (\sqrt{3} + i)t - i = 0$ and $\sqrt{3}z + (\sqrt{3} + i)\lambda - \sqrt{3}i = 0$ respectively $\forall$ t, $\lambda \in R$. Let for some z_1 and z_2, $\arg(z_1) = \frac{\pi}{4}$ and $\arg(z_2) = -\frac{3\pi}{4}$, find the area of the triangle ABC (where C is point 0 + i)

62. If α and β are any two complex numbers, show that $\left|\alpha+\sqrt{\alpha^2-\beta^2}\right|+\left|\alpha-\sqrt{\alpha^2-\beta^2}\right|=|\alpha+\beta|+|\alpha-\beta|$ for any two complex numbers z_1 and z_2.

63. Prove that $\cot^2\frac{\pi}{2n+1}, \cot^2\frac{2\pi}{2n+1}, \ldots, \cot^2\frac{n\pi}{2n+1}$ are the roots of the equation

$$x^n - \frac{{}^{2n+1}C_3}{{}^{2n+1}C_1}x^{n-1} + \frac{{}^{2n+1}C_5}{{}^{2n+1}C_1}x^{n-2} - \ldots = 0.$$

And hence show that $\lim_{n\to\infty}\left(\sum_{r=1}^{n}\frac{1}{r^2}\right)=\frac{\pi^2}{6}$

64. A function f is defined by $f(z) = (4 + i)z^2 + \alpha z + \gamma$ for all complex numbers z, where α and γ are complex numbers. If f(1) and f(i) are both real, then the smallest possible value of $|\alpha| + |\gamma|$ equals

65. If we define $A^{-1} = \frac{A^T}{f(A)}$ where $f(A) = |A^T|$ and A^{-1} exists then prove that $A = \underbrace{\text{adj adj adj} \ldots\ldots \text{adj}}_{\text{n times}} A$, n is the order of the matrix. Also find the value of n

66. If A is a square matrix of order 5 and $9A^{-1} = 4A^T$ find the number of digits in |adj (adj (adj A))| (where A^{-1}, A^T and adj(A) denotes the inverse, transpose and adjoint of matrix A respectively)
(log 3 = 0.477, log 2 = 0.303)

67. If $0 \le [x] < 2$, $-1 \le [y] < 1$ and $1 \le [z] < 3$, (where [.] denotes the greatest integer function) then find the maximum value of the determinant $\begin{vmatrix}[x]+1 & [y] & [z]\\ [x] & [y]+1 & [z]\\ [x] & [y] & [z]+1\end{vmatrix}$

68. Area of triangle whose vertices are (a, a^2), (b, b^2), (c, c^2) is $\frac{1}{2}$, and area of another triangle whose vertices are (p, p^2), (q, q^2) and (r, r^2) is 4, find the value of $\begin{vmatrix} (1+ap)^2 & (1+bp)^2 & (1+cp)^2 \\ (1+aq)^2 & (1+bq)^2 & (1+cq)^2 \\ (1+ar)^2 & (1+br)^2 & (1+cr)^2 \end{vmatrix}$

69. Let m and p be two integers such that $m \geq p + 2$.

If $\Delta(m, p) = \begin{vmatrix} {}^{m}C_{p} & {}^{m}C_{p+1} & {}^{m}C_{p+2} \\ {}^{m+1}C_{p} & {}^{m+1}C_{p+1} & {}^{m+1}C_{p+2} \\ {}^{m+2}C_{p} & {}^{m+2}C_{p+1} & {}^{m+2}C_{p+2} \end{vmatrix}$, show that $\Delta(m, p) = \frac{{}^{m+2}C_{3}}{{}^{p+2}C_{3}} \Delta(m-1, p-1)$. Hence or otherwise, prove that $\Delta(m, p) = \frac{{}^{m+2}C_{3} \cdot {}^{m+1}C_{3} \cdots\cdots {}^{m-p+3}C_{3}}{{}^{p+2}C_{3} \cdot {}^{p+1}C_{3} \cdots\cdots {}^{3}C_{3}}$

70. In a triangle ABC, D and E are points on BC and AC respectively, such that $\frac{BD}{DC} = m$ and $\frac{AE}{EC} = n$. Let P be the point of intersection of AD and BE. Find $\frac{BP}{PE}$ using the vector method.

71. Let the area of a given triangle ABC be Δ. Points A_1, B_1 and C_1 are the mid points of the sides BC, CA and AB respectively. Point A_2 is the mid point of CA_1. Lines C_1A_1 and AA_2 meet the median BB_1 at points E and D respectively. If Δ_1 be the area of the quadrilateral A_1A_2DE, using vectors or otherwise find the value of $\frac{\Delta_1}{\Delta}$

72. Consider the given equation $(x + iy)^{2002} = x - iy$ if the number of ordered pairs (x, y) is N then sum of digits of N is _____

73. If $\alpha, \beta, \gamma, \delta$ are the smallest positive angles in ascending order of magnitude which have their sines equal to the positive quantity k, find the value of $4\sin\frac{\alpha}{2} + 3\sin\frac{\beta}{2} + 2\sin\frac{\gamma}{2} + \sin\frac{\delta}{2}$ (In terms of k)

74. Let $f(x) = \tan^{-1}(\cot x - 2\cot 2x)$ and $\sum_{r=1}^{5} f(r) = a - b\pi$ where $a, b \in N$, then the value of b is _____

75. Let $\alpha = \sin^{-1}\left(\frac{36}{85}\right)$, $\beta = \cos^{-1}\frac{4}{5}$, $\gamma = \tan^{-1}\left(\frac{8}{15}\right)$ and $A = \frac{\cot\alpha + \cot\beta + \cot\gamma}{\cot\alpha \cdot \cot\beta \cdot \cot\gamma}$, $B = \tan\alpha \cdot \tan\beta + \tan\beta \cdot \tan\gamma + \tan\gamma \cdot \tan\alpha$ then $A + B$ is _____

76. Tangents are drawn to the incircle of ΔABC, which are parallel to its sides. Let p, q, r be the length of the tangents opposite to a, b, c respectively, find the value of $\frac{p}{a} + \frac{q}{b} + \frac{r}{c}$

77. Solve the equation

$\cos(\pi 3^x) - 2\cos^2(\pi 3^x) + 2\cos(4\pi 3^x) - \cos(7\pi 3^x)$
$= \sin(\pi 3^x) + 2\sin^2(\pi 3^x) - 2\sin(4\pi 3^x) + 2\sin(\pi 3^{x+1}) - \sin(7\pi 3^x)$.

78. In the ΔABC a similar triangle $A'B'C'$ is inscribed so that $B'C' = \lambda BC$. If $B'C'$ is inclined at anle θ with BC, prove that $\lambda \cos\theta = \frac{1}{2}$

79. A circle of radius R is tangent to the hypotenuse, the incircle and one leg of an isosceles right angle triangle with inradius $r = 1+\sin\frac{\pi}{8}$. If the value of $Rr = \frac{a+\sqrt{a}}{b}$, where $\sqrt{a}$ is surd then find the value of $(a+b)$

80. Prove that if the lengths of the sides of a triangle form an arithmetic progression, then the centre of the circle inscribed in this triangle and the point of intersection of its medians lie on a straight line parallel to that side of the triangle which is of medium length

TEST YOURSELF

MCQ (SINGLE OPTION CORRECT)

1. $f: \{1, 2, 3, 4\} \to \{1, 4, 9, 16\}$ and $g: \{1, 4, 9, 16\} \to \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}\right\}$ are two bijective functions such that $x_1 > x_2 \Rightarrow f(x_1) < f(x_2), g(x_1) > g(x_2)$, then $f^{-1}\left(g^{-1}\left(\frac{1}{2}\right)\right)$ is equal to

(A) 1
(B) 4
(C) 16
(D) none of these

2. Let $f(x) = \frac{\log x}{x}$, and solution of $f(x) = k$ be denoted by $g(k)$ {where $k < 0$}, $g: (-\infty, 0) \to (0, 1)$, then $y = g(x)$ will be

(A) bijective function
(B) many one function
(C) periodic function
(D) into function

3. If $f(x) = (x + 2)e^{\ln(x+2)}$ and $g(x) = \frac{e^{\left(\frac{-1}{\log_e^x}\right)}}{x} - \frac{2x}{e^{-\ln 2}}$ then the smallest positive integer in the range of $h(x) = f(x) + g(x)$ is

(A) 2
(B) 3
(C) 6
(D) 7

4. The domain of the function $f(x) = \sqrt{1 - \sqrt{2 - \sqrt{3 - \sin 2x}}}$ is

(A) $\left[2n\pi - \frac{\pi}{2}, 2n\pi + \frac{\pi}{6}\right]$
(B) $\left[n\pi - \frac{\pi}{2}, \pi + \frac{\pi}{6}\right]$
(C) R
(D) $\left[n\pi - \frac{\pi}{6}, n\pi + \frac{\pi}{2}\right]$

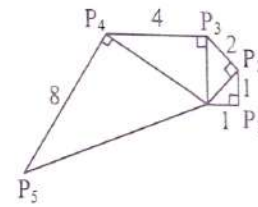
5. If a function satisfies $f(x + 1) + f(x - 1) = \sqrt{2}\, f(x)$, then a period of $f(x)$ can be

(A) 2
(B) 4
(C) 6
(D) 8

6. If $f(x)$ satisfies $x + |f(x)| = 2f(x)$, then $f^{-1}(x)$ satisfies

(A) $3x + |f^{-1}(x)| = 2f^{-1}(x)$
(B) $x + |f^{-1}(x)| = 2f^{-1}(x)$
(C) $f^{-1}(x) - |x| = 2x$
(D) $3x - |f^{-1}(x)| = 2f^{-1}(x)$

7. Let $\{P_n\}$ be a sequence of points determined as in the figure. Thus $|AP_1| = 1$, $|P_nP_{n+1}| = 2^{n-1}$, and angle AP_nP_{n+1} is a right angle. $\lim_{n\to\infty} \angle P_nAP_{n+1}$ equals

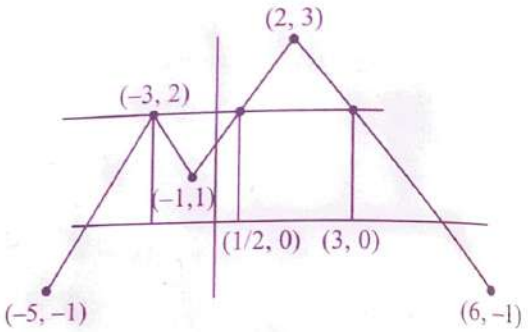


(A) $\frac{\pi}{6}$
(B) $\frac{\pi}{4}$
(C) $\frac{\pi}{3}$
(D) $\frac{5\pi}{12}$

8. Let $f(t) = |t| + |t-1|\ \forall\ t \in R$ and $g(x) = \begin{cases} \max.(f(t)) & , \ x-1 \le t \le x \\ 3-x & , \ 1 < x \le 2 \end{cases}$, $0 \le x \le 1$. Number of points where g(x) is non-derivable in [0, 2], is
(A) 0
(B) 1
(C) 2
(D) 3

9. f(x) is a polynomial function and $(f(\alpha)^2 + (f'(\alpha))^2 = 0$ then find $\lim_{x \to \alpha} \frac{f(x)}{f'(x)}\left[\frac{f'(x)}{f(x)}\right]$ where [.] denotes greatest integer function.
(A) 0
(B) 1
(C) –1
(D) none of these

10. The graph of y = f(x) is shown, then the number of solutions of f(f(x)) = 2 will be
(A) 1
(B) 2
(C) 3
(D) 4



11. A straight line cuts a rectangular figure of length 8 and breadth 6 units respectively in two parts such that their areas are equal. If s is sum of perimeters of both parts then range of s is
(A) $[34, 38]$
(B) $[28, 38]$
(C) $[40, 44]$
(D) $[40, 48]$

12. Number of points of non-differentiability of the function
$g(x) = [x^2]\{\cos^2 4x\} + \{x^2\}[\cos^2 4x] + x^2 \sin^2 4x + [x^2][\cos^2 4x] + \{x^2\}\{\cos^2 4x\}$ in (– 50, 50), (where [.] denotes the greatest integer function and {.} denotes the fractional part of function), is equal to
(A) 98
(B) 99
(C) 100
(D) 0

13. If f(x) is monotonically increasing function for all $x \in R$, such that $f'(x) > 0$ and $f^{-1}(x)$ exists then
(A) $\frac{f^{-1}(x_1) + f^{-1}(x_2) + f^{-1}(x_3)}{3} < f^{-1}\left(\frac{x_1 + x_2 + x_3}{3}\right)$
(B) $\frac{f^{-1}(x_1) + f^{-1}(x_2) + f^{-1}(x_3)}{3} > f^{-1}\left(\frac{x_1 + x_2 + x_3}{3}\right)$
(C) $\frac{f^{-1}(x_1) + f^{-1}(x_2) + f^{-1}(x_3)}{3} = f^{-1}\left(\frac{x_1 + x_2 + x_3}{3}\right)$
(D) none of these

14. $\log_{x-1}(4x - 5) \ge 2$, then the exhaustive set of x is
(A) $\left(2, 3+\sqrt{3}\right)$
(B) $\left(\frac{5}{4}, 3-\sqrt{3}\right) \cup \left(2, 3+\sqrt{3}\right)$
(C) $\left(\frac{5}{4}, 2\right) \cup (2, \infty)$
(D) $\left(\frac{5}{4}, 3+\sqrt{3}\right) - \{2\}$

15. If $g(x) = \max|y^2 - xy|$ $(0 \le y \le 1)$ then the minimum value of $g(x)$ (for real x) is

(A) $\frac{1}{4}$

(B) $3-\sqrt{8}$

(C) $3+\sqrt{8}$

(D) $\frac{1}{2}$

16. If a, b, c are positive constants a > b, then the maximum value of r, given by $\frac{c^4}{r^2} = \frac{a^2}{\sin^2\theta} + \frac{b^2}{\cos^2\theta}$, must be

(A) $\frac{c^2}{a-b}$

(B) $\frac{c^2}{a+b}$

(C) $\frac{c^2}{\sin^2\theta}$

(D) $\frac{c^2}{\sqrt{ab}}$

17. Number of solution of equation $e^{-x} = \frac{k}{x}$, $x > 0$ is given by

(A) 0 ; if k = 1

(B) 1 ; if k = 2

(C) 2 ; if k = 3

(D) 3 ; if k = 4

18. If $f(x) = \sqrt{ax^2 + bx}$, then for how many real values 'a' there is atleast one positive value of 'b' for which the domain of f and the range of f are the same set

(A) 3

(B) 1

(C) 2

(D) none of these

19. If $f(x)$ is polynomial function such that $f(x)+f'(x)+f''(x)+f'''(x) = x^3$ and $g(x) = \int \frac{f(x)}{x^3}dx$ and $g(1) = 1$, then $g(e)$ is

(A) $e+3$

(B) $e-3$

(C) $1+e$

(D) $1-e$

20. $I = \int x^{27}(6x^2+5x+4)(x^2+x+1)^6\,dx = f(x)+c$ then f(x) is equal to

(A) $\frac{(x^6+x^5+x^4)^7}{7}$

(B) $\frac{(6x^5+5x^4+4x^3)^7}{7}$

(C) $\frac{(6x^6+5x^5+4x^4)^7}{7}$

(D) $\frac{(x^5+x^4+x^3)^7}{7}$

21. $\int e^{x^4}(x+x^3+2x^5)e^{x^2}dx$ is equal to

(A) $\frac{1}{2}xe^{x^2}e^{x^4}+c$

(B) $\frac{1}{2}x^2e^{x^4}+c$

(C) $\frac{1}{2}e^{x^2}e^{x^4}+c$

(D) $\frac{1}{2}x^2e^{x^2}e^{x^4}+c$

22. $\int \frac{\sin^3 x + \sin^5 x}{\cos^2 x + \cos^4 x}\,dx$

(A) $\sin x - \frac{2}{\sin x} - 6\tan^{-1}(\sin x) + c$

(B) $6\tan^{-1}(\cos x) + 2\sec x - \cos x + c$

(C) $6\tan^{-1}(\sin x) - \frac{2}{\cos x} - \cos x + c$

(D) $6\tan^{-1}(\cos x) - \frac{2}{\cos x} - \sin x + c$

23. $\int \frac{x^2(x\sec^2 x + \tan x)}{(x\tan x + 1)^2}$

(A) $\frac{-x}{(x\tan x + 1)} + 2\ln|x\sin x + \cos x| + C$

(B) $\frac{-x^2}{(x\tan x + 1)^2} + 2\ln|x\sin x + \cos x| + C$

(C) $\frac{-x^2}{(x\tan x + 1)} + 2\ln|x\sin x + \cos x| + C$

(D) none of these

24. $\int \frac{x^2(1 - \ln x)}{(\ln x)^4 - x^4}dx$ is equal to

(A) $\frac{1}{2}\ln\left(\frac{x}{\ln x}\right) - \frac{1}{4}\ln(\ln^2 x - x^2) + c$

(B) $\frac{1}{4}\ln\left(\frac{\ln x - x}{\ln x + x}\right) - \frac{1}{2}\tan^{-1}\left(\frac{\ln x}{x}\right) + c$

(C) $\frac{1}{4}\ln\left(\frac{\ln x + x}{\ln x - x}\right) + \frac{1}{2}\tan^{-1}\left(\frac{\ln x}{x}\right) + c$

(D) $\frac{1}{4}\ln\left(\frac{\ln x - x}{\ln x + x}\right) + \tan^{-1}\left(\frac{\ln x}{x}\right) + c$

25. If $\int \frac{2x^7 + 3x^2}{x^{10} - 2x^5 + 1}dx = \frac{ax^3 + bx^8}{x^{10} - 2x^5 + 1} + c$. Then

(A) $a = 1$

(B) $a = -1$

(C) $b = 1$

(D) none of these

26. $\int \frac{e^x + e^{-x} - 1}{(e^x \sin x + \cos x)(\cos x - \sin x \cdot e^{-x})}dx$ is equal to

(A) $\ln\left[\frac{e^x \sin x - \cos x}{e^x \cos x + \sin x}\right] + c$

(B) $\ln\left[\frac{e^x \sin x + \cos x}{e^x \cos x - \sin x}\right] + c$

(C) $\ln\left[\frac{e^x \sin x + 2\cos x}{e^x \cos x - \sin x}\right] - c$

(D) $\left[\frac{e^x \sin x + \cos x}{e^x \cos x - \sin x}\right] + c$

27. $\int e^x(x + \ln x)(x + 1)dx$ is equal to

(A) $xe^x(x + \ln x + 1) + c$

(B) $xe^x(\ln x - 1) + c$

(C) $xe^x(x - \ln x - 1) + c$

(D) $xe^x(x + \ln x - 1) + c$

28. If α, β $(\beta > \alpha)$ are the roots of $f(x) \equiv ax^2 + bx + c = 0$ and $f(x)$ is an even function, then $\int_{\alpha}^{\beta} \frac{e^{f\left(\frac{f(x)}{x-\alpha}\right)}dx}{e^{f\left(\frac{f(x)}{x-\alpha}\right)} + e^{f\left(\frac{f(x)}{x-\beta}\right)}}$ is equal to

(A) $\left|\frac{b}{a}\right|$

(B) $\left|\frac{b}{2a}\right|$

(C) $\frac{\sqrt{b^2-4ac}}{|2a|}$

(D) none of these

29. $f^2(x) = \int_0^x \frac{f(t)\cdot\sec^2 t}{1+\tan t}dt$, then $f(x)$ is

(A) $\frac{1}{2}\ln(1+\cot x)$

(B) $\frac{1}{2}\ln(1+\tan x)$

(C) $\frac{1}{2}\ln(1+\sin x)$

(D) $\frac{1}{2}\ln(1-\tan x)$

30. The value of the integral $\int_{-10}^{0} \frac{\left|\frac{2[x]}{[x]-3x}\right|}{\left(\frac{2[x]}{3x-[x]}\right)}dx$, where [.] denotes the greatest integer function, is.

(A) $\frac{28}{3}$

(B) 0

(C) 10

(D) – 10

31. $\int_0^1 e^{\tan^{-1}(x)}dx + \int_1^{e^{\pi/4}} \tan(\ln x)dx$ is equal to

(A) $e^{\frac{\pi}{2}}$

(B) $e^{\frac{\pi}{4}}$

(C) $e^{\frac{\pi}{4}} - 1$

(D) $e^{\frac{\pi}{2}} - 1$

32. Let $f : R \to R$ be a continuous function and $f(x) = f(2x)$ is true $\forall\ x \in R$. If $f(1) = 3$, then the value of $\int_{-1}^{1} f(f(f(x)))dx$ is equal to

(A) 3f(0)

(B) 0

(C) 3f(3)

(D) 6

33. The value of the constant $a > 0$ such that $\int_0^a [\tan^{-1}\sqrt{x}]dx = \int_0^a [\cot^{-1}\sqrt{x}]dx$, where [.] denotes the greatest integer function, is

(A) $\frac{2(3+\cos 4)}{(1-\cos 4)}$

(B) $\frac{2(3-\cos 4)}{(1+\cos 4)}$

(C) $\frac{2(3-\cos 4)}{(1-\cos 4)}$

(D) $\frac{2(3+\cos 4)}{(1+\cos 4)}$

34. Let R be the region in the 1st quadrant bounded by $f(x)=\frac{9}{25}x+b$; $y = f^{-1}(x)$ and the co-ordinate axes, with area = 49 sq. units, then b is equal to

(A) $\frac{18}{5}$ (B) $\frac{22}{5}$

(C) $\frac{28}{5}$ (D) $\frac{56}{5}$

35. If A denotes the area bounded by $f(x)=\left|\frac{\sin x+\cos x}{x}\right|$, x–axis, $x=\pi$ and $x = 3\pi$, then

(A) $1 < A < 2$ (B) $0 < A < 2$

(C) $2 < A < 3$ (D) none of these

36. $f:[0,\infty)\longrightarrow[0,\infty), f(x)$ is a function such that area bounded between the curve, x-axis and $x = t$ is $\frac{1}{2}$ of the area bounded by triangle having $(0,0)$, $(t,0)$ & $(t,f(t))$ as its vertices. $f(x)$ can be

(A) x (B) x^2

(C) x^3 (D) $\sqrt{x}$

37. The area bounded by the identity curve in the first quadrant by $y = 0$ and $x = \sin^{-1}(a^4 + 1) + \cos^{-1}(a^4 + 1) - \tan^{-1}(a^4 + 1)$ is

(A) $\frac{\pi^2}{8}-\frac{a^2}{4}$ (B) $\frac{\pi^2}{32}$

(C) $\frac{\pi^2}{16}$ (D) $\frac{\pi^2}{8}-\frac{a^2}{4}$

38. The area bounded by the two branches of curve $(y - x)^2 = x^3$ and the straight line $x = 1$ is

(A) $\frac{1}{5}$ sq. units (B) $\frac{3}{5}$ sq. unit

(C) $\frac{4}{5}$ sq. unit (D) $\frac{8}{4}$ sq. unit

39. Solution of the differential equation $\frac{dy}{dx}+\frac{1}{x}\tan y=\frac{1}{x^2}\tan y\sin y$ is

(A) $2x=\sin y(1+2x)$ (B) $2x=\sin y(1+cx^2)$

(C) $2x+\sin y(1+cx^2)=0$ (D) none of these

40. A function f(x) defined on $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ with real values and has a primitive F(x) such that $f(x) + \cos x \cdot F(x) = \frac{\sin 2x}{(1+\sin x)^2}$, then f(x) is

(A) $\frac{2\cos x}{(1+\sin x)^2}+ce^{-\sin x}\cos x$ (B) $\frac{2\cos x}{(1-\sin x)^2}+ce^{-\sin x}\cos x$

(C) $\frac{-2\cos x}{(1+\sin x)^2}-ce^{-\sin x}\cos x$ (D) $\frac{-2\cos x}{(1-\sin x)^2}-ce^{-\sin x}\cos x$

41. The order of differential equation of family of circles touching two given circles externally is
(A) 1
(B) 2
(C) 3
(D) none of these

42. If $x\,dy - y\log_e y\,dx + 2yx^3 \sin x\,dx + y x^4 \cos x\,dx = 0$, then solution is
(A) $y = e^{x(c-x^2\sin x)}$
(B) $y = e^{x^2(c-x\sin x)}$
(C) $y = e^{x(c+x^2\sin x)}$
(D) $y = e^{x^2(c+x\sin x)}$

43. If differential equation of family of curves $y = (\sin^{-1} x)^2 + A\cos^{-1} x + B$ (where A and B are arbitrary constants) is $(p - x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = q$ then $(p + q)$ is equal to
(A) 1
(B) 2
(C) 3
(D) 4

44. The solution of differential equatin $(2xy^4e^y + 2xy^3 + y)dx + (x^2y^4e^y - x^2y^2 - 3x)dy = 0$ is
(A) $x^2e^y + \frac{x^2}{y} + \frac{x}{y^3} = c$
(B) $x^2e^y - \frac{x^2}{y} + \frac{x}{y^3} = c$
(C) $x^2e^y + \frac{x^2}{y} - \frac{x}{y^3} = c$
(D) $x^2e^y - \frac{x^2}{y} - \frac{x}{y^3} = c$

45. A pair of perpendicular lines passing through P(1, 4) intersect x axis at Q and R, then locus of incentre of ΔPQR is
(A) $x^2 + y^2 - 2x - 8y + 17 = 0$
(B) $x^2 + y^2 + 2x - 8y + 17 = 0$
(C) $x^2 - y^2 - 2x - 8y + 17 = 0$
(D) $x^2 + y^2 - 2x + 8y + 17 = 0$

46. P is a point inside the triangle ABC. Lines are drawn through P, parallel to sides of triangle. The three resulting triangles with the vertex at P have areas 4, 9 and 49 sq. units. The area of triangle ABC is
(A) $2\sqrt{3}$
(B) 12
(C) 24
(D) 144

47. A(3, 4), B(0, 0) and C(3, 0) are vertices of ΔABC off P is a point inside ΔABC such that $d(P, BC) \le \min\{d(P, AB), d(P, AC)\}$ then the maximum of d(P, BC) is {d(P, BC), represents distance between P and BC}
(A) 1
(B) $\frac{1}{2}$
(C) 2
(D) none of these

48. Triangle formed by the lines $x + y = 0$, $x - y = 0$ and $\ell x + my = 1$. If ℓ and m vary subject to the condition $\ell^2 + m^2 = 1$ then locus of its circumcentre is
(A) $(x^2 - y^2)^2 = x^2 + y^2$
(B) $(x^2 + y^2)^2 = x^2 - y^2$
(C) $(x^2 + y^2) = 4x^2y^2$
(D) $(x^2 - y^2)^2 = (x^2 + y^2)^2$

49. If the vertices P and Q of a triangle PQR are given by (2, 5) and (4, –11) respectively, and the point R moves along the line N : $9x - 7y + 4 = 0$, then the locus of the centroid of the triangle PQR is a straightl line parallel to
(A) PQ
(B) QR
(C) RP
(D) N

50. A line passes through (1, 0). The slope of the line, for which its intercept between $y = x - 2$ and $y = -x + 2$ subtends a right angle at the origin is

(A) $\pm\frac{2}{3}$ (B) $\pm\frac{3}{2}$

(C) ± 1 (D) undefined

51. Consider a circle, $x^2 + y^2 = 1$ and point $P(1, \sqrt{3})$. PAB is a secant drawn from P intersecting circle in A and B (distinct) then range of |PA| + |PB| is

(A) $[2, 2\sqrt{3}]$ (B) $(2\sqrt{3}, 4]$

(C) $(\sqrt{3}, 4]$ (D) none of these

52. The radical centre of three circles described on the three sides $x + y = 5$; $2x + y - 9 = 0$ and $x - 2y + 3 = 0$ of a triangle as diameters is

(A) (4, 4) (B) (3, 3)

(C) (3, 4) (D) (4, 1)

53. Consider circles $S : x^2 + y^2 = 1$ and $S' : (x - 4)^2 + (y - 0)^2 = 9$. Now a circle is drawn touching S, S' externally and also their direct common tangent then its radius is

(A) $3(4 + 2\sqrt{3})$ (B) $\frac{3}{4}(4 - 2\sqrt{3})$

(C) $4 - 2\sqrt{3}$ (D) none of these

54. The chord of contact of tangent from a point A to a circle $x^2 + y^2 = a^2$ passes through a point B. If p and q are the lengths of tangents from A and B on the given circle, then AB is equal to

(A) $\sqrt{p^2 + q^2}$ (B) $\sqrt{p^2 - q^2}$

(C) $2|p + q|$ (D) $2|p - q|$

55. P, Q, R and S are 4 points on tangent at P to $y^2 = 4ax$. Such that PQ : QR : RS ≡ 2 : 5 : 7. Perpendiculars from Q, R and S on double ordinate of P cuts parabola at M, N and K respectively. (Where Q, R, S are on the same side of P on tangents at P). If RN = 3, then $\left(\frac{SK}{QM}\right)$ is equal to

(A) 4 (B) 9

(C) 49 (D) 144

56. From a fixed point A, three normals are drawn to the parabola $y^2 = 4ax$ at the points P, Q and R. Two circles C_1 and C_2 be drawn on AP and AQ as diameter. If slope of the common chord of the circles C_1 and C_2 be m_1 and the slope of the tangent to the parabola at the point R be m_2, then $m_1 \times m_2$ is equal to

(A) 1/2 (B) 2

(C) –1/2 (D) –2

57. Through the vertex O of the parabola, $y^2 = 4ax$ two chords OP and OQ are drawn and the circles on OP and OQ as diameters intersect in R. If θ_1, θ_2 and ϕ are the angles made with the axis by the tangents at P and Q on the parabola and by QR, then the value of $\cot\theta_1 + \cot\theta_2$ is

(A) $-2\tan\phi$ (B) $-2\tan(\pi - \phi)$

(C) 0 (D) $2\cot\phi$

58. A normal is drawn to the parabola $y^2 = 9x$ at the point P(4, 6). A circle is described on SP as diameter; where S is the focus. The length of the intercept made by the circle on the normal at point P is

(A) 17/4 (B) 15/4

(C) 4 (D) 5

59. A trapezium is inscribed in the parabola $y^2 = 4x$, such that its diagonal pass through the point (1, 0) and each has length $\frac{25}{4}$. If the area of the trapezium be P, then 4P is equal to

(A) 70
(B) 71
(C) 80
(D) 75

60. The length of the shortest path that begins at the point (–1, 1), touches the x-axis and then ends at a point on the parabola $(x - y)^2 = 2(x + y - 4)$, is

(A) $3\sqrt{2}$
(B) 5
(C) $4\sqrt{10}$
(D) 13

61. If CF be the perpendicular from the centre C of the ellipse $\frac{x^2}{12} + \frac{y^2}{8} = 1$ on the tangent at any point P and G is the point where the normal at P meets the major axis, then the value of (CF·PG) equals to

(A) 5
(B) 6
(C) 8
(D) none of these

62. Consider the ellipse $\frac{x^2}{f(k^2+2k+5)} + \frac{y^2}{f(k+11)} = 1$ and f(x) is a positive decreasing function, then the value of k for which major axis to be x-axis is

(A) $k \in (-7, -5)$
(B) $k \in (-5, -3)$
(C) $k \in (-3, 2)$
(D) none of these

63. The set of values of k for which it is possible to draw the chord $y = \sqrt{k}x + 1$ to the curve $x^2 + 2xy + (2 + \sin^2\beta)y^2 = 1$, which subtends a right angle at the origin for some value of β is

(A) [2, 3]
(B) [0, 1]
(C) [1, 3]
(D) [0, 3]

64. A series of concentric ellipse $E_1, E_2, \ldots, E_n$ are drawn such that E_n touches the extremities of the major axis of E_{n-1} and the foci of E_n coincide with the extremities of minor axis of E_{n-1}. If the eccentricity of the ellipse is independent of n, the value of the eccentricity, is

(A) $\frac{\sqrt{5}}{3}$
(B) $\frac{\sqrt{5}-1}{2}$
(C) $\frac{\sqrt{5}+1}{2}$
(D) $\frac{1}{\sqrt{5}}$

65. If the curves $\frac{x^2}{4} + y^2 = 1$, $\frac{x^2}{a^2} + y^2 = 1$ for suitable value of 'a' cut on 4 concyclic points, the equation of circle passing through these 4 points is

(A) $x^2 + y^2 = 2$
(B) $x^2 + y^2 = 4$
(C) $x^2 + y^2 = 1$
(D) none of these

66. The locus of the centre of a variable circle touching two circles of radius r_1, r_2 externally, which also touch each other externaly, is a conic. The eccentricity of the conic if $\frac{r_1}{r_2} = 3 + 2\sqrt{2}$ is

(A) 1
(B) $\sqrt{2}$
(C) $\frac{1}{2}$
(D) $2\sqrt{2}$

67. The exhaustive set of values of p^2 such that there exists a tangent to the ellipse $x^2 + p^2y^2 = p^2$ such that the portion of the tangent intercepted by the hyperbola $p^2x^2 - y^2 = 1$ subtends a right angle at the centre of the curves is

(A) $\left[\frac{\sqrt{5}+1}{2}, 2\right]$ (B) $(1, 2]$

(C) $\left[\frac{\sqrt{5}-1}{2}, 1\right)$ (D) $\left[\frac{\sqrt{5}-1}{2}, 1\right) \cup \left(1, \frac{\sqrt{5}+1}{2}\right]$

68. The number of integral points on the hyperbola $x^2 - y^2 = (2000)^2$ are (an integral point is a point both of whose co-ordinates are integer)

(A) 98 (B) 96

(C) 48 (D) 24

69. P(x, y) satisfies $x^2 + y^2 = 1$ and let maximum value of $x^2 + 4xy + y^2$ is λ, then number of tangent(s)/asymptote(s) drawn from point $(\lambda, 1)$ to the hyperbola $(x-2)^2 - y^2 = 1$ is

(A) 1 (B) 2

(C) 0 (D) 4

70. If the plane $x = y = z$ intersect the plane $b^2x + (2 - 4a)y + z = 1$, $a^2x + (1 - 2b)y + z = -1$, then the all possible values of a and b are

(A) $a = 1, b = 1$ (B) $a = 1, b = 2$

(C) $a = 2, b = 1$ (D) $a = 2, b = 2$

71. The lines $r = a + \lambda(b \times c)$ and $r = b + \mu(c \times a)$ will intersect if

(A) $a \times c = b \times c$ (B) $a \cdot c = b \cdot c$

(C) $b \times a = c \times a$ (D) none of these

72. Coordinate of a point Q such that PQ, where $P \equiv (2, 1, 6)$ is perpendicular to the plane determined by (2, 1, 0), (5, 0, 1) and (4, 1, 1) and mid-point of PQ lies on it, is

(A) (6, 5, –2) (B) (–6, –5, 2)

(C) (6, –5, 2) (D) (–6, 5, –2)

73. Let a, b, $c \in R$ such that $abc = p$ and $qa - b = 0$, where p and q are fixed positive numbers, then the minimum distance of the point (a, b, c) from origin in the three dimensional coordinate system, is

(A) $\sqrt{3}\left(\frac{p(q^2+1)}{2q}\right)^{1/3}$ (B) $\sqrt{3}\left(\frac{p(q^2+1)}{q}\right)^{1/3}$

(C) $\sqrt{3}(p)^{1/3}$ (D) $\sqrt{2}\left(\frac{p}{q}\right)^{1/2}$

74. The line $\frac{x}{k} = \frac{y}{2} = \frac{z}{-1}$ makes an isosceles triangle with planes $2x + y + 3z - 1 = 0$ and $x + 2y - 3z - 1 = 0$, then k is

(A) –1 (B) –2

(C) –3 (D) –4

75. The number of integral solutions of equation $y^2(5x^2 + 1) = 25(2x^2 + 13)$ where $x, y \in I$

(A) 2 (B) 4

(C) 8 (D) Infinite

76. Consider two quadratic expressions $f(x) = ax^2 + bx + c$ and $g(x) = ax^2 + px + q$, $(b \neq p)$ such that their discriminants are equal. If $f(x) = g(x)$ has a root $x = \alpha$, then
(A) α will be A.M. of the roots of $f(x) = 0$ and $g(x) = 0$
(B) α will be A.M. of the roots of $f(x) = 0$
(C) α will be A.M. of the roots of $g(x) = 0$
(D) none of these

77. Let $\alpha, \beta \in R$. If α, β^2 be the roots of quadratic equation $x^2 - px + 1 = 0$ and α^2, β be the roots of quadratic equation $x^2 - qx + 8 = 0$, then the value of 'r' if $\frac{r}{8}$ be arithmetic mean of p and q, is
(A) $\frac{83}{8}$
(B) $\frac{83}{4}$
(C) 83/2
(D) 83

78. Complete set of real values of 'a' for which the equation $x^4 - 2ax^2 + x + a^2 - a = 0$ has all it's roots real
(A) $\left[\frac{3}{4}, \infty\right)$
(B) $[1, \infty)$
(C) $[2, \infty)$
(D) $[0, \infty)$

79. Number of values of 'a' for which the curves $y = ax^2 + ax + \frac{1}{24}$ and $x = ay^2 + ay + \frac{1}{24}$ touch each other, is
(A) 2
(B) 3
(C) 4
(D) 0

80. Let $S = \frac{1^2}{1.3} + \frac{2^2}{3.5} + \frac{3^2}{5.7} + \ldots + \frac{500^2}{999.1001}$. If where [.] denotes the greatest integer function then the number of total divisors of [S] is
(A) 1
(B) 2
(C) 3
(D) 4

81. A sequence of real numbers $a_1, a_2, a_3 \ldots$ is such that $a_1 = 0$, $|a_2| = |a_1 + 1|$, $|a_3| = |a_2 + 1| \ldots$ and $|a_n| = |a_{n-1} + 1|$, the arithmetic mean $\frac{a_1 + a_2 + a_3 + \ldots + a_n}{n}$ is
(A) $\geq -\frac{1}{3}$
(B) $\leq -\frac{1}{2}$
(C) ≥ 1
(D) none of these

82. If one root of $Ax^3 + Bx^2 + Cx + D = 0$, $A \neq 0$ is the arithmetic mean of other two roots, then relation $2B^3 + k_1ABC + k_2A^2D = 0$ holds good. The value of $(k_1 + k_2)$ is equal to
(A) 36
(B) 20
(C) 18
(D) 15

83. Let x, y, z be real numbers such that 3x, 4y and 5z form a G.P. while x, y, z form H.P. If the value of $\left(\frac{x}{z} + \frac{z}{x}\right)$ can be expressed as a lowest rational $\frac{p}{q}$, then $(p + q)$ has the value equal to
(A) 29
(B) 39
(C) 49
(D) 59

84. The value of $\sum_{m=0}^{2014}\sum_{n=0}^{m}\left({}^{2014}C_m \cdot {}^{m}C_n\right)$ is

(A) 2014 (B) 2^{2014}
(C) 3^{2014} (D) 3^{2013}

85. Value of $\dfrac{\sum_{k=0}^{r} {}^{n}C_k \, {}^{n-2k}C_{r-k}}{\sum_{k=r}^{n} {}^{n}C_k \, {}^{2k}C_{2r}\left(\frac{3}{4}\right)^{n-k}\left(\frac{1}{2}\right)^{2k-2r}}$ $(n \geq 2r)$ is

(A) 1/2 (B) 2
(C) 1 (D) none of these

86. The sum of the series $\sum_{r=1}^{3n-1}\dfrac{(-1)^{r-1}\,r}{{}^{3n}C_r}$ is (where n is an even natural number)

(A) 0 (B) $\dfrac{3n}{3n+1}$
(C) $\dfrac{3n+1}{3n+2}$ (D) $\dfrac{3n}{3n+2}$

87. $\sum_{r=1}^{\infty}\dfrac{r^3+(r^2+1)^2}{(r^4+r^2+1)(r^2+r)}$ is equal to

(A) $\dfrac{3}{2}$ (B) 1
(C) 2 (D) infinite

88. Number of ways 3 girls and 5 boys can be seated around a circular table such that no two girls sit together and not more than two boys sit between two girls

(A) $4! \times 3!$ (B) $4! \times 3!$
(C) $4! \times {}^{5}C_3 \times 3!$ (D) $4! \times ({}^{5}C_3 - 5) \times 3!$

89. In a plane, there are two families of lines $y = x + r$, $y = -x + r$, where $r \in \{0, 1, 2, 3, 4\}$. The number of squares of diagonals of length 2 units formed by the lines is

(A) 9 (B) 16
(C) 25 (D) none of these

90. The number of ways in which the squares of a 8×8 chess board can be painted red or blue so that each 2×2 square has two red and two blue squares is

(A) 2^9 (B) $2^9 - 1$
(C) $2^9 - 2$ (D) none of these

91. First nine natural numbers are to be divided in three groups G_1, G_2, G_3 containing 3 numbers each. In how many different ways this can be done if sum of the numbers in each group is odd

(A) 5880 (B) 120
(C) 360 (D) none of these

92. There are 5 balls of different colours and 5 boxes of colours same as those of the balls. The number of ways in which the balls could be placed such that exactly 2 balls go to the box of its own colour is

(A) 31 (B) 30
(C) 20 (D) none of these

93. An octopus has one sock and one shoe for each of its eight legs. In how many different orders can the octopus put on its socks and shoes, assuming that, on each leg, the sock must be put on before the shoe

(A) $\frac{16}{(8!)^2 2!}$ (B) $\frac{16!}{(8!)^2}$

(C) $\frac{16!}{2^8}$ (d) none of these

94. Suppose there are 8 white and 2 red balls in a packet, each time one ball is drawn and replaced by a white one. Then probability of drawing out all the red balls just in the 4^{th} draw is:

(A) 0.0534 (B) 0.0434

(C) 0.12 (D) none of these

95. Four dice are thrown simultaneously, given that 4 and 2 has appeared on any two of them, the probability that 3 has appeared on the remaining dice is

(A) $\frac{{}^4P_2.36}{1296}$ (B) $\frac{12}{6^4-2.5^4-4^4}$

(C) $\frac{12}{6^4-4^4}$ (D) $\frac{6}{151}$

96. Consider a function $f:\{1,2,3,.....13\}\to\{1,2,3.....9\}$. Given that function is surjective and non-decreasing, the probability that $f(7)=4$ is

(A) $\frac{{}^6C_3}{{}^{12}C_4}$ (B) $\frac{{}^6C_1}{{}^{12}C_4}$

(C) $\frac{{}^6C_4}{{}^{12}C_4}$ (D) $\frac{{}^6C_3\,{}^6C_1}{{}^{12}C_4}$

97. 3 different numbers are selected random from the set A = {1, 2, 3, ... 10}. Then the probability that the product of two numbers equal to the third is $\frac{p}{q}$, where p and q are relatively prime positive integer, then the value of (p + q) is

(A) 39 (B) 40

(C) 41 (D) 42

98. $\left|z^3+\frac{1}{z^3}\right|\le 2$, then range of $\left|z+\frac{1}{z}\right|$

(A) [0, 2] (B) (0, 2]

(C) [–1, 2] (D) none of these

99. Let $A(\omega^2)$, B $(2i\omega)$ and C (–4) be three points lying on the Argand plane. Now a point P is taken on the circumcircle of the triangle ABC such that PA · BC = PC · AB, (where P, A, B, C are in order). If z is the complex number associated with the mid-point of PB, then the value of $|z|^2$ is, (ω is a non-real cube root of unity)

(A) 0 (B) 1

(C) 2 (D) 3

100. Consider a line $z(i-1)+\bar{z}(i+1)=0$ in the argand plane and a point $z_1 = 2 + 3i$ then the reflection of z_1 in the given line is

(A) 2 – 3i (B) –2 + 3i

(C) 3 + 2i (D) none of these

101. Consider a circle represented by $\left|z-\left(3\sqrt{3}+3i\right)\right|=2$. Let, centre 'C' and origin 'O' are joined and $\overline{OC}$ is rotated about 'O' by an angle $\frac{2\pi}{3}$, in anti-clockwise direction, then equation of circle is

(A) $\left|z-\left(\sqrt{15}+\sqrt{15}i\right)\right|=2$

(B) $\left|z-\sqrt{30}i\right|=2$

(C) $\left|z-\sqrt{30}\right|=2$

(D) none of these

102. If z_1 and z_2 satisfy $2|z+3|=|Rez|$ and $\arg\frac{(z+3)}{(i+1)}=\frac{\pi}{2}$, then $\arg\left(\frac{z_1+3}{z_2+3}\right)$ is equal to

(A) 0

(B) $\pm\frac{\pi}{2}$

(C) $\pm\pi$

(D) none of these

103. A point p(x, y) undergoes the successive transformations about the lines $y = x$, $y = 0$ and $x = 0$. The initial and final points form a matrix 'M' such that trace (M) = 0. If $z = x + iy$ is a complex number then total number of elements in a set A which satisfy

$A = F \cap G \cap H$

$F = \left|\text{adj}(\text{adj } M)\right| = -1$

where $G = \text{Im}\left(z-\frac{1}{2}\right) > 1$ is

$H = \left|\text{Re}(z-2)\right| - \text{Im}(z) = 0$

(A) infinite

(B) zero

(C) two

(D) one

104. An $n \times n$ matrix is formed using only 0, 1 and –1 as its elements. Number of such matrices which are skew symmetric is

(A) $\frac{n(n-1)}{2}$

(B) $(n-1)^2$

(C) $2^{\frac{n(n-1)}{2}}$

(D) $3^{\frac{n(n-1)}{2}}$

105. Matrix A is given by $A=\begin{bmatrix}6 & 11\\ 2 & 4\end{bmatrix}$, then the determinant of $A^{2005} - 6A^{2004}$ is

(A) 2^{2006}

(B) $(-11)\,2^{2005}$

(C) $-2^{2005}\cdot 7$

(D) $(-9)\,2^{2004}$

106. If A is a square matrix of order n then $\underbrace{(\text{adj adj adj} \ldots\ldots \text{adj A})}_{(n-1)\text{ times}} \cdot \underbrace{(\text{adj adj adj} \ldots\ldots \text{adj A})}_{n\text{ times}}$ is equal to

(A) $\left|A^{-1}\right|^{(n-1)^2}$

(B) $\left|A^{-1}\right|^{(n-1)^{(n-1)}}$

(C) $\frac{1}{n}|A|^{(n-1)^{(n-1)}}\cdot I_n$

(D) $|A|^{(n-1)^{(n-1)}}\cdot I_n$

107. The number of all the matrices $A = [a_{ij}]$, $1 \le i, j \le 4$ such that $a_{ij} = \pm 1$ and $\sum_{i=1}^{4} a_{ij} = \sum_{j=1}^{4} a_{ij} = 0$ is

(A) 90

(B) 72

(C) 18

(D) none of these

108. If A is an idempotent matrix satisfying, $(I - 0.4A)^{-1} = I - \alpha A$ where I is the unit matrix of the same order as that of A, then the value of α is equal to
(A) 2/5
(B) 2/3
(C) –2/3
(D) 1/2

109. If [x] stands for the greatest integer less than or equal to x, then in order that the set of equations x–3y=4; 5x + y = 2; $[2\pi]x - [e]y = [2a]$ may be consistent then 'a' should lie in
(A) $\left[3, \frac{7}{2}\right)$
(B) $\left(3, \frac{7}{3}\right)$
(C) $\left(3, \frac{7}{3}\right]$
(D) none of these

110. Let a, b, c, be such that $(b + c) \neq 0$ and $\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n+2}b & (-1)^n c \end{vmatrix}^T = 0$. Then the value of n is
(A) zero
(B) any even integer
(C) any odd integer
(D) any integer

111. Let, $|A| = \begin{vmatrix} b & c \\ 4 & b \end{vmatrix}$; b, c, $\in$ I. If |A| is a perfect square, then $\left[|b| + |\text{Det}(A)|^{\frac{1}{2}}\right]$ is always divisible by
(A) 2
(B) 3
(C) 4
(D) none of these

112. A square PQRS of side length p is folded along the diagonal PR so that the point Q reaches of Q′ and planes PRQ′ and PRS are perpendicular to one another, the shortest distance between PQ′ and RS is $\frac{kp}{\sqrt{6}}$, then k is equal to
(A) 2
(B) 0
(C) 21
(D) $\frac{\sqrt{1218}}{2}$

113. Let $\vec{a}, \vec{b}, \vec{c}$ are any three vectors. If $\vec{a} \times \vec{b} = \vec{c}$, $\vec{b} \times \vec{c} = \vec{a}$, $\vec{c} \times \vec{a} = \vec{b}$. If vectors $\vec{a}, \vec{b}, \vec{c}$ are forming a right handed system. Also, $\vec{x} = \vec{a} + \vec{b}$, $\vec{y} = 2\vec{a} - \vec{b}$, then point of intersection of straight line $\vec{r} \times \vec{x} = \vec{y} \times \vec{x}$, $\vec{r} \times \vec{y} = \vec{x} \times \vec{y}$ is
(A) $2\vec{b}$
(B) $3\vec{b}$
(C) $3\vec{a}$
(D) $2\vec{a}$

114. Let $\vec{a}$ and $\vec{b}$ are two non-collinear vectors such that $|\vec{a}| = 1$. The angle of a triangle whose two sides are represented by the vectors $\sqrt{3}(\vec{a} \times \vec{b})$ and $\vec{b} - (\vec{a} \cdot \vec{b})\vec{a}$ are
(A) $\frac{\pi}{2}, \frac{\pi}{4}, \frac{\pi}{4}$
(B) $\frac{\pi}{2}, \frac{\pi}{3}, \frac{\pi}{6}$
(C) $\frac{\pi}{2}, \frac{5\pi}{12}, \frac{\pi}{12}$
(D) none of these

115. Let $\vec{a}$ and $\vec{b}$ are two mutually perpendicular unit vectors and $\vec{c}$ is a unit vector inclined at an angle θ to both $\vec{a}$ and $\vec{b}$. If $\vec{c} = x\vec{a} + x\vec{b} + y(\vec{a} \times \vec{b})$, where $x, y \in R$, then exhaustive range of θ is

(A) $\left[\frac{\pi}{2}, \frac{3\pi}{4}\right]$ (B) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$

(C) $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$ (D) none of these

116. A variable point P is on the circle $x^2 + y^2 = 1$ on XY-plane. From point P perpendicular PN is drawn to the line L: $x - 2 = y - 2 = -z$, then minimum length PN drawn is

(A) $\sqrt{\frac{9}{2} - 2\sqrt{2}}$ (B) $\sqrt{3 - 4\sqrt{2}}$

(C) $\frac{2\sqrt{2} - 1}{\sqrt{3}}$ (D) $\frac{2\sqrt{2} - 1}{3}$

117. $\sum_{r=1}^{8} \tan(rA)\tan((r+1)A)$ where $A = 36^\circ$ is

(A) $-10 - \tan A$ (B) $-10 + \tan A$

(C) -10 (D) -9

118. If $\theta = \frac{2\pi}{2009}$, then $\cos\theta \cos 2\theta \cos 3\theta \ldots \cos 1004\theta$ is equal to

(A) 0 (B) $\frac{1}{2^{2008}}$

(C) $\frac{1}{2^{1004}}$ (D) $-\frac{1}{2^{1004}}$

119. If $A = \sqrt{\sin 2 - \sin\sqrt{3}}$, $B = \sqrt{\cos 2 - \cos\sqrt{3}}$, then which of the following statement is true?

(A) A and B both are real numbers and $A > B$ (B) A and B both are real numbers and $A < B$

(C) exactly one of A and B is not real number (D) both A and B are not real numbers

120. If $u_n = \sin(n\theta)\sec^n\theta$, $v_n = \cos(n\theta)\sec^n\theta$, $n \in N$, $n \neq 1$, then $\frac{v_n - v_{n-1}}{u_{n-1}} + \frac{1}{n}\frac{u_n}{v_n}$ is equal to

(A) $-\cot\theta + \frac{1}{n}\tan(n\theta)$ (B) $\cot\theta + \frac{1}{n}\tan(n\theta)$

(C) $\tan\theta + \frac{1}{n}\tan(n\theta)$ (D) $-\tan\theta + \frac{\tan(n\theta)}{n}$

121. The number of solution of the equation $\log_{\sqrt{6}\sin x}(7\cos x + 8) = 2$ in the interval $[0, 2\pi]$ is

(A) 4 (B) 3

(C) 2 (D) 0

122. Sum of the series $\sum_{r=1}^{88} (-1)^{r+1} \frac{1}{\sin^2(r+1)^\circ - \sin^2 1^\circ}$ is equal to

(A) $\frac{\cot 2^\circ}{\sin 2^\circ}$ (B) $\frac{-\cot 2^\circ}{\sin 2^\circ}$

(C) $\cot 2^\circ$ (D $\frac{\cot 2^\circ}{\sin^2 2^\circ}$

123. The set of values of x for which $|\sin^{-1}\sin x| > \cos^{-1}\cos x + \tan^{-1}\tan x$, is

(A) $\left(\pi, \frac{3\pi}{2}\right)$ (B) $(0, \pi)$

(C) $(\pi, 2\pi)$ (D) $\left(\frac{3\pi}{2}, 2\pi\right)$

124. If $3\sin x + 4\cos(\alpha x) = 7$ has atleast one solution, then the possible values of α is

(A) $\frac{4m}{4n-1}, m, n \in z$ (B) $\frac{4m}{4n+1}, m, n \in z$

(C) $\frac{4n+1}{4m}, m, n \in z$ (D) none of these

125. The product of common differences of all possible A.P. which are made from values of 'x' satisfying $\cos^2\left(\frac{1}{2}\lambda x\right) + \cos^2\left(\frac{1}{2}\mu x\right) = 1$

(A) $\frac{4\pi^2}{\lambda^2 - \mu^2}$ (B) $\frac{4\pi}{\lambda + \mu}$

(C) $\frac{2\pi^2}{\lambda^2 - \mu^2}$ (D) none of these

126. The sum of n terms of the series $\tan^{-1}3 + \tan^{-1}9/2 + \tan^{-1}33/4 + \tan^{-1}\frac{129}{8} + \ldots$ is $\frac{(2n-1)\pi}{a} - \tan^{-1}2^n$; then a is

(A) 1 (B) 2

(C) 3 (D) 4

127. The number of solution of equation $\sin\sin^{-1}([x]) + \cos^{-1}\cos x = 1$ is (where [.] denotes the greatest integer function)

(A) one (B) two

(C) three (D) none of these

128. The equation $(1+k)\frac{\cos x \cos(2x-\alpha)}{\cos(x-\alpha)} = 1 + k\cos 2x$ has no repeated root in its domain of definition, if

(A) $|k| \le |\operatorname{cosec}\alpha|, k \ne \pm 1$ (B) $|k| \ge |\operatorname{cosec}\alpha|, k \ne \pm 1$

(C) $|k| \le |\operatorname{cosec}\alpha|, k \ne -1$ (D) $|k| \ge |\operatorname{cosec}\alpha|, k \ne -1$

129. Let L and M be the respective intersection of the internal and external angle bisector of the triangle ABC at C and side AB produced. If CL = CM, then the value of $\sqrt{(a^2 + b^2)}$ is

(A) R (B) 2R

(C) 3R (D) 4R

130. In a right–angled triangle, the acute angle α, β are satisfying $\tan\alpha(1 + \tan\alpha) + \tan\beta(1 + \tan\beta) = 4$. If hypotenuse is of length d, then area of the triangle is

(A) $\frac{\sqrt{3}}{4}d^2$ (B) $\frac{\sqrt{3}}{8}d^2$

(C) $\frac{d^2}{4}$ (D) d^2

131. The altitude from A, B, C of ΔABC to opposite sides are respectively 24, 18, 72/5, if Δ is area of ΔABC, then $(\Delta)^{1/3}$ is equal to

(A) 2 (B) 4
(C) 6 (D) 8

132. Triangle ABC is isosceles with AB = AC and $\angle CAB = 106^\circ$. Point M is the interior of the triangle so that $\angle MBA = 7^\circ$ and $\angle MAB = 23^\circ$. The number of degrees in $\angle AMC$ is equal to

(A) 87° (B) 67°
(C) 74° (D) 83°

133. There were 3 players from town B in a cricket camp. Every player in the camp bowls one over to every other player. The number of overs that players from town A bowled among themselves exceeded by 98 to the number of overs bowled when player from town A and B are involved. The number of players from town A is

(A) 9 (B) 16
(C) 14 (D) 11

134. The centre of the circle S = 0 lies on the line $2x - 2y + 9 = 0$ and S = 0 cut orthogonally the circle $x^2 + y^2 = 4$, then S = 0 passes through 2 fixed points and their coordinates are

(A) $(-4, 9), \left(\frac{1}{2}, -\frac{1}{2}\right)$ (B) $(-4, 4), \left(-\frac{1}{2}, \frac{1}{2}\right)$
(C) $(4, -4), \left(\frac{1}{2}, \frac{1}{2}\right)$ (D) $(-4, -4), \left(-\frac{1}{2}, -\frac{1}{2}\right)$

135. If α and β are the roots of the equation $x^2 - (p - q)x - p = 0$ then find the value of

$$\frac{\alpha^3+\alpha^2-\alpha-1}{\alpha^3+\alpha^2+\alpha(q-2)-q}+\frac{\beta^3+3\beta^2+3\beta+1}{\beta^3+3\beta^2+\beta(q+2)+q}$$

(A) 0 (B) 1
(C) 2 (D) 3

136. The number of values of x satisfying the equation $\log_{(x-4)}(x-2)-\log_{(x-2)}\left(x^2-8x+16\right)=\frac{\left(\sqrt{2}\right)^{\log_2 4}}{2}$

(A) 1 (B) 2
(C) 3 (D) 4

137. Consider all $2^n - 1$ non-empty subsets of the set {1, 2, 3, ..., n}. For every such subset, we find the product of the reciprocals of each of its elements. The sum of all these products is

(A) n (B) n + 1
(C) $1/n^n$ (D) none of these

138. Three particles which are at (0, 1), (0, 2) and (0, 3) initially starts moving simultaneously with the same speed along positive y-axis. If a particle reaches (0, 4) or (0, 0) or hits any other particle then it reverses its direction along y-axis. How many particle particle collisions takes place before each of the particle reaches back to its initial position for the first time?

(A) 5 (B) 6
(C) 7 (D) none of these

139. The orthogonal trajectory of straight lines passing through a fixed point (a, b) is

(A) circle (B) line
(C) ellipse (D) none of these

140. Consider the equation $\frac{xy}{x+y} = 2^4 3^5 5^4$ then the number of positive integral solutions of equation are

(A) 140 (B) 891
(C) 72 (D) none of these

TEST YOURSELF

ASSERTION AND REASONING TYPE

141. Let f be a twice differentiable real valued function satisfying $f'(x) \neq 0$, $f(x) + f''(x) = -xg(x)f'(x)$ where $g(x) > 0 \ \forall \ x > 0$. If $f(0) = -3$ and $f'(0) = 4$ then

Statement –1: $|f(x)| \leq 5 \ \forall \ x > 0$

and

Statement –2: $(f(x))^2 + (f'(x))^2$ is decreasing $\forall \ x > 0$.

(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement -1 is True, Statement -2 is False

(D) Statement -1 is False, Statement -2 is True

142. Statement-1: $\lim\limits_{x\to\infty} x - x^2 \log\left(1+\frac{1}{x}\right) = \frac{1}{2}$

and

Statement-2: $\lim\limits_{x\to 0} \frac{\log(1+x)-x}{x^2} = -\frac{1}{2}$

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

143. Statement-1: Let f(x) be differentiable on the interval $(0, \infty)$ such that $f(1) = 1$, and $\lim\limits_{t\to x} \frac{t^2 f(x) - x^2 f(t)}{t-x} = 1$ for each $x > 0$. Then f(x) is

and

Statement-2: Differential equation of f(x) is linear

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

144. Statement-1: Consider an ellipse $ax^2 + 2hxy + by^2 = 1$ then its eccentricity (e) is $e^2 = \frac{\sqrt{(a-b)^2+4h^2}}{2(ab-h^2)}\left[(a+b) - \sqrt{(a-b)^2+4h^2}\right]$

and

Statement-2: Any ellipse can be transformed to the principal axis as $\frac{x'^2}{\alpha^2} + \frac{y'^2}{\beta^2} = 1$ then associated eccentricity is $e^2 = \frac{\alpha^2+\beta^2}{\alpha^2}$

(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.

(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.

(C) Statement 1 is true and Statement 2 is false.

(D) Statement 1 is false and Statement 2 is true.

145. Statement 1: The curve $y = -x^2/2 + x + 1$ is symmetric with respect to the line $x = 1$.

and

Statement 2: A parabola is symmetric about its axis.

(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.
(C) Statement 1 is true and Statement 2 is false.
(D) Statement 1 is false and Statement 2 is true.

146. Let $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ be the quadratic equations such that $\Delta_1 = b_1^2 - 4a_1c_1$ and $\Delta_2 = b_2^2 - 4a_2c_2$.

Statement 1: Atleast one of the given equations have imaginary roots if and only if $\Delta_1 + \Delta_2 < 0$.

and

Statement 2: If $\Delta_1 + \Delta_2 < 0$, then atleast one of the Δ_1 and Δ_2 is negative.

(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1
(B) Statement -1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement -1 is True, Statement -2 is False
(D) Statement -1 is False, Statement -2 is True

147. Let P(z) be a point on the argand plane such that $z\bar{z} + a\bar{z} + \bar{a}z + b = 0$, where $a \in C$ and $b \in R - \{-1\}$ and $|a|^2 - b > 0$.

Statement –1: If $\arg(z + 1)(\bar{z} - b) = \frac{\pi}{2}$, then Im(a) is zero.

and

Statement –2: If $\text{Im}(a) > 0$ and $0 < \arg\left(\frac{z - x_1}{z - x_2}\right) < \frac{\pi}{2}$, $x_1, x_2 \in R$ then $x_1 < x_2$.

(A) Statement -1 is True, Statement -2 is True; Statement-2 **is** a correct explanation for Statement-1
(B) Statement -1 is True, Statement -2 is True; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement -1 is True, Statement -2 is False
(D) Statement -1 is False, Statement -2 is True

148. Let $\alpha, \beta, \gamma > 0$ and $\alpha + \beta + \gamma = \pi/2$.

Statement 1: $\left|\tan\alpha\tan\beta - \frac{a!}{6}\right| + \left|\tan\beta\tan\gamma - \frac{b!}{2}\right| + \left|\tan\gamma\tan\alpha - \frac{c!}{3}\right| \le 0$, where n! = 1.2 ..... .n, then $\tan\alpha\tan\beta$, $\tan\beta\tan\gamma$, $\tan\gamma\tan\alpha$ are in A.P.

and

Statement 2: $\tan\alpha\tan\beta + \tan\beta\tan\gamma + \tan\gamma\tan\alpha = 1$.

(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.
(C) Statement 1 is true and Statement 2 is false.
(D) Statement 1 is false and Statement 2 is true.

149. Statement-1: If A = diag(2, 4, –3) and B = diag(4, 1, 2) then, $\det(AB^{-1}) = -3$

and

Statement-2: $(a_{11}, a_{22}, \ldots. a_{nn})$ is a diagonal matrix then $A^{-1} = \text{diag}\left(\frac{1}{a_{11}}, \frac{1}{a_{22}}, \ldots. \frac{1}{a_{nn}}\right)$. If $a_{ii} \neq 0$

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1
(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True

150. The equations of two straight lines are $\frac{x-1}{2}=\frac{y+3}{1}=\frac{z-2}{-3}$ and $\frac{x-2}{1}=\frac{y-1}{-3}=\frac{z+3}{2}$

Statement-1: The given lines are coplanar
and
Statement-2: The equations $2r - s = 1, r + 3s = 4, 3r + 2s = 5$ are consistent
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(B) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True

MCQ (MULTIPLE OPTIONS CORRECT)

151. If f be periodic function such that the equation $f(x) = 1 - \cos \pi x$ has a unique solution then period of $f(x)$ can be
(A) π
(B) e
(C) $\sqrt{3}$
(D) 1

152. Let $f(x)=\frac{1}{x}$, $g(x)=\frac{1}{4x^2-1}$ and $h(x)=\frac{5x}{x+2}$ three functions and k(x) = h(f(g(x)). If the domain and range of k(x) are R – A and R – B respectively where R is set of real numbers and $A = \{a_1, a_2, a_3, a_4 ..... a_n\}$, then

(A) $n+\sum_{i=1}^{n} a_i = 5$
(B) $n+\sum_{i=1}^{n} a_i = 10$
(C) number of integers in set B is 5
(D) number of integers in set B is 7

153. Let $f(x)=\begin{cases}-3+|x| & , \ -\infty < x < 1 \\ a+|2-x| & , \ 1 \le x < \infty\end{cases}$ and $g(x)=\begin{cases}2-|-x| & , \ -\infty < x < 2 \\ -b+\text{sgn}(x) & , \ 2 \le x < \infty\end{cases}$, where sgn(x) denotes signum function of x. If h(x) = f(x) + g(x) is discontinuous at exactly one point, then
(A) a = –3, b = 0
(B) a = –3, b = 1
(C) a = 2, b = 1
(D) a = 0, b = 1

154. Let $f(x) = 1 + e^{\ln(\ln x)} \cdot \ln(k^2 + 25)$ and $g(x)=\frac{1}{|x|-1}$. If $\lim_{x\to\infty}(f(x))^{g(x)} = k\left(2\sin^2\alpha + 3\cos\beta + 5\right)$, $k > 0$ and $\alpha, \beta \in R$, then which of the following is/are correct?

(A) the value of k is equal to 5
(B) the value of $\frac{\sin^{10}\alpha+\cos^5\beta}{\sin^2\alpha+\cos\beta}$ is equal to 1
(C) $\cos^2\beta + \sin^4\alpha = 2$
(D) $\sin^2\alpha > \cos\beta$

155. Let f(x) x a function such that $f'(x) = \frac{1}{x^2+f^2(x)}$ for all $x \ge 1$ and $f(1) = 1$

(A) $f(a) \le 1 + \pi/4$ for $a \ge 1$
(B) $f(a) = 1 - \pi/4$ for $a \ge 1$
(C) area enclosed by $y = f(x) \le [\max f(a)]x^2$ where $[\cdot]$ denotes the greatest integer function $x^2 + y^2 \le 2$ and $y \ge 0$ is $\pi/4 + 1/3$
(D) area enclosed by $y = f(x) \le [\max f(a)]x^2$ where $[\cdot]$ denotes the greatest integer function $x^2 + y^2 \le 2$ and $y \ge 0$ is $\pi/4 + 1/6$

156. If f(x) is a non-increasing function such that $f(x)+f''(x)\geq 0\ \forall x\in R$ & $f(0)=f'(0)=0$ then f(x) is not strictly decreasing in

(A) $[-4,-2]$ (B) $[-2,2]$

(C) $[3,4]$ (D) $[2,4]$

157. If $\int_a^x\left(\left(\frac{f(t)-f(a)}{(t-a)^2}\right)-\frac{f'(a)}{(t-a)}\right)dt$ has no critical points, then $f(x)$ can be

(A) $x+\sin x$ (B) x^3+3x+2

(C) $x^2+\sin x$ (D) $x^2-\sin x$

158. If $\int\frac{\cos x}{x}\left(x^2\ln x+1\right)dx = f(x)+c$; $f(1)=\cos 1$, and $\int f(x)\sec^2 x\,dx = g(x)+c'$; $g(1)=0$ and $L=\lim_{x\to 0^+} g(x)$, then

(A) $f(x)=\cos x\,(1+\ln x)-x\sin x\ln x$ (B) $g(x)=\frac{x\ln x}{\cos x}$

(C) $L=0$ (D) $\lim_{x\to\infty}\frac{g(x)}{x^2}=0$

159. If $\int(\sin 3\theta+\sin\theta)e^{\sin\theta}\cos\theta\,d\theta = (A\sin^3\theta+B\cos^2\theta+C\sin\theta+D\cos\theta+E)e^{\sin\theta}+F$, then

(A) $B=12$ (B) $D=0$

(C) $B=-12$ (D) none of these

160. Let $f: R^+\to R$ defined as $f(x)=e^x+\ln x$ and $g=f^{-1}$ then correct statement(s) is/are

(A) $g''(e)=\frac{1-e}{(1+e)^3}$ (B) $g''(e)=\frac{e-1}{(1+e)^3}$

(C) $g'(e)=e+1$ (D) $g'(e)=\frac{1}{e+1}$

161. If $\int\cos(2016x)\cdot\sin^{2014}x\,dx = f(x)+c$; $f(0)=0$, then

(A) $f(x)$ is odd (B) $\int_\pi^{2\pi} f(x)dx=0$

(C) $\int_0^\pi f(x)dx=0$ (D) $\lim_{x\to 0}\frac{2015f(x)}{x^{2015}}=1$

162. $\int_0^{3\pi/2}\max\left(\sin^{-1}(\sin x),\tan^{-1}(\tan x)\right)dx$

(A) $\frac{\pi^2}{8}$ (B) $\frac{\pi^2}{4}$

(C) $\frac{3\pi^2}{8}$ (D) $\frac{\pi^2}{2}$

163. If $f(x) = \int_a^x [f(x)]^{-1} dx$ and $\int_a^1 [f(x)]^{-1} dx = \sqrt{2}$, then

(A) $f(2) = 2$

(B) $f'(2) = 1/2$

(C) $f^{-1}(2) = 2$

(D) $\int_0^1 f(x)dx = \sqrt{2}$

164. A function $y = f(x)$ satisfying the differential equation $\frac{dy}{dx} \cdot \sin x - y\cos x + \frac{\sin^2 x}{x^2} = 0$ is such that, $y \to 0$ as $x \to \infty$, then the statement which is correct is

(A) $\lim_{x\to 0} f(x) = 1$

(B) $\int_0^{\pi/2} f(x)dx < \frac{\pi}{2}$

(C) $\int_0^{\pi/2} f(x)dx > 1$

(D) $f(x)$ is an odd function

165. Given a differentiable function $f: R^+ \to R$; where $f(x) = (x-1)(\ln x - 1) + \int_1^x f(x)dx$. Find $f(x)$

(A) $y = \ln x + ce^x$

(B) $y = \ln x + ce^{-x}$

(C) $ye^{-x} = \ln x\, e^{-x} + c$

(D) none of these

166. The equation $\cos x = [5 + \sin x] - [\cos x + 3]$ has ((where [.] denotes the greatest integer less than or equal to x)

(A) has no solution in $\left(0, \frac{\pi}{2}\right)$

(B) has no solution in $\left[\frac{\pi}{2}, \pi\right]$

(C) has no solution in $\left[\pi, \frac{3\pi}{2}\right]$

(D) has exactly one solution in $\left[0, \frac{\pi}{2}\right]$

167. Sides of a rhombus are parallel $x + y - 1 = 0$ and $7x - y - 5 = 0$. It is given that the diagonals of the rhombus intersect at (1, 3) and one vertex A lies on the line $y = 2x$, then the coordinates of A can be

(A) $\left(\frac{8}{5}, \frac{16}{5}\right)$

(B) $\left(\frac{7}{15}, \frac{14}{15}\right)$

(C) $\left(\frac{6}{5}, \frac{12}{5}\right)$

(D) $\left(\frac{4}{15}, \frac{8}{15}\right)$

168. Sides of ΔABC are in AP. If $a <$ minimum $\{b, c\}$, then $\cos A$ may be equal to

(A) $\frac{4b-3c}{2b}$

(B) $\frac{3c-4b}{2c}$

(C) $\frac{4c-3b}{2b}$

(D) $\frac{4c-3b}{2c}$

169. If inside a big circle exactly 24 small circles, each of radius 2, can be drawn in such a way that each small circle touches the big circle and also touches both its adjacent small circles, then radius of the big circle is

(A) $2\left(1+\operatorname{cosec}\frac{\pi}{24}\right)$

(B) $\left(1+\tan\frac{\pi}{24}\Big/\cos\frac{\pi}{24}\right)$

(C) $2\left(1+\operatorname{cosec}\frac{\pi}{12}\right)$

(D) $\frac{2\left(\sin\frac{\pi}{48}+\cos\frac{\pi}{48}\right)^2}{\sin\frac{\pi}{24}}$

170. If the conics whose equations are
$S_1 : (\sin^2\theta)x^2 + (2h\tan\theta)xy + (\cos^2\theta)y^2 + 32x + 16y + 19 = 0$
$S_2 : (\cos^2\theta)x^2 - (2h'\cot\theta)xy + (\sin^2\theta)y^2 + 16x + 32y + 19 = 0$
intersect in four concyclic points, where $\theta \in \left[0, \frac{\pi}{2}\right]$, then the correct statement(s) can be

(A) $h + h' = 0$
(B) $h - h' = 0$
(C) $\theta = \frac{\pi}{4}$
(D) none of these

171. Tangent is drawn at any point (x_1, y_1) other than vertex on the parabola $y^2 = 4ax$. If tangents are drawn from any point on this tangent to the circle $x^2 + y^2 = a^2$ such that all the chords of contact pass through a fixed point (x_2, y_2) then

(A) x_1, a, x_2 are in G.P.
(B) $\frac{y_1}{2}$, a, y_2 are in G.P.
(C) -4, $\frac{y_1}{y_2}$, $\frac{x_1}{x_2}$ are in G.P.
(D) $x_1x_2 + y_1y_2 = a^2$.

172. A circle 'S' is described on the focal chord of the parabola $y^2 = 4x$ as diameter. If the focal chord is inclined at an angle of 45° with axis of x, then which of the following is/are true
(A) radius of the circle is 4
(B) centre of the circle is (3, 2)
(C) the line $x + 1 = 0$ touches the circle
(D) the circle $x^2 + y^2 + 2x - 6y + 3 = 0$ is orthogonal to 'S'

173. A focal chord of parabola $y^2 = 8(x + 2)$ is inclined at an angle of 60° with positive x-axis and intersects the parabola at P and Q. Let perpendicular bisector of the chord PQ intersects the x-axis at R, then the distance of R from focus is

(A) $\frac{8}{3}$
(B) $\frac{16\sqrt{3}}{3}$
(C) $\frac{16}{3}$
(D) $8\sqrt{3}$

174. Let α and $f(\alpha)$ be the eccentricity of the ellipse $\frac{x^2}{3b^2 - 2a^2} + \frac{y^2}{2b^2 - a^2} = 1$, $(3b^2 > 2a^2)$ and $\frac{x^2}{2b^2 - a^2} + \frac{y^2}{b^2} = 1$ $(2b^2 > a^2)$ respectively, then

(A) $f(\alpha) = \frac{\alpha}{\sqrt{1-\alpha^2}}$, $b \in R - \{0\}$
(B) $\int_0^{1/2} ffff(\alpha)\,d\alpha = \frac{1}{4}$
(C) $\int e^{\alpha}(f(\alpha) - f''(\alpha))\,d\alpha = e^{\alpha}\left[\frac{1}{(1-\alpha^2)^{3/2}} + \frac{\alpha}{\sqrt{1-\alpha^2}}\right] + c$
(D) $f(\alpha) = \frac{\alpha}{2\sqrt{1-\alpha^2}}$, $b \in R - \{0\}$

175. A straight line through a point $P(\alpha, 2)$, $(\alpha \neq 0)$ meets the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ at A and D & meets axes at B and C such that PA, PB, PC and PD are in GP; then 'α' can lie in interval
(A) $[7, \infty)$
(B) $(-12, -8)$
(C) $(-5, 0)$
(D) $(10, \infty)$

176. The director circle of a hyperbola is $x^2+y^2-4y=0$. One end of the transverse axis is $(2,0)$ then a focus is

(A) $\left(\sqrt{3},2-\sqrt{3}\right)$ (B) $\left(-\sqrt{3},2+\sqrt{3}\right)$

(C) $\left(\sqrt{6},2-\sqrt{6}\right)$ (D) $\left(-\sqrt{6},2+\sqrt{6}\right)$

177. If tangent of any member of family of hyperbola $xy = 4\sin^2\theta$; $\theta \in (0, 2\pi) - \{\pi\}$ is not a normal to member of family of circles $x^2 + y^2 - 2x - 2y + \mu = 0$, where μ is any real parameter then θ belongs to

(A) $\left(\frac{5\pi}{6}, \frac{7\pi}{6}\right)$ (B) $\left(0, \frac{\pi}{6}\right)$

(C) $\left(\frac{11\pi}{6}, 2\pi\right)$ (D) $\left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$

178. Circles are drawn on chords of the rectangular hyperbola $xy = c^2$ parallel to the line $y = x$ as diameters. All such circles pass through two fixed points whose coordinates are

(A) (c, c) (B) (c, –c)

(C) (–c, c) (D) (–c, –c)

179. Let PM be the perpendicular from the point P (1, 2, 3) to xy plane, If $\overrightarrow{OP}$ makes an angle O with the positive direction of z-axis and $\overrightarrow{OM}$ makes an angle ϕ with the positive direction of x –axis, where 0 is the origin and O and ϕ are acute angles, then

(A) $\tan\theta = \frac{\sqrt{5}}{3}$ (B) $\sin\theta \sin\phi = \frac{2}{\sqrt{14}}$

(C) $\tan\phi = 2$ (D) $\cos\theta \cos\phi = \frac{1}{\sqrt{14}}$

180. Let $6a < 4b < 3c$ then the equation $(2x - a)(3x - b) + (3x - b)(4x - c) + (4x - c)(2x - a) = 0$ has

(A) both roots real (B) both roots imaginary

(C) one root lies between $\left(\frac{a}{2}, \frac{b}{3}\right)$ (D) other root lies between $\left(\frac{b}{3}, \frac{c}{4}\right)$

181. Given $ax^2 + bx + c \geq 0$, $bx^2 + cx + a \geq a$ and $cx^2 + ax + b \geq 0$, where $a \neq b \neq c$ and $a, b, c \in R$. Now $\frac{a^2+b^2+c^2}{ab+bc+ca}$ can not take values

(A) $\frac{2}{3}$ (B) $\frac{1}{2}$

(C) $\frac{3}{2}$ (D) $\frac{9}{2}$

182. Consider the equation $\sqrt{1+\frac{3\cos 2x}{\sqrt{\sin^4 x+4\cos^2 x}+\sqrt{\cos^4 x+4\sin^2 x}}} = \sqrt{2}\sin^{-1}(\sin 2x)$ $x \in [-\pi, \pi)$

(A) number of roots of this equation is 2

(B) number of roots of this equation is 4

(C) one root lies in the interval $\left(\frac{\pi}{8}, \frac{\pi}{6}\right)$

(D) if $x \in [0, n\pi]$ $n \in N$, number of roots will never be odd

183. $P(x)$ is a polynomial function satisfying $xP(x-n) = (x-1)P(x)$ for some $n \in N$ and for all $x \in R$, then $P(x)$ may be

(A) $x^2 + 2x$ (B) $2x$

(C) x^3 (D) $-2x$

184. Let $(a_1, a_2, a_3 ..... a_n)$ be a permutation of $(1,2,3.....n)$ for which $a_1 > a_2 > a_3 > ..... > a_{\frac{n}{2}}$ and $a_{\frac{n}{2}} < a_{\frac{n}{2}+1} < a_{\frac{n}{2}+2} < ... < a_n$ for n as an even positive integer. Also $a_1 > a_2 > a_3 ..... > a_{\frac{n+1}{2}}$ and $a_{\frac{n+1}{2}} < a_{\frac{n+1}{2}+1} < ... < a_n$ for n as an odd positive integers. Let the total number of permutation of n be $p(n)$, if $200 < p(n) < 500$, then values of n is/are

(A) 10 (B) 11

(C) 12 (D) 13

185. If $1 + (1+x) + (1+x)^2 + ... + (1+x)^n = \sum_{k=0}^{n} a_k x^k$, then

(A) $\sum_{k=0}^{n} a_k = 2^{n+1}$ (B) $a_{n-2} = \frac{n(n+1)}{2}$

(C) $a_p > a_{p-1}$ for $p < \frac{n}{2}$ (D) $(a_9)^2 - (a_8)^2 = {}^{n+2}C_{10}\,({}^{n+1}C_{10} - {}^{n+1}C_9)$

186. Let $a(n) = 1 - \frac{1}{2} + \frac{1}{3} - ..... + (-1)^{n-1}\frac{1}{n}$, then

(A) $\frac{1}{n+1} + \frac{1}{n+2} + ... + \frac{1}{2n} = a(2n)$ (B) $a(2n) < 1\ \forall\ n$

(C) $a(2n) \geq 0.5\ \forall\ n$ (D) $a(2n) \geq 2\ \forall\ n$

187. Let the sequence $a_1, a_2, a_3 ..... a_n$ forms a G.P. If the area bounded by the parabola $y^2 = 4a_n x$ and $y^2 = 4a_n(a_n - x)$ be A_n then

(A) $A_1, A_2, A_3 .....$ are in A.P. (B) $A_1, A_2, A_3 .....$ are in G.P.

(C) $A_n = \frac{4\sqrt{2}}{3}a_n^2$ (D) $A_n = \frac{2\sqrt{2}}{3}a_n^2$

188. n tennis players take part in a tournament in which each of them plays exactly one game with each of the others. If x_i and y_i denote the number of victories and losses respectively of the i^{th} player for $i = 1, 2, ..., n$ then

(A) $x_i + y_i = n - 1\ \forall\ i = 1, 2, 3, ..., n$ (B) $\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i$

(C) $x_i + y_i = n^2 - n\ \forall\ i = 1, 2, ..., n$ (D) $\sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} y_i^2$

189. If H_n denotes the number of ways in which n horses participating in a race can reach the finishing line, when one or more horses can reach the finishing line at the same time. Then

(A) $H_2 = 3$ (B) $H_3 = 13$

(C) $H_4 = 75$ (D) $H_3 = 10$

190. If n lines are drawn in a plane such that no two are parallel and no three are concurrent then the number of regions the plane is divided is denoted by P_n then
(A) $P_5 = 6$
(B) $P_7 = 29$
(C) $P_9 = 48$
(D) $P_{10} = 56$

191. 10 married couples are sitting in a room. The number of ways in which 6 people can be selected so that
(A) they do not contain any couples is 95
(B) they contain at least one couple is 210
(C) they contain exactly one couple is 80
(D) they comprise of 3 men & 3 women and contain exactly two couples is 60

192. Number of natural numbers between 100 and 1000 which can be formed using the digits 1, 2, 3, 4, 5, 6, 7 if each digit is used at most once, is also equal to
(A) $^{10}C_5$
(B) the number of triangles that can be formed with 12 points in a plane with exactly 5 of them being collinear
(C) the number of ways in which 4 alike apples can be distributed among 10 childern if each child gets at most one apple
(D) the number of ways of distributing 16 identical fruits among 5 children if each child gets not less than 2 fruits

193. If n different objects are distributed among n + 2 persons, then
(A) Probability that exactly 2 persons will get nothing is $\frac{(n+1)!}{2.(n+2)^{n-1}}$
(B) Probability that exactly 3 persons will not get anything is $\frac{^{n+2}C_3(n-1)(^nC_2)(n-2)!}{(n+2)^n}$
(C) Probability that exactly 3 persons will not get anything is $\frac{n.(n-1)^2.(n+1)}{12(n+2)^{n-1}}$
(D) Probability that exactly 2 persons will get nothing is $\frac{n^2.(n-1)^2}{(n+2)^{n-1}}$

194. A consignment of 15 record players contain 4 defectives. The record players are selected at random, one by one and examined. The one examined is not put back. Then
(A) probability of getting exactly 3 defectives in the examination of 8 record players is $\frac{^4C_3\ ^{11}C_5}{^{15}C_8}$
(B) probability that 9th one examined is the last defective is $\frac{8}{197}$
(C) probability that 9th examined record player is defective, given that there are 3 defectives in first 8 players examined is $\frac{1}{7}$
(D) probability that 9th one examined is the last defective is $\frac{8}{195}$

195. If $A_1, A_2, A_3, \ldots, A_{1006}$ be independent events such that $P(A_i) = \frac{1}{2i}(i = 1, 2, 3, \ldots, 1006)$ and probability that none of these of the events occurs be $\frac{\alpha!}{2^{\alpha}(\beta!)^2}$, then
(A) β is of form $4k + 2, k \in I$
(B) $\alpha = 2\beta$
(C) β is a composite number
(D) α is of form $4k, k \in I$

196. From an urn containing x white and y–x black balls, n balls have been lost, then probability of drawing a white ball

(A) before the loss is $\frac{x}{y}$

(B) after the loss is $\frac{x}{y}$ for $n = 1$

(C) after the loss is $\frac{x}{y}$ for $n = 2$

(D) after the loss is $\frac{x}{y}$ for $n = 3$

197. Let z_1, z_2, z_3 be three complex numbers such that $|z_1| = |z_2| = |z_3| = 1$ and $\frac{z_1^2}{z_2 z_3} + \frac{z_2^2}{z_1 z_3} + \frac{z_3^2}{z_1 z_2} + 1 = 0$, then $|z_1 + z_2 + z_3|$ can take the value equal to

(A) 1

(B) 2

(C) 3

(D) 4

198. Equation of a straight line on the complex plane passing through a point P denoting the complex number α and perpendicular to the vector $\overrightarrow{OP}$ where 'O' is the origin can be written as

(A) $\text{Im}\left(\frac{z-\alpha}{\alpha}\right) = 0$

(B) $\text{Re}\left(\frac{z-\alpha}{\alpha}\right) = 0$

(C) $\text{Re}(\bar{\alpha} z) = 0$

(D) $\bar{\alpha} z + \alpha \bar{z} - 2|\alpha|^2 = 0$

199. Let u, v, w, z be complex number such that $|u| < 1$, $|v| = 1$ and $w = \frac{v(u-z)}{\bar{u}z - 1}$ then

(A) $|w| \le 1 \Rightarrow |z| \le 1$

(B) $|w| \ge 1 \Rightarrow |z| \ge 1$

(C) $|w| \le 1 \Rightarrow |z| \ge 1$

(D) $|w| \ge 1 \Rightarrow |z| \le 1$

200. Let $A_\alpha = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then

(A) $A_{\alpha+\beta} = A_\alpha A_\beta$

(B) $A_\alpha^{-1} = A_{-\alpha}$

(C) $A_\alpha^{-1} = -A_{-\alpha}$

(D) $A_\alpha^2 = -1$

201. If $f(x) = \begin{vmatrix} a^{-x} & e^{x\ln a} & x^2 \\ a^{-3x} & e^{3x\ln a} & x^4 \\ a^{-5x} & e^{5x\ln a} & 1 \end{vmatrix}$, then

(A) graph of g(x) is symmetrical about the origin

(B) graph of g(x) is symmetrical about the y-axis

(C) $\left.\frac{d^4 f(x)}{dx^4}\right|_{x=0} = 0$

(D) $g(x) = f(x)\ln\left(\frac{a-x}{a+x}\right)$

202. Let $\hat{\alpha}$, $\hat{\beta}$ and $\hat{\gamma}$ be the unit vectors such that $\hat{\alpha}$ and $\hat{\beta}$ are mutually perpendicular and $\hat{\gamma}$ is equally inclined to $\hat{\alpha}$ and $\hat{\beta}$ at an angle θ. If $\hat{\gamma} = x\hat{\alpha} + y\hat{\beta} + z(\hat{\alpha} \times \hat{\beta})$, then

(A) $z^2 = 1 - 2x^2$

(B) $z^2 = 1 - 2y^2$

(C) $z^2 = 1 - x^2 - y^2$

(D) $x^2 = y^2$

203. A vector $(\vec{d})$ is equally inclined to three vectors $\vec{a} = \hat{i} - \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} + \hat{j}$ and $\vec{c} = 3\hat{j} - 2\hat{k}$. Let $\vec{x}$, $\vec{y}$, $\vec{z}$ be three vectors in the plane of $\vec{a}, \vec{b}; \vec{b}, \vec{c}; \vec{c}, \vec{a}$ respectively, then

(A) $\vec{x} \cdot \vec{d} = 14$
(B) $\vec{y} \cdot \vec{d} = 3$
(C) $\vec{z} \cdot \vec{d} = 0$
(D) $\vec{r} \cdot \vec{d} = 0$ (Where $\vec{r} = \lambda\vec{x} + \mu\vec{y} + \delta\vec{z}$)

204. If there are three square matrix A, B, C of same order satisfying the equation $A^2 = A^{-1}$ and let $B = A^{2^n}$ and $C = A^{2^{(n-2)}}$, then which of the following statements are true?

(A) $\det.(B - C) = 0$
(B) $(B + C)(B - C) = 0$
(C) B must be equal to C
(D) none of these

205. Which of the following is/are positive?

(A) $\log_{\sin 1} \tan 1$
(B) $\log_{\cos 1}(1 + \tan 3)$
(C) $\log_{\log_{10} 5}(\cos\theta + \sec\theta)$
(D) $\log_{\tan 15^\circ}(2\sin 18^\circ)$

206. The solution of the equation $|x-1|^{\log_3 x^2 - 2\log_x 9} = (x-1)^7$ is/are

(A) 2
(B) 81
(C) $\frac{1}{\sqrt{3}}$
(D) 3

207. For $x \in \left(0, \frac{\pi}{2}\right) - \left\{\frac{\pi}{4}\right\}$. Find the values of x which can satisfy the inequality $\log_{\sqrt{2}\sin x} \cos^2 x > 2$

(A) $\left(\sin^{-1}\left(\frac{1}{\sqrt{3}}\right), \frac{\pi}{3}\right)$
(B) $\left(\sin^{-1}\left(\frac{1}{\sqrt{3}}\right), \frac{\pi}{4}\right)$
(C) $\left(\sin^{-1}\left(\frac{1}{\sqrt{3}}\right), \sin^{-1}\left(\sqrt{\frac{2}{5}}\right)\right)$
(D) $\left(0, \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)\right)$

208. The incircle of ΔABC touches side BC at D. Then difference between BD and CD equals

(A) $\left|4R\sin\frac{A}{2}\sin\frac{B-C}{2}\right|$
(B) $\left|4R\cos\frac{A}{2}\sin\frac{B-C}{2}\right|$
(C) $|b - c|$
(D) $\left|\frac{b-c}{2}\right|$

209. Solution(s) of the equation $x^{\left(\frac{\log_{10} x + 7}{4}\right)} = 10^{\log_{10} x + 1}$, is/are

(A) $\frac{1}{10000}$
(B) $\frac{1}{10}$
(C) 10
(D) 100

210. The direction cosines of the lines connected by the relations $l + m + n = 0$, $2lm + 2ln - mn = 0$ is/are

(A) $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$
(B) $\left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}\right)$
(C) $\left(\frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}\right)$
(D) $\left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

211. If $f: R \to \left[-\frac{\pi}{4}, \frac{\pi}{2}\right)$ is a function defined by $f(x) = \tan^{-1}\left(x^4 - x^2 - \frac{7}{4} + \tan^{-1}\alpha\right)$ and f is surjective, then

(A) $\cos^{-1}\left(\frac{1-\alpha^2}{1+\alpha^2}\right) = 2$

(B) $\alpha + \frac{1}{\alpha} = 2\ \text{cosec}\ 2$

(C) $\sin^{-1}\left(\frac{2\alpha}{\alpha^2+1}\right) = \pi - 2$

(D) $\tan^{-1}\left(\frac{2\alpha}{\alpha^2-1}\right) = 2 - \pi$

212. Let $f(x) = \begin{cases} 1+x & ; \ 0 \le x \le 2 \\ 3-x & ; \ 2 < x \le 3 \end{cases}$, then which of the following are INCORRECT?

(A) f(f(x)) = 2 has 2 solutions

(B) range of f(f(x)) is [0, 3] – {2}

(C) f(f(x)) = [x] has 3 solutions

(D) $\int_0^3 f(f(x))dx = 3$

213. Consider, $f(x) = \min\left\{x^3 - 1, -\frac{1}{4}(|x-2| + |x+2|), 7 - x^3\right\}$ p and q denote number of points where f(x) is discontinuous and non-derivable in [–2, 3] respectively, then

(A) p = 0

(B) p = 1

(C) q = 0

(D) q = 1

214. If $f: R \to R$ is a function such that $|f(x) - f(y)| \le |\sin x - \sin y| \forall x, y \in R$, then $f(x)$ is

(A) surjective

(B) one-one

(C) many one

(D) periodic

215. Let $f(x) = 30 - 2x - x^3$ and $f_n(x) = \text{fofof}_{(n\ \text{times})}....f(x)$, if $f_{2014}(x) < f_{2013}(-x)$, then x can not be equal to

(A) 4

(B) 5

(C) 6

(D) 7

216. If the function $f(x) = \left(\frac{\sqrt{a+4}}{1-a} - 1\right) x^5 - 3x + \log 5$ decreases for all real x, then a lies in

(A) $(1, \infty)$

(B) $\left(-\infty, \frac{3-\sqrt{21}}{2}\right) \cup \left(\frac{3+\sqrt{21}}{2}, \infty\right)$

(C) $(-4, -2)$

(D) $\left(-4, \frac{3-\sqrt{21}}{2}\right) \cup (1, \infty)$

217. If $f(x) = \begin{cases} \alpha + \frac{\sin[x]}{x}, & x > 0 \\ 2, & x = 0 \\ \beta + \left[\frac{\sin x - x}{x^3}\right], & x < 0 \end{cases}$ (where [.] denotes the greatest integer function) if f(x) is continuous at x = 0, then β is equal to

(A) $\alpha = 2$

(B) $\alpha = 3$

(C) $\beta = 3$

(D) $\beta = 2$

218. Let $f(x) = ab\sin x + b\sqrt{1-a^2}\cos x + c$, where $|a| < 1, b > 0$ then
(A) maximum value of $f(x)$ is b if $c = 0$
(B) difference of maximum and minimum value of $f(x)$ is $2b$
(C) $f(x) = c$ if $x = -\cos^{-1}a$
(D) $f(x) = c$ if $x = \cos^{-1}a$

219. If $f(x) = \begin{cases} 0 & , \ x \neq 0 \\ 1 & , \ x = 0 \end{cases}$, then which of the following option(s) is/are correct?
(A) $f(f(x))$ is continuous $\forall\ x \in R$
(B) $\lim\limits_{x\to a}\left(f\left(f(x)\right)\right) = 1 \ \forall\ a \in R$
(C) $\lim\limits_{x\to a}\left(f\left(f(x)\right)\right)$ does not exist for exactly one value of a
(D) the graph of $y = f(f(x))$ is identical with the graph of $y = \text{sgn}(|x|)$

220. If $\lim\limits_{x\to\infty}\left(8x^3 + mx^2\right)^{1/3} - nx$ exists and is equal to 1, then
(A) $m = 12$
(B) $m = 6$
(C) $n = 4$
(D) $n = 2$

221. If $f(x)$ is a continuous function such that $\lim\limits_{x\to 0}\left(\left[\dfrac{f(x)}{x^3}\right]\right) = 4$, then which of the following option(s) is/are correct? (where [.] denotes the greatest integer function)
(A) $\lim\limits_{x\to 0}\left(\left[\dfrac{f(x)}{x^2}\right]\right) = 0$
(B) $\lim\limits_{x\to 0}\left[\dfrac{f(x)}{x^2}\right]$ does not exist
(C) $\lim\limits_{x\to 0}\left[\dfrac{f(x)}{x}\right] = 0$
(D) $\lim\limits_{x\to 0}\left[\dfrac{f(x)}{x}\right]$ does not exist

222. The maximum values attended by $\dfrac{x^4 - x^2}{x^6 + 2x^3 - 1}$ over real number $x > 1$ is
(A) $\dfrac{1+\sqrt{5}}{2}$
(B) $\dfrac{2}{\sqrt{5}-1}$
(C) $\dfrac{1}{6}$
(D) a rational numbers

223. Let $f(x) = \cot x - \tan x - 2\tan 2x - 8\cot 8x$, $x \neq \dfrac{n\pi}{8}$, $n \in I$ and $g(x) = x^3 + 6x - 1$, one more function $h(x)$ is defined as $h : R - \left\{\dfrac{n\pi}{8}, n \in I\right\} \to R$, $h(x) = f(x) + g(x)$, then identify the correct statement
(A) $h''\left(\dfrac{\pi}{24}\right) = \dfrac{\pi}{4}$
(B) $h(x)$ is odd function
(C) $h(x)$ is increasing in the domain
(D) if the equation $h(x) = \lambda$ has a solution in $(0, 3)$ then the number of integral values of λ is 7

224. $\int \frac{dx}{\left(1+\sqrt{x}\right)^8} = \frac{-\left(1+\sqrt{x}\right)^{-\lambda}}{3} + \frac{2}{7}\left(1+\sqrt{x}\right)^{-\mu} + c$, then

(A) $\lambda = 6$
(B) $\lambda = 5$
(C) $\mu = 8$
(D) $\mu = 7$

225. $\int \left(\sin 2\theta + \frac{\sin 4\theta}{2}\right) \cdot \cos\theta \cdot e^{\sin\theta} d\theta = (P \sin^3 Q + Q\cos^2\theta + R\sin\theta + S\cos\theta + T)\, e^{\sin\theta} + \lambda$, then

(A) $P = -4$
(B) $Q = -12$
(C) $R = -20$
(D) none of these

226. If $\int_0^x \left(x - [x] - \frac{1}{2}\right) dx = f(x).g(x)$ where $[x]$ and $\{x\}$ are integral and fractional parts of x, respectively

(A) $f(x) = \frac{\{x\}}{2}$
(B) $g(x) = (\{x\} - 1)$
(C) $f(x) = \frac{[x]}{2}$
(D) $g(x) = ([x] - 1)$

227. $f(x) = \int \frac{x^8 + 4}{x^4 - 2x^2 + 2} dx$ and $f(0) = 0$, then

(A) f(x) is an odd function
(B) f(x) has range R
(C) f(x) has at least one rela root
(D) f(x) is a monotonic function

228. If $\int \frac{(x-1)dx}{x^2\sqrt{2x^2 - 2x + 1}} = \frac{\sqrt{f(x)}}{g(x)} + c$ then

(A) $f(x) = 2x^2 - 2x + 1$
(B) $g(x) = x + 1$
(C) $f(x) = \sqrt{2x^2 - 2x}$
(D) $g(x) = x$

229. Let $I_n = \int_0^{\frac{\pi}{2}} \cos^n x\, dx$, $n \geq 2$ is natural number, then

(A) $I_{n-2} > I_n$
(B) $n(I_{n-2} - I_n) = I_{n-2}$
(C) $I_n - I_{n-1} = \frac{n}{n-1}$
(D) $I_4 = \frac{3\pi}{8}$

230. The integral $\int_{\tan^{-1}\lambda}^{\cot^{-1}\lambda} \frac{\tan x}{\tan x + \cot x} dx$, $\forall\, \lambda \in R$ can not take the value,

(A) $-\frac{\pi}{4}$
(B) $-\frac{\pi}{2}$
(C) $\frac{\pi}{4}$
(D) $\frac{3\pi}{4}$

231. The curve $y = a\sqrt{x} + bx$ passes through the point (1,2) and the area enclosed by the curve, the x-axis and the line $x = 4$ is 8 sq. units. Then

(A) $a + 2b = 1$
(B) $2a + b = 1$
(C) $a = -1, b = 3$
(D) $a = 3, b = -1$

232. If A be the area bounded by $f(x)=\frac{\sin x}{x}$, $x=0, x=\frac{\pi}{2}$ and x-axis, then which of following is/are true

(A) $1<A<\frac{\pi}{2}$

(B) $\left(\frac{1+\sqrt{2}}{2}\right)<A<\left(\frac{\pi+2\sqrt{2}}{4}\right)$

(C) $A>\frac{\pi+2\sqrt{2}}{4}$

(D) $\frac{2}{\pi}<A<1$

233. Area bounded by curve $y^2 = x$ and $x = 4$ is divided into 4 equal parts by the lines $x = a$ and $y = b$ then.

(A) Area of each part $=\frac{8}{3}$

(B) $b = 0$

(C) $a=\sqrt{2}$

(D) $a=(16)^{1/3}$

234. If y(x) is the solution of the differential equation $\frac{dy}{dx}=\frac{y}{2y\ln y+y-x}$ and $y(0) = 1$, $y > 0$, then which of the following is/are true?

(A) $x = y^2 \ln y$

(B) $\lim_{y\to 0^+} x = 0$

(C) $\int_0^1 x\,dy=-\frac{1}{4}$

(D) $\lim_{y\to 1} x = 0$

235. Let f(x) be a differentiable function on the interval $(0, \infty)$ such that $f(1) = 1$ and $\lim_{z\to x}\frac{z^2f(x)-x^2f(z)}{z-x}=1$, then

(A) $f(2)=\frac{1}{2}$

(B) $f(2)=\frac{17}{6}$

(C) $\lim_{x\to 0^+} xf(x)=\frac{1}{3}$

(D) $\lim_{x\to 0^+} xf(x)=1$

236. The differential equation for the family of curves $y = c \sin x$ can be given by

(A) $\left(\frac{dy}{dx}\right)^2 = y^2\cot^2 x$

(B) $\left(\frac{dy}{dx}\right)^2-\left(\sec x\frac{dy}{dx}\right)^2+y^2=0$

(C) $\left(\frac{dy}{dx}\right)^2 = \tan^2 x$

(D) $\frac{dy}{dx} = y\cot x$

237. $0 < c < b < a$ and the roots α, β of the equation $cx^2 + bx + a = 0$ are imaginary, then

(A) $\frac{|\alpha|+|\beta|}{2}=|\alpha|\,|\beta|$

(B) $\frac{1}{|\alpha|}=\frac{1}{|\beta|}$

(C) $\frac{1}{|\alpha|}+\frac{1}{|\beta|}<2$

(D) none of these

238. Let $\frac{dy}{dx}+y=f(x)$, where y is continuous function of x with $y(0) = 1$ and $f(x)=\begin{cases}e^{-x}, & 0\le x\le 2\\ e^{-2}, & x>2\end{cases}$ then

(A) $y(1) = 2e^{-1}$

(B) $y'(1) = -e^{-1}$

(C) $y(3) = -2e^{-3}$

(D) $y'(3) = -2e^{-3}$

239. Let $y = f(x)$ be a real valued function satisfying $x\frac{dy}{dx} = x^2 + y - 2$, $f(1) = 1$ then

(A) $f(x)$ is minimum at $x = 1$
(B) $f(x)$ is maximum at $x = 1$
(C) $f(3) = 5$
(D) $f(2) = 3$

240. If the slope of one line of the pair of lines $4x^2 + axy + y^2 = 0$ is four times the slope of the other line, then a is

(A) 10
(B) -10
(C) 5
(D) -5

241. Equation of the line which is parallel to $x + y - 4 = 0$ and at distance $3\sqrt{2}$ from (1, 2) is

(A) $x + y + 3 = 0$
(B) $x + y - 3 = 0$
(C) $x + y + 9 = 0$
(D) $x + y - 9 = 0$

242. Equation of line passing through (2, 7) and making equal angles with the lines $12x + y = 13$ and $x + 12y = 17$, is

(A) $x + y - 9 = 0$
(B) $2x + y - 11 = 0$
(C) $x - y + 5 = 0$
(D) $x - 2y + 12 = 0$

243. Triangle ABC is equilateral D is on AB and E is on AC, such that DE is tangent to the circle, then which is true

(A) $\frac{AD}{DB} + \frac{AE}{CE} = 1$
(B) $\frac{AD}{DB} + \frac{AE}{CE} = 2$
(C) $BC + DE = BD + CE$
(D) $BD + DE = BC + CE$

244. Consider the circle C_1: $x^2 + y^2 - 4x + 6y + 8 = 0$; C_2: $x^2 + y^2 - 10x - 6y + 14 = 0$. Which of the following statements hold good in respect of C_1 and C_2

(A) C_1 and C_2 are orthogonal
(B) C_1 and C_2 touch each other
(C) radical axis between C_1 and C_2 is also one of their common tangent
(D) middle point of the line joining the centre of C_1 and C_2 lies on their radical axis

245. If the tangent to the ellipse $x^2 + 4y^2 = 16$ at the point $R(\alpha)$ is passing through the centre of the circle $(x - 4)^2 + (y - 2)^2 = 20$, then α equal to

(A) 0
(B) $\pi/4$
(C) $\pi/6$
(D) $\pi/2$

246. The equations of the common tangents to the parabola $y = x^2$ and $y = -(x - 2)^2$ is/are

(A) $y = 4(x - 1)$
(B) $y = 0$
(C) $y = -4(x - 1)$
(D) $y = -30x - 50$

247. Equation of common tangent to the parabola $y^2 = 4x$ and $x^2 = 32y$ is $AX + BY + C = 0$ (where A, B, C are coprime), then

(A) $ABC = 8$
(B) $\frac{C^3}{A^3B^3} = 8$
(C) $\frac{A^2}{BC} = \frac{1}{8}$
(D) none of these

248. For A point R on the ellipse the circle with SR as diameter. S is focus of the given ellipse, intersect the auxiliary circle of ellipse at A and B, then which of the following is/are correct

(A) segment AB subtenes acute angle on point R
(B) A, B coincide
(C) the circle with diameter SR, is tangent to auxiliary circle
(D) there are exactly 4 possibilities of point R in all positions. For given focus, so that SR will not subtend 90° at any point on major axis

249. If the ellipse $\frac{x^2}{4}+y^2=1$ meets the ellipse $x^2+\frac{y^2}{a^2}=1$ in four distinct points and $a = b^2 - 5b + 7$ then b can take values
(A) $(-\infty, 0)$
(B) $[4, 5]$
(C) $[2, 3]$
(D) $(0, \infty)$

250. Circles are drawn on chords of rectangular hyperbola $xy = 4$ which are parallel to the line $y = x$ as diameters. All such circles pass through two fixed points which are
(A) (2, 2)
(B) (2, –2)
(C) (–2, 2)
(D) (–2, –2)

251. The normal at one extremity of latus rectum in the first quadrant of ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $a > b > 0$, meets the rectangular hyperbola $xy = 9$ at points P and Q. Then
(A) if P is $\left(6, \frac{3}{2}\right) \Rightarrow$ Q is $\left(-\frac{3\sqrt{2}}{2}, -3\sqrt{2}\right)$
(B) eccentricity of hyperbola is $\sqrt{2}$
(C) if P is $\left(6, \frac{3}{2}\right) \Rightarrow$ Q is $\left(-\frac{3e}{2}, -\frac{6}{e}\right)$ where e is eccentiricy of ellipse
(D) if O is origin, then product of slopes of OP and OQ is positive

252. If $ax^2 - bx + c = 0$ have two distinct root lying in the interval (0, 1) a, b, $c \in N$, then the value of $\log_5(abc)$ is greater than
(A) 1
(B) 2
(C) 3
(D) 4

253. If both roots of $(2a - 4)9^x - (2a - 3)3^x + 1 = 0$ are non-negative, then 'a' can be
(A) $\frac{9}{4}$
(B) $\frac{13}{4}$
(C) $\frac{11}{5}$
(D) $\frac{13}{6}$

254. Consider a four digit number having a, b, c, d as its digits. If a, b, c are in G.P. not necessarily in the same order b, d, a and b, c, d form an increasing A.P., then
(A) $a + b + c + d = 18$
(B) $a + b + d = 17$
(C) $abcd = x^2 - y^2$; $x \in N$, $y \in N$
(D) $abcd = x^2 + y^2$; $x \in N$, $y \in N$

255. If p(x) is a polynomial with rational coefficient so that for all $|x| \le 1$; $p(x) = p\left(\frac{-x+\sqrt{3-3x^2}}{2}\right)$, then
(A) $p(x) = a_0 + a_1(3x - 4x^3)^1 + a_2(3x - 4x^3)^2 + \ldots + (3x - 4x^3)^k\, k(x)$ where k(x) is a polynomial with rational coefficient
(B) $p'(x) = a_1 \phi'(x) + 2a_2 \phi(x) \phi'(x) + \ldots + k[\phi(x)]^{k-1} \phi'(x)\, k(x + [\phi(x)]^k\, k'(x)$ where f(x) is a polynomial with rational coefficient and $\phi(x) = 3x - 4x^3$
(C) can't be predicted
(D) none of these

256. For distinct complex numbers $z_1, z_2, \ldots, z_n$, the value of $\frac{|z_2 - z_1|^2 + |z_3 - z_2|^2 + \ldots + |z_n - z_{n-1}|^2}{|z_n - z_1|^2}$ can not be less than

(A) $\frac{1}{n-1}$ (B) $\frac{1}{n}$

(C) $\frac{1}{n+1}$ (D) $\frac{1}{n-2}$

257. For the equation $x + y + z + \omega = 19$, the number of positive integral solutions is equal to
(A) the number of ways in which 15 identical things can be distributed among 4 persons
(B) the number of ways in which 19 identical things can be distributed among 4 persons
(C) coefficient of x^{19} in $(x^0 + x^1 + x^2 + \ldots + x^{19})^4$
(D) coefficient of x^{19} in $(x + x^2 + x^3 + \ldots + x^{19})^4$

258. The sum of the coefficient in the expansion of $(1 + ax^3 - 2x^5)^n$ is
(A) positive, when $a < 1$, and $n = 2k$, $k \in N$
(B) negative, when $a < 1$, and $n = 2k + 1$, $k \in N$
(C) positive, when $a > 1$, and $n \in N$
(D) zero, when $a = 1$

259. If $S_n = \frac{1}{1\cdot(1+3)} + \frac{3}{(1+3)(1+3+9)} + \frac{9}{(1+3+9+27)} + \ldots..$ n terms, then

(A) $S_\infty = \frac{2}{3}$ (B) $S_\infty = \frac{5}{9}$

(C) $S_4 = \frac{40}{121}$ (D) $S_5 = \frac{121}{364}$

260. Let $X = \sum_{r=0}^{n}\left(\frac{(-1)^r}{r+1}\right)$, then $|X - \ln 2|$ will be

(A) greater than $\frac{1}{2(n+1)}$ (B) greater than $\frac{1}{2(n+2)}$

(C) less than $\frac{1}{2(n+1)}$ (D) less than $\frac{1}{2(n+2)}$

261. Consider the expansion $(1 - 5x + 6x^2)^{-1}$, then $(|x| < 1)$
(A) coefficient of $x^n = 3^n - 2^n$ (B) coefficient of $x^n = 3^{n+1} - 2^{n+1}$
(C) coefficient of $x^{n-1} = 3^n - 2^n$ (D) coefficient of $x^{n-1} = 3^{n-1} - 2^{n-1}$

262. The value of ${}^{1000}C_{50} + 2\,{}^{999}C_{49} + 3\,{}^{998}C_{48} + \ldots.. + 51\,{}^{950}C_0$ is equal to

(A) ${}^{1002}C_{49}$ (B) ${}^{1002}C_{50}$

(C) ${}^{1002}C_{952}$ (D) ${}^{1002}C_{953}$

263. If $P = n(n^2 - 1^2)(n^2 - 2^2)(n^2 - 3^2) \ldots (n^2 - r^2)$, $n > r$, $n \in N$, then P is divisible by
(A) $(2r + 2)!$ (B) $(2r - 1)!$
(C) $(2r + 1)!$ (D) none of these

264. Consider the equation $z^n + z^{n+1} + z^{n+2} + ... + z^{2n} = 0$, if the non-zero roots of the equation are plotted on the Argand plane to form a polygon and L_n represents the length of the longest side of this polygon, then

(A) L_5 is $\frac{\sqrt{3}}{2}$

(B) L_5 is $\sqrt{3}$

(C) $\lim_{n\to\infty} \frac{L_n}{n}$ is 0

(D) $\lim_{n\to\infty} nL_n$ is 4π

265. If $|z_1|^3 + |z_2|^3 + |z_3|^3 - 3|z_1z_2z_3| = 0$, then

(A) maximum area of triangle formed by z_1, z_2, z_3 is $\frac{3\sqrt{3}}{4}|z_1|^2$

(B) orthocentre of triangle formed by z_1, z_2, z_3 is $z_1 + z_2 + z_3$

(C) if $z_1 + z_2 + z_3 = 0$, then orthocentre of triangle formed by z_1, z_2, z_3 is origin

(D) $2\arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right) = \arg\left(\frac{z_2}{z_3}\right)$

266. If $z = -2 + 2\sqrt{3}i$ then $z^{2n} + 2^{2n}z^n + 2^{4n}$ be equal to

(A) 2^{2n}

(B) 0, n is not a multiple of 3

(C) $3\cdot 2^{4n}$, n is multiple of 3

(D) none of these

267. Number of ways to select three numbers from first 'n' whole number such that they are in A.P. are

(A) $\frac{n(n-2)}{4}$ if n is even

(B) $\frac{n^2-1}{4}$ if n is odd

(C) $\frac{n^2-1}{4}$ if n is even

(D) $\frac{n^2}{4}$ if n is odd

268. Number of ways in which 100 people can be divided into 50 couples are

(A) $\frac{100!}{2^{50}\cdot 50!}$

(B) $\frac{100!}{50!}$

(C) $50!\left(\frac{51}{2}\right)\left(\frac{53}{2}\right).....\left(\frac{99}{2}\right)$

(D) $\frac{50!}{2^{25}}\left(\frac{51}{2}\right)\left(\frac{53}{2}\right).....\left(\frac{99}{2}\right)$

269. If 2050! is divisible by 17^n, then n can be

(A) 125

(B) 126

(C) 127

(D) 128

270. If p and q are chosen randomly from the set {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} with replacement, then the probability that the roots of the equation $x^2 + px + q = 0$

(A) are real is 33/50

(B) are imaginary is 19/50

(C) are real and equal is 3/100

(D) are real and distinct is 3/5

271. The probability that a student passes at least in one of the three examinations A, B, C is 0.75; the probability that he passes in atleast two of the exams is 0.5 and the probability he passes exactly two of the exams is 0.4. If α, β, γ are respectively the probabilities of the student passing in A, B, C, then

(A) $\alpha + \beta + \gamma = \frac{17}{20}$

(B) $\alpha + \beta + \gamma = \frac{27}{20}$

(C) $\alpha\beta\gamma = \frac{1}{10}$

(D) $\alpha^2 + \beta^2 + \gamma^2 = 1$

272. Let $\vec{a}=2\hat{i}-\hat{j}+\hat{k}$, $\vec{b}=\hat{i}+2\hat{j}-\hat{k}$ and $\vec{c}=\hat{i}+\hat{j}-2\hat{k}$ be three vectors. A vector in the plane of $\vec{b}$ and $\vec{c}$ whose projection on $\vec{a}$ is of magnitude $\sqrt{\frac{2}{3}}$ is

(A) $2\hat{i}+3\hat{j}-3\hat{k}$
(B) $2\hat{i}+3\hat{j}+3\hat{k}$
(C) $-2\hat{i}-\hat{j}+5\hat{k}$
(D) $2\hat{i}-\hat{j}+5\hat{k}$

273. Let unit vectors $\vec{a}$ and $\vec{b}$ are perpendicular and unit vector $\vec{c}$ be inclined at an angle θ to both $\vec{a}$ and $\vec{b}$. If $\vec{c}=\alpha\vec{a}+\beta\vec{b}+\gamma(\vec{a}\times\vec{b})$, then

(A) $\alpha=\beta$
(B) $1-2\alpha^2=\gamma^2$
(C) $\alpha^2=\frac{1+\cos 2\theta}{2}$
(D) $\alpha^2-\beta^2=\gamma^2$

274. If $\bar{a}$ and $\bar{b}$ are two vectors and angle between $\bar{a}$ and $\bar{b}$ is θ, then

(A) $|\bar{a}\times\bar{b}|^2+(\bar{a}\cdot\bar{b})^2=|\bar{a}|^2|\bar{b}|^2$
(B) $|\bar{a}\times\bar{b}|=(\bar{a}\cdot\bar{b})$, if $\theta=\frac{\pi}{4}$
(C) $\bar{a}\times\bar{b}=(\bar{a}\cdot\bar{b})\hat{n}$, ($\hat{n}$ is normal unit vector), if $\theta=\frac{\pi}{4}$
(D) $(\bar{a}\times\bar{b})\cdot(\bar{a}+\bar{b})=0$

275. Consider the points $A(\vec{o})$, $B(3\hat{i}+f(t)\hat{j}+\hat{k})$ and $C(2\hat{i}+f'(t)\hat{j}+2\hat{k})$ and another point D are coplanar for all $t\in R$

(A) if D is $(\hat{i})$ then $f(x)=ce^{2x}$, $c\in R$
(B) if D is $(\hat{i})$ then $f(x)=c\ln(2x)$, $c\in R$
(C) if D is $(\hat{i}+\hat{j})$ then $f(x)=2+ce^{2x}$, $x\in R$
(D) if D is $(\hat{i}+\hat{j})$ then $f(x)=2+c\ln(2x)$, $c\in R$

276. If ABCD is a tetrahedron, volume of tetrahedron ABCD is $\frac{1}{6}$. Also $\angle ACB=45^\circ$ and $AD+BC+\frac{AC}{\sqrt{2}}=3$. Then which of the following is correct?

(A) $AD=1$
(B) $AC=\sqrt{2}$
(C) $DC=\sqrt{3}$
(D) $AD=\sqrt{2}$

277. Let a plane pass through origin and is parallel to the line $\frac{x-1}{2}=\frac{y+3}{-1}=\frac{z+1}{-2}$ such that the distance between plane and the line is $\frac{5}{3}$. Then the equation of the plane is

(A) $x-2y+2z=0$
(B) $x-2y-2z=0$
(C) $2x+2y+z=0$
(D) $x+y+z=0$

278. Three mutually perpendicular lines are drawn from the point (1, 2, −1). If one of the lines is perpendicular to the x-axis and the direction ratios of the second line are (1, 2, −1), then the equation(s) of the third line is/are

(A) $\frac{x-1}{5}=\frac{y-2}{-2}=\frac{z+1}{1}$
(B) $\frac{x+4}{5}=\frac{y-4}{-2}=\frac{z+2}{1}$
(C) $\vec{r}=6\hat{i}+\lambda(5\hat{i}-2\hat{j}+\hat{k})$
(D) $\frac{x-1}{-5}=\frac{y-2}{2}=\frac{z+1}{-1}$

279. Let $f(x)=\frac{3}{x-2}+\frac{4}{x-3}+\frac{5}{x-4}$, then $f(x)=0$ has

(A) 2 roots in (2, 3)
(B) 2 roots in (3, 4)
(C) exactly one root in (2, 3)
(D) exactly one root in (3, 4)

280. If the line $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ intersects the line $3\beta^2x + 3(1 - 2\alpha)y + z = 3 = -\frac{1}{2}\left(6\alpha^2x+3(1-2\beta)y+2z\right)$. Then point $(\alpha, \beta, 1)$ lie on the plane

(A) $2x - y + z = 4$
(B) $x + y - z = 2$
(C) $x - 2y = 0$
(D) $2x - y = 0$

281. $\cot 12^\circ \cdot \cot 24^\circ \cdot \cot 28^\circ \cdot \cot 32^\circ \cdot \cot 48^\circ \cdot \cot 88^\circ$ is equal to

(A) $\tan 45^\circ$
(B) 2
(C) $2 \tan 15^\circ \cdot \tan 45^\circ \cdot \tan 75^\circ$
(D) $\tan 15^\circ \cdot \tan 45^\circ \cdot \tan 75^\circ$

282. The triangle formed by the normal to the curve $f(x) = x^2 - ax + 2a$ at the point (2, 4) and the co-ordinate axes lies in second quadrant if its area is 2 sq. units then a can be

(A) 2
(B) 17/4
(C) 5
(D) 19/4

283. Which of the following triangles ABC have the altitude from A, angle bisector of $\angle B$ and median through C concurrent

(A) $A = B = C$
(B) $B=\frac{\pi}{6}, C=\frac{\pi}{3}, \frac{c}{a}=3$
(C) $B=\frac{\pi}{6}, C=\frac{\pi}{3}, \frac{a}{c}=3$
(D) $B=\frac{\pi}{3}, C=\frac{\pi}{6}, \frac{a}{c}=3$

284. Let A, B, C be $n \times n$ real matrices and are pair wise commutative and $ABC = O_n$ and if $\lambda = \det(A^3 + B^3 + C^3) \cdot \det(A + B + C)$ then

(A) $\lambda > 0$
(B) $\lambda < 0$
(C) $\lambda = 0$
(D) $\lambda \in (-\infty, \infty) - \{0\}$

285. If $\cos \beta$ is geometric mean between $\sin \alpha$ and $\cos \alpha$, where $0 < \alpha, \beta < \frac{\pi}{2}$, then $\cos 2\beta$ is equal to

(A) $-2\sin^2\left(\frac{\pi}{4}-\alpha\right)$
(B) $-2\cos^2\left(\frac{\pi}{4}+\alpha\right)$
(C) $2\sin^2\left(\frac{\pi}{4}+\alpha\right)$
(D) $2\cos^2\left(\frac{\pi}{4}-\alpha\right)$

286. The inequality $4 \sin 3x + 5 \geq 4 \cos 2x + 5 \sin x$ is true for x belongs to

(A) $\left[-\pi, \frac{3\pi}{2}\right]$
(B) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
(C) $\left[\frac{5\pi}{8}, \frac{13\pi}{8}\right]$
(D) $\left[\frac{23\pi}{14}, \frac{41\pi}{14}\right]$

287. The possible value(s) of 'θ' satisfying the equation $\sin^2 \theta \tan \theta + \cos^2 \theta \cot \theta - \sin 2\theta = 1 + \tan \theta + \cot \theta + \cot \theta$, where $\theta \in [0, \pi]$ is/are

(A) $\pi/4$
(B) $7\pi/6$
(C) $\frac{7\pi}{12}$
(D) $\frac{11\pi}{12}$

288. Three equal circles each of radius r touch one another externally. The radius of the circle touching all the three given circles is

(A) $\left(2-\sqrt{3}\right)r$
(B) $\frac{2+\sqrt{3}}{\sqrt{3}}r$
(C) $\frac{2-\sqrt{3}}{\sqrt{3}}r$
(D) $\left(2-\sqrt{3}\right)r$

289. Let x_1, x_2, x_3, x_4 be four non-zero numbers satisfying the equation $\tan^{-1}\frac{a}{x}+\tan^{-1}\frac{b}{x}+\tan^{-1}\frac{c}{x}+\tan^{-1}\frac{d}{x}=\frac{\pi}{2}$.

Then

(A) $x_1 + x_2 + x_3 + x_4 = a + b + c + d$

(B) $\frac{1}{x_1}+\frac{1}{x_2}+\frac{1}{x_3}+\frac{1}{x_4}=0$

(C) $x_1x_2x_3x_4 = abcd$

(D) $(x_2 + x_3 + x_4)(x_3 + x_4 + x_1)(x_4 + x_1 + x_2)(x_1 + x_2 + x_3) = abcd$

290. Given an isosceles triangle with equal sides of length b and base angle $\alpha < \frac{\pi}{4}$. Then

(A) $R = \frac{b}{2}\operatorname{cosec}\alpha$

(B) $\Delta = 2b^2 \sin 2\alpha$

(C) $r = \frac{b\sin 2\alpha}{4(1+\cos\alpha)}$

(D) none of these

291. The solution of the equation $\sin 2x + \sin 4x = 2 \sin 3x$ is

(A) $x = \frac{n\pi}{3}$

(B) $x = n\pi$

(C) $x = 2n\pi$

(D) none of these

292. If $g(x)=\begin{vmatrix} a^{-x} & e^{x\log_e a} & x^2 \\ a^{-3x} & e^{3x\log_e a} & x^4 \\ a^{-5x} & e^{5x\log_e a} & 1 \end{vmatrix}$, then

(A) graph of g(x) is symmetrical about origin

(B) graph of g(x) is symmetrical about y-axis

(C) $\left.\frac{d^4g(x)}{dx^4}\right|_{x=0} = 0$

(D) $f(x) = g(x)\log\left(\frac{a-x}{a+x}\right)$ is an odd function

293. If $f(x)=\begin{vmatrix} (1-x)^a & (1+x)^b & 1 \\ 1 & (1-x)^a & (1+x)^b \\ (1+x)^b & 1 & (1-x)^a \end{vmatrix}$, then

(A) constant term of f(x) = a + b

(B) constant term of f(x) = 0

(C) coefficient of x in f(x) is zero

(D) none of these

294. Let $\alpha = \frac{\pi}{5}$ and $A = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$, then $B = A + A^2 + A^3 + A^4$

(A) singular

(B) non-singular

(C) skew symmetric

(D) $|B| = 1$

295. If $A = (a_{ij})_{n\times n}$ and f is a function defined as $f(A) = \{f(a_{ij})\}_{n\times n}$. Let $A = \begin{bmatrix} \frac{\pi}{2}-\theta & \theta \\ -\theta & \frac{\pi}{2}-\theta \end{bmatrix}$. Then

(A) sin A is invertible

(B) sin A = cos A

(C) sin A is orthogonal

(D) sin (2A) = 2 sin A cos A

296. If $\left(a^{\log_b x}\right)^2 - 5x^{\log_b a} + 6 = 0$, where $a > 0, b > 0$ and $ab \neq 1$, then the value of x can be equal to

(A) $b^{\log_a 2}$
(B) $3^{\log_a b}$
(C) $2^{\log_b a}$
(D) $a^{\log_b 3}$

297. If $\alpha = \log_4 5$ and $\beta = \log_5 6$, then

(A) $\alpha\beta = \log_4 6$
(B) $\alpha\beta = \log_6 4$
(C) $\frac{1}{2\alpha\beta - 1} = \log_3 2$
(D) $\frac{1}{2\alpha\beta - 1} = \log_2 3$

298. If x_1 and x_2 are positive numbers between 0 and 1, then which of the following is/are true?

(A) $\sin\left(\frac{x_1 + x_2}{2}\right) \leq \frac{\sin x_1 + \sin x_2}{2}$
(B) $\tan\left(\frac{x_1 + x_2}{2}\right) \leq \frac{\tan x_1 + \tan x_2}{2}$
(C) $\log\left(\frac{x_1 + x_2}{2}\right) \leq \frac{\log x_1 + \log x_2}{2}$
(D) $\left(\frac{x_1 + x_2}{2}\right)^2 \leq \frac{x_1^2 + x_2^2}{2}$

299. The solution(s) of the equation $2\log_3(x-2) + \log_3(x-4)^2 = 0$, is/are

(A) $3 + \sqrt{2}$
(B) $3 - \sqrt{2}$
(C) 3
(D) 5

300. The sum of the series $\sum_{r=0}^{\infty}\left(\log_3\left(e^{4/3}\right)^r\right)^{-1}$ is

(A) $\log_e 81$
(B) $8\log_e 2^3$
(C) $8\log_e 3^2$
(D) $4\log_e 3$

COMPREHENSIONS

COMPREHENSION–I

Read the paragraph carefully and answer the following questions:
Let f, g, h be real-valued functions defined on R. Consider $f(x) = -1 + 4\sin^2 x$, $g(x) = 2x - 1$ and $h(x) = 2x + 1$.

301. The range of function $y = \frac{h(f(x))}{g(f(x))}$ is equal to

(A) $\left[\frac{1}{3}, \frac{7}{5}\right]$
(B) $(-\infty, 1) \cup \left[\frac{7}{5}, \infty\right)$
(C) $\left[1, \frac{7}{5}\right]$
(D) $\left(-\infty, \frac{1}{3}\right] \cup \left[\frac{7}{5}, \infty\right)$

302. If $\lim_{x \to -\frac{1}{2}} \frac{f\left(\sin^{-1} kx\right) - 3}{h(x)}$ exists and have value equal to ℓ then $|k| + |\ell|$ is equal to

(A) 6
(B) 8
(C) 10
(d) 12

COMPREHENSION–II

Read the paragraph carefully and answer the following questions:
Definite integral of any discontinous or non–differenciable function is normally solved by the property $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$ wherer $c \in (a, b)$ is the point of discontinuity or non-differenciability

303. $A = \int_1^{\infty} [\text{cosec}^{-1}x]dx$ is equal to (where [.] denote the greatest integer function)

(A) cosec 1 – 1
(B) 1
(C) 1 – sin 1
(D) none of these

304. $B = \int_1^{100} [\sec^{-1}x]dx$ is equal to (where [.] denote the greatest integer function)

(A) sec 1
(B) 100 – sec 1
(C) 99 – sec 1
(D) none of these

COMPREHENSION–III

Read the paragraph carefully and answer the following questions:
Let $f : R \to R$, $y = f(x)$, $f(0) = 0$, $f'(x) > 0$ and $f''(x) > 0$. Three points $A(\alpha, f(\alpha))$, $B(\beta, f(\beta))$, $C(\gamma, f(\gamma))$ on $y = f(x)$ such that $0 < \alpha < \beta < \gamma$.

305. Which of the following is false

(A) $\alpha f(\beta) > \beta(f(\alpha))$
(B) $\alpha f(\beta) < \beta f(\alpha)$
(C) $\gamma f(\beta) < \beta(f(\gamma))$
(D) $\gamma (f(\alpha)) < \alpha f(\gamma)$

306. Which of following is true

(A) $\frac{f(\alpha)+f(\beta)}{2} < f\left(\frac{\alpha+\beta}{2}\right)$
(B) $\frac{f(\alpha)+f(\beta)}{2} > f\left(\frac{\alpha+\beta}{2}\right)$
(C) $\frac{f(\alpha)+f(\beta)}{2} = f\left(\frac{\alpha+\beta}{2}\right)$
(D) $\frac{2f(\alpha)+f(\beta)}{3} < f\left(\frac{2\alpha+\beta}{3}\right)$

COMPREHENSION–IV

Read the paragraph carefully and answer the following questions:
A line PQ parallel to the diagonal BD of a square ABCD with side length 'a' unit is drawn at a distance x from the vertex A, where $x \in [0, \sqrt{2}\,a]$ cuts the adjacent sides at P and Q. Let f(x) be the area of the segment of a square cut off by PQ, with A as one of the vertex.

307. Let $g(x) = f^{-1}(x)$, then the domain of g(x) is

(A) $x \in [0, \sqrt{2}\,a]$
(B) $x \in [0, 2a^2]$
(C) $x \in [\sqrt{2}\,a, a^2]$
(D) $x \in [0, a^2]$

308. For a = 2, the domain of the function $\phi(x) = \sqrt{f^{-1}(x) - f(x)}$ is/are

(A) $x \in [0, 1]$
(B) $x \in [0, 2]$
(C) $x \in [1, 2]$
(D) $x \in [2, \infty)$

COMPREHENSION–V

Read the paragraph carefully and answer the following questions:

It is known that $\sqrt{\tan x}+\sqrt{\cot x}=\begin{cases}\dfrac{\sqrt{\sin x}}{\sqrt{\cos x}}+\dfrac{\sqrt{\cos x}}{\sqrt{\sin x}} & \text{if } 0<x<\dfrac{\pi}{2}\\ \dfrac{\sqrt{-\sin x}}{\sqrt{-\cos x}}+\dfrac{\sqrt{-\cos x}}{\sqrt{-\sin x}} & \text{if } \pi<x<\dfrac{3\pi}{2}\end{cases}$

309. The value of integral $I=\int\left(\sqrt{\tan x}+\sqrt{\cot x}\right)dx$, where $x\in\left(0,\frac{\pi}{2}\right)\cup\left(\pi,\frac{3\pi}{2}\right)$

(A) $\sqrt{2}\tan^{-1}\left(\dfrac{\sqrt{\tan x}-\sqrt{\cot x}}{\sqrt{2}}\right)+c$

(B) $\sqrt{2}\tan^{-1}\left(\dfrac{\sqrt{\tan x}+\sqrt{\cot x}}{\sqrt{2}}\right)+c$

(C) $-\sqrt{2}\tan^{-1}\left(\dfrac{\sqrt{\tan x}-\sqrt{\cot x}}{\sqrt{2}}\right)+c$

(D) $-\sqrt{2}\tan^{-1}\left(\dfrac{\sqrt{\tan x}+\sqrt{\cot x}}{\sqrt{2}}\right)+c$

310. The value of integral $I=\int\left(\sqrt{\tan x}+\sqrt{\cot x}\right)dx$, where $x\in\left(0,\frac{\pi}{2}\right)$

(A) $\sqrt{2}\sin^{-1}(\cos x-\sin x)+c$

(B) $\sqrt{2}\sin^{-1}(\sin x-\cos x)+c$

(C) $\sqrt{2}\sin^{-1}(\sin x+\cos x)+c$

(D) $-\sqrt{2}\sin^{-1}(\sin x+\cos x)+c$

COMPREHENSION–VI

Read the paragraph carefully and answer the following questions:

Let $f(x)=\left|\,|x|-1\right|-1\right|$ and $g(x)=\left\{\max f(t):x\le t\le x+1,x\in R\right\}$.

311. The graph of $g(x)$ is symmetrical about

(A) $x=\dfrac{1}{2}$

(B) $x=\dfrac{-1}{2}$

(C) $x=0$

(D) $x=-1$

312. The number of points of non differentiability of $g(x)$ is

(A) 7

(B) 8

(C) 9

(D) 5

COMPREHENSION–VII

Read the paragraph carefully and answer the following questions:

Let $h(x) = f(x) - g(x)$ where $f(x) = \sin^4\pi x$ and $g(x) = \ln x$. Let $x_0, x_1, x_2, \ldots, x_{n+1}$ be the roots of $f(x) = g(x)$ in increasing order.

313. Then the absolute area enclosed by $y = f(x)$ and $y = g(x)$ is given by

(A) $\displaystyle\sum_{r=0}^{n}\int_{x_r}^{x_{r+1}}(-1)^r h(x)dx$

(B) $\displaystyle\sum_{r=0}^{n}\int_{x_r}^{x_{r+1}}(-1)^{r+1} h(x)dx$

(C) $\displaystyle 2\sum_{r=0}^{n}\int_{x_r}^{x_{r+1}}(-1)^r h(x)dx$

(D) $\displaystyle\frac{1}{2}\sum_{r=0}^{n}\int_{x_r}^{x_{r+1}}(-1)^{r+1} h(x)dx$

314. In the above question the value of n is

(A) 1

(B) 2

(C) 3

(D) 4

COMPREHENSION–VIII

Read the paragraph carefully and answer the following questions:
A certain radioactive material is known to decay at a rate proportional to the amount present. Initially there is 50 kg of the material present and after two hours it is observed that the material has lost 10 percent of its original mass.

315. The expression for the mass of the material remaining at any time t
(A) $N = 50e^{(1/2)(\ln(0.9))t}$ (B) $N = 50e^{-(1/4)(\ln 9)t}$
(C) $N = 50e^{-(\ln(0.9))t}$ (D) none of these

316. The time at which the material has decayed to one half of its initial mass is
(A) $\frac{\ln 1/2}{(1/2)\ln 9}$ hr (B) $\frac{\ln 2}{(1/2)\ln(0.9)}$ hr
(C) $\frac{\ln 1/2}{(1/2)(\ln(0.9))}$ hr (D) none of these

COMPREHENSION–IX

Read the paragraph carefully and answer the following questions:
A straight line passing through O(0, 0) cuts the lines $x = \alpha$, $y = \beta$ and $x + y = 8$ at A, B and C respectively such that $OA \cdot OB \cdot OC = 48\sqrt{2}$ and $f(\alpha, \beta) \le 0$. Where $f(x, y) = \left|\frac{y}{x} - \frac{3}{2}\right| + (3x - 2y)^6 + \sqrt{ex + 2y - 2e - 6}$

317. The point of intersection of lines $x = \alpha$ and $y = \beta$ is
(A) (3, 2) (B) (2, 3)
(C) (3, 4) (D) (e, 3)

318. OA + OB + OC equals
(A) $2\sqrt{2}$ (B) $4\sqrt{2}$
(C) $7\sqrt{2}$ (D) $9\sqrt{2}$

COMPREHENSION–X

Read the paragraph carefully and answer the following questions:
For each positive real number k, let C_k denotes the circle with centre at origin and radius k units. On a circle C_k, particle α moves k units in the counter–clock wise direction. After completing its motion on C_k, the particle moves on to the circle $C_{k+\gamma}$, in some well defined manner, where $\gamma > 0$. The motion of the particle continues in this manner.

Now answer the following questions based on the above comprehension:

319. Let $k \in I^+$ and $\gamma = 1$, particle moves in the **radial direction** from circle C_k to C_{k+1}. If particle starts from the point P(–1, 0) then
(A) It will cross the x–axis again at (3, 0) (B) It will cross the x–axis again at (4, 0)
(C) It will cross the +ve y–axis at (0, 4) (D) It will cross the +ve y–axis at (0, 6).

320. Let $k \in I^+$, $\gamma \in R$, particle moves **tangentially** from the circle C_k to $C_{k+\gamma}$, such that length of tangent is equal to k units itself. If particle starts from the point Q(1, 0), then
(A) The particle will cross +ve x–axis again for $2\sqrt{2} < x < 4$
(B) The particle will cross +ve x–axis again at $x = 2\sqrt{2}$
(C) The particle will cross x–axis again at $x = 4$
(D) The particle will cross x–axis again at $x = 3$

COMPREHENSION–XI

Read the paragraph carefully and answer the following questions:

Consider the following lines:

$L_1 : x - y - 1 = 0$

$L_2 : x + y - 5 = 0$

$L_3 : y - 4 = 0$

Let L_1 is axis to a parabola, L_2 is tangent at the vertex to this parabola and L_3 is another tangent to this parabola at some point P.

Let 'C' be the circle circumscribing the triangle formed by tangent and normal at point P and axis of parabola. The tangent and normals at the extremities of latus rectum of this parabola form a quadrilateral ABCD.

321. The equation of the circle 'C' is

(A) $x^2 + y^2 - 2x - 31 = 0$

(B) $x^2 + y^2 - 2y - 31 = 0$

(C) $x^2 + y^2 - 2x - 2y - 31 = 0$

(D) $x^2 + y^2 + 2x + 2y = 31$

322. The area of the quadrilateral ABCD is

(A) 16

(B) 8

(C) 64

(D) 32

COMPREHENSION–XII

Read the paragraph carefully and answer the following questions:

A coplanar beam of light emerging from a point source have equation $\lambda x - y + 2(1 + \lambda) = 0$, $\lambda \in R$. the rays of the beam strike an elliptical surface and get reflected. The reflected rays form another convergent beam having equation $\mu x - y + 2(1 - \mu) = 0$, $\mu \in R$. Further it is found that the foot of the perpendicular from the point (2, 2) upon any tangent to the ellipse lies on the circle $x^2 + y^2 - 4y - 5 = 0$.

323. The eccentricity of the ellipse is equal to

(A) $\frac{1}{3}$

(B) $\frac{1}{\sqrt{3}}$

(C) $\frac{2}{3}$

(D) $\frac{1}{2}$

324. The area of the largest triangle that an incident ray and the corresponding reflected ray can enclose with the axis of the ellipse is equal to

(A) $4\sqrt{5}$

(B) $2\sqrt{5}$

(C) $\sqrt{5}$

(D) none of these

COMPREHENSION–XIII

Read the paragraph carefully and answer the following questions:

A ray of light emancipating from the point source P(1, –3, 2) and traveling parallel to the line $\frac{x-2}{1} = \frac{y}{3} = \frac{z+1}{2}$ is incident on the plane $x + y - 3z = 0$ at the point Q. After reflecting from the plane the ray travels along the line QR. It is also known that the reflected ray, the incident ray and the normal to the plane at the point of incident are in the same plane. Then

325. The equation of the line QR is

(A) $\frac{x-12}{15} = \frac{y-22}{37} = \frac{z-4}{10}$

(B) $\frac{x-3}{3} = \frac{y-15}{7} = \frac{z-6}{2}$

(C) $\frac{x-3}{15} = \frac{y-6}{37} = \frac{z-3}{10}$

(D) $\frac{x+3}{3} = \frac{y+6}{7} = \frac{z+3}{2}$

326. The equation of the plane PAR is

(A) $5x + 2y - z + 3 = 0$

(B) $11x - 5y + 2z = 30$

(C) $5x - y - z = 6$

(D) $x - y + z = 6$

COMPREHENSION–XIV

Read the paragraph carefully and answer the following questions:
Let $f(x) = 4x^4 - 24x^3 + 31x^2 + 6x - 8$ be a polynomial function suppose $f(x) = 0$ has four roots $\alpha < \beta < \gamma < \delta$. If sum of two of its roots vanishes. Then

327. The value of the expression $\delta^{\beta} + \frac{1}{\delta^{\alpha}} + \delta^{\gamma} + \gamma^{\delta}$ is

(A) 36
(B) 35
(C) 20
(D) 16

328. $\int_{2\alpha}^{2\beta} \frac{x^{\delta+1} - 5x^{\gamma+1} + 2\beta|x| - 1}{x^2 + 4\beta|x| + 1} dx$ is

(A) $\ell n2$
(B) $2\ell n2$
(C) $\frac{1}{2}\ell n2$
(D) none of these

COMPREHENSION–XV

Read the paragraph carefully and answer the following questions:
Let $f(x) = x^2 + bx + c \ \forall \ x \in R$, $(b, c \in R)$ attains it's least value at $x = -1$, and the graph of $f(x)$ cuts y-axis at $y = 2$ now answer the following

329. The least value of $f(x) \ \forall \ x \in R$ is

(A) -1
(B) 0
(C) 1
(D) $\frac{3}{2}$

330. If $f(x) = a$ has two distinct real roots, then complete set of values of 'a' is

(A) $(1, \infty)$.
(B) $(-2, -1)$
(C) $(0, 1)$
(D) $(1, 2)$

COMPREHENSION–XVI

Read the paragraph carefully and answer the following questions:

If r^{th} term of a series can be written as $a_r = f(r) - f(r-1)$, then $S_n = \sum_{r=1}^{n} a_r = f(n) - f(0)$ and $S_\infty = \lim_{n\to\infty} S_n$, then

331. Value of $\sum_{r=1}^{\infty} \frac{(4r-1)5^r}{r^2 + r}$ is

(A) 5
(B) -5
(C) 10
(D) none of these

332. Value of $(\cos 1 - 1)\sum_{1}^{50} \sin n + \sin 1 \sum_{1}^{50} \cos n$ is equal to

(A) $\sin 50°$
(B) $\sin 50° - \sin 1°$
(C) $\sin 51° - \sin 1°$
(D) 1/2

COMPREHENSION–XVII

Read the following passages and answer the following questions

Let f(n) denotes the number of different ways in which the positive integer n can be expressed as the sum of 1's and 2's. For example, f(4) = 5, since 4 = 2 + 2 = 2 + 1 + 1 = 1 + 2 + 1 = 1 + 1 + 2 = 1 + 1 + 1 + 1. Note that order of 1's and 2's are important.

333. The value of f(6) is
(A) 12
(B) 13
(C) 18
(D) 21

334. The value of f(f(6)) is
(A) 400
(B) 350
(C) 377
(D) none of these

COMPREHENSION–XVIII

Read the paragraph carefully and answer the following questions:

A player 'A' plays a game against a machine. At each round he deposits one rupee in a slot and then flips a coin which has a probability 'P' of showing a head. If he flipped coin shows head, he gets back the rupee he deposited and one more rupee from the machine, else he loses his rupee. Let 'A' starts with 10 rupee coins and q = 1 – p (the probability of showing a tail), then

335. The probability that he will be drained out with all of his rupee coins exactly at the 12^{th} round is
(A) q^{11}
(B) $1 - q^{11}$
(C) $pq^{10} + q^{11}$
(D) 0

336. The probability that he is left with no money by the 14^{th} round or earlier is
(A) $q^{10}(1 + 10pq + 65p^2q^2)$
(B) $q^{14}(p^2q + 36pq + 7)$
(C) $q^{12} + 3pq^{13} + 3p^{13}q + p^{12}$
(D) $1 - {}^{10}C_1pq^{11} - {}^{10}C_2p^2q^{12}$

COMPREHENSION–XIX

Read the paragraph carefully and answer the following questions:

In argand plane consider $A(z_1)$, $B(z_2)$ and $C(z_3)$ as distinct complex numbers lying on the curve $|z| = \sqrt{3}$.

337. If a root of $z_1z^2 + z_2z + z_3 = 0$ has modulus unity then z_1, z_2, z_3 are in
(A) A.P.
(B) G.P.
(C) H.P.
(D) all of above

338. If $z_1z^2 + z_2z + z_3 = 0$ and $z_2z^2 + z_3z + z_1 = 0$ each have a common root having modulus unity then ΔABC is
(A) isosceles
(B) equilateral
(C) isosceles right angled
(D) scalene

COMPREHENSION–XX

Read the paragraph carefully and answer the following questions:

There exists a matrix Q such that $PQP^T = N$, where $P = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Given N is a diagonal matrix of form N = diag. (n_1, n_2, n_3) where n_1, n_2, n_3 are three values of n satisfying the equation $\det.(P - nI) = 0$, $n_1 < n_2 < n_3$.

(**Note:** I is an identity matrix of order 3×3)

339. The value of det.(adj N) is equal to (Note: adj M denotes the adjoint of a square matrix M)

(A) 4 (B) $\frac{1}{4}$

(C) $\frac{1}{9}$ (D) 9

340. The trace of matrix P^{2012} is equal to (Note: the trace of a matrix is the sum of its diagonal entries)

(A) $3^{2011}+2$ (B) 3^{2012}

(C) $3^{2012}+2$ (D) 3^{2011}

COMPREHENSION–XXI

Read the paragraph carefully and answer the following questions:

Let f(x), g(x) and h(x) be function of degree more than 5 and $\Delta(x)$

$$\Delta(x)=\begin{vmatrix} xf(x)+f'(\alpha) & xg(x)+g'(\alpha) & xh(x)+h'(\alpha) \\ -2f(\alpha)+3f'(\alpha) & -2g(\alpha)+3g'(\alpha) & -2h(\alpha)+3h'(\alpha) \\ f'(\alpha) & g'(\alpha) & h'(\alpha) \end{vmatrix}$$

341. If f(x), g(x), h(x) are even function $\Delta(x)$ will not give remainder 0 when divided by

(A) $x^2+\alpha x$ (B) $x^2-\alpha^2$

(C) $x(x+\alpha)^2$ (D) none of these

342. Let $\alpha = 1$ and $f(x) = x^5 + 2x^4$, $g(x) = x^7 - x^8$, $h(x) = 1 + x - x^9$ then remainder when, $\Delta(x)$ is divided by $x^3 - 1$

(A) 0 (B) $176(x-1)$

(C) $58(x-1)$ (D) $-176(x-1)$

COMPREHENSION–XXII

Read the paragraph carefully and answer the following questions:

Let ABC be an acute angle with orthocenter H. D, E and F are feet of perpendicular from A, B and C respectively, on opposite sides. Let R be circumradius of ΔABC.

Given $AH \cdot BH \cdot CH = 12$ and $AH^2 + BH^2 + CH^2 = 10$

343. Value of $\frac{\cos A \cos B \cos C}{\cos^2 A + \cos^2 B + \cos^2 C}$

(A) $\frac{3}{5}$ (B) $\frac{3}{10}$

(C) 3 (D) none of these

344. Value of $HD \cdot HE \cdot HF$ is

(A) $\frac{9}{4}$ (B) $\frac{9}{2}$

(C) $\frac{9}{16}$ (D) none of these

COMPREHENSION–XXIII

Read the paragraph carefully and answer the following questions:

Consider the equation $\int_0^x \left(t^2-8t+13\right)dt = x\sin\left(\frac{a}{x}\right)$

345. If x takes the values for which the equation has a solution, then the number of values of $a \in [0, 100]$ is
(A) 2
(B) 1
(C) 5
(D) 3

346. One of the solutions of $|y - \cos\alpha| < x$, where x and a are values that satisfy the given equation is
(A) $y \in [-5, 7]$
(B) $y \in [-7, 5]$
(C) $y \in [5, 7]$
(D) none of these

COMPREHENSION–XXIV

Read the paragraph carefully and answer the following questions:
Let z be a complex number satisfying $z^2 + 2z\lambda + 1 = 0$; where λ is a parameter which can take any real value.

347. The roots of this equation lie on a certain circle if
(A) $-1 < \lambda < 1$
(B) $\lambda > 1$
(C) $\lambda < 1$
(D) none of these

348. One root lies inside the unit circle and one outside if
(A) $-1 < \lambda < 1$
(B) $\lambda > 1$
(C) $\lambda < 1$
(D) none of these

COMPREHENSION–XXV

Read the paragraph carefully and answer the following questions:

Consider the locus of the complex number z in the Argand plane given by $\text{Re}(z) - 2 = |z - 7 + 2i|$. Let $P(z_1)$, $Q(z_2)$ be two complex numbers satisfying the given locus and also satisfying $\arg\left(\frac{z_1 - (2+\alpha i)}{z_2 - (2+\alpha i)}\right) = \frac{\pi}{2} (\alpha \in R)$.

349. The minimum value of PQ is
(A) 4
(B) 12
(C) 10
(D) 8

350. The minimum value of PR.QR where R represents the point (7, –2) is
(A) 25
(B) 12
(C) 10
(D) $\sqrt{50}$

COMPREHENSION–XXVI

Read the paragraph carefully and answer the following questions:
Let ABC be a triangle with $\angle A = 90°$ and S be its circumcircle. Let S_1 be the circle touching the rays AB, AC and the circle S internally. Further let S_2 be the circle touching the rays AB, AC and circle S externally. If r_1, r_2 be the radii of the circles S_1 and S_2 respectively and a, b, c are the sides opposite to angle A, B, C of ΔABC respectively.

351. r_1 equals
(A) $a + b - c$
(B) $b + c - a$
(C) $a + c - b$
(D) $a + b + c$

352. r_2 equals
(A) $a + b - c$
(B) $b + c - a$
(C) $a + c - b$
(D) $a + b + c$

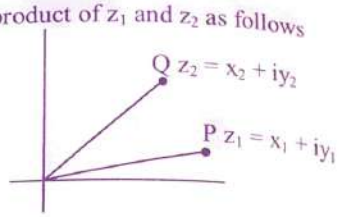
COMPREHENSION–XXVII

Read the paragraph carefully and answer the following questions:

For non–zero complex numbers z_1 and z_2 define the dot product and the cross product of z_1 and z_2 as follows

Dot product $z_1 \cdot z_2 = |z_1|\,|z_2| \cos\theta = x_1x_2 + y_1y_2 = \text{Re}\,(\bar{z}_1 z_2)$

Cross product $z_1 \times z_2 = |z_1|\,|z_2| \sin\theta = x_1y_2 - x_2y_1 = \text{im}\,(\bar{z}_1 z_2)$



353. $z_1z_2 + i(z_1 \times z_2)$ equals

(A) z_1z_2 (B) $z_1\bar{z}_2$

(C) $\bar{z}_1z_2$ (D) $\bar{z}_1\bar{z}_2$

354. $z_1 \cdot z_2 = = 0$ is equivalent to say

(A) $\frac{z_2}{z_1}$ is purely imaginary (B) $\frac{z_1}{z_2}$ is purely real

(C) $z_1 = 0$ or $z_2 = 0$ (D) none of these

COMPREHENSION–XXVIII

Read the paragraph carefully and answer the following questions:

The logarithmic function is defined as, if $\log_a x = y \Rightarrow (a)^y = x$ where $x > 0$, $a > 0$ and $a \neq 1$. Properties of logarithmic function is defined as $a^{\log_a x} = x$

355. The expression $q^{\left(\sqrt{6+\sqrt{6+\sqrt{6+\ldots\ldots\infty}}}\right)^{\log_3 p}}$ $^{\frac{\log_q(\log_q r)}{(\log_q p)}}$ simplified to

(A) p (B) r

(C) $\frac{p}{q}$ (D) $\frac{p}{r}$

356. If a, b are co-prime numbers and satisfying $\left(2+\sqrt{3}\right)^{\frac{1}{\log_a(2-\sqrt{3})}+\frac{1}{\log_b\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)}} = \frac{1}{12}$, then (a + b) can be is equal to

(A) 13 (B) 5

(C) 7 (D) 8

TEST YOURSELF

NUMERICAL BASED PROBLEMS

(One Integer Value Correct Type)

357. Let $f(x) = 30 - 2x - x^3$ and $f_n(x) = \text{fof}\underset{(n\text{ times})}{\ldots}f(x)$, then find the number of positive integral values of x which satisfies $f_{2014}(x) < f_{2013}(-x)$ is ____

358. Let S_1 and S_2 be two circles of unit radius touching each other externally and the line L is one of the direct common tangent. A circle C_1 with radius r_1 is such that it touches circles S_1, S_2 and the line L. Similarly circles C_2, C_3, C_4, C_5, ..., C_n are the circles of radius $r_2, r_3, r_4, \ldots, r_n$ where circle C_r touches circle S_1, C_{r-1} and one of the direct common tangents between them. If $\lim\limits_{n\to\infty} \dfrac{\frac{1}{r_1}+\frac{1}{r_2}+\ldots+\frac{1}{r_n}}{n^3} = \dfrac{p}{q}$ where p and q are relatively prime, then find the value of $(p + q)$ ____

359. Let $f(x) = a_1 \sin x + a_2 \sin 2x + \ldots + a_n \sin nx$, where $a_i \in R$ and $n \in N$. If $|f(x)| \le |\sin x|\ \forall\ x$, then the maximum value of $|a_1 + 2a_2 + \ldots + na_n|$ is ____

360. Let $y = x^4 + 9x^3 + ax^2 + 9x + 4$ contain four collinear points then find the maximum value of $\left[\dfrac{a}{5}\right]$ ____ (where [.] denotes the greatest integer function)

361. If $\int \dfrac{(1+x^2)}{3-4\sqrt{2}x+4x^4}dx = \dfrac{a}{x+b} + c\tan^{-1}(x+d) + e$, then $a + b + c + d$ is equal to ____

362. If $\int (\sin 101x)\cdot \sin^{99} x\,dx$ is equal to $\dfrac{\sin(100x)(\sin x)^{100}}{25\times K} + C$ then K is equal to ____

363. The value of integral $\int\limits_{1/e}^{\tan x} \dfrac{t}{1+t^2}dt + \int\limits_{1/e}^{\cot x} \dfrac{dt}{t(1+t^2)}$ is equal to ____

364. $\dfrac{3}{4}\lim\limits_{n\to\infty}\left(\dfrac{\sum\limits_{r=1}^{n}\frac{1}{\sqrt{r}}\sum\limits_{r=1}^{n}\sqrt{r}}{\sum\limits_{r=1}^{n} r}\right)$ is equal to ____

365. Area enclosed by the solution set of $[x]\cdot[y] = 2$ ____ (where [.] denotes the greatest integer function)

366. If the system of equations $3x + 4y + 1 = 0$, $2x + 5y - 2 = 0$, and $3x^2 + 8xy + 10y^2 + 2x - 8y + 5 + \lambda = 0$ admits unique solution then $[|\lambda|]$ is equal to ____

367. If one side of square ABCD is on the line $y = 2x - 17$ and the other two vertices lie on parabola $y = x^2$; the minimum area of the square is A; then $\left[\sqrt{A}\right]$ is equal to ____ (where [.] denotes the greatest integer function)

368. The line $x = c$ cuts a triangle with vertices (0, 0), (1, 1) and (9, 1) into 2 regions. For the areas of two regions to be same, C must be equal to _____

369. Given a circle $(x + 4)^2 + (y - 2)^2 = 25$. Another circle is drawn passing through (–4, 2) and touching the given circle internally at the point A(–4, 7). AB is a chord of length 8 units of the larger circle intersecting the other circle at the point C. Then AC will be _____

370. If the locus of the focus of the variable parabola C_1, which always touch the curve $C_2 \equiv x^2 = -y$ at the origin, is $x^2 + y^2 = 2ay^4$, (given that C_1 and C_2 have the same latus rectum and they have no other point of intersection). Then the value of a is _____

371. The tangents to the parabola at the extremities of a common chord PQ of the circle $x^2 + y^2 = 5$ and the parabola $y^2 = 4x$ intersects at point T. A square PQRS is constructing on this chord lying inside the parabola, then $\left[\frac{(TR)^2 + (TS)^2}{9}\right]$ is equal to _____ (Where [.] represent greatest integer function)

372. If the equation of the curve on reflection of the ellipse $\frac{(x-4)^2}{16} + \frac{(y-3)^2}{9} = 1$ about the line $x - y - 2 = 0$ is $16x^2 + 9y^2 + k_1x - 36y + k_2 = 0$, then $(k_1 + k_2)/66$ is equal to _____

373. On an ellipse $\frac{x^2}{64} + \frac{y^2}{9} = 1$, tangents drawn at $P_1, P_2, P_3 \ldots P_n$ intersects the major axis at $T_1, T_2, T_3 \ldots T_n$ respectively. if the value of $\sum_{i=1}^{n} \frac{\text{Area}(\Delta P_i T_i S) \cdot \text{Area}(\Delta P_i T_i S')}{(P_i T_i)^2} = 18$, then n is equal to _____

374. The parabola $y = 4 - x^2$ has vertex P. It intersects x-axis at A and B. If the parabola is translated from its initial position to a new position by moving its vertex along the line $y = x + 4$, so that it intersects x-axis at B and C, then abscissa of C will be

375. Let e be the eccentricity of a hyperbola and f(e) be the eccentricity of its conjugate hyperbola then $\int_1^3 \underbrace{f\,f\,f\cdots f}_{n \text{ times}}(e)de$ is equal to _____ (where n is even)

376. The plane denoted by $\pi_1 : 4x + 7y + 4z + 81 = 0$ is rotated through a right angle about its line of intersection with the plane $\pi_2 : 5x + 3y + 10z = 25$. If the plane in its new position be denoted as π, and the distance of this plane from origin is $\sqrt{k}$, where $k \in N$, then $(k - 12)/100$ is _____

377. A rubber cube $C = \{(x, y, z) \mid 0 \le x, y, z \le 1\}$ is cut by a sharp knife along the plane $x = y$, $y = z$ and $z = x$. If no piece is moved until all three cuts are made, then the number of pieces is _____

378. P is a point on the plane $ax + by + cz = d$. A point Q is taken on the line OP such OP. $OQ = d^2$, then the locus of Q satisfies $\frac{d(ax + by + cz)}{x^2 + y^2 + z^2}$ is equal to _____

379. Let $A(\alpha, 0)$, $B(\beta, 0)$, $C(\gamma, 0)$, $D(\delta, 0)$ and α, β are the roots of equation $ax^2 + 2hx + b = 0$ while γ, δ are the those of $a_1x^2 + 2h_1x + b_1 = 0$ if C and D divides AB in the ratio $\lambda : 1$ and $\mu : 1$ respectively also ab_1, hh_1, a_1b are in A.P., then $\lambda + \mu$ is equal to _____

380. Let the values of 'a' for which the roots of the equation $(x - a)(x - a - 1) = 0$ lie between the roots of the equation $(x + a)(x + a^2 - 2) = 0$ be $a \in (-\infty, p) \cup \left(q, \frac{-1+\sqrt{5}}{2}\right)$. Find the value of $q - p$ _____

381. $\frac{\sum_{r=0}^{24} {}^{100}C_{4r} \, {}^{100}C_{4r+2}}{\sum_{r=1}^{25} {}^{200}C_{8r-6}}$ is equal to _____

382. If the number of non-negative integral solutions of $3x + y + z \le 30$ is N, where N is a four-digit integer 'abba', then the value of $b - a$ is _____

383. 6 boys and 5 girls are to be seated around a table. If A be the number of ways of seating them without any restriction and B be the number of ways of seating them such that a particular girl is adjacent to two particular boys then $\left[\frac{A}{7B}\right]$ is _____ (where [.] denotes the greatest integer function)

384. Let z be a complex number satisfying $|z - 3| \le |z - 1|$, $|z - 3| \le |z - 5|$, $|z - i| \le |z + i|$ and $|z - i| \le |z - 5i|$. The the area of region in which z lies is A square units, where A is equal to _____

385. Let $A_1, A_2, \ldots, A_n$ be the vertices of a regular polygon of n sides in a circle of radius unity and $a = |A_1A_2|^2 + |A_2A_3|^2 + \ldots |A_1A_n|^2$, $b = |A_1A_2|\,|A_1A_3| \ldots |A_1A_n|$, then $\frac{a}{b}$ is equal to _____

386. The value of $\begin{vmatrix} 2x_1y_1 & x_1y_2 + x_2y_1 & x_1y_3 + x_3y_1 \\ x_1y_2 + x_2y_1 & 2x_2y_2 & x_2y_3 + x_3y_2 \\ x_1y_3 + x_3y_1 & x_2y_3 + x_3y_2 & 2x_3y_3 \end{vmatrix}$ is _____

387. The number of points in the rectangle $\{(x, y) | -12 \le x \le 12$ and $-3 \le y \le 3\}$ which lie on the curve $y = x + \sin x$ and at which the tangent to the curve is parallel to the x-axis is _____

388. If $a = \frac{2\pi}{1999}$ and $\cos a \cos 2a \cos 3a \ldots \cos 999a = \frac{1}{k}$, then $\frac{\log_2 k}{999}$ is equal to _____

(Numerical Based Answer Type)
(In decimal notation, truncated / rounded off to the second decimal place)

389. Given a right angle triangle ABC with $\angle BAC = 90°$ and $AB = 3$, $AC = \sqrt{3}$. Let I be the incentre and let D and E be the intersections of BI and CI with AC and AB respectively. The value of $\frac{BI^2 + DI^2}{CI^2 + EI^2}$ is equal to _____

390. The number of ordered pairs (x, y) satisfying the equations $\sin x \cos y = 1$ and $x^2 + y^2 \le 9\pi^2$ is k, then the value of 26/k is _____

391. If p_1, p_2, p_3 are the altitudes of a triangle which circumscribe a circle of diameter $\frac{4}{3}$ units, then the least value of $p_1 + p_2 + p_3$ is equal to _____

392. Number of solution(s) of the equation $\frac{\sin x}{\cos 3x}+\frac{\sin 3x}{\cos 9x}+\frac{\sin 9x}{\cos 27x}=0$ is the interval $\left(0,\frac{\pi}{4}\right)$ is _____

393. The minimum area bounded by the function y= f(x) and $y = \alpha x + 9$ ($\alpha \in R$) where f satisfies the relation $f(x + y) = f(x) + f(y) + y\sqrt{f(x)}$ $\forall x, y \in R$ and $f'(0) = 0$ is 9A, value of A is _____

394. If f is continuous function satisfying $f(f(x)) = 1 + x \; \forall \; x \in R$. then f'(1) is equal to _____

395. The longest geometric progression that can be obtained from the set (100, 101, ..., 1000) has the number of terms equal to _____

396. If there are 3 square matrix A, B, C of same order satisfying the equation $A^2 = A^{-1}$, and let $B = A^{2^n}$ and $C = A^{2^{(n-2)}}$ then det(B – C) is equal to _____

397. N is the set of all natural number and R is the set of all real numbers. A function f: N → R is given by $f(n)=\frac{4n+\sqrt{4n^2-1}}{\sqrt{2n-1}+\sqrt{2n+1}}$, the value of $\left[\frac{\sum_{r=1}^{40} f(r)}{40}\right]$ is k, then $\frac{k}{2}$ is equal to _____ (where [.] denotes the greatest integer function)

398. $y = f(x)$ is differentiable function and $g(x)=f\left(x-x^2\right)$. If $y=g(x)$ has local maxima at $x=\frac{1}{2}$ but the absolute maximum exists at some other points, then minimum number of solution of $g'(x)=0$ is p, then $\frac{p}{2}$ is equal to _____

399. If f(x) is a differentiable function for all $x \in R$ such that f(x) has fundamental period 2. $f(x) = 0$ has exactly two solution in [0, 2], also $f(0) \neq 0$. If minimum number of zeros of $h(x) = f'(x) \cos x - f(x) \sin x$ in [0, 99) is 120 + k, then $\frac{k}{2}$ is _____

400. Divide the unit square into 9 equal squares by two pairs of lines parallel to the sides. Now remove the central square. Treat the remaining 8 squares the same way and repeat the process n times. If Δ represents the area of all removed squares, such that $\frac{2}{5}<\Delta<\frac{1}{2}$, then the value of n is _____

401. A curve y = f(x) is such that $f(x) \geq 0$ and $f(0) = 0$ and bounds a curvelinear trapezoid with the base [0, x] whose area is proportional to (n + 1)th power of f(x). If $f(1) = 1$, then $\{f(10)\}^n$ is k, then the value of $\frac{k}{4}$ is _____

402. Six points (x_i, y_i), i = 1, 2, ..., 6 are taken on the circle $x^2 + y^2 = 4$ such that $\sum_{i=1}^{6} x_i = 8$ and $\sum_{i=1}^{6} y_i = 4$. The line segment joining orthocentre of a triangle made by any three points and the centroid of the triangle made by other three points passes through a fixed points (h, k), then $\frac{h+k}{2}$ is _____

403. Let $a = 3^{1/223} + 1$ and for all $n \geq 3$ let $f(n) = {}^{n}c_0 \cdot a^{n-1} - {}^{n}c_1 a^{n-2} + {}^{n}c_2 a^{n-3} - \ldots + (-1)^{n-1}\,{}^{n}c_{n-1} \cdot a^0$ if the value of $f(2007) + f(2008) = 3^k$ where $k \in N$, then k is ____

404. If $\dfrac{\left(1^4+\frac{1}{4}\right)\left(3^4+\frac{1}{4}\right)\ldots\left\{(2n-1)^4+\frac{1}{4}\right\}}{\left(2^4+\frac{1}{4}\right)\left(4^4+\frac{1}{4}\right)\ldots\left\{(2n)^4+\frac{1}{4}\right\}} = \dfrac{1}{k_1n^2+k_2n+k_3}$, then $\dfrac{k_1-k_2+k_3}{2}$ is equal to ____

405. Let the numbers tan x, cos x and sec x are the first three terms of an A.P. If the nth terms of the progression is cot x, then the value of n ____

406. If $x + y + z = 1$ and $0 \leq x, y, z \leq 1$, then the maximum value of expression $27(xy + yz + zx - 2xyz)$ is equal to ____

407. Die A has 2 white and 4 red faces whereas die B has 4 white and 2 red faces. A coin is flipped once. If it shows a head, the game continues by throwing die A, if it shows tail, then die B is to be used. If the probability that die A is used is $\dfrac{32}{33}$ when it is given that red turns up every time in first n throws. Then find the value of n ____

408. f is any function from A to A where $A = \{1, 2, \ldots 10\}$. The discrete points on the graph of f(x) are joined with line segments the probability that f is onto and the resulting graph has only one local minima and no local maxima is $\dfrac{2^k-2}{10^{10}}$ then k is ____

409. The set of natural numbers is divided into array of rows and columns in the form of matrices as $A_1 = [1]$, $A_2 = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$, $A_3 = \begin{bmatrix} 6 & 7 & 8 \\ 9 & 10 & 11 \\ 12 & 13 & 14 \end{bmatrix}$ and so on. If the trace of A_{10} be λ, then unit digit of λ is ____

410. Let A and B be two 3×3 invertible matrices such that A is an idempotent matrix and $\det.(\text{adj } B) = \det.(A)^{12} + \det.(A - A^T)^{13} + \det.(\text{adj } A - I)^{14} + L$ where

$L = \lim\limits_{x\to 0} \dfrac{\int_0^{\sin^2 x} \ln\left(1+t^3\right)dt}{\int_0^{x} \left(e^t-1-t\right)(\tan t - t)(1-\cos t)\,dt}$. Then find the absolute value of det. B ____

411. If a is a real constant and A, B, C are variable angles and $\sqrt{a^2-b^2}\tan A + a\tan B + \sqrt{a^2+b^2}\tan C = ma$. If the least value of $\tan^2 A + \tan^2 B + \tan^2 C$ is 27, then the value of $|m|$ is equal to ____

412. In ΔABC, circumradius is 3 and Inradius is 1.5 units. If the value of $a\cot^2 A + b^2\cot^3 b + C^3\cot^4 C$ is $m\sqrt{n}$ where m and n are prime numbers, then find the value of $\left[\dfrac{(m+n)}{5}\right]$ ____

413. A circle C is tangent to the x and y-axis in first quadrant at the points P and Q respectively. BC and AD are parallel tangents to the circle with slope –1. If the points A and B are on the y–axis while C and D are on x-axis and area of figure ABCD is $900\sqrt{2}$ sq. units. The radius of the circle is k then $\frac{k}{3}$ is equal to ____

414. Let the volume of the parallelopiped whose coterminous edges are represented by vectors $\vec{a}$, $\vec{a}\times\vec{b}$ and $\vec{a}\times(\vec{a}\times\vec{b})$ is $\frac{p}{q}$. The angle between $\vec{a}$ and $\vec{b}$ is θ. Vectors $\vec{a}$ and $\vec{b}$ satisfy the relation $\int_{-(a\cdot b)^2}^{|a\times b|^2} 2t\,dt = 0$.

Then the least value of $\frac{p+q}{2}$ is ____

415. If $\sum_{r=1}^{n}\left(\frac{\tan 2^{r-1}}{\cos 2^r}\right) = \tan p^n - \tan q$. Then the value of $(p+q)$ ____

416. If $x, y \in R^+$ and $\log_{10}(2x) + \log_{10} y = 2$ and $\log_{10} x^2 - \log_{10}(2y) = 4$ and $x + y = m/n$, where m and n are relatively prime, the value $\frac{m-3n^6}{2}$ is ____

417. In an equilateral triangle ex–centre opposite vertex A is (2, –4) and equation of side BC is $x + y - 2 = 0$, if co–ordinates of vertex A is (α, β), then $\frac{2\alpha-\beta}{5}$ is ____

418. Let 'a' be a positive constant number consider two curves $C_1 : y = e^x$, $C_2 = y = e^{a-x}$. Let S be the area of the part surrounding by C_1, C_2 and the y-axis. Then $\lim_{a\to 0}\frac{S}{a^2}$ is equal to ____

419. If eccentricity of conjugate hyperbola of the given hyperbola $\left|\sqrt{(x-1)^2+(y-2)^2} - \sqrt{(x-5)^2+(y-5)^2}\right| = 3$ is equal to ____

420. A ray emanating from the point (0, 6) is incident on the ellipse $25x^2 + 16y^2 = 1600$ at the point P with ordinate 5. After reflection, ray cuts the y-axis at B. Find the length of PB is ____

421. If difference of radii of largest and smallest circle passing through the focus of parabola $y^2 - 4x = 0$ and is contained in it (circles touch parabola in at least one point) is equal to ____

422. One of the roots of equation $2000x^6 + 100x^5 + 10x^3 + x - 2 = 0$ is of the form $\frac{m+\sqrt{n}}{r}$, where m is non-zero integer, and r and n are relatively prime natural number, then the value of $(m + n + r)$ is ____

423. The line which contains all points (x, y, z) which are of the form $(x, y, z) = (2, -2, 5) + \lambda(1, -3, 2)$ intersects the plane $2x - 3y + 4z = 163$ at P and intersects the YZ plane at Q. If the distance PQ is $a\sqrt{b}$ where $a, b \in N$ and $a > 3$ then $(a + b)$ equals ____

424. If $x > 0$, then the minimum value of $x^{1000} + x^{900} + x^{90} + x^6 + \frac{1996}{x}$ is ____

425. If a, b, c are determined by throw of a single dice thrice, given $a^2+b^2+c^2 \le ab+bc+ca$, the probability that a point (a,b,c) determined in 3 dimensional co-ordinate system lies inside the tetrahedron formed by co-ordinate planes and $x+y+z=12$ is ____

426. Let 'a' is the result of tossing a black die and 'b' is the result of tossing a white die. The number of ordered pairs (a, b) such that there exists real numbers x, y, z for which $x + y + z = a$ and $xy + yz + zx = b$, is ____

427. The solution of the equation $\log_2 a \cdot \log_3 a \cdot \log_5 a = \log_2 a \cdot \log 3\, a + \log_3 a + \log_5 a + \log_5 a \cdot \log_2 a$ ____

428. Consider a ΔABC with $A(\bar{a})$, $B(\bar{b})$, $C(\bar{c})$ where $\bar{a}, \bar{b}, \bar{c}$ are position vectors of A, B, C respectively. If ΔABC is an equilateral triangle, $P(\bar{p})$ is any point inside it. $\overline{PD}$, $\overline{PE}$, $\overline{PF}$ are drawn perpendicular to sides $\overline{BC}$, $\overline{CA}$, $\overline{AB}$ respectively. If $D(\bar{d})$, $E(\bar{e})$, $F(\bar{f})$ lie on the sides $\overline{BC}$, $\overline{CA}$, $\overline{AB}$ respectively $|\bar{d}-\bar{p}|+|\bar{e}-\bar{p}|+|\bar{f}-\bar{p}| = k|\bar{c}-\bar{a}|$, then $\frac{k}{\sqrt{3}}$ is ____

429. If r_1 and r_2 are the radii of smallest and largest circle which passes through (5, 6) and touches the circle $(x - 2)^2 + y^2 = 4$, then $r_1 r_2$ is ____

430. The number of different rational numbers $\frac{p}{q}$, where $p, q \in \{1, 2, 3, 4, 5, 6\}$ is ____

431. If $A = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$, then if $A^{100} - I = \lambda B$, where $B = \begin{bmatrix} 0 & 0 \\ k & 0 \end{bmatrix}$ then λ is equal to ____

432. If the sum of all solution of the equation $\left(x^{\log_{10} 4}\right)^2 - 4^{\log_{10} x} - 2 = 0$ is $\left(a^{\log_b c}\right)$ a, b, c $\in$ N, then the value of $\frac{(a+b+c)}{5}$ is ____

433. If $b \ge a > 1$, then find the value of $\left(\sqrt{\log_a \sqrt[4]{ab} + \log_b \sqrt[4]{ab}} - \sqrt{\log_a \sqrt[4]{\frac{b}{a}} + \log_b \sqrt[4]{\frac{a}{b}}}\right) \cdot \sqrt{\log_a b}$ ____

434. If the side length of regular pentagon is a and diagonals length is b then the value of $\left(\frac{a^2}{b^2} + \frac{b^2}{a^2}\right)$, is ____

435. A matrix 'M' describes the reflection of a point (x, y) about the line $y = \left(2-\sqrt{3}\right)x$. The matrix 'M' also satisfies the relation AM – MA = I, where A is a non-singular matrix. If A^{-1} is a symmetric matrix with entries x and y, then x + y is equal to ____

436. If the complex number $z = x + iy$ moves in argand plane such that $1 \le |x + iy| \le 2$ and $x - y \le 0$ where x, y $\in$ R determines a region with area k then value of the definite integral $\int_0^{2k/3} \{\sin x\}\, dx$ is equal to ____

(where $\{k\}$ represents fractional part of k)

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MATCH THE FOLLOWING

MM-I. If a, b, c are real roots of the cubic equation $f(x) = 0$ such that $(x-1)^2$ is a factor of $f(x) + 2$ and $(x+1)^2$ is a factor of $f(x) - 2$, then there are some information given in the List–I whose values are given in List–II below

LIST–I		LIST–II	
(I)	Coefficient of x^2 in $f(x)$	(P)	3
(II)	Value of $\lvert ab + bc + ca \rvert$	(Q)	2
(III)	If α is a root of $f'(x) = 0$, then $\lvert\alpha\rvert$	(R)	0
(IV)	Value of $\frac{2(f(4)+8)}{3(f(3)+f(2))}$	(S)	4
		(T)	1
		(U)	5

437. Which of the following is the only CORRECT combination?
(A) (I), (T) (B) (II), (R)
(C) (I), (Q) (D) (II), (P)

438. Which of the following is the only CORRECT combination?
(A) (III), (P) (B) (III), (T)
(C) (IV), (S) (D) (IV), (U)

MM-II. $f(x)$ is polynomial function of degree '6', which satisfies $\lim_{x\to 0}\left(1+\frac{f(x)}{x^3}\right)^{\frac{1}{x}} = e^2$ and has local maximum at $x = 1$ and local minimum at $x = 0$ and $x = 2$. Now, some information given in the List–I, whose values are given in List–II below

LIST–I		LIST–II	
(I)	The coefficient of x^6	(P)	0
(II)	The coefficient of x^5	(Q)	2
(III)	The coefficient of x^4	(R)	$-\frac{12}{5}$
(IV)	The coefficient of x^3	(S)	$\frac{2}{3}$
		(T)	$\frac{1}{3}$
		(U)	3

439. Which of the following is the only CORRECT combination?
(A) (I), (T) (B) (II), (U)
(C) (I), (S) (D) (II), (P)

440. Which of the following is the only INCORRECT combination?
(A) (III), (Q) (B) (IV), (P)
(C) (III), (U) (D) (II), (R)

MM-III. Match the following List–I with List–II

LIST–I		LIST–II	
(I)	$\int \tan\left(2\tan^{-1}\sqrt{\frac{\sqrt{1+\sqrt{x}}-1}{\sqrt{1+\sqrt{x}}+1}}\right)dx$	(P)	$\frac{4}{3}x^{\frac{3}{4}}+C$
(II)	$\int \cot\left(2\tan^{-1}\sqrt{\frac{\sqrt{1+\sqrt{x}}-\sqrt[4]{x}}{\sqrt{1+\sqrt{x}}+\sqrt[4]{x}}}\right)dx$	(Q)	$\frac{4}{5}x^{\frac{5}{4}}+C$
(III)	$\int\left(\frac{1-\tan\left(\frac{1}{2}\sin^{-1}\left(\frac{1-\sqrt{x}}{1+\sqrt{x}}\right)\right)}{1+\tan\left(\frac{1}{2}\sin^{-1}\left(\frac{1-\sqrt{x}}{1+\sqrt{x}}\right)\right)}\right)dx$	(R)	$\frac{2}{3}x^{\frac{3}{4}}+C$
(IV)	$\int\sqrt{x}\tan\left(2\tan^{-1}\left(\frac{\sqrt{1+\sqrt{x}+1}-\sqrt{\sqrt{1+\sqrt{x}}-1}}{\sqrt{\sqrt{1+\sqrt{x}}+1}+\sqrt{\sqrt{1+\sqrt{x}}-1}}\right)\right)dx$	(S)	$\frac{2}{5}x^{\frac{5}{4}}+C$
		(T)	$2\sqrt{1+\sqrt{1+x^2}}+C$
		(U)	$2\sqrt{x^2+2}+C$

441. Which of the following is the only CORRECT combination?
(A) (I), (P)
(B) (II), (Q)
(C) (III), (R)
(D) (IV), (S)

442. Which of the following is the only CORRECT combination?
(A) (I), (Q)
(B) (II), (R)
(C) (III), (P)
(D) (IV), (S)

MM-IV. From a point P, two tangents PA and PB are drawn to the parabola $y^2 = 4x$, where A and B lie on the parabola. S being the focus of the parabola and $P \equiv (2, 3)$ and r is the radius of circumcircle PAB and α is abscissa of image of S about tangent at A. There are some questions given in the List–I whose values are given in List–II

LIST–I		LIST–II	
(I)	Area of ΔPAB	(P)	$\sqrt{11}$
(II)	(SA)(SB)	(Q)	9
(III)	r	(R)	$\frac{1}{2}$
(IV)	SP	(S)	5
		(T)	10
		(U)	18

443. Which of the following is the only INCORRECT combination?
(A) (IV), (P)
(B) (III), (U)
(C) (II), (T)
(D) (II), (U)

444. Which of the following is the only CORRECT combination?
(A) (I), (T) (B) (I), (R)
(C) (II), (S) (D) (III), (R)

MM-V. Let the letters of the word PARACETAMOL be arranged in the ways specified in List–I. List–II contains statements about the number of ways these arrangements are possible.

LIST–I		LIST–II	
(I)	"PARA" occurs in the rearranged word	(P)	$\in(2^{15}, 3^{10})$
(II)	"P" occurs before "A", "A" occurs before "R" and "R" occurs before "A"	(Q)	$\in(2^{20}, 3^{13})$
(III)	"A" does not occur in the middle 5 positions	(R)	8!
(IV)	The rearranged word ends and begins in a vowel	(S)	${}^{6}C_{3} \cdot 8!$
		(T)	$3 \times 6! \times {}^{11}C_{6}$
		(U)	$\frac{10!}{3}$

445. Which of the following is the only CORRECT combination?
(A) (I), (R), (Q) (B) (II), (T), (Q)
(C) (III), (U), (P) (D) (IV), (U), (Q)

446. Which of the following is the only CORRECT combination?
(A) (I), (R), (P) (B) (III), (S), (P)
(C) (IV), (T), (Q) (D) (II), (U), (Q)

MM-VI. "n" whole numbers are chosen randomly and multiplied. Let probability that last digit is k be P(k)

LIST–I (k)		LIST–II P(k)	
(I)	1, 3, 7 or 9	(P)	$\left(\frac{4}{5}\right)^{n} - \left(\frac{2}{5}\right)^{n}$
(II)	2, 4, 6 or 8	(Q)	$\left(\frac{4}{5}\right)^{n}$
(III)	5	(R)	$\left(\frac{2}{5}\right)^{n}$
(IV)	0	(S)	$\frac{5^{n} - 4^{n}}{10^{n}}$
		(T)	$\frac{10^{n} - 8^{n} - 5^{n} + 4^{n}}{10^{n}}$
		(U)	$\frac{5^{n} + 4^{n}}{10^{n}}$

447. Which of the following is the only CORRECT combination?
(A) (II), (S)
(C) (III), (Q) (B) (I), (R)
(D) (IV), (U)

448. Which of the following is the only INCORRECT combination?
(A) (II), (P)
(B) (IV), (T)
(C) (III), (Q)
(D) (I), (P)

MM-VII. In a triangle AOB, R and Q are then points on the sides OB and AB respectively such that $3OR = 2RB$ and $2AQ = 3QB$. Let OQ and AR intersect at the point P (where O is origin). Moreover area of triangle AOB is $\frac{19}{2}$ sq. units. List-I contains some expressions and List-II contains, then values

LIST–I		LIST–II	
(I)	$\frac{AP}{PR}$ is	(P)	10 : 9
(II)	$\frac{OP}{PQ}$ is	(Q)	15 : 4
(III)	Area of triangle OPA is	(R)	19 : 4
(IV)	Area of quadrilateral PQBR	(S)	3
		(T)	6
		(U)	$3\sqrt{3}$

449. Which of the following is the only CORRECT combination?
(A) (I), (Q)
(B) (II), (Q)
(C) (III), (S)
(D) (IV), (T)

450. Which of the following is the only CORRECT combination?
(A) (I), (P)
(B) (II), (P)
(C) (III), (T)
(D) (IV), (T)

451. In the following, [.] and {.} are greatest integer function and fractional part function respectively. Match the functions in Column–I with the properties in Column–II and indicate your answer by darkening the appropriate bubbles in the 4 × 5 matrix given in ORS

Column – I		Column – II	
(A)	$f_1(x) = \left[\frac{4x}{\pi}\right] \operatorname{sgn}\left(x^2 - x + 1\right)$	(p)	discontinuous at more than 3 points but less than 6 points in [–2, 2]
(B)	$f_2(x) = \cos^{-1}\left(\operatorname{sng}\left(\cos\frac{2x-1}{2}\pi\right)\right)$	(q)	non derivable at more than two points but atmost 5 points in [–2, 2]
(C)	$f_3(x) = \max.\left(\{x+1\}, \{5-x\}\right)$	(r)	range contains atleast one integer but not more than seven and no irrational value in [–2, 2]
(D)	$f_4(x) = \sqrt{x^2} + [x]^2$	(s)	many one but not even function in [–2, 2]
		(t)	neither odd nor periodic in [–2, 2]

452. Consider set S of first 18 natural numbers. Five different numbers a_1, a_2, a_3, a_4 and a_5 are to be selected from set S under the refractions mentioned in column–I given below. The numbers of ways (T) such a selection can be made are mentioned in column–II. Match the entries in column–I to the correct entry in column–II

Column – I		Column – II	
(A)	$a_{i+1} - a_i \geq 2\ \forall\ i = 1, 2, 3, 4$	(p)	$50 \leq T \leq 300$
(B)	$a_{i+1} - a_i \geq 3\ \forall\ i = 1, 2, 3, 4$	(q)	$250 \leq T \leq 800$
(C)	$a_2 - a_1 \geq 2, a_3 - a_2 \geq 3, a_4 - a_3 \geq 4, a_5 - a_4 \geq 5$	(r)	$700 \leq T \leq 2500$
(D)	$a_2 - a_1 \geq 2, a_3 - a_2 \geq 2, a_4 - a_3 \geq 3, a_5 - a_4 \geq 3$	(s)	$2000 \leq T \leq 3500$
		(t)	$0 \leq T \leq 80$

453. If incentre of ΔDEF and circumcentre of ΔABC (formed by joining ex-centres of ΔDEF) be the points (10, 15) and (30, 10) respectively. If $D \equiv (8, 9)$, then

	Column – I		Column – II
(A)	Circumradius of ΔABC is equal to	(p)	$5\sqrt{10}$
(B)	The distance of orthocentre of ΔABC from vertex A is equal to	(q)	25
(C)	The value of sin B sin C is equal to	(r)	30
(D)	The ordinate of vertex A is equal to	(s)	$\frac{7}{5\sqrt{10}}$
		(t)	10

454. A man has 3 coins A, B and C, A is unbiased, the probability that a head will result when B is tossed is 2/3, the probability that a head will result when C is tossed is 1/3. One of the coin chosen at random and is tossed 3 times, giving a total of two heads and one tail.

	Column – I		Column – II
(A)	The probability that the chosen coin was A	(p)	32/75
(B)	The probability that the chosen coin was B	(q)	1/2
(C)	The probability that the fourth toss will result in head	(r)	209/1296
(D)	The probability that the fourth toss will result in tail	(s)	9/25
		(t)	215/1296

455. Consider the matrix $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} a & 6 \\ 0 & 1 \end{bmatrix}$ and let P be any orthogonal matrix and $Q = PAP^T$ and $R = P^TQ^KP$ also $S = PBP^T$ and $T = P^TS^KP$

	Column I		Column II
(A)	If we vary K from 1 to 'n', then first row first column elements of R will form	(p)	G.P. with common ratio a
(B)	If we vary k from 1 to n, then the 2^{nd} row; 2^{nd} column elements of R will form.	(q)	A.P with common difference 2
(C)	If we vary k from 1 to n, then first row first column elements of T will form	(r)	G.P. with common ratio '6'
(D)	If we vary k from 3 to n, then the first row 2^{nd} column element of T will represent the sum of	(s)	A.P with common difference –2
		(t)	A.P with common difference –7

456. Match the following

	Column – I		Column – II
(A)	Let f(x) be a non negative continuous function $\forall\ x \geq 0$, $f(0) = 0$. If $f'(x) \leq f(x)\sin x\ \forall\ x \geq 0$, then the value of $f\left(\frac{3\pi}{2}\right)$ is	(p)	4
(B)	If $a(x^2 + 1) + 12x + 6 - 3x^2 < \sin^{-1}\sin(100) + \cos^{-1}\cos(100)$ $\forall\ x \in R$; $a \in I$ then a can not be equal to	(q)	3
(C)	If two events A and B are such that $P(A^C) = 0.35$, $P(B) = 0.4$ and $P(A \cap B) = 0.15$, then $20P\left(\frac{B}{A \cup B^C}\right)$ is	(r)	0
(D)	The number of elements in the range of the function $f(x) = \sqrt{\cos^{-1}(4x^3 - 3x) - \frac{\pi}{2}} + \frac{(2x+3)!}{\sqrt{x+2}}$ is / are	(s)	5
		(t)	8

457. Match the following:

	Column – I		Column – II
(A)	The number of solution of $\frac{x}{2}+\frac{\sin x}{\cos x}=\frac{\pi}{4}$ in $[-\pi, \pi]$	(p)	1
(B)	The number of solution of equation $\sin^{-1}\left(\lvert x^2-1\rvert\right)+\cos^{-1}\left(\lvert 2x^2-5\rvert\right)=\frac{\pi}{2}$	(q)	0
(C)	The number of solution of $x^4-2x^2\sin^2\frac{\pi}{2}x+1=0$	(r)	3
(D)	The number of solution of $x^2+2x+2\sec^2\pi x+\tan^2\pi x=0$	(s)	4
		(t)	2

458. Match the following column-I with column-II

	Column – I		Column – II
(A)	A parallelopiped is formed by planes drawn through the points (1, 2, 3) and (9, 8, 5) parallel to the coordinate planes, then length of edge of this rectangular parallelopiped can be (8, 6, 2)	(p)	2
(B)	The distance between the point (1, 1, 1) on the line $\frac{x-1}{3}=\frac{y-1}{4}=\frac{z-1}{12}$ is 13, then point P is $(\alpha, \beta\ \gamma)$ where $\frac{\alpha}{2}$ is	(q)	4
(C)	Consider planes $x-y+z=5$ and $2x+y-3z=4$, then unit vertor parallal to intersection of planes is $\frac{1}{\sqrt{38}}\left(-2\hat{i}+q\hat{j}+p\hat{k}\right)$ where $\lvert p\rvert+\lvert q\rvert+2$ is	(r)	6
(D)	The three vectors $\hat{i}+\hat{j}$, $\hat{j}+\hat{k}$, $\hat{k}+\hat{i}$ taken two at a time form three planes. The three unit vectors drawn perpendicular to these three planes form a parallelopiped of volume $\frac{\alpha}{\beta\sqrt{3}}$, then $\alpha+\beta-3$ is	(s)	8
		(t)	10

459. Match the column:

	Column – I		Column – II
(A)	If $\left(x+\frac{1}{x}+x^2+\frac{1}{x^2}\right)^9=a_0x^{-18}+a_1x^{-17}+\ldots..+a_{36}x^{18}$, then $(a_0+a_2+\ldots..+a_{36})$ is equal to	(p)	2^{24}
(B)	If $(1+x+x^2+x^3)^{13}=a_0+a_1x+a_2x^2+\ldots..+a_{39}x^{39}$, then $(a_0+a_4+a_8+a_{12}+\ldots..+a_{36})$ is	(q)	$(18\times2^{19}+1)$
(C)	${}^{20}C_2+2.{}^{20}C_3+3.{}^{20}C_4+\ldots..+19.{}^{20}C_{20}$ is equal to	(r)	(2^8+2^{17})
(D)	${}^{20}C_1+2.{}^{20}C_2+3.{}^{20}C_3+4.{}^{20}C_4+\ldots..+10.{}^{20}C_{10}$ is equal to	(s)	$20(2^{18}+{}^{19}C_{10})$
		(t)	$(9\times2^{20}+1)$

460. Matrix matching

	Column – I		Column – II
(A)	If x, y, z are 3 positive numbers in A.P., then $\frac{3x+y}{2y-x}+\frac{3z+y}{2y-z}$ is $\geq$	(p)	3
(B)	If x_1, x_2, x_3 are real numbers in A.P. which are also roots of $x^3-x^2+\alpha x+\beta=0$, then the value of $9\alpha+27\beta$ is	(q)	5
(C)	If the roots of $x^4-16x^3+86x^2-176x+105=0$ are in A.P. whose common difference is d, then the value of $2d^2+4$ is	(r)	2
(D)	If $a_1, a_2, a_3, \ldots..$ are in G.P. having r as common ratio such that $\sum_{k=1}^{n}a_{2k+2}\neq0$, then the number of possible values of r is	(s)	12
		(t)	8

Answer Q.461, Q.462 and Q.463 by appropriately matching the information given in the three columns of the following table.

Column–I (Trigonometric function), Column–II (Maximum value) and Column–III (Minimum value)

Column–I		Column–II		Column–III	
(I)	$f(x)=\frac{7+6\tan x-\tan^2 x}{\left(1+\tan^2 x\right)}$ for all values of $x \neq \frac{\pi}{2}$	(i)	$\frac{1}{4}$	(P)	-4
(II)	$f(x)=5\cos x+3\cos\left(x+\frac{\pi}{3}\right)+3$ for all values of x	(ii)	1	(Q)	-2
(III)	$f(x)=\sin^6 x+\cos^6 x$ for all values of x	(iii)	8	(R)	$-\frac{1}{4}$
(IV)	$f(x)=\cos x\cdot\cos\left(\frac{\pi}{3}+x\right)\cos\left(\frac{\pi}{3}-x\right)$ for all values of x	(iv)	10	(S)	$\frac{1}{4}$

461. Which of the following option is the **CORRECT** combination?
(A) (I) (iii) (Q) (B) (II) (iv) (R)
(C) (III) (i) (S) (D) (IV) (ii) (P)

462. Which of the following options is the only **CORRECT** combination?
(A) (I) (ii) (R) (B) (II) (iii) (Q)
(C) (III) (iv) (S) (D) (IV) (i) (R)

463. Which of the following options is the only **INCORRECT** combination?
(A) (I) (iii) (Q) (B) (II) (iv) (P)
(C) (III) (ii) (S) (D) (IV) (i) (S)

Answer Q.464, Q.465 and Q.466 by appropriately matching the information given in the three columns of the following table.

Column–I (Complex equation), Column–II (Greatest value of $|z|$) and Column–III (Least value of $|z|$)

Column–I		Column–II	Column–III
(I)	$\left\|z-\frac{1}{z}\right\|=2$	(i) 8	(P) $-1+\sqrt{2}$
(II)	$\left\|z+\frac{2}{z}\right\|=4$	(ii) $2+\sqrt{6}$	(Q) $-3+2\sqrt{3}$
(III)	$\left\|z-\frac{3}{z}\right\|=6$	(iii) $3+2\sqrt{3}$	(R) 1
(IV)	$\left\|z-\frac{8}{z}\right\|=7$	(iv) $1+\sqrt{2}$	(S) $-2+\sqrt{6}$

464. Which of the following option is the **CORRECT** combination?
(A) (I) (ii) (S) (B) (II) (iii) (Q)
(C) (III) (iv) (P) (D) (IV) (i) (R)

465. Which of the following options is the only **CORRECT** combination?
(A) (I) (iv) (P) (B) (II) (iii) (S)
(C) (III) (i) (Q) (D) (IV) (ii) (R)

466. Which of the following options is the only **INCORRECT** combination?
(A) (I) (iv) (P)
(B) (II) (ii) (S)
(C) (III) (iii) (Q)
(D) (IV) (i) (P)

Answer Q.467, Q.468 and Q.469 by appropriately matching the information given in the three columns of the following table.

Number of tangent and their equation common match the following column(s)

Column–I		Column–II	Column–III
(I)	Let $S_1 = x^2 + y^2 - 2x = 0$ and $S_2 = x^2 + y^2 - 12x + 32 = 0$	(i) number of common tangent is 4	(P) equation of common tangent is $x = 0$
(II)	Let $S_1 = x^2 + y^2 - 2x = 0$ and $S_2 = 5x^2 + 5y^2 - x - 12y - 24 = 0$	(ii) number of common tangent is 3	(Q) equation of common tangent is $3x - 4y - 8 = 0$
(III)	Let $S_1 = x^2 + y^2 - 12x + 20 = 0$ $S_2 = x^2 + y^2 - 6x = 0$	(iii) number of common tangent is 2	(R) equation of common tangent is $4y + 3x - 8 = 0$
(IV)	Let $S_1 = x^2 + y^2 - 6x = 0$ and $S_2 = x^2 + y^2 - 2x = 0$	(iv) number of common tangtent is 1	(S) equation of common tangent is $2\sqrt{2}y - x + 12 = 0$

467. Which of the following combination is **CORRECT?**
(A) (I) (ii) (R)
(B) (I) (i) (S)
(C) (I) (i) (Q)
(D) (I) (ii) (Q)

468. Which of the following combination is **CORRECT?**
(A) (II) (ii) ((R)
(B) (II) (iii) (Q)
(C) (II) (iii) (R)
(D) (II) (iv) (Q)

469. Which of the following combination is **INCORRECT?**
(A) (III) (iii) (Q)
(B) (III) (iii) (S)
(C) (I) (ii) (R)
(D) (IV) (iv) (P)

Answer Q.470, Q.471 and Q.472 by appropriately matching the information given in the three columns of the following table.

Let equation P ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $a > b$

Column–I	Column–II	Column–III
(I) P(ϕ) be any point on ellipse and P′ be the corresponding point on auxiliary circle, then B is one end of minor axis and C is centre of ellipse	(i) x coordinate of incentre of Δ PSS′ is ae cos ϕ	(P) Ratio of area of ΔCBP and CPP′ is $\frac{b}{a-b}\lvert\sin\phi\rvert$
(II) P be any point on the ellipse whose radii makes angle ϕ with x-axis if θ be eccentric angle of P	(ii) $\alpha + \beta + \gamma + \delta$ is odd multiple of π	(Q) $\tan\theta = \frac{a}{b}\tan\phi$
(III) If P be any point on ellipse whose eccentric angle ϕ and eccentricity e and S and S′ be there foci	(iii) Area if ΔCBP′ is $\frac{1}{2}ab\lvert\sin\phi\cos\phi\rvert$	(R) y coordinate of ΔPSS′ is $\frac{be\sin\phi}{1+e}$
(IV) In general four normals can be drawn to an ellipse from any point and if α, β, γ, δ are eccentric angle of these four co-normal points then	(iv) length OP = $\frac{ab}{\sqrt{a^2\sin^2\phi + b^2\cos^2\phi}}$	(S) conormal points lies on a fixed curve that passess through (h, k) $(a^2 - b^2)xy + b^2kx - a^2hy = 0$

470. Which of the following option is the only **CORRECT** combination?
(A) (I) (ii) (R) (B) (IV) (ii) (S)
(C) (III) (iv) (P) (D) (IV) (ii) (Q)

471. Which of the following option is the only **CORRECT** combination?
(A) (II) (iv) (Q) (B) (IV) (ii) (S)
(C) (III) (iv) (P) (D) (IV) (ii) (Q)

472. Which of the following option is the only **INCORRECT** combination?
(A) (II) (iv) (Q) (B) (IV) (ii) (S)
(C) (I) (iii) (P) (D) (III) (i) (R)

Answer Q.473, Q.474 and Q.475 by appropriately matching the information given in the three columns of the following table.

Column-I has family of curves; Column-II has differential equation of the curves in Column-I; Column-III has orthogonal trajectories of the curves in Column-I

Column–I	Column–II	Column–III
(I) $x^2 + y^2 - 2ay = 0$	(i) $\frac{dy}{dx} = \cot y$	(P) $\sin y + c \cdot e^{-x} = 0$
(II) $y^2 = 4(x - a)$	(ii) $\frac{dy}{dx} = \frac{2xy}{x^2 - y^2}$	(Q) $x^2 + y^2 + cx = 0$
(III) $\cos y = a \cdot e^{-x}$	(iii) $\frac{dy}{dx} = 1 + \tan^2 x$	(R) $x + 2y + \sin 2x = c$
(IV) $y = \tan x + c$	(iv) $\frac{dy}{dx} - \frac{2}{y} = 0$	(S) $y = c \cdot e^{-x/2}$

473. Which of the following are **CORRECT** combinations?
(A) (I) (ii) (R) (B) (I) (ii) (Q)
(C) (II) (iv) (P) (D) (III) (ii) (P)

474. Which of the following are **CORRECT** combinations?
(A) (IV) (iii) (R) (B) (IV) (ii) (R)
(C) (III) (i) (P) (D) (III) (ii) (R)

475. Which of the following are **INCORRECT** combinations?
(A) (I) (ii) (Q) (B) (IV) (iii) (R)
(C) (II) (iv) (S) (D) (III) (i) (P)

Answer Q.476, Q.477 and Q.478 by appropriately matching the information given in the three columns of the following table.

Let $f(x) = x^2 - 5x + 6$, Column-I contains information about f(x), Column-II contains information about f(|x|) and Column-III contains information about |f(|x|)|

Column–I	Column–II	Column–III
(I) Vertex of $f(x) = \left(\frac{5}{2}, -\frac{1}{4}\right)$	(i) f(|x|) is non differentiable at x = 0	(P) |f(|x|)| = 1 has 8 real and distinct solution
(II) Vertex of $f(x) = \left(-\frac{1}{4}, \frac{5}{2}\right)$	(ii) f(|x|) is differentiable at x = 0	(Q) |f(|x|)| = 1 has 4 real and distinct solution
(III) f(x) increasing in $\left[\frac{5}{2}, \infty\right)$	(iii) f(|x|) = 1 has 4 real and distinct solutions	(R) |f(|x|)| is non differentiable at 5 points
(IV) f(x) decreasing in $\left[\frac{5}{2}, \infty\right)$	(iv) f(|x|) = 1 has 2 real and distinct solutions	(S) |f(|x|)| is non differentiable at 6 points

476. Which of the following options is the only **CORRECT** combination?
(A) (IV) (i) (S)
(B) (I) (ii) (R)
(C) (III) (iv) (P)
(D) (I) (iii) (Q)

477. Which of the following options is the only **CORRECT** combination?
(A) (III) (iii) (R)
(B) (I) (i) (P)
(C) (IV) (iv) (S)
(D) (II) (ii) (Q)

478. Which of the following options is the only **INCORRECT** combination?
(A) (I) (i) (Q)
(B) (III) (iii) (R)
(C) (II) (ii) (S)
(D) (III) (i) (Q)

ANSWERS

SUBJECTIVE

1. $n = 3$

3. $f(x) = 1 - x^2$

4. 4

5. 1

6. -1

7. 2

8. 2

9. 20

10. $S_{min} = S_1\left(1+\frac{\pi}{4}\right)+\frac{4v}{\sqrt{S_1}}$ $S_{max} = S_1\left(1+\frac{\pi}{4}\right)+2v\left(1+\frac{1}{S_1}\right)$

11. Max and min value of expression $y = f(x)$ are 46 and -6

12. 4

13. $\sqrt{3}\ln\left(\frac{t-\sqrt{3}}{t+\sqrt{3}}\right)+2\tan^{-1}t+c$ (where $t = \sqrt{\frac{2-\sin x}{2+\sin x}}$)

14. $6\cot^{-1}\left(\sqrt{\frac{1+4x^{1/3}}{x^{1/3}}}\right)+c$

15. $-6\left(\frac{R^3}{3}+\frac{3R^2}{2}+3R+\log R\right)$, where $R = \left(x+\frac{1}{x}+1\right)^{1/6}-1$

16. $\sqrt{t^2+2t-3}-\log\left(t+1+\sqrt{t^2+2t-3}\right)+\sqrt{3}\sin^{-1}\left(\frac{t+8}{t+2}\right)$ where $t = x+\frac{1}{x}$

17. $\frac{e}{2}$

18. 3

19. $\frac{3\pi}{2}$

20. $\frac{\pi}{2}\ln a$

21. $\frac{1}{6m^3(e^{2k}-1)}$

22. 4

24. 5

26. $1-\frac{KT^2}{2}$

27. 63

28. $x - 7y + 17 = 0$ or $7x + y - 31 = 0$

29. 400 sq. units

30. $6x^2 - 12y^2 + 14xy + 17y - 20x = 0$

31. 2

32. 9

33. a^2

34. $x^2\left(1-\frac{c^2}{a^2}\right)+y^2 = a^2-c^2$

35. 8

37. $AB = \frac{18a}{17}$

38. $e = \frac{1}{2}$

39. $\sqrt{7} : \sqrt{18}$

40. 3

41. $81\left[\left(x - \frac{2h}{3}\right)^2 + \left(y - \frac{2k}{3}\right)^2\right] = 16b^2$

44. 2

45. 2

46. 2

47. $-{}^8C_5\left(\frac{1}{2}\right)^5$

48. 6

49. $\frac{2^n\left(3^{10n} - 2^{10n}\right)}{3^{10n}\left(3^n - 2^n\right)}$

50. $S = \frac{A(1-A^n)}{(1-A)^3} - \frac{nA^{n-1}}{(1-A)^2} - \frac{n(n+1)A^{n+1}}{(1-A)}$

51. ${}^{2n-1}C_n$

52. 5

53. 189

55. ${}^{10}P_6$

56. $\frac{{}^{6n+2}C_2 - 3(3n-1) - 1}{3!}$

57. $\frac{3}{4}$

58. 11

60. $\frac{n \cdot 3^{n-1}}{3^n - 1}$

61. $\frac{\sqrt{3}+2}{2}$

64. $|\alpha| + |\gamma| = \sqrt{2}$

65. $n = 2$

66. 56 digits

67. 4

68. 16

70. $\frac{BP}{PE} = \frac{(n+1)m}{n}$

71. $\frac{11}{56}$

72. 6

73. $2\sqrt{1+k}$

74. 5

75. 2

76. 1

77. $\log_3\left(\frac{n}{4} + (-1)^n \frac{\pi}{16} - \frac{\pi}{16}\right)$

79. 6

MEGACOSM COGNITIONS PRIVATE LIMITED

OBJECTIVE

1.	D	2.	A	3.	D	4.	C
5.	D	6.	D	7.	C	8.	B
9.	B	10.	C	11.	D	12.	D
13.	A	14.	B	15.	B	16.	B
17.	A	18.	C	19.	B	20.	A
21.	D	22.	B	23.	C	24.	B
25.	A	26.	B	27.	D	28.	C
29.	B	30.	A	31.	B	32.	D
33.	A	34.	C	35.	B	36.	C
37.	B	38.	C	39.	B	40.	C
41.	A	42.	A	43.	C	44.	A
45.	C	46.	D	47.	A	48.	A
49.	D	50.	D	51.	B	52.	B
53.	B	54.	A	55.	C	56.	A
57.	A	58.	B	59.	D	60.	A
61.	C	62.	C	63.	A	64.	B
65.	C	66.	B	67.	A	68.	A
69.	A	70.	C	71.	B	72.	A
73.	A	74.	D	75.	B	76.	A
77.	D	78.	A	79.	C	80.	D
81.	D	82.	C	83.	C	84.	C
85.	C	86.	D	87.	A	88.	D
89.	A	90.	C	91.	C	92.	C
93.	C	94.	B	95.	D	96.	D
97.	C	98.	A	99.	A	100.	C
101.	D	102.	C	103.	D	104.	D
105.	B	106.	D	107.	A	108.	C
109.	A	110.	B	111.	A	112.	A
113.	C	114.	B	115.	B	116.	C
117.	C	118.	C	119.	D	120.	D
121.	C	122.	A	123.	D	124.	B
125.	A	126.	D	127.	D	128.	C
129.	B	130.	C	131.	C	132.	D
133.	C	134.	B	135.	B	136.	C
137.	A	138.	B	139.	A	140.	B
141.	A	142.	A	143.	A	144.	C
145.	A	146.	A	147.	C	148.	D
149.	A	150.	A				

151.	A, B, C	152.	A, D	153.	A, C, D	154.	A, B, C
155.	A, D	156.	B, C, D	157.	C, D	158.	B, C
159.	B, C	160.	A, D	161.	A, B, C, D	162.	C
163.	A, B, C	164.	A, B, C	165.	A, C	166.	B, C, D
167.	A, C	168.	A, D	169.	A, D	170.	B, C
171.	B, C, D	172.	A, B, C	173.	C	174.	A, B, C
175.	A, B, D	176.	C, D	177.	A, B, C	178.	A, D
179.	A, B, C	180.	A, C, D	181.	A, B, D	182.	B, C, D
183.	B, D	184.	B, C	185.	B, C, D	186.	A, B, C
187.	B, C	188.	A, B, D	189.	A, B, C	190.	A, B, D
191.	B, C, D	192.	B, C, D	193.	A, B	194.	A, C, D
195.	A, B, C, D	196.	A, B, C, D	197.	A, B	198.	B, D

199.	A, B	200.	A, B	201.	A, C	202.	A, B, C, D
203.	C, D	204.	A, B, C	205.	B, D	206.	A, B
207.	B, C	208.	A, C	209.	A, C	210.	B, C
211.	A, B, C	212.	A, B, C, D	213.	A, D	214.	C, D
215.	A, B, C, D	216.	A, C, D	217.	A, C	218.	A, B, C
219.	B, D	220.	A, D	221.	B, C	222.	C, D
223.	A, C	224.	A, D	225.	A, B, C	226.	A, B
227.	A, B, C, D	228.	A, D	229.	A, B	230.	A, B, D
231.	A, D	232.	A, B	233.	A, B, C	234.	B, C, D
235.	B, C	236.	A, B, D	237.	B, C	238.	A, B, D
239.	A, C	240.	C, D	241.	A, D	242.	A, C
243.	A, C	244.	B, C	245.	A, D	246.	A, B
247.	A, B, C	248.	B, D	249.	A, B	250.	A, D
251.	B, C, D	252.	A, B	253.	A, C, D	254.	A, C
255.	A, B	256.	A, B, C	257.	A, D	258.	A, B, C, D
259.	A, C, D	260.	B, C	261.	B, C	262.	B, C
263.	B, C	264.	A, C, D	265.	A, B, C	266.	B, C
267.	B, C	268.	A, C	269.	A, B, C	270.	B, C
271.	B, C	272.	A, C	273.	A, B, C	274.	A, B, C, D
275.	A, C	276.	A, B, C	277.	A, C	278.	A, D
279.	C, D	280.	A, B, C	281.	A, D	282.	B, C
283.	A, B, D	284.	A, C	285.	A, B	286.	A, B, C, D
287.	C, D	288.	B, C	289.	B, C, D	290.	A, C
291.	A, C	292.	A, C	293.	B, C	294.	B, C
295.	A, C	296.	A, B	297.	A, C	298.	B, D
299.	A, C	300.	A, D				

301.	D	302.	C	303.	A	304.	B
305.	B	306.	B	307.	D	308.	A
309.	A	310.	B	311.	B	312.	A
313.	A	314.	B	315.	A	316.	C
317.	B	318.	D	319.	B	320.	A
321.	A	322.	C	323.	C	324.	B
325.	A	326.	B	327.	A	328.	B
329.	C	330.	A	331.	D	332.	C
333.	B	334.	C	335.	D	336.	A
337.	B	338.	B	339.	D	340.	C
341.	D	342.	B	343.	B	344.	A
345.	D	346.	B	347.	A	348.	B
349.	C	350.	A	351.	B	352.	D
353.	C	354.	C	355.	B	356.	C

357.	2	358.	4	359.	1	360.	6
361.	0	362.	4	363.	1	364.	2
365.	4	366.	1	367.	8	368.	3
369.	4	370.	8	371.	8	372.	2
373.	8	374.	8	375.	4	376.	2
377.	6	378.	1	379.	0	380.	2
381.	1	382.	8	383.	6	384.	6
385.	2	386.	0	387.	0	388.	1

389.	3.00	390.	2.00	391.	6.00	392.	6.00
393.	8.00	394.	1.00	395.	6.00	396.	0.00
397.	4.50	398.	2.50	399.	3.50	400.	5.00
401.	2.50	402.	1.50	403.	9.00	404.	2.50
405.	5.00	406.	7.00	407.	5.00	408.	9.00
409.	5.00	410.	5.00	411.	9.00	412.	3.00
413.	5.00	414.	1.50	415.	3.00	416.	4.50
417.	2.40	418.	0.25	419.	1.25	420.	13.00
421.	3.50	422.	200.00	423.	23.00	424.	2000.00
425.	0.50	426.	21.00	427.	30.00	428.	0.50
429.	10.25	430.	23.00	431.	100.00	432.	3.20
433.	2.00	434.	3.00	435.	2.00	436.	2.00

437.	D	438.	B	439.	C	440.	C
441.	B	442.	A	443.	A	444.	B
445.	D	446.	A	447.	B	448.	C
449.	C	450.	B				

451.	(A) → (p, q, r, s, t)	(B) → (p, q, s, t)	(C) → (p, t)	(D) → (p, q, s, t)
452.	(A) → (r, s)	(B) → (p, q)	(C) → (p, t)	(D) → (q, r)
453.	(A) → (q)	(B) → (p)	(C) → (s)	(D) → (r)
454.	(A) → (s)	(B) → (p)	(C) → (q)	(D) → (q)
455.	(A) → (q)	(B) → (s)	(C) → (p)	(D) → (p)
456.	(A) → (r)	(B) → (p, q, r, s)	(C) → (p)	(D) → (q)
457.	(A) → (r)	(B) → (t)	(C) → (t)	(D) → (q)
458.	(A) → (p, r, s)	(B) → (p)	(C) → (t)	(D) → (q)
459.	(A) → (r)	(B) → (p)	(C) → (q, t)	(D) → (s)
460.	(A) → (t)	(B) → (r)	(C) → (s)	(D) → (p)

461.	A	462.	D	463.	D	464.	D
465.	A	466.	D	467.	C	468.	D
469.	A	470.	B	471.	A	472.	C
473.	B	474.	C	475.	B	476.	D
477.	A	478.	C				

MATHEMATICS

HINTS & SOLUTIONS

SUBJECTIVE

1. $g(x) = 2 + \left[\sin^{-1}\left(\frac{2x}{x^2+1}\right)\right] = 2 + -2 \text{ or } -1 \text{ or } 0 \text{ or } 1$

So, range of $g(x) = \{0, 1, 2, 3\}$ for $f(g(x)) < 0 \ \forall \ x \in R$

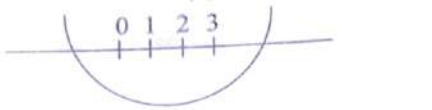
$\Rightarrow f(0) < 0$ and $f(3) < 0$

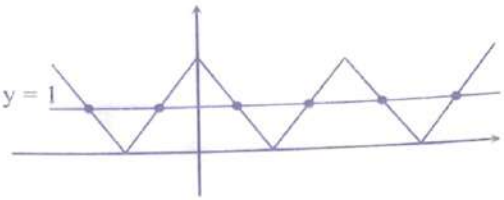
$\Rightarrow a < 2$ and $a > \frac{7}{5} \Rightarrow a \in \left(\frac{7}{5}, 2\right)$

So, $k_1 = \frac{7}{5}$, $k_2 = 2$ thus $[5k_1 - 2k_2] = 1$

Now, we have, $\underbrace{f\left(f\left(.....\left(f(x)\right)\right).....\right)}_{n \text{ times}} = 1$

Where $f(x) = |x - 2|$, clearly $n = 3$





2. $f'(x) = -(2 \cos(\sin x) \sin \sin(x) \cos x + 2 \sin(\cos x) \cos(\cos x) \sin x) \le 0$

$\Rightarrow$ Range is $\left[f\left(\frac{\pi}{2}\right), f(0)\right]$ as $f(x)$ is periodic with period $\frac{\pi}{2}$

3. $f(1 - x) = 2x - x^4 - x^2 f(x)$

Replace x by $1 - x$ ..... (1)

$(1-x)^2 f(1-x) + f(x)$

$= 2(1-x) - (1-x)^4$

$\Rightarrow f(x) = 2(1-x) - (1-x)^4 - (1-x)^2 f(1-x)$

$= 2(1-x) - (1-x)^4 - (1-x)^2\{2x - x^4 - x^2 f(x)\}$

On solving $f(x) = \frac{2(1-x) - (1-x)^4 - 2x(1-x)^2 + x^4(1-x)^2}{1-(1-x)^2 x^2}$

$= \frac{(1-x)\left\{2 - (1-x)^3 - 2x(1-x) + x^4(1-x)\right\}}{\{1-(1-x)x\}\{1+(1-x)x\}}$

$= \frac{(1-x)\left\{2 - \left(1 - 3x + 3x^2 - x^3\right) - 2x + 2x^2 + x^4 - x^5\right\}}{\left(1 - x + x^2\right)\left(1 + x - x^2\right)} = 1 - x^2$

4. Let $x + y = \alpha$ and $x - y = \beta \Rightarrow x = \frac{\alpha+\beta}{2}$ and $y = \frac{\alpha-\beta}{2}$

Given equation $\Rightarrow \beta f(\alpha) - \alpha f(\beta) = (\alpha^2 - \beta^2)\alpha\beta \Rightarrow \frac{f(\alpha)}{\alpha} - \alpha^2 = \frac{f(\beta)}{\beta} - \beta^2$

Hence, $\frac{f(x)}{x} - x^2 = k$ (k is some real constant)

$\Rightarrow f(x) = x^3 + kx$

$f(1) = -2 \Rightarrow k = -3$, $f(x) = x^3 - 3x$

$f'(x) = 3(x - 1)(x + 1)$ on the interval $x \in \left[-\sqrt{3}, \sqrt{3}\right]$ extreme value are -2 and 2

5. $f'(x) = -3x^2 + 2x - 1$

$D = 4 - 12 = -8 < 0$

$$g(x) = \begin{cases} -x^3 + x^2 - x + 1, & 0 \le x \le 1 \\ x - 1, & 1 < x \le 2 \end{cases}$$

$$\lim_{x \to 1^-} g(g(x)) = \lim_{x \to 1^-} g(0^+) = 1$$

$$\lim_{x \to 1} g(g(x)) = \lim_{x \to 1} g(0^+) = 1$$

$$\therefore \lim_{x \to 1} g(g(x)) = 1.$$

6. $$\alpha = \lim_{x \to 0} \frac{1 - \left\{1 - \frac{(1-\cos x)^2}{2!} + \frac{(1-\cos x)^4}{4!} + \ldots\right\}}{x^4} \Rightarrow \lim_{x \to 0} \frac{\frac{(1-\cos x)^2}{2!}\left\{1 - \frac{(1-\cos x)^2}{2 \cdot 3!} + \ldots\right\}}{x^4}$$

$$\Rightarrow \lim_{x \to 0} \frac{\left(2\sin^2 \frac{x}{2}\right)^2}{2x^4}\left[1 - \frac{(1-\cos x)^2}{2 \cdot 3!} \ldots\right] \Rightarrow \lim_{x \to 0}\left[1 - \frac{(1-\cos x)^2}{2 \cdot 3!} + \ldots\right] \Rightarrow \alpha = \frac{1}{8}$$

$$\beta = [2\alpha - 1] = \left[2 \times \frac{1}{8} - 1\right] = \left[-\frac{3}{4}\right] = -1$$

Now, $\beta = -1$, $\beta + 2 = -1 + 2 = 1$

The quadratic expression is $x^2 - (-1 + 1)x + (-1 \times (-1 \times 1)$

$f(x) = x^2 - 1 \Rightarrow f'(x) = 2x - 0 = 0 \Rightarrow x = 0$

Minimum value is (-1)

7. $$f(x) = \left[\frac{x^2 + x + 1}{x^2 + 1}\right] + \left[\frac{4x^2 + x + 2}{2x^2 + 1}\right] + \left[\frac{9x^2 + x + 3}{3x^2 + 1}\right] + \ldots + \left[\frac{n^2x^2 + x + n}{nx^2 + 1}\right]$$

$$= \left[1 + \frac{x}{x^2 + 1}\right] + \left[2 + \frac{x}{2x^2 + 1}\right] + \left[3 + \frac{x}{3x^2 + 1}\right] + \ldots + \left[n + \frac{x}{nx^2 + 1}\right]$$

$$= 1 + 2 + \ldots + n + \left[\frac{x}{x^2 + 1}\right] + \left[\frac{x}{2x^2 + 1}\right] + \left[\frac{x}{3x^2 + 1}\right] + \ldots + \left[\frac{x}{nx^2 + 1}\right]$$

For $x > 0$, $\frac{x}{x^2 + 1} = \frac{1}{x + \frac{1}{x}} < \frac{1}{2}$

$$\frac{x}{2x^2 + 1} = \frac{1}{2x + \frac{1}{x}} < \frac{1}{2\sqrt{2}}$$

$$\vdots \qquad \vdots \qquad \vdots$$

$$\frac{x}{nx^2 + 1} = \frac{1}{nx + \frac{1}{x}} < \frac{1}{2\sqrt{n}}$$

$$\therefore \left[\frac{x}{x^2 + 1}\right] = \left[\frac{x}{2x^2 + 1}\right] = \ldots = \left[\frac{x}{nx^2 + 1}\right] = 0 \Rightarrow f(x) = \frac{n(n+1)}{2}$$

Now, $$\lim_{n \to \infty}\left(\frac{f(x) - n}{(f(x))^2 - \frac{n^3(n+2)}{4}}\right) = \lim_{n \to \infty}\left(\frac{\frac{n(n+1)}{2} - n}{\frac{n^2(n+1)^2}{4} - \frac{n^3(n+2)}{4}}\right)$$

$$= \lim_{n\to\infty}\left(\frac{\frac{n^2-n+1}{2}}{\frac{n^2\left(n^2+2n+1\right)}{4}-\frac{n^3(n+2)}{4}}\right) = \lim_{n\to\infty}\left(\frac{\frac{n^2-n+1}{2}}{\frac{n^2}{4}}\right) = 2$$

8. Let $1+p=y$

$\left(y^{1/6}-1\right)x^2+\left(y^{1/3}-1\right)x+\left(y^{1/9}-1\right)=0$

Divide by $y-1$

$\frac{y^{1/6}-1}{y-1}x^2+\frac{y^{1/3}-1}{y-1}x+\frac{y^{1/9}-1}{y-1}=0$

Taking limit as $y\to 1 \Rightarrow p\to 0$ then, we get $\frac{1}{6}x^2+\frac{1}{3}x+\frac{1}{9}=0$

$3x^2+6x+2=0 \begin{cases}\alpha(p)\\ \beta(p)\end{cases}$

Sum of roots $= -\frac{6}{3}=-2$

$\lim_{p\to 0^+}\left[\alpha(p)+\beta(p)\right]=-2$

$\therefore \lim_{p\to 0^+}\left|\left[\alpha(p)+\beta(p)\right]\right| = 2$

9. Let us assume that maximum value of function occurs at $x=c$

$\Rightarrow f'(c)=0$, now applying mean value theorem on $f'(x)$

$\Rightarrow \frac{f'(c)-f'(0)}{c-0}=f''(\alpha)$, where $\alpha\in(0,c)$

$\Rightarrow -f'(0)=c\,f''(\alpha) \Rightarrow |f'(0)|=c|f''(\alpha)| \Rightarrow |f'(0)|\le 5c$ ..... (1)

Similarly $\frac{f'(4)-f'(c)}{4-c}=f''(\beta)$, where $\beta\in(c,4) \Rightarrow |f'(4)|\le 5(4-c)$ ..... (2)

Adding equation (1) and (2), we get $|f'(0)|+|f'(4)|\le 20$

10. Clearly base area, $S_1=\ell b$ and volume, $V=\ell bh+\frac{1}{2}\pi\left(\frac{b}{2}\right)^2\ell = S_1h+\frac{\pi}{8}bS_1$

$\Rightarrow b=\frac{S_1}{\ell}$, $\frac{V}{S_1}=h+\frac{\pi b}{8} \Rightarrow h=\frac{V}{S_1}-\frac{\pi b}{8}$

Now surface area $S=\ell b+2\ell h+2hb+\pi\left(\frac{b}{2}\right)^2+\pi\left(\frac{b}{2}\right)\ell$

Putting the values of h and b we get $S=S_1\left(1+\frac{\pi}{4}\right)+2V\left(\frac{\ell}{S_1}+\frac{1}{\ell}\right)$

Now $\frac{ds}{d\ell}=2V\left(\frac{1}{S_1}-\frac{1}{\ell^2}\right)=\frac{2V}{S_1\ell^2}\left(\ell^2-S_1\right)=\frac{2V}{S_1\ell^2}\left(\ell-\sqrt{S_1}\right)\left(\ell+\sqrt{S_1}\right)$. Since $\ell\in[1,S_1]$ and S_1 we have to look for maximum and minimum values of S under the given restriction. Clearly $\ell=\sqrt{S_1}$ is the point of minima of S.

Also $S|_{\ell=1}=S|_{\ell=S_1}$

$\Rightarrow S_{min} = S\big|_{t=\sqrt{S_1}} = S_1\left(1+\frac{\pi}{4}\right)+\frac{4V}{\sqrt{S_1}}$ and $S_{max} = S\big|_{t=1 \text{ or } S_1} = S_1\left(1+\frac{\pi}{4}\right)+2V\left(1+\frac{1}{S_1}\right)$

11. We have $\frac{dy}{dx} = 6t^2 - 30t + 24 = 6(t-1)(t-4)$ and $\frac{dx}{dt} = 6t - 18 = 6(t-3)$. Thus

$\frac{dy}{dx} = \frac{(t-1)(t-4)}{(t-3)}$, which indicates that t = 1, 3 and 4 are the critical points of y = f(x).

Now $\frac{d^2y}{dx^2} = \frac{d}{dt}\left(\frac{dy}{dx}\right).\frac{dt}{dx} = \frac{t^2-6t+11}{(t-3)^2} \times \frac{1}{6(t-3)}$

at t = 1, $\frac{d^2y}{dx^2} < 0 \Rightarrow$ t = 1 is a point of local maxima

at t = 4, $\frac{d^2y}{dx^2} > 0 \Rightarrow$ t = 4 is a point of local minima

at t = 3, $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ are not defined and changes its sign

$\frac{d^2y}{dx^2}$ is unknown in the vicinity of t = 3, thus t = 3 is a point of neither maxima nor minima.

Finally for t = 1, y = 21, t = 4, y = –6

t = 0, y = 10, t = 6, y = 46

Thus maximum and minimum values of expression y = f(x) are 46 and –6 respectively.

12. We have $f'(x) = -1 - 12x^2 < 0$

f(x) is decreasing

Now $f(f(x)) = 1 - f(x) - 4f(x)^3$

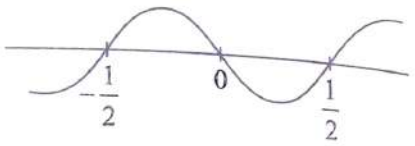
Inequality reduces to $f(f(x) > f(1-2x)$

$\Rightarrow f(x) < 1 - 2x$

$\Rightarrow 1 - x - 4x^3 < 1 - 2x \Rightarrow 4x^3 - x > 0$

$\Rightarrow x(4x^2 - 1) > 0$

Hence, number of integral values is 4



13. Let $1 + \sin x = y$

$I = \int \frac{1}{y}\sqrt{\frac{3-y}{1+y}}dy$

again let $\frac{3-y}{1+y} = t^2$

$I = 8\int \frac{t^2}{(t^2-3)(t^2+1)}dt = \sqrt{3}\ln\frac{t-\sqrt{3}}{t+\sqrt{3}} + 2\tan^{-1}t + c$

14. Put $x^{1/3} = t^2$

$\Rightarrow$ Given integration $= 6\int \frac{dt}{(1+5t^2)\sqrt{1+4t^2}}$.

Let $4 + \frac{1}{t^2} = u^2 \Rightarrow I = \int -\frac{6du}{u^2+1} = 6\cot^{-1}\sqrt{4+\frac{1}{x^{1/3}}} + c$

15. $I=\int\frac{x^{7/6}\left(x^{-7/6}-x^{5/6}\right)dx}{x^{7/6}x^{1/3}\left(x^2+x+1\right)^{1/2}-x^{7/6}x^{1/2}\left(x^2+x+1\right)^{1/3}}$

$=\int\frac{\left(1-x^2\right)dx}{x^{3/2}\left(x^2+x+1\right)^{1/2}-x^{5/3}\left(x^2+x+1\right)^{1/3}}=\int\frac{-\left(1-\frac{1}{x^2}\right)dx}{\left(x+\frac{1}{x}+1\right)^{1/2}-\left(x+\frac{1}{x}+1\right)^{1/3}}$

$=\int\frac{-dt}{(t+1)^{1/2}-(t+1)^{1/3}}\quad\left(\text{putting}\quad x+\frac{1}{x}=t\right)$

Substituting $(t+1)=z^6$

$=-6\int\frac{z^5dz}{z^3-z^2}=-6\int\frac{z^3}{(z-1)}dz\quad(\text{put } z-1=R)=-6\int\frac{(R+1)^3\,dR}{R}=-6\int\left(R^2+3R+3+\frac{1}{R}\right)\,dR$

$=-6\left(\frac{R^3}{3}+\frac{3R^2}{2}+3R+\log R\right)+C$, where $R=\left(x+\frac{1}{x}+1\right)^{1/6}-1$.

16. $I=\int\frac{\left(x^2-1\right)\sqrt{x^4+2x^3-x^2+2x+1}}{x^2\left(x^2+2x+1\right)}dx=\int\frac{\left(1-\frac{1}{x^2}\right)\sqrt{x^2\left(x^2+2x-1+\frac{2}{x}+\frac{1}{x^2}\right)}}{x\left(x+2+\frac{1}{x}\right)}dx$

Substituting $x+\frac{1}{x}=t$

$\int\frac{\sqrt{t^2-2+2t-1}}{t+2}dt=\int\frac{\sqrt{t^2+2t-3}}{t+2}dt=\int\frac{\left(t^2+2t-3\right)}{(t+2)\sqrt{t^2+2t-3}}dt=\int\frac{t(t+2)-3}{(t+2)\sqrt{t^2+2t-3}}dt$

$=\int\frac{t\ dt}{\sqrt{t^2+2t-3}}-\int\frac{3}{(t+2)\sqrt{t^2+2t-3}}dt$

I_1 $\qquad$ I_2

$I_1=\int\frac{t\ dt}{\sqrt{(t+1)^2-4}}$ $\qquad$ putting $t+1=z$

$=\int\frac{(z-1)\ dz}{\sqrt{z^2-2^2}}=\int\frac{zdz}{\sqrt{z^2-2^2}}-\int\frac{dz}{\sqrt{z^2-2^2}}=\sqrt{z^2-2^2}-\log\left(z-\sqrt{z^2-4}\right)$

$=\sqrt{t^2+2t-3}-\log\left(t+1+\sqrt{t^2+2t-3}\right)$

$I_2=\sqrt{3}\int\frac{dR}{R^2\frac{1}{R}\sqrt{\left(\frac{1}{R}-2\right)^2+2\left(\frac{1}{R}-2\right)-3}}\qquad\left(\text{by substituting}\quad t+2=\frac{1}{R}\right)$

$=3\int\frac{dR}{\sqrt{\left(\frac{2}{3}\right)^2-\left(R+\frac{1}{3}\right)^2}}=\sqrt{3}\sin^{-1}\left(\frac{R+\frac{1}{3}}{\frac{2}{3}}\right)+C=\sqrt{3}\sin^{-1}\left(\frac{t+5}{2(t+2)}\right)+C$

$\therefore I=\sqrt{t^2+2t-3}-\log\left(t+1+\sqrt{t^2+2t-3}\right)+\sqrt{3}\sin^{-1}\left(\frac{t+5}{2(t+2)}\right)$

17. Take $t = xe^x$

$$I = \int_0^e \frac{\sin^2(t)dt}{\sin^2(t)+\sin^2(e-t)}$$

$$I = \int_0^e \frac{\sin^2(e-t)dt}{\sin^2(e-t)+\sin^2(t)}$$

$$2I = \int_0^e 1\,dt$$

$$I = \frac{e}{2}$$

18. Let $f(r) = \int_0^{\pi/2} x^r \sin x dx$

Now, $\int_0^{\pi/2} x^r \cos x dx = \frac{x^{r+1}}{r+1}\cos x\Big|_0^{\pi/2} + \int_0^{\pi/2} \frac{x^{r+1}}{r+1}\sin x dx$

$$\Rightarrow \int_0^{\pi/2} x^r \cos x dx = \frac{f(r+1)}{(r+1)}$$

So, we have $\lim_{r\to\infty} \frac{r^C f(r)(r+1)}{f(r+1)} = L$

Now, consider $f(r) < \int_0^{\pi/2} x^r dx = \frac{\left(\frac{\pi}{2}\right)^{r+1}}{r+1}$

Also as $\sin x > \frac{2x}{\pi}\ \forall\ x \in \left(0, \frac{\pi}{2}\right)$

$$f(r) > \int_0^{\pi/2} \frac{2}{\pi} x^{r+1} dx = \frac{\left(\frac{\pi}{2}\right)^{r+1}}{r+2}$$

$$\Rightarrow \frac{r}{r+2} < r\left(\frac{2}{\pi}\right)^{r+1} f(r) < \frac{r}{r+1}$$

Hence, $\lim_{r\to\infty} \frac{f(r)}{f(r+1)} = \lim_{r\to\infty} \frac{r\left(\frac{2}{\pi}\right)^{r+1} f(r)}{(r+1)\left(\frac{2}{\pi}\right)^{r+2} f(r+1)} \times \frac{2(r+1)}{\pi r} = \frac{2}{\pi}$

Now, $\lim_{r\to\infty} r^C (r+1)\frac{f(r)}{f(r+1)} = L$

$$\Rightarrow \lim_{r\to\infty} \frac{2}{\pi} r^C (r+1) = L$$

For positive L we should have $C = -1$

and $L = \frac{2}{\pi}$

19. $I_N = \int_0^{\pi} x^3 \log_e(\sin x)\,dx = \int_0^{\pi} (\pi - x)^3 \log_e(\sin x)\,dx$

$= \frac{3\pi}{2}\int_0^{\pi} x^2 \log_e \sqrt{2}\,dx + \frac{3\pi}{2}\int_0^{\pi} x^2 \log_e(\sin x)\,dx$

$= \frac{3\pi}{2} I_D \Rightarrow \frac{I_N}{I_D} = \frac{3\pi}{2}$

20. $I(a) = \int_0^{\infty}\left(\frac{\tan^{-1}(ax) - \tan^{-1} x}{x}\right)dx$

$\Rightarrow I'(a) = \int_0^{\infty} \frac{1}{x}\cdot\left(\frac{x}{1+a^2x^2}\right)dx$

$\Rightarrow I'(a) = \int_0^{\infty} \frac{1}{a^2x^2+1}\,dx$

$I'(a) = \frac{\pi}{2a}$

$I(a) = \frac{\pi}{2}\ell na$

21. Solving the two equations,
$m^2x^2 = (e^{-kr})\,x$

$x_1 = 0,\ x_2 = \frac{e^{-kr}}{m^2}$

So, $A_r = \int_0^{x_2}\left(e^{-\frac{kr}{2}}\sqrt{x} - mx\right)dx$

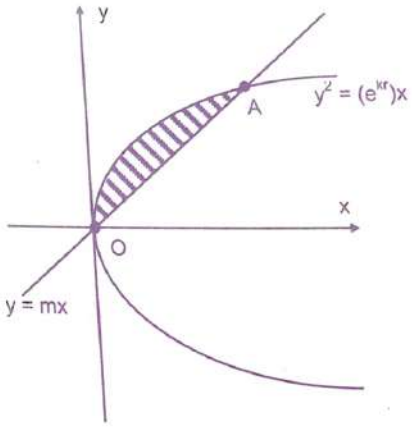
$= \frac{2}{3}e^{-kr/2}x_2^{3/2} - m\frac{x_2^2}{2}$

$= \frac{2}{3}e^{-kr/2}\frac{e^{-3kr/2}}{m^3} - \frac{m}{2}\frac{e^{-2kr}}{m^4} = \frac{e^{-2kr}}{6m^3}$

Now, $\frac{A_{r+1}}{A_r} = \frac{e^{-2k(r+1)}}{e^{-2kr}} = e^{-2k} = \text{constant}$

So, $A_1, A_2, A_3, \ldots$ are in G.P.

Sum to infinite terms $= A_1\frac{1}{1-e^{-2k}} = \frac{e^{-2k}}{6m^3}\times\frac{e^{2k}}{e^{2k}-1} = \frac{1}{6m^3(e^{2k}-1)}$



22. $\Rightarrow A(x) = 2x.e^{-2^{(4x-2)}}.x^4$ let area be A(x)

$A'(x) = 2e^{-2^{(4x-2)}}\left(1 - 2^{4x}x^4\right)$

Maxima occurs at $x = \frac{1}{2^x}$

and $f(x)\max = \frac{2}{e^{1/4}}\frac{1}{2^x}$, $\sum_{x=1}^{\infty}\frac{2}{e^{1/4}}\frac{1}{2^x} = \frac{2}{e^{1/4}}$

23. $2 \le D(P, L_1) + D(P, L_2) \le 4$

$$2 \le \frac{|x-y|}{\sqrt{2}} + \frac{|x+y|}{\sqrt{2}} \le 4$$

$2\sqrt{2} \le |x-y| + |x+y| \le 4\sqrt{2}$

When $y - x < 0, y + x > 0$

$\therefore \sqrt{2} \le x \le 2\sqrt{2}$

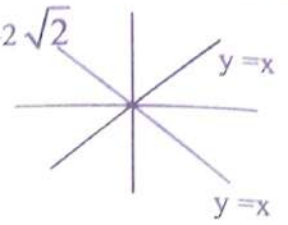
$y - x > 0, y + x > 0$

$\sqrt{2} \le y \le 2\sqrt{2}$

Similarly $-2\sqrt{2} \le x \le -\sqrt{2}$

$-2\sqrt{2} \le y \le -\sqrt{2}$

Area $= S = \left[4\sqrt{2}\right) \le (2\sqrt{2})^2 = 24$



24. Putting $x^2 + y^2 = t$

$$\Rightarrow \frac{y}{x}\cdot\frac{dy}{dx} = \frac{1}{2x}\cdot\frac{dt}{dx} - 1 = \frac{1-2t}{t+1}$$

$$\Rightarrow 2x\,dx = \frac{t+1}{2-t}dt$$

On integration, we get $2x^2 + y^2 + 3\log(2 - x^2 - y^2) = c$

25. Given equation is $\int_0^x t f(x-t)dt = \int_0^x f(t)dt + \sin x + \cos x - x - 1$

Apply Leibnitz's rule of differentiation, we have $\int_0^x t f'(x-t)dt + xf(x-x) = f(x) + \cos x - \sin x - 1$

$$\Rightarrow \left[\because \int t f'(x-t)dt = t\int f'(x-1)dt - \int \frac{dt}{dt}\int f'(x-t) = t f(x-t) - \int f(x-t)dt\right]$$

$$\Rightarrow \left[t f(x-t)\right]_0^x - \int_0^x f(x-t)dt + xf(0) = f(x) + \cos x - \sin x - 1$$

$$\Rightarrow xf(0) - \int_0^x f(x-t)dt + xf(0) = f(x) + \cos x - \sin x - 1$$

$$2xf(0) - \int_0^x f(x-t)dt = f(x) + \cos x - \sin x - 1$$

Again applying Leibnitz's rule

$$\Rightarrow 2f(0) - f(0) = f'(x) = \sin x - \cos x \Rightarrow \frac{dy}{dx} = \sin x + \cos x + f(0)$$

$\Rightarrow y = -\cos x + \sin x + f(0)\cdot x$

$\Rightarrow f(x) = \sin x - \cos x + kx$; where $k = f(0)$

26. $$\frac{dV(t)}{dt} = -K(T-t) \Rightarrow V(t) = \frac{K(T-t)^2}{2} + C$$

$$\text{At } t = 0,\ V = I \Rightarrow C = I - \frac{KT^2}{2}$$

$$\therefore \text{Scrap value } V(T) \Rightarrow V(t = T) = C = I - \frac{KT^2}{2}$$

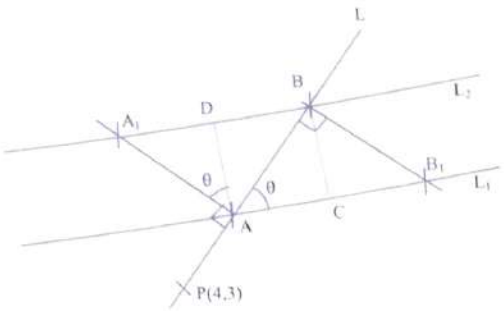
27. Distance of P from BC $= R\cos A = \left|\frac{4.2-3.3+3}{5}\right|$ and distance of H from BC

$= 2R\cos B\cos C = \left|\frac{4.1-3.2+3}{5}\right|$

$\Rightarrow \cos A\cos B\cos C = \frac{1}{25R^2}$ and $\sqrt{R^2-8R^2\cos A\cos B\cos C} = PH = \sqrt{2}$. Hence $R^2 = \frac{58}{25}$

28. The given lines (L_1 and L_2) are parallel and distance between them (BC or AD) is;

$= \frac{15-5}{5} = 2$ units.



Let $\angle BAC = \theta$

$\therefore AB_1 = A_1B = 2(\tan\theta + \cot\theta)$

Now area of parallelogram AA_1BB_1 is

$\Delta = AB_1 \cdot BC = 4(\tan\theta + \cot\theta)$

Clearly Δ is least for $\theta = \pi/4$

Let slope of AB be m

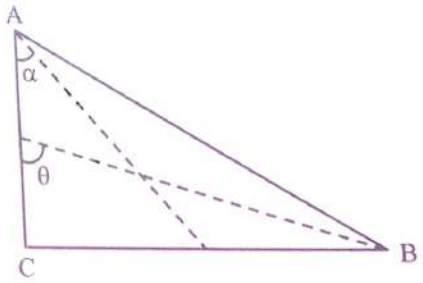
$\Rightarrow 1 = \left|\frac{m+3/4}{1-\frac{3m}{4}}\right| \Rightarrow 4m+3 = \pm(4-3m)$

$\Rightarrow$ m = 1/7 or –7

$\Rightarrow$ Equation of 'L' is ; $x - 7y + 17 = 0$ or $7x + y - 31 = 0$.

29. $\tan\theta = \frac{2a}{b}$, $\tan\alpha = \frac{a}{2b}$

$\Rightarrow \tan(\theta-\alpha) = \frac{\frac{2a}{b}-\frac{a}{2b}}{1+\frac{a^2}{b^2}} = \frac{3ab}{2(a^2+b^2)} = \frac{1}{3}$



Now, $\sqrt{a^2+b^2} = 60$ and $\frac{1}{2}ab = 400$ sq. units

30. Let the parametric coordinates of B

$(h + r\cos\theta, k + r\sin\theta)$

It will satisfy the equation $3x^2 + 7xy - 6y^2 = 0$

$\Rightarrow 3(h + r\cos\theta)^2 + 7(h + r\cos\theta)(k + r\sin\theta) - 6(k + 2\sin\theta)^2 = 0$

$\Rightarrow r^2\{3\cos^2\theta + 7\cos\theta\sin\theta - 6\sin^2\theta\} + r\{6h\cos\theta + 7h\sin\theta + 7k\cos\theta - 12k\sin\theta\}$
$+ 3h^2 + 7hk - 6k^2 = 0$

Let the roots of this equation be r_1 and r_2

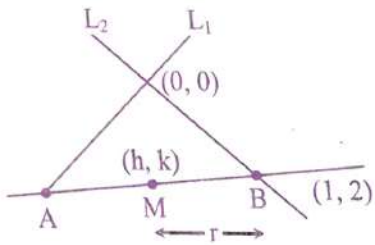
$\because$ r_1 and r_2 are in opposite direction

$r_1 = -r_2$

$\sqrt{r_1 + r_2} = 0$

$\sin\theta(7h - 12k) + \cos\theta(6h + 7k) = 0$

$\tan\theta = -\left(\frac{7k+6h}{7h-12k}\right)$



$\Rightarrow \frac{2-k}{1-h} = -\left(\frac{7k+6h}{7h-12k}\right)$

$\Rightarrow 14h - 7hk - 24k + 12k^2 = -7k - 6h + 7hk + 6h^2$

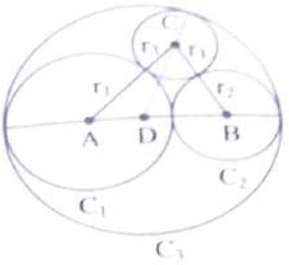
$\Rightarrow 20h - 17k - 14hk - 6h^2 + 12k^2 = 0$

$\Rightarrow 6h^2 - 12k^2 + 14hk + 17k - 20h = 0$
$\Rightarrow 6x^2 - 12y^2 + 14xy + 17y - 20x = 0$

31. $\cos(\angle CDA) = -\cos(\angle CDB)$

$$\Rightarrow \frac{AD^2 + DC^2 - AC^2}{2 \cdot AD \cdot DC} = -\frac{BD^2 + DC^2 - BC^2}{2 \cdot BD \cdot DC}$$

$$\frac{r_2^2 + (r_1 + r_2 - r_3)^2 - (r_1 + r_3)^2}{2r_2(r_1 + r_2 - r_3)} = -\frac{r_1^2 + (r_1 + r_2 - r_3)^2 - (r_2 + r_3)^2}{2r_1(r_1 + r_2 - r_3)}$$



Now, r_1, r_2, r_3 are in A.P.

$\Rightarrow 2r_2 = r_1 + r_3$

$$\Rightarrow \frac{r_2^2 + (2r_1 - r_2)^2 - (2r_2)^2}{2r_2(2r_1 - r_2)} = -\frac{r_1^2 + (2r_1 - r_2)^2 - (3r_2 - r_1)^2}{2r_1(2r_1 - r_2)}$$

$$\Rightarrow \left(\frac{r_1}{r_2}\right)^3 = 2$$

32. For lines to be chord of circle, its distance from the centre must be less than the radius

For L_1: $\left|\frac{\lambda - 24}{\sqrt{13}}\right| < 2$

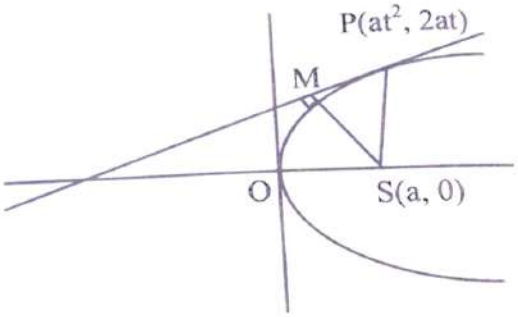
$\Rightarrow \lambda \in \left(24 - 2\sqrt{13}, 24 + 2\sqrt{13}\right)$

$\Rightarrow \lambda \in \{17, 18, \ldots\ldots, 31\}$ ($\because \lambda \in z$)

Similarly for L_2: $\lambda \in \{11, 12, \ldots\ldots, 25\}$

33. Let $P \equiv (at^2, 2at)$

Tangent at P is $ty = x + at^2$



$$\Rightarrow SM = \frac{a(1+t^2)}{\sqrt{1+t^2}} = a\sqrt{1+t^2}$$

and $SP = a(1 + t^2)$

$$\Rightarrow PM = \sqrt{SP^2 - SM^2} = at\sqrt{1+t^2}$$

$\Rightarrow$ Area of triangle SPM

$= \frac{1}{2} SM \times PM = \frac{1}{2} a^2 t(1+t^2)$ which is an increasing functions of t

$$\Rightarrow \text{Area}_{(max)} = (A)_{t=1} = \frac{1}{2} a^2 (1+1) = a^2$$

34. $$x^2\left(1 - \frac{c^2}{a^2}\right) + y^2 = a^2 - c^2$$

35. $t_1 + t_2 = 0$

$$2\left(t_1^2 + t_2^2\right) + 4\left(t_1 + t_2\right) - 4 = 0$$

$\Rightarrow t_1 t_2 = -1$

$$L_{AB} = \sqrt{(t_1 - t_2)^2 + \left(2\left(t_1^2 - t_2^2\right) + 4(t_1 - t_2)\right)^2} = (t_1 - t_2)\sqrt{1 + \left(2(t_1 + t_2) + 4\right)^2} = 2\sqrt{17}$$

$L^2 = 68$

36. Let any point P on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ be (x_1, y_1), then

$NQ = PS = a + ex_1$

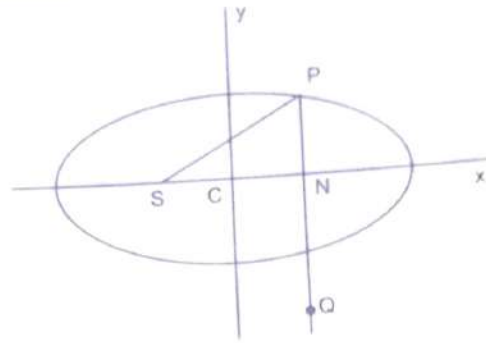
Let the coordinate of Q be (x, y), then

$x = CN = x_1$

And $y = NQ = -(a + ex_1)$

Putting the value of x_1 from (1) in (2), the required locus is $y = -(a + ex) \Rightarrow y + a + ex = 0$

Taking S′ instead of S the locus of Q will be $y + a - ex = 0$.



37. $\because e = \frac{4}{5}$

$\therefore 1-\frac{b^2}{a^2}=\frac{16}{25} \Rightarrow \frac{b^2}{a^2}=\frac{9}{25} \Rightarrow \frac{b}{a}=\frac{3}{5} \Rightarrow b=\frac{3a}{5}$

Now, radius of the circle $r = a - \lambda$ were $(\lambda, 0)$ is centre of circle

Also, $r = AC = b \sin\theta$

$a - \lambda = b \sin\theta$ where $\lambda = a\cos\theta$

$a(1 - \cos\theta) = b \sin\theta$

$\Rightarrow a^2(1 - \cos\theta)^2 = b^2(1 - \cos^2\theta)$

$\Rightarrow a^2(1 - \cos\theta) = b^2(1 + \cos\theta)$

$\Rightarrow \frac{1-\cos\theta}{1+\cos\theta}=\frac{b^2}{a^2}=\frac{9}{25} \Rightarrow 16 = 34\cos\theta \Rightarrow \cos\theta=\frac{8}{17}$

Diameter $AB = 2b \sin\theta = 2\cdot\frac{3a}{5}\cdot\frac{15}{17}=\frac{18a}{17}$

38. Clearly the parabola should pass through (1, 0) and (−1, 0). Let directrix of this parabola be $x\cos\theta + y\sin\theta = 2$. If M(h, k) be the focus of this parabola, then distance of (±1, 0) from 'M' and from the directrix should be same

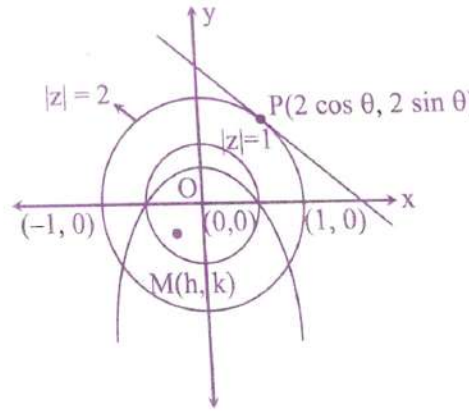
$\Rightarrow (h - 1)^2 + k^2 = (\cos\theta - 2)^2$ ..... (1)

and $(h + 1)^2 + k^2 = (\cos\theta + 2)^2$ ..... (2)

Now equation (2) − (1) $\Rightarrow \cos\theta = \frac{h}{2}$ ..... (3)

Also equation (2) + (1)

$\Rightarrow (h^2 + k^2 + 1) = (\cos^2\theta + 4)$ ..... (4)



$\therefore$ From (3) and (4), we get $h^2 + k^2 + 1 = 4 + \frac{h^2}{4} \Rightarrow \frac{3h^2}{4} + k^2 = 3$

Hence locus of focus M(h, k) is $\frac{x^2}{4}+\frac{y^2}{3}=1$

Also we know that area of the quadrilateral formed by the tangents at the ends of the latus-rectum is $\frac{2a^2}{e}$ (where e is eccentricity of ellipse)

$\therefore$ Required area $= \frac{2(4)}{\frac{1}{2}} = 16$ (square units) (As $e^2 = 1 - \frac{3}{4} = \frac{1}{4} \Rightarrow e = \frac{1}{2}$)

39. $S_1P = S_2P \Rightarrow a - e\alpha = E\alpha - \frac{a}{2}$. Also, $\alpha = \frac{ae + \frac{a}{2}E}{2}$

Eliminating α we get, $E^2 + 3eE + (2e^2 - 6) = 0$

Now, as $e = \sqrt{1 - \frac{b^2}{a^2}} = \frac{1}{\sqrt{7}}$, $E = \frac{5}{\sqrt{7}}$

$$\Rightarrow \frac{a}{c} = \frac{1}{\sqrt{E^2 - 1}} = \frac{\sqrt{7}}{\sqrt{18}}$$

40. $\frac{SN}{SP} = \frac{S'N}{S'P}$ as PN is angle bisector of focal radii

Also, $\frac{SN}{SP} = e = \sqrt{2}$

$$\Rightarrow 6\left(\frac{SP}{SN}\right)^2 = 6 \cdot \frac{1}{2} = 3$$

$$\left(\frac{SN}{SP} = \left|\frac{SN - S'N}{SP - S'P}\right| = \frac{SS'}{|SP - S'P|} = e\right)$$

41. Let the circle be $(x - h)^2 + (y - k)^2 = r^2$, r–variable

Its intersection with $x^2 - y^2 = \frac{16}{9}b^2$ is obtained by putting

$$y^2 = x^2 - \frac{16}{9}b^2$$

$$\therefore (x-h)^2 + (y-k)^2 = r^2$$

$$\Rightarrow x^2 + y^2 - 2hx + h^2 + k^2 - r^2 = 2ky$$

$$\Rightarrow x^2 + x^2 - \frac{16}{9}b^2 - 2hx + h^2 + k^2 - r^2 = 2k\sqrt{x^2 - \frac{16}{9}b^2}$$

$$\Rightarrow \left[2x^2 - 2hx + \left(h^2 + k^2 - r^2 - \frac{16}{9}b^2\right)\right]^2 = 4k^2\left(x^2 - \frac{16}{9}b^2\right)$$

$\Rightarrow 4x^4 - 8hx^3 + \ldots\ldots = 0$

$\therefore$ Above gives the abscissa of the four points of intersection

$$\therefore \sum x_1 = \frac{8h}{4} = 2h$$

$\Rightarrow x_1 + x_2 + x_3 + x_4 = 2h$

Similarly, $y_1 + y_2 + y_3 + y_4 = 2k$

Now, if (α, β) be the centroid of ΔABC, then

$3\alpha = x_1 + x_2 + x_3$, $3\beta = y_1 + y_2 + y_3$

$\therefore x_4 = 2h - 3\alpha$, $y_4 = 2k - 3\beta$

But, (x_4, y_4) lies on $x^2 - y^2 = \frac{16}{9}b^2$

$$\Rightarrow (2h - 3\alpha)^2 + (2k - 3\beta)^2 = \frac{16}{9}b^2$$

$$\Rightarrow 81\left[\left(x - \frac{2h}{3}\right)^2 + \left(y - \frac{2k}{3}\right)^2\right] = 16b^2$$

43. Let the point be $P \equiv (\alpha, \beta, \gamma)$.

Hence, $\frac{\alpha}{a}+\frac{\beta}{b}+\frac{\gamma}{c}=1$ ..... (1)

Equation of the plane perpendicular to OP is $\alpha x + \beta y + \gamma z = d$

But it passes through $P(\alpha, \beta, \gamma)$, we have $d = \alpha^2 + \beta^2 + \gamma^2$

Hence equation of plane through P and perpendicular to OP is $\alpha x + \beta y + \gamma z = \alpha^2 + \beta^2 + \gamma^2$

$\Rightarrow OA = \frac{\alpha^2+\beta^2+\gamma^2}{\alpha}$, $OB = \frac{\alpha^2+\beta^2+\gamma^2}{\beta}$, $OC = \frac{\alpha^2+\beta^2+\gamma^2}{\gamma}$

So, the planes through A, B, C parallel to the planes YOZ, ZOX, XOY intersect in the point Q whose coordinates are $x = \frac{\alpha^2+\beta^2+\gamma^2}{\alpha}$, $y = \frac{\alpha^2+\beta^2+\gamma^2}{\beta}$, $z = \frac{\alpha^2+\beta^2+\gamma^2}{\gamma}$ with the help of (1)

$\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{\alpha^2+\beta^2+\gamma^2}$ and $\frac{1}{ax}+\frac{1}{by}+\frac{1}{cz}=\frac{\frac{\alpha}{a}+\frac{\beta}{b}+\frac{\gamma}{c}}{\alpha^2+\beta^2+\gamma^2}$

Hence, required locus is $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{ax}+\frac{1}{by}+\frac{1}{cz}$

44. Distance from (0, 0, 0), $D=\sqrt{x^2+y^2+z^2}=\sqrt{x^2+y^2+\frac{2}{xy}}$

$D^2 = x^2 + y^2 + \frac{2}{xy} \geq 2xy + \frac{2}{xy} \geq 4$

$\therefore D = 2$

45. Put $\sqrt{x^2+11}=t$ so that $\sqrt{x^2+\sqrt{x^2+11}}+\sqrt{x^2-\sqrt{x^2+11}}=4$ can be written as

$\sqrt{t^2+t-11}+\sqrt{t^2-t-11}$ (1)

We have also have the identity

$(t^2 + t - 11) - (t^2 - t - 11) = 2t$ ..... (2)

Dividing (1) by (2), we get

$\sqrt{t^2+t-11}-\sqrt{t^2-t-11}=\frac{t}{2}$ ..... (3)

Adding (2) and (3), we get $2\left(\sqrt{t^2+t-11}\right)=4+\frac{t}{2} \Rightarrow t^2+t-11=4+t+\frac{t^2}{16} \Rightarrow t=4 \Rightarrow x=\pm\sqrt{5}$

46. We have $P(x)=\frac{5}{3}-6x-9x^2=-(3x+1)^2+\frac{8}{3}$

$\Rightarrow P_{max}\left(x=-\frac{1}{3}\right)=\frac{8}{3}$

Similarly, $Q(y)=-4y^2+4y+\frac{13}{2}=(2y-1)^2+\frac{15}{2}$

$\Rightarrow Q_{max}\left(y=\frac{1}{2}\right)=\frac{15}{2}$

Now, $P_{max}\left(x=\frac{1}{3}\right)\times Q_{max}\left(y=\frac{1}{2}\right)=\frac{8}{3}\times\frac{15}{2}=20$

$\therefore (x, y) \equiv \left(-\frac{1}{3}, \frac{1}{2}\right)$

47. Let the roots be $\alpha_1, \alpha_2, \ldots, \alpha_8$

$\Rightarrow \alpha_1, \alpha_2, \ldots, \alpha_8 = 4$

$\alpha_1, \alpha_2, \ldots, \alpha_8 = \frac{1}{2^8}$

$\Rightarrow (\alpha_1, \alpha_2, \ldots, \alpha_8)^{1/8}$

$= \frac{1}{2} = \frac{\alpha_1 + \alpha_2 + \ldots + \alpha_8}{8}$

$\Rightarrow$ A.M. = G.M.

$\Rightarrow$ all the roots are equal to $\frac{1}{2}$

$\Rightarrow a_1 = -{}^8C_7\left(\frac{1}{2}\right)^7 = -\frac{1}{24}$

$a_2 = {}^8C_6\left(\frac{1}{2}\right)^6 = -\frac{7}{24}$

$a_3 = -{}^8C_5\left(\frac{1}{2}\right)^5$

48. The ratio must be rational. Suppose $r = \frac{h}{k}$.

If there are more than six terms than the first term must be divisible by k^6 and last by h^6.
If $k \geq 3$, then $h > k$ so $h \geq 4$ but $h^6 > 1000$ so $k < 3$.
$\Rightarrow k = 2$ and $h = 3$ so the first term must be a multiple of 64.
The smallest is 128, but then we get only six terms 128, 192, 288, 432, 648, 972.

49. $S = \sum_{r=0}^{n}(-1)^r \cdot {}^nC_r \frac{1}{3^r} + \sum_{r=0}^{n}(-1)^r \cdot {}^nC_r \left(\frac{5}{9}\right)^r + \ldots\ldots \sum_{r=0}^{n}(-1)^r \cdot {}^nC_r \left(\frac{19}{27}\right)^r$ ..... upto 10 terms

$= \left(1-\frac{1}{3}\right)^n + \left(1-\frac{5}{9}\right)^n + \ldots\ldots$ upto 10 terms

$= \left(\frac{2}{3}\right)^n + \left(\frac{2}{3}\right)^{2n} + \left(\frac{2}{3}\right)^{3n}$ ..... upto 10 terms

$= \left(\frac{2}{3}\right)^n \frac{\left(1-\left(\frac{2}{3}\right)^{10n}\right)}{1-\left(\frac{2}{3}\right)^n} = \frac{2^n\left(3^{10n}-2^{10n}\right)}{3^{10n}\left(3^n-2^n\right)}$

50. $\sum_{r=1}^{n} \frac{r(r+1)}{2} \left[\log_{1/2}\sqrt{4x-x^2}\right]^r = \frac{1}{2}\sum_{r=1}^{n} r(r+1)\ A^r$ where $A = \log_{1/2}\sqrt{4x-x^2}$

$= \frac{1}{2}\{(1.2)A + (2.3)A^2 + (3.4)A^3 + \ldots + n(n+1)A^n\}$

$S = \frac{1}{2}S_1$ (say) ...(1)

$S_1 = 2A + 6A^2 + 12A^3 + \ldots\ldots + n(n+1)A^n$

$A\,S_1 = \quad 2A^2 + 6A^3 + \ldots + (n-1)n\,A^n + n(n+1)A^{n+1}$

On subtracting, we get

$(1-A)S_1 = 2A + 4A^2 + 6A^3 + \ldots\ldots + 2nA^n - n(n+1)A^{n+1}$

$(1-A)S_1 = 2(A + 2A^2 + 3A^3 + \ldots\ldots + nA^n) - n(n+1)A^{n+1}$

$(1-A)S_1 = 2S_2 - n(n+1)A^{n+1}$ (say)

$S_2 = A + 2A^2 + 3A^3 + \ldots + nA^n$ ...(2)

$AS_2 = \quad A^2 + 2A^3 + \ldots + (n-1)A^n + nA^{n+1}$

$\therefore (1-A)S_2 = A + A^2 + \ldots + A^n - nA^{n+1}$

$$S_2 = \frac{A(1-A^n)}{(1-A)^2} - \frac{nA^{n+1}}{(1-A)}$$

From (2), $S_1 = \dfrac{2A(1-A^n)}{(1-A)^3} - \dfrac{2nA^{n+1}}{(1-A)} - \dfrac{n(n+1)A^{n+1}}{(1-A)}$

From (1), $S = \dfrac{A(1-A^n)}{(1-A)^3} - \dfrac{nA^{n+1}}{(1-A)^2} - \dfrac{n(n+1)A^{n+1}}{(1-A)}$

51. Let the required value be S

$$\sum_{i=0}^{n}\sum_{j=0}^{n}(-1)^{i+j-1}\,{}^nC_i\,{}^nC_j = \sum_{i=0}^{n}(-1)^{2i-1}\left({}^nC_i\right)^2 + 2S$$

$$\Rightarrow \qquad 0 \qquad = -\sum_{i=0}^{n}\left({}^nC_i\right)^2 + 2S$$

$$\Rightarrow S = {}^{2n-1}C_{n-1} = {}^{2n-1}C_n$$

52. $2\cos x = \tan x + \sec x$

$2\cos^2 x = 1 + \sin x$

$2(1-\sin x)(1+\sin x) = 1 + \sin x$

Either $1 + \sin x = 0$ or $\sin x = \dfrac{1}{2}$

$\sin x = -1$

$$\tan x = \frac{1}{\sqrt{3}}$$

$$d = \sec x - \cos x = \frac{1-\cos^2 x}{\cos x} = \frac{1}{2\sqrt{3}}$$

$$t_n = a + (n-1)d = \frac{1}{\sqrt{3}} + (n-1)\frac{1}{2\sqrt{3}} = \sqrt{3} = \cot x$$

$$\Rightarrow 1 + \frac{n-1}{2} = 3$$

$$\Rightarrow n = 5$$

53. $\log_6(abc) = 6$

$\Rightarrow (abc) = 6^6$

let $a = \dfrac{b}{r}$ and $c = br$

$$\Rightarrow b = 36 \text{ and } a = \frac{36}{r}$$

$\Rightarrow r = 2, 3, 4, 6, 9, 12, 18$ also, $36\left(1-\dfrac{1}{r}\right)$ is a perfect cube.

$\Rightarrow 36\left(1-\dfrac{1}{r}\right)$ is a perfect cube for $r = 4 \Rightarrow a + b + c = 36 + 9 + 144 = 189.$

54. Let number of children of John and Mary is y and number of children of John and his first wife is x
Hence, number of children of Mary and his first husband is $(x + 1)$
$\Rightarrow x + x + 1 + y = 24$
$y = 23 - 2x$
Total number of fights between two children is ${}^{24}C_2 = 276$
Now total number of fights between two children of same parents
$= {}^{x+1}C_2 + {}^{x}C_2 + {}^{y}C_2 = ({}^{x}C_1 + {}^{x}C_2) + {}^{x}C_2 + {}^{y}C_2$
$= ({}^{x}C_1 + {}^{x}C_2) + {}^{x}C_2 + {}^{23-2x}C_2 = 3x^2 - 45x + 253$
Hence, the total number of fights subject to the condition that any two children of same parents do not fight
$N = 276 - (3x^2 - 45x + 253) = 23 - 3x^2 + 45x$
$\frac{dN}{dx} = -6x + 45$

For max or min of N, $\frac{dN}{dx} = 0$
$\Rightarrow x = 7.5$
$\Rightarrow x = 7$ (since $x \in I$)
$\frac{d^2N}{dx^2} < 0$ at $x = 7$
$\therefore$ N will be maximum when $x = 7$
$N = 23 - 3.7^2 + 45.7$
$= 23 - 147 + 315 = 191.$

55. Let the six persons P_1, P_2, P_3, P_4, P_5 and P_6 be seated.
$\underset{x_1}{_}\ P_1\ \underset{x_2}{_}\ P_2\ \underset{x_3}{_}\ P_3\ \underset{x_4}{_}\ P_4\ \underset{x_5}{_}\ P_5\ \underset{x_6}{_}\ P_6\ \underset{x_7}{_}$
x_1 is the number of seats to left of P_1, x_2 is the number of seats in between P_1 and P_2, ..., x_7 is the number of seats to right of P_6 such that $x_1 + x_2 + x_3 + ... + x_7 = 9$.

i.e., coefficient of x^9 in $\left(1 + x + x^2 + ...\right)^2 \left(x + x^2 + x^3 + ...\right)^5 = {}^{15-6+1}C_6 = {}^{10}C_6$

$\because$ 6 persons can be arranged in 6! ways
$\therefore$ ${}^{10}P_6$ is the answer

56. $$\text{Number of ways} = \frac{\text{Coefficient of } x^{6n} \text{ in } \left(1 + x + x^2 + ... + x^{6n-1}\right)^3 - R}{3!}$$
where R = number of cases to be rejected $= 3(3n - 1) + 1$
$$\Rightarrow \text{Required number of ways} = \frac{{}^{6n+2}C_2 - 3(3n-1) - 1}{3!}$$

57. Let us suppose that the events
A = The rose bush has been watered.
A′ = The rose bush has not been watered.
B = The rose bush has been withered. From the question it is given that
$$P(A') = \frac{2}{3}\ P\left(\frac{B}{A}\right) = \frac{1}{2},\ P\left(\frac{B}{A'}\right) = \frac{3}{4}\ P(A) = \frac{1}{3}$$
From Baye's theorem,
$$P\left(\frac{A'}{B}\right) = \frac{P(A'B)}{P(A'B) + P(AB)} = \frac{P(A')P(B, A')}{P(A')P(B, A') + P(A)P(B, A)} = \frac{\frac{2}{3} \times \frac{3}{4}}{\frac{2}{3} \times \frac{3}{4} + \frac{1}{3} \times \frac{1}{2}} = \frac{3}{4}$$

58. Let A be the event that target is hit
E_1, E_2 be the events to select target P and Q respectively

Then $P(E_1)=\frac{9}{10}$, $P(E_2)=\frac{1}{10}$

$P\left(\frac{A}{E_1}\right)=1-\left(\frac{1}{3}\right)^x$, $P\left(\frac{A}{E_2}\right)=1-\left(\frac{1}{3}\right)^{20-x}$ where x number of shells used to select target P

Then $P(A)=P(E_1)\cdot P\left(\frac{A}{E_1}\right)+P(E_2)\cdot P\left(\frac{A}{E_2}\right)=\frac{9}{10}\left(1-\left(\frac{1}{3}\right)^x\right)+\frac{1}{10}\left(1-\left(\frac{1}{3}\right)^{20-x}\right)$

$\frac{dP(A)}{dx}=\frac{9}{10}\times-\left(\frac{1}{3}\right)^x\times-\log_e 3+\frac{1}{10}\times\left(\frac{1}{3}\right)^{20-x}\times1\times-\log_e 3=0$

$\frac{\log_e 3}{10}(9.3^{-x}-3^{x-20})=0$

$3^{2-x}=3^{x-20}$, $x=11$

59. HH ..... m heads $\rightarrow$ Let it is one unit say X
$\rightarrow$ X ..... rest n places
On rest n places either H or T may come for which probability for each place is 1

$\Rightarrow$ Probability for above case $=\frac{1}{2^m}\cdot1\cdot1\cdot1$ (n times)

$=\frac{1}{2^m}$ ..... (1)

$\Rightarrow$ Now further to avoid repetition TX comes together and rest are n – 1 places

$\Rightarrow$ Probability $={}^nC_1\cdot\frac{1}{2^{m+1}}\cdot(1)^{n-1}=\frac{n}{2^{m+1}}$ ..... (2)

From equation (1) and (2), we get $\frac{n+2}{2^{m+1}}$ is the required probability

60. $z^n-1=(z-1)(z-z_1)(z-z_2)\ldots..(z-z_{n-1})$
Differentiating w.r.t. x

$$\frac{nz^{n-1}}{z^n-1}=\frac{1}{z-1}+\frac{1}{z-2}+.....+\frac{1}{z-z_{n-1}}$$

61. $z_1: z+(\sqrt{3}+i)t-i=0$ and $\arg(z_1)=\frac{\pi}{4}\Rightarrow Re(z_1)=\frac{\sqrt{3}}{2}(\sqrt{3}+1)$

and $z_2=z+\left(\frac{\sqrt{3}+i}{\sqrt{3}}\right)\lambda-i=0\Rightarrow Re(z_2)=-\frac{\sqrt{3}+1}{2}$

Hence area of $\Delta ABC=\frac{1}{2}\times1\times\left(\frac{\sqrt{3}+1}{2}\right)(\sqrt{3}+1)=\frac{\sqrt{3}+2}{2}$

62. We have $|z_1+z_2|^2+|z_1-z_2|^2=2|z_1|^2+2|z_2|^2$

Let $z_1=\alpha+\sqrt{\alpha^2-\beta^2}$

$z_2=\alpha-\sqrt{\alpha^2-\beta^2}$

LHS $|z_1|+|z_2|$

Now $(|z_1|+|z_2|)^2=\left(|z_1|^2+|z_2|^2\right]+2|z_1||z_2|$

$$\Rightarrow \frac{1}{2}\left[|z_1+z_2|^2+|z_1+z_2|^2\right]+2|z_1z_2|$$

$$\Rightarrow \frac{1}{2}\left[4|\alpha|^2+4|\alpha|^2+4|\alpha^2-\beta^2|\right]$$

$$\Rightarrow 2|\alpha|^2+2|\alpha^2-\beta^2|+2|\beta|^2$$

$$2|\alpha|^2+2|\beta^2|+2|\alpha+\beta||\alpha-\beta|$$

$$|\alpha+\beta|^2+|\alpha-\beta|^2+2|\alpha-\beta||\alpha-\beta|$$

$$(|\alpha+\beta|+|\alpha-\beta|)^2$$

Hence $|z_1|+|z_2|=|\alpha+\beta|+|\alpha-\beta|$

63. From Demoivre's Theorem, we know that

$\sin(2n+1)\alpha = {}^{2n+1}C_1(\cos\alpha)^{2n}\sin\alpha - {}^{2n+1}C_3(\cos\alpha)^{2n-2}\sin^3\alpha + \ldots + (-1)^n\sin^{2n-1}\alpha$

$\Rightarrow \sin(2n+1)\alpha = \sin^{2n+1}\alpha\left[{}^{2n+1}C_1\cot^{2n}\alpha - {}^{2n+1}C_3\cot^{2n-2}\alpha + {}^{2n+1}C_5\cot^{2n-4}\alpha - \ldots\right]$

It follows that for $\alpha = \frac{\pi}{2n+1}, \frac{2\pi}{2n+1}, \ldots, \frac{n\pi}{2n+1}$

$\Rightarrow {}^{2n+1}C_1\cot^{2n}\alpha - {}^{2n+1}C_3\cot^{2n-2}\alpha + {}^{2n+1}C_5\cot^{2n-4}\alpha - \ldots = 0$

$\Rightarrow \cot^2\frac{\pi}{2n+1}, \cot^2\frac{2\pi}{2n+1}, \ldots, \cot^2\frac{n\pi}{2n+1}$ are the roots of the equation

${}^{2n+1}C_1x^n - {}^{2n+1}C_3x^{n-1} + \ldots = 0$

If $0<\theta<\frac{\pi}{2}$

$\sin\theta<\theta<\tan\theta$

$\Rightarrow \cot^2\theta < \frac{1}{\theta^2} < \operatorname{cosec}^2\theta$

Now, $\cot^2\frac{\pi}{2n+1}+\cot^2\frac{2\pi}{2n+1}+\ldots\ldots+\cot^2\frac{n\pi}{2n+1} < \left(\frac{2n+1}{\pi}\right)^2+\left(\frac{2n+1}{2\pi}\right)^2+\left(\frac{2n+1}{3\pi}\right)^2+\ldots+\left(\frac{2n+1}{n\pi}\right)^2$

$< \operatorname{cosec}^2\frac{\pi}{2n+1}+\operatorname{cosec}^2\frac{2\pi}{2n+1}+\ldots\ldots+\operatorname{cosec}^2\frac{n\pi}{2n+1}$

$\Rightarrow \frac{n(2n-1)}{3} < \frac{(2n+1)^2}{\pi^2}\left(1+\frac{1}{2^2}+\frac{1}{3^2}+\ldots+\frac{1}{n^2}\right) < \frac{2n(n+1)}{3}$

$\left(\text{since } \cot^2\frac{\pi}{2n+1}, \cot^2\frac{2\pi}{2n+1}, \ldots\ldots, \cot^2\frac{n\pi}{2n+1} \text{ are the roots of given equation}\right)$

$\Rightarrow \left(1-\frac{1}{2n+1}\right)\left(1-\frac{2}{2n+1}\right)\frac{\pi^2}{6} < \left(1+\frac{1}{2^2}+\frac{1}{3^2}+\ldots\ldots+\frac{1}{n^2}\right) < \left(1-\frac{1}{2n+1}\right)\left(1+\frac{1}{2n+1}\right)\frac{\pi^2}{6}$

Taking limit $n\to\infty$ and applying Sandwitch theorem, we get the desired result.

64. Let $\alpha = a + ib$ and $\gamma = c + id$ where $a, b, c, d \in R$

We have to minimize $\sqrt{a^2+b^2}+\sqrt{c^2+d^2}$

Now $f(z) = (4+i)z^2 + z(a+ib) + (c+id)$

$f(1) = 4 + i + a + ib + c + id$ is real

Or $(4 + a + c)\, i(1 + b + d)$ is real

Hence $b + d + 1 = 0$ ..... (1)

$f(i) = -(4+i) + i(a+ib) + (c+id)$ is real

$f(i) = -4 - b + c + i(a + d - 1)$ is real

$a + d = 1$..... (2)

From equation (1) and (2), we get $a - b = 2$ ..... (3)

Hence there is no restriction on 'c'. Let $c = 0$

Hence $|\alpha|+|\gamma| = \sqrt{a^2+b^2}+\sqrt{d^2} = \sqrt{4+2ab}+|d| \geq \sqrt{4+2ab} \geq \sqrt{2}$

With equality if $d = 0$; $a = 1$ and $b = -1$

$\therefore$ The minimum value of $|\alpha|+|\gamma| = \sqrt{2}$

65. Given $A^{-1} = \dfrac{A^T}{f(A)}$

$\Rightarrow A^T = f(A)A^{-1}$

$f(A) = |A^T| = f(A)^n |A^{-1}|$ ..... (1)

But $\left|A^{-1}\right| = \dfrac{1}{|A|}$

From (1) $f(A) = f(A)^n \dfrac{1}{|A|}$

$\Rightarrow |A| = f(A)^{n-1}$

$|\text{adj } A| = |A|^{n-1} = \left[f(A)^{n-1}\right]^{n-1} = f(A)^{(n-1)^2}$ ..... (2)

$|\text{adj A adj A}| = |A|^{(n-1)^2}$ ..... (3)

But from definition $|A| = |A^T| = f(A)$

Hence, from (2) and (3) A = adj A

$|A| = f(A)^{n-1} = |A^T| = f(A)$

$\Rightarrow n - 1 = 1$

$\Rightarrow n = 2$

66. $9A^{-1} = 4A^T \Rightarrow \left|9A^{-1}\right| = \left|4A^T\right| \Rightarrow 9^5 \times \dfrac{1}{|A|} = 4^5|A| \Rightarrow |A| = \left(\dfrac{3}{2}\right)^5$

$|\text{adj adj adj } A| = |A|^{4^3} = \left(\dfrac{3}{2}\right)^{320}$

No. of digits in $|\text{adj adj adj } A| = \left[\log\left(\dfrac{3}{2}\right)^{320}\right]+1 = 56$

(Where [.] denotes greatest integer function).

67. $\begin{vmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ [x] & [y] & [z]+1 \end{vmatrix}$ $R_1 \to R_1 - R_3, R_2 \to R_2 - R_3$ on solving [x] + [y] + [z] + 1, taking maximum value, we get 4

68. $\Delta = \begin{vmatrix} 1 & 2a & a^2 \\ 1 & 2b & b^2 \\ 1 & 2c & c^2 \end{vmatrix} \cdot \begin{vmatrix} 1 & p & p^2 \\ 1 & q & q^2 \\ 1 & r & r^2 \end{vmatrix}$

$= 2 \cdot 2\Delta_1 \cdot 2\Delta_2 = 8\Delta_1\Delta_2 = 8 \cdot \dfrac{1}{2} \cdot 4 = 16$

69. Since $r \cdot {}^nC_r = n \cdot {}^{n-1}C_{r-1} \Rightarrow {}^nC_r = \frac{n}{r} \cdot {}^{n-1}C_{r-1}$

Now, $\Delta(m, p) = \begin{vmatrix} \frac{m}{p} \cdot {}^{m-1}C_{p-1} & \frac{m}{p+1} \cdot {}^{m-1}C_p & \frac{m}{p+2} \cdot {}^{m-1}C_{p+1} \\ \frac{m+1}{p} \cdot {}^{m}C_{p-1} & \frac{m+1}{p+1} \cdot {}^{m}C_p & \frac{m+1}{p+2} \cdot {}^{m}C_{p+1} \\ \frac{m+2}{p} \cdot {}^{m+1}C_{p-1} & \frac{m+2}{p+1} \cdot {}^{m+1}C_p & \frac{m+2}{p+2} \cdot {}^{m+1}C_{p+1} \end{vmatrix}$

Taking m common from R_1, $(m+1)$ from R_2, $(m+2)$ from R_3, $\frac{1}{p}$ from C_1, $\frac{1}{p+1}$ from C_2 and $\frac{1}{p+2}$ from C_3, we get, $\Delta(m, p) = \frac{m(m+1)(m+2)}{p(p+1)(p+2)} \begin{vmatrix} {}^{m-1}C_{p-1} & {}^{m-1}C_p & {}^{m-1}C_{p+1} \\ {}^{m}C_{p-1} & {}^{m}C_p & {}^{m}C_{p+1} \\ {}^{m+1}C_{p-1} & {}^{m+1}C_p & {}^{m+1}C_{p+1} \end{vmatrix}$

$= \frac{{}^{m+2}C_3}{{}^{p+2}C_3} \Delta(m-1, p-1) = \frac{{}^{m+2}C_3}{{}^{p+2}C_3} \cdot \frac{{}^{m+1}C_3}{{}^{p+1}C_3} \Delta(m-2, p-2)$

$= \frac{{}^{m+2}C_3}{{}^{p+2}C_3} \cdot \frac{{}^{m+1}C_3}{{}^{p+1}C_3} \cdot \frac{{}^{m}C_3}{{}^{p}C_3} \Delta(m-3, p-3)$

$= \frac{{}^{m+2}C_3}{{}^{p+2}C_3} \cdot \frac{{}^{m+1}C_3}{{}^{p+1}C_3} \cdots\cdots\cdots \frac{{}^{m-p+3}C_3}{{}^{3}C_3} \Delta(m-p, 0)$ ...(1)

now, $\Delta(m-p, 0) = \begin{vmatrix} {}^{m-p}C_0 & {}^{m-p}C_1 & {}^{m-p}C_2 \\ {}^{m-p+1}C_0 & {}^{m-p+1}C_1 & {}^{m-p+1}C_2 \\ {}^{m-p+2}C_0 & {}^{m-p+2}C_1 & {}^{m-p+2}C_2 \end{vmatrix}$

$= \begin{vmatrix} 1 & m-p & \frac{(m-p)(m-p-1)}{2} \\ 1 & m-p+1 & \frac{(m-p+1)(m-p)}{2} \\ 1 & m-p+2 & \frac{(m-p+2)(m-p+1)}{2} \end{vmatrix}$

$= \begin{vmatrix} 1 & m-p & \frac{(m-p)(m-p-1)}{2} \\ 0 & 1 & m-p \\ 0 & 1 & m-p+1 \end{vmatrix}$ (applying $R_3 \to R_3 - R_1$ & $R_2 \to R_2 - R_1$)

$= 1$

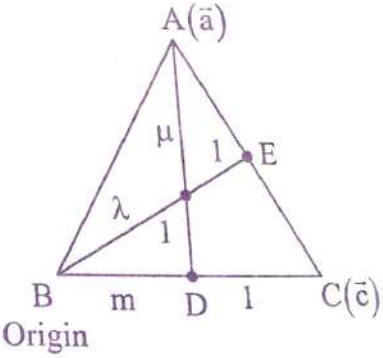
Hence from (1), $\Delta(m, p) = \frac{{}^{m+2}C_3 \cdot {}^{m+1}C_3 \cdots\cdots\cdots {}^{m-p+3}C_3}{{}^{p+2}C_3 \cdot {}^{p+1}C_3 \cdots\cdots\cdots {}^{3}C_3}$.

70. Let vertices B, C, A are origin $(\vec{O})$, $\vec{c}$ and $\vec{a}$, then

$\overrightarrow{OD} = \frac{m\vec{c}}{m+1}$, $\overrightarrow{DE} = \frac{n\vec{c} + \vec{a}}{n+1}$

Let $\frac{BP}{PE} = \lambda$ and $\frac{AP}{PD} = \mu$

P.V. of P is $= \lambda \frac{\overrightarrow{OE} + 1.\vec{O}}{\lambda + 1} = \frac{\lambda}{\lambda + 1} \overrightarrow{OE}$



Also P.V. of P is $= \dfrac{\mu\overrightarrow{OD}+\overrightarrow{OA}}{\mu+1}$

Thus $\dfrac{\lambda}{\lambda+1}\overrightarrow{OE} = \mu\dfrac{\overrightarrow{OD}+\overrightarrow{OA}}{\mu+1}$

$\dfrac{\lambda}{\lambda+1}\dfrac{n\vec{c}+\vec{a}}{n+1} = \dfrac{\mu}{\mu+1}\dfrac{m}{m+1}\vec{c}+\dfrac{1}{\mu+1}\vec{a}$

Compare coefficient of $\vec{c}$ and $\vec{a}$

$\dfrac{\lambda}{(\lambda+1)(n+1)} = \dfrac{1}{\mu+1}, \dfrac{\lambda}{(\lambda+1)}\dfrac{n}{(n+1)}\dfrac{m+1}{m} = \dfrac{\mu}{\mu+1}$

Adding above 2 equation $\dfrac{\lambda}{(\lambda+1)(n+1)}\left(\dfrac{n(m+1)}{m}+1\right)=1$

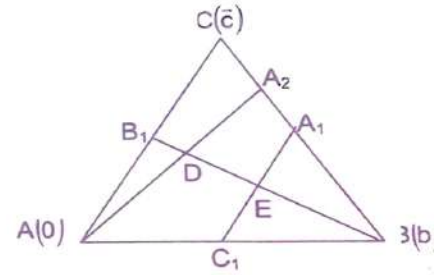
$\dfrac{\lambda}{\lambda+1} = \dfrac{n+1}{\dfrac{n(m+1)}{m}+1}$ ..... (1)

$\dfrac{1}{\lambda+1} = 1-\dfrac{\lambda}{\lambda+1} = 1-\dfrac{n+1}{\dfrac{n(m+1)}{m}+1}$ ..... (2)

Dividing equation (1) and (2), we get $\dfrac{\lambda}{1} = \dfrac{(n+1)m}{n}$

Hence, $\dfrac{BP}{PE} = \dfrac{(n+1)m}{n}$

71. Let the position vectors of A, B, and C be $\vec{0}$, $\vec{b}$ and $\vec{c}$ respectively. We have ,



$\overrightarrow{AC_1} = \dfrac{\vec{b}}{2}$, $\overrightarrow{AB_1} = \dfrac{\vec{c}}{2}$, $\overrightarrow{AA_1} = \dfrac{\vec{b}+\vec{c}}{2}$, $\overrightarrow{AA_2} = \dfrac{3\vec{c}+\vec{b}}{4}$

Equation of lines BB_1, AA_2 and C_1A_1 are respectively $\vec{r} = \vec{b}+\lambda_1\left(\dfrac{\vec{c}}{2}-\vec{b}\right)$,

$\vec{r} = \lambda_2\dfrac{3\vec{c}+\vec{b}}{4}$ and $\vec{r} = \dfrac{\vec{b}}{2}+\lambda_3\left(\dfrac{\vec{c}}{2}\right)$

For point D, we have $\vec{b}+\lambda_1\left(\dfrac{\vec{c}}{2}-\vec{b}\right) = \lambda_2\left(\dfrac{3\vec{c}+\vec{b}}{4}\right)$

$\Rightarrow \vec{b}\left(1-\lambda_1-\dfrac{\lambda_2}{4}\right)+\dfrac{\vec{c}}{4}(2\lambda_1-3\lambda_2) = \vec{0} \Rightarrow \lambda_1 = \dfrac{6}{7}, \lambda_2 = \dfrac{4}{7} \Rightarrow \overrightarrow{AD} = \dfrac{3\vec{c}+\vec{b}}{7}$

For point E we have $\vec{b}+\lambda_1\left(\dfrac{\vec{c}}{2}-\vec{b}\right) = \dfrac{\vec{b}}{2}+\dfrac{\lambda_3}{2}\vec{c}$

$\Rightarrow \vec{b}\left(\dfrac{1}{2}-\lambda_1\right)+\dfrac{\vec{c}}{2}(\lambda_1-\lambda_3) = \vec{0} \Rightarrow \lambda_1 = \lambda_3 = \dfrac{1}{2} \Rightarrow \overrightarrow{AE} = \dfrac{2\vec{b}+\vec{c}}{4}$

Now, $\overrightarrow{EA_2} = \dfrac{3\vec{c}+\vec{b}-2\vec{b}-\vec{c}}{4} = \dfrac{2\vec{c}-\vec{b}}{4}$, $\overrightarrow{DA_1} = \dfrac{\vec{b}+\vec{c}}{2}-\dfrac{3\vec{c}+\vec{b}}{7} = \dfrac{5\vec{b}+\vec{c}}{14}$

Area of the quadrilateral $EA_1A_2D = \dfrac{1}{2}\left|\overrightarrow{EA_2}\times\overrightarrow{DA_1}\right|$

$$= \frac{1}{112}\left|(2\vec{c}-\vec{b})\times(5\vec{b}+\vec{c})\right| = \frac{1}{112}\left|10\vec{c}\times\vec{b}-\vec{b}\times\vec{c}\right|$$

$$= \frac{11}{112}\left|\vec{c}\times\vec{b}\right| = \frac{11}{56}\cdot\frac{1}{2}\left|\vec{c}\times\vec{b}\right| = \frac{11}{56}\text{ area of }\Delta ABC.$$

Thus required ratio is $\frac{11}{56}$

72. $(z)^{2002} = \bar{z}$

$|z|^{2002} = |\bar{z}| = |z| \Rightarrow |z|\,[|z|^{2001} - 1] = 0$

$|z| = 0 \Rightarrow (0, 0)$

$|z|^{2001} = 1 \Rightarrow |z| = 1$

$z^{2002} = \bar{z} \Rightarrow z^{2003} = z\bar{z} = 1$

$\Rightarrow z^{2003} = 1 \Rightarrow$ 2003 distinct solution

So, total (x, y) pairs are 2004

73. Given $\alpha < \beta < \gamma < \delta$ also $\sin\alpha = \sin\beta = \sin\gamma = \sin\delta = k$

and $\alpha, \beta, \gamma, \delta$ are smallest positive angles

$\therefore \beta = \pi - \alpha, \gamma = 2\pi + \alpha, \delta = 3\pi - \alpha$

as $\sin\beta = \sin\alpha$ and $\beta > \alpha$

$\sin\beta = \sin\gamma$ and $\gamma > \beta$

$\sin\gamma = \sin\delta$ and $\delta > \gamma$

Putting these values in the given expansion, we have given expression

$$= 2\left(\sin\frac{\alpha}{2} + \cos\frac{\alpha}{2}\right) = 2\sqrt{1+\sin\alpha} = 2\sqrt{1+k}$$

74. $f(x) = \tan^{-1}(\tan x)$

$s = 1 + 2 - \pi + 3 - \pi + 4 - \pi + 5 - 2\pi = 15 - 5\pi$

$a = 15, b = 5$

75. $$\alpha+\beta+\gamma = \tan^{-1}\left(\frac{36}{77}\right) + \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{8}{15}\right) = \frac{\pi}{2}$$

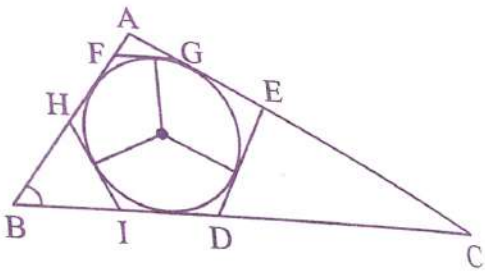
$$\Rightarrow \sum\cot\alpha = \prod\cot\alpha \text{ and hence } A = \frac{\sum\cot\alpha}{\prod\cot\alpha} = 1$$

and $\sum\tan\alpha\tan\beta = 1 \Rightarrow A + B = 2$

76. $FG = p$

$DE = q$

$HI = r$



In ΔAFG $\frac{P}{\sin A} = \frac{AG}{\sin B} = \frac{AF}{\sin C}$

$\Rightarrow AG = \frac{bp}{a}, AF = \frac{cp}{a}$

Now, incircle of ΔABC is the ex–circle opposite to A for ΔAFG the semi perimeter of ΔAFG is

$$S_{\Delta AFG} = \frac{1}{2}\left(p + \frac{bp}{a} + \frac{cp}{a}\right) = \frac{Sp}{a}$$

$$\Rightarrow r = S_{\Delta AFG}\tan\frac{A}{2} = \frac{Sp}{a}\tan\frac{A}{2}$$

Since $r = (s-a)\tan\frac{A}{2}$

$\Rightarrow s-a=\frac{sp}{a} \Rightarrow \frac{p}{a}=1-\frac{a}{s}$

Similarly $\frac{q}{b}=1-\frac{b}{s} \Rightarrow \frac{r}{c}=1-\frac{c}{s}$

77. Let $\pi 3^x=\theta$ then:

$\Rightarrow \cos\theta-2\cos^2\theta+2\cos4\theta-\cos7\theta=\sin\theta+2\sin^2\theta-2\sin4\theta+2\sin3\theta-\sin7\theta$

$\Rightarrow \cos\theta-\cos7\theta+\sin7\theta-\sin\theta+2(\cos4\theta+\sin4\theta)=2+2\sin3\theta$

$\Rightarrow 2\sin4\theta.\sin3\theta+2\cos4\theta.\sin3\theta+2(\cos4\theta+\sin\theta)=2+2\sin3\theta$

$\Rightarrow \sin4\theta(\sin3\theta+1)+\cos4\theta(\sin3\theta+1)-(1+\sin3\theta)=0$

$\Rightarrow (\sin3\theta+1)(\sin4\theta+\cos4\theta-1)=0$

$\Rightarrow \sin3\theta=-1 \Rightarrow 3\theta=2n\pi-\frac{\pi}{2}$

$\Rightarrow 3\pi 3^x=2n\pi-\frac{\pi}{2}$

$\Rightarrow 3^{x+1}=2n-\frac{1}{2}=\frac{4n-1}{2}$

$\Rightarrow x+1=\log_3\frac{4n-1}{2}$

$\Rightarrow x=\log_3\frac{4n-1}{2}-1$

And $\sin4\theta+\cos4\theta=1$

$\Rightarrow \sin\left(4\theta+\frac{\pi}{4}\right)=\sin\frac{\pi}{4}$

$\Rightarrow 4\theta+\frac{\pi}{4}=n\pi+(-1)^n\frac{\pi}{4}$

$\Rightarrow 4\pi 3^x=n\pi+(-1)^n\frac{\pi}{4}-\frac{\pi}{4}$

$\Rightarrow 3^x=\frac{n}{4}+(-1)^n\frac{\pi}{16}-\frac{\pi}{16} \Rightarrow x=\log_3\left(\frac{n}{4}+(-1)^n\frac{\pi}{16}-\frac{\pi}{16}\right)$.

78. $\Delta ABC \sim \Delta A'B'C'$

In $\Delta AC'B'$: $\angle AB'C'=\theta+C$

$\frac{AC'}{\sin(\theta+C)}=\frac{B'C'}{\sin A}=\frac{\lambda BC}{\sin A}=2\lambda R$

In $\Delta BA'C'$: $\angle BA'C'=C-\theta$

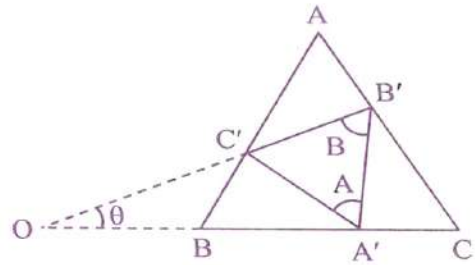
$\frac{BC'}{\sin(C-\theta)}=\frac{A'C'}{\sin B}=2\lambda R$

$AC'=2\lambda R\sin(\theta+C)$

$BC'=2\lambda R\sin(C-\theta)$

$C=AC'+BC'=2\lambda R\,2\sin C\cos\theta$

$\lambda\cos\theta=\frac{1}{2}$ (H.P.)



79. $R=1+\sin\theta,\ \theta=\frac{\pi}{8}$

$\Rightarrow c_1c_2=R+r$

$\therefore \sin\theta = \dfrac{R-r}{R+r}$

$R\sin\theta + r\sin\theta = R - r$

$R(1-\sin\theta) = r(1+\sin\theta)$ (Put $R = 1 + \sin\theta$)

$\Rightarrow (1+\sin\theta)(1-\sin\theta) = r(1+\sin\theta)$

$\Rightarrow r = 1 - \sin\theta$

$\Rightarrow Rr = 1 - \sin^2\theta = \cos^2\theta$

$\Rightarrow \cos 2\theta = \dfrac{1}{2}(1+\cos\theta) = \dfrac{1}{2}\left(1+\cos\dfrac{\pi}{4}\right) = \dfrac{2+\sqrt{2}}{4}$

$\Rightarrow a = 2, b = 4$

80. Let a, b and c be the lengths of the sides of the triangle ABC, where $a < b < c$, the centre of the inscribed circle and M the point of intersection of the medians. Since a, b and c form M arithmetic progression, we have $b - a = c - b$. or $a + c = 2b$. Therefore, the perimeter of the triangle ABC.

$2p = 3b$, or $p = \dfrac{3}{2}b$.

let us show that the points O and M are equidistant from the side AC.

We have $|00_1| = r = \dfrac{S}{p} = \dfrac{2S}{3b}$

Where r is the radius of the inscribed circle, S is the area of the triangle and P is its semi perimeter, we find

$|MM_1| = \dfrac{1}{3}h_b = \dfrac{1}{3}\dfrac{2S}{b} = \dfrac{2S}{3b}$

Consequently, $|OO_1| = |MM_1|$ and the straight line OM is parallel to the side AC.

2. $y = g(x)$ is inverse of $f(x)$

3. $h(x) = f(x) + g(x) = (x+2)e^{\ln(x+2)} + \frac{e^{\left(\frac{-1}{\log_x^e}\right)}}{x} - \frac{2x}{e^{-\ln 2}}$

$= (x+2)^2 + \frac{1}{x^2} - 2xe^{\ln 2} = x^2 + 4x + 4 + \frac{1}{x^2} - 4x$

$= x^2 + \frac{1}{x^2} + 4 \Rightarrow h(x) \geq 6 \; \forall \; x \in R^+ - \{1\}$

4. $3 - \sin 2x \geq 0 \Rightarrow x \in R$

$2 - \sqrt{3 - \sin 2x} \geq 0$

$\Rightarrow \sin 2x \geq -1$

$\Rightarrow x \in R$

$1 - \sqrt{2 - \sqrt{3 - \sin 2x}} \geq 0$

$\Rightarrow \sin 2x \leq 2 \Rightarrow x \in R$

5. By replacing $x = x + 1$ and $x = x - 2$, we get

$f(x+2) + f(x) = \sqrt{2} f(x+1)$ ..... (1)

$f(x) + f(x-2) = \sqrt{2} f(x-1)$ ..... (2)

From equation (1) + (2) gives $f(x+2) + f(x-2) + 2f(x)$

$= \sqrt{2}\left[f(x+1) + f(x-1)\right] - \sqrt{2}\sqrt{2} f(x)$

$\therefore f(x+2) + f(x-2) = 0$

On replacing x by $x + 2$, we get $f(x+4) + f(x) = 0$

Finally replacing x by $x + 4$, we get

$f(x+8) + f(x+4) = 0$

$f(x+8) = -f(x+4) = f(x) \; \forall \; x$

$\therefore f(x)$ is periodic with period 8

6. $f(x) = \frac{x}{3}, x < 0 \Rightarrow f^{-1}(x) = 3x, x < 0$

$x, x \geq 0 \qquad x \; x \geq 0$

7. Consider a general triangle

$(AP_{n+1})^2 - (AP_n)^2 = (P_nP_{n+1})^2 = (2^{n-1})^2$

Put $n = 1, 2, 3, ..., n$

$(AP_2)^2 - (A)^2 = (P_1P_2)^2 = 1$

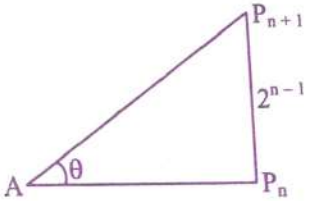
$(AP_3)^2 - (AP_2)^2 = (P_2P_3)^2 = 2^2$

$(AP_4)^2 - (AP_3)^2 = (P_3P_4)^2 = (2^2)^2$

$\vdots \qquad \vdots \qquad \vdots \qquad \vdots$

$\underline{(AP_{n+1})^2 - (AP_n)^2 = (P_nP_{n+1})^2 = \left(2^{n-1}\right)^2}$

$(AP_{n+1})^2 - \underbrace{(AP_1)^2}_{=1} = \underbrace{1 + 2^2 + \left(2^2\right)^2 + \left(2^3\right)^2 + ... + \left(2^{n-1}\right)^2}_{\text{G.P. with common ratio } 2^2} = \frac{2^{2n} - 1}{2^2 - 1} = \frac{4^n - 1}{3}$

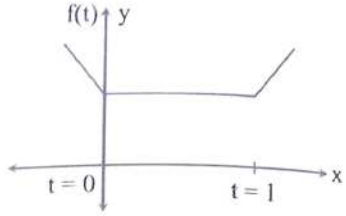


$$\therefore (AP_{n+1})^2 = \frac{4^n - 1}{3} + 1 = \frac{4^n + 2}{3} \Rightarrow (AP_{n+1}) = \sqrt{\frac{4^n + 2}{3}}$$

$$\text{Now, } \sin\theta = \frac{|P_n P_{n+1}|}{|AP_{n+1}|} = \frac{2^{n-1}}{\sqrt{\frac{4^n + 2}{3}}} = \frac{2^n}{\sqrt{4^n + 2}} \frac{\sqrt{3}}{2}$$

$$\text{As } n \to \infty, \frac{2^n}{\sqrt{4^n + 2}} = 1 \Rightarrow \sin\theta = \frac{\sqrt{3}}{2} \text{ as } n \to \infty \Rightarrow \theta = \frac{\pi}{3}$$

8. Given $f(t) = \begin{cases} 1 - 2t & , \ t < 0 \\ 1 & , \ 0 \le t \le 1 \\ 2t - 1 & , \ t > 1 \end{cases}$



$$\Rightarrow g(x) = \begin{cases} f(x-1) & , \ 0 \le x \le 1 \\ 3 - x & , \ 1 < x \le 2 \end{cases}$$

$$\Rightarrow g(x) = \begin{cases} 3 - 2x & , \ 0 \le x \le 1 \\ 3 - x & , \ 1 < x \le 2 \end{cases}$$

The function g(x) is discontinuous at x = 1, as $\lim_{x \to 1} g(x)$ does not exist

So, g(x) is non-derivable at x = 1

9. Clearly α is repeated root of f(x)=0

$$\lim_{x \to \alpha} \frac{f(x)}{f'(x)} \left(\frac{f'(x)}{f(x)} - \left\{ \frac{f'(x)}{f(x)} \right\} \right) \Rightarrow \lim_{x \to \alpha} \left(1 - \frac{f(x)}{f'(x)} \left\{ \frac{f'(x)}{f(x)} \right\} \right) = 1 - 0 = 1$$

$$\left(\lim_{x \to \alpha} \frac{f(x)}{f'(x)} = 0 \ \& \ \left\{ \frac{f'(x)}{f(x)} \right\} \text{is bounded function} \right)$$

10. Let $f(x) = t \Rightarrow f(t) = 2$ so $t = -3, \frac{1}{2}, 3$

Now, $f(x) = -3 \to$ no solution

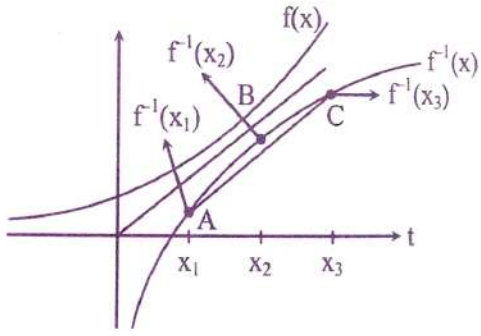
$f(x) = \frac{1}{2} \to 2$ solutions

$f(x) = 3 \to 1$ solution

11. Maximum occurs when figure is cut along diagonal and minimum when we cut it parallel to smaller side

12. On simplifying, $[x^2]\cos^2 4x + \{x^2\}\cos^2 4x + x^2 \sin 4x = x^2 \cos^2 4x + x^2 \sin^2 4x$
$= x^2$, which is always differentiable

13. For f(x) monotonically increasing $f'(x) > 0$ and given $f''(x) > 0$ and $f^{-1}(x)$ exist, then it is clearly visible from graph that average value of $f^{-1}(x_1)$, $f^{-1}(x_2)$ inside the ΔABC as like centroid, but $f^{-1}\frac{(x_1 + x_2 + x_3)}{3}$ will lie on the curve.



14. $x-1>0$ and $4x-5>0, x-1\neq 1$

$\Rightarrow x\in\left(\frac{5}{4},\infty\right)-\{2\}$

Case-1: $x-1>1\Rightarrow x>2\Rightarrow 4x-5\geq(x-1)^2$

$\Rightarrow x\in\left(3-\sqrt{3},3+\sqrt{3}\right)$

Thus $x\in\left(2,3+\sqrt{3}\right)$

Case-2: $1<x<2\Rightarrow 4x-5\leq(x-1)^2$

$\Rightarrow x\in\left(-\infty,3-\sqrt{3}\right)\cup\left(3+\sqrt{3},\infty\right)$

$\Rightarrow x\in\left(\frac{5}{4},3-\sqrt{3}\right)$

15. $y^2-xy=\left(y-\frac{x}{2}\right)^2-\frac{x^2}{4}\Rightarrow y^2-xy$ is decreasing for $y\leq\frac{x}{2}$ and increasing for $y\geq\frac{x}{2}$.

Thus the largest value of y^2-xy must be at $y=0,\ \frac{x}{2}$ or 1.

The values are $0,\ \frac{x^2}{4},\ |1-x|\Rightarrow$ for $x\notin(0,1)\ g(x)\geq\frac{1}{4}$

and for $x\in(0,1)\ g(x)=\max\left(\frac{x^2}{4},\ 1-x\right)$

also, $x^2+4x-4=0\Rightarrow x=-2\pm\sqrt{8}\Rightarrow g(x)=1-x$ for $x\leq\sqrt{8}-2$ and $\frac{x^2}{4}$ for $x\geq\sqrt{8}-2$

$\Rightarrow g(x)$ is minimum at $x=\sqrt{8}-2$ and minimum value is $3-\sqrt{8}$

16. We have $r^2=\frac{c^4}{a^2\cos ec^2\theta+b^2\sec^2\theta}=\frac{c^4}{(a\cot\theta-b\tan\theta)^2+(a+b)^2}$

$\Rightarrow$ maximum value of $r^2=\frac{c^4}{(a+b)^2}$ {when denominator is minimized}

$\Rightarrow$ maximum value of $r=\frac{c^2}{(a+b)}$

17. Given equation can be written as $f(x)=\frac{x}{e^x}=k$

$f'(x)=\frac{(1-x)}{e^x}$

$\Rightarrow$ Maximum $f(x)=\frac{1}{e}$

18. Obviously 'a' can be equal to 0
When $a<0$

Domain of f(x) is $\left[0,-\frac{b}{a}\right]\Rightarrow$ Range is $\left[0,\sqrt{\frac{-b^2}{4a}}\right]\Rightarrow-\frac{b}{a}=\sqrt{\frac{-b^2}{-4a}}$

$\Rightarrow a=-4$
$a>0$ is not possible

19. $g(x)=\int 1-\frac{f'(x)+f''(x)+f'''(x)}{f(x)+f'(x)+f''(x)+f'''(x)}dx$

$=x-\ln|f(x)+f'(x)+f''(x)+f'''(x)|+C$

$=x-3\ln|x|+C$

20. $I=\int(x^4)^6(x^3)(6x^2+5x+4)(x^2+x+1)^6\,dx$

$I=\int(6x^5+5x^4+4x^3)(x^6+x^5+x^4)^6\,dx$

Put $x^6+x^5+x^4=t$

$(6x^5+5x^4+4x^3)\,dx=dt$

$I=\int t^6dt$

$I=\frac{t^7}{7}+c=\frac{(x^6+x^5+x^4)^7}{7}+c$

21. Put $x^2=t$

$I=\frac{1}{2}\int e^t\left[te^{t^2}+(e^{t^2}+2t^2e^{t^2})\right]dt$

22. $\int\frac{\sin^3x+\sin^5x}{\cos^2x+\cos^4x}dx=\int\frac{(\sin^2x+\sin^4x)}{(\cos^2x+\cos^4x)}\sin x\,dx=-\int\frac{(1-t^2)+(1-t^2)^2}{t^2+t^4}dt$ (Taking $\cos x=t$)

$=-\int\frac{1-t^2+1+t^4-2t^2}{t^2(t^2+1)}dt=-\int\frac{t^4-3t^2+2}{t^2(t^2+1)}dt$ ..... (1)

Let $t^2=y\Rightarrow\frac{y^2-3y+2}{y(y+1)}=\left(1+\frac{2}{y}-\frac{6}{y+1}\right)=\left(1+\frac{2}{t^2}-\frac{6}{(t^2+1)}\right)$ (Using partial fraction)

$\therefore\ I=-\int 1\,dt-2\int\frac{dt}{t^2}+6\int\frac{dt}{(t^2+1)}$

$=-t-2\cdot\frac{-1}{t}+6\tan^{-1}(t)+c$

$=-\cos x+\frac{2}{\cos x}+6\tan^{-1}(\cos x)+c$

23. $I=\int x^2\frac{(x\sec^2x+\tan x)}{(x\tan x+1)^2}dx$

$I=x^2\left(\frac{-1}{(x\tan x+1)}\right)-\int 2x\cdot\left(\frac{-1}{x\tan x+1}\right)dx$

$I=\frac{-x^2}{x\tan x+1}+\int\frac{2x\cos x}{x\sin x+\cos x}dx$

$I=\frac{-x^2}{x\tan x+1}+2\ln|x\sin x+\cos x|+C$.

24. $I = \int \frac{\left(\frac{1-\ln x}{x^2}\right)}{\left(\frac{\ln x}{x}\right)^4 - 1} dx$

Let $t = \frac{\ln x}{x} \Rightarrow dt = \frac{1-\ln x}{x^2} dx$

25. $\int \frac{2x^7 + 3x^2}{x^{10} - 2x^5 + 1} dx = \int \frac{x^6\left(2x + \frac{3}{x^4}\right)}{x^6\left(x^4 - \frac{2}{x} + \frac{1}{x^6}\right)} dx = \int \frac{2x + \frac{3}{x^4}}{\left(x^2 - \frac{1}{x^3}\right)^2} dx$

Let $x^2 - \frac{1}{x^3} = t \Rightarrow \left(2x + \frac{3}{x^4}\right) dx = dt$

$= \int \frac{dt}{t^2} = -\frac{1}{t} + c = -\frac{x^3}{x^5 - 1} + c = -\frac{x^3(x^5-1)}{(x^5-1)^2} + c = \frac{x^3 - x^8}{(x^5-1)^2} + c$

26. $\int \frac{e^x + e^{-x} - 1}{\left(e^x \sin x + \cos x\right)\left(\cos x - \sin x \cdot e^{-x}\right)} - dx$

$I = \int \frac{e^{2x} - e^x + 1}{\left(e^x \sin x + \cos x\right)\left(e^x \cos x - \sin x\right)} dx$

$= \int \frac{\left(e^x \cos x - \sin x\right)\left(e^x(\sin x + \cos x) - \sin x\right) - \left(e^x \sin x + \cos x\right)\left[e^x(\cos x - \sin x) - \cos x\right]}{\left(e^x \sin x + \cos x\right)\left(e^x \cos x - \sin x\right)} dx$

$= \ln\left[\frac{e^x \sin x + \cos x}{e^x \cos x - \sin x}\right] + c$

27. $\int \ln\left(x e^x\right) \cdot d\left(x e^x\right)$

$= xe^x(x + \ln x) - xe^x + c = xe^x(x + \ln x - 1) + c$

28. $I = \int_\alpha^\beta \frac{e^{f\left(\frac{a(x-\alpha)(x-\beta)}{x-\alpha}\right)}}{e^{f\left(\frac{a(a-\alpha)(x-\beta)}{(x-\alpha)}\right)} + e^{f\left(\frac{a(x-\alpha)(x-\beta)}{(x-\beta)}\right)}} dx$

$= \int_\alpha^\beta \frac{e^{f(a(x-\beta))}}{e^{f(a(x-\beta))} + e^{f(a(x-\alpha))}} dx \quad \ldots (1)$

$= \int_\alpha^\beta \frac{e^{f(a(\alpha+\beta-x-\beta))}}{e^{f(a(\alpha+\beta-x-\beta))} + e^{f(a(\alpha+\beta-x-\alpha))}} dx$

$I = \int_\alpha^\beta \frac{e^{f(a(x-\alpha))}}{e^{f(a(x-\alpha))} + e^{f(a(x-\beta))}} dx \quad \ldots (2)$

$2I = \int_\alpha^\beta dx \Rightarrow I = \frac{|\alpha - \beta|}{2} = \frac{\sqrt{b^2 - 4ac}}{2|a|}.$

29. $2f'(x)f(x) = \frac{f(x)\sec^2 x}{1+\tan x}$

$\int 2f'(x)dx = \int \frac{\sec^2 x}{1+\tan x}dx$

$2f(x) = \ell n(1+\tan x)+c \quad \{\because f(x) = 0 \text{ at } x = 0 \Rightarrow c = 0\}$

$f(x) = \frac{1}{2}\ell n(1+\tan x)$

30. Let $f(x) = \frac{2[x]}{3x-[x]}$

It is clear that $f(x)$ is not defined if $x = 0$ and $3x = [x]$

$\Rightarrow$ in $(-10, 0)$, f is not defined at $x = -1/3$.

Case I: $x \in (-10, -1/3)$

$[x] < 0$ and $3x - [x] < 0$

$\Rightarrow \frac{[x]}{3x-[x]} > 0$

$\therefore \left|\frac{2[x]}{3x-[x]}\right| = \frac{2[x]}{3x-[x]} \Rightarrow \int_{-10}^{-1/3} 1.dx = -\frac{1}{3}+10 = \frac{29}{3}$

Case II: $x \in \left(-\frac{1}{3}, 0\right)$

$[x] < 0$ and $3x - [x] > 0$

$\Rightarrow \frac{[x]}{2x-[x]} < 0 \quad \therefore \left|\frac{2[x]}{3x-[x]}\right| = -\left(\frac{2[x]}{3x-[x]}\right) \Rightarrow \int_{-1/3}^{0} (-1).dx = -\left(0+\frac{1}{3}\right) = -\frac{1}{3}$

$\therefore \int_{-10}^{0} \frac{|f(x)|}{f(x)}dx = \int_{-10}^{-1/3} \frac{|f(x)|}{f(x)}dx + \int_{-1/3}^{0} \frac{|f(x)|}{f(x)}dx = \frac{29}{3} - \frac{1}{3} = \frac{28}{3}$

31. $\int_a^b f(x)dx + \int_{f(a)}^{f(b)} f^{-1}(x)dx = bf(b) - af(a)$

32. $f(2x) = f(x) = f(x/2) = f(x/2^2) = \ldots = f(x/2^n)$

So, when $n \to \infty$

$\Rightarrow f(2x) = f(0)$ ($\because f(x)$ is continuous function)

$\Rightarrow f(x)$ is constant function.

$\Rightarrow f(x) = f(1)$

$= 3, \int_{-1}^{1} f(f(f(x)))dx = \int_{-1}^{1} 3dx = 6.$

33. The given integrals can be equal only if $a > \tan^2 1$

$\therefore \int_0^a [\tan^{-1}\sqrt{x}]dx = \int_0^{\tan^2 1} [\tan^{-1}\sqrt{x}]dx + \int_{\tan^2 1}^{a} [\tan^{-1}\sqrt{x}]dx = 0 + 1\int_{\tan^2 1}^{a} dx = a - \tan^2 1$

$\int_0^a [\cot^{-1}\sqrt{x}]dx = \int_{\tan^2 1}^{a} [\cot^{-1}\sqrt{x}]dx + 0 = 1(\cot^2 1 - 0)$

$\therefore a - \tan^2 1 = \cot^2 1$

$\Rightarrow a = \tan^2 1 + \cot^2 1$

$= \frac{\sin^2 1}{\cos^2 1} + \frac{\cos^2 1}{\sin^2 1}$

$= \frac{1-\cos 2}{1+\cos 2} + \frac{1+\cos 2}{1-\cos 2} = \frac{(1+\cos 2)^2 + (1-\cos 2)^2}{\sin^2 2}$

$= \frac{2(1+\cos^2 2)}{\sin^2 2}$

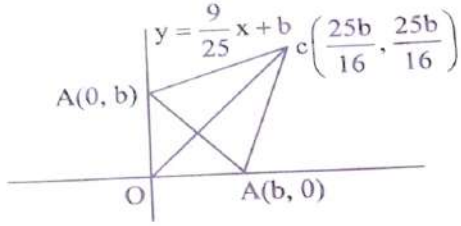
$= \frac{2(2+2\cos^2 2)}{2\sin^2 2}$

$= \frac{2(3+\cos 4)}{1-\cos 4}.$

34. $\text{Area} = \frac{1}{2} AA' \times OC$

$= \frac{1}{2} b\sqrt{2} \cdot \frac{25}{16} \cdot b\sqrt{2} = 49$

$\Rightarrow b = \frac{28}{5}$



35. $\frac{2\sqrt{2}}{2\pi} < \int_{\pi}^{2\pi} \frac{|\sin x + \cos x|}{x} dx < \frac{2\sqrt{2}}{\pi}$ ...(i)

$\because \pi < x < 2\pi$

$\Rightarrow \frac{1}{2\pi} < \frac{1}{x} < \frac{1}{\pi}$

$\frac{2\sqrt{2}}{3\pi} < \int_{2\pi}^{3\pi} \frac{|\sin x + \cos x|}{x} dx < \frac{2\sqrt{2}}{2\pi}$ ....(ii)

Adding equation (i) and (ii), we get

$\frac{5\sqrt{2}}{3\pi} < A < \frac{3\sqrt{2}}{\pi}$

$0.75 < A < 1.3$

36. $\int_0^t f(x)dx = \frac{1}{4} f(t)$

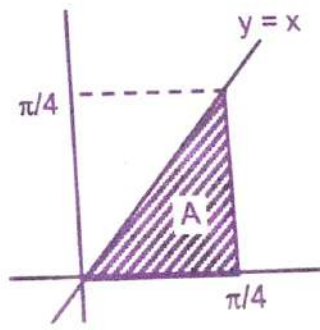
$\Rightarrow \int_0^t f(x)dx = Kt^4 \Rightarrow f(x) = x^3$

37. $x = \sin^{-1}(a^4 + 1) + \cos^{-1}(a^4 + 1) - \tan^{-1}(a^4 + 1)$

the functions are defined in their domain of $a = 0$

$\Rightarrow x = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$

$\therefore A = \frac{1}{2}\left(\frac{\pi}{4}\right)\left(\frac{\pi}{4}\right) = \frac{\pi^2}{32}$

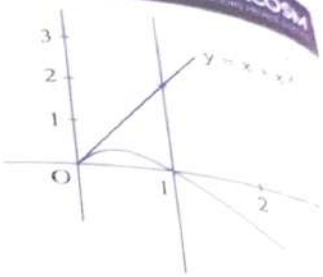


38. $(y-x)^2 = x^3$ where $x \geq 0$

$y - x = \pm x^{\frac{3}{2}}$

$y = x + x^{\frac{3}{2}}$

$y = x - x^{\frac{3}{2}}$



Required area $\int_0^1 \left\{\left(x + x^{\frac{3}{2}}\right) - \left(x - x^{\frac{3}{2}}\right)\right\} dx$

$\Rightarrow 2\int_0^1 x^{\frac{3}{2}} dx = 2\left[\frac{x^{\frac{5}{2}}}{\frac{5}{2}}\right]_0^1 = \frac{4}{5}$ sq. unit

39. Take cosec $y = t$, we get $\frac{dt}{dx} - \frac{t}{x} = -\frac{1}{x^2}$

$\frac{t}{x} = \int -\frac{1}{x^2} dx$

$\Rightarrow 2x = \sin y(1 + x^2)$

40. Given $\int f(x)dx = F(x) + c$

$\therefore f(x) = F'(x)$ ..... (1)

Now, given $f(x) + \cos x \cdot F(x) = \frac{\sin 2x}{(1+\sin x)^2}$

$\because$ From equation (1), we get $f(x) = F'(x)$

$\therefore F'(x) + \cos x \cdot F(x) = \frac{\sin 2x}{(1+\sin x)^2}$

$\therefore \frac{dy}{dx} + (\cos x)y = \frac{\sin 2x}{(1+\sin x)^2}$

$\therefore I.F. = e^{\int \cos x dx} = e^{\sin x}$

$\therefore y \cdot e^{\sin x} = \int \frac{e^{\sin x} \cdot \sin 2x}{(1+\sin x)^2} dx = 2\int \frac{e^{\sin x} \cdot \sin x \cos x}{(1+\sin x)^2} dx$

Put $\sin x = t$

$\therefore y \cdot e^{\sin x} = 2\int \frac{e^t \cdot t\, dt}{(1+t)^2} = 2\int e^t \frac{(1+t-1)}{(1+t)^2} dt$

$= 2\int e^t \left(\frac{1}{t+1} - \frac{1}{(t+1)^2}\right) dt = 2 \cdot \frac{e^t}{(t+1)} + c = \frac{2e^{\sin x}}{1+\sin x} + c$

$\therefore y = \frac{2}{(1+\sin x)} + c\, e^{-\sin x}$

$\Rightarrow f(x) = F'(x) = \frac{-2\cos x}{(1+\sin x)^2} - c\, e^{-\sin x} \cos x$

41. Locus of the centre of the circle is a hyperbola.
If (α, β) is centre of circle then we can say
$\beta = f(\alpha)$ and $r = f(\alpha, \beta)$ where r is radius.
Hence the given family is a one–parameter family of circles.

42. $x\,dy - y\log_e y\,dx + x^2y(2x\sin x + x^2\cos x)\,dx = 0$

$$\frac{xdy - y\log_e ydx}{yx^2} + (2x\sin x + x^2\cos x)dx = 0$$

$$\int d(x^2\sin x) + \int d\left(\frac{\log_e y}{x}\right) = 0 + c$$

$$x^2\sin x + \frac{\log_e y}{x} = c$$

$$\frac{\log_e y}{x} = c - x^2\sin x$$

$$\log_e y = x(c - x^2\sin x)$$

$$y = e^{x(c-x^2\sin x)}$$

44. $(2xy^4e^y + 2xy^3 + y)dx + (x^2y^4e^y - x^2y^2 - 3x)dy = 0$

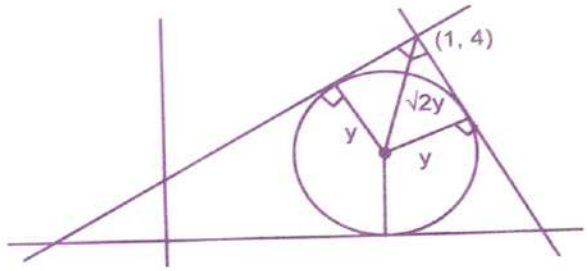
Dividing both the sodes by y^4, we get $\left\{2xe^y + \left(\frac{2x}{y}\right) + \left(\frac{1}{y^3}\right)\right\}dx + \left(x^2e^y - \frac{x^2}{y^2} - \frac{3x}{y^4}\right)dy = 0$

$$\Rightarrow \left(2xe^y dx + x^2e^y dy\right) + \left(2x\frac{dx}{y} - \frac{x^2}{y^2}dy\right) + \left(\frac{dx}{y^3} - \frac{3xdy}{y^4}\right) = 0$$

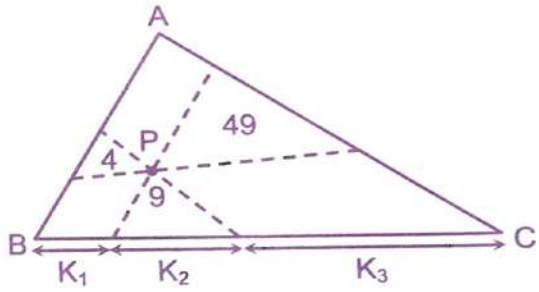
$$\Rightarrow d\left(x^2e^y\right) + d\left(\frac{x^2}{y}\right) + d\left(\frac{x}{y^3}\right) = 0$$

$$\therefore\ x^2e^y + \frac{x^2}{y} + \frac{x}{y^3} = c$$

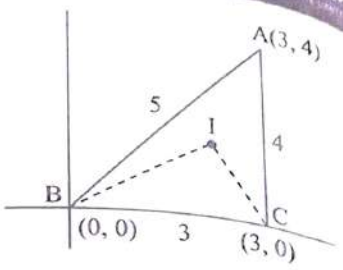
45. $(x-1)^2 + (y-4)^2 = \left(y\sqrt{2}\right)^2$
$x^2 - 2x + 1 + y^2 - 8y + 16 = 2y^2$



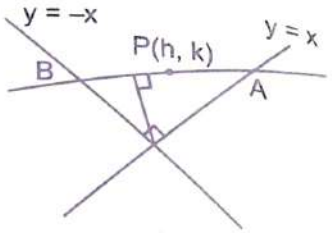
46. Since all 4 triangles (including ABC) are similar
$4 : 9 : 49 \equiv K_1^2 : K_2^2 : K_3^2$
$\Rightarrow K_1 : K_2 : K_3 \equiv 2 : 3 : 7$
2K, 3K, 7K (say)
So side of ΔABC is $12K$ $\frac{4}{\text{Area}} = \left(\frac{2K}{12K}\right)^2$
$\Rightarrow$ Area = 144



47. $\because$ $d(P, BC) \le \min\{d(P, AB), d(P, AC)\}$
$\Rightarrow d(P, BC) \le d(P, AB)$ and $d(P, BC) \le d(P, AC)$
$\Rightarrow d(P, BC)$ is maximum when
$\Rightarrow d(P, BC) = d(P, AB) = d(P, AC)$
$\Rightarrow$ P is the incentre of ΔABC
$\Rightarrow$ Maximum value of $d(P, BC)$
= y-coordinate of the incentre of ΔABC
$= \dfrac{3\times4+4\times0+5\times0}{3+4+5} = 1$



48. Let A = (t, t) and B = $(\alpha, -\alpha)$ then $t + \alpha = 2h$
$t - \alpha = 2k$
$\Rightarrow t = h + k$, $\alpha = h - k$
Now find equation of AB and then perpendicular dist = 1



49. R(x, y) lies along the line N : $9x - 7y + 4 = 0$
$\Rightarrow R\left(a, \dfrac{4+9a}{7}\right)$ and centroid of ΔPQR is (h, k)

$h = \dfrac{2+4+a}{3} = \dfrac{6+a}{3}$ ..... (1)

$k = \dfrac{5-11+\left(\dfrac{4+9a}{7}\right)}{3} = \dfrac{35-77+4+9a}{21}$

$= -\dfrac{38+9a}{21}$ ..... (2)

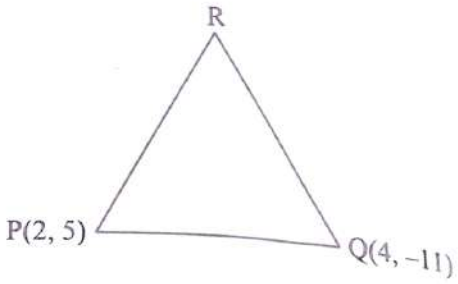
Now equating the value of a from equation (1) and (2)

$\Rightarrow 3h - 6 = \left(\dfrac{21k+38}{9}\right)$

$\Rightarrow 27h - 54 + 21k + 38$

$\Rightarrow 27h - 21k = 92$

$\Rightarrow 9x - 7y - \dfrac{92}{3} = 0$ which is parallel to N



50. Let slope of the line = m

$\therefore$ Equation of the line is $y = m(x-1) \Rightarrow \left(\dfrac{mx-y}{m}\right) = 1$ ..... (1)

Equation of pair of lines is $(x+y-2)(x-y-2) = 0$

$\Rightarrow x^2 - y^2 - 4x + 4 = 0$ ..... (2)

From equation (1) and (2), we get $x^2 - y^2 - 4x\left(\dfrac{mx-y}{m}\right) + 4\left(\dfrac{mx-y}{m}\right)^2 = 0$

$\Rightarrow x^2 + \left(\dfrac{4}{m^2} - 1\right)y^2 + \left(\dfrac{-4}{m}\right)xy = 0$ ..... (3)

Coefficient of x^2 + Coefficient of $y^2 = 0$

$\therefore$ Slope is undefined

51. $\frac{|PA| + |PB|}{2} \geq [|PA| \cdot |PB|]^{1/2}$

$|PA| + |PB| \geq 2|PT| = 2\sqrt{3}$

Maximum length occurs when PAB passes through centre

i.e. $|PA| + |PB| = 4$ (Maximum)

So, range is $\left(2\sqrt{3}, 4\right]$

52. Since the radical centre of three circles described on the sides of a triangles as diameters is the othocentre of the triangle.

Radical centre = orthocentre

Given sides are

$x + y - 5 = 0$ ...(1)

$2x + y - 9 = 0$ ...(2)

$x - 2y + 3 = 0$ ...(3)

since equation (2) and (3) are perpendicular the point of intersect of is (3, 3) the orthocentre of triangle.

Hence radical centre is (3, 3).

54. Equation of chord of contact of (x_1, y_1) its passes through (x_2, y_2)

Hence $x_1x_2 + y_1y_2 = a^2$

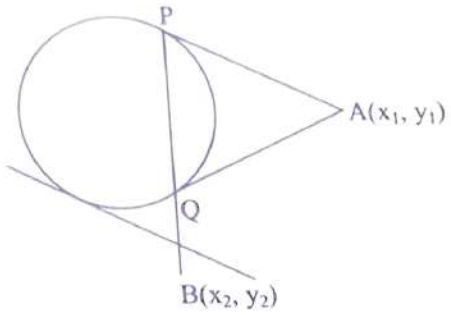
Now $(AB)^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$ ..... (1)

$= x_1^2 + x_2^2 + y_1^2 + y_2^2 - 2(x_1x_2 + y_1y_2)$

$= \left(x_1^2 + y_1^2\right) + \left(x_2^2 + y_2^2\right) - 2a^2$

$= \left(x_1^2 + y_1^2 - a^2\right) + \left(x_2^2 + y_2^2 - a^2\right)$

$AB = \sqrt{p^2 + q^2}$



55. On transforming, parabola is,

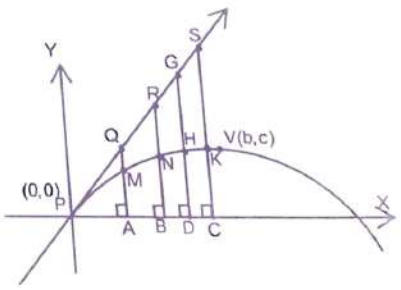
$(x - b)^2 = -4a(y - c)$

and $(b^2 = 4ac)$

$PS \equiv y = \frac{b}{2a}x$

For H(x, y), $GH = \frac{x^2}{4a}$

$\Rightarrow QM : RN : SK \equiv (2)^2 : (2 + 5)^2 = (2 + 5 + 7)^2$



56. Let $A \equiv (\alpha, \beta)$ the equation of the normal to the parabola at $(at^2, 2at)$ is

$y + tx = 2at + at^3$

As it is drawn from (α, β)

So, $at^3 + (2a - \alpha)t - \beta = 0$

$\Rightarrow t_1 + t_2 + t_3 = 0$ ...(1)

So, we have

$P \equiv \left(at_1^2, 2at_1\right)$, $Q \equiv \left(at_2^2, 2at_2\right)$ and $R \equiv \left(at_3^2, 2at_3\right)$

$C_1 \equiv$ Circle drawn on AP as diameter $= (x - \alpha)(x - at_1^2) + (y - \beta)(y - 2at_1) = 0.$

$C_2 \equiv$ Circle drawn on AQ as diameter $= (x - \alpha)(x - at_2^2) + (y - \beta)(y - 2at_2) = 0.$

Common chord $\equiv (t_1 + t_2)x + 2y - \alpha(t_1 + t_2) - 2\beta = 0$ $(\text{as } t_1 \neq t_2)$

(from (1))

$\therefore m_1 = -\left(\frac{t_1 + t_2}{2}\right) = \frac{t_3}{2}$

Tangent to the parabola at R

$t_3 y = x + a t_3^2$

$\Rightarrow m_2 = \frac{1}{t_3}$

So, $m_1 \times m_2 = \frac{1}{2}$

58. $y - 6 = -\frac{6}{9/2}(x - y)$

$3y + 4x - 34 = 0$

Radius of circle $= \frac{SP}{2} = \frac{25}{8}$

Lencth of perpendicular from centre on the normal is $\left|\frac{3\times 3 + 4\times\frac{25}{8} - 34}{5}\right| = \frac{5}{2}$

$\therefore$ Lencth of intercept $= 2\sqrt{\left(\frac{25}{8}\right)^2 - \left(\frac{5}{2}\right)^2} = \frac{15}{4}$

60. Required length is OP + OV where $P \equiv (-1, 1)$, $V \equiv (2, 2)$ i.e. vertex and O is origin.

61. Length of perpendicular from C(0, 0) to the tangent at $P\left(2\sqrt{3}\cos\theta, 2\sqrt{2}\sin\theta\right)$ is

$CF = \left|\frac{-1}{\sqrt{\frac{\cos^2\theta}{12} + \frac{\sin^2\theta}{8}}}\right|$ equation of normal at P is

$\frac{2\sqrt{3}x}{\cos\theta} - \frac{2\sqrt{2}y}{\sin\theta} = 12 - 8$

which meets the major axis at $G\left(\frac{2}{\sqrt{3}}\cos\theta, 0\right)$

$\Rightarrow CF \times PG = 8$

62. $f(k^2 + 2k + 5) > f(k + 11)$

$\Rightarrow k^2 + 2k + 5 < k + 11$

$\Rightarrow k \in (-3, 2)$

63. Combined equation of pair of lines through the origin joining the points of intersection of line $y = \sqrt{k}x + 1$ with the given curve is

$x^2 + 2xy + \left(2 + \sin^2\beta\right)y^2 - \left(y - \sqrt{k}x\right)^2 = 0$

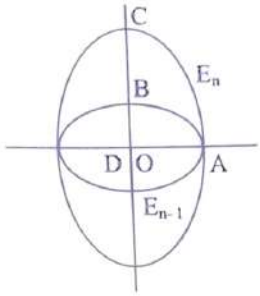
For the chord to subtend a right angle at the origin $(1 - k) + (2 + \sin^2\beta - 1) = 0$

[As sum of the coefficients of x^2 and $y^2 = 0$]

$\Rightarrow \sin^2\beta = k - 2$

$\Rightarrow 0 \le k - 2 \le 1 \Rightarrow 2 \le k \le 3$

64. The figure shows two ellipse E_{n-1} and E_n
The eccentricity is given to be independent of n, implies the ratio of minor axis to the major axis, is same for all the ellipse, for ellipse E_{n-1} let minor axis = b for ellipse E_n, we have



Minor axis = a and major axis = $\frac{DB}{\sqrt{5}e} = \frac{b}{e}$ ($\because$ B is focus of E_n)

Assuming e to be the eccentricity. Thus, we have

$\frac{b}{a} = \frac{a}{b/e} \Rightarrow e = \frac{b^2}{a^2} = 1 - e^2 \Rightarrow e^2 + e - 1 = 0 \Rightarrow e = \frac{\sqrt{5}-1}{2}$

65. Let equation of the circle be $\left(\frac{x^2}{4} + y^2 - 1\right) + \lambda\left(\frac{x^2}{a^2} + y^2 - 1\right) = 0$

$\therefore x^2\left(\frac{a^2 + 4\lambda}{4a^2(1+\lambda)}\right) + y^2 = 1$

So circle is $x^2 + y^2 = 1$.

66. The curve is a hyperbola and $e = \frac{r_1 + r_2}{r_1 - r_2} = \sqrt{2}$.

67. Equation of tangent at point P(p cos θ, sin θ) is $\frac{x}{p}\cos\theta + \frac{y}{1}\sin\theta = 1$

Let, it cut the hyperbola at A and B
Homogenizing the hyperbola $p^2x^2 - y^2 = 1$ with the help of the above equation, we get

$p^2x^2 - y^2 = \left(\frac{x}{p}\cos\theta + y\sin\theta\right)^2$

This is a pair of straight lines OA and OB

Given $\angle AOB = \frac{\pi}{2}$

$\Rightarrow$ Coefficient of x^2 + coefficient of $y^2 = 0$

$\Rightarrow p^2 - \frac{\cos^2\theta}{p^2} - 1 - \sin^2\theta = 0$

$\Rightarrow p^2 - \frac{\cos^2\theta}{p^2} - 1 - 1 + \cos^2\theta = 0$

$\Rightarrow \cos^2\theta = \frac{p^2(2-p^2)}{p^2-1}$

Now, $0 \le \cos^2\theta \le 1$

$\Rightarrow 0 \le \frac{p^2(2-p^2)}{p^2-1} \le 1 \Rightarrow p^2 \in \left[\frac{\sqrt{5}+1}{2}, 2\right]$

68. $x^2 - y^2 = 2000^2 = (2^4 \cdot 5^3)^2 \Rightarrow (x+y)(x-y) = 2^8 \cdot 5^6$
for $y = 0$, $x = \pm 2000$.
Let $(x > 0, y > 0)$ x + y and x – y are both even.

$\Rightarrow$ Number of integral values of x, $y = \frac{7\times7-1}{2} = 24$

$\Rightarrow$ number of integral points = 24 × 4 + 2 = 98.

69. Let $x = \cos\theta$ and $y = \sin\theta$
$\Rightarrow x^2 + 4xy + y^2 = 1 + 2\sin 2\theta \le 3$
As (3, 1) lies on an asymptote

70. $x = y = z = r$
If both lines intersect, then intersection of first line with both the planes is same. i.e.
$$\frac{1}{b^2 + (2-4a) + 1} = \frac{-1}{a^2 + 1 - 2b + 1}$$
$\Rightarrow (b-1)^2 + (a-2)^2 = 0 \Rightarrow a = 2$ and $b = 1$

71. The given lines ($r = a + \lambda(b \times c)$ and $r = b + \mu(c \times a)$) pass through points a and b respectively and are parallel to vector $b \times c$ and $c \times a$ respectively.
$\therefore$ They intersect if $a - b$, $b \times c$ and $c \times a$ are coplanar.
So, $(a-b) \cdot [(b \times c) \times (c \times a)] = 0$
$\Rightarrow (a-b) \cdot \{[bca]c - [bcc]a\} = 0$
$\Rightarrow ((a-b) \cdot c)[bca] = 0$
$\Rightarrow a \cdot c - b \cdot c = 0 \Rightarrow a \cdot c = b \cdot c$

72. Equation of plane passing through (2, 1, 0) is $a(x-2) + b(y-1) + c(z-0) = 0$
It also passes through (5, 0, 1) and (4, 1, 1)
$\Rightarrow 3a - b + c = 0$ and $2a + 0\cdot b + c = 0$
On solving, we get $\frac{a}{-1} = \frac{b}{-1} = \frac{c}{2}$
$\therefore$ Equation of plane is $-(x-2) - (y-1) + 2(z-0) = 0$
$\Rightarrow -x + 2 - y + 1 + 2z = 0$
$\Rightarrow x + y - 2z = 3$ ..... (1)
Let the coordinate of $Q(\alpha, \beta, \gamma)$
Equation of line PQ is $\frac{x-2}{1} = \frac{y-1}{1} = \frac{z-6}{-2}$
Since mid-point of P and Q is $\left(\frac{\alpha+2}{2}, \frac{\beta+1}{2}, \frac{\gamma+6}{2}\right)$ which lies in line PQ
$$\Rightarrow \frac{\frac{\alpha+2}{2} - 2}{1} = \frac{\frac{\beta+1}{2} - 1}{1} = \frac{\frac{\gamma+6}{2} - 6}{-2}$$
$$= \frac{1\cdot\left(\frac{\alpha+2}{2} - 2\right) + 1\cdot\left(\frac{\beta+1}{2} - 1\right) - 2\cdot\left(\frac{\gamma+6}{2} - 6\right)}{1.1 + 1\cdot 1 + (-2)(-2)} = 2$$
Since, $\left(\frac{\alpha+2}{2}\right) + 1\cdot\left(\frac{\beta+1}{2}\right) - 2\left(\frac{\gamma+6}{2}\right) = 3$
$\Rightarrow \alpha = 6, \beta = 5, \gamma = -2$
$\Rightarrow Q \equiv (6, 5, -2)$

73. Given $b = qa$, $p = abc = qa^2c \Rightarrow a^2c = \frac{p}{q}$

Now, $a^2 + b^2 + c^2 = (1+q^2)a^2 + c^2 = \frac{(1+q^2)a^2}{2} + \frac{(1+q^2)a^2}{2} + c^2 \ge 3\left[\frac{(1+q^2)^2}{4}a^4c^2\right]^{\frac{1}{3}} = 3\left[\frac{(1+q^2)p}{2q}\right]^{\frac{2}{3}}$

74. Clearly, $\frac{2x+y+3z-1}{\sqrt{14}} = \pm\frac{(x+2y-3z-1)}{\sqrt{14}}$

$3x + 3y - 2 = 0$ and $x - y + 6z = 0$

$\therefore k = -2$, solve $k, k = -4$

75. $y^2(5x^2 + 1) = 25(2x^2 + 13)$, $x, y \in I \Rightarrow y = \pm 5\sqrt{\frac{2x^2+13}{5x^2+1}}$

For integral solutions of x and y we have $2x^2 + 13 = 5x^2 + 1 \Rightarrow 3x^2 = 12 \Rightarrow x = \pm 2$

$x = 2 \Rightarrow y = \pm 5\sqrt{\frac{8+13}{20+1}} = \pm 5$

$x = -2 \Rightarrow y = \pm 5$ and $2x^2 + 13 < 5x^2 + 1$ for $x > 2$

$\therefore \frac{2x^2+13}{5x^2+1}$ is fractional part for $x > 2$

$\therefore (2, \pm 5)$ and $(-2, \pm 5)$ are the only 4 Integral solutions.

76. $a\alpha^2 + b\alpha + c = a\alpha^2 + p\alpha + q$

$\therefore \alpha = \frac{q-c}{b-p}$

A.M. of roots of $f(x) = 0$ and $g(x) = 0$ is $= \frac{-b/a - p/a}{4} = -\frac{b+p}{4a}$ (Number of roots is 4)

Since discriminants are equal

$b^2 - 4ac = p^2 - 4aq$

$b^2 - p^2 = 4ac - 4aq$

$\frac{b+p}{4a} = \frac{c-q}{b-p} = -\alpha$

$\therefore \alpha$ = A.M. of roots of $f(x) = 0$ and $g(x) = 0$

77. For the equation $x^2 - px + 1 = 0$, the product of roots, $\alpha\beta^2 = 1$ and for the equation $x^2 - qx + 8 = 0$, the product of roots $\alpha^2\beta = 8$

Hence, $(\alpha\beta^2)(\alpha^2\beta) = 8$

$\Rightarrow \alpha^3\beta^3 = 8$

$\Rightarrow \alpha\beta = 2$

$\therefore$ From $\alpha\beta^2 = 1$, we have $\beta = \frac{1}{2}$ and from $\alpha^2 \cdot \beta = 8$, we have $\alpha = 4$

Hence, from sum of roots $= -\frac{b}{a}$,

relation, we have $p = \alpha + \beta^2 = 4 + \frac{1}{4} = \frac{17}{4}$

and $q = \alpha^2 + \beta = 16 + \frac{1}{2} = \frac{33}{2}$

$\because \frac{r}{8}$ is arithmetic mean of p and q

$\therefore \frac{r}{8} = \frac{p+q}{2}$

$\Rightarrow r = 4(p+q)\left(\frac{17}{4} + \frac{33}{2}\right) = 17 + 66 = 83$

78. $a = \dfrac{2x^2 + 1 \pm (2x - 1)}{2}$

$\Rightarrow a = x^2 + x$

$\Rightarrow a \geq -\dfrac{1}{4}$

Also, $a = x^2 - x + 1$

$\Rightarrow a \geq \dfrac{3}{4}$ hence, $a \in \left[\dfrac{3}{4}, \infty\right)$

79. Subtracting the equations $y - x = a(x - y)(x + y) + a(x - y)$

$\Rightarrow y = x$ or $x = -\dfrac{1}{a} - y - 1$

Substituting in second curve we get $ay^2 + (a-1)y + \dfrac{1}{24} = 0$

or $ay^2 + (a+1)y + \dfrac{25}{24} + \dfrac{1}{a} = 0$ for curves to touch each other either any one of the quadratics have equal roots or they have a common root, finding the values of a we get

$a = \dfrac{2}{3}, \dfrac{3}{2}, \dfrac{13 \pm \sqrt{601}}{12}$

80. $T_r = \dfrac{r^2}{(2r-1)(2r+1)} = \dfrac{1}{4}\left(\dfrac{r}{2r-1} + \dfrac{r}{2r+1}\right)$

$\therefore S = \dfrac{1}{4}\sum_{r=1}^{500}\left(\dfrac{r}{2r-1} + \dfrac{r}{2r+1}\right)$

$\Rightarrow 4S = \sum_{r=1}^{500}\left(\dfrac{r}{2r-1} + \dfrac{r}{2r+1}\right)$

$= \dfrac{1}{1} + \left(\dfrac{1}{3} + \dfrac{2}{3}\right) + \left(\dfrac{2}{5} + \dfrac{3}{5}\right) + \left(\dfrac{3}{7} + \dfrac{4}{7}\right) + \ldots\ldots + \dfrac{500}{999} + \dfrac{500}{1001}$

$\because \dfrac{r}{2r+1} + \dfrac{r+1}{2r+1} = 1$

Combining all such terms, we get $4S = 1 + 499 + \dfrac{500}{1001} = \dfrac{501000}{1001}$

$S = \dfrac{125250}{1001} \Rightarrow [S] = 125$

Number of divisors = 4

81. We have $a_n^2 = a_{n-1}^2 + 1 + 2a_{n-1}$

Putting $n = 2, 3, 4, \ldots, (n+1)$ and adding we get $a_{n+1}^2 = n + 2(a_1 + a_2 + \ldots + a_n)$

82. It means that roots are in A.P. $Ax^3 + Bx^2 + (Cx + D) = 0$

Let roots are $a - d, a, a + d \Rightarrow 3a = -\dfrac{B}{A}$, $a = -\dfrac{B}{3A}$

Here a is a root of above cubic so it must satisfy the equation

$A\left(-\dfrac{B}{3A}\right)^3 + B\left(-\dfrac{B}{3A}\right)^2 + C\left(-\dfrac{B}{3A}\right) + D = 0$. Solving $2B^3 - 9ABC + 27A^2D = 0$

$k_1 = -9, k = 27 \Rightarrow k_1 + k_2 = 18$

83. 3x, 4y, 5z form a G.P.

$16y^2 = 15xz$

x, y, z form H.P.

$$y = \frac{2xz}{x+z}$$

$$16 \cdot 4 \cdot \frac{x^2z^2}{(x+z)^2} = 15xz$$

$$64xz = 15(x^2 + z^2 + 2xz)$$

$$64 = 15\left(\frac{x}{z} + \frac{z}{x} + 2\right)$$

$$\frac{64}{15} - 2 = \frac{x}{z} + \frac{z}{x} = \frac{p}{q}$$

$$\frac{p}{q} = \frac{34}{15} \Rightarrow p + q = 49$$

84. $$\sum_{m=0}^{2014}\sum_{n=0}^{m}\left({}^{2014}C_m \cdot {}^{m}C_n\right)$$

$$= \sum_{m=0}^{2014} {}^{2014}C_m \left({}^{m}C_0 + {}^{m}C_1 + \ldots\ldots + {}^{m}C_m\right)$$

$$= \sum_{m=0}^{2014} {}^{2014}C_m \left(2^m\right) = {}^{2014}C_0\left(2^0\right) + {}^{2014}C_1\left(2^1\right) + {}^{2014}C_2\left(2^2\right) + \ldots\ldots + {}^{2014}C_{2014}\left(2^{2014}\right) = 3^{2014}$$

85. Coefficient of x^{2r} in $(1 + x + x^2)^n$

= coefficient of x^{2r} in $(1 + x(1 + x))^n$

$$= \text{coefficient of } x^{2r} \text{ in } \left(\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}\right)^n$$

coefficient of x^{2r} in $(1 + x(1+ x))^n$

$$= \sum_{k=0}^{r} {}^{n}C_{r+k}\,{}^{r+k}C_{r-k} = \sum_{k=0}^{r} \frac{n!}{(r+k)!(n-r-k)!}\frac{(r+k!)}{(2k)!(r-k)!} = \sum_{k=0}^{n} {}^{n}C_{2k}\,{}^{n-2k}C_{r-k}$$

$$\text{coefficient of } x^{2r} \text{ in } \left(\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}\right)^n = x^{2r} \text{ in } \sum_{k=0}^{n} {}^{n}C_k \left(x + \frac{1}{2}\right)^{2k}\left(\frac{3}{4}\right)^{n-k}$$

$$= \sum_{k=r}^{n} {}^{n}C_k\,{}^{2k}C_{2r}\left(\frac{1}{2}\right)^{2(k-r)}\left(\frac{3}{4}\right)^{n-k} \Rightarrow \text{Ratio of both expansions} = 1$$

86. We have $$\frac{1}{{}^{3n}C_r} = \frac{(3n-r)!\,r!}{(3n)!} = \frac{3n+1}{3n+2}\left[\frac{\{(3n+1-r)+(r+1)\}\{(3n-r)!\,r!\}}{(3n+1)!}\right]$$

$$= \frac{3n+1}{3n+2}\left[\frac{(3n+1-r)!\,r! + (3n-r)!\,(r+1)!}{(3n+1)!}\right]$$

$$= \left[\frac{1}{{}^{3n+1}C_r} + \frac{1}{{}^{3n+1}C_{r+1}}\right]\frac{3n+1}{3n+2}.$$

$$\text{So, } \sum_{r=1}^{3n-1}\frac{(-1)^{r-1}r}{{}^{3n}C_r}=\sum_{r=1}^{3n-1}(-1)^{r-1}r\left[\frac{1}{{}^{3n+1}C_r}+\frac{1}{{}^{3n+1}C_{r+1}}\right]\left(\frac{3n+1}{3n+2}\right)$$

$$=\left[\left(\frac{1}{C_1}+\frac{1}{C_2}\right)-2\left(\frac{1}{C_2}+\frac{1}{C_3}\right)+3\left(\frac{1}{C_3}+\frac{1}{C_4}\right)-\ldots+(3n-1)\left(\frac{1}{C_{3n-1}}+\frac{1}{C_{3n}}\right)\right]\left(\frac{3n+1}{3n+2}\right)$$

[Where $C_r = {}^{3n+1}C_r$]

$$=\left(\frac{3n+1}{3n+2}\right)\left[\frac{1}{C_1}-\frac{1}{C_2}+\frac{1}{C_3}-\frac{1}{C_4}+\ldots+\frac{1}{C_{3n-1}}-\frac{1}{C_{3n}}+\frac{3n}{C_{3n}}\right]$$

$$=\frac{3n+1}{3n+2}\cdot\frac{3n}{3n+1}=\frac{3n}{3n+2}.$$

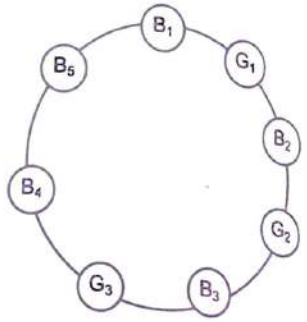
87. $$\sum\frac{r^4+r^3+2r^2+1}{(r^4+r^2+1)(r^2+r)}=\sum\frac{r}{r^4+r^2+1}+\sum\frac{1}{r(r+1)}=\frac{1}{2}\sum\frac{2r}{(r^2-r+1)(r^2+r+1)}+\sum\frac{(r+1)-r}{r(r+1)}$$

$$=\frac{1}{2}\sum\left(\frac{1}{r^2-r+1}-\frac{1}{r^2+r+1}\right)+\sum\frac{1}{r}-\frac{1}{(r+1)}$$

Applying V_n method

$$=\frac{1}{2}+1=\frac{3}{2}$$

88. Total number of ways of arranging 3 girls and 5 boys such that no two girls sit together = $4! \times {}^5C_3 \times 3!$
We have to subtract the ways where more than two boys sit between two girls i.e.
Number of such ways = $5 \times 3! \times 4!$
$\therefore$ Required number of ways = $4! \times ({}^5C_3 - 5) \times 3!$.



89. Total 16 squares of unit diagonal length are obtained from the intersection of two given families of straight lines. Out of these, squares of 2 unit diagonal length are $3 \times 3 = 9$

90. Number of ways when the squares are alternating colour in first column is 2^8.
Number of ways when the squares in the first column are not alternating colour = $2^8 - 2$
$\Rightarrow$ Total number of ways = $2^9 - 2$.

91. Odd number 1, 3, 5, 7, 9 $\Rightarrow$ even number 2, 4, 6, 8
If sum of numbers in each group is odd then it is possible only when one group should have 3 odd numbers and remaining two groups should have 2 even and 1 odd number

This can be done in ${}^3C_1 \cdot {}^5C_3 \cdot {}^2C_1 \cdot \frac{4!\,2!}{2!\,2!\,2!} = 360$ ways

92. Number of required ways = $ {}^5C_2 \times 3!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}\right] = 20$

93. Number of ways octopus can put on his socks and shoes is equal to number of sequares of $a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, b_1, b_2, b_3, b_4, b_5, b_6\ b_7, b_8$ such that a_i occurs before $b_i\ \forall\ i = 1, 2, 3, \ldots\ldots, 8$

Number of such sequences = $\frac{16!}{(2!)^8}$

94.

	1st	2nd	3rd	4th	Probability
case1: –	W	W	R	R	$\frac{8}{10}\times\frac{8}{10}\times\frac{2}{10}\times\frac{1}{10}$
case2: –	W	R	W	R	$\frac{8}{10}\times\frac{2}{10}\times\frac{9}{10}\times\frac{1}{10}$
case3: –	R	W	W	R	$\frac{2}{10}\times\frac{9}{10}\times\frac{9}{10}\times\frac{1}{10}$

Probability = 0.0434

95. The Sample Space reduces to $6^4-2.5^4+4^4=302$. Favourable outcomes are all possible orientations of 4,2,3,3 i.e. $\frac{4!}{2!}=12$. Required probability is $\frac{6}{151}$

96. If the function is non-decreasing then, $f(1)\le f(2)\le f(3).....\le f(13)$, all the values of set co-domain are to be images of some numbers of domain. This implies that exactly 4 equalities should hold in the inequality $f(1)\le f(2)\le f(3).....\le f(13)$, which can happen in ${}^{12}C_4$ ways. However, if $f(7)=4$ exactly 3 equalities should hold before $f(7)$ and exactly one equality should hold after $f(7)$. This can occur in ${}^6C_3\,{}^6C_1$ ways.

Therefore required probability is $\frac{{}^6C_3\,{}^6C_1}{{}^{12}C_4}$.

97. It product of two numbers is equal to third number, then possibilities are (2, 3, 6), (2, 4, 8), (2, 5, 10)

$\therefore$ Probability $=\frac{3}{{}^{10}C_3}=\frac{1}{40}$

98. $\left(z+\frac{1}{z}\right)^3=z^3+\frac{1}{z^3}+3\left(z+\frac{1}{z}\right)$

Let $z+\frac{1}{z}=a$

$$\left|z+\frac{1}{z}\right|^3=\left|z^3+\frac{1}{z^3}+3\left(z+\frac{1}{z}\right)\right|\le\left|z^3+\frac{1}{z^3}\right|+3\left|z+\frac{1}{z}\right|$$

$a^3\le 2+3a$

$a^3-3a-2\le 0$

$(a-2)(a+1)^2\le 0$

$a=\left|z+\frac{1}{z}\right|\ge 0$

Range of a is [0, 2]

99. Applying Rotation formula at P and B we get,

$$\frac{p+4}{p-\omega^2}\times\frac{2i\omega-\omega^2}{2i\omega+4}=-1\Rightarrow p\left(2i\omega-\omega^2+2i\omega+4\right)=\omega^2\left(2i\omega+4\right)-4\left(2i\omega-\omega^2\right)$$

$$\Rightarrow p=\frac{2i-8i\omega+8\omega^2}{4i\omega-\omega^2+4}=\frac{2\left(i-4i\omega+4\omega^2\right)}{\left(-4\omega^2-i+4i\omega\right)}\times i\omega=-2i\omega\text{. So, }z=\frac{p+b}{2}=0$$

$\Rightarrow |z|^2=0$

100. The given line can be written as $y = x$. Hence reflection is $3 + 2i$

101. Centre becomes $\left(-3\sqrt{3}+3i\right)$ and equation is $\left|z-\left(-3\sqrt{3}+3i\right)\right|=2$

102. Distance of z from $(-3, 0)$ is half that from imaginary axis
$\Rightarrow$ z moves on an ellipse with focus $(-3, 0)$ and corresponding directrix as imaginary axis z_1, z_2 are extremities of focal chord of contact of tangents from $(0, 3)$.

$\because \arg\dfrac{(z+3)}{3i+3}=\dfrac{\pi}{2}$

$\therefore \arg\dfrac{z_1+3}{z_2+3}=\pm\pi$

103. Since P and Q form a matrix 'm' such that trace (M) = 0

$\therefore M=\begin{bmatrix} x & y \\ -y & -x \end{bmatrix}$

F: $|\text{adj (adj M)}| = -1 \Rightarrow |M|^{(2-1)^2} = -1 \Rightarrow y^2 - x^2 = -1$

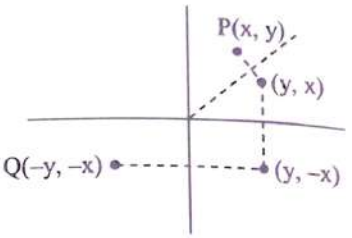
$\Rightarrow x^2 - y^2 = 1 \Rightarrow$ G: $I_m\left(z-\dfrac{1}{2}\right)>1$

$\Rightarrow I_m\left(x+iy-\dfrac{1}{2}\right)>1$

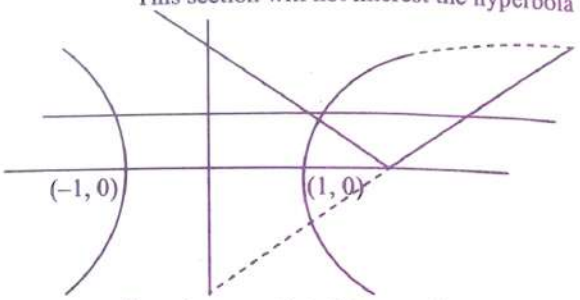
$\Rightarrow I_m\left[x+i\left(y-\dfrac{1}{2}\right)\right]>1$

$\Rightarrow y-\dfrac{1}{2}>1 \Rightarrow y>\dfrac{1}{2}+1 \Rightarrow y>\dfrac{3}{2}$

H: $|\text{Re}(z-2)| - I_m(z) = 0$
$\Rightarrow |\text{Re}(x+iy-2)| - I_m(z) = 0$
$\Rightarrow |x-2| - y = 0$
$\Rightarrow y = |x-2|$



This section will not interest the hyperbola



So only one point of intersection

104. Diagonal elements $\rightarrow$ all O's. For each member in upper Δ, lower Δ member is fixed.

So, number of such matrices is given by $3^{\frac{n(n-1)}{2}}$
$3 \rightarrow$ because each element can be 0, 1 or –1
$\dfrac{n(n-1)}{2} \rightarrow$ because total elements (leaving diagonal) is upper Δ is

$= (n-1) + (n-2) + \ldots\ldots + 1 = \dfrac{n(n-1)}{2}$

105. $|A^{2005} - 6A^{2004}| = |A|^{2004}\,|A - 6I| = 2^{2004} \cdot \begin{vmatrix} 0 & 11 \\ 2 & -2 \end{vmatrix} = (-22)\,2^{2004} = (-)\,2 \times 11 \times 2^{2004} = (-11)(2)^{2005}.$

106. As we know $A(\text{adj } A) = |A|I_n$
$\text{adj } A(\text{adj adj } A) = |\text{adj } A|I_n = |A|^{n-1} \cdot I_n$

$\text{adj adj } A(\text{adj adj adj } A) = |\text{adj adj } A| \cdot I_n = |A|^{(n-1)^2} \cdot I_n$

$\Rightarrow \underbrace{(\text{adj adj adj} \ldots\ldots \text{adj } A)}_{(n-1)\text{ times}} \cdot \underbrace{(\text{adj adj adj} \ldots\ldots \text{adj } A)}_{n\text{ times}} = |A|^{(n-1)^{(n-1)}} \cdot I_n$

107. The first row can be filled in $\frac{4!}{2!\,2!} = 6$ ways.

Then the first column can be filed in $\frac{3!}{2!} = 3$ ways.

A		B	
C		D	

Let A, B, C and D are four elements which are same
$\Rightarrow$ remaining elements can be filled in only one way
$\Rightarrow$ number of such matrices $= 6 \cdot 3$.
If D is different from A, B, C, then other elements can be filled in 4 ways
$\Rightarrow$ total matrices $= 6 \cdot 3 + 6 \cdot 3 \cdot 4 = 90$.

108. Given $A^2 = A$

$I = (I - 0.4A)(I - \alpha A)$

$= I - I\alpha A - 0.4AI + 0.4\alpha A^2$

$= I - A\alpha - 0.4A + 0.4\alpha A$

$= I - A(0.4 + \alpha) + 0.4\alpha A$

Hence $0.4\alpha = 0.4 + \alpha \Rightarrow \alpha = -\frac{2}{3}$

109. As $[2\pi] = 6$ and $[e] = 2$

So that the three equations are consistence if $[2a] = 6$. This gives $6 \le 2a < 7$

Or $3 \le a < \frac{7}{2}$

110. $\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + (-1)^n \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ a & -b & c \end{vmatrix}^T = 0 \Rightarrow \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + (-1)^n \begin{vmatrix} a+1 & b-1 & a \\ b+1 & b-1 & -b \\ c-1 & c+1 & c \end{vmatrix} = 0$

$\Rightarrow \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} \left(1 + (-1)^n\right) = 0$ which is possible only when n is even.

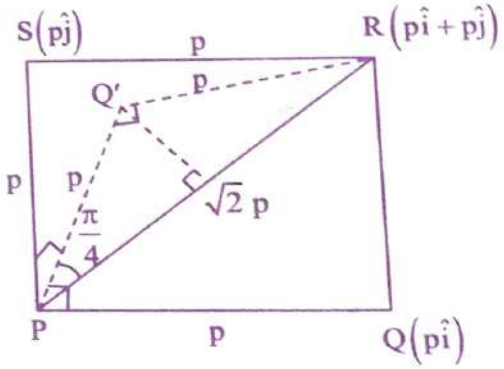
111. $\sqrt{b^2 - 4c} + |b|$ is an even number

112. Taking 'P' as origin

$\therefore$ p.v. of $Q = p\hat{i}$

p.v. of $R = p\hat{i} + p\hat{j}$

sp.v. of $S = p\hat{j}$



Equation of $PQ' = \frac{p}{2}\hat{i} + \frac{p}{2}\hat{j} + \frac{p}{\sqrt{2}}\hat{k} = \left[\hat{i} + \hat{j} + \sqrt{2}\,\hat{k}\right]\frac{p}{2}$

Equation of $\overline{SR} = p\hat{i} + p\hat{j} + \lambda\left(\hat{i}\right)$

$\therefore$ Shortest distance between $\overline{SR}$ and $\overline{PQ}$

$$= \frac{\left(p\hat{i} + p\hat{j}\right)\left(k - \sqrt{2}\,\hat{i}\right)}{\sqrt{1 + \left(\sqrt{2}\right)^2}}$$

$$= \frac{\sqrt{2}\,p}{\sqrt{3}} = \frac{kp}{\sqrt{6}} \Rightarrow sk = 2$$

MEGACOSM

113. Given $\vec{r}\times\vec{x}=\vec{y}\times\vec{x}$

$\Rightarrow (\vec{r}-\vec{y})\times\vec{x}=0$

$\Rightarrow \vec{r}=\vec{y}+\lambda\vec{x}$ ..... (1)

Also, $\vec{r}\times\vec{y}=\vec{x}\times\vec{y}$

$\Rightarrow (\vec{r}-\vec{x})\times\vec{y}=0$

$\Rightarrow \vec{r}=\vec{x}+\mu\vec{y}=0$ ..... (2)

Solving (1) and (2), we get $y+\lambda\vec{x}=\vec{x}+\mu\vec{y}$

$\Rightarrow 2\vec{a}-\vec{b}+\lambda(\vec{a}+\vec{b})=\vec{a}+\vec{b}+\mu(2\vec{a}-\vec{b})$

$\Rightarrow 2+\lambda=1+2\mu$ ..... (3)

Also, $\lambda-1=1-\mu$ ..... (4)

$\therefore \mu=1, \lambda=1$

$\therefore$ Point of intersection $= 3\vec{a}$

114. $\vec{r}_1=\sqrt{3}(\vec{a}\times\vec{b}),\ \vec{r}_2=\vec{b}-(\vec{a}\cdot\vec{b})\vec{a}$

Clearly $\vec{r}_1\perp\vec{r}_2$ as $\vec{r}_2$ is coplaner with $\vec{a}+\vec{b}$ and $\vec{r}_1$ is at right angles to the plane $\vec{a}$ and $\vec{b}$

$|\vec{r}_1|=\sqrt{3}|\vec{a}\times\vec{b}|\Rightarrow|\vec{r}_1|^2=3(\vec{a}\times\vec{b})\cdot(\vec{a}\times\vec{b})$

$=3\{(\vec{a}\cdot\vec{a})(\vec{b}\cdot\vec{b})-(\vec{a}\cdot\vec{b})^2\}=3(|\vec{b}|^2-(\vec{a}\cdot\vec{b})^2)$

Also $|\vec{r}_2|^2=(\vec{b}-(\vec{a}\cdot\vec{b})\vec{a})\cdot(\vec{b}-(\vec{a}\cdot\vec{b})\vec{a})$

$=|\vec{b}|^2-2(\vec{a}\cdot\vec{b})^2+(\vec{a}\cdot\vec{b})^2(\vec{a}\cdot\vec{a})=|\vec{b}|^2-(\vec{a}\cdot\vec{b})^2$

$\Rightarrow \dfrac{|\vec{r}_1|}{|\vec{r}_2|}=\sqrt{3}$

$\Rightarrow$ Angles are $\dfrac{\pi}{2},\ \dfrac{\pi}{3},\ \dfrac{\pi}{6}$

115. $\vec{c}=x\vec{a}+x\vec{b}+y(\vec{a}\times\vec{b})$

$\vec{a}\cdot\vec{c}=\vec{b}\cdot\vec{c}=\cos\theta\Rightarrow\vec{a}\cdot\vec{c}=x|\vec{b}|^2=\cos\theta\Rightarrow\vec{c}=\cos\theta(\vec{a}+\vec{b})+y(\vec{a}\times\vec{b})$

Now $|\vec{c}|^2=1=(\cos\theta(\vec{a}+\vec{b})+y(\vec{a}\times\vec{b}))\cdot(\cos\theta(\vec{a}+\vec{b})+y(\vec{a}\times\vec{b}))$

$\Rightarrow 1=\cos^2\theta(\vec{a}+\vec{b})\cdot(\vec{a}+\vec{b})+y^2(\vec{a}\times\vec{b})\cdot(\vec{a}\times\vec{b})$

$\Rightarrow 1=\cos^2\theta(|\vec{a}|^2+|\vec{b}|^2)+y^2|\vec{a}|^2|\vec{b}|^2$

$\Rightarrow 1=2\cos^2\theta+y^2\Rightarrow\cos^2\theta=\dfrac{1}{2}-\dfrac{y^2}{2}\Rightarrow 0\le\cos^2\theta\le\dfrac{1}{2}$

$\Rightarrow -\dfrac{1}{\sqrt{2}}\le\cos\theta\le\dfrac{1}{\sqrt{2}}\Rightarrow\theta\in\left(\dfrac{\pi}{4},\ \dfrac{3\pi}{4}\right).$

116. Let P $\equiv$ (cosθ, sinθ, 0) point M on line (2, 2, 0)

$\Rightarrow \overrightarrow{PM}=(\cos\theta-2)\hat{i}+(\sin\theta-2)\hat{j}$

vector in the direction of line L N $=\hat{i}+\hat{j}-\hat{k}$

$\Rightarrow |\overrightarrow{MN}|=\dfrac{\cos\theta-2}{\sqrt{3}}+\dfrac{\sin\theta-2}{\sqrt{3}}$

$\Rightarrow PN^2 = (\cos\theta - 2)^2 + (\sin\theta - 2)^2 = \left(\frac{\cos\theta + \sin\theta - 4}{\sqrt{3}}\right)^2 = l$

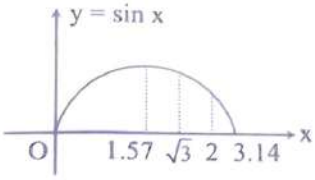
$\Rightarrow \frac{dl}{d\theta} = 0 \Rightarrow \tan\theta = 1 \Rightarrow \theta = \frac{\pi}{4}, \frac{5\pi}{4}$

$\Rightarrow$ Minimum $PN = \frac{2\sqrt{2}-1}{\sqrt{3}}$

117. $\sum_{r=1}^{8} \tan(rA)\tan((r+1)A) = \sum_{r=1}^{8}\left(\frac{\tan(r+1)A - \tan(rA) - \tan A}{\tan A}\right) = \frac{\tan 9A - 9\tan A}{\tan A} = -10$

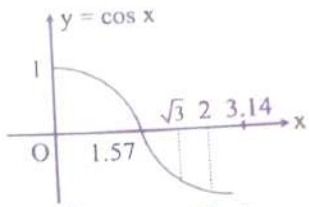
118. Let $P = \cos\theta \cos 2\theta \cos 3\theta \ldots \cos 1004\theta$
$Q = \sin\theta \sin 2\theta \sin 3\theta \ldots \sin 1004\theta$
Then $2^{1004}PQ = \sin 2\theta \sin 4\theta \ldots \sin 2008\theta$
$= (\sin 2\theta \sin 4\theta \ldots \sin 1004\theta)[\sin(2\pi - 1003\theta)\sin(2\pi - 1001\theta) \ldots \sin(2\pi - \theta)]$
$= (\sin 2\theta \sin 4\theta \ldots \sin 1004\theta)[-\sin 1003\theta][-\sin 1001\theta]\ldots[-\sin\theta] = Q \Rightarrow P = \frac{1}{2^{1004}}$.

119. $A^2 = \sin 2 - \sin\sqrt{3}$



$\sin\sqrt{3} > \sin 2$, $A^2 < 0$

$B^2 = \cos 2 - \cos\sqrt{3}$



$\cos\sqrt{3} > \cos 2$, $B^2 < 0$

Both A and B are not real numbers

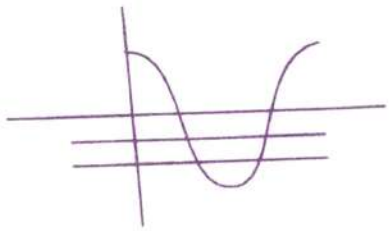
120. $\frac{v_n - v_{n-1}}{u_{n-1}} = \frac{\cos n\theta \cdot \sec^n\theta - \cos(n-1)\theta \cdot \sec^{n-1}\theta}{\sin(n-1)\theta \cdot \sec^{n-1}\theta}$

$= \frac{\cos[(n-1)\theta + \theta] - \cos(n-1)\theta \cdot \cos\theta}{\cos\theta \cdot \sin(n-1)\theta}$

$= \frac{\cos(n-1)\theta\cos\theta - \sin(n-1)\theta \cdot \sin\theta - \cos(n-1)\theta \cdot \cos\theta}{\cos\theta \cdot \sin(n-1)\theta} = -\tan\theta$

$\frac{v_n - v_{n-1}}{u_{n-1}} + \frac{1}{n}\frac{u_n}{v_n} = -\tan\theta + \frac{1}{n} \cdot \tan(n\theta)$

121. $\sqrt{6}\sin x > 0$, $\sqrt{6}\sin x \neq 1$
$7\cos x + 8 > 0$
then $7\cos x + 8 = 6\sin^2 x$
$6\cos^2 x + 7\cos x + 2 = 0$
$6\cos^2 x + 4\cos x + 3\cos x + 2 = 0$
$2\cos x(3\cos x + 2) + 1(3\cos x + 2) = 0$
$\Rightarrow \cos x = -\frac{2}{3}, -\frac{1}{2}$



No solution in 4th quadrant
$\therefore$ only two solutions

122. $\sum_{r=1}^{88}(-1)^{r+1}\frac{1}{\sin^2(r+1)^\circ-\sin^2 1^\circ}$

$=\left(\frac{1}{\sin 1^\circ\sin 3^\circ}+\frac{1}{\sin 3^\circ\sin 5^\circ}+\ldots+\frac{1}{\sin 87^\circ\sin 89^\circ}\right)-\left(\frac{1}{\sin 2^\circ\sin 4^\circ}+\frac{1}{\sin 4^\circ\sin 6^\circ}+\ldots+\frac{1}{\sin 88^\circ\sin 90^\circ}\right)$

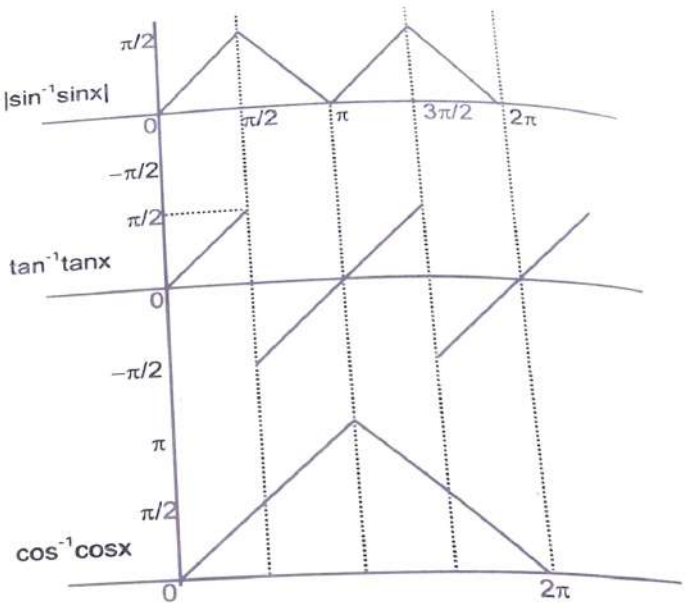
$=\frac{1}{\sin 2^\circ}[(\cot 1^\circ-\cot 3^\circ)+(\cot 3^\circ-\cot 5^\circ)+\ldots+(\cot 87^\circ-\cot 89^\circ)]$

$-[(\cot 2^\circ-\cot 4^\circ)+(\cot 4^\circ-\cot 6^\circ)+\ldots+(\cot 88^\circ-\cot 90^\circ)]$

$=\frac{1}{\sin 2^\circ}(2\cot 2^\circ-\cot 2^\circ)=\frac{\cot 2^\circ}{\sin 2^\circ}$

123. The graph of $|\sin^{-1}\sin x|$

Clearly, $|\sin^{-1}\sin x| > \cos^{-1}\cos x + \tan^{-1}\tan x$ when $x \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)\cup\left(\frac{3\pi}{2}, 2\pi\right)$.



124. $3\sin x + 4\cos(\alpha x) = 7$

It is true at only extreme values

$\Rightarrow \sin x = 1 ; \cos(\alpha x) = 1$

$x = (4n+1)\frac{\pi}{2}$; $\alpha x = 2m\pi$

$\Rightarrow \alpha = \frac{2m\pi}{(4n+1)\frac{\pi}{2}} \Rightarrow \alpha = \frac{4m}{4n+1}$

125. $\frac{1+\cos(\lambda x)}{2}+\frac{1+\cos(\mu x)}{2}=1$

$\cos(\lambda x)+\cos(\mu x)=0$

$2\cos\left(\frac{\lambda+\mu)x}{2}\right).\cos\left(\frac{(\lambda-\mu)x}{2}\right)=0$

$\frac{(\lambda+\mu)x}{2}=(2n+1)\frac{\pi}{2}$ or $\frac{(\lambda-\mu)x}{2}=(2m+1)\frac{\pi}{2}$

$x=\frac{(2n+1)\pi}{\lambda+\mu}$ or $x=\frac{(2m+1)\pi}{\lambda-\mu}$

Common difference $\frac{2\pi}{\lambda+\mu}$ and common difference $\frac{2\pi}{\lambda-\mu}$

Product $\frac{4\pi^2}{\lambda^2-\mu^2}$

126. We have $\tan^{-1}3+\tan^{-1}\left(\frac{9}{2}\right)+\tan^{-1}\left(\frac{33}{4}\right)+\tan^{-1}\left(\frac{129}{8}\right)+\ldots$ to n terms

$=\frac{n\pi}{2}-\left[\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{2}{9}+\ldots\right]$

$=\frac{n\pi}{2}-\left[\tan^{-1}\left(\frac{2-1}{1+1\cdot2}\right)+\tan^{-1}\left(\frac{2^2-2^1}{1+2^2\times2^1}\right)+\ldots+\tan^{-1}\frac{2^n-2^{n-1}}{1+2^{2n-1}\cdot1}\right]=\frac{(2n-1)\pi}{4}-\tan^{-1}2^n \Rightarrow a=4.$

127. $\sin\sin^{-1}[x]=[x]$ if $-1\le[x]\le1$ i.e. $-1\le x<2$

$\cos^{-1}\cos x=x$ if $0\le x\le\pi$

$\Rightarrow$ Given equation becomes $[x]+x=1$ where $0\le x<2$

$[x]=1-x \qquad 0\le x<2$

No solution

Number of solution is zero

128. We have $\frac{1}{\cos(x-\alpha)}\{(1+k)\cos x\cos(2x-\alpha)-(1+k\cos2x)\cos(x-\alpha)\}=0$

or $(1+k)[\cos(3x-\alpha)+\cos(x-\alpha)]-\cos(x-\alpha)-k[\cos(3x-\alpha)+\cos(x+\alpha)]=0$

or $\cos(3x-\alpha)-\cos(x-\alpha)+k\{\cos(x-\alpha)-\cos(x+\alpha)\}=0$

or $\sin x\{k\sin\alpha-\sin(2x-\alpha)\}=0$

Hence, $\sin x=0$ or $\sin(2x-\alpha)=k\sin\alpha$

That is, $x=n\pi$

The other equation has roots if $|k\sin\alpha|\le1 \Rightarrow |k|\le|\operatorname{cosec}\alpha|$

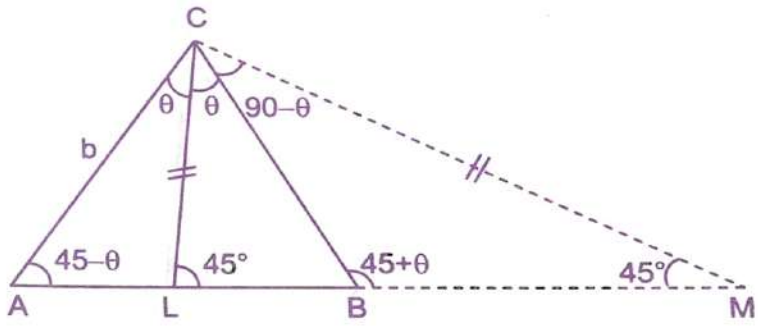
If other equation has repeated roots, then $\sin(2n\pi-\alpha)=k\sin\alpha \Rightarrow k=-1$.

129. $a^2+b^2=4R^2$

$[\sin^2(45-\theta)+\sin^2(135-\theta)]$

$=4R^2[\sin^2(45-\theta)+\cos^2(45-\theta)=4R^2$

$\therefore \sqrt{(a^2+b^2)}=2R$



130. As $\alpha+\beta=90°$

$\Rightarrow \tan\alpha+\tan^2\alpha+\cot\alpha+\cot^2\alpha=4 \Rightarrow x+\frac{1}{x}+x^2+\frac{1}{x^2}=4$; let $\tan\alpha=x$

As $x+\frac{1}{x}\ge2$ for $x>0$ and $x+\frac{1}{x}=2 \Rightarrow x=1$

$\Rightarrow \alpha=\beta=\frac{\pi}{4}$. Hence area of isosceles right-angled triangle $=\frac{d^2}{4}$

131. $\Delta = \frac{1}{2} \times a \times 24 \Rightarrow a = \frac{\Delta}{12}$, $b = \frac{\Delta}{9}$, $c = \frac{5\Delta}{36}$

$s = \frac{1}{2}(a+b+c) = \frac{\Delta}{s}$

So $\Delta^2 = s(s-a)(s-b)(s-c)$

$\Delta = 6^3 \Rightarrow (\Delta)^{\frac{1}{3}} = 6$

132. In ΔAMB

$\frac{x}{\sin 7^\circ} = \frac{a}{\sin 150^\circ}$

$x = 2a \sin 7^\circ$

Using cosine rule in ΔAMC,

$y^2 - x^2 + a^2 - 2ax \cos 83^\circ$

$= 4a^2 \sin^2 7^\circ + a^2 - 4a^2 \sin^2 7^\circ$

$y^2 = a^2$

$\Rightarrow y = a$

Hence, $\angle AMC = 83^\circ$

134. Let $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$

Since centre of this circle $(-g, -f)$ lie on ...(1)

$2x - 2y + 9 = 0$

$-2g + 2f + 9 = 0$

and the circle cuts orthogonally to the circle ...(2)

$x^2 + y^2 - 4 = 0$ ...(3)

$2g \times 0 + 2f \times 0 = c - 4$

$c = 4$

substituting the value of g and c from (3) and (2) in (1) then

$x^2 + y^2 + (2f + 9)x + 2fy + 4 = 0$

$(x^2 + y^2 + 9x + 4) + 2f(x + y) = 0$

Required two fixed points are the points of intersection of circle

$x^2 + y^2 + 9x + 4 = 0$ and $x + y = 0$

Solving we get $(-4, 4)$, $\left(-\frac{1}{2}, \frac{1}{2}\right)$.

135. $\frac{\alpha(\alpha^2-1)+1(\alpha^2-1)}{\alpha^3+\alpha^2+\alpha q-2\alpha-q} + \frac{(\beta+1)^3}{\beta^3+3\beta^2+\beta q+2\beta+q}$

$\frac{(\alpha+1)(\alpha-1)(\alpha+1)}{\alpha(\alpha^2-1)+\alpha(\alpha-1)+q(\alpha-1)} + \frac{(\beta+1)^3}{\beta(\beta^2+2\beta+q)+1(\beta^2+2\beta+q)}$

$\frac{(\alpha+1)^2}{\alpha^2+2\alpha+q+1-1} + \frac{(\beta+1)^2}{\beta^2+2\beta+q+1-1}$

$\frac{(\alpha+1)^2}{(\alpha+1)^2-(\alpha+1)(\beta+1)} + \frac{(\beta+1)^2}{(\beta+1)^2-(\alpha+1)(\beta+1)} = \frac{\alpha+1}{\alpha-\beta} + \frac{\beta+1}{\beta-\alpha} = 1$

136. $\log_{(x-4)}(x-2) - \log_{(x-2)}(x-4)^2 = 1$

$\Rightarrow t - \frac{2}{t} = 1 \Rightarrow t^2 - t - 2 = 0 \Rightarrow t = 2, -1$

$\Rightarrow \log_{(x-4)}(x-2)=2$ or $\log_{(x-4)}(x-2)=-1$

$\Rightarrow x^2-9x+18=0$ $\quad (x-2)(x-4)=1$

$\Rightarrow x=6,3$ $\quad x^2-6x+7=0$

$(x=3$ rejected) $\quad x=3\pm\sqrt{2}$ $(x=3-\sqrt{2}$ rejected)

137. Each of the products required in the problem is one of the term in the expansion of following product

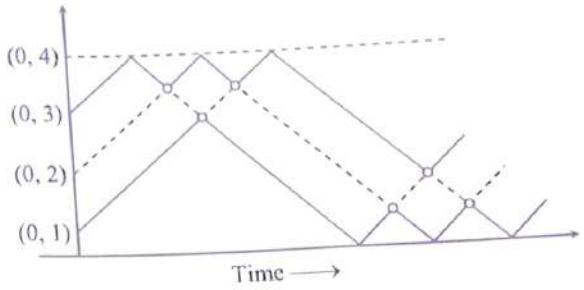
$$(1+1)\left(1+\frac{1}{2}\right)\left(1+\frac{1}{3}\right)\cdots\left(1+\frac{1}{n}\right)-1$$

Which is $2\times\frac{3}{2}\times\frac{4}{3}\times\frac{5}{4}\cdots\frac{(n+1)}{n}-1=n$

Hence, required sum of the products $= n$

138. Motion of the three particles can be visulaised four following graph

Clearly number of particle-particle collisions before ostipulated time is 6



139. Equation of family of straight lines passing through (a, b) is

$y-b=m(x-a)$ ..... (1)

m being the parameter

From equation (1), we get $\frac{dy}{dx}=m$

$\therefore\ y-b=\frac{dy}{dx}(x-a)$ ..... (2)

This is the differential equation of the given family of straight lines (1). Replace $\frac{dy}{dx}$ by $-\left(\frac{dx}{dy}\right)$, the differential equation of required orthogonal trajectories is given by $y-b=-\frac{dx}{dy}(x-a)$

$\Rightarrow (x-a)dx+(y-b)dy=0$

$\Rightarrow (x-a)^2+(y-b)^2=c$

Where, c being an arbitrary constant. The required locus is circle.

140. Let, $2^4\cdot3^5\cdot5^4=N$

$\frac{xy}{x+y}=N\Rightarrow(x-N)(y-N)=N^2$

$[x-N=X,\ y-N=Y]$

$X\cdot Y=N^2=2^8\cdot3^{10}\cdot5^8$

Number of solutions $=9\times11\times9=891$

141. $\frac{d(f(x))^2+(f'(x))^2}{dx}=2f'(x)(f(x)+f''(x))=-2xg(x)(f'(x))^2<0\ \forall\ x>0$

$\therefore (f(x))^2<25-(f'(x))^2<25\ \therefore\ |f(x)|<5$

142. $\lim_{x\to 0}\frac{\log(1+x)-x}{x^2} \Rightarrow \lim_{x\to 0}\frac{\frac{1}{1+x}-1}{2x}=\frac{-x}{(1+x)2x}=-\frac{1}{2}$ and $\lim_{x\to\infty} x-x^2\log\left(1+\frac{1}{x}\right)$

Let $x=\frac{1}{t} \Rightarrow \lim_{t\to 0}\frac{1}{t}-\frac{1}{t^2}\log(1+t) \Rightarrow \lim_{t\to 0}\frac{t-\log(1+t)}{t^2}=\frac{1}{2}$

143. $\lim_{t\to x}\frac{t^2f(x)-x^2f(t)}{t-x}=1$

$\Rightarrow x^2f'(x)-2xf(x)+1=0$

$\Rightarrow \frac{x^2f'(x)-2xf(x)}{(x^2)^2}+\frac{1}{x^4}=0$

$\Rightarrow \frac{d}{dx}\left(\frac{f(x)}{x^2}\right)=-\frac{1}{x^4}$

$\Rightarrow f(x)=cx^2+\frac{1}{3x}$, also $f(1)=1 \Rightarrow c=\frac{2}{3}$

$\therefore f(x)=\frac{2}{3}x^2+\frac{1}{3x}$

144. If e is the eccentricity then, $e^2=\frac{\alpha^2-\beta^2}{\alpha^2}$

Now, we know, $\frac{1}{\alpha^2}+\frac{1}{\beta^2}=a+b \Rightarrow \frac{1}{\alpha^2\beta^2}=ab-h^2$

$\Rightarrow \alpha^2+\beta^2=\frac{a+b}{ab-h^2}$, $\alpha^2\beta^2=\frac{1}{ab-h^2}$

$\therefore \alpha^2-\beta^2=\sqrt{\frac{(a+b)^2-4(ab-h^2)}{(ab-h)^2}}$

$\frac{1}{\alpha^2}-\frac{1}{\beta^2}=\mp\sqrt{(a+b)^2-4(ab-h^2)}$

For an ellipse $\frac{2}{\alpha^2}=a+b-\sqrt{(a+b)^2-4(ab-h^2)}$

So, $e^2=\frac{\sqrt{(a-b)^2+4h^2}}{2(ab-h^2)}\left[(a+b)-\sqrt{(a-b)^2+4h^2}\right]$

145. $y=-\frac{1}{2}(x-1)^2+\frac{3}{2}$

$\Rightarrow$ Axis of the parabola is $x=1$ and any parabola is symmetric about its axis.

146. Given that $\Delta_1=b^2_1-4a_1c_1$ and $\Delta_2=b^2_2-4a_2c_2$

$\Delta_1+\Delta_2<0 \Rightarrow \Delta_1$ and Δ_2 both are less than zero or one of them is less than zero

$\begin{pmatrix}\Delta_1<\Delta_2 \,\&\, \Delta_2<0\\ \text{or } \Delta_2<\Delta_1 \,\&\, \Delta_1<0\end{pmatrix}$

So at least one of them is negative.

$\therefore$ Statement – I is true and statement – II is the correct explanation.

148. Statement 2: $\alpha+\beta=\frac{\pi}{2}-\gamma$, $\frac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta}=\frac{1}{\tan\gamma} \Rightarrow \sum \tan\alpha\tan\beta = 1$

$\therefore$ Statement 2: is true.

Statement 1: $\tan\alpha\tan\beta=\frac{a!}{6}$, $\tan\beta\tan\gamma=\frac{b!}{2}$ and $\tan\alpha\tan\gamma=\frac{c!}{3}$

$\frac{a!}{6}+\frac{b!}{2}+\frac{c!}{3}=1 \Rightarrow a!=1, b!=1, c!=1 \Rightarrow \tan\alpha\tan\beta, \tan\gamma\tan\alpha$ and $\tan\beta\tan\gamma$ are in A.P.

$\therefore$ Statement 1: is false.

149. Statement-2: $\begin{bmatrix} a_{11} & 0 & 0 & \ldots \\ 0 & a_{22} & \ldots & \\ \ldots & \ldots & 0 & a_{nn} \end{bmatrix}\begin{bmatrix} \frac{1}{a_{11}} & 0 & 0 & \ldots \\ \ldots & \frac{1}{a_{22}} & \ldots & \end{bmatrix} = I$. If $a_{ii} \neq 0$

150. Any point on the first line is $(2r+1, r-3, -3r+2)$
Any point on the second line is $(s+2, -3s+1, 2s-3)$
Two lines will intersect (be coplanar) if $2r+1=s+2$, $r-3=-3s+1$ and $-3r+2=2s-3$ are consistent
i.e. $2r-s=1$, $r+3s=4$ and $3r+2s=5$ are consistent. Condition for this is

$$\begin{vmatrix} 2 & -1 & -1 \\ 1 & 3 & -4 \\ 3 & 2 & -5 \end{vmatrix} = 0$$, which is true.

So, statement–1 and statement–2 are true and statement–2 is a correct explanation for statement–1

151. Period cannot be rational number as solution is unique.

152. $k(x) = h(g(f(x))) = \frac{5x^2}{8-x^2}$

Domain of $k(x) = R - \{0, \pm 2, \pm 2\sqrt{2}\}$
Range of $k(x)$ is $(-\infty, -5) \cup (0, \infty) - \{5\}$
or $R - ([-5, 0] \cup \{5\})$
Hence, B is $[-5, 0] \cup \{5\}$

153. $h(x) = f(x) + g(x) = \begin{cases} -1 & , \ -\infty < x < 1 \\ a+4-2x & , \ 1 \le x < 2 \\ a-b-1+x & , \ 2 \le x < \infty \end{cases}$

We must have either $a = -3, b \neq 1$ or $b = 1, a \neq -3$

154. As $\lim\limits_{x\to 1}\left(1+\ln x\cdot\ln\left(k^2+25\right)\right)^{\frac{1}{|x|-1}} = \lim\limits_{x\to 1}\left(\left(1+\ln x\cdot\ln\left(k^2+25\right)\right)\right)^{\frac{1}{x-1}}$

$= e^{\lim\limits_{x\to 1}\frac{\ln x\ln\left(k^2+25\right)}{x-1}} = e^{\ln\left(k^2+25\right)} = k^2+25$

$\therefore k^2 = 25 = k(2\sin^2\alpha + 3\cos\beta + 5)$

$\Rightarrow 2\sin^2\alpha + 3\cos\beta + 5 = k + \frac{25}{k}$

$\because \frac{k+\frac{25}{k}}{2} \ge \sqrt{k\cdot\frac{25}{k}} \Rightarrow k+\frac{25}{k} \ge 10$

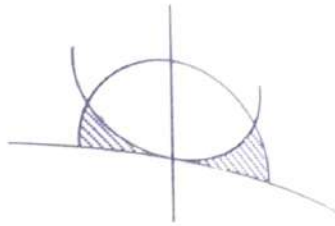
But L.H.S. ≤ 10

So, both L.H.S. and R.H.S. should be equal to 10

$\Rightarrow \sin^2\alpha = 1$ and $\cos\beta = 1$

Hence, $\dfrac{\sin^{10}\alpha+\cos^5\beta}{\sin^2\alpha+\cos\beta} = \dfrac{1+1}{1+1} = 1$

155. $f'(x) > 0 \Rightarrow f(x)$ is an increasing function for $x \geq 1$



$$\therefore \frac{1}{x^2+f^2(x)} = f'(x) \leq \frac{1}{x^2+1}$$

$$\int_1^a f'(x)\,dx \leq \int_1^a \frac{1}{x^2+1}\,dx$$

$\therefore f(a) - f(1) \leq \tan^{-1}x]_1^a = \tan^{-1}a - \pi/4$

$f(a) \leq 1 + \tan^{-1}a - \pi/4 \leq 1 + \pi/4$

max of $f(a) = 1 + \pi/4$

[max of $f(a)$] $= 1$

$$2\left(\int_0^1 x^2dx + \int_0^{\sqrt{2}} \sqrt{2-x^2}\,dx\right) = \frac{\pi}{4}+\frac{1}{6}$$

156. $f'(x)\leq 0 \Rightarrow f(x)f'(x)+f'(x)f''(x)\leq 0$

$$\frac{d}{dx}\left(\left(f(x)\right)^2+\left(f'(x)\right)^2\right)\leq 0 \Rightarrow \left(f^2(x)\right)+\left(f'(x)\right)^2 \leq f^2(0)+\left(f(0)\right)^2 \quad \forall x \geq 0$$

$$\Rightarrow f(x) = 0 \forall x \geq 0$$

157. $$f(x) = \int_a^x \left(\frac{f(t)-f(a)}{(t-a)^2} - \frac{f'(a)}{(t-a)}\right)dt$$

$$f'(x) = \frac{f(x)-f(a)}{(x-a)^2} - \frac{f'(a)}{(x-a)} \neq 0$$

Using LMVT we get $\dfrac{f(x)-f(a)}{(x-a)^2} - \dfrac{f'(a)}{(x-a)} = \dfrac{f''(a)}{2}$

158. $$\int \cos x\left(x\ln x + \frac{1}{x}\right)dx = \int \underset{\downarrow v}{\cos x}\left(\underset{\downarrow u}{x\ln x}\right)dx + \int \underset{\downarrow u'}{\cos x}\cdot\underset{\downarrow v'}{\frac{1}{x}}\,dx$$

$$= x\ln x\cdot\sin x - \int(\ln x+1)\sin x\,dx + \cos x\ln x + \int \sin x\cdot\ln x\,dx$$

$$= x\ln x\sin x + \cos x + \cos x\ln x + c = f(x) + c$$

$$\text{Now } \int \frac{\cos x(1+\ln x)+x\ln x\sin x}{\cos^2 x}dx = \frac{x\ln x}{\cos x}+c$$

$$\text{Now } L = \lim_{x\to 0^+}\frac{x\ln x}{\cos x} = 0 \text{ and } \lim_{x\to\infty}\frac{g(x)}{x^2} = \lim_{x\to\infty}\frac{\ln x}{x\cos x}, \text{ when } x = (2n+1)\frac{\pi}{2},\ n\to\infty, \frac{\ln x}{x\cos x} \text{ is undefined}$$

159. $I = \int (3\sin\theta - 4\sin^3\theta)e^{\sin\theta}\cos\theta d\theta = (A\sin^3\theta - B\sin^2\theta + C\sin\theta + D\cos\theta + B + E)e^{\sin\theta} + F$

Taking $\sin\theta = t$

$I = \int (3t - 4t^3)e^t dt = \underbrace{(At^3 - Bt^2 + Ct + B + E)}_{f(t)}e^t + F$

$D = 0$ as $f(t)$ is a polynomial in t.

$(3t - 4t^3)e^t = (At^3 - Bt^2 + Ct + B + E)e^t + (3At^2 - 2Bt + C)e^t$

$A = -4$

$3A = B$

$\Rightarrow B = -12$

160. $f(g(x)) = x$

$f'(g(x))g'(x) = 1$

$$g'(x) = \frac{1}{f'(g(x))}$$

$$g'(e) = \frac{1}{f'(g(e))} = \frac{1}{f'(1)} = \frac{1}{e+1}$$

$$g''(x) = \frac{-1}{f'(g(e))^2} f''(g(x))g'(x)$$

$$g''(e) = \frac{-1}{(f'(1))^2} f''(1)\cdot g'(e)$$

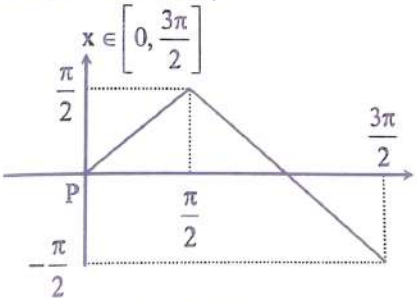
161. $\int \cos(2015x + x)\sin^{2014} x dx$

$$= \int \underset{\downarrow}{\cos 2015x}\cdot\cos x\cdot\underset{\downarrow}{\sin^{2014} x}\, dx - \int \sin 2015x\cdot\sin^{2015} x\, dx$$

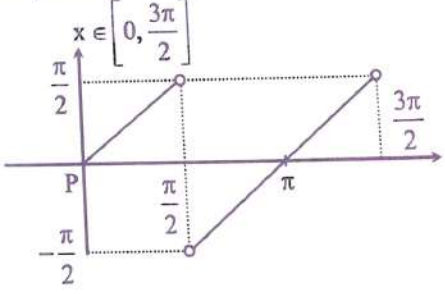
$$= \frac{\cos 2015x\cdot\sin^{2015} x}{2015} + \frac{2015}{2015}\int \sin 2015x\cdot\sin^{2015} x\, dx - \int \sin 2015x\cdot\sin^{2015} x\, dx$$

$$= \frac{\cos 2015x\cdot\sin^{2015} x}{2015} + c$$

162. Graph of $\sin^{-1}(\sin x)$



Graph of $\tan^{-1}(\tan x)$



Clearly area bounded by

$\max((\sin^{-1}(\sin x), \tan^{-1}(\tan x))$ in $x \in \left(0, \frac{3\pi}{2}\right)$ is

$$\frac{1}{2}\left(\frac{\pi}{2}\right)(\pi) + \frac{1}{2}\left(\frac{\pi}{2}\right)\left(\frac{\pi}{2}\right) = \frac{3\pi^2}{8}$$

163. $f(x) = \int_a^x \frac{1}{f'(x)}dx \Rightarrow f'(x) = \frac{1}{f'(x)} \cdot 1 - 0 \Rightarrow f(x)f'(x) = 1$

$\Rightarrow \int f(x)f'(x)dx = \int 1.dx \Rightarrow \frac{(f(x))^2}{2} = x + c \ldots(1)$

Given $\int_a^1 [f(x)]^{-1}dx = \sqrt{2} \Rightarrow f(1) = \sqrt{2}.$

From (1), $\frac{(f(1))^2}{2} = 1 + c \Rightarrow c = 0$

$f(x) = \pm \sqrt{2x}$

But $f(1) = \sqrt{2} \Rightarrow f(x) = \sqrt{2x} \Rightarrow f(2) = 2.$

$f'(x) = \frac{1}{\sqrt{2x}} \Rightarrow f'(2) = \frac{1}{2}$

$\int_0^1 f(x)dx = \frac{2^{3/2}}{3}$

$f^{-1}(x) = \frac{x^2}{2}$

$\Rightarrow f^{-1}(2) = 2.$

164. $\frac{dy}{dx} - y\left(\frac{\cos x}{\sin x}\right) = -\frac{\sin x}{x^2}$

$\Rightarrow \frac{1}{\sin x}\frac{dy}{dx} - y\cot x \operatorname{cosec} x = -\frac{1}{x^2}$

$\Rightarrow y = \frac{\sin x}{x} + c\sin x$

As $y \to 0$ and $x \to \infty$

$\Rightarrow c = 0$

$\Rightarrow y = \frac{\sin x}{x}$

165. Given integral equation is

$$y = (x-1)(\ln x - 1) + \int_1^x f(x)dx \quad \ldots\ldots (1)$$

Differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = (x-1)\left(\frac{1}{x}\right) + (\ln x - 1) + f(x)$$

$$\Rightarrow \frac{dy}{dx} = 1 - \frac{1}{x} + \ln x - 1 + y$$

$$\Rightarrow \frac{dy}{dx} - y = \left(\ln x - \frac{1}{x}\right) \quad \ldots\ldots (2)$$

Which is a linear differential equation with I.F. $= e^{\int(-1)dn} = e^{-x}$

$\therefore$ Solution of (2) is given by $y \cdot e^{-x} = -\int e^{-x}\left(\frac{1}{x} - \ln x\right)dx$

Let $(-x) = t$

$\Rightarrow -dx = dt$

$\Rightarrow ye^t = \int e^t\left(\ln(-t)+\frac{1}{t}\right)dt = ye^t = e^t\ln(-t)$

$\Rightarrow y = \ln(-t) + ce^{-t}$

$\Rightarrow y = \ln x + ce^x$

166. $\cos x = (5 + [\sin x]) - ([\cos x] + 3) = 2 + [\sin x] - [\cos x]$

$-1 \le \cos x \le 1$

$\Rightarrow -1 \le 2 + [\sin x] - [\cos x] \le 1$

$\Rightarrow -3 \le [\sin x] - [\cos x] \le -1$

$\Rightarrow [\sin x] - [\cos x] \ge -3$ is always true

$[\sin x] - [\cos x] \le -1$ is true only when $[\sin x] = 0, [\cos x] = 1$ or $[\sin x] = -1, [\cos x] = 0$

Only solution are $x = 0, 2\pi, 4\pi, \ldots$ etc

167. Equation of bisector is given by $\frac{x+y-1}{\sqrt{2}} = \pm\left(\frac{7x-y-5}{5\sqrt{2}}\right)$

or $2x - 6y = 0; 6x + 2y = 5$

Line parallel to this and passing through $(1, 3)$ are $x - 3y + 8 = 0$ and $3x + y - 6 = 0$

Solving these lines with $y = 2x$, we get $\left(\frac{8}{5}, \frac{16}{5}\right)$ and $\left(\frac{6}{5}, \frac{12}{5}\right)$

168. a is the least side

$\therefore$ order of AP can be b, c, a or c, b, a

Case I : If $2c = a + b$

$\cos A = \frac{b^2+c^2-a^2}{2bc}$

$= \frac{b^2+c^2-(2c-b)^2}{2bc} = \frac{4b-3c}{2b}$

Case II : If $2b = a + c$

$\cos A = \frac{b^2+c^2-(2b-c)^2}{2bc} = \frac{4c-3b}{2c}$

169. $\sin\frac{\pi}{24} = \frac{2}{R-2} \Rightarrow R = 2\left(1+\operatorname{cosec}\frac{\pi}{24}\right)$

(A) is true $\because$ $\frac{\left(\sin\frac{\alpha}{2}+\cos\frac{\alpha}{2}\right)^2}{\sin A} = 1+\operatorname{cosec}\alpha$

(D) is true.

170. Curve through the intersection of S_1 and S_2 is given by $S_1 + \lambda S_2 = 0$

$\Rightarrow x^2(\sin^2\theta + \lambda\cos^2\theta) + 2(h\tan\theta - \lambda h'\cot\theta)xy + (\cos^2\theta + \lambda\sin^2\theta)y^2$

$+ (32 + 16\lambda)x + (16 + 32\lambda)y + 19(1 + \lambda) = 0$

The above equation will represent a circle if $\sin^2\theta + \lambda\cos^2\theta = \cos^2\theta + \lambda\sin^2\theta$

$\Rightarrow (1 - \lambda)(\sin^2\theta - \cos^2\theta) = 0 \Rightarrow \lambda = 1$ or $\theta = \frac{\pi}{4}$

$h\tan\theta - \lambda h'\cot\theta = 0$

$\Rightarrow h\tan\theta = \lambda h'\cot\theta$ which is satisfied if $\lambda = 1$ and $\theta = \frac{\pi}{4} \Rightarrow h = h'$.

171. Let $(x_1, y_1) \equiv (at^2, 2at)$

Tangent at this point is $ty = x + at^2$

Any point on this tangent is $\left(h, \frac{h+at^2}{t}\right)$

Chord of contact of this point with respect to the circle $x^2 + y^2 = a^2$ is

$hx + \left(\frac{h+at^2}{t}\right)y = a^2$ or $\left(aty - a^2\right) + h\left(x + \frac{y}{t}\right) = 0$

Which is a family of straight lines passing through point of intersection of

$ty - a = 0$ and $x + \frac{y}{t} = 0$

So, the fixed point is $\left(-\frac{a}{t^2}, \frac{a}{t}\right)$

$\therefore x_2 = -\frac{a}{t^2}, y_2 = \frac{a}{t}$

Clearly, $x_1x_2 = -a^2, y_1y_2 = 2a^2$

Also, $\frac{x_1}{x_2} = -t^4 \quad \frac{y_1}{y_2} = 2t^2$

$\Rightarrow 4\frac{x_1}{x_2} + \left(\frac{y_1}{y_2}\right)^2 = 0$

172. Circle on PQ as diameter

$\left(x - t_1^2\right)\left(x - t_2^2\right) + (y - 2t_1)(y - 2t_2) = 0$

$x^2 + y^2 - \left(t_1^2 + t_2^2\right)x - 2(t_1 + t_2)y + t_1^2t_2^2 + 4t_1t_2 = 0$

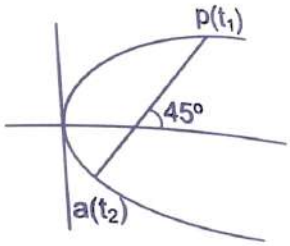
Put $\frac{t_1 + t_2}{2} = 1$ and $t_1t_2 = -1$

Hence equation of circle become

$x^2 + y^2 - 6x - 4y - 3 = 0$ ..... (1)

With centre (3, 2) and radius 4

Equation (1) is not orthogonal to $x^2 + y^2 + 2x - 6y + 3 = 0$



173. Shifting the origin at vertex equation of perpendicular bisector of the focal chord is

$y - 2(t_1 + t_2) = -\frac{1}{\sqrt{3}}\left(x - \left(t_1^2 + t_2^2\right)\right)$; where $t_1 + t_2 = \frac{2}{\sqrt{3}}$

Putting $y = 0$

R is $t_1^2 + t_2^2 + \left(2\sqrt{3}\right)(t_1 + t_2) = \frac{22}{3}$

$\therefore RS = \frac{16}{3}$

174. $f(\alpha) = \frac{\alpha}{\sqrt{1-\alpha^2}}$

$\int e^{\alpha}\left(f(\alpha) - f''(\alpha)\right)d\alpha = e^{\alpha}\left(f(\alpha) - f'(\alpha)\right) = e^{\alpha}\left(\frac{\alpha}{\sqrt{1-\alpha^2}} + \frac{1}{\sqrt{\left(1-\alpha^2\right)^3}}\right) + c$

$ffff(\alpha) = \frac{\alpha}{\sqrt{1-4\alpha^2}} \Rightarrow \int_0^{1/2} ffff(\alpha)d\alpha = \frac{1}{4}$

175. Any point on the line through P(α, 2) is (α + r cos θ, 2 + r sin θ) and meets the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ at A and D

$$\Rightarrow \frac{(\alpha+r\cos\theta)^2}{9}+\frac{(2+r\sin\theta)^2}{4}=1$$

$$\Rightarrow (4\cos^2\theta+9\sin^2\theta)r^2+4(2\alpha\cos\theta+9\sin\theta)r+4\alpha^2=0$$

$$\Rightarrow PA\cdot PD=\frac{4\alpha^2}{4\cos^2\theta+9\sin^2\theta}$$

Again solving with $xy = 0$

$$\Rightarrow PB\cdot PC=\frac{2\alpha}{\sin\theta\cos\theta}$$

$$\Rightarrow \frac{4\alpha^2}{4\cos^2\theta+9\sin^2\theta}=\frac{2\alpha}{\sin\theta\cos\theta}$$

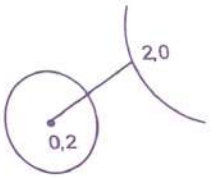
$$\Rightarrow 2\alpha\sin 2\theta+5\cos 2\theta=13$$

$$\Rightarrow -1\le\frac{13}{\sqrt{4\alpha^2+25}}\le 1 \Rightarrow |\alpha|\ge 6$$

176. Radius of director circle $=\sqrt{a^2-b^2}=2$

$$\left(2\sqrt{2}\right)^2-b^2=4$$

$$b^2=4 \qquad e=\frac{\sqrt{3}}{\sqrt{2}}$$



$\therefore$ Focus lies on the line $\frac{x-0}{\frac{-1}{\sqrt{2}}}=\frac{y-2}{\frac{1}{\sqrt{2}}}$ at a distance ae units from centre.

$$\therefore \quad \frac{x-0}{-\frac{1}{\sqrt{2}}}=\frac{y-2}{\frac{1}{\sqrt{2}}}=\pm 2\sqrt{3}$$

$$(x,y)=\left(\sqrt{6},2-\sqrt{6}\right) \text{ or } \left(-\sqrt{6},2+\sqrt{6}\right)$$

177. Any tangent to $xy = 4\sin^2\theta$ is $y = mx \pm 4\sin\theta\sqrt{-m}$.

If it is normal to any circle of given family it will pass through the centre of the circle i.e. (1, 1)

$$1 = m \pm 4\sin\theta\sqrt{-m}$$

$$\Rightarrow m^2+2[8\sin^2\theta-1]m+1=0$$

for non real roots $D < 0$

$$\Rightarrow \sin^2\theta<\frac{1}{4} \text{ or } -\frac{1}{2}<\sin<\frac{1}{2}.$$

$$\therefore \theta\in\left(0,\frac{\pi}{6}\right)\cup\left(\frac{5\pi}{6},\frac{7\pi}{6}\right)\cup\left(\frac{11\pi}{6},2\pi\right).$$

178. $\frac{1}{t_1t_2}=-1;\ (x-xt_1)(x-ct_2)+\left(y-\frac{c}{t_1}\right)\left(y-\frac{c}{t_2}\right)=1 \ [t_1t_2=-1]$

$$\Rightarrow \left(x^2+y^2-2c\right)-(t_1+t_2)(x-y)=0$$

$$\Rightarrow S-\lambda L=0$$

Hence, we get (c, c) and (–c, –c)

179. If P(x, y, z) then the figure

$x = r\sin\theta\sin\phi$
$y = r\sin\theta\sin\phi$
$z = r\cos\theta$
$\Rightarrow 1 = r\sin\theta\cos\phi$
$2 = r\sin\theta\sin\phi$
$3 = r\cos\theta$
$\Rightarrow 1^2 + 2^2 + 3^3 = r^2$
$\Rightarrow r = \pm\sqrt{14}$
$\sin\theta\cos\phi = \frac{1}{\sqrt{14}}$ ($r = \sqrt{14} \because \theta$ & ϕ are acute)

$\sin\theta\sin\phi = \frac{2}{\sqrt{14}}$

$\cos\theta = \frac{3}{\sqrt{14}}$

$\therefore \frac{\sin\theta\sin\phi}{\sin\theta\cos\phi} = \frac{2}{1} \Rightarrow \tan\phi = 2$, Also $\tan\theta = \frac{\sqrt{5}}{3}$

180. Let $f(x) = (2x - a)(3x - b) + (3x - b)(4x - c) + (4x - c)(2x - a) = 0$

$f\left(\frac{a}{2}\right) = \left(\frac{3a}{2} - b\right)(2a - c) > 0$ since $\frac{a}{2} < \frac{b}{3} < \frac{c}{4}$

$f\left(\frac{b}{3}\right) = \left(\frac{4b}{3} - c\right)\left(\frac{2b}{3} - a\right) < 0$

Since $f\left(\frac{a}{2}\right)f\left(\frac{b}{3}\right) < 0$

$\Rightarrow$ one real root between $\left(\frac{a}{2}, \frac{b}{3}\right)$

$\Rightarrow$ Both roots are real
$\therefore$ Equation is quadratic in nature

Also, $f\left(\frac{c}{4}\right) > 0$

$f\left(\frac{b}{3}\right)f\left(\frac{c}{4}\right) < 0 \Rightarrow$ other between $\left(\frac{b}{3}, \frac{c}{4}\right)$

181. Given $a \neq b \neq c$, $a, b, c \in R$
Now $ax^2 + bx + c \geq 0 \Rightarrow b^2 - 4ac \leq 0$ and $a > 0$ ...(1)
$bx^2 + cx + a \geq 0 \Rightarrow c^2 - 4ab \leq 0$ and $b > 0$ ...(2)
$cx^2 + ax + b \geq 0 \Rightarrow a^2 - 4bc \leq 0$ and $c > 0$ ...(3)
Equality can not hold simultaneously ($\because a \neq b \neq c$)

$\therefore \frac{a^2 + b^2 + c^2}{ab + bc + ca} < 4$

and since $a \neq b \neq c$
$\Rightarrow (a - b)^2 + (b - c)^2 + (c - a)^2 > 0$

$\Rightarrow \frac{a^2 + b^2 + c^2}{ab + bc + ca} > 1$

$\Rightarrow \frac{a^2 + b^2 + c^2}{ab + bc + ca} \in (1, 4)$

182. $\sqrt{\sin^4 x + 4\cos^2 x} + \sqrt{\cos^4 x + 4\sin^2 x}$

$= \sqrt{\sin^4 x - 4\sin^2 x + 4} + \sqrt{\cos^4 x + 4 - 4\cos^2 x}$

$= 2 - \sin^2 x + 2 - \cos^2 x = 3$

equation reduces to

$= \sqrt{1+\cos 2x} = \sqrt{2}\sin^{-1}(\sin 2x)$

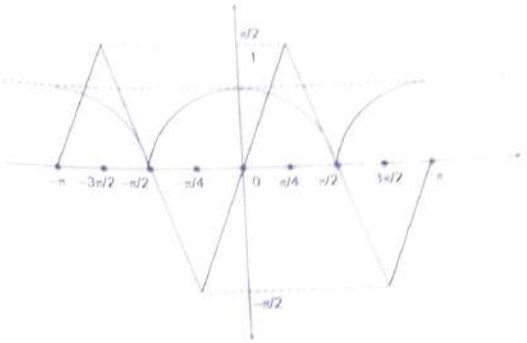
$|\cos x| = \sin^{-1}(\sin 2x)$

$f(\pi/8) = |\cos \pi/8| - \sin^{-1}\sin \pi/4$

$= \cos\pi/8 - \pi/4 = \sqrt{\frac{1+\cos\frac{\pi}{4}}{2}} - \pi/4 > 0$

$f(\pi/6) = \left|\cos\frac{\pi}{6}\right| - \sin^{-1}\frac{\pi}{3} = \frac{\sqrt{3}}{2} - \frac{\pi}{3} < 0$

so one root exists between $(\pi/8, \pi/6)$.



183. $xP(x-n) = (x-1)P(x)$

$\Rightarrow P(1-n) = 0,\ P(0) = 0$ and $(n-1)P(n) = 0$

$\therefore$ either $n = 1$ or $P(n) = 0$

If $n = 1$, then $xP(x-1) = (x-1)P(x)$

i.e. $\frac{P(x)}{P(x-1)} = \frac{x}{x-1}$ clearly $P(x) = kx$ is a possible solution

Thus $P(x)$ may be equal to $2x, -2x$

184. $a_{n/2}$ and $a_{(n+1)/2}$ are both 1, Note that the number of ways to choose numbers left of $a_{n/2}$ and $a_{\frac{n+1}{2}}\left(a_1 > a_2 > a_3 \ldots a_{\frac{n}{2}}; a_{\frac{n+1}{2}}\right)$ is same as that of right. We started with 9

$n = 9 \Rightarrow {}^8C_4 = 70(\text{rejected})$

$n = 10 \Rightarrow {}^9C_4 = 126(\text{rejected})$

$n = 11 \Rightarrow {}^{10}C_5 = 252$

$n = 12 \Rightarrow {}^{11}C_5 = 462$

$n = 13 \Rightarrow {}^{12}C_6 = 924(\text{rejected})$

185. Coefficient of $x^k = a_k = {}^{n+1}C_{k+1}$

$a_{n-2} = {}^{n+1}C_{n-1} = \frac{n(n+1)}{2}$

$\sum_{k=0}^{n} a_k = {}^{n+1}C_1 + {}^{n+1}C_2 + \ldots + {}^{n+1}C_{n+1} = 2^{n+1} - 1$

$a_p > a_{p-1} \Rightarrow p < \frac{n}{2}$

$(a_9)^2 - (a_8)^2 = {}^{n+2}C_{10}({}^{n+1}C_{10} - {}^{n+1}C_9)$

186. $\frac{1}{n+1} + \frac{1}{n+2} + \ldots + \frac{1}{2n} = \left(1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{2n}\right) - \left(1 + \frac{1}{2} + \ldots + \frac{1}{n}\right)$

$= \left(1 + \frac{1}{3} + \ldots + \frac{1}{2n-1}\right) + \frac{1}{2}\left(1 + \frac{1}{2} + \ldots + \frac{1}{n}\right) - \left(1 + \frac{1}{2} + \ldots + \frac{1}{n}\right)$

$= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} \ldots \frac{1}{2n-1} - \frac{1}{2n} = a(2n)$

Also $a(2n) < \frac{\overbrace{\frac{1}{n}+\frac{1}{n}+\ldots+\frac{1}{n}}^{n \text{ times}}}{} = 1$

$a(2n) \geq \frac{1}{2n}+\frac{1}{2n}+\ldots+\frac{1}{2n} \geq \frac{1}{2}$

187. $A_n = 2\int_0^{\frac{a_n}{2}} 2\sqrt{a_n}\sqrt{x}dx + 2\int_{\frac{a_n}{2}}^{a_n} 2\sqrt{a_n}\sqrt{a_n - x}dx$

$= 4\sqrt{a_n}\frac{2}{3}\left(x^{3/2}\right)_0^{\frac{a_n}{2}} - 4\sqrt{a_n}\frac{2}{3}\left((a_n - x)^{3/2}\right)_{\frac{a_n}{2}}^{a_n}$

$A_n = \frac{4\sqrt{2}}{3}a_n^2$

As $a_1, a_2 \ldots\ldots a_n$ are in G.P.

$\Rightarrow a_1^2, a_2^2 \ldots\ldots a_n^2$ are in G.P.

$\Rightarrow A_1, A_2 \ldots\ldots A_n$ are in G.P.

188. Since each tennis player played $(n-1)$ games, $x_i + y_i = n - 1 \ \forall$ i. Altogether there are as many victories as losses, hence $x_1 + x_2 + \ldots\ldots + x_n = y_1 + y_2 \ldots\ldots + y_n$

We have $x_1^2 + x_2^2 + \ldots\ldots + x_n^2 - y_1^2 - y_2^2 - y_3^2 \ldots\ldots - y_n^2$

$= \left(x_1^2 - y_1^2\right) + \left(x_2^2 - y_2^2\right) + \ldots\ldots + \left(x_n^2 - y_n^2\right)$

$= (x_1 + y_1)(x_1 - y_1) + (x_2 + y_2)(x_2 - y_2) + \ldots\ldots + (x_n + y_n)(x_n - y_n)$

$= (n-1)(x_1 - y_1 + x_2 - y_2 + \ldots\ldots + x_n - y_n)$

$= (n-1)((x_1 + x_2 + \ldots\ldots + x_n) - (y_1 + y_2 + \ldots\ldots + y_n)) = 0$

Hence, option (C) is also correct

189. The sequence of two horses reaching the finishing line can be 2 or 1 + 1

$\Rightarrow$ Number of ways = 1 + 2! = 3

The sequence of three horses reaching the finishing line and number of ways are shown below

$3 \Rightarrow 1$ way

$2 + 1 \Rightarrow {}^3C_2 = 3$ ways

$1 + 2 \Rightarrow {}^3C_1 = 3$ ways

$1 + 1 + 1$

$\Rightarrow 3! = 6$

Hence, $H_3 = 1 + 3 + 3 + 6 = 13$

The sequence of four horses reading the finishing line and number of ways are shown below

$4 \Rightarrow 1$ ways

$3 + 1 \Rightarrow {}^4C_3$ ways

$1 + 3 \Rightarrow {}^4C_1$ ways

$2 + 2 \Rightarrow {}^4C_2$

$2 + 1 + 1 \Rightarrow \left(\frac{4!}{2!1!1!2!}\right) \times 3!$ ways

$1 + 1 + 1 + 1$

$\Rightarrow 4!$ ways

Hence, $H_4 = 1 + 4 + 4 + 6 + 36 + 24 = 75$

190. Blank plane is one region

Every line adds are region to the plane

Every point of intersection in plane adds one more region

Hence, $P_n = 1 + {}^nC_2 + n = \frac{n^2+n+2}{2}$

191. Any selection of 6 has to contain at least one couple as there are only 5 couples.

Total number of ways of selecting 6 out of $10 = {}^{10}C_6 = \frac{10\times9\times8\times7}{4\times3\times2} = 210$

Exactly one couple $= {}^5C_1 \times 2^4 = 80$ (choose couple and then 1 partner out of the remaining 4 couples)
Exactly 2 couple and 3 men and 3 women $= {}^5C_2 \times {}^3C_1 \times {}^2C_1 = 60$ (choose 2 couples and couple from which to choose man and couple from which to choose woman)

192. Number of natural numbers between 100 and 1000 which can be formed with the digits 1, 2, 3, 4, 5, 6, 7 if each digit is used atmost once $= 7 \times 6 \times 5 = 210$
Option (B) $= {}^{12}C_3 - {}^5C_3 = 210$
Option (C) $= {}^{10}C_4 \times 1 = 210$
Option (D) $= {}^{10}C_4 = 210$

193. Probability that exactly 2 persons will get nothing $= {}^{n+2}C_2 \frac{n!}{(n+2)^n} = \frac{(n+2)(n+1)...n!}{2.(n+2)^n} = \frac{(n+1)!}{2(n+2)^{n-1}}$

Probability that exactly 3 persons will get nothing $= {}^{n+2}C_3 . \frac{{}^{n-1}C_1 . {}^nC_2 . (n-2)!}{(n+2)^n}$

194. Probability that 9th one examined is the last defective $= \frac{{}^4C_3 \times {}^{11}C_5}{{}^{15}C_8} \times \frac{1}{7} = \frac{8}{195}$

195. Required probability $= \left(1-\frac{1}{2}\right)\left(1-\frac{1}{4}\right)\left(1-\frac{1}{6}\right)\left(1-\frac{1}{8}\right).....\left(1-\frac{1}{2012}\right) = \frac{\lfloor 2012}{2^{2012}\left(\lfloor 1006\right)^2}$

196. Let w be the event that the ball drawn is white

Then $P(w) = \frac{x}{y}$

When, ball is lost $P(w) = \frac{x}{y}\frac{x-1}{y-1} + \frac{y-x}{y}\frac{x}{y-1} = \frac{x}{y}$

When 2 balls is lost $P(w) = \frac{{}^xC_2}{{}^yC_2}\frac{x-2}{y-2} + \frac{{}^xC_1\ {}^{y-x}C_1}{{}^yC_2}\frac{x-1}{y-2} + \frac{y-{}^xC_2}{{}^yC_2}\frac{x}{y-2} = \frac{x}{y}$

Similarly for $n = 3$, $P(w) = \frac{x}{y}$

197. Given $|z_1| = |z_2| = |z_3| = 1 \Rightarrow z_1 = \frac{1}{\bar{z}_1}$ etc.

Also $\frac{z_1^2}{z_2z_3} + \frac{z_2^2}{z_1z_3} + \frac{z_3^2}{z_1z_2} + 1 = 0 \Rightarrow (z_1)^3 + (z_2)^3 + (z_3)^3 + z_1z_2z_3 = 0$

$\Rightarrow (z_1)^3 + (z_2)^3 + (z_3)^3 - 3z_1z_2z_3 = -4z_1z_2z_3$

$(z_1+z_2+z_3)\left[(z_1)^2 + (z_2)^2 + (z_3)^2 - \sum z_1z_2\right] = -4z_1z_2z_3$

$\sum z_1\left[\left(\sum z_1\right)^2 - 3\sum z_1z_2\right] = -4z_1z_2z_3$

Let $z_1 + z_2 + z_3 = z \Rightarrow \bar{z}_1 + \bar{z}_2 + \bar{z}_3 = \bar{z}$

$$z\left[z^2 - 3\sum z_1 z_2\right] = -4z_1 z_2 z_3$$

$$z^3 = 3z\sum z_1 z_2 - 4z_1 z_2 z_3$$

$$z^3 = z_1 z_2 z_3\left[3z\left(\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right) - 4\right] = z_1 z_2 z_3\left[3z\left(\bar{z}_1 + \bar{z}_2 + \bar{z}_3\right) - 4\right]$$

$z^3 = z_1 z_2 z_3[3|z|^2 - 4]$

$\therefore |z|^3 = |3|z|^2 - 4|$ ..... (1)

Now if $|z| \geq \frac{2}{\sqrt{3}}$

Then $|z|^3 = 3|z|^2 - 4 \Rightarrow |z|^3 - 3|z|^2 + 4 = 0$

$\Rightarrow |z|^2(|z| - 2) - |z|(|z| - 2) - 2(|z| - 2) = 0 \Rightarrow (|z| - 2)(|z|^2 - |z| - 2) = 0$

$\Rightarrow (|z| - 2)(|z| - 2)(|z| - 2)(|z| + 1) = 0$

$\Rightarrow |z| = 2$ or $|z| = -1$ (rejected)

Now if $0 < |z| < \frac{2}{\sqrt{3}}$, then equation (1) becomes

$|z|^3 = 4 - 3|z|^2 \Rightarrow |z|^3 + 3|z|^2 - 4 = 0$

$\Rightarrow |z|^2(|z| - 1) + 4|z|(|z| - 1) + 4(|z| - 1) = 0$

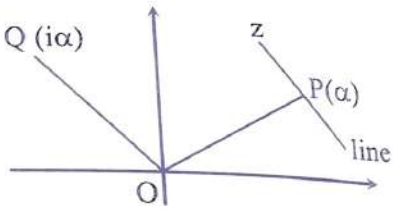
$\Rightarrow (|z| - 1)(|z|^2 + 4|z| + 4) = 0$

$\Rightarrow (|z| - 1)(|z| + 2)^2 = 0$

$\Rightarrow |z| = +1$ or $|z| = -2$ (rejected)

Hence $|z| = \{1, 2\}$ where $|z| = |z_1 + z_2 + z_3|$

198. Required line is passing through $P(\alpha)$ and parallel to the vector $\overrightarrow{OQ}$



Hence $z = \alpha + i\lambda\alpha$, $\lambda \in R$

$\frac{z-\alpha}{\alpha}$ = purely imaginary

$$\Rightarrow \text{Re}\left(\frac{z-\alpha}{\alpha}\right) = 0$$

$$\Rightarrow \text{Re}\left((z-\alpha)\bar{\alpha}\right) = 0$$

$$\Rightarrow \text{Re}\left(z\bar{\alpha} - |\bar{\alpha}|\right) = 0$$

Also $\frac{z-\alpha}{\alpha} + \frac{\bar{z}-\bar{\alpha}}{\bar{\alpha}} = 0$

$$\bar{\alpha}(z-\alpha) + \alpha(\bar{z}-\bar{\alpha}) = 0$$

$$\bar{\alpha}z + \alpha\bar{z} - 2|\alpha|^2 = 0$$

199. $|w| = \frac{|v||u-z|}{|\bar{u}z-1|} = \frac{|u-z|}{|\bar{u}z-1|}$

Let, $|w| \leq 1$

$\Rightarrow |u-z| \leq |\bar{u}z-1|$

$\Rightarrow (u-z)(\bar{u}-\bar{z}) \leq (u\bar{z}-1)(\bar{u}z-1)$

$\Rightarrow (|u|^2 - 1)(|z|^2 - 1) \geq 1$

$\Rightarrow |z|^2 - 1 \leq 0$

200. $A_\alpha A_\beta = A_{\alpha+\beta}$

$$A_0 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$A_\alpha A_{-\alpha} = A_{\alpha-\alpha} = A_0 = I$

$\Rightarrow A_\alpha^{-1} = A_{-\alpha}$

201. $f(-x) = -f(x)$

202. $\hat{\alpha}\cdot\hat{\gamma} = x(\hat{\alpha}\cdot\hat{\alpha}) + y(\hat{\alpha}\cdot\hat{\beta}) + z\,\hat{\alpha}(\hat{\alpha}\times\hat{\beta})$

$\Rightarrow x = \cos\theta$, similarly $y = \cos\theta$

$\hat{\gamma}\cdot(\hat{\alpha}\times\hat{\beta}) = z(\hat{\alpha}\times\hat{\beta}).(\hat{\alpha}\times\hat{\beta})$

$\Rightarrow z = \left[\hat{\alpha}\hat{\beta}\hat{\gamma}\right]$

$$\left[\hat{\alpha}\hat{\beta}\hat{\gamma}\right]^2 = \begin{vmatrix} 1 & 0 & \cos\theta \\ 0 & 1 & \cos\theta \\ \cos\theta & \cos\theta & 1 \end{vmatrix} = 1 - 2\cos^2\theta.$$

203. Since $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = 0$

Also, $\vec{d}\cdot\vec{a} = \vec{d}\cdot\vec{b} = \vec{d}\cdot\vec{c} = 0$

$\therefore\ \vec{d}\cdot\vec{x} = \vec{d}\cdot\vec{y} = \vec{d}\cdot\vec{z} = 0$

$\therefore\ \vec{d}\cdot\vec{r} = 0$

204. $B = A^{2^n} = A^{2\cdot 2^{n-1}} = \left(A^2\right)^{2^{n-1}} = \left(A^{-1}\right)^{2^{n-1}} = \left(A^{2^{n-1}}\right)^{-1} = \left(A^{2\cdot 2^{n-2}}\right) = \left(\left(A^2\right)^{2^{n-2}}\right)^{-1}$

$$= \left(\left(A^{-1}\right)^{-1}\right)^{2^{n-2}} = A^{2^{(n-2)}}$$

So $B = C$

$(B - C) = \text{null} \Rightarrow (B + C)(B - C) = 0 \Rightarrow \det.(B - C) = 0$

$\Rightarrow$ A, B, C are cottect answer

205. (A) $\tan 1 > 1$ and $\sin 1 < 1$, then $\log_{\sin 1}\tan 1 < 0$

(B) $1 + \tan 3 < 1$ and $\cos 1 < 1$, then $\log_{\cos 1}(1 + \tan 3) > 0$

(C) $\cos\theta + \sec\theta > 2$ and $\log_{10} 5 < 1$, then $\log_{\tan_{10}5}(\cos\theta + \sec\theta) < 0$

(D) $2\sin 18^\circ < 1$ and $\tan 15^\circ < 1$, then $\log_{\tan_{15^\circ}} 2\sin 18^\circ > 0$

206. $\because\ x - 1 > 0$

Taking log both sides $\left(\log_3 x^2 - 2\log_x 9\right)\log(x-1) = 7\log(x-1)$

$\Rightarrow x = 81, \dfrac{1}{\sqrt{3}}, 2$

$\therefore\ x \neq \dfrac{1}{\sqrt{3}}$

$\therefore\ x = 2, 81$

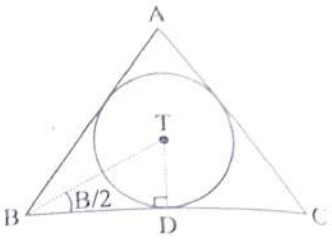
207. For $x \in \left(0, \frac{\pi}{4}\right)\sqrt{2}\sin x < 1$

$\Rightarrow \cos^2 x < 2\sin^2 x$

$\Rightarrow \sin^2 x > \frac{1}{3} \Rightarrow \sin x > \frac{1}{\sqrt{3}}$

$\Rightarrow x \in \left(\sin^{-1}\left(\frac{1}{\sqrt{3}}\right), \frac{\pi}{4}\right)$

208. $|BD - CD| = \left|r\left(\cot\frac{B}{2} - \cot\frac{C}{2}\right)\right|$



$= r\left|\frac{\sin\left(\frac{B-C}{2}\right)}{\sin\frac{B}{2}\sin\frac{C}{2}}\right| = \left|4R\sin\frac{A}{2}\sin\frac{B-C}{2}\right|$

$= \left|4R\cos\frac{B+C}{2}\sin\frac{B-C}{2}\right|$

$|2R(\sin B - \sin C)| = |b - c|$

$\tan\frac{B}{2} = \frac{r}{BD}$

$BD = r\cot\frac{B}{2}$

$CD = r\cot\frac{C}{2}$

Using $r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

209. $x^{\left(\frac{\log_{10}x+7}{4}\right)} = 10^{\log_{10}x+1}$

Taking log both sides on base, we get

$\left(\frac{\log_{10}x+7}{4}\right)\log_{10}x = (\log_{10}x+1)\log_{10}10$

Put, $\log_{10}x = t$

$\therefore (t+7)t = 4t + 4$

$\Rightarrow t = -4, 1$

$x = 10^{-4}, 10$

210. Given lines are $\ell + m + n = 0$ ..... (1)

$2\ell m + 2\ell n - mn = 0$ ..... (2)

Putting the value of n from (1) in (2), we get $2\ell m + 2\ell\{-(\ell+m)\} - m\{-(\ell+m)\} = 0 \because n = -(\ell+m)$

$\Rightarrow 2\ell m - 2\ell^2 - 2\ell m + m\ell + m^2 = 0 \Rightarrow m^2 + m\ell - 2\ell^2 = 0$

$\Rightarrow \left(\frac{m}{\ell}\right)^2 + \left(\frac{m}{\ell}\right) - 2 = 0$

$\Rightarrow \frac{m}{\ell} = -\frac{1 \pm \sqrt{1+8}}{2} = -\frac{1 \pm \sqrt{9}}{2} = -\frac{1 \pm 3}{2} \Rightarrow \frac{m}{\ell} = -2, 1$

If $\frac{m}{\ell}=-2$ if $\frac{m}{\ell}=1$. Then $n=-(\ell-2\ell)=\ell \Rightarrow n=-(2\ell)$

$\Rightarrow \frac{m}{-2}=\frac{\ell}{1}=\frac{n}{1} \Rightarrow \frac{m}{1}=\frac{\ell}{1}=\frac{n}{-2}$

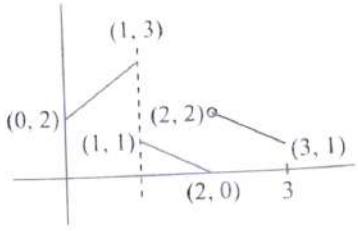
$\therefore$ Direction cosines are $-\left(\frac{1}{\sqrt{6}},-\frac{2}{\sqrt{6}},\frac{1}{\sqrt{6}}\right);\left(\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}},-\frac{2}{\sqrt{6}}\right)$

211. $f(x)=\tan^{-1}\left(x^4-x^2+\frac{1}{4}-2+\tan^{-1}\alpha\right)=\tan^{-1}\left(\left(x^2-\frac{1}{2}\right)^2+\tan^{-1}\alpha-2\right)$ for f to be surjective

$\tan^{-1}\alpha-2=-1$

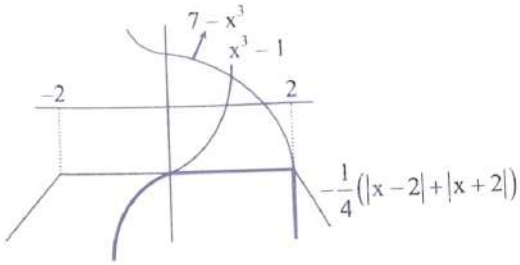
So, $\alpha=\tan 1$

212. $f(f(x))=\begin{cases}x+2 & ; \ 0\le x\le 1\\ 2-x & ; \ 1<x\le 2\\ 4-x & ; \ 2<x\le 3\end{cases}$



213. $f(x)=\begin{cases}x^3-1 & \text{for } x<0\\ -\frac{1}{4}(|x-2|+|x+2|) & \text{for } 0\le x\le 2\\ 7-x^3 & \text{for } x>2\end{cases}$

f(x) is continuous every where
but non–derivable at x = 2 only
$\therefore$ p = 0 and q = 1



214. $|f(x)-f(y)|\le 2 \Rightarrow f(x)$ is bounded and for $y=x+2n\pi,\ n\in 1$ we get

$|f(x)-f(x+2n\pi)|\le 0 \Rightarrow f(x)=f(x+2n\pi)$

215. $f_{2014}(x)<f_{2013}(-x) \Rightarrow f(x)>-x$

$\Rightarrow x^3+x-30<0 \Rightarrow (x-3)(x^2+3x+10)<0$

$\Rightarrow x<3$

216. $f'(x)=5\left(\frac{\sqrt{a+4}}{1-a}-1\right)x^4-3$

If f(x) is decreasing for all x

$\left(\frac{\sqrt{a+4}}{1-a}-1\right)$ must be negative.

If $a>1$ then $\left(\frac{\sqrt{a+4}}{1-a}-1\right)$ must be negative, if $a\in(-4,1)$ then on squaring the inequality

$\frac{\sqrt{a+4}}{(1-a)}<1 \Rightarrow \frac{a+4}{(1-a)^2}<1$

$\Rightarrow a^2-3a-3>0$

$\Rightarrow a \in \left(-\infty, \frac{3-\sqrt{21}}{2}\right) \cup \left(\frac{3+\sqrt{21}}{2}, \infty\right)$

Since $a \in (-4, 1)$ was assumed, $\therefore$ we get $a \in \left(-4, \frac{3-\sqrt{21}}{2}\right)$

Combining the solutions of two cases, we get $a \in \left(-4, \frac{3-\sqrt{21}}{2}\right) \cup (1, \infty)$

217. $f(0^+) = \alpha$ as $\sin[x] = 0$ as $x \to 0^+$

$f(0^-) = \beta - 1$ as $\frac{\sin x - x}{x^3} = -\frac{1}{6} + \frac{x}{4!} + \frac{x^2}{5!} + \ldots\ldots$

$f(0) = 2 \Rightarrow \alpha = 2; \beta = 3$

218. $f(x) = \sqrt{a^2b^2 + b^2 - b^2a^2}\sin(x + \alpha) + c = b\sin(x + \alpha) + c$ where $\tan\alpha = \frac{b\sqrt{1-a^2}}{ab} = \frac{\sqrt{1-a^2}}{a}$.

$f(x)_{max} - f(x)_{min} = c + b - (c - b) = 2b$

Also, $\alpha = \cos^{-1}a$

at $x = -\cos^{-1}a$

$f(x) = c$.

219. $f(x) = \begin{cases} 0 & , \ x \neq 0 \\ 1 & , \ x = 0 \end{cases}$

$f(f(x)) = \begin{cases} 0 & , \ x = 0 \\ 1 & , \ x \neq 0 \end{cases}$

220. $\lim\limits_{x\to\infty}\left(8x^3 + mx^2\right)^{1/3} - nx = \lim\limits_{x\to\infty}(2x)\left(1 + \frac{m}{8x}\right)^{1/3} - nx = \lim\limits_{x\to\infty}(2x)\left(1 + \frac{m}{24x}\right) - nx$

$= \lim\limits_{x\to\infty}(2-n)x + \frac{m}{12}$

For limit to exists $2 - n = 0$

$\Rightarrow n = 2$ and $\frac{m}{12} = 1$

$\Rightarrow m = 12$

$\therefore \frac{m}{n} = \frac{12}{2} = 6$

221. $f'(0^+) > 0$ and $f'(0^-) < 0 \Rightarrow \lim\limits_{x\to0^+}\left(\left[\frac{f(x)}{x^2}\right]\right) = 0, \ \lim\limits_{x\to0^-}\left(\left[\frac{f(x)}{x^2}\right]\right) = -1$

$\lim\limits_{x\to0^+}\left(\left[\frac{f(x)}{x}\right]\right) = 0, \ \lim\limits_{x\to0^-}\left(\left[\frac{f(x)}{x}\right]\right) = 0$

222. $\frac{x^4 - x^3}{x^6 + 2x^3 - 1} = \frac{x - \frac{1}{x}}{x^3 - \frac{1}{x^3} + 2} = \frac{x - \frac{1}{x}}{\left(x - \frac{1}{x}\right)^3 + 2 + 3\left(x - \frac{1}{x}\right)} \leq \frac{x - \frac{1}{x}}{3\left(x - \frac{1}{x}\right) + 3\left(x - \frac{1}{x}\right)} = \frac{1}{6}$

223. Simplify $f(x) = 0 \Rightarrow h(x) = g(x)$

$\Rightarrow h'(x) = 3x^2 + 6 \Rightarrow h''(x) = 6x \Rightarrow h''\left(\frac{\pi}{24}\right) = \frac{\pi}{4}$ and $h'(x) > 0 \Rightarrow h(x)$ is increasing

224. $x = t^2 \Rightarrow dx = 2tdt$

$$\therefore I = \int \frac{2tdt}{(1+t)^8} = 2\left[\int \frac{dt}{(t+1)^7} - \frac{dt}{(t+1)^8}\right] = -\frac{1}{3}(1+t)^{-6} + \frac{2}{7}(1+t)^{-7}$$

225. $I = \int \left(4\sin\theta - 4\sin^3\theta\right) e^{\sin\theta} \cdot \cos\theta \cdot d\theta$

Put, $\sin\theta = t$

$\cos\theta\, d\theta = dt$

$I = 4(-t^3 + 3t^2 - 5t + 5)\, e^t$

$\therefore P = -4, Q = -12, R = -20$

226. $$I = \int_0^{[x]+\{x\}} \left(\{x\} - \frac{1}{2}\right) dx = \int_0^{[x]} \left(\{x\} - \frac{1}{2}\right) dx + \int_{[x]}^{[x]+\{x\}} \left(\{x\} - \frac{1}{2}\right) dx$$

$$= [x]\int_0^1 \left(\{x\} - \frac{1}{2}\right) dx + \int_0^{\{x\}} \left(\{x\} - \frac{1}{2}\right) dx = [x]\int_0^1 \left(x - \frac{1}{2}\right) dx + \int_0^{\{x\}} \left(x - \frac{1}{2}\right) dx$$

$$= [x]\left(\frac{x^2}{2} - \frac{x}{2}\right)_0^1 + \left(\frac{x^2}{2} - \frac{x}{2}\right)_0^{\{x\}} = 0 + \frac{\{x\}(\{x\}-1)}{2} = \frac{\{x\}}{2}(\{x\}-1)$$

227. $$f(x) = \int \frac{x^8 + 4}{x^4 - 2x^2 + 2} dx \text{ , } f(0) = 0$$

$$= \int \frac{x^8 + 4 + 4x^4 - 4x^4}{x^4 - 2x^2 + 2} dx = \int \frac{\left(x^4 + 2x^2 + 2\right)\left(x^4 - 2x^2 + 2\right)}{\left(x^4 - 2x^2 + 2\right)} dx = \frac{x^5}{5} + \frac{2}{3}x^3 + 2x + c$$

$f(0) = 0 + c = 0 \Rightarrow c = 0$

$$\therefore f(x) = \frac{x^5}{5} + \frac{2x^3}{3} + 2x$$

$\therefore f(x)$ is an odd function, its range is R, has at least one real root, is monotonic

228. $$I = \int \frac{x-1}{x^3\sqrt{2 - \frac{2}{x} + \frac{1}{x^2}}} dx$$

Put $2 - \frac{2}{x} + \frac{1}{x^2} = t^2$

$$\Rightarrow \left(\frac{2}{x^2} - \frac{2}{x^3}\right) dx = 2t\, dt \Rightarrow I = \int \frac{t}{t} dt$$

$I = t + c$

$$I = \sqrt{2 - \frac{2}{x} + \frac{1}{x^2}} + c$$

$$I = \frac{\sqrt{2x^2 - 2x + 1}}{x} + c$$

229. $I_n = \int_0^{\frac{\pi}{2}} \cos^n x\,dx$

$= \int_0^{\frac{\pi}{2}} \cos^{n-1} x.\cos x\,dx = \left[\cos^{n-1} x \sin x\right]_0^{\pi/2} + \int_0^{\pi/2} (n-1)\cos^{n-2} x\,(\sin x)\sin x\,dx$

$= (n-1)\int_0^{\pi/2} \cos^{n-2} x\,(1-\cos^2 x)\,dx$

$I_n = (n-1)\,I_{n-2} - (n-1)I_n$

$nI_n = (n-1)\,I_{n-2}$

$n(I_{n-2} - I_n) = I_{n-2}$

Clearly $I_{n-2} > I_n$

Also for $0 < x < \frac{\pi}{2}$

$0 < \cos x < 1$

So, $\cos^n x < \cos^{n-1} x$

$\Rightarrow I_n < I_{n-1}$

$I_4 = \int_0^{\frac{\pi}{2}} \cos^4 x\,dx = \frac{1}{4}\int_0^{\frac{\pi}{2}} (2\cos^2 x)^2\,dx = \frac{1}{4}\int_0^{\frac{\pi}{2}} (1+\cos 2x)^2\,dx$

$= \frac{1}{4}\int_0^{\frac{\pi}{2}} \left(1 + 2\cos 2x + \frac{1+\cos 4x}{2}\right)dx = \frac{1}{4}\left[x + \sin 2x + \frac{1}{2}x + \frac{1}{8}\sin 4x\right]_0^{\frac{\pi}{2}} = \frac{3\pi}{16}$

230. $I = \int_{\tan^{-1}\lambda}^{\cot^{-1}\lambda} \frac{\tan x}{\tan x + \cot x}\,dx$ ... (1)

$= \int_{\tan^{-1}\lambda}^{\cot^{-1}\lambda} \frac{\cot x}{\tan x + \cot x}\,dx$ ... (2)

$2I = \cot^{-1}\lambda - \tan^{-1}\lambda \Rightarrow I = \frac{\pi}{4} - \tan^{-1}\lambda$

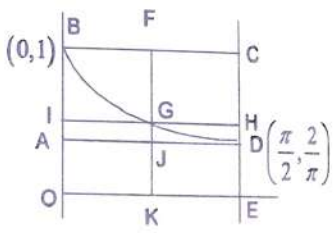
Now, $\frac{-\pi}{2} < \tan^{-1}\lambda < \frac{\pi}{2}\ -\frac{\pi}{2} < -\tan^{-1}\lambda < \frac{\pi}{2} \Rightarrow -\frac{\pi}{4} < I < \frac{3\pi}{4}$

231. $a + b = 2$ and $\int_0^4 \left(a\sqrt{x} + bx\right)dx = 8$ implies $2a + 3b = 3$. Solving $a = 3$ and $b = -1$

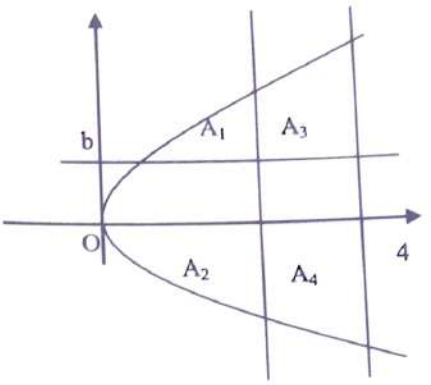
232. $\text{Ar}(OADE) < A < \text{Ar}(OBCE)$

$\text{Ar}(OIGK) + \text{Ar}(KJDE) < A < \text{Ar}(OBFK) + \text{Ar}(KGHE)$

(K is mid-point of OE)



233. Total area $= 2\int_0^4 \sqrt{x}\,dx = \frac{32}{3}$



Area of each part = 8/3

$A_3 = A_4 \Rightarrow \int_a^4 \left(\sqrt{x} - b\right)dx =$

$\int_a^4 \left(b + \sqrt{x}\right)dx = \frac{8}{3} \Rightarrow b = 0$

$\int_a^4 \sqrt{x}\,dx = \frac{8}{3} \Rightarrow a^3 = 16$

234. $\frac{dx}{dy} = \frac{2y\ln y + y - x}{y}$

$\Rightarrow \frac{dx}{dy} + \frac{1}{y}x = 2\ln y + 1$

Integrating factor

$\therefore x \cdot y = y^2 \ln y + c$

$y(0) = 1 \therefore c = 0$

$\Rightarrow x = y \ln y$

235. Using L Hospital rule

$\lim_{z \to x} \frac{2zf(x) - x^2 f'(z)}{1} = 1$

$\Rightarrow 2xf(x) - x^2 f'(x) = 1$

$\Rightarrow f'(x) - \frac{2f(x)}{x} = -\frac{1}{x^2}$

This a linear differential equation integrating factor $= \frac{1}{x^2}$

$\Rightarrow \frac{f(x)}{x^2} = \int -\frac{1}{x^4}dx \Rightarrow f(x) = \frac{1}{3x} + cx^2$

$\because f(1) = 1 \therefore c = \frac{2}{3}$

$\therefore f(x) = \frac{1}{3x} + \frac{2}{3}x^2$

236. $y = c \sin x$ ...(1)

$\frac{dy}{dx} = c\cos x$ ...(2)

from (2)

$\left(\frac{dy}{dx}\right)^2 = c^2 \cos^2 x$ ...(3)

Putting $c = \frac{y}{\sin x}$ from (1), $\left(\frac{dy}{dx}\right)^2 = y^2 \cot^2 x$

Eliminating c from (1) and (2), $\frac{dy}{dx} = y\cot x$.

Finding value of c from (2) and eliminating with help of (3), we get

$\left(\frac{dy}{dx}\right)^2 - \left(\sec x \frac{dy}{dx}\right)^2 + y^2 = 0$.

237. Since roots are imaginary. So, discriminant < 0

$\alpha = \frac{-b + i\sqrt{4ac - b^2}}{2c}$, $\beta = \frac{-b - i\sqrt{4ac - b^2}}{2c}$, $|\alpha| = |\beta| = \sqrt{\frac{b^2}{4c^2} + \frac{4ac - b^2}{4c^2}} = \sqrt{\frac{a}{c}} > 1$

238. $e^x \cdot \frac{dy}{dx} + y \cdot e^x = e^x f(x)$

$\Rightarrow y \cdot e^x = \int e^x f(x)dx + c$

$\Rightarrow y \cdot e^x = \int e^x \cdot e^{-x} dx + c, 0 \le x \le 2$

$\Rightarrow y \cdot e^x = x + c \Rightarrow c = 1$

$\Rightarrow y = (x + 1)e^{-x}\ 0 \le x \le 2$

Similarly, $y = e^{-2} + ce^{-x}$, $x > 2$ as y is continuous

$\Rightarrow \lim_{x \to 2}\left(\frac{x+1}{e^x}\right) = \lim_{x \to 2}\left(e^{-2} + ce^{-x}\right)$

$\Rightarrow c = 2$

$\Rightarrow y = \begin{cases} (x+1)e^{-x} & , \ 0 \le x \le 2 \\ e^{-2} + 2e^{-x} & , \ x > 2 \end{cases}$

239. $\frac{xdy - ydx}{x^2} = \left(1 - \frac{2}{x^2}\right)dx$

$\frac{y}{x} = x + \frac{2}{x} + c$

$y = x^2 + 2 + cx$

$f(1) = 1 \Rightarrow c = -2$

$f(x) = x^2 - 2x + 2$

240. $\left(\frac{y}{x}\right)^2 + a\left(\frac{y}{x}\right) + 4 = 0$

Let $\frac{y}{x} = m$

$\Rightarrow m^2 + am + 4 = 0$

$\Rightarrow$ Roots of the euation are $\alpha, 4\alpha$

$\Rightarrow 4\alpha^2 = 4 \Rightarrow \alpha = \pm 1$

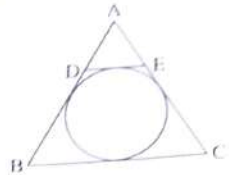
$\Rightarrow 5\alpha = -a \Rightarrow a = \pm 5$

241. Let the equation of line is $x + y + k = 0$

$\Rightarrow \left|\frac{k+3}{\sqrt{2}}\right| = 3\sqrt{2} \Rightarrow k + 3 = \pm 6 \Rightarrow k = 3, -9$

242. Lines making equal angles with two lines say, $L_1 = 0$ and $L_2 = 0$ are parallel to the angle bisector of $L_1 = 0$ and $L_2 = 0$

243. Let sides of Δ is a
Let BD = x, AD = a – x
ω' CE = y, and AE = a – y
By Pitotes theorem for circumscribed quadrilateral BDEC
BC + DE = BD + CE
$\Rightarrow$ DE = x + y – a
$\therefore$ $DE^2 = (x + y - a)^2$
$= x^2 + y^2 + a^2 + 2xy - 2ax - 2ay$
By cosine rule on ΔADE ..... (1)
$DE^2 = (a - x)^2 + (a - y)^2 - 2(a - x)(a - y)\cos 60^\circ$
i.e., $DE^2 = x^2 + y^2 + a^2 - ay - ax - xy$ ..... (2)



From equation (1) and (2), we get $a = \frac{3xy}{x+y}$

Substituting this value of a for AD and AE, we have $AD = a - x = x\left(\frac{2y-x}{x+y}\right)$

$\Rightarrow \frac{AD}{DB} = \frac{2y-x}{x+y}$ (As x = DB) $\Rightarrow AE = a - y = y\left(\frac{2x-y}{x+y}\right)$

$\Rightarrow \frac{AE}{CE} = \frac{2x-y}{x+y}$ (y = EC) $\Rightarrow \frac{AD}{DB} + \frac{AE}{CE} = 1$

244. C_1(centre) (2, –3), $r_1 = \sqrt{5} \Rightarrow C_2$ (centre) (5, 3), $r_2 = 2\sqrt{5}$
$C_1C_2 = r_1 + r_2$. Hence, circle touch each other externally and B and C are correct

245. Equation of tangent $\frac{x\cos\alpha}{4} + \frac{y\sin\alpha}{2} = 1$ passing through (4, 2)
So $\sin\alpha + \cos\alpha = 1$
$\Rightarrow \alpha = \frac{n\pi}{2}, n \in Z$

246. Equation of tangent to $x^2 = y$ is $y = mx - \frac{1}{4}m^2$ ..... (1)

Equation of tangent to $(x - 2)^2 = -y$ is $y = m(x-2) + \frac{1}{4}m^2$ ..... (2)

Equation (1) and (2) are identical $\Rightarrow$ m = 0 or 4.
$\therefore$ Common tangents are y = 0 and y = 4x – 4 = 4(x – 1)

247. The equation of tangent on parabola $y^2 = 4x$

$$y = mx + \frac{1}{m}$$

This line also be tangent to the parabola $x^2 = 32y$ then meets parabola in two coincident points so,

$x^2 = 32\left(mx + \frac{1}{m}\right)$ has equal roots.

$(-32m)^2 = 4.1.\left(-\frac{32}{m}\right)$

$m^3 = -\frac{1}{8}$

$m = -\frac{1}{2}$

Hence equation of common tangent is x + 2y + 4 = 0

Given equation is

$Ax + By + C = 0$

A.B.C = 8

$$\frac{C^3}{A^3B^3} = \frac{4^3}{1^3 2^3} = 8$$

$$\frac{A^2}{B \cdot C} = \frac{1^3}{2 \cdot 4} = \frac{1}{8}$$

248. Circle with diameter as SR will touch the auxiliary circle

249. Both the ellipse have their centre at (0, 0). The major axis of first is along x-axis and in case the two ellipses meet in four distinct points, then the major axis of second ellipse should be along y-axis. This will be possible if $a^2 > 1$ or $b^2 - 5b + 7 > 1$

$\Rightarrow b^2 - 5b + 6 > 0$

$(b-2)(b-3) > 0$

$\therefore b < 2$ or $b > 3$

$\therefore b \notin [2, 3]$.

250. Let $P\left(2t, \frac{2}{t_1}\right)$ and $Q\left(2t_2, \frac{2}{t_2}\right)$

$M_{PQ} = 1 \Rightarrow t_2 = -\frac{1}{t_1}$

$\therefore Q\left(-\frac{2}{t_1}, -2t_1\right)$

Circle with PQ as diameter is $(x - 2t_1)\left(x + \frac{2}{t_1}\right) + \left(y - \frac{2}{t_1}\right)(y + 2t_1) = 0$

$\Rightarrow \left(x^2 + y^2 - 8\right) + \left(\frac{2}{t_1} - 2t_1\right)(x - y) = 0$

$\therefore$ Intersection point is of $x^2 + y^2 - 8 = 0$ and $x - y = 0$

251. $L\left(ae, \frac{b^2}{a}\right)$, N_L: $ax \sec\theta - by \operatorname{cosec}\theta = a^2 - b^2$

$ae = a\cos\theta, \ \frac{b^2}{a} = b\sin\theta \Rightarrow e = \cos\theta, \ \frac{b}{a} = \sin\theta$

Let S: $xy = 9$

Solving N_L with S, we get $ax\sec\theta - \frac{9b}{x}\operatorname{cosec}\theta = a^2 - b^2$

$\Rightarrow \frac{ax}{e} - \frac{9b}{x} \cdot \frac{a}{b} = a^2 - b^2$

$\Rightarrow x^2 - ae^3x - 9e = 0 \Rightarrow x_1x_2 = -9e$

If $x_1 = 6$, $x_2 = -\frac{3e}{2}$ where e is eccentricity of ellipse

$m_{OP} \cdot m_{OQ} = \frac{81}{(x_1x_2)^2} > 0$

252. $\alpha+\beta=\frac{b}{a},\ \alpha\cdot\beta=\frac{c}{a}$

Now, $0<1-\alpha<1$ and $0<1-\beta<1$

How $\frac{1}{4}\geq\alpha(1-\alpha)>0$ also $\frac{1}{4}\geq\beta(1-\beta)>0$

$\therefore\ 0<\alpha\beta(1-\beta)(1-\alpha)<\frac{1}{16}$ (α, β is distinct)

$\Rightarrow 0<\frac{c}{a}\left(1-\frac{b}{a}+\frac{c}{a}\right)<\frac{1}{16}\Rightarrow 0<c(1-b+c)<\frac{a^2}{16}$

$\because\ c(a-b+c)=N\ \therefore\ \frac{a^2}{16}>1\Rightarrow a\geq 5$

Also, $b^2-4ac\geq 0\Rightarrow b^2\geq 20c\Rightarrow b\geq 5$ and mini value of $c=1$

$\therefore\ abc\geq 25\Rightarrow \log_5(abc\geq 2$

253. Let $3^x=t\ \because\ x\geq 0\Rightarrow t\geq 1$

$\therefore\ (2a-4)t^2-(2a-3)t+1=0$

$\alpha\cdot\beta\geq 1\Rightarrow\frac{1}{2a-4}\geq 1\Rightarrow\frac{5-2a}{2a-4}\geq 0\Rightarrow a\in\left(2,\frac{5}{2}\right)$

254. Let the digits are a_1-3d, a_1-d, a_1, a_1+d

$\therefore\ (a_1-d)^2=a_1(a_1-3d)$

$\Rightarrow d=5a_1$ not possible as $d<a$

$\therefore\ a_1^2=(a_1-d)(a_1+3d)$

$\Rightarrow d=\frac{2a_1}{3}$

If $a_1=3$, $d=2$

$\therefore$ Number is 9135

255. $p\left(\frac{-x+\sqrt{3-3x^2}}{2}\right)$ for $|x|\leq 1$ ..... (1)

Let $x=0$, $p(0)$, $p(0)=p\left(\frac{\sqrt{3}}{2}\right)$

Which shows $p(x)-p(0)$ is divisible by $x\left(x-\frac{\sqrt{3}}{2}\right)$

Since, $p(x)-p(0)$ has rational coefficient and $\frac{\sqrt{3}}{2}$ is one of the root, $-\frac{\sqrt{3}}{2}$ is too a root of $p(x)-p(0)$

Thus $x\left(x-\frac{\sqrt{3}}{2}\right)\left(x+\frac{\sqrt{3}}{2}\right)=x^3-\frac{3}{4}x=\frac{4x^3-3x}{4}$ is factor of $p(x)-p(0)$

$p(x)=p(0)+(3x-4x^3)p_1(x)$ ..... (2) $|x|\leq 1$

$p(x)=p\left(\frac{-x+\sqrt{3-3x^2}}{2}\right)$

$3\left(\frac{-x+\sqrt{3-3x^2}}{2}\right)-4\left(\frac{-x+\sqrt{3-3x^2}}{2}\right)=3x-4x^3$

$$p_1(x) = p_1\left(\frac{-x+\sqrt{3-3x^2}}{2}\right)$$

$p_1(x) = p_1(0) + (3x - 4x^3)p_2(x)$
$p(x) = p(0) + (3x - 4x^3)\{p_1(0)\} + (3x - 4x^3)p_2(x)$
$p(x) = p(0) + (3x - 4x^3)p_1(0) + (3x - 4x^3)^2p_2(x)$
In general $p(x) = a_0 + a_1(3x - 4x^3) + a_2(3x - 4x^3)^2 + \ldots\ldots + (3x - 4x^3)^k \cdot k(x)$, where $k(x)$ is polynomial with rational coefficient

256. $z_n - z_1 = z_n - z_{n-1} + \ldots + z_3 - z_2 + z_2 - z_1$

$\Rightarrow |z_n - z_1| \le |z_n - z_{n-1}| + |z_{n-1} - z_{n-2}| + \ldots + |z_3 - z_2| + |z_2 - z_1|$

$$\Rightarrow \frac{|z_2 - z_1|^2 + |z_3 - z_2|^2 + \ldots + |z_n - z_{n-1}|^2}{(n-1)} \ge \left(\frac{|z_n - z_{n-1}| + |z_{n-1} - z_{n-2}| + \ldots + |z_2 - z_1|}{(n-1)}\right)^2$$

$$\ge \frac{|z_n - z_1|^2}{(n-1)^2}$$

$$\Rightarrow \frac{|z_2 - z_1|^2 + |z_3 - z_2|^2 + \ldots + |z_n - z_{n-1}|^2}{|z_n - z_1|^2} \ge \frac{1}{n-1}$$

257. $x + y + z + \omega = 19$, $x, y, z, \omega \ge 1$ has solution as coefficient of x^{19} in $(x + x^2 + \ldots + x^{19})^4$ in turn is same as coeff. of x^{15} in $(1 + x + \ldots + x^{15})$.

258. Sum of the coefficient in the expansion of $(1 + ax^3 - 2x^5)^n$ is obtained when we put $x = 1$.

$$\therefore S = (1 + a - 2)^n = (a - 1)^n = \begin{cases} +ve, & \text{when } a < 1 \quad , \quad n = 2k \\ +ve, & \text{when } a > 1 \quad , \quad n \in N \\ -ve, & \text{when } a < 1 \quad , \quad n = 2k+1 \\ 0, & a = 1 \end{cases}$$

259. $$T_k = \frac{3^{k-1}}{\left(1+3+3^2 \ldots\ldots 3^{k-1}\right)\left(1+3+3^2+\ldots\ldots 3^k\right)} = \frac{4.3^{k-1}}{\left(3^k-1\right)\left(3^{k+1}-1\right)}$$

$$T_k = \frac{2}{3}\left[\frac{1}{3^k-1} - \frac{1}{3^{k+1}-1}\right]$$

$$\Rightarrow S_n = \frac{2}{3}\left[\frac{1}{2} - \frac{1}{3^{n+1}-1}\right]$$

$$\therefore S_\infty = \frac{2}{3}$$

260. $$1 - x + x^2 - x^3 + \ldots + (-x)^n = \frac{1+(-1)^n x^{n+1}}{1+x}$$

$$\Rightarrow \int_0^1 \left(1 - x + x^2 - x^3 + \ldots + (-x)^n\right)dx = \int_0^1 \left(\frac{1}{1+x}\right)dx + (-1)^n \int_0^1 \left(\frac{x^{n+1}}{1+x}\right)dx$$

$$\Rightarrow |X - \ln 2| = \int_0^1 \left(\frac{x^{n+1}}{1+x}\right)dx$$

$$\text{Where } \frac{1}{2(n+2)} < \int_0^1 \left(\frac{x^{n+1}}{1+x}\right)dx < \frac{1}{2(n+1)}$$

261. $(1-5x+6x^2)=((1-2x)(1-3x))^{-1}$

$=\frac{1}{(1-2x)(1-3x)}=\frac{3}{1-3x}-\frac{2}{1-2x}$

$=3\{1+(3x)+(3x)^2+(3x)^3\ldots..\}-2\{1+(2x)+(2x)^2\ldots..\}$

Coefficient of x^n is $(3.3^n-2.2^n)=3^{n+1}-2^{n+1}$

262. Given ${}^{1000}C_{950}+2\,{}^{999}C_{949}+3\,{}^{998}C_{948}+\ldots..+51\,{}^{950}C_{950}$

Coefficient of x^{950} in $\left[(1+x)^{950}+2(1+x)^{999}+3(1+x)^{998}+\ldots..+51(1+x)^{950}\right]$

Let $S=(1+x)^{950}+2(1+x)^{999}+\ldots..+51(1+x)^{950}$

Then $S=(1+x)^{951}\left\{\frac{(1+x)^{51}-1}{x^2}\right\}-51\left\{\frac{(1+x)^{950}}{x}\right\}$

So required sum = coefficient of x^{950} in S

= coefficient of x^{952} in $\left[(1+x)^{1002}-(1+x)^{951}\right]$ – coefficient of x^{951} in $\left[51(1+x)^{950}\right]$

$={}^{1002}C_{952}$

263. P = (n – r) …. (n – 2)(n – 1) n(n + 1)(n + 2) … (n + r) is multiplication of 2r + 1 consecutive integers which is divisible by $\underline{|2r+1}$

264. $L_n=2\sin\frac{2\pi}{n+1}$ since non-zero roots are $(n+1)^{th}$ roots of unity except 1

265. Condition implies $|z_1|=|z_2|=|z_3|$

266. $z=4\left(\frac{-1+i\sqrt{3}}{2}\right)=4\omega$

Now $z^{2n}+2^{2n}\cdot z^n+2^{4n}$

$(4\omega)^{2n}+2^{2n}(4\omega)^n+2^{4n}$

$\Rightarrow 2^{4n}\left[\omega^{2n}+\omega^n+1\right]$

$\Rightarrow 0,\ n\notin 3I$ and $2^{4n}\cdot 3,\ n\in 3I$

267. If n is odd $\Rightarrow 2\cdot{}^{\frac{n+1}{2}}C_2$

If n is even $\Rightarrow {}^{\frac{n}{2}+1}C_2+{}^{\frac{n}{2}}C_2$

268. Number of groups $=\frac{100!}{50!2^{50}}$

269. $E_{17}(2050!)=\left[\frac{2050}{17}\right]+\left[\frac{2050}{17^2}\right]+\ldots=120+7+0+0+\ldots$

so in 2050! exponents of prime 17 are 127.

So n can be 125, 126 and 127 so answers are (A, B, C).

270. For real roots $p^2\geq 4q$ p q

Possible selections are as follows

1	1
2	1, 2
3	1, 2, 3, 4
4	1, 2, ..., 6
5	1 9
6	1 10
7	1 10
8	1 10
9	1 10
10	
Total = 62	

$\Rightarrow$ Probability for real roots $= \frac{62}{100} = \frac{31}{50}$

$\Rightarrow$ For equal and real roots $p^2 = 4q$

$\Rightarrow (p, q) = (2, 1), (4, 4), (6, 9)$

Probability $= \frac{3}{100}$

$\Rightarrow$ Probability for imaginary roots $= 1 - \frac{3}{50} = \frac{19}{50}$

$\Rightarrow$ Probability for real and distinct roots

$= \frac{62}{100} - \frac{3}{100} = \frac{59}{100}$

271. We have $\frac{3}{4} = 1 - (1-\alpha)(1-\beta)(1-\gamma) = \alpha + \beta + \gamma - (\alpha\beta + \beta\gamma + \alpha\gamma) + \alpha\beta\gamma,$

$\frac{1}{2} = \alpha\beta\gamma + \alpha\beta(1-\gamma) + \beta\gamma(-\alpha) + \gamma\alpha(1-\beta) = \alpha\beta + \beta\gamma + \gamma\alpha - 2\alpha\beta\gamma,$

$\frac{2}{5} = \alpha\beta(1-\gamma) + \beta\gamma(1-\alpha) + \gamma\alpha(1-\beta) = \alpha\beta + \beta\gamma + \gamma\alpha - 3\alpha\beta\gamma$

$\Rightarrow \alpha\beta\gamma = \frac{1}{2} - \frac{2}{5} = \frac{5-4}{10} = \frac{1}{10} \Rightarrow \alpha\beta + \beta\gamma + \gamma\alpha = \frac{1}{2} + \frac{1}{5} = \frac{7}{10}$

$\Rightarrow \alpha + \beta + \gamma = \frac{3}{4} + \frac{7}{10} - \frac{1}{10} = \frac{3}{4} + \frac{3}{5} = \frac{27}{20}$

$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) = \left(\frac{27}{20}\right)^2 - \frac{7}{5} \neq 1.$

272. Let $\vec{r}$ be the vector in the plane of $\vec{b}$ and $\vec{c}$

$\therefore \vec{r} = \left(\hat{i} + 2\hat{j} - \hat{k}\right) + \lambda\left(\hat{i} + \hat{j} - 2\hat{k}\right) = (1+\lambda)\hat{i} + (2+\lambda)\hat{j} - (1+2\lambda)\hat{k}$

Now projection of $\vec{r}$ on $\vec{a} = \frac{\vec{r}\cdot\vec{a}}{|\vec{a}|}$

$\Rightarrow -(1+\lambda) = \pm 2$

$\Rightarrow \lambda = 1, -3$

$\therefore \vec{r} = 2\hat{i} + 3\hat{j} - 3\hat{k}$ and $-2\hat{i} - \hat{j} + 5\hat{k}$

273. $|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$

$\cos\theta = \frac{\vec{a}\cdot\vec{c}}{|\vec{a}|\cdot|\vec{c}|} = \frac{\vec{b}\cdot\vec{c}}{|\vec{b}|\cdot|\vec{c}|} = \vec{a}\cdot\vec{c} = \vec{b}\cdot\vec{c}$

$\vec{a}\cdot\vec{b} = 0$

$\vec{a}\cdot\vec{c} = \alpha\cdot\vec{a}\cdot\vec{a} + \beta\cdot\vec{a}\cdot\vec{b} + \gamma\cdot\vec{a}\cdot\left(\vec{a}\times\vec{b}\right) = \alpha = \cos\theta$

Similarly, $\vec{b}\cdot\vec{c} = \cos\theta = \beta$

$1 = \vec{c}\cdot\vec{c} = 2\alpha^2 + \gamma^2|\vec{a}\times\vec{b}|^2$

$= 2\alpha^2 + \gamma^2\left[|\vec{a}|^2\cdot|\vec{b}|^2 - (\vec{a}\cdot\vec{b})^2\right] = 2\alpha^2 + \gamma^2$

274. $\bar{a}\times\bar{b} = |\bar{a}||\bar{b}|\sin\theta\,\hat{n}$

$\Rightarrow |\bar{a}\times\bar{b}| = |\bar{a}|\,|\bar{b}|\sin\theta \Rightarrow \sin\theta = \dfrac{|\bar{a}\times\bar{b}|}{|a||b|}$ ... (1)

$\bar{a}\cdot\bar{b} = |\bar{a}|\,|\bar{b}|\cos\theta \Rightarrow \cos\theta = \dfrac{\bar{a}\cdot\bar{b}}{|\bar{a}||\bar{b}|}$ ... (2)

From (1) and (2),

$\sin^2\theta + \cos^2\theta = 1 \Rightarrow |\bar{a}\times\bar{b}|^2 + (\bar{a}\cdot\bar{b})^2 = |\bar{a}|^2|\bar{b}|^2$

If $\theta = \dfrac{\pi}{4}$, then $\sin\theta = \cos\theta = \dfrac{1}{\sqrt{2}}$

$\therefore |\bar{a}\times\bar{b}| = \dfrac{|\bar{a}||\bar{b}|}{\sqrt{2}}$ and $\bar{a}\cdot\bar{b} = \dfrac{|\bar{a}||\bar{b}|}{\sqrt{2}}$

$\therefore |\bar{a}\times\bar{b}| = \bar{a}\cdot\bar{b}$

$\therefore \bar{a}\times\bar{b} = |\bar{a}||\bar{b}|\sin\theta\,\hat{n} = \dfrac{|\bar{a}||\bar{b}|}{\sqrt{2}}\hat{n}$

or $\bar{a}\times\bar{b} = (\bar{a}\cdot\bar{b})\hat{n}$.

275. Four point A, B, C, D are coplanar if three vector $\overrightarrow{AB}$, $\overrightarrow{AC}$, $\overrightarrow{AD}$ are coplanar or STP of there vectors is 0

$\overrightarrow{AB}\cdot(\overrightarrow{AC}\times\overrightarrow{AD}) = 0$

$\Rightarrow \begin{vmatrix} 1 & 0 & 0 \\ 3 & f(t) & 1 \\ 2 & f'(t) & 2 \end{vmatrix} = 0 \ \forall\, t \in R$

$\Rightarrow f'(t) = 2f(t)$

$\Rightarrow f(t) = ce^{2t}$

Similarly for $\bar{B}(\hat{i}+\hat{j})$ differential equation will be $f'(t) - 2f(t) = -4$

On solving $f(x) = 2 + ce^{2x}$, $c \in R$

276. Volumn of ABCD $= \dfrac{1}{6}\left[\overline{AD}\ \overline{BC}\ \overline{AC}\right] = V$

$\dfrac{1}{6} = V \le \dfrac{1}{3}\left|\overline{AD}\right|\left(\dfrac{1}{2}\left|\overline{BC}\right|\cdot\left|\overline{AC}\right|\cdot\sin 45^\circ\right)$ $\left(\left|\overline{AD}\right| = AD, \left|\overline{BC}\right| = BC, \left|\overline{AC}\right| = AC\right)$

So, $AD\cdot BC.\dfrac{AC}{\sqrt{2}} \ge 1$

$3 = AD + BC + \dfrac{AC}{\sqrt{2}} \ge 3\sqrt{AD\cdot BC\cdot\dfrac{AC}{\sqrt{2}}} \ge 3$ equality holds if and only if $AD = BC = \dfrac{AC}{\sqrt{2}} = 1$

So, AD is perpendicular to the face ABC

$DC = \sqrt{AD^2 + AC^2} = \sqrt{3}$

277. Let the equation of plane be $\ell x + my + nz = 0$ where ℓ, m, n be direction cosines

So, $\ell^2 + m^2 + n^2 = 1$ ..... (1)

Since, plane is parallel to line $\dfrac{x-1}{2} = \dfrac{y+3}{-1} = \dfrac{z+1}{-2}$

So, $2\ell - m - 2n = 0$ ..... (2)

Perpendicular distance of point (1, –3S, –1) from plane $\ell x + my + nz = 0 = 0$ is

$$\left|\frac{\ell - 3m - n}{\sqrt{\ell^2 + m^2 + n^2}}\right| = \frac{5}{3} \quad \ldots\ldots (3)$$

From equation (1), (2) and (3) equation of plane as $x - 2y + 2z = 0$ or $2x + 2y + z = 0$

278. Let the direction ratios of the two lines be $(0, m_1, n_1)$ and (l_2, m_2, n_2) so that $2m_1 - n_1 = 0$, $l_2 + 2m_2 - n_2 = 0$ and $m_1m_2 + n_1n_2 = 0$

$\Rightarrow \frac{l_1}{0} = \frac{m_1}{1} = \frac{n_1}{2} \Rightarrow \frac{l_2}{5} = \frac{m_2}{-2} = \frac{n_2}{1}$. Hence equations of the required line are

$\frac{x-1}{5} = \frac{y-2}{-2} = \frac{z+1}{1} = \lambda$. For $\lambda = -1$ and $\lambda = 1$, we get (B) and (C).

279. $f(x) = \frac{3(x-3)(x-4) + 4(x-2)(x-4) + 5(x-2)(x-3)}{(x-2)(x-3)(x-4)} = \frac{g(x)}{(x-2)(x-3)(x-4)}$

Root of $f(x) = 0$ are the roots of $g(x) = 0$ so one root in between 2 and 3 and another between 3 and 4.

280. Let $\frac{x}{1} = \frac{y}{2} = \frac{z}{3} = k \Rightarrow x = k, y = 2k, z = 3k$

Put in $3\beta^2x + 3(1 - 2\alpha)y + z = 3 \Rightarrow 3\beta^2k + 3(1 - 2\alpha)2k + 3k = 3$

$k(3\beta^2 + 6(1 - 2\alpha) + 3) = 3$

$$k = \frac{3}{3\beta^2 + 6(1-2\alpha) + 3} \quad \ldots\ldots (1)$$

Put in $6\alpha^2x + 3(1 - 2\beta)y + 2z = -6$

$6\alpha^2k + 3(1 - 2\beta)2k + 2.3k = -6$

$k(6\alpha^2 + 6(1 - 2\beta) + 6) = -6$

$$k = \frac{-6}{6\alpha^2 + 6(1-2\beta) + 6} \quad \ldots\ldots (2)$$

From equation (1) and (2), we get $\frac{3}{3\beta^2 + 6(1-2\alpha) + 3} = \frac{-6}{6\alpha^2 + 6(1-2\beta) + 6}$

$\Rightarrow 2(\beta - 1)^2 + 3(\alpha - 2)^2 = 0 \Rightarrow \alpha = 2, \beta = 1$

281. $\cot 12^\circ \cdot \cot 24^\circ \cdot \cot 48^\circ (\cot 28^\circ \cdot \cot(60^\circ - 28^\circ) \cot(60^\circ + 28^\circ)) = (\cot 12^\circ \cdot \cot 48^\circ)(\cot 24^\circ \cdot \cot 84^\circ)$

$$= \frac{\cot 36^\circ}{\cot 72^\circ} \times \frac{\cot 72^\circ}{\cot 36^\circ} = 1$$

282. $f'(x) = 2x - a$

At (2, 4)

$f'(x) = 4 - a$

equation of normal at (2, 4) is

$$(y - 4) = -\frac{1}{(4-a)}(x - 2).$$

Let point of intersection with x and y axis be A and B respectively then

$$A \equiv (-4a + 18, 0), B \equiv \left(0, \frac{4a-18}{a-4}\right)$$

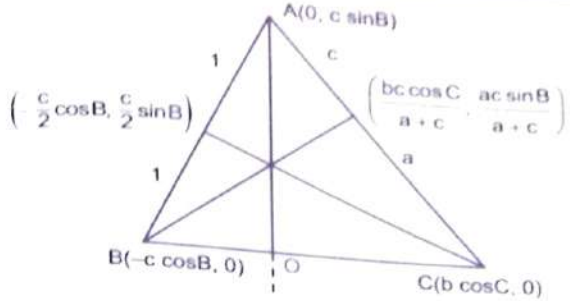
Hence $a > \frac{9}{2}$ as

$$\therefore \text{Area of triangle} = \frac{1}{2}(4a - 18)\frac{(4a-18)}{(a-4)} = 2$$

$\Rightarrow (4a - 17)(a - 5) = 0$

$\Rightarrow a = 5$ or $\frac{17}{4}$.

283.



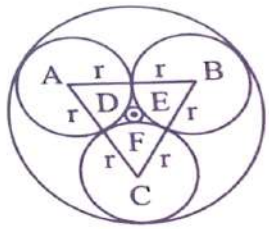
284. $A^3 + B^3 + C^3 = A^3 + B^3 + C^3 - 3ABC = (A + B + C)(A^2 + B^2 + C^2 - AB - BC - CA)$
$= (A + B + C)(A + \omega B + \omega^2 C)(A + \omega^2 B + \omega C)$ {ω = cube root of unity}
$= (A + B + C)(A + \omega B + \omega^2 C)\overline{(A + \omega B + \omega^2 C)}$
then $\det(A^3 + B^3 + C^3) \cdot \det(A + B + C) = \det(A + B + C)^2 \cdot \det(A + \omega B + \omega^2 C)\,\overline{\det(A + \omega B + \omega^2 C)}$
$= (\det(A + B + C))^2 \cdot |\det(A + \omega B + \omega^2 C)|^2 \geq 0.$

285. $2 \sin\alpha \cos\alpha = 2\cos^2\beta \Rightarrow \sin 2\alpha = 1 + \cos 2\beta$
$\therefore \cos 2\beta = -(1 - \sin 2\alpha) = -\left(1 - \cos\left(\frac{\pi}{2} - 2\alpha\right)\right) = -2\sin^2\left(\frac{\pi}{4} - \alpha\right) = -2\cos^2\left(\frac{\pi}{4} + \alpha\right)$

286. $4 \sin 3x + 5 \geq 4 \cos 2x + 5 \sin x$
$\Rightarrow 4(3 \sin x - 4 \sin^3 x) + 5 \geq 4(1 - 2\sin^2 x) + 5 \sin x$
$\Rightarrow 16 \sin^3 x - 8 \sin^2 x - 7 \sin x - 1 \leq 0$
$\Rightarrow (4 \sin x + 1)^2 (\sin x - 1) \leq 0$
$\Rightarrow x \in R$

287. $\sin^2\theta \tan\theta + \cos^2\theta \cot\theta - \sin 2\theta = 1 + \tan\theta + \cot\theta$
$\Rightarrow \tan\theta(1 - \sin^2\theta) + \cot\theta(1 - \cos^2\theta) + 1 + \sin 2\theta = 0$
$\Rightarrow 2 \sin\theta \cos\theta + 1 + \sin 2\theta = 0$
$\Rightarrow \sin 2\theta = -\frac{1}{2}$

288. Let r_1 and r_2 be radii of circles touching the given circles externally and internally respectively

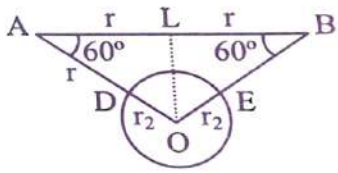


In Δ OBL

$\cos B = \frac{LB}{OB} = \frac{r}{r_2 + r}$

$\frac{\sqrt{3}}{2} = \frac{r}{r_2 + r} \Rightarrow r_2 + r = \frac{2r}{\sqrt{3}}$

$r_2 = \left(\frac{2 - \sqrt{3}}{\sqrt{3}}\right) r$



Also $r_2 = r_1 + 2r$, $r_2 = \left(\frac{2 - \sqrt{3}}{\sqrt{3}}\right) + 2r = \left(\frac{2 + \sqrt{3}}{\sqrt{3}}\right) r$

289. $\tan^{-1}\frac{a}{x} + \tan^{-1}\frac{b}{x} = \frac{\pi}{2} - \left(\tan^{-1}\frac{c}{x} + \tan^{-1}\frac{d}{x}\right) \Rightarrow \tan^{-1}\frac{\frac{a}{x} + \frac{b}{x}}{1 - \frac{ab}{x^2}} = \frac{\pi}{2} - \tan^{-1}\frac{\frac{c}{x} + \frac{d}{x}}{1 - \frac{cd}{x^2}}$

$\Rightarrow \tan^{-1}\frac{(a+b)x}{x^2-ab} = \cot^{-1}\frac{(c+d)x}{x^2-cd} \Rightarrow \frac{(a+b)x}{x^2-ab} = \frac{x^2-cd}{(c+d)x}$

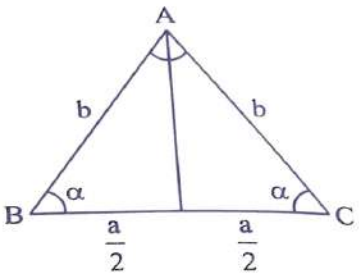
$\Rightarrow x^4 - (ab + ac + ad + bc + bd + cd)x^2 + abcd = 0$

its roots are x_1, x_2, x_3, x_4

$\therefore \sum x_1 = 0,\ \sum x_1x_2 = -\sum ab,\ \sum x_1x_2x_3 = 0$ and $x_1x_2x_3x_4 = abcd$

$\therefore \sum \frac{1}{x_1} = 0$ and $(x_2 + x_3 + x_4)(x_3 + x_4 + x_1)(x_4 + x_1 + x_2)(x_1 + x_2 + x_3) = x_1x_2x_3x_4$.

290. $\angle A = 180^\circ - 2\alpha$



By sine rule $\frac{a}{\sin A} = 2R,\ R = \frac{a}{2\sin(180^\circ - 2\alpha^2)} = \frac{a}{2}\operatorname{cosec} 2\alpha$

Also, $\frac{a}{2b} = \cos\alpha,\ R = b\cos\alpha \times \operatorname{cosec} 2\alpha = \frac{b}{2}\operatorname{cosec}\alpha$

Also, $\Delta = \frac{1}{2}b^2\sin(180^\circ - 2\alpha) = \frac{1}{2}b^2\sin 2\alpha$

Also, $r = \frac{\Delta}{s} = \frac{\Delta}{a+2b} = \frac{\Delta}{2b\cos\alpha + 2b} = \frac{b^2\sin 2\alpha}{4b(1+\cos\alpha)}$

$r = \frac{b\sin 2\alpha}{4(1+\cos\alpha)}$

291. $(\sin 2x + \sin 4x) - 2\sin 3x = 0 \Rightarrow 2\sin 3x \cdot \cos x - 2\sin 3x = 0 \Rightarrow 2\sin 3x(\cos x - 1) = 0$

Either, $\sin 3x = 0$ or $\cos x = 1$

$3x = n\pi$

$x = \frac{n\pi}{3}$

$x = 2n\pi$

$n \in I$.

292. $g(x) = \begin{vmatrix} a^{-x} & a^{x} & x^2 \\ a^{-3x} & a^{3x} & x^4 \\ a^{-5x} & a^{5x} & 1 \end{vmatrix} \Rightarrow g(-x) = \begin{vmatrix} a^{x} & a^{-x} & x^2 \\ a^{3x} & a^{-3x} & x^4 \\ a^{5x} & a^{-5x} & 1 \end{vmatrix} = -\begin{vmatrix} a^{-x} & a^{x} & x^2 \\ a^{-3x} & a^{3x} & x^4 \\ a^{-5x} & a^{5x} & 1 \end{vmatrix}$

$g(-x) = -g(x) \Rightarrow g(x) + g(-x) = 0$

$\therefore$ g(x) is an odd function

Also $g_4(x)$ is an odd function (where $g_4(x)$ is fourth derivative of g(x))

$g_4(x) + g_4(-x) = 0$

$g_4(0) = -g_4(0) \Rightarrow g_4(0) = 0$

293. Let, $f(x) = A + Bx + Cx^2 + Dx^3 + \ldots..$

Now, $f(0) = 0$ and $f'(0) = 0$

294. $A^2 = \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix},\ A^3 = \begin{bmatrix} \cos 3\alpha & \sin 3\alpha \\ -\sin 3\alpha & \cos 3\alpha \end{bmatrix},\ A^4 = \begin{bmatrix} \cos 4\alpha & \sin 4\alpha \\ -\sin 4\alpha & \cos 4\alpha \end{bmatrix}$

$\cos\alpha + \cos 2\alpha + \cos 3\alpha + \cos 4\alpha$

$= \cos\alpha + \cos 2\alpha + \cos(\pi - 2\alpha) + \cos(\pi - \alpha)\ (\because\ 5\alpha = \pi)$

$= \cos\alpha + \cos 2\alpha - \cos 2\alpha - \cos\alpha = 0$

$\sin\alpha + \sin 2\alpha + \sin 3\alpha + \sin 4\alpha$

$\Rightarrow \sin\alpha + \sin 2\alpha + \sin(\pi - 2\alpha) + \sin(\pi - \alpha)$

$\Rightarrow 2(\sin\alpha + \sin 2\alpha) \Rightarrow 4\sin\frac{3\alpha}{2}\cos\frac{\alpha}{2} \Rightarrow 4\sin\left(\frac{3\pi}{10}\right)\cos\left(\frac{\pi}{10}\right)$

$\Rightarrow 4\sin\left(\frac{\pi}{2} - \frac{\pi}{5}\right)\cos\frac{\pi}{10} = 4\cos\frac{\pi}{5}\cos\frac{\pi}{10} = a$

$B = \begin{bmatrix} 0 & a \\ -a & 0 \end{bmatrix}$ $\therefore$ B is skew symmetric

$|B| = a^2 = 16\cos^2\frac{\pi}{5}\cos^2\frac{\pi}{10} > 0$ $\therefore$ B is non-singular

295. $\sin A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ and $\cos A = \begin{bmatrix} \sin\theta & \cos\theta \\ \cos\theta & \sin\theta \end{bmatrix}$

$\therefore |\sin A| = \cos^2\theta + \sin^2\theta = 1$

Hence, sin A is invertible

Also, $(\sin A)\times(\sin A)^T = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1$

Hence, sin A is orthogonal

Also, $2\sin A\cos A = 2\begin{bmatrix} -\cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} \sin\theta & \cos\theta \\ \cos\theta & \sin\theta \end{bmatrix}$

$= 2\begin{bmatrix} \sin 2\theta & 1 \\ \cos 2\theta & 0 \end{bmatrix} \neq \sin 2A$

296. Let $x^{\log_b a} = y$

$y^2 - 5y + 6 = 0$

$y = 2, y = 3$

$\Rightarrow x = b^{\log_a 2}$, $x = 3^{\log_a b}$

297. $\alpha\beta = \log_4 5 \cdot \log_5 6$

$= \frac{\log_e 5}{\log_e 4} \cdot \frac{\log_e 6}{\log_e 5} = \log_4 6$

$\frac{1}{2\alpha\beta - 1} = \frac{1}{2\log_{2^2} 6 - 1}$

$= \frac{1}{\log_2 3} = \log_3 2$

298. The sine curve chord going two points on the curve lies below the curve, hence (A) cannot be true. In log curve also, the same pattern is following but in tan curve and x^2 curves opposite pattern is followed.

299. $\Rightarrow \log((x-2)|x-4|) = 0$

$\Rightarrow (x-2)|x-4| = 1$

Case (I): $x - 4 > 0$

$\Rightarrow (x-2)(x-4) = 1$

$\Rightarrow x = 3 \pm \sqrt{2}$

$\because x \neq 3 - \sqrt{2}$

$\therefore x = 3 + \sqrt{2}$

Case (II): If, $x - 4 < 0$

$\Rightarrow -(x-2)(x-4) = 1$

$\Rightarrow x = 3$

300. $\sum_{r=0}^{\infty}\left(\log_3\left(e^{4/3}\right)^r\right)^{-1} = \sum_{r=0}^{\infty}\left(\frac{4}{3}\right)^r (\log_3 e)^{-1}$

$= \frac{1}{\log_3 e}\sum_{r=0}^{\infty}\left(\frac{4}{3}\right)^{-r}$

$= \log_e 3\left[1+\frac{3}{4}+\left(\frac{3}{4}\right)^2+\ldots+\infty\right]$

$= \log_e 3\left[\frac{1}{1-3/4}\right] = 4\log_e 3 = \log_e 81$

301. $y = \frac{h(f(x))}{g(f(x))} = \frac{2(4\sin^2 x-1)+1}{2(4\sin^2 x-1)-1} = \frac{8\sin^2 x-1}{8\sin^2 x-3} = 1+\frac{2}{8\sin^2 x-3}$

Hence, range is $\left(-\infty, \frac{1}{3}\right]\cup\left[\frac{7}{5}, \infty\right)$

302. $\lim_{x\to -\frac{1}{2}} \frac{f(\sin^{-1} kx)-3}{g(x)} = \lim_{x\to -\frac{1}{2}} \frac{4\sin^2(\sin^{-1} kx)-1-3}{2x-1}$

$= \lim_{x\to -\frac{1}{2}} \frac{4k^2x^2-4}{2x+1}$

Here, Nr must be zero, $k^2\ 0 \Rightarrow k = \pm 2$

$= \lim_{x\to -\frac{1}{2}} \frac{4(4x^2-1)}{2x+1} = \lim_{x\to -\frac{1}{2}} 4(2x-1) = -8$

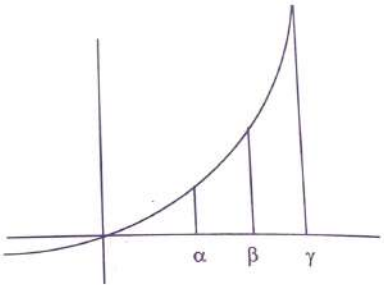
So, $|k| + |l| = 2 + 8 = 10$

305. Consider $g(x) = \frac{f(x)}{x}$

$g'(x) = \frac{f'(x)x-f(x)}{x^2} = \frac{f'(x)}{x^2}\left(x-\frac{f(x)}{f'(x)}\right)$ as $\frac{f'(x)}{x^2} > 0$

Let $h(x) = x - \frac{f(x)}{f'(x)}$

$h'(x) = 1 - \frac{f'(x)^2 - f(x)f''(x)}{f'(x)^2} = \frac{f(x)f''(x)}{f'(x)^2} > 0$



$\therefore x - \frac{f(x)}{f'(x)}$ is increasing function

$x - \frac{f(x)}{f'(x)} > 0 - \frac{f(0)}{f'(0)} = 0$

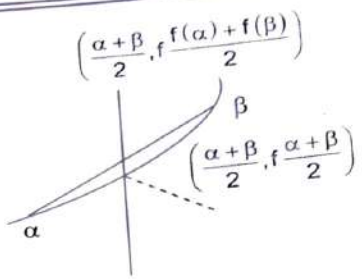
$\therefore g'(x) > 0.$

$\therefore \frac{f(\gamma)}{\gamma} > \frac{f(\beta)}{\beta} > \frac{f(\alpha)}{\alpha}$

Satisfies A, C, D

306. As the function is concave up

$$\frac{f(\alpha)+f(\beta)}{2} > f\left(\frac{\alpha+\beta}{2}\right)$$



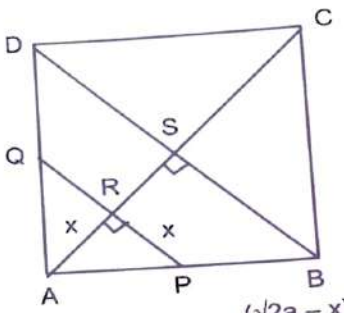
307.-308. Two cases are possible for PQ w.r.t. diagonal OA

Case –I: When $x = AR \le AS$ i.e. $x \in \left[0, \frac{a}{\sqrt{2}}\right]$

Triangle ARP is an isosceles right angle triangle,

Hence $AP = \sqrt{2}\,x = AQ$

$\Rightarrow$ Area of segment i.e. $f(x) = \frac{1}{2}(\sqrt{2}x)^2 = x^2,\ x \in \left[0, \frac{a}{\sqrt{2}}\right]$



Case –II: When $x = AR \ge AS$

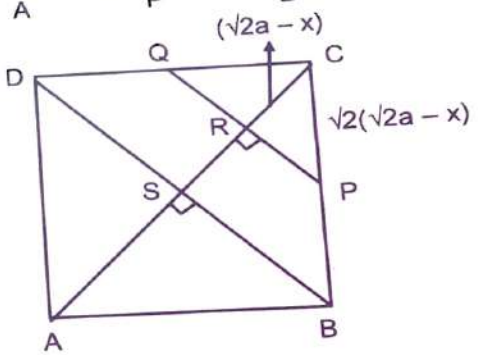
i.e. $x \in \left[\frac{a}{\sqrt{2}}, \sqrt{2}a\right]$

$\Rightarrow CP = CQ = \sqrt{2}\,(\sqrt{2}\,a - x)$

Area of segment $f(x) = a^2 - (\sqrt{2}a - x)^2$

$x \in \left[\frac{a}{\sqrt{2}}, \sqrt{2}a\right]$

i.e. $f: [0, \sqrt{2}\,a] \to [0, a^2]$



$$f(x) = \begin{cases} x^2, & 0 \le x \le \frac{a}{\sqrt{2}} \\ a^2 - (\sqrt{2}a - x)^2, & \frac{a}{\sqrt{2}} \le x \le a \end{cases}$$

When $0 \le x \le \frac{a}{\sqrt{2}}$, $f(x) = x^2$, $0 \le f(x) \le \frac{a^2}{2}$

Put $x = f^{-1}(x) \Rightarrow (f^{-1}(x))^2 = x$, $0 \le x \le \frac{a^2}{2}$, $0 \le f^{-1}(x) \le \frac{a}{\sqrt{2}}$

$\Rightarrow f^{-1}(x) = \sqrt{x}$, $0 \le x \le \frac{a^2}{2}$

When $\frac{a}{\sqrt{2}} \le x \le \sqrt{2}\,a$, $f(x) = a^2 - (\sqrt{2}a - x)^2$, $\frac{a^2}{2} \le f(x) \le a^2$

Put $x = f^{-1}(x)$, $x = a^2 - (\sqrt{2}a - f^{-1}(x))^2$

$\frac{a^2}{2} \le x \le a^2$, $\frac{a}{\sqrt{2}} \le f^{-1}(x) \le \sqrt{2}\,a$

$\Rightarrow (\sqrt{2}a - f^{-1}(x))^2 = a^2 - x$

$\Rightarrow \sqrt{2}a - f^{-1}(x) = \pm\sqrt{a^2 - x}$

$\Rightarrow f^{-1}(x) = \sqrt{2}a \mp \sqrt{a^2 - x}$

Neglecting positive sign since $\frac{a}{\sqrt{2}} \le f^{-1}(x) \le \sqrt{2}\,a$

$\Rightarrow f^{-1}(x) = \sqrt{2}a - \sqrt{a^2 - x}$

$$f^{-1}(x) = \begin{cases} \sqrt{x}, & 0 \le x \le \frac{a^2}{2} \\ \sqrt{2}a - \sqrt{a^2 - x}, & \frac{a^2}{2} \le x \le a^2 \end{cases}$$

Hence domain of $f^{-1}(x)$ is $x \in [0, a^2]$

For $a = 2$,

$$f(x) = \begin{cases} x^2, & 0 \le x \le \sqrt{2} \\ 4 - (2\sqrt{2} - x)^2, & \sqrt{2} \le x \le 2 \end{cases}$$

And $f^{-1}(x) = \begin{cases} \sqrt{x}, & 0 \le x \le 2 \\ 2\sqrt{2} - \sqrt{4 - x}, & 2 \le x \le 4 \end{cases}$

Domain of $\sqrt{f^{-1}(x) - f(x)}$ is when $f^{-1}(x) - f(x) \ge 0$

Case –I: $0 \le x \le \sqrt{2}$

$\sqrt{x} - x^2 \ge 0 \Rightarrow \sqrt{x}\,(1 - x^{3/2}) \ge 0$

$\Rightarrow 0 \le x \le 1$.

Case –II: $\sqrt{2} \le x \le 2$

$f^{-1}(x) = \sqrt{x}$ i.e. $f^{-1}(x) \in \left[2^{1/4}, \sqrt{2}\right]$

And $f(x) = 4 - (2\sqrt{2} - x)^2$, $f(x) \in [2, 8(\sqrt{2} - 1)]$

Hence $f^{-1}(x) - f(x) \ge 0\ \forall\ x \in \phi$.

Hence domain of $\phi(x) = \sqrt{f^{-1}(x) - f(x)}$ is $x \in [0, 1]$ for $a = 2$

The equation $f(x) - f^{-1}(x) = 0$ has exactly three solutions

$$f(x) = \begin{cases} x^2, & 0 \le x \le \frac{a}{\sqrt{2}} \\ a^2 - (\sqrt{2}a - x)^2, & \frac{a}{\sqrt{2}} \le x \le a \end{cases}$$

$$f^{-1}(x) = \begin{cases} \sqrt{x}, & 0 \le x \le \frac{a^2}{2} \\ \sqrt{2}a - \sqrt{a^2 - x}, & \frac{a^2}{2} \le x \le a^2 \end{cases}$$

Exactly three solutions exist if $\frac{a^2}{2} = \frac{a}{\sqrt{2}}$ i.e. $a = \sqrt{2}$

309.-310. For interval $x \in \left(0, \frac{\pi}{2}\right)$

$$I = \int \left(\sqrt{\tan x} + \sqrt{\cot x}\right) dx = \int \left(\frac{\sqrt{\sin x}}{\sqrt{\cos x}} + \frac{\sqrt{\cos x}}{\sqrt{\sin x}}\right) dx$$

$$= \int \frac{\sin x + \cos x}{\sqrt{\sin x \cos x}}\, dx = \int \sqrt{2}\, \frac{(\sin x + \cos x)}{\sqrt{\sin 2x}}\, dx$$

$$= \sqrt{2} \int \frac{\sin x + \cos x}{\sqrt{1 - (\sin x - \cos x)^2}}\, dx = \sqrt{2} \sin^{-1}(\sin x - \cos x) + c$$

Put $\sin x - \cos x = t$

$I = \sqrt{2}\sin^{-1}(\sin x - \cos x) + c \quad x \in \left(0, \frac{\pi}{2}\right)$

$I = \sqrt{2}\tan^{-1}\left(\frac{\sin x - \cos x}{\sqrt{1-(\sin x - \cos x)^2}}\right) + c$

$I = \sqrt{2}\tan^{-1}\left(\frac{\sin x - \cos x}{\sqrt{2}\sqrt{\sin x \cos x}}\right) + c$

$I = \sqrt{2}\tan^{-1}\left(\frac{\sqrt{\tan x} - \sqrt{\cot x}}{\sqrt{2}}\right) + c$

$x \in \left(0, \frac{\pi}{2}\right)$ for interval $x \in \left(\pi, \frac{3\pi}{2}\right)$

$I = \int(\sqrt{\tan x} + \sqrt{\cot x})\,dx = -\int \frac{\sin x + \cos x}{\sqrt{\sin x \cos x}}\,dx$

$I = -\sqrt{2}\sin^{-1}(\sin x - \cos x) + c$

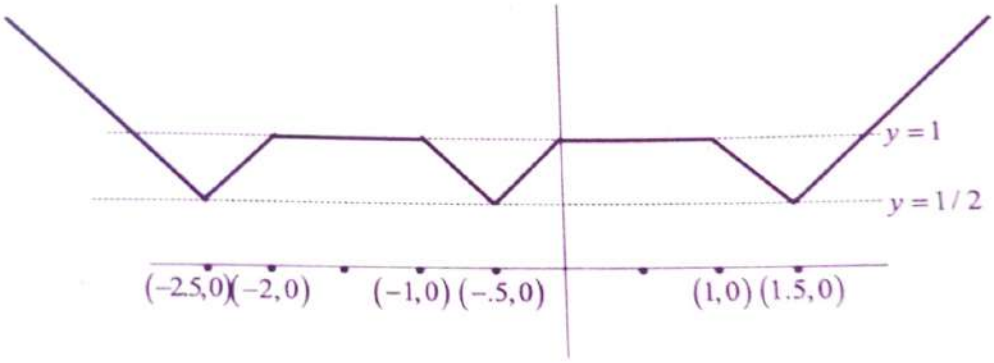
$I = \sqrt{2}\sin^{-1}(\cos x - \sin x) + c \quad x \in \left(\pi, \frac{3\pi}{2}\right)$

$I = \sqrt{2}\tan^{-1}\left(\frac{\cos x - \sin x}{\sqrt{1-(\cos x - \sin x)^2}}\right) + c$

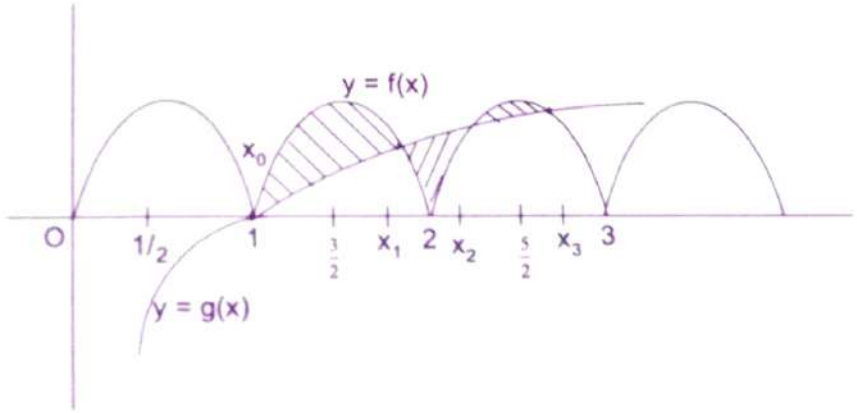
$I = \sqrt{2}\tan^{-1}\left(\frac{\cos x - \sin x}{\sqrt{2}\sqrt{\sin x \cos x}}\right) + c$

$I = \sqrt{2}\tan^{-1}\left(\frac{\sqrt{\tan x} - \sqrt{\cot x}}{\sqrt{2}}\right) + c \quad x \in \left(\pi, \frac{3\pi}{2}\right) \because \sqrt{\tan x} = \frac{\sqrt{-\sin x}}{\sqrt{-\cos x}},\ \sqrt{\cot x} = \frac{\sqrt{-\cos x}}{\sqrt{-\sin x}}$

312. $y = g(x)$



313.



314. $x_{n+1} = x_3 \Rightarrow n = 2$

317. $f(\alpha, \beta) \le 0$

$$\left|\frac{\beta}{\alpha} - \frac{3}{2}\right| + (3\alpha - 2\beta)^6 + \sqrt{e\alpha + 2\beta - 2e - 6} \le 0$$

$$\frac{\beta}{\alpha} - \frac{3}{2} = 0 \Rightarrow \beta = \frac{3\alpha}{2}$$

$e\alpha + 2\beta - 2e - 6 = 0$

$2(\beta - 3) + e(\alpha - 2) = 0$

$\alpha = 2, \beta = 3$

318. Let equation of straight line $\frac{x}{\cos\theta} = \frac{y}{\sin\theta} = r$

Let $OA = r_1$, $OB = r_2$, $OC = r_3$ then $A(r_1 \cos\theta, r_1 \sin\theta)$

$B(r_2 \cos\theta, r_2 \sin\theta)$

$C(r_3 \cos\theta, r_3 \sin\theta)$

$\therefore$ A lies on $x = 2$, $r_1 \cos\theta = 2$

$$\Rightarrow r_1 = \frac{2}{\cos\theta} \qquad \ldots\ldots (1)$$

B lies on $y = 3$, $r_2 \sin\theta = 3$, $r_2 = \frac{3}{\sin\theta}$ $\qquad \ldots\ldots (2)$

and C lines on $x + y = 8$

$r_3 \cos\theta + r_3 \sin\theta = 8$

$\therefore\ r_1 r_2 r_3 = 48\sqrt{2}$

$$\left(\frac{2}{\cos\theta}\right)\left(\frac{3}{\sin\theta}\right)\left(\frac{8}{\sin\theta + \cos\theta}\right) = 48\sqrt{2}$$

$$\sqrt{2}\cos\theta\sin\theta(\cos\theta + \sin\theta) = 1$$

$$\sin 2\theta \sin\left(\theta, \frac{\pi}{4}\right) = 1$$

$\therefore$ Both $\sin 2\theta$ and $\sin\left(\theta + \frac{\pi}{4}\right)$ should be equal to 1, so $\theta - \frac{\pi}{4}$ then $r_1 = 2\sqrt{2}$, $r_2 = 3\sqrt{2}$, $r_3 = \frac{8}{\sqrt{2}} = 4\sqrt{2}$

Hence, $OA + OB + OC = r_1 + r_2 + r_3 = 9\sqrt{2}$

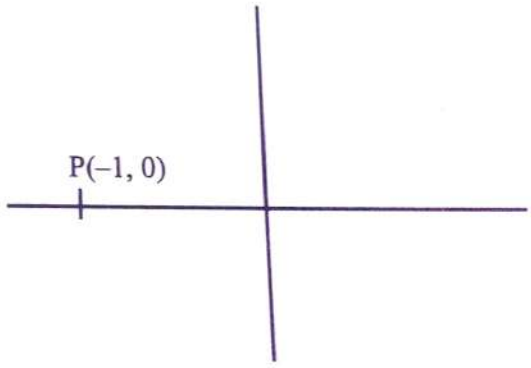
319. Particle moves 1 radian on each circle starts from P(–1, 0).

To cross x-axis again should cover π radians, hence will be on 4^{th} circle

$\Rightarrow$ point (4, 0)

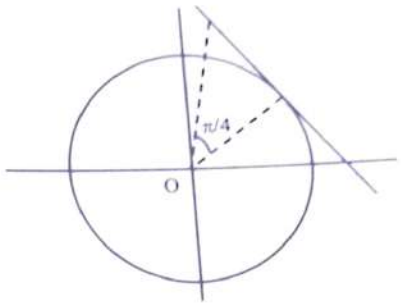
and to cross positive y-axis $\frac{3\pi}{2}$ radian i.e. 5^{th} circle

$\Rightarrow$ point (0, 5).



320. Angular distance travelled from starting of motion on one circle to reach to other is $\left(1+\frac{\pi}{4}\right)$.

$\Rightarrow$ on 4th circle particle will cross the positive x-axis again
$\Rightarrow x \in (2\sqrt{2}, 4)$.



321.-322. Foot of perpendicular from focus S(h, k) on the tangent $y = 4$ is (h, 4) which lies on line $x + y - 5 = 0$
$h = 1 \therefore$ focus lies on the line $x - y - 1 = 0 \therefore k = 0$
Also, R is (5, 4) and Q is (–3, –4) (S is mid-point of QR)
Centre of the circle C lies at focus in radius = RS
Let side of square be K then $\frac{K}{\sqrt{2}} = 4\sqrt{2}$

323.-324. $\lambda x - y + 2(1 + \lambda) = 0 \Rightarrow \lambda(x + 2) - (y - 2) = 0$ pass through (–2, 2)
$\mu x - y + 2(1 - \mu) = 0 \Rightarrow \mu(x - 2) - (y - 2) = 0$ pass through (2, 2)
Clearly these represent the foci of ellipse, so $2ae = 4$
The circle $x^2 + y^2 - 4y - 5 = 0 \Rightarrow x^2 + (y - 2)^2 = 9$ represents auxiliary circle thus $a^2 = 9 \Rightarrow e = \frac{2}{3}$ and $b^2 = 5$

323. The eccentricity $e = \frac{2}{3}$

324. Area of the largest triangle = $abe = 2\sqrt{5}$

325. The equations of PQ are $\frac{x-1}{1} = \frac{y+3}{3} = \frac{z-2}{2}$
Any point on PQ has co-ordinates $(1 + r, -3 + 3r, 2 + 2r)$
It is Q if $1 + r + (-3 + 3r) - 3(2 + 2r) = 0 \Rightarrow r = -4$
$\therefore Q \equiv (-3, -15, -6)$
Let $P' \equiv (\alpha, \beta, \gamma)$ be the image of P in the plane. Then the middle point of PP′ is $\left(\frac{1+\alpha}{2}, \frac{-3+\beta}{2}, \frac{\gamma+2}{2}\right)$ which is on the plane

So, $\frac{1+\alpha}{2} + \frac{-3+\beta}{2} - 3 \cdot \frac{\gamma+2}{2} = 0$

$\Rightarrow \alpha + \beta - 3r = 8$

Also, PP′ is perpendicular to the plane

So, $\frac{\alpha-1}{1} = \frac{\beta+3}{1} = \frac{\gamma-2}{-3} = \frac{\alpha-1+\beta+3-3(\gamma-2)}{1+1-3(-3)} = \frac{16}{11}$

$\therefore \alpha = \frac{27}{11}, \beta = -\frac{17}{11}, \gamma = -\frac{26}{11}.$

So, $P' \equiv \left(\frac{27}{11}, -\frac{17}{11}, \frac{26}{11}\right)$

Clearly, P′, Q, R are on the same line. So, QR has the equations $\frac{x+3}{15} = \frac{y+15}{37} = \frac{z+6}{10}$

$\Rightarrow \frac{x+3-15}{15} = \frac{y+15-37}{37} = \frac{z+6-10}{10} \Rightarrow \frac{x-12}{15} = \frac{y-22}{37} = \frac{z-4}{10}$

326. Three points on the plane are $P \equiv (1, -3, 2)$, $Q \equiv (-3, -15, -6)$ and $P' \equiv \left(\frac{27}{11}, -\frac{17}{11}, -\frac{26}{11}\right)$. So, the equation of the plane is

$$\begin{vmatrix} x-1 & y+3 & z-2 \\ -3-1 & -15+3 & -6-2 \\ \frac{27}{11}-1 & -\frac{17}{11}+3 & -\frac{26}{11}-2 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x-1 & y+3 & z-2 \\ -4 & -12 & -8 \\ 16 & 16 & -48 \end{vmatrix} = 0 \text{ or } 11x - 5y + 2z = 30$$

327. Sum of all four roots = 6
Sum of two roots = 0
$\Rightarrow$ Sum of other two roots = 6
The factors corresponding to there roots are of the type
$x^2 - 0(x) + p$ and $x^2 - 6x + q$ i.e. $x^2 + p$ and $x^2 - 6x + q$
Thus we have $4x^2 - 24x^3 + 31x^2 + 6x - 8 \equiv 4(x^2 + p)(x^2 - 6x + q)$
Equating like powers of x

$$31 = 4(p+q)\, p+q = \frac{31}{4}$$

$$6 = 4(-6p) \Rightarrow p = -\frac{1}{4}$$

$\therefore\ q = 8$

$\therefore$ Equation $f(x) = 0$ can be written as $\left(x^2 - \frac{1}{4}\right)\left(x^2 - 6x + \delta\right) = 0$

$$x = \pm\frac{1}{2},\ x = 2, 4$$

$$\therefore\ \alpha = -\frac{1}{2},\ \beta = \frac{1}{2},\ \gamma = 2,\ \delta = 4$$

$$\therefore\ \delta^{\beta} + \frac{1}{\delta^{\alpha}} + \delta^{\gamma} + \gamma^{\delta} = 2 + 2 + 16 + 16 = 36$$

328. $\displaystyle\int_{2\alpha}^{2\beta} \frac{x^{\delta+1} - 5x^{\gamma+1} + 2\beta|x| + 1}{x^2 + 4\beta|x| + 1}\,dx$ (Putting α, β, γ and δ)

$$= \int_{-1}^{1} \frac{x^5 - 5x^3 + |x| + 1}{x^2 + 2|x| + 1}\,dx$$

$\because\ x^5 - 5x^3$ are odd function

$$= \int_{-1}^{1} \frac{|x|+1}{(|x|+1)^2}\,dx = 2\int_{0}^{1} \frac{dx}{(|x|+1)} = 2\int_{0}^{1} \frac{dx}{1+x}$$

$$= 2\ell n|(1+x)|$$

$$= 2\ell n 2$$

329.-330. $f(x) = x^2 + bx + c\ \forall\ x \in R$

Least value at $-\frac{b}{2} = -1$

$\Rightarrow b = 2$
Graph of f(x) cuts y-axis, when $x = 0$
$\Rightarrow c = 2 \Rightarrow f(x) = x^2 + 2x + 2$
Now, $f(-2) + f(0) + f(1) = 9$
Also, $x^2 + 2x + 2 = a$
$\Rightarrow a \in (1, \infty)$

331. $a_r = \dfrac{5^{r+1}}{r+1} - \dfrac{5^r}{r}$

$S_n = f(n) - f(0) = \dfrac{5^{n+1}}{n+1} - 5$
$S_\infty \to \infty$

332. $a_n = \sin(n+1) - \sin n$
$S_n = \sin(n+1) - \sin 1$
$S_{50} = \sin 51 - \sin 1.$

333. $f(n) = 2f(n-1) - f(n-2) + f(n-4)\ \forall\ n \geq 5.$
Now $f(1) = 1$, $f(2) = 2$, $f(3) = 3$, $f(4) = 5$, $f(5) = 2 \times 5 - 3 + 1 = 8$, $f(6) = 2 \times 8 - 5 + 2 = 13$

334. $f(7) = 2 \times 13 - 8 + 3 = 21$
Similarly $f(8) = 34$, $f(9) = 55$, $f(10) = 89$
$f(11) = 144$, $f(12) = 233$, $f(13) = 377$

335. A coin can be drawn only at even numbered round. Therefore A will not be drained out at the 11^{th} round.

336. He gets exactly L head in the first '10' rounds and then exactly one head at the next two rounds
$\therefore$ Probability in this case is ${}^{10}C_1 pq^9 \cdot {}^2C_1 \cdot pq \cdot q^2 = 20p^2q^{12}$
Therefore required probability $= 65p^2q^{12}$

337.-338. Let z_α, z_β be roots of the equation $z_1z^2 + z_2z + z_3 = 0$ such that $|z_\alpha| = 1$then
$(z_\alpha + z_\beta)^2 = z_\alpha z_\beta$

$$\left(\frac{-z_2}{z_1}\right)^2 = \frac{z_3}{z_1}$$

$\Rightarrow z_2^2 = z_1z_3$
Hence z_1, z_2, z_3 are in G.P.
Since $z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$
$\Rightarrow \Delta ABC$ is equilateral.
Also $PA^2 + PB^2 + PC^2$
$= 3|z_p|^2 - (1 + \omega + \omega^2)z_1\,\bar{z} - (1 + \omega + \omega^2)z_1z + |z_1|^2 + |z_1|^2 + |z_1|^2 = 6r^2 = 18.$

339.-340. $P = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$; $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$|P - nI| = 0$

$$\begin{vmatrix} 1-n & 2 & 0 \\ 2 & 1-n & 0 \\ 0 & 0 & 1-n \end{vmatrix} = 0\ ;\ (1-n)((1-n)^2 - 4) = 0$$

$n = 1$; $1 - n = \pm 2$; $n = -1$, $n = 3$
$n_1 = -1$; $n_2 = 1$, $n_3 = 3$

$N = \text{diag}(n_1, n_2, n_3)$
$|N| = |n_1 n_2 n_3|$
$|\text{adj } N| = |N|^2 = (n_1 n_2 n_3)^2 = 9$
$Q^T = Q + \alpha I$ ..... (1) (Now take its transpose)
$Q = Q^T + \alpha I$ ..... (2)
Adding equation (1) and (2), we get
$Q^T + Q = Q + Q^T + 2\alpha I;\ 2\alpha I = 0;\ \alpha = 0$

$$P = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix};\ T_r(P) = 3$$

$$P^2 = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 4 & 0 \\ 4 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 5 & 4 & 0 \\ 4 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 13 & 14 & 0 \\ 14 & 13 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$T_r(P^2) = 11;\ T_r(P^3) = 27$

$$T_r\left(P^n\right) = \begin{cases} 3^n + 2 & \text{if n is even} \\ 3^n & \text{if n is odd} \end{cases}$$

341-342. By applying $R_1 \to R_1 - R_3,\ R_2 \to R_2 - 3R_3$

$$\Delta(x) = -2x\begin{vmatrix} f(x) & g(x) & h(x) \\ f(\alpha) & g(\alpha) & h(\alpha) \\ f'(\alpha) & g'(\alpha) & h'(\alpha) \end{vmatrix}$$

Clearly $\Delta(\alpha) = 0,\ \Delta'(\alpha) = 0$

341. As f, g, h are function $\Delta(-\alpha) = 0,\ \Delta'(-\alpha) = 0$
Means $\Delta(x)$ has factors $x, (x-\alpha)^2, (x-\alpha)^2$

342. As $\Delta(x)$ is divided by $x^2 - 1$ remainder will be $ax + b$ (linear)
$\Delta(x) = (x^2 - 1)\,Q(x) + ax + b$
$\Delta(1) = a + b = 0$
$\Delta(-1) = -a + b = -352$
$ax + b = 176(x - 1)$

343.-344. $AH = 2R\cos A$
$AH \cdot BH \cdot CH = 12$
$\Rightarrow 8R^3 \cos A \cos B \cos C = 12$ ..... (1)
$\Rightarrow AH^2 + BH^2 + CH^2 = 10$
$4R^2(\cos^2 A + \cos^2 B + \cos^2 C) = 10$ ..... (2)
Also we know for a triangle $\sum \cos^2 A = 1 - 2\prod \cos A$

$$\Rightarrow \frac{5}{2R^2} = 1 - 2 \cdot \frac{3}{2R^3}$$
$$\Rightarrow 2R^3 - 5R - 6 = 0$$
$$\Rightarrow R = 2$$

343. $$\frac{\cos A \cos B \cos C}{\cos^2 A + \cos^2 B + \cos^2 C} = \frac{3/2R^3}{5/2R^2} = \frac{3}{5R} = \frac{3}{10}$$

344. $HD \cdot HE \cdot HF = \prod (2R\cos B\cos C)$

$= 8R^3(\cos A \cos B \cos C)^2$

$= 12 \cdot \frac{3}{2R^3} = \frac{9}{4}$

345.-346. Given that $\int_0^x \left(t^2 - 8t + 13\right)dt = x\sin\left(\frac{a}{x}\right)$ RHS shows that $x \neq 0$

Integrating LHS, we get $\left[\frac{t^2}{3} - 4t^2 + 13t\right]_0^x = x\sin\left(\frac{a}{x}\right)$

or $\sin\left(\frac{a}{x}\right) = \left(\frac{1}{3}\right)\left[x^2 - 12x + 39\right] = \frac{1}{3}(x-6)^2 + 1$

But $\sin\left(\frac{a}{x}\right) \leq 1$

So $\sin\left(\frac{a}{x}\right) = 1$ which is possible only for $x = 6$

Then we have $\sin\left(\frac{a}{6}\right) = 1$

$\frac{a}{6} = 2n\pi + \frac{\pi}{2}$

Hence $x = 6$, $a = 12n\pi + 3\pi$, $n \in Z$ for $a \in [10, 100]$

There are exactly three values of $a = 3\pi$, 15π and 27π

$|y - \cos a| < x$

$|y + 1| < 6$

$y \in [-7, 5]$

347.-348. $z = -\lambda \pm \sqrt{\lambda^2 - 1}$

Case – I Where $\lambda \in (-1, 1)$; we have $\lambda^2 < 1 \Rightarrow \lambda^2 - 1 < 0$

$\Rightarrow z = -\lambda \pm i\sqrt{1-\lambda^2}$; or $x = -\lambda$, $y = \pm\sqrt{1-\lambda^2}$

$\Rightarrow y^2 = 1 - x^2$

$\Rightarrow x^2 + y^2 = 1$

Case – II $\lambda > 1$

$\Rightarrow \lambda^2 - 1 > 0$

$\Rightarrow z = -\lambda \pm \sqrt{\lambda^2 - 1}$

or; $x = -\lambda \pm \sqrt{\lambda^2 - 1}$; $y = 0$

$\Rightarrow$ roots are $\left(-\lambda + \sqrt{\lambda^2 - 1}, 0\right), \left(-\lambda - \sqrt{\lambda^2 - 1}, 0\right)$

One root lies inside the unit circle and other root lies outside the unit circle.

Case – III Where λ is very large, then $z = -\lambda - \sqrt{\lambda^2 - 1} \approx -2\lambda$

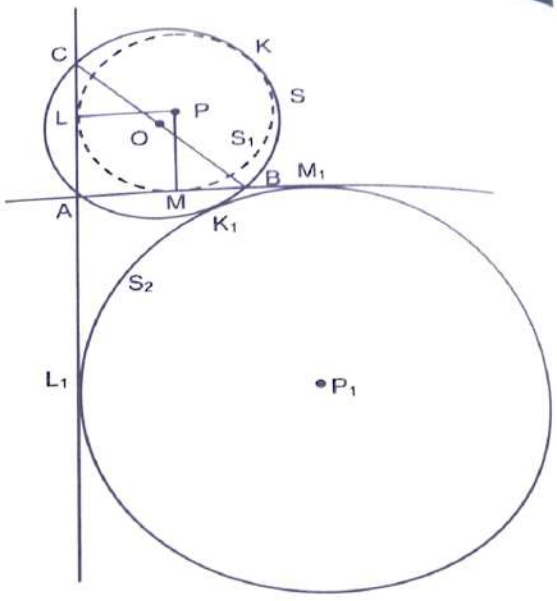
and $z = -\lambda + \sqrt{\lambda^2 - 1} = \frac{\left(-\lambda + \sqrt{\lambda^2 - 1}\right)\left(-\lambda - \sqrt{\lambda^2 - 1}\right)}{\left(-\lambda - \sqrt{\lambda^2 - 1}\right)}$

$= \frac{1}{-\lambda - \sqrt{\lambda^2 - 1}} = -\frac{1}{2\lambda}$

351.-352.

Let S_1 touch the circle S at K, the rays AB and AC at M and L respectively.
We have $PL = PM = PK = r_1$
and $R = OK = OP + r_1$ where P is the centre of S_1.
R is the circumradius of ΔABC, also AMPL is a square of side r_1
$BM = AB - AM = c - r_1$
$LC = AC - AL = b - r_2$
$PB^2 = PM^2 + MB^2 = r_1^2 + (c - r_1)^2$
$PC^2 = PL^2 + LC^2 = r_1^2 + (b - r_1)^2$
By Apollonius theorem $PB^2 + PC^2 = 2(PO^2 + CO^2)$
i.e. $r_1^2 + (c - r_1)^2 + r_1^2 + (b - r_1)^2 = 2((R - r_1)^2 + R^2)$

putting $R = \frac{a}{2}$, $a^2 = b^2 + c^2$
we get $r_1 = b + c - a$, $r_2 = b + c + a$.
Hence $r_1 r_2 = (b + c - a)(b + c + a)$
$= 2bc = 4\left(\frac{bc}{2}\right) = 4$area of ΔABC.



353. $z_1 \cdot z_2 + i(z_1 \times z_2) = |z_1||z_2|\cos\theta + i|z_1||z_2|\sin\theta$
$= |z_1||z_2|e^{i\theta}$ where $\theta = \beta - \alpha = \left(|z_1|e^{-i\alpha}\right)\left(|z_2|e^{i\beta}\right) = z_2\bar{z}_1$

354. $z_1 \cdot z_2 = |z_1||z_2|\cos\theta = 0$
$Re(\bar{z}_1 z_2) = 0$
$\bar{z}_1 z_2 = ik$
$\frac{z_2}{z_1}(\bar{z}\ \bar{z}_1) = ik \Rightarrow \frac{z_2}{z_1} = ik$
Also even if $z_1 = 0$ or $z_2 = 0$
$z_1 \cdot z_2 = 0$

355. Clearly, $p^{\frac{\log_q(\log_q r)}{(\log_q p)}} = p^{\log_p(\log_q r)} = \log_q r$

and let $y = \sqrt{6\sqrt{6\sqrt{6+.....\infty}}}$, $y > 0$
$\Rightarrow y = \sqrt{6+y}$
$\Rightarrow y^2 = 6 + y$
$\Rightarrow y^2 - y - 0$
$\Rightarrow (y - 3)(y + 2) = 0$
But $y > 0$, so $y = 3$

$\therefore$ Given expression $\log_3(\log_q r) = q^{3^{\log_3(\log_q r)}} = q^{(\log_q r)} = r$

356. As, $\frac{1}{\log_a\left(2-\sqrt{3}\right)} + \frac{1}{\log_b\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)} = \log_{2-\sqrt{3}} a + \log_{\frac{\sqrt{3}-1}{\sqrt{3}+1}} b$

$= \log_{2-\sqrt{3}} a + \log_{2-\sqrt{3}} b = \log_{2-\sqrt{3}}(ab)$

Now, $\left(2+\sqrt{3}\right)^{\log_{2+\sqrt{3}}(ab)} = \frac{1}{12}$

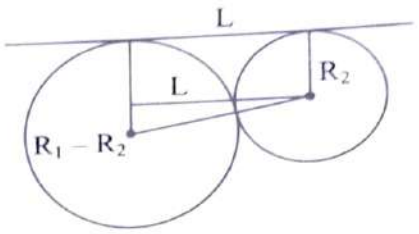
$\Rightarrow \left(2+\sqrt{3}\right)^{\log_{2+\sqrt{3}}\left(\frac{1}{ab}\right)} = \frac{1}{12} \Rightarrow \frac{1}{ab} = \frac{1}{12} \Rightarrow ab = 12$

357. $f_{2014}(x) < f_{2013}(-x) \Rightarrow f(x) > -x$

$\Rightarrow x^3 + x - 30 < 0 \Rightarrow (x-3)\left(x^2+3x+10\right) < 0 \Rightarrow x < 3$

358. Note that lengths of the direct common tangent on two circles, tangent to each other

$L = \sqrt{(R_1+R_2)^2 - (R_1-R_2)^2} = 2\sqrt{R_1R_2}$



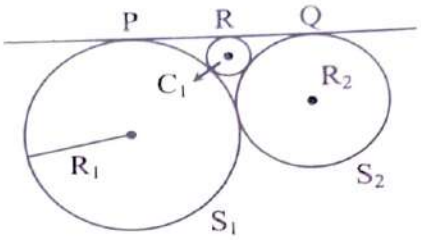
Now between C_1 and S_1, $PR = 2\sqrt{r_1R_1}$

and $PQ = 2\sqrt{r_1R_2}$

but $PQ = PR + RQ$

$2\sqrt{R_1R_2} = 2\sqrt{r_1R_1} + 2\sqrt{r_1R_2}$

$1 = \frac{\sqrt{r_1}}{\sqrt{R_2}} + \frac{\sqrt{r_1}}{\sqrt{R_1}}$ (Dividing by $\sqrt{R_1R_2}$)



$\frac{1}{\sqrt{r_1}} = \frac{1}{\sqrt{R_1}} + \frac{1}{\sqrt{R_2}}$ ..... (1)

But $R_1 = R_2 = 1$ (Given)

$\frac{1}{\sqrt{r_1}} = 2 \Rightarrow r_1 = \frac{1}{2^2}$

Again for next circle, $\frac{1}{\sqrt{r_2}} = \frac{1}{\sqrt{R_1}} + \frac{1}{\sqrt{r_1}} = 1 + 2 = 3$ [From (1), we have $R_1 \to r_1$]

$r_2 = \frac{1}{3^2}$

Parallely $r_n = \frac{1}{(n+1)^2}$

$$\lim_{x\to\infty} \frac{\sum\left(\frac{1}{r_1} + \frac{1}{r_2} + \ldots + \frac{1}{r_n}\right)}{n^3} = \lim_{x\to\infty} \frac{2^2+3^2+\ldots+(n+1)^2}{n^3} = \lim_{x\to\infty} \frac{\sum(n+1)^2 - 1}{n^3} = \frac{1}{3}$$

359. $|a_1 \sin x + a_2 \sin 2x + \ldots + a_n \sin nx| \le |\sin x|$

Let $f(x) = a_1 \sin x + a_2 \sin 2x + \ldots + a_n \sin nx$

Given $|f(x)| \le |\sin x|$, $f(0) = 0$

$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \Rightarrow f'(0) = \lim_{h\to 0} \frac{f(h)}{h}$ also $\left|\frac{f(h)}{h}\right| \le \left|\frac{\sin h}{h}\right| \Rightarrow |f'(0)| \le 1$

$\Rightarrow |a_1 + 2a_2 + \ldots + na_n| \le 1$

360. $\frac{dy}{dx} = 4x^3 + 27x^2 + 2ax + 9$

$\Rightarrow y'' = 12x^2 + 54x + 2a$ should have real and distinct roots

$\Rightarrow D > 0 \Rightarrow a < \frac{243}{8}$

361. $I=\frac{1}{4}\int\frac{(1+x^2)}{x^4-\sqrt{2}x+\frac{3}{4}}dx=\frac{1}{4}\int\frac{(1+x^2)}{\left(x^4-x^2+\frac{1}{4}\right)+x^2-\sqrt{2}x+\left(\frac{1}{\sqrt{2}}\right)^2}dx$

$$=\frac{1}{4}\int\frac{(1+x^2)}{\left(x^2-\sqrt{2}x+\frac{1}{2}\right)\left(x^2+\sqrt{2}x+\frac{3}{2}\right)}dx$$

$$=-\frac{1}{8}\cdot\frac{1}{\left(x-\frac{1}{\sqrt{2}}\right)}+\frac{1}{8}\cdot\tan^{-1}\left[x+\frac{1}{\sqrt{2}}\right]+e$$

362. $I=\int\sin(100x+x)\cdot\sin^{99}x\,dx=\int\{(\sin 100x)\cos x+\cos 100x\cdot\sin x\}\sin^{99}x\,dx$

$$=\int\sin 100x\cdot\cos x\cdot\sin^{99}x\,dx+\int\cos 100x\cdot\sin^{100}x\,dx$$

$$=\frac{\sin(100x)(\sin x)^{100}}{100}-\frac{100}{100}\int\cos(100x)\cdot(\sin x)^{100}dx+\int\cos 100x\cdot\sin^{100}x\,dx$$

$$=\frac{\sin(100x)(\sin x)^{100}}{100}+C$$

363. $I=\int_{1/e}^{\tan x}\frac{t}{1+t^2}dt+\int_{1/e}^{\cot x}\frac{dt}{t(1+t^2)}$. Put $t=\frac{1}{z}\Rightarrow dt=\frac{-1}{z^2}dz$

$$I=\int_{1/e}^{\tan x}\frac{t}{1+t^2}dt+\int_{e}^{\tan x}\frac{-\frac{1}{z^2}}{\frac{1}{z}\left(1+\frac{1}{z^2}\right)}dz$$

$$\Rightarrow I=\int_{1/e}^{\tan x}\frac{t}{1+t^2}dt-\int_{e}^{\tan x}\frac{z}{z^2+1}dz$$

$$\Rightarrow I=\int_{1/e}^{\tan x}\frac{t}{1+t^2}dt+\int_{\tan x}^{e}\frac{t}{1+t^2}dt$$

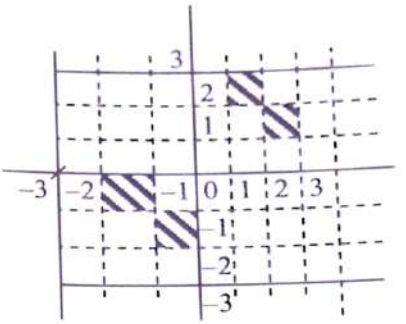
$$\Rightarrow I=\int_{1/e}^{e}\frac{t}{1+t^2}dt=\frac{1}{2}\ln(1+t^2)\Big|_{1/e}^{e}$$

$$=\frac{1}{2}\left[\ln(1+e^2)-\ln\left(1+\frac{1}{e^2}\right)\right]$$

$$\Rightarrow I=\frac{1}{2}[2]=1$$

364. $\lim_{n\to\infty}\frac{\left(\frac{1}{n}\sum_{r=1}^{n}\sqrt{\frac{n}{r}}\right)\left(\frac{1}{n}\sum_{r=1}^{n}\sqrt{\frac{r}{n}}\right)}{\left(\frac{1}{n}\sum_{r=1}^{n}\frac{r}{n}\right)}=\frac{\int_0^1\frac{dx}{\sqrt{x}}\int_0^1\sqrt{x}dx}{\int_0^1 x\,dx}=\frac{8}{3}$

365. $[x]\cdot[y] = 2$
$\Rightarrow [x] = 2$ and $[y] = 1$
or $[x] = -2$ and $[y] = -1$
or $[x] = 1$ and $[y] = 2$
or $[x] = -1$ and $[y] = -2$



366. $3x^2 + 8xy + 10y^2 + 2x - 8y + 5 + \lambda = 0$
$\Rightarrow x(3x + 4y + 1) + y(4x + 10y - 4) + x - 4y + 5 + \lambda = 0$
$\Rightarrow x - 4y + 5 + \lambda = 0$
$3x + 4y + 1 = 0$
$2x + 5y - 2 = 0$

$x - 4y + 5 + \lambda = 0$ has solution $\begin{vmatrix} 3 & 4 & 1 \\ 2 & 5 & -2 \\ 1 & -4 & 5+\lambda \end{vmatrix} = 0$. Solve $\lambda = \frac{10}{7}$

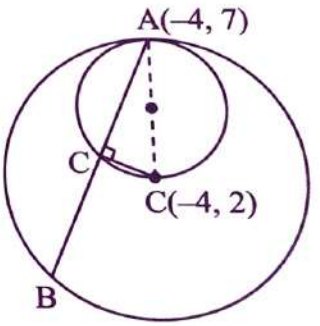
367. Assuming that AB is on the line $y = 2x - 17$ and the coordinates of the other two vertices on the parabola are $C(x_1, y_1)$ and $D(x_2, y_2)$
Then CD is on a line L whose equation is $y = 2x + b$. Solving with parabola, we get $x^2 = 2x + b$
$\Rightarrow x_{1,2} = 1 \pm \sqrt{b+1}$. Assuming there the length of one side of square is a
$\Rightarrow a^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2 = 5(x_1 - x_2)^2 = 20(b + 1)$ ..... (1)
Let us select a point $(6, -5)$ on the line $y = 2x - 17$, clearly the distance from the point to the line $y = 2x + b$ is a

$\Rightarrow a = \frac{|17 + b|}{\sqrt{5}}$ ..... (2)

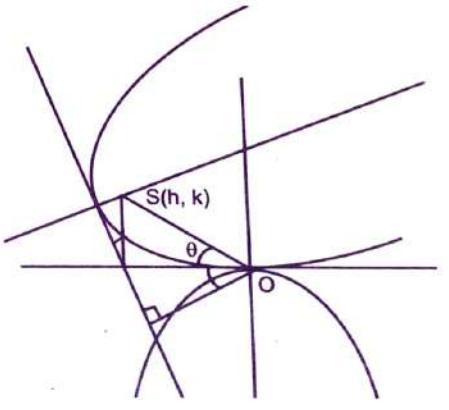
From equation (1) and (2), we get $b_1 = 3$, $b_2 = 63$, so $a^2 = 80$ or $a^2 = 1280 \Rightarrow a^2 = 80$

368. $\frac{1}{2}\left(1 - \frac{c}{9}\right)(9 - c) = 2$

369. Let O be $(-4, 2)$
OC is perpendicular to AC as OA is diameter
$\Rightarrow$ C bisects chord AB
As O is centre of bigger circle
$\Rightarrow$ AC = 4 units

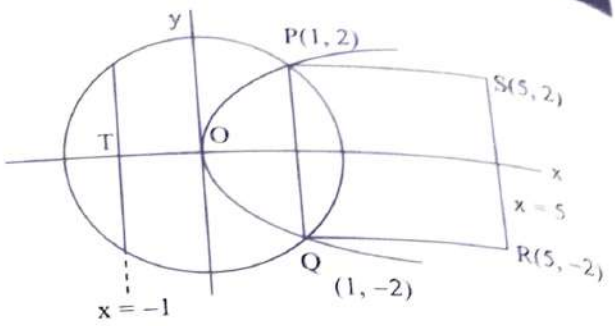


370. Clearly required locus is $16y^4 = x^2 + y^2$.



371. The points of intersection the circle and the parabola are P(1, 2) Q(1, –2)
The equation of common chord is $x = 1$, which is the latus rectum of the parabola.
$\therefore$ The tangent at the extremities of PQ intersect on the directrix $x = -1$, coordinate of T is $(-1, 0)$.
Since length of AB = 4, sides of the square PQRS are of length 4, coordinate of R are (+5, –2) and S are (+5, 2).
$(TR)^2 = (TS)^2 = (5 + 1)^2 + 4 = 40$

$$\left[\frac{(TR)^2 + (TS)^2}{9}\right] = \left[\frac{80}{9}\right] = 8$$



372. $\frac{(x-4)^2}{16} + \frac{(y-3)^2}{9} = 1$
any point P on the curve $(4(1 + \cos\theta), 3(1 + \sin\theta))$.
The reflection of the point P about the line $x - y - 2 = 0$ can be found using

$$\frac{h-4(1+\cos\theta)}{1} = \frac{k-3(1+\sin\theta)}{-1} = \frac{-2(4(1+\cos\theta)-3(1+\sin\theta)-2)}{2} = -(4\cos\theta - 3\sin\theta)+1$$

$h = 4(1 + \cos\theta) - 4\cos\theta + 3\sin\theta + 1 = 4 + 3\sin\theta + 1$
and $k = 3(1 + \sin\theta) + 4\cos\theta - 3\sin\theta = 3 + 4\cos\theta - 1$

$$\left(\frac{h-5}{3}\right)^2 + \left(\frac{k-2}{4}\right)^2 = 1$$

$\therefore$ Equation of image $16(x - 5)^2 + 9(y - 2)^2 = 144$
$16x^2 + 9y^2 - 160x - 36y + 292 = 0$
$k_1 = -160$, $k_2 = 292$
$\Rightarrow k_1 + k_2 = 132$

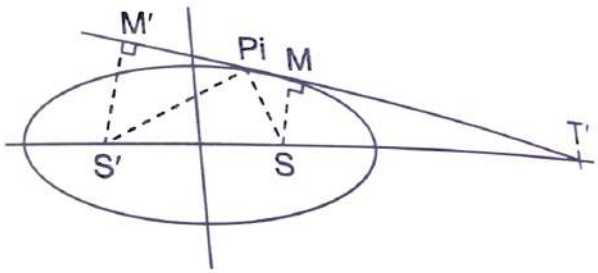
373. Area of $\Delta P_1ST_1 = \frac{1}{2}P_1T_1 \times SM$

Area of $\Delta P_1T_1S' = \frac{1}{2}P_1T_1 \times S'M'$

$$\frac{\text{Area}(\Delta P_1T_1S)\cdot\text{Area}(\Delta P_1T_1S')}{(P_1T_1)^2} = \frac{1}{4}SM\cdot S'M' = \frac{b^2}{4}$$

$$\sum_{i=1}^{n}\frac{\text{Area}(\Delta PiTiS)\cdot\text{Area}(\Delta PiTiS')}{(PiTi)^2} = \frac{nb^2}{4}$$

$$\frac{nb^2}{4} = 18 \Rightarrow n = \frac{4\times 18}{9} = 8 \Rightarrow n = 8$$



374. Let vertex of parabola $(h, h + 4)$
$\therefore$ Equation is $(x - h)^2 = -(y - h - 4)$
It passes through (2, 0)
Hence, $h = 0, 5$

375. $f(e) = \frac{e}{\sqrt{e^2 - 1}} \Rightarrow ff(e) = e$

$\Rightarrow f\,f\,f\ldots f(e) = \dfrac{e}{\sqrt{e^2-1}}$, where n is odd

e, where n is even

$\Rightarrow \int_1^3 f\,f\cdots f(e) = 2\sqrt{2}$ if n is odd

4 if n is even

376. $4x + 7y + 4z + 81 = 0$

$5x + 3y + 10z = 25$ ..... (1)

..... (2)

Equation of plane passing through intersection of (1) and (2)

$(4x + 7y + 4z + 81) + \lambda(5x + 3y + 10z - 25) = 0$

$\Rightarrow (4 + 5\lambda)k + (7 + 3\lambda)y + (4 + 10\lambda)z + 81 - 25\lambda = 0$ ..... (3)

Plane (3) perpendicular to (1)

So, $4(4 + 5\lambda) + 7(7 + 3\lambda) + 4(4 + 10\lambda) = 0$

$\Rightarrow \lambda = -1$

Putting $\lambda = -1$ in (3)

$-x + 4y - 6z + 106 = 0$ ..... (4)

$\therefore$ distance of (4) from (0, 0, 0)

$$\sqrt{k} = \frac{106}{\sqrt{1+16+36}}$$

$\Rightarrow k = 212$

377. The cut $x = y$ separates the cube into points with $x < y$ and those with $x > y$

Similarly for the other cuts

So, number of pieces equals to the number of ways of arrangements of x, yand z, which is $3! = 6$

378. Let $P(x_1, y_1, z_1)$ be a point on $ax + by + cz = d$. Then

$ax_1 + by_1 + cz_1 = d$ ...(1)

Let OP = r. Then direction cosines of OP are $\dfrac{x_1}{r}, \dfrac{y_1}{r}, \dfrac{z_1}{r}$

Equation of lien OP is $\dfrac{x-0}{\frac{x_1}{r}} = \dfrac{y-0}{\frac{y_1}{r}} = \dfrac{z-0}{\frac{z_1}{r}}$

Let $Q(\alpha, \beta, r)$ be a point on OP such that $OQ = \lambda$. Then coordinates of

$$Q\left(\frac{x_1\lambda}{r}, \frac{y_1\lambda}{r}, \frac{z_1\lambda}{r}\right).$$

But coordinate of Q be (α, β, r)

$$x_1 = \frac{\alpha r}{\lambda}, y_1 = \frac{\beta r}{\lambda}, z_1 = \frac{r^2}{\lambda}$$

$\Rightarrow ax_1 + by_1 + cz_1 = d$

$$\frac{r}{\lambda}(a\alpha + b\beta + cr) = d \quad \ldots(ii)$$

$OP.OQ = d^2$ ...(iii)

$(a\alpha + b\beta + c\gamma) = r = d\lambda$

$(a\alpha + b\beta + c\gamma) = r\lambda = d\lambda^2$

$(a\alpha + b\beta + c\gamma) = d^2 = d\lambda^2$

$(a\alpha + b\beta + c\gamma) = d = \lambda^2 = \alpha^2 + \beta^2 + \gamma^2$

locus of (α, β, γ) is $(ax + by + cz)\,d = x^2 + y^2 + z^2$

$$\frac{(ax+by+cz)d}{x^2+y^2+z^2} = 1$$

379. $\alpha+\beta=-\frac{2h}{a}$

$\alpha\beta=\frac{b}{a}$

$\gamma+\delta=-\frac{2h_1}{a_1}$

$\gamma\delta=\frac{b_1}{a_1}$

given $\gamma=\frac{\lambda\beta+\alpha}{\lambda+1}\Rightarrow\lambda=\frac{\gamma-\alpha}{\beta-\gamma}$

similarly $\mu=\frac{\delta-\alpha}{\beta-\delta}$

Now $\lambda+\mu=\frac{(\gamma-\alpha)(\beta-\delta)+(\beta-\gamma)(\delta-\alpha)}{(\beta-\gamma)(\beta-\delta)}$

$=\frac{\frac{2}{aa_1}(2hh_1-a_1b-ab_1)}{(\beta-\gamma)(\beta-\delta)}=0$

380. Roots of $(x-a)(x-a-1)=0$ are $a, a+1$
Roots of $(x+a)(x+a^2-2)$ are $-a, -a^2+2$
Since the roots of $(x-a)(x-a-1)=0$ lies between roots of $(x+a)(x+a^2-2)=0$
$\Rightarrow f(a)=(a+a)(a^2-2+a)<0=(2a)(a^2+a-2)<0=(2a)(a-1)(a+2)<0$
$\Rightarrow a\in(-\infty,-2)\cup(0,1)$

$f(a+1)=(2a+1)[a+1-2+a^2]=(2a+1)(a^2+a-1)=(2a+1)\left(a+\frac{1+\sqrt{5}}{2}\right)\left(a+\frac{1-\sqrt{5}}{2}\right)$

$a\in\left(-\infty,-\frac{1+\sqrt{5}}{2}\right)\cup\left(-\frac{1}{2},-\frac{1}{2}+\frac{\sqrt{5}}{2}\right)\Rightarrow a\in(-\infty,-2)\cup\left(0,\frac{-1+\sqrt{5}}{2}\right)$

$\Rightarrow p=-2, q=0$
$\Rightarrow q-p=2$

381. $S={}^{100}C_0\,{}^{100}C_2+{}^{100}C_4\,{}^{100}C_6+\ldots+{}^{100}C_{96}\,{}^{100}C_{98}$

$=$ Coefficient of x^0 in $(1+x)^{100}\frac{(1+x)^{100}}{x^2}+\frac{(1+x)^{100}}{x^4}\frac{(1+x)^{100}}{x^6}+\ldots+\frac{(1+x)^{100}}{x^{96}}\frac{(1+x)^{100}}{x^{98}}$

$=$ Coefficient of x^0 in $\frac{(1+x)^{200}(1-x^{200})}{x^{194}(1-x^8)}$

$=$ Coefficient of x^{194} in $(1+x)^{200}(1-x^8)^{-1}={}^{200}C_2+{}^{200}C_{10}+\ldots+{}^{200}C_{194}$.

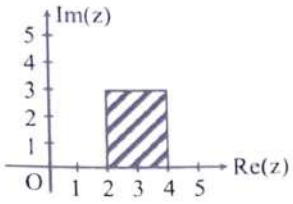
382. Number of solution $=\sum_{x=0}^{10}{}^{30-3x+2}C_2=\frac{1}{2}(1.2+4.5+\ldots..+25.26+28.29+31.32)=1991$

$\Rightarrow a=1, b=9\ \therefore\ b-a=8$

383. $A=(11-1)!=10!$
$B=(11-3)!2=2\cdot8!$ (Consider the girl and the 2 boys as a single unit)

$\frac{A}{7B}=\frac{10!}{2.7.8!}=\frac{45}{7}$

384. Shown in the figure



385. Let us assume that O is the centre of the polygon and $z_0, z_1, \ldots, z_{n-1}$ represent the affixes of $A_1, A_2, \ldots, A_n$, such that

$z_0 = 1, z_2 = \alpha, z_2 = \alpha^2, \ldots, z_{n-1} = \alpha^{n-1}$, where $\alpha = e^{\frac{i2\pi}{n}}$

Now, $|A_1A_r|^2 = |\alpha^r - 1|^2 = |1 - \alpha^r|^2$

$$= \left|1 - \cos\frac{2r\pi}{n} - i\sin\frac{2r\pi}{n}\right|^2 = \left(1 - \cos\frac{2r\pi}{n}\right)^2 + \left(\sin\frac{2r\pi}{n}\right)^2 = 2 - 2\cos\frac{2r\pi}{n}$$

$$\therefore a = \sum_{r=2}^{n} |A_1A_r|^2 = \sum_{r=1}^{n}\left(2 - 2\cos\frac{r\pi}{n}\right)$$

$$= 2(n-1) - 2\left\{\cos\frac{2\pi}{n} + \cos\frac{4\pi}{n} + \ldots + \cos\frac{2(n-1)\pi}{n}\right\}$$

$= 2(n-1) - 2$ Real part of $(\alpha + \alpha^2 + \ldots + \alpha^{n-1})$

$= 2(n-1) - 2(-1)$ [Since $\{1 + \alpha + \alpha^2 + \ldots + \alpha^{n-1} = 0\}$]

$\therefore |A_1A_2|^2 + |A_2A_3|^2 + \ldots |A_1A_n|^2 = 2n.$

Also, $b = |A_1A_2|\,|A_1A_3| \ldots |A_1A_n|$

$= |1 - \alpha|\,|1 - \alpha^2|\,|1 - \alpha^3| \ldots |1 - \alpha^{n-1}| = |(1-\alpha)(1-\alpha^2)(1-\alpha^3)\ldots(1-\alpha^{n-1})|$

Since, $1, \alpha, \alpha^2, \ldots, \alpha^{n-1}$ are the roots of $z^n - 1 = 0$

$\Rightarrow (z-1)(z-\alpha)(z-\alpha^2)\ldots(z-\alpha^{n-1}) = z^n - 1$

$$\Rightarrow (z-\alpha)(z-\alpha^2)\ldots(z-\alpha^{n-1}) = \frac{z^n - 1}{z-1} = 1 + z + z^2 + \ldots + z^{n-1}$$

Substituting $z = 1$, we have

$(1-\alpha)(1-\alpha^2)\ldots(1-\alpha^{n-1}) = n$

$$\therefore |1-\alpha|\,|1-\alpha^2|\ldots|1-\alpha^{n-1}| = n. \Rightarrow \frac{a}{b} = 2$$

386. $$\Delta = \begin{vmatrix} x_1 & y_1 & 0 \\ x_2 & y_2 & 0 \\ x_3 & y_3 & 0 \end{vmatrix}\begin{vmatrix} y_1 & x_1 & 0 \\ y_2 & x_2 & 0 \\ y_3 & x_3 & 0 \end{vmatrix} = 0$$

387. Curve is $y = x + \sin x$, $y' = 1 + \cos x$

Derivative is 0 at π and $-\pi$, but it's y co-ordinate i.e. outside the rectangle $-3 \le y \le 3$ and $-12 \le x \le 12$. So, number of points as per the given condition is 0

388. $$\frac{1}{k} = \cos a \cos 2a \cos 3a \ldots \cos 999a$$

let $p = \sin a \sin 2a \ldots \sin 999a$

$$\therefore \frac{2^{999}p}{k} = (2\sin a \cos a)(2\sin 2a \cos 2a)\ldots(2\cos 999a \sin 999a)$$

$$\frac{2^{999}p}{k} = 2\sin 2a \sin 4a \ldots \sin 1998a$$

$= (\sin 2a \sin 4a \ldots \sin 998a)(-\sin(2\pi - 1000a))(-\sin(2\pi - 1002a)) \ldots (-\sin(2\pi - 1998a))$

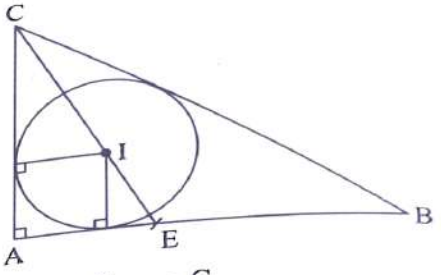
$\Rightarrow (\sin 2a \sin 4a \ldots \sin 1998a)(\sin 999a \sin 997a \ldots \sin a) = p \Rightarrow 2^{999} = k.$

389. $\sin\frac{B}{2}=\frac{r}{BI}$

$\cos\frac{B}{2}=\frac{r}{DI}$

$\sin\frac{C}{2}=\frac{r}{CI}$

$\cos\frac{C}{2}=\frac{r}{EI}$



Hence, $\frac{BI^2+DI^2}{CI^2+EI^2}=\frac{\left(\frac{r}{\sin B/2}\right)^2+\left(\frac{r}{\cos B/2}\right)^2}{\left(\frac{r}{\sin C/2}\right)^2+\left(\frac{r}{\cos C/2}\right)^2}=\frac{\sin^2\frac{C}{2}\cdot\cos^2\frac{C}{2}}{\sin^2\frac{B}{2}\cdot\cos^2\frac{B}{2}}=\frac{\sin^2 C}{\sin^2 B}=\frac{AB^2}{AC^2}=\frac{9}{3}=3$

390. $\sin x\cos y=1 \Leftrightarrow \sin x=1$ and $\cos y=1$

or $\sin x=-1$ and $\cos y=-1$

Case 1: $\sin x=\cos y=1 \Rightarrow x=2n\pi+\pi/2$; m, n $\in$ I

$y=2m\pi$.

Now, $x^2+y^2\le 9\pi^2$

$\Rightarrow (4n+1)^2\frac{\pi^2}{4}+4m^2\pi^2\le 9\pi^2$

$\Rightarrow (4n+1)^2+(4m)^2\le 36$

when $m=0, -6\le 4n+1\le 6$

$\frac{-7}{4}\le n\le\frac{5}{4}$

$\therefore n=-1, 0, 1.$

When $m=\pm 1, n=-1, 0$

$\therefore$ 7 solutions.

Case 2: when $\sin x=\cos y=-1$

$\Rightarrow x=2p\pi-\pi/2, y=22\pi+\pi,$ where p, q $\in$ I

$\therefore (4p-1)^2+(4q+2)^2\le 36$

When $q=-1, p=-1, 0, 1$

When $q=0, p=-1, 0, 1$

$\therefore$ 6 solutions.

Thus, total number of solutions = 7 + 6 = 13 pairs are possible.

391. $p_1=\frac{2\Delta}{a}, p_2=\frac{2\Delta}{b}, p_3=\frac{2\Delta}{c}$

$$p_1+p_2+p_3\ge 3(p_1p_2p_3)^{1/3}=3\left\{\frac{(2\Delta)^3}{abc}\right\}^{1/3}=6\Delta\left(\frac{1}{abc}\right)^{1/3}$$

$$=6rs\left(\frac{1}{abc}\right)^{1/3}=6r\frac{a+b+c}{2}\left(\frac{1}{abc}\right)^{1/3}$$

$$=9r\frac{(a+b+c)}{3}\left(\frac{1}{abc}\right)^{1/3}\ge 9r$$

$$\therefore p_1+p_2+p_3\ge 9\times\frac{2}{3}=6$$

392. $\frac{\sin x}{\cos 3x}+\frac{\sin 3x}{\cos 9x}+\frac{\sin 9x}{\cos 27x}=0$

or $\frac{2\sin x\cos x}{2\cos 3x\cos x}+\frac{2\sin 3x\cos 3x}{2\cos 9x\cdot\cos 3x}+\frac{2\sin 9x\cos 9x}{2\cos 27x\cos 9x}=0$

or $\frac{\sin(3x-x)}{2\cos 3x\cos x}+\frac{\sin(9x-3x)}{2\cos 9x\cdot\cos 3x}+\frac{\sin(27x-9x)}{2\cos 27x\cdot\cos 9x}=0$

or $(\tan 3x-\tan x)+(\tan 9x-\tan 3x)+(\tan 27x-\tan 9x)=0$

$\tan x=\tan 27x$

$27x=n\pi+x,\ x\in I$

$x=\frac{n\pi}{26},\ n\in I$

$x=\frac{\pi}{26},\frac{2\pi}{26},\frac{3\pi}{26},\frac{4\pi}{26},\frac{5\pi}{26},\frac{6\pi}{26}$

393. $f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x+0)}{h}$

$\Rightarrow \lim_{h\to 0}\frac{f(x)+f(h)+h\sqrt{f(x)}-f(x)-f(0)-0\sqrt{f(x)}}{h}\Rightarrow \lim_{h\to 0}\left(\frac{f(h)-f(0)}{h-0}\right)+\sqrt{f(x)}$

$f'(x)=\sqrt{f(x)}$

$\int\frac{f'(x)}{\sqrt{f(x)}}dx=\int dx$

$2\sqrt{f(x)}=x+c$

$f(x)=\frac{x^2}{4}$

When $\alpha=0$ area is minimum

Required minimum area $=2\int_0^9 2\sqrt{y}\,dy\Rightarrow 4\left(\frac{y^{3/2}}{3/2}\right)_0^9=72$ sq. unit

394. Consider $g(x)=f(x)-x-\frac{1}{2}$

for any $x=a$

$g(a)=f(a)-a-\frac{1}{2}$

$g(f(a))=f(f(a))-f(a)-\frac{1}{2}=1+a-f(a)-\frac{1}{2}=\frac{1}{2}+a-f(a)$

$\Rightarrow g(f(a))=-g(a)$

$\Rightarrow$ Between any point a and f(a) there lies a root of g(x) ...(1)

$g(f(x))=-g(x)\Rightarrow g(f(x))+g(x)=0$

$\Rightarrow g(f(f(x))+g(f(x))=0$

$\Rightarrow g(f(f(x))=g(x)$

$\Rightarrow g(1+x)=g(x)$

$\Rightarrow$ g is periodic function with period 1 ...(2)

Considering any n, n + 1 interval, we can see from (1) and (2) that g(x) is identically zero

$\Rightarrow g(x)=0\ \Rightarrow f(x)=x+\frac{1}{2}$

395. The ratio must be rational.

Suppose $r = \frac{h}{k}$

If there are more than six terms than the first term must be divisible by k^6 and last by h^6

If $k \geq 3$, then $h > k$ so $h \geq 4$ but $h^6 > 1000$ so $k < 3$

$\Rightarrow k = 2$ and $h = 3$ so the first term must be a multiple of 64

The smallest is 128, but then we get only six terms 128, 192, 288, 432, 648, 972

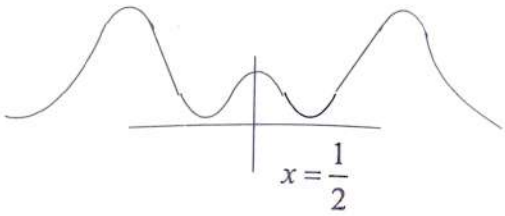
396. $B = A^{2^n} = \left(A^2\right)^{2^{n-1}} = \left(A^{-1}\right)^{2^{n-1}} = \left(A^{2\cdot 2^{n-2}}\right)^{-1} = \left(A^2\right)^{-2^{n-2}} = \left(A^{-1}\right)^{-2^{n-2}} = A^{2^{n-2}}$

$B = C$

$\det(B - C) = 0$

397. $f(n) = \frac{(2n+1)^{3/2} - (2n-1)^{3/2}}{2}$

398. Shown in the figure



399. $h(x) = \frac{d}{dx}\left(f(x).\cos x\right)$

first find the minimum number of zeroes of $\left(f(x).\cos x\right) = 0$

$f(x) = 0$ has minimum 98 roots in [0, 99)

$\cos x = 0$ has 31 roots in [0, 99)

maximum common possible root is only 1

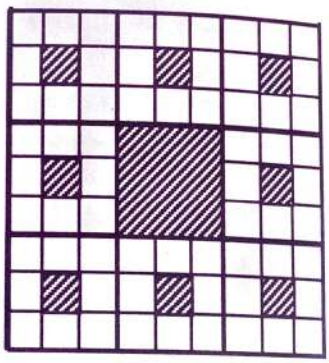
hence minimum number of roots of $f(x)\cos x = 0$ is 128.

Hence $\frac{d}{dx}\left(f(x)\cos x\right) = 0$ has minimum 127 roots.

400. After the first step 8 squares are left each with side length $\frac{1}{3}$

After two steps $8.9 - 8 = 8(9 - 1) = 8^2$ squares are left with side length $\frac{1}{3^2}$

After 3 steps $8^2.9 - 8^2 = 8^2(9 - 1) = 8^3$ squares are left with side length $\frac{1}{3^3}$



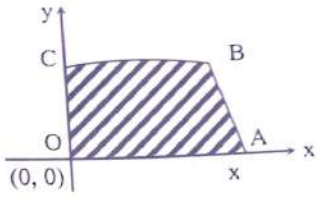
After n steps 8^n squares are left and each has side length $\frac{1}{3^n}$

The sum of the areas of the squares that are left is $8^n\left(\frac{1}{3^n}\right)^2 = \left(\frac{8}{9}\right)^n$

So the sum of the areas of the removed squares $1 - \left(\frac{8}{9}\right)^n$

Since, $\frac{2}{5} < 1 - \left(\frac{8}{9}\right)^n < \frac{1}{2} \Rightarrow n = 5$

401. Area of curvelinear trapezoid OABC $= \int_0^x f(x)\,dx$



According to $= \int_0^x f(x)\,dx = k\{f(x)\}^{n+1}$

Where k is constant of proportionality

Differentiating both sides w.r.t. x

$f(x) = k(n+1)[f(x)]^n f'(x)$

or $\{f(x)^{n-1}\} f'(x) = \dfrac{1}{k(n+1)}$

Integrating both sides w.r.t. x

$$\frac{\{f(x)\}^n}{n} = \frac{x}{k(n+1)} + c$$

Putting $x = 0$, $\dfrac{\{f(0)\}^n}{n} = 0 + c$

$\Rightarrow O = 0 + c\ [\because f(0) = 0]$

$$\therefore \frac{\{f(x)\}^n}{n} = \frac{x}{k(n+1)} \Rightarrow \{f(x)\}^n = \frac{nx}{k(n+1)}$$

Again, putting $x = 1$, then $\{f(1)\}^n = \dfrac{n}{k(n+1)} = 1$

From equation (1) $\{f(x)\}^n = x$, $\{f(10)\}^n = 10$

402. Let $\sum_{i=1}^{6} x_i = \alpha$ and $\sum_{i=1}^{6} y_i = \beta$

Let O be the orthocentre of the triangle made by (x_1, y_1), (x_2, y_2) and (x_3, y_3)

$\Rightarrow$ O is $(x_1 + x_2 + x_3, y_1 + y_2 + y_3) \equiv (\alpha_1, \beta_1)$

Similarly let G be the centroid of the triangle made by other three points

$$\Rightarrow \text{G is } \left(\frac{x_4 + x_5 + x_6}{3}, \frac{y_4 + y_5 + y_6}{3}\right) \Rightarrow \text{G is } \left(\frac{\alpha - \alpha_1}{3}, \frac{\beta - \beta_1}{3}\right)$$

The point dividing OG in the ratio 3 : 1 is $\left(\dfrac{\alpha}{4}, \dfrac{\beta}{4}\right) \equiv (2, 1) \Rightarrow h + k = 3$

403. $f(n) = {}^nc_0 a^{n-1} - {}^nc_1 a^{n-2} + {}^nc_2 a^{n-3} - \ldots\ldots + (-1)^{n-1}\,{}^nc_{n-1} a^0$

$$= \frac{1}{a}\cdot\left({}^nc_0 a^n - {}^nc_1 a^{n-1} + {}^nc_2 a^{n-2} + \ldots\ldots + (-1)^{n-1} {}^nc_{n-1} a\right) = \frac{1}{a}\left((a-1)^n - (-1)^n \cdot {}^nc_n\right) = \frac{1}{a}\left(\frac{1}{3^{223}} - (-1)^n\right)$$

$$f(x) = \frac{3^{x/223} - (-1)^n}{(3^{1/223} + 1)}$$

$$f(2007) = \frac{(3^{2007/223} + 1)}{(3^{1/223} + 1)}$$

$$f(2008) = \frac{(3^{2008/223} - 1)}{(3^{1/223} + 1)}$$

$f(2007) + f(2008) = 3^9$ then $k = 9$

404. $m^4+\frac{1}{4}=\left(m^2+m+\frac{1}{2}\right)\left(m^2-m+\frac{1}{2}\right)$

$$\therefore \prod_{k=1}^{n}\frac{(2k-1)^4+\frac{1}{4}}{(2k)^4+\frac{1}{4}}=\prod_{k=1}^{n}\frac{\left((2k-1)^2+(2k-1)+\frac{1}{2}\right)\left((2k-1)^2-(2k-1)+\frac{1}{2}\right)}{\left((2k)^2+2k+\frac{1}{2}\right)\left((2k)^2+2k+\frac{1}{2}\right)}=\frac{1}{8n^2+4n+1}$$

$\therefore k_1=8, k_2=4, k_3=1, k_1-k_2+k_3=8-4+1=5$

405. $2\cos x=\tan x+\sec x$

$2\cos^2 x=1+\sin x$

$2(1-\sin x)(1+\sin x)=1+\sin x$

Either $1+\sin x=0$ or $\sin x=1/2$

$\sin x=-1,\ \tan x=\frac{1}{\sqrt{3}}$

$$d=\sec x-\cos x=\frac{1-\cos^2 x}{\cos x}=\frac{1}{2\sqrt{3}}$$

$$t_n=a+(n-1)d=\frac{1}{\sqrt{3}}+(n-1)\frac{1}{2\sqrt{3}}=\sqrt{3}=\cot x$$

$$\Rightarrow 1+\frac{n-1}{2}=3$$

$\Rightarrow n=5$

406. $(1-2x)(1-2y)(1-2z)=1-2\sum x+4\sum xy-8xyz=4\sum xy-8xyz-1$

$$\therefore \sum xy-2xyz=\frac{1+(1-2x)(1-2y)(1-2z)}{4}$$

$$\frac{\sum(1-2x)}{3}\geq\{(1-2x)(1-2y)(1-2z)\}^{1/3}$$

$$\Rightarrow (1-2x)(1-2y)(1-2z)\leq\frac{1}{27}$$

$$\Rightarrow xy+yz+zx-2xyz\leq\frac{7}{27}$$

Equality holds for $x=y=z=\frac{1}{3}$

$\therefore$ Maximum value $=\frac{7}{27}$

407. $P\left(\frac{R}{A}\right)=\left(\frac{2}{3}\right)^n,\ P\left(\frac{R}{B}\right)=\left(\frac{1}{3}\right)^n$, in n throws

$$P(R)=\frac{1}{2}\left(\frac{2}{3}\right)^n+\frac{1}{2}\left(\frac{1}{3}\right)^n, \text{ in n throws}$$

$$P\left(\frac{A}{R}\right)=\frac{P\left(\frac{R}{A}\right)P(A)}{P(R)}=\frac{\frac{1}{2}\left(\frac{2}{3}\right)^n}{\frac{1}{2}\left(\frac{2}{3}\right)^n+\frac{1}{2}\left(\frac{1}{3}\right)^n}=\frac{2^n}{2^n+1}=\frac{32}{33}\Rightarrow n=5$$

408. Total number of function satisfying the given condition is ${}^9C_1 + {}^9C_2 + \ldots {}^9C_8 = 2^9 - 2$

$\Rightarrow$ Probability $= \dfrac{2^9 - 2}{10^{10}}$

409. First element of matrix $A_{10} = 286$
(10^{th} element of the sequence 1, 2, 6, 15, ...)
Trace of $A_{10} = 286 + 297 + 308 + 319 + \ldots + 385 = 3055$

410. $A^2 = A$
$|A|^2 = |A|$; $|A| = 1$ or 0 (Not possible)
$|A| = 1$
$A - A^T$ will skew symmetric so $|A - A^T| = 0$
$A^2 = A$
Premultiply by A^{-1}, $A^{-1}A^2 = A^{-1}A$
$A = I$ so adj $A = I$
$|\text{adj } A - I| = 0$

$$L = \lim_{x\to 0} \frac{\int_0^{\sin^2 x} \ln\left(1+t^3\right)dt}{\int_0^{x}\left(e^t - 1 - t\right)(\tan t - t)(1-\cos t)\,dt}$$

$$L = \lim_{x\to 0} \frac{\ln\left(1+\sin^6 x\right)2\sin x\cos x}{\left(e^x - 1 - x\right)(\tan x - x)(1-\cos x)}$$

$$= \lim_{x\to 0} \frac{\dfrac{\ln\left(1+\sin^6 x\right)}{\sin^6 x}\cdot\dfrac{\sin^6 x}{x^6}\cdot x^6 2\dfrac{\sin x}{x}\cos x\cdot x}{\left(1+x+\dfrac{x^2}{2!}+\ldots..-1-x\right)\left(x+\dfrac{x^3}{3}+\dfrac{2x^5}{15}+\ldots..-x\right)\left(1-1+\dfrac{x^2}{2!}+\ldots.\right)}$$

$$= \lim_{x\to 0} \frac{2x^7}{\left(\dfrac{x^2}{2!}\right)\left(\dfrac{x^3}{3}\right)\left(\dfrac{x^2}{2!}\right)} = 24$$

$|\text{adj } B| = 1 + 0 + 0 + 24 = 25$
$\Rightarrow |B|^2 = 25$
$\Rightarrow |B| = 5$

411. The given relation can be written as $\left(\left(\sqrt{a^2-b^2}\right)\hat{i} + a\hat{j} + \left(\sqrt{a^2+b^2}\right)\hat{k}\right)\cdot\left(\tan A\hat{i} + \tan B\hat{j} + \tan C\hat{k}\right) = ma$

$\Rightarrow \left(\sqrt{a^2-b^2+a^2+a^2+b^2}\right)\sqrt{\left(\tan^2 A+\tan^2 B+\tan^2 C\right)}\cos\theta = ma$

$\Rightarrow \sqrt{3}a\sqrt{\left(\tan^2 A+\tan^2 B+\tan^2 C\right)}\cos\theta = ma$

$\Rightarrow \tan^2A + \tan^2B + \tan^2C = \left(\dfrac{m}{\sqrt{3}}\sec\theta\right)^2$

but $\dfrac{m^2}{3}\sec^2\theta \geq \dfrac{m^2}{3}$

$\therefore \tan^2A + \tan^2B + \tan^2C \geq \dfrac{m^2}{3} \quad \Rightarrow \dfrac{m^2}{3} = 27$

$\Rightarrow m^2 = 81$
$\Rightarrow |m| = 9$

412. Given R : r = 3 : 1.5 = 2
$\Rightarrow \Delta ABC$ must be equilateral

So $a = b = c = 2R\sin\frac{\pi}{3} = R\sqrt{3}$ (By sine rule)

Now, $a\cot^2 A + b^2\cot^3 B + c^3\cot^4 C = R\sqrt{3}\left(\frac{1}{\sqrt{3}}\right)^2 + \left(R\sqrt{3}\right)^2\left(\frac{1}{\sqrt{3}}\right)^3 + \left(R\sqrt{3}\right)^3\left(\frac{1}{\sqrt{3}}\right)^4$

$= \frac{R}{\sqrt{3}} + \frac{R^2}{\sqrt{3}} + \frac{R^3}{\sqrt{3}} = \frac{3+3^2+3^3}{\sqrt{3}} = \frac{39}{\sqrt{3}} = 13\sqrt{3} = m\sqrt{n}$

Hence, (m + n) = 16

413. ABCD is a trapezium and its area = $\frac{1}{2}(a+b)h$

Where $\frac{a+b}{2}$ = EF (mid parallel)

Area = 2r(EF) ..... (1)

Now equation of EF is $y = -x + c$

From equation (1) passes through (r, r)

$\Rightarrow c = 2r$

$x + y = 2r$, hence E = (2r, 0) and F(0, 2r)

$\therefore$ EF $= \sqrt{4r^2 + 4r^2} = 2\sqrt{2}r$

$\therefore$ Area ABCD $= 2r \cdot 2\sqrt{2}r = 4\sqrt{2}r^2$

$\therefore 4\sqrt{2}\,r^2 = 900\sqrt{2}$, $r^2 = 225$

$\Rightarrow r = 15$

414. As $\vec{a}\cdot\vec{b} = \cos\theta$, $|\vec{a}\times\vec{b}| = \sin\theta$

$$\int_{-(a\cdot b)^2}^{|a\times b|^2} 2t\,dt = \int_{-\cos^2\theta}^{\sin^2\theta} 2t\,dt = 0 = \sin^4\theta - \cos^4\theta = 0$$

$\Rightarrow \sin^2\theta - \cos^2\theta = 0$

$\tan^2\theta = 1$

$\theta = n\pi \pm \frac{\pi}{4}$

We know that $\vec{a}\times(\vec{a}\times\vec{b}) = (\vec{a}\cdot\vec{b})\vec{a} - (\vec{a}\cdot\vec{a})\vec{b} = \cos\theta\,\vec{a} - \vec{b}$

Volume of parallelepiped whose edges are $A(\vec{a})$, $B(\vec{a}\times\vec{b})$ and $C(\vec{a}\times(\vec{a}\times\vec{b}))$

$= \left|\left[\vec{A}\,\vec{B}\,\vec{C}\right]\right| = \left|\left[\vec{C}\,\vec{A}\,\vec{B}\right]\right| = \left|\vec{C}\cdot(\vec{A}\times\vec{B})\right|$

On solving, we get $\sin^2\theta = \frac{1}{2} = \frac{p}{q}$

So p + q = 3

415. $$\sum_{r=1}^{n}\frac{\sin\left(2^r - 2^{r-1}\right)}{\cos 2^r\cos 2^{r-1}} = \sum_{r=1}^{n}\left(\tan 2^r - \tan 2^{r-1}\right) = \tan 2^n - \tan 1$$

416. $\log_{10} 2xy = 2$

$2xy = 100$

$xy = 50$ ..... (1)

$$\log_{10}\frac{x^2}{2y}=4$$

$$\frac{x^2}{2y}=10^4$$

$$x\cdot\frac{x^2}{2\cdot 10^4}=50 \qquad \text{..... (2)}$$

$$x^3=10^6$$

$$x=10^2$$

$$y=\frac{1}{2}$$

$$x+y=100+\frac{1}{2}=\frac{201}{2}=\frac{m}{n}$$

$$m=201,\ n=2$$

$$m-3n^6=201-3(2)^6=201-192=9$$

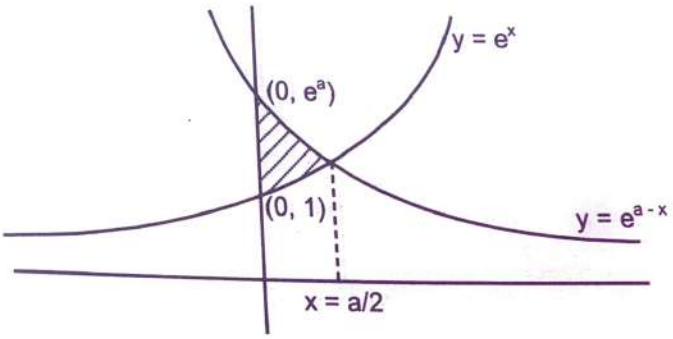
417. Vertex A will be mirror image of ex–centre about $x+y-2=0$

$$\Rightarrow \frac{\alpha-2}{1}=\frac{\beta+4}{1}=\frac{-2(2-4-2)}{2}=4 \Rightarrow \alpha=6 \text{ and } \beta=0$$

418. $$S=\int_0^{a/2}(e^{a-x}-e^x)\,dx=[-e^{a-x}-e^x]_0^{a/2}=[(-e^{a/2}-e^{a/2})-(-e^a-1)]$$

$$S=(-2e^{a/2}+e^a+1)$$

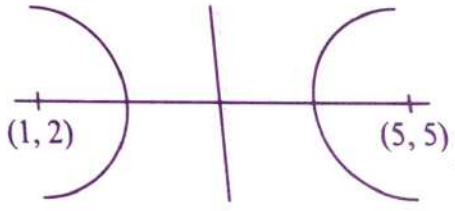
$$S=e^a-2e^{a/2}+1=(e^{a/2}-1)^2$$



$$\lim_{a\to 0}\frac{S}{a^2}=\lim_{a\to 0}\frac{(e^{a/2}-1)^2}{4\times\frac{a^2}{4}}=\lim_{a\to 0}\frac{1}{4}\left(\frac{e^{a/2}-1}{\frac{a}{2}}\right)^2=\frac{1}{4}=0.25$$

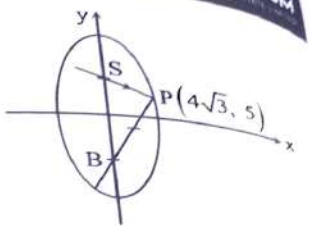
419. $2ae=5$ and $2a=3$

$$\Rightarrow e=\frac{5}{3} \Rightarrow \frac{1}{e^2}+\frac{1}{(e')^2}=1 \Rightarrow e'=\frac{5}{4}$$



420. Given ellipse is $\frac{x^2}{64}+\frac{y^2}{100}=1$

Clearly (0, 6) is one of the foci of given ellipse
According to reflection property. A ray passing through focus will pass through other B (other focus) after reflection from ellipse. So, B is on y-axis which is also a focus i.e., B(0, –6)
By distance formula BP = 13



421. Clearly minimum radius = $\frac{1}{2}$. For maximum radius r; Equation of circle touching parabola is $(x-r-1)^2+y^2=r^2$, solving with $y^2=4x$ and taking D = 0

$\Rightarrow r=4 \Rightarrow r_{max}-r_{min}=4-\frac{1}{2} \Rightarrow \frac{7}{2}=3.50$

422. $2000x^6+100x^5+10x^3+x-2=0$

$\Rightarrow 2000x^6+\frac{x\left((10x^2)^3-1\right)}{10x^2-1}-2=0$

$\Rightarrow \frac{x(1000x^6-1)}{10x^2-1}=-2(1000x^6-1)$

$\therefore 1000x^6-1=0$ or $\frac{x}{10x^2-1}=-2$

$\Rightarrow x=-(10x^2-1)$

$\Rightarrow x^2=\frac{1}{10}$ which is not possible

$\Rightarrow 20x^2+x-2=0$

$\Rightarrow x=\frac{-1\pm\sqrt{161}}{40}$, m = –1, n = 161, r = 40

$\therefore (m+n+r)=200$

423. Equation of the line is $\frac{x-2}{1}=\frac{y+2}{-3}=\frac{z-5}{2}=\lambda$ ..... (1)

Hence any point on the line (1) can be taken as $\Rightarrow x=\lambda+2$
$y=-(3\lambda+2)$
$z=(2\lambda+5)$ for some λ point lies on the plane
$2x-3y+4z=163$ ..... (2)
$2(\lambda+2)+3(3\lambda+2)+4(2\lambda+5)=163$
This gives $19\lambda=133 \Rightarrow \lambda=7$
Hence, P ≡ (9, –23, 19)
Also, (1) intersect YZ plane i.e. $x=0 \Rightarrow \lambda+2=0$, hence $\lambda=-2$
$\therefore$ Q(0, 4, 1). So, $PQ=\sqrt{9^2+27^2+18^2}=9\sqrt{1+3^2+2^2}=9\sqrt{14}$
$\therefore a=9$ and $b=14$
Hence, a + b = 23

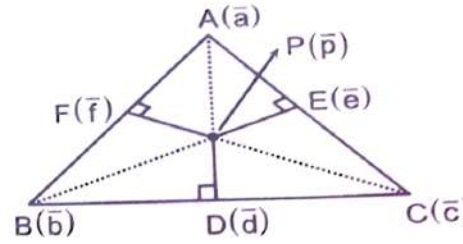
424. $\frac{x^{1000}+x^{900}+x^{90}+x^6+\frac{1}{x}+.....+\frac{1}{x}}{2000} \geq 1$. Hence $x^{1000}+x^{900}+x^{90}+x^6+\frac{1996}{x} \geq 2000$

425. For real numbers a,b,c $(a-b)^2+(b-c)^2+(c-a)^2 \geq 0 \Rightarrow a^2+b^2+c^2 \geq ab+bc+ca$ and equality holds iff $a=b=c$. Therefore Sample Space $(a,b,c) \in \{(1,1,1),.....,(6,6,6)\}$ of which 3 cases satisfy $a+b+c<12$ to lie inside the specified tetrahedron. Required probability $\frac{1}{2}=0.50$

426. For x, y, $z \in R$
$x^2+y^2+z^2-xy-yz-zx \geq 0$
$\Rightarrow (x+y+z)^2 \geq 3(xy+yz+zx)$
$\Rightarrow a^2 \geq 3b$
a = 1, no value of b is possible
a = 2 b = 1
a = 3 b = 1, 2, 3
a = 4 b = 1, 2, 3, 4
a = 5 b = 1, 2, 3, 4, 5, 6
a = 6 b = 1, 2, 3, 4, 5, 6
Hence, required number of ordered pairs (a, b) = 21

427. $a>0$
$$\frac{1}{\log_5 a}+\frac{1}{\log_2 a}+\frac{1}{\log_3 a}=\log_a 30$$
$\Rightarrow a=30$

428. $Ar(\Delta ABC)=Ar(\Delta BPC)+Ar(\Delta BPA)+Ar(\Delta APC)$
$$\Rightarrow \frac{\sqrt{3}}{4}|\bar{c}-\bar{a}|^2$$
$$=\frac{1}{2}\times|\bar{c}-\bar{a}|\times\left[|\bar{d}-\bar{p}|+|\bar{f}-\bar{p}|+|\bar{e}-\bar{p}|\right]$$
$$\Rightarrow k=\frac{\sqrt{3}}{2}$$



429. $(x-2)^2+y^2=4$
$2r_2=\sqrt{9+36}+2=\sqrt{45}+2$
$2r_1=\sqrt{45}-2$
$\therefore 4r_1r_2=45-4=41 \Rightarrow r_1r_2=\frac{41}{4}$.

430. Distinct numbers of the form $\frac{p}{q}$ are 36 minus the number of numbers where greatest common divisor of p and q is more than 1. Number of such numbers is 23.

431. $$A^2=\begin{bmatrix}1&0\\k&1\end{bmatrix}\begin{bmatrix}1&0\\k&1\end{bmatrix}=\begin{bmatrix}1&0\\2k&1\end{bmatrix}$$
$$A^3=\begin{bmatrix}1&0\\2k&1\end{bmatrix}\begin{bmatrix}1&0\\k&1\end{bmatrix}=\begin{bmatrix}1&0\\3k&1\end{bmatrix}$$
$$A^{100}=\begin{bmatrix}1&0\\100k&1\end{bmatrix}$$
So, $\lambda=100$

432. $\left(4^{\log_{10} x}\right)^2 - \left(4^{\log_{10} x}\right) - 2 = 0$.

Let $4^{\log_{10} x} = t$

$t^2 - t - 2 = 0$

$(t - 2)(t + 1) = 0$

$4^{\log_{10} x} = 2$

$\log_{10} x = \log_4 2$

$x = 10^{\log_4 2}$

Now, $a^{\log_b c} = 10^{\log_4 2}$

$a = 10, b = 4, c = 2$

$a + b + c = 16$.

433. Let $x = \sqrt{\log_a \sqrt[4]{ab} + \log_b \sqrt[4]{ab}} = \sqrt{\frac{1}{4}(\log_a (ab) + \log_b (ab))} = \sqrt{\frac{1}{4}(2 + \log_a b + \log_b a)}$

$= \frac{1}{2}\left(\sqrt{2 + \log_a b + \frac{1}{\log_a b}}\right) = \frac{1}{2}\left(\sqrt{|\log_a b|} + \frac{1}{\sqrt{|\log_a b|}}\right)$

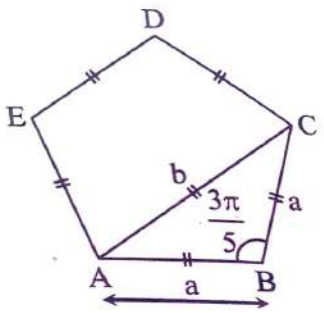
Similarly, $y = \sqrt{\log_a \sqrt[4]{\frac{b}{a}} + \log_b \sqrt[4]{\frac{a}{b}}} = \frac{\left|\sqrt{|\log_a b|} - \frac{1}{\sqrt{|\log_a b|}}\right|}{2}$

$\because\ b \geq a > 1$

$\therefore\ x - y = \frac{1}{2}\left\{\sqrt{|\log_a b|} + \frac{1}{\sqrt{|\log_a b|}} - \sqrt{|\log_a b|} + \frac{1}{\sqrt{|\log_a b|}}\right\};\ x - y = \frac{1}{\sqrt{|\log_a b|}}$

$\therefore\ 2^{(x-y)\sqrt{\log_a b}} = 2$

434. Using consine rule, we get $\cos 108^\circ = \frac{2a^2 - b^2}{2a^2} = \left(1 - \frac{b^2}{2a^2}\right)$



$\Rightarrow \frac{b^2}{a^2} = 2\left(1 - \cos\frac{3\pi}{5}\right) = 2\left(1 + \frac{\sqrt{5} - 1}{4}\right)$

$= 2\left(\frac{3 + \sqrt{5}}{4}\right) = \frac{3 + \sqrt{5}}{2}$

So, $\left(\frac{a^2}{b^2} + \frac{b^2}{a^2}\right) = \frac{2}{3 + \sqrt{5}} + \frac{3 + \sqrt{5}}{2}$

$= \frac{2\left(3 - \sqrt{5}\right)}{4} + \frac{3 + \sqrt{5}}{2} = \left(\frac{3 - \sqrt{5}}{2}\right) + \left(\frac{3 + \sqrt{5}}{2}\right) = 3$

435. Given, $AM - MA = I$

We can write $AM - MA = AA^{-1} \Rightarrow AM = MA + AA^{-1} \Rightarrow AM = A(M + A^{-1})$

$\Rightarrow \det(AM) = \det\{A(M + A^{-1})\}$

$\Rightarrow \det(A)\det(M) = \det(A)\det(M + A^{-1})$

$\Rightarrow \det(M) = \det(M + A^{-1})$ ..... (1)

Similarly, $\det(M) = \det(M - A^{-1})$ ..... (2)

From (1) and (2), we get

$\det(M - A^{-1}) = \det(M + A^{-1})$ ..... (3)

Now, since M describes the reflection of p(x, y) about the line $y = \left(2 - \sqrt{3}\right)x$

$\therefore M = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$ where $\theta = 15^\circ$

$\therefore M = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix}$

From equation (3), we get $\det(M - A^{-1}) = \det(M + A^{-1})$

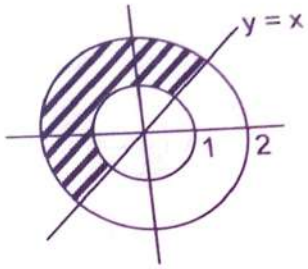
Since A^{-1} is a symmetrics matrix with entries x and y

$\therefore A^{-1} = \begin{bmatrix} x & y \\ y & x \end{bmatrix}$

Using relation (3), we get $x = 2$ and $y = 0$

436. $k = \frac{\pi(2^2 - 1^2)}{2} = \frac{3\pi}{2}$

$\int_0^{\pi} \{\sin x\}\, dx = \int_0^{\pi} \sin x dx = 2$



437-438. Let, $f(x) = Ax^3 + Bx^2 + Cx + D$

Since $(x - 1)^2$ is a factor of $f(x) + 2 \Rightarrow f(1) + 2 = 0$

$\Rightarrow A + B + C + D + 2 = 0$ ..... (1)

Also $(x + 1)^2$ is a factor of $f(x) - 2 \Rightarrow f(-1) - 2 = 0$

$\Rightarrow -A + B - C + D - 2 = 0$ ..... (2)

From equation (1), we get $D = -A - B - C - 2$

$\therefore f(x) = Ax^3 + Bx^2 + Cx - A - B - C - 2$

$\Rightarrow f(x) + 2 = A(x^3 - 1) + B(x^2 - 1) + C(x - 1)$

Since $(x - 1)^2$ is a factor of $f(x) + 2$

$\Rightarrow f'(1) = 0$

$\Rightarrow 3A + 2B + C = 0$ ..... (3)

Similarly $(x + 1)^2$ is a factor $f(x) - 2$

$\Rightarrow f'(-1) = 0$

$\Rightarrow 3A - 2B + C = 0$ ..... (4)

From equation (1), (2), (3) and (4), we get

$A = 1, B = D = 0, C = -3$

$\Rightarrow f(x) = x^3 - 3x$

439-440. $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6$

Appling the condition, we get $f(x) = 2x^4 - \frac{12}{5}x^5 + \frac{2}{3}x^6$

441-442. Let $\sqrt{x} = \tan^2\theta \Rightarrow dx = 4\tan^3\theta \sec^2\theta\, d\theta$

$\sqrt{1 + \sqrt{x}} = \sec\theta$

(I) $I = \int \tan\left(2\tan^{-1}\sqrt{\frac{\sec\theta - 1}{\sec\theta + 1}}\right)dx = \int \tan\left(2\tan^{-1}\left(\tan\frac{\theta}{2}\right)\right) = \int \tan\theta\, dx$

$= \int \tan\theta \cdot 4\tan^3\theta\sec^2\theta d\theta = \frac{4}{5}\tan^5\theta + C = \frac{4}{5}\left(x^{\frac{5}{4}}\right) + C$

(II) $\int \cot\left(2\tan^{-1}\sqrt{\frac{\sec\theta-\tan\theta}{\sec\theta+\tan\theta}}\right)dx$

$= \int \cot\left(2\tan^{-1}\left(\frac{1-\sin\theta}{\sin\theta}\right)\right)dx = \int \cot\left(2\tan^{-1}\tan\left(\frac{\pi}{4}-\frac{\theta}{2}\right)\right)dx$

$= \int \tan\theta\,dx = \int \tan\theta\cdot 4\tan^3\theta\sec^2\theta\,d\theta = \frac{4}{5}x^{\frac{5}{4}}+C$

(III) $\frac{1}{2}\sin^{-1}\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right)dx = \frac{\pi}{4}-\theta$

$I = \int \frac{1-\tan\left(\frac{\pi}{4}-\theta\right)}{1+\tan\left(\frac{\pi}{4}-\theta\right)}\cdot 4\tan^3\theta\sec^2\theta\,d\theta = \int \tan\left(\frac{\pi}{4}-\left(\frac{\pi}{4}-\theta\right)\right)\cdot 4\tan^3\theta\sec^2\theta d\theta$

$= \int 4\tan^4\theta\sec^2\theta d\theta = \frac{4}{5}x^{\frac{5}{4}}+C$

(IV) $I = \int \tan^2\theta\cdot\tan\left(2\tan^{-1}\left(\frac{\sqrt{\sec\theta+1}-\sqrt{\sec\theta-1}}{\sqrt{\sec\theta+1}+\sqrt{\sec\theta-1}}\right)\right)dx$

$= \int \tan^2\theta\cdot\tan\left(\frac{\pi}{4}-\theta\right)dx = \int \tan\theta\cdot 4\tan^3\theta\sec^2\theta d\theta = \frac{\pi}{4}x^{\frac{5}{4}}+C$

443-444. $t = 1, 2$

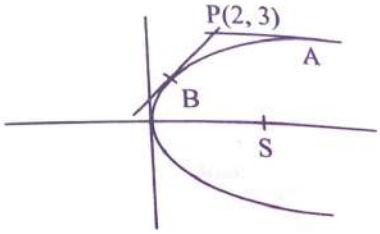
$A \equiv (4, 4), B = (1, 2)$

$\Delta(PAB) = \frac{1}{2}$ and $SA = 5, SB = 2$

$\therefore SA\cdot SB = 10$

or using property $SA\cdot SB = (SP)^2$

and image of focus (1, 0) about any of the tangent lies on the directrix



445-446. (I) 8!

(II) $3\times 6!\times {}^{11}C_6$

(III) ${}^6C_3\cdot 8!$

(IV) $9! + {}^2C_1\cdot 2!\cdot\frac{9!}{2!} + 2!\cdot\frac{9!}{3!}$

447-448. Units digit will be 1, 3, 7 or 9 when all chosen numbers have 1, 3, 7 or 9 at units place, so probability is

$\left(\frac{4}{10}\right)^n$

$P(2, 4, 6 \text{ or } 8)$

$= P(k \neq 0, 5) - P(1, 3, 7 \text{ or } 9)$

$= \left(\frac{8}{10}\right)^n - \left(\frac{4}{10}\right)^n$

$P(5) = P(1, 3, 5, 7 \text{ or } 9) - P(1, 3, 7 \text{ or } 9)$

$= \frac{5^n - 4^n}{10^n}$

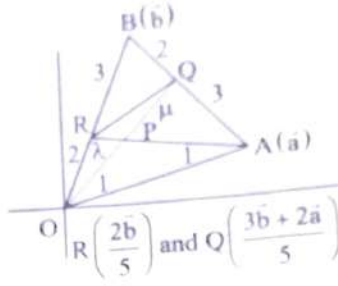
$P(0) = 1 - P(1, 3, 7 \text{ or } 9) - P(2, 4, 6 \text{ or } 8) - P(5)$

449-450. For position vector P

$$\frac{\left(\frac{3\vec{b}+2\vec{a}}{5}\right)}{\mu+1}=\frac{\lambda\vec{a}+\frac{2\vec{b}}{5}}{\lambda+1}$$

$$\Rightarrow \frac{\lambda}{\lambda+1}=\frac{2\mu}{5(\mu+1)} \text{ and } \frac{3\mu}{5(\mu+1)}=\frac{2}{5(\lambda+1)}$$

$$\Rightarrow \mu=\frac{9}{10} \text{ and } \lambda=\frac{4}{15}$$



$$\text{Ar}(\Delta OPA)=\frac{1}{2}|\overrightarrow{OP}\times\overrightarrow{OA}|=\frac{1}{2}\left[\frac{2}{19}(3\vec{b}+2\vec{a})\times\vec{a}\right]=\frac{3}{19}|\vec{b}\times\vec{a}|=3$$

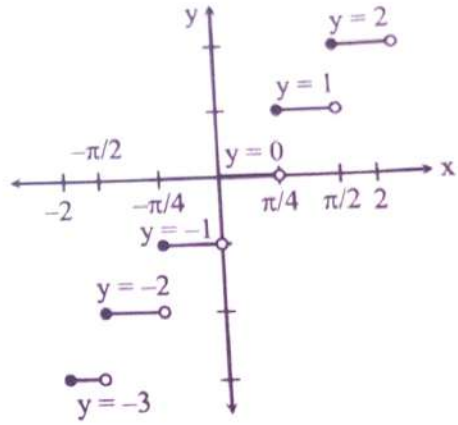
$$\text{Ar}(PQBR)=\frac{1}{2}|\overrightarrow{OQ}\times\overrightarrow{OB}-\overrightarrow{OP}\times\overrightarrow{OR}|=\frac{1}{2}\left[\left(\frac{3\vec{b}+2\vec{a}}{5}\right)\times\vec{b}-\frac{2}{19}(3\vec{b}+2\vec{a})\times\frac{2\vec{b}}{5}\right]=\frac{3}{19}|\vec{a}\times\vec{b}|$$

451. (A) $f_1(x)=\left[\frac{4x}{\pi}\right]\text{sgn}\left(x^2-x+1\right)$

Clearly, x^2-x+1

$=\left(x-\frac{1}{2}\right)^2+\frac{3}{4}$ which is $\geq\frac{3}{4}\Rightarrow \text{sgn}(x^2-x+1)=1$

$$\Rightarrow f_1(x)=\left[\frac{4x}{\pi}\right]=\begin{cases}-3 & ; \ -2\leq x<-\frac{\pi}{2}\\ -2 & ; \ -\frac{\pi}{2}\leq x<-\frac{\pi}{4}\\ -1 & ; \ -\frac{\pi}{4}\leq x<0\\ 0 & ; \ 0\leq x<\frac{\pi}{4}\\ 1 & ; \ \frac{\pi}{4}\leq x<\frac{\pi}{2}\\ 2 & ; \ \frac{\pi}{2}\leq x\leq 2\end{cases}$$



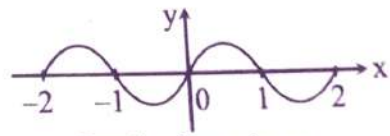
Clearly f_1 is discontinuous and hence non-derivable at 5 points viz. $x=-\frac{\pi}{2},-\frac{\pi}{4},0,\frac{\pi}{4},\frac{\pi}{2}$

Also $R_{f_1}=\{-3,-2,-1,0,1,2\}$

(B) $f_2(x)=\cos^{-1}\left(\text{sgn}\left(\cos\frac{2x-1}{2},\pi\right)\right)$;

$\because \cos\left(\frac{2x-1}{2}\right)\pi=\cos\left(x\pi-\frac{\pi}{2}\right)=\sin\pi x$

$\Rightarrow f_2(x)=\cos^{-1}(\text{sgn}(\sin\pi x))$

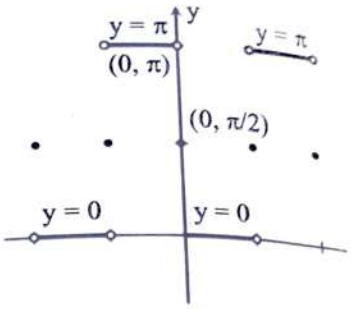


Graph of $y=\sin\pi x$

$$\therefore \begin{cases} 0 & ; \ x=-2 \\ 1 & ; \ -2<x<-1 \\ 0 & ; \ x=-1 \\ -1 & ; \ -1<x<0 \\ 0 & ; \ x=0 \\ 1 & ; \ 0<x<1 \\ 0 & ; \ x=1 \\ -1 & ; \ 1<x<2 \\ 0 & ; \ x=2 \end{cases} \Rightarrow f_2(x)=\cos^{-1}(\text{sgn}(\sin\pi x))=\begin{cases} \frac{\pi}{2} & ; \ x=-2 \\ 0 & ; \ -2<x<-1 \\ \frac{\pi}{2} & ; \ x=-1 \\ \pi & ; \ -1<x<0 \\ \frac{\pi}{2} & ; \ x=0 \\ 0 & ; \ 0<x<1 \\ \frac{\pi}{2} & ; \ x=1 \\ \pi & ; \ 1<x<2 \\ \frac{\pi}{2} & ; \ x=2 \end{cases}$$

Clearly, $f_2(x)$ is distinuous and non-derivable at 5 points viz. $x=-2,-1,0,1,2$

Also, $R_{f_2}=\left\{0,\frac{\pi}{2},\pi\right\}$

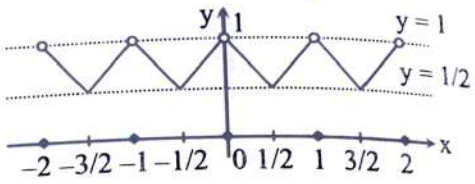


(C) $f_3(x)=\max(\{x\},\{-x\})$

Clearly, f_3 is discontinuous at 5 points viz. $x=-2,-1,0,1,2$ and non-derivable at 9 point viz.

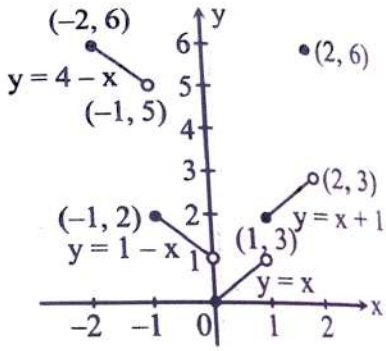
$$x=-2,-\frac{3}{2},-1,-\frac{1}{2},0,\frac{1}{2},1,\frac{3}{2},2$$

Also $R_{f_3}=\{0\}\cup\left[\frac{1}{2},1\right)$



(D) $f_4(x)=\sqrt{x^2}+[x]^2$

$$=|x|+[x]^2=\begin{cases} -x+4 & ; \ -2\le x-1 \\ -x+1 & ; \ -1\le x<0 \\ x & ; \ 0\le x<1 \\ x+1 & ; \ 1\le x<2 \\ 6 & ; \ x=2 \end{cases}$$



Clearly, f_4 is discontinuous and non-derivable at 4 points viz. $x=-1,0,1,2$

Also, $f(-2)=f(2)=6$

but $f\left(-\frac{1}{2}\right)=\frac{3}{2}$

and $f\left(\frac{1}{2}\right)=\frac{1}{2}$

$\Rightarrow f_4$ is many one but not even

452. Let number of numbers before a_1 is x_0 and number of numbers between a_{i+1} and $a_i = x_i$ for $i = 1, 2, 3, 4$ and number of number after $a_5 = x_5$

Now consider the equation: $x_0 + x_1 + x_2 + x_3 + x_4 + x_5 = 13$

(A) if $a_{i+1} - a_i \geq 2 \; \forall \; i = 1, 2, 3, 4$, then number of such selection

= number of integral solutions of $x_0 + x_1 + x_2 + x_3 + x_4 + x_5 = 13$ under the restriction $x_0, x_5 \geq 0$ $x_1, x_2, x_3, x_4 \geq 1$

= number of non-negative integral solutions of $\sum_{i=0}^{5} x_i = 9 = {}^{9+6-1}C_5 = {}^{14}C_5 = 2002$

(B) is equal to integral solutions of $\sum_{i=0}^{n} x_i = 13$, $x_0, x_5 \geq 0$, $x_1, x_2, x_3, x_4 \geq 2$

= non negative integral solution of $\sum_{i=0}^{n} x_i = 5$

$= {}^{5+6-1}C_5 = {}^{10}C_5 = 252$

(C) is equal to integral solutions of $\sum_{i=0}^{n} x_i = 13$, $x_0, x_5 \geq 0$, $x_1 \geq 1$, $x_2 \geq 2$, $x_3 \geq 3$, $x_4 \geq 4$

= non negative integral solutions of $\sum_{i=0}^{n} x_i = 3 = {}^{3+6-1}C_5 = {}^{8}C_5 = 56$

(D) is equal to integral solutions of $\sum_{i=0}^{n} x_i = 13 \Rightarrow x_1, x_5 \geq 0$, $x_1, x_2 \geq 1$, $x_3, x_4 \geq 2$

= non negative integral solutions of $\sum_{i=0}^{n} x_i = 7 = {}^{7+6-1}C_5 = {}^{12}C_5 = 792$

453. $\angle DGC = \angle DPC = B$ and $CD \perp PG \Rightarrow PD = DG$

Clearly incentre (10, 15) of ΔDEF is orthocentre of ΔABC and mirror image of orthocentre about any side of triangle, lies on its circumcircle

$\therefore$ Mirror image of (10, 15) about side BC i.e. $x + 3y - 35 = 0$ is (6, 3)

$\therefore$ Radius of circumcircle of $\Delta ABC = 25$ units

$\therefore$ Equation of circumcircle is $(x-30)^2 + (y-10)^2 = 25^2$ ... (1)

As equation of line BC is $x + 3y - 35 = 0$ ... (2)

And equation of line DP is $3x - y - 15 = 0$ ... (3)

P is orthocentre.

Solving (1), (2) and (3), we get points A, B and C as (15, 30); (50, – 5) and (5, 10).

$\therefore$ $AP = 5\sqrt{10}$ units.

$$\sin B \sin C = \frac{bc}{4R^2} = \frac{35\sqrt{2} \times 10\sqrt{5}}{4 \times 25 \times 25} = \frac{7}{50}\sqrt{10} \text{ units.}$$

454. Let E be the event of getting 2 head and 1 tail

$P(A) = P(B)$

$= P(C) = \frac{1}{3}$

Using Bayes' theorem $P(A/E) = \frac{9}{25}$,

$P(B/E) = \frac{32}{75}$

455. $R = P^T Q^K P = P^T (PAP^T)^K P = P^T PAP^T PAP^T \ldots\ldots PAP^T P$ (n times)

$= A^K$ as $PP^T = 1$ as P is orthogonal

$$A^K = \begin{bmatrix} 1+2k & -4k \\ k & 1-2k \end{bmatrix}$$

$$T = P^T S^K P = B^K$$

$$B^K = \begin{bmatrix} a^k & \frac{6(a^k-1)}{a-1} \\ 0 & 1 \end{bmatrix}$$

456. (A) Let $g(x) = f(x)\, e^{\cos x}$

$\because$ $g(x)$ is a decreasing function

$\therefore$ $g(x) \le g(0) = 0$ i.e. $f(x)\, e^{\cos x} \le 0$ but $f(x) \ge 0$, $e^{\cos x} > 0 \Rightarrow f(x) = 0$

$\therefore$ $f\left(\frac{3\pi}{2}\right) = 0.$

(B) $(a-3)x^2 + 12x + 6 + a - (\sin^{-1}\sin 100 + \cos^{-1}\cos 100) < 0 \Rightarrow a < 3,\ D < 0$

$144 - 4(a+6)(a-3) < 0 \Rightarrow a < -9$ or $a > 6$.

(C) $P\left(\frac{B}{A \cup B^C}\right) = \frac{P(A \cap B)}{P(A \cup B^C)} = \frac{1}{5}.$

(D) $\cos^{-1}(4x^3 - 3x) - \frac{\pi}{2} \ge 0 \Rightarrow -1 \le 4x^3 - 3x \le 0$ i.e. $x \in \left[-4, -\frac{\sqrt{3}}{2}\right] \cup \left[0, \frac{\sqrt{3}}{2}\right]$

also $\frac{(2x+3)!}{\sqrt{x+2}}$ is defined $\forall\ x > -2$ and $(2x+3) \in W$

$x = -1, 0, \frac{1}{2}.$

457. (A) $\frac{x}{2} + \tan x = \frac{\pi}{4}$

From graph, the equation has 3 solutions in $[-\pi, \pi]$

(B) $\sin^{-1}|x^2 - 1| + \cos^{-1}|2x^2 - 5| = \frac{\pi}{2}$

$\Rightarrow |x^2 - 1| = |2x^2 - 5|$

$\Rightarrow x^2 = 2 \Rightarrow x = \pm\sqrt{2}$ (Two solutions)

(C) $x^4 - 2x^2 \sin^2 \frac{\pi x}{2} + 1 = 0$

$\Rightarrow \left(x^2 - \sin^2 \frac{\pi x}{2}\right)^2 + 1 - \sin^4 \frac{\pi x}{2} = 0$

$\Rightarrow x = (2n+1),\ n \in I$ and $x^2 = \sin^2 \frac{\pi x}{2} = 1$

$\Rightarrow x = \pm 1$ is the solution

(D) Let $y = x^2 + 2x + 2\sec^2 \pi x + \tan^2 \pi x$

$\Rightarrow y = (x+1)^2 + (2\sec^2 \pi x - 1) + \tan^2 \pi x > 0 \Rightarrow$ no solution

458. (A) $x = 9;\ x = 1$

$y = 8;\ y = 2$

$z = 5,\ z = 3$

Possible edges are 8, 6, 2

(B) $(3k+1, 4k+1, 12k+1)$ is the point

So, $9k^2 + 16k^2 + 144k^2 = 169$

$\Rightarrow k = 1$ and point is $(4, 5, 13)$

(C) $\hat{a} = \pm\frac{\hat{n}_1 \times \hat{n}_2}{|\hat{n}_1 \times \hat{n}_2|} = \pm\left(\frac{2\hat{i} + 5\hat{j} + 3\hat{k}}{\sqrt{38}}\right)$

(D) $(\hat{i}+\hat{j})\times(\hat{j}+\hat{k}) = \hat{i}-\hat{j}\times\hat{k}$

$\Rightarrow$ Unit vector perpendicular as to the plane of similarly other two unit vector are

$\hat{i}+\hat{j}$ and $\hat{j}+\hat{k}$ is $\frac{1}{\sqrt{3}}(\hat{i}-\hat{j}+\hat{k})$

$\frac{1}{\sqrt{3}}(\hat{i}+\hat{j}-\hat{k})$ and $\frac{1}{\sqrt{3}}(-\hat{i}+\hat{j}+\hat{k})$

$$v=[\hat{n}_1\ \hat{n}_2\ \hat{n}_3]=\frac{1}{3\sqrt{3}}\begin{vmatrix}1 & -1 & 1\\ 1 & 1 & -1\\ -1 & 1 & 1\end{vmatrix}=\frac{4}{3\sqrt{3}}$$

459. (A) $\left(x+\frac{1}{x}+x^2+\frac{1}{x^2}\right)^9 = a_0x^{-18}+a_1x^{-17}+.....+a_{36}x^{18}$

Put $x = 1$ and -1, and odd

We get, $(a_0 + a_2 + ... + a_{36}) = (2^8 + 2^{17})$

(B) $(1 + x + x^2 + x^3)^{13} = a_0 + a_1x + a_2x^2 + ... + a_{39}x^{39}$

Put fourth roots of unity, 1, α, α^2, α^3 and add together

We get, $(a_0 + a_4 + a_8 + a_{12} + ... + a_{36}) = \frac{2^{26}}{4} = 2^{24}$

(C) ${}^{20}C_2 + 2.{}^{20}C_3 + 3.{}^{20}C_4 + ...+ 19.{}^{20}C_{20}$

Let $(1 + x)^{20} = {}^{20}C_0 + {}^{20}C_1x + {}^{20}C_2x^2 + ... + {}^{20}C_{20}x^{20}$

divide by x, and differentiae both sides, $({}^{20}C_2 + 2.{}^{20}C_3 + 19.{}^{20}C_{20}) = (18 \times 2^{19}) + 1$

(D) ${}^{20}C_1 + 2.{}^{20}C_2 + 3.{}^{20}C_3 + 4.{}^{20}C_4 +.....+ 10.{}^{20}C_{10}$

$$=\sum_{r=0}^{10} r\ {}^{20}C_r=\sum_{r=1}^{10}20\ {}^{19}C_{r-1}$$

$= 20(2^{18} + {}^{19}C_{10})$

460. (A) $\frac{3x+x+z/2}{x+z-x}+\frac{3x+x+z/2}{x+z-z}=\frac{7}{2}\left(\frac{x}{z}+\frac{z}{x}\right)+1$

$\frac{x}{z}+\frac{z}{x}\geq 2.$

Expression is $\geq 7 + 1 = 8$

(B) $x_1+x_2+x_2=1$

$x_1+x_3=2x_2$

$\Rightarrow x_2=\frac{1}{3}\Rightarrow\frac{1}{27}-\frac{1}{9}+\frac{\alpha}{3}+\beta=0$

$\Rightarrow 1-3+9\alpha+27\beta=9\alpha+27\beta=2$

(C) Let the roots be $\alpha - 3\beta$, $\alpha - \beta$, $\alpha + \beta$, $\alpha + 3\beta$

Sum of roots, $4\alpha = 16$

$\Rightarrow \alpha = 4$

Also, $\beta^2 = 1$ or $\frac{|5|}{9}.\ \beta^2 = 1$

$\Rightarrow \beta = \pm 1$

Roots are 1, 3, 5, 7, or 7, 5, 3, 1

$\therefore$ d = 2 or d = –2

$\therefore 2d^2 + 4 = 12$

(D) $a_1 + a_3 + a_5 + ... + a_{2n-1} = a_4 + a_6 + a_8 + ... + a_{2n+2} = r^3(a_1 + a_3 + a_5 + ... + a_{2n-1})$

$r^3 = 1$

$\Rightarrow r = 1, \omega, \omega^2$

461.-463. (I) $f(x) = 7\cos^2 x + 6\sin x\cos x - \sin^2 x$

$= 7\left(\frac{1+\cos 2x}{2}\right) + 3\sin 2x - \left(\frac{1-\cos 2x}{2}\right)$

$= 3\sin 2x + 4\cos 2x + 3$

$\Rightarrow -2 \le f(x) \le 8$

(II) $f(x) = 5\cos x + 3\cos\left(x + \frac{\pi}{2}\right) + 3$

$= \frac{13}{2}\cos x - \frac{3\sqrt{3}}{2}\sin x + 3 \Rightarrow -4 \le f(x) \le 10$

(III) $f(x) = (\sin^2 x + \cos^2 x)^3 - 3\sin^2 x\cos^2 x\,(\sin^2 x + \cos^2 x)$

$= 1 - 3\sin^2 x\cos^2 x$

$= 1 - \frac{3}{4}\sin^2 2x$

$\Rightarrow 0 \le \sin^2 x \le 1$

$\Rightarrow \frac{1}{4} \le f(x) \le 1$

(IV) $f(x) = \frac{1}{4}\cos 3x$

$\Rightarrow -\frac{1}{4} \le f(x) \le \frac{1}{4}$

464.-466. If $\left|z \pm \frac{a}{z}\right| = b$ where $a > 0$ and $b > 0$, then $\frac{-b+\sqrt{b^2+4a}}{2} \le |z| \le \frac{b+\sqrt{b^2+4a}}{2}$

(I) $a = 1$ and $b = 2 \Rightarrow -1+\sqrt{2} \le |z| \le 1+\sqrt{2}$

(II) $a = 2$ and $b = 4 \Rightarrow -2+\sqrt{6} \le |z| \le 2+\sqrt{6}$

(III) $a = 3$ and $b = 6 \Rightarrow -3+2\sqrt{3} \le |z| \le 3+2\sqrt{3}$

(IV) $a = 8, b = 7 \Rightarrow 1 \le |z| \le 8$

467. $C_1C_2 > r_1 + r_2$ number of common tangent is 4 for internal tangent

Let point divides centers in ration $r_1 : r_2$ then point will be $\left(\frac{8}{3}, 0\right)$

Tangent drawn from point will be $3x - 4y - 8 = 0$

468. For circle $x^2 + y^2 - 2x = 0$ and line $3x - 4y - 8 = 0$ point of contact $\left(\frac{8}{5}, -\frac{4}{5}\right)$

Circle $5(x^2 + y^2) - x - 12y - 24 = 0$ touches at same point

469. $|r_1 - r_2| < C_1C_2 < r_1 + r_2$

470. Equation of normal at $P(x_1, y_1)$ is $(a^2 - b^2)x_1y_1 + b^2yx_1 - a^2xy_1 = 0$

But the point T(h, k) lies on it $\Rightarrow (a^2 - b^2)x_1y_1 + b^2kx_1 - a^2hy_1 = 0$

Similarly $(a^2 - b^2)\,x_2y_2 + b^2kx_2 - a^2hy_2 = 0$

Hence, P, Q, R, S satisfy the curve $(a^2 - b^2)xy + b^2kx - a^2hy = 0$

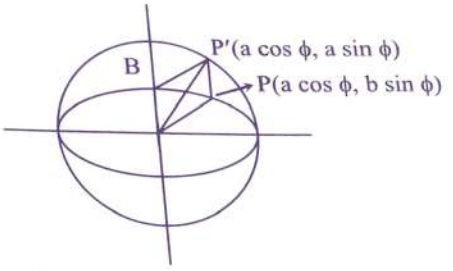
471.-472. Area of $\Delta CBP'$

$= \frac{1}{2} CB \times (\perp \text{ distance of } P' \text{ from minor area})$

Area of $\Delta CPP'$

$= \frac{1}{2} PP' \times (\perp \text{ distance of } PP' \text{ from minor area})$

$= \frac{1}{2}(a-b)|\sin\phi||\cos\phi| \quad \therefore \quad \frac{\Delta CBP'}{\Delta CPP'} = \frac{b}{a-b}|\text{cosec}\,\phi|$



473. $x^2 + y^2 - 2ay = 0 \Rightarrow \frac{dy}{dx} = \frac{2xy}{x^2 - y^2}$

For orthogonal trajectory $\Rightarrow \frac{dy}{dx} = \frac{y^2 - x^2}{2xy} \Rightarrow x^2 + y^2 = cx = 0$

474. $y^2 = 4(x - a) \Rightarrow \frac{dy}{dx} = \frac{2}{y}$

For orthogonal trajectory $\Rightarrow \frac{dy}{dx} = -\frac{y}{2} \Rightarrow y = c \cdot e^{-x/2}$

475. $\cos y = a \cdot e^{-x} \Rightarrow \frac{dy}{dx} = \cot y$

For orthogonal trajectory $\frac{dy}{dx} = -\tan y \Rightarrow \sin y = c \cdot e^{-x}$

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